

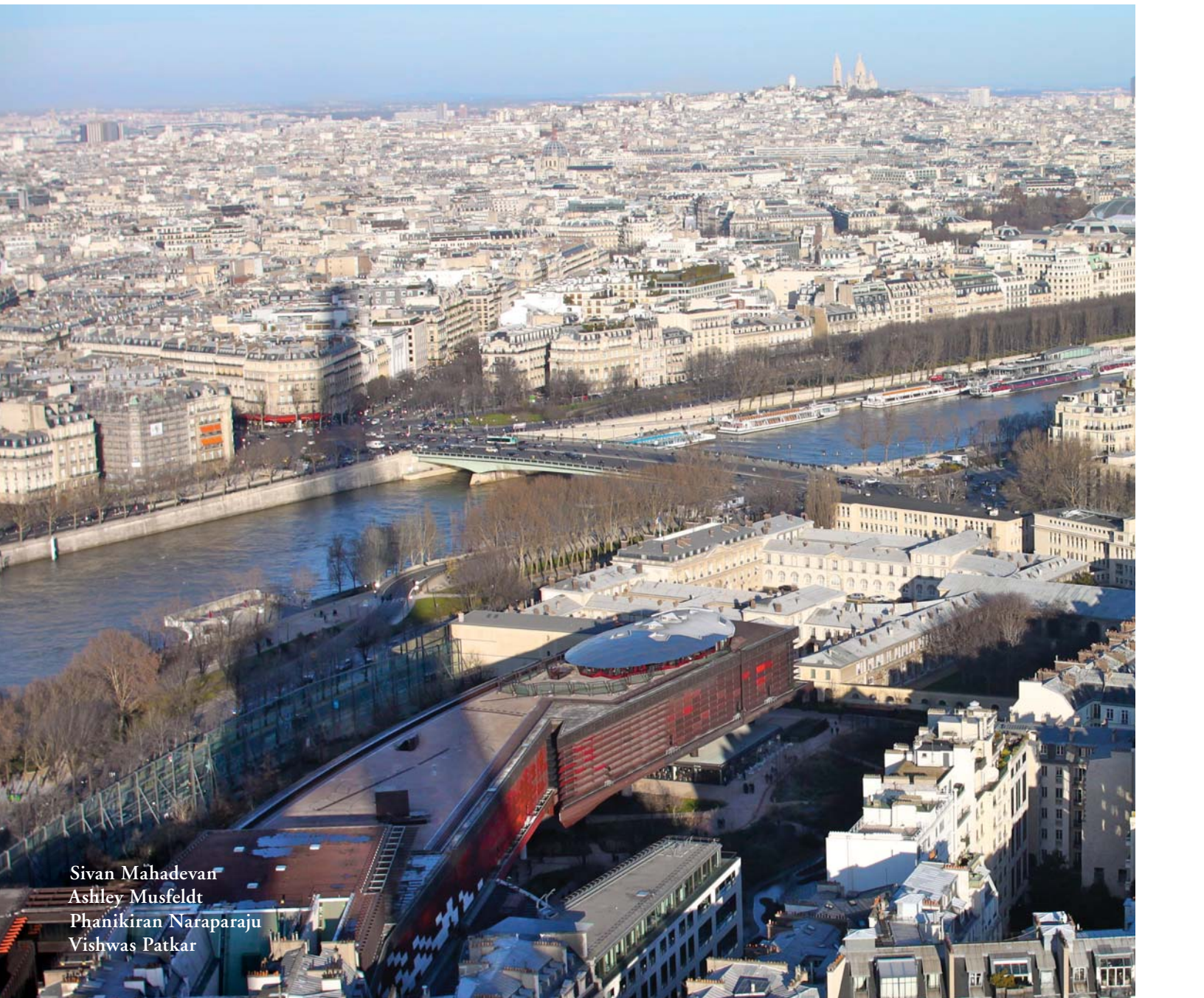
RESEARCH

Morgan Stanley

Credit Derivatives Insights

SIXTH EDITION 2012

*Handbook of Credit
Derivatives and Structured
Credit Strategies*



Sivan Mahadevan
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CREDIT DERIVATIVES INSIGHTS SIXTH EDITION—PROGRESS IN A NEW WORLD

We published the first edition of our credit derivatives and structured credit handbooks in 2004 under separate covers, with the goal of covering these markets both functionally and strategically. Our handbooks evolved over the years as the markets changed, innovation succeeded or failed and standardization progressed. In the past several years, we have witnessed an enormous amount of all three experiences. Consequently, we are taking a fresh look at a new market, with an updated handbook that encompasses the entire credit derivatives market; from single name CDS to portfolio forms of credit derivatives.

Our sixth edition includes 40 chapters divided into six sections. The first section provides market primers for credit default swaps, CDS and bond options, tranches, baskets, and concepts like recovery risk and forward credit spreads. Next, we discuss market themes, including systematic and tail risk credit hedging strategies, credit valuation in a low yield world, bank credit challenges, and the performance of tranches over time. In the third section, we focus on valuation including interest rate, credit curve and basis dynamics, as well as our newly developed macro credit models. In the fourth section, we focus specifically on credit derivatives usage by bank loan hedgers, the market's original and most important buyers of protection, covering many Basel regulation related themes that drive these flows. In the fifth section, we discuss the market for customized credit solutions, including how the market evolved prior to the financial crisis, its performance since then, and the evolution of customization today. In the sixth section, we focus on newer areas of credit derivatives markets including developed market sovereigns and U.S. municipalities. Finally, the glossary provides definitions for over 200 terms used in the credit derivatives markets.

We hope Morgan Stanley clients find this handbook useful, and we welcome any feedback that will help us improve future editions.

Morgan Stanley

SIXTH EDITION

2012

Credit Derivatives Insights

Handbook of Credit Derivatives and
Structured Credit Strategies

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March 29, 2012

INTRODUCTION

1

SECTION A. GETTING STARTED: INSTRUMENTS AND PRIMERS

1	THE CDS LIFECYCLE – A MARKET PRIMER	6
2	THE CREDIT VOLATILITY CULTURE – AN OPTIONS PRIMER	19
3	TRANCHE PRIMER – A TALE OF TWO MARKETS	32
4	BASKETS GO BACK TO THE FUTURE	48
5	WHAT HAPPENS IN A DEFAULT?	52
6	BASIS BASICS IN A NORMALIZED WORLD	64
7	TRADING RECOVERY RISK – THE MISSING LINK	69
8	FLOATING A NEW IDEA – CMCDS	72
9	CORPORATE BOND OPTIONS – FROM MACRO TO MICRO	75

SECTION B. MARKET THEMES

10	TRACKING THE CREDIT CYCLE – A HISTORY OF TRANCHE RETURNS	82
11	THE CREDIT HEDGER'S MANUAL	92
12	UNDERSTANDING CREDIT IN A LOW YIELD WORLD	101
13	US BANK CREDIT – REGULATORY AND MARKET DYNAMICS	111

SECTION C. VALUATION

14	LIBOR METRICS	122
15	RATES AND THE CREDIT CULTURE	124
16	COLLATERAL – FULL FAITH OR CREDIT?	127
17	A CREDIT-RATES BOOSTER SHOT	130
18	LEARNING CURVES	134
19	TRADING THE CDS BASIS TODAY	141
20	INTRODUCING THE CDX IG MACRO CREDIT MODEL	150
21	INTRODUCING THE ITRAXX MAIN MACRO CREDIT MODEL	154
22	FALLING FORWARD	159
23	TWO TALES ABOUT TWO TAILS	165
24	TRADING TAILS IN CREDIT-EQUITY	170

SECTION D. CREDIT PORTFOLIO MANAGEMENT

25	BANK LOAN HEDGING TODAY – CPM PART I	178
26	BASEL II BASED HEDGING – CPM PART II	182
27	SECURITIZATION AS A SOLUTION – CPM PART IV	186
28	BANK LOAN HEDGING DYNAMICS – CPM UPDATE	190
29	BASEL UPDATE AND STRUCTURED CREDIT IMPLICATIONS	193

SECTION E. THE MARKET FOR CUSTOMIZED CREDIT

30	DISTRESS, IN AN UNLIKELY PLACE	198
31	GETTING A HANDLE ON DISTRESSED MEZZ	202

32	RESTRUCTURING RENAISSANCE	206
33	FROM DISTRESSED TO JUST STRESSED	209
34	DISTRESSED CSOS DECOUPLE AND STABILIZE	213
35	THE MARKET FOR CUSTOMIZED CREDIT SOLUTIONS	216

SECTION F. SOVEREIGNS AND MUNIS

36	MUNICIPAL CDS – THE NEXT STAGE	222
37	MUNI MANIA	232
38	MCDX: CREDIT RISK OR RISK PREMIUM?	236
39	MUNIS VS. CORPORATES IN RECESSION: CDS CONSIDERATIONS	240
40	SOVEREIGN CDS MARKETS – A CORPORATE PERSPECTIVE	243

SECTION G. GLOSSARY	250
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Introduction

Credit derivatives and structured credit instruments significantly reshaped the corporate credit markets over the past dozen years, and these derivatives markets have gone through their own cultural revolution over the past several years. The natural question to ask, and one that we are frequently asked to opine on, concerns the future direction of derivatives within the world of credit. We have given the topic a lot of thought both during and post the financial crisis, having shared those thoughts quite often via our research publications. Much has changed in the credit derivatives market in the past three years, and going forward we think usage will be a key area of change as well.

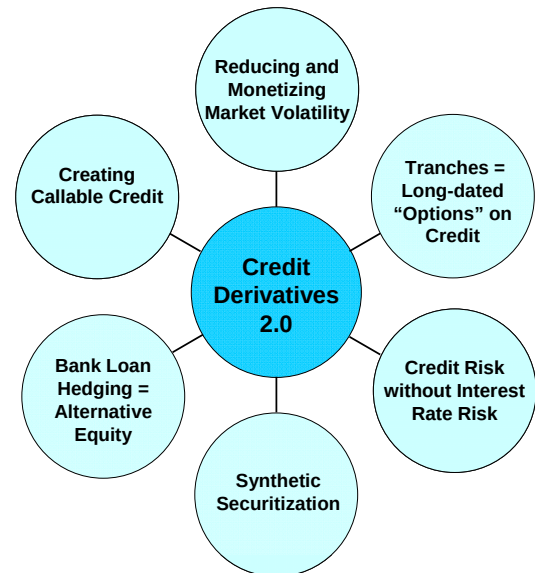
Credit derivatives referencing corporate credit, a vast and fast-growing market for much of the decade preceding the financial crisis, was exposed to daunting challenges during the crisis, including the lowest valuations on record for investment-grade credit, unprecedented market volatility, counterparty risk stemming from the failure of financial institutions, and a default cycle that matched the peaks of previous cycles. Corporate credit derivatives were in the center of the storm for the same reasons that investment-grade credit was; contagion from the housing and mortgage crisis that resulted in the worst US recession since the 1930s, with particular pain inflicted on financials.

Credit derivatives markets were in their teen years prior to the crisis: important to the market, serving a useful purpose and too big to ignore, but rapid physical growth did not imply that adulthood was around the corner. Trading processes, counterparty and margin standards, central clearing, and some form of automated trading are natural next steps, and courtesy of a variety of regulatory and legislative reforms, we are now in this transition process.

Three years ago, credit derivatives markets were indeed at a crossroads and under intense focus from the media and regulators. Ultimately though, the process of reform has resulted in a world where credit derivatives are indeed a necessary part of the solution rather than being the source of the problem. This is a significant change, and to borrow a term from the venerable world of the Internet that emerged about a decade ago, we are calling this next stage of the market Credit Derivatives 2.0. We expect that, much like the business of using the Internet, the business of using credit derivatives will grow stronger with this transformation, as we learn lessons from the past and build better standards and infrastructure for the future. Valuations have healed, a good amount of flow has returned, and much high-level regulation has been written (Basel 3 and Dodd-Frank), which, in our view sets the stage quite nicely for the rite of passage into adulthood.

exhibit 1

Credit Derivatives 2.0



Source: Morgan Stanley Research

IS INNOVATION OR REFORM A GOOD THING?

Credit derivatives were created to help protect lenders (mainly banks) from the credit risk of the corporations and sovereigns to whom they lend. But a market with only buyers (of protection) is by definition not a market, so the other side of the trade needed to be developed via those willing to take credit risk synthetically. A good balance between buyers and sellers as well as an equal playing field will be best for credit markets going forward as well.

The market requires innovators as well, without whom we will be stuck trying to implement tomorrow's ideas using yesterday's technology. In most industries, innovation is welcomed as a driver of growth, product improvement and even secular changes in products and services. Innovation is not guaranteed to succeed, and not all ideas are good ones, with the technology sector giving us distinct examples of both good and bad. And just because innovation has not had a 100% success rate, this is not a valid reason to dismiss it, as spinning the proverbial wheels is a necessary part of the process. Within credit, some forms of innovation have essentially represented convenient forms of leverage and regulatory arbitrage, which clearly got punished during the credit cycle, and the market can learn from that experience. But there are other ideas out there which are important to pursue, as an industry too comfortable with the status quo will not advance.

In 2009, the credit derivatives markets moved quickly to standardize the operational aspects of CDS, creating a world of fungible contracts and facilities to deal with credit events on a mass scale. This system has been tested and has largely proven successful. In 2010, financial regulation in the form of Basel-based capital rules and Dodd-Frank reform effectively legislated trading practices that formalize the regulation of counterparties, clearing and eventual electronic trading. Bank capital guidelines have been adjusted as well. Next, the market will move into implementation stage. This is a big step forward, and investors will need to get operationally up to speed.

Once we have much of this in place, we expect fertile ground for new credit opportunities given the benefits of standardization. Equities have always benefited from standardization in the form of exchange trading of massively liquid, fungible securities. There is nothing new here, but the explosive growth of portfolio trading within equities is something that has changed the structure of the market over the past decade, particularly in the US.

WHAT ARE THE ELEMENTS OF CREDIT DERIVATIVES 2.0?

There are, of course, many challenges ahead for credit derivatives. Much of the long-credit flow in structured credit prior to the financial crisis was with financial institutions themselves. Credit ratings and regulatory capital treatment were clearly motivators, to a point where they dominated the demand more than actual valuations. Given much more conservative approaches going forward, we do not expect financial institutions to represent as large a percentage of even a smaller market than before. However, we do see the usage scale being tipped more toward hedge funds and traditional asset managers. Hedge funds have always been active in the market, looking for relative value in index tranches, and convex type opportunities from both the long and short side within bespokes.

However, what is new is interest among more traditional asset managers in using tranches for convexity and tail plays. To some degree, this complements asset managers' interest in credit options. In terms of other challenges, we see potential roadblocks from legislation as investors attempt to get set up to trade derivatives in the new world.

Despite these challenges, we see a number of key areas for growth and opportunity, the best of which are those that serve some purpose in the market, and for which there is organic demand. We highlight these credit derivatives 2.0 themes as follows:

Protecting against or monetizing market volatility: We have focused for several years now on the potential for growth in credit options, and market volumes and liquidity have blossomed recently, thanks in part to high levels of market volatility. We believe the credit options markets

have reached critical mass now, and we expect options usage, from hedging to yield generation and upside plays, to be part of the culture of Credit Derivatives 2.0. With many investors living in a mark-to-market world within credit, protecting against or monetizing mark-to-market volatility is an essential task, much like it is in nearly every other asset class.

Callable credit: Related to the topic of monetizing market volatility, liquidity in the credit options markets will likely make it easier for investors to create callable forms of credit risk, in either corporate bond or synthetic form. We see growth here given the general acceptance of call-away risk in many investment mandates within fixed income and equities (analogous to buy-writing or overwriting in equities). This becomes another avenue for yield generation that does not necessarily involve leverage.

Tranches are long-dated “options” on credit: As the market for “traditional” structured credit goes through its restructuring, we see tranches as long-dated “options” on both spreads and defaults. In a world where long-dated options in equities lack liquidity and are expensive, and where they do not yet exist in credit options space, tranches are the natural solution. They are unfortunately more complex than simple calls and puts on spreads, since it is harder to predict exactly how much tranche spreads will widen or tighten relative to underlying markets than it is to predict how they will behave if there are defaults. But tranches remain attractively priced in many cases for market hedges, upside plays and yield generation opportunities.

Credit risk without interest rate risk: While interest rates in the developed world could certainly stay low for a while, the risks that they rise significantly owing to either positive news on growth (and the associated reduction in monetary and fiscal stimulus) or negative news on debt burdens is not to be ignored. Total returns on corporate bonds clearly suffer in this scenario, even though credit risk itself may not change much, or even fall. In those scenarios, we believe investors will seek out credit risk with reduced levels of interest rate risk, and synthetic credit through CDS and portfolios fits this bill quite elegantly. We also expect to see credit/interest rate hybrid products proliferate in a rising rate environment.

Synthetic securitization: While securitization suffered the most during the credit crisis, and markets have yet to fully recover, it is clear that regulators and legislators expect securitization markets to function and be part of the banking system going forward. For markets like consumer finance and commercial real estate, securitization is essential as the market for “whole loans” will not be sufficiently robust to transfer enough credit risk in the system. We expect that any significant return of issuance in funded synthetic CSOs will follow the lead set by the

broader securitization markets, with much focus on shorter-dated and thicker tranches than before. We expect high yield to be part of this world as well, perhaps more actively than in the last cycle, as securitization technology is well suited for asset classes that have high degrees of default risk through the cycle.

Bank loan hedging – the equity alternatives trade: As banks think about ways to raise additional capital or extend their lending books, regulatory-capital based hedging solutions will be an area where flows will occur. This is nothing new, as much Basel I based hedging occurred prior to the financial crisis. But the Basel II/III trade for most banks is to buy first-loss or junior mezzanine protection, forms of credit risk that are generally not efficient to hold on balance sheets. This “alternative equity” needs to find a home, and we expect to see much activity around this risk transfer trade.

THE SIXTH EDITION – PROGRESS IN A NEW WORLD

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Our sixth edition includes 40 chapters divided into 6 sections. The first section provides market primers for credit default swaps, CDS and bond options, tranches, baskets, and concepts like recovery risk and forward credit spreads. Next, we discuss market themes, including systematic and tail risk credit hedging strategies, credit valuation in a low yield world, bank credit challenges, and the performance of tranches over time. In the third section, we focus on valuation including interest rate, credit curve and basis dynamics, as well as our newly developed macro credit models. In the fourth section, we focus specifically on credit derivatives usage by bank loan hedgers, the market’s original and most important buyers of protection, covering many Basel regulation-related themes that drive these flows. In the fifth section, we discuss the market for customized credit solutions, including how the market evolved prior to the financial crisis, its performance since then, and the evolution of customization today. In the sixth section, we focus on newer areas of credit derivatives markets including developed market sovereigns and U.S. municipalities. Finally, the glossary provides definitions for over 200 terms used in the credit derivatives markets.

We hope Morgan Stanley clients find this handbook useful, and we welcome any feedback that will help us improve future editions.

We acknowledge the contributions of Prateek Saxena to this handbook.

Section A

Getting Started: Instruments and Primers

The CDS Lifecycle – A Market Primer

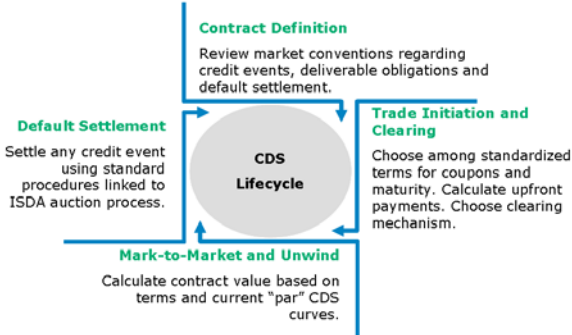
April 23, 2010

Sivan Mahadevan
Ashley Musfeldt
Phanikiran Naraparaju

The credit default swap (CDS) markets have entered their third decade of life, moving from non-standardized bi-lateral contracts in the late 1990s to instruments with standardized terms and trading and default settlement conventions today. Going forward, much of the focus and development will most likely be on clearing and potentially exchange trading. While both the contract and trading conventions have evolved several times over this nearly 15 year process, 2009 was a watershed year for the CDS market: We saw a number of changes aimed at standardizing CDS around the world starting with the Big Bang, which was a global protocol standardizing CDS around the world, with some US-specific changes, followed by the Small Bang in Europe and then some changes in Asia as well.

exhibit 1

The CDS Lifecycle



Source: Morgan Stanley

The changes in 2009, while varied, all related to contract and trading standardization. The push toward centralized clearing and trading over the medium term are the most significant events of late, and as of the writing of this document, the clearing end of the process moves forward. The objective of all these changes is to create greater fungibility of contracts traded at different points in time with different counterparties, thereby facilitating the netting of contracts in addition to improving market transparency and risk management.

We have written much material about the CDS markets over the years, and in this primer we attempt to provide a comprehensive overview of the credit derivatives market, from single name corporate, municipal and sovereign CDS to the standardized indices in these markets. The approach we take follows the lifecycle of a CDS trade, from contract definition to trade inception to valuation during the trade to ultimately credit event and trade termination. These stages are summarized as follows:

Understanding credit events and deliverables. We summarize credit events and deliverables for various types of CDS contracts across continents.

Initiating the trade. In these sections we summarize trading conventions and how these can impact pricing and valuation. We also look at the important role that different coupons have on CDS trades, and how fixed coupons are affected by CDS curve shape. We also provide an update on the clearinghouse, one of the more important developments in the CDS market today.

During the trade. Here we discuss mark-to-market concerns and the impact of succession events on the CDS contract.

Default or termination. We address the most frequently asked questions about the auction settlement process, which is the standard in CDS contracts today.

We highlight that while this chapter is comprehensive from a CDS lifecycle perspective, there are some CDS topics that we don't delve into as deeply, such as recovery swaps, and some of the finer nuances of the municipal and sovereign markets. In these cases we have written much more comprehensive publications on these topics, which should serve as additional references for the CDS markets (see Chapters 7, 36, and 40).

WHAT IS A CREDIT DEFAULT SWAP?

Before we can delve too deeply into the life cycle stages, we first must define a credit default swap (CDS). A single name CDS is an OTC contract between the seller and the buyer of protection against the risk of default on a set of debt obligations issued by a specified reference entity. The protection buyer pays a periodic premium (typically quarterly) over the life of the contract and is, in turn, covered for the period. A CDS is triggered if, during the term of protection, an event that materially affects the cash flows of the reference debt obligation takes place. For example, the reference entity files for bankruptcy, is dissolved or becomes insolvent. Other credit events can include failure to pay, restructuring, obligation acceleration, repudiation, and moratorium, all of which we cover in greater detail in this chapter.

A CDS contract specifies the precise name of the legal entity on which it provides default protection. Given the possibility of existence of several legal entities associated with a company, a default by one of them may not be tantamount to a default on the specific CDS. Therefore, it is important to know the exact name of the legal entity and the seniority of the capital structure covered by the CDS. On a related topic, changes in ownership of the reference entity's bonds or loans

can also result in a change in the reference entity covered by the CDS contract. This can be influenced by a succession event, a topic we will cover in greater detail later in the chapter.

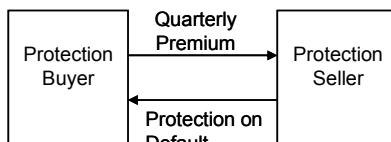
CASHFLOW MECHANICS

If a credit event is triggered, the protection buyer receives the equivalent of 100% of the par less the recovery – the price typically determined for the relevant deliverable obligations in an auction protocol. It is this probability weighted expected loss that the CDS premium strives to capture.

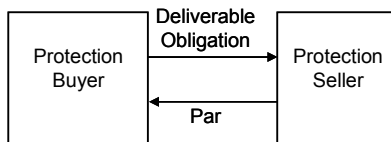
exhibit 2

CDS Mechanics

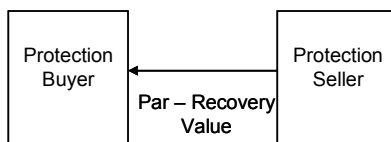
CDS Cashflows before Maturity/Default



Physical Settlement in Case of Default



Cash Settlement in Case of Default

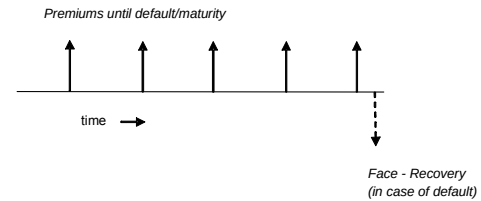


Source: Morgan Stanley

All CDS contracts now trade with an upfront payment which is paid at trade inception, and a standard coupon, which is paid quarterly. Cash flow dates are standardized – the 20th of March, June, September, and December of every year. If an investor enters a transaction in between the payment dates, the protection seller would make a payment of accrued premium to the protection buyer, to reflect the fact that although the protection buyer would pay premium for the full quarter on the next payment date, the protection is only for part of the quarter. Exhibit 3 shows simplified cash flows of a CDS contract – essentially two types of cash flows: the coupon coming in for the protection seller and the loss payout if a credit event occurs. The PV of the annuity stream should be the same as the expected loss on the contract, which is equivalent to the probability of default multiplied by the loss severity in the event of default.

exhibit 3

CDS Cashflows



Source: Morgan Stanley

Based on the probability of default and the loss given default, one can calculate the risky PV, as the CDS premium reflects the expected cost of providing the protection in a risk neutral sense. First, we need to estimate the probability of default and expected loss given default. The following equations summarize the pricing approach:

$$PV \text{ of CDS Spread} = PV \text{ of Expected Default Loss}$$

$$\text{Expected Default Loss} = LGD * \text{Probability of Default}$$

Where LGD stands for expected loss given default and equals

$$\text{Protection Notional} * (1 - \text{Estimated Recovery Rate}).$$

We make some further simplifying assumptions to better understand CDS pricing. First, we assume that we have a CDS spanning only one period, with the premium paid at the end of the period (see Exhibit 3). We also assume that a default can happen only at the end of the period. In case of default, the protection seller pays for the loss on the bond, i.e., par minus recovery. Now, we can calculate the implied probability of default from the given CDS spreads, using the equation mentioned earlier:

$$s1 \cdot (1 - p1) = p1 \cdot (1 - R)$$

$$p1 = \frac{s1}{s1 + 1 - R} \approx \frac{s1}{1 - R}$$

Next, we extend the model to two periods. Similar to one-period calculations, we can use the present value of the CDS spread and the expected loss in the event of default to get the implied probability of default in the second period, as shown in the two-period probability tree. The following equation summarizes this calculation:

$$\frac{s2 \cdot (1 - p1)}{1 + r \cdot t} + \frac{s2 \cdot (1 - p1) \cdot (1 - p2)}{(1 + r \cdot t)^2} = \frac{(1 - R) \cdot p1}{1 + r \cdot t} + \frac{(1 - R) \cdot (1 - p1) \cdot p2}{(1 + r \cdot t)^2}$$

$\underbrace{\hspace{10em}}$
PV of Spread

$\underbrace{\hspace{10em}}$
PV of Default

Since we know all of the variables other than p_2 , we can calculate it from this equation. While these calculations can clearly get quite onerous once we move beyond just one or two periods, most market participants use either internal models or CDSW on Bloomberg, which is the standard used when calculating upfronts and other costs at trade inception and unwinds.

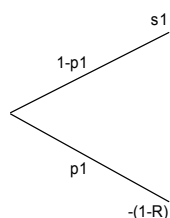
exhibit 4

Determining Default Probabilities

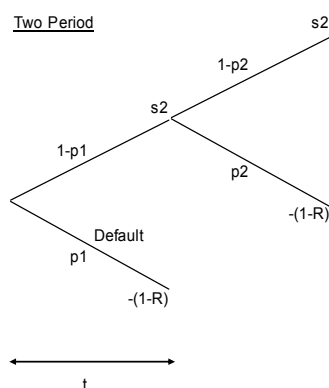
Assumptions

s1	CDS spread for single period maturity
s2	CDS spread for two period maturity
p1	probability of default in the first period
p2	probability of default in the second period
R	recovery rate
t	time period
r	riskfree rate

Single Period



Two Period



Source: Morgan Stanley

A NOTE ON THE INDICES

In prior introductory publications, we generally tended to separate single name CDS and the standard corporate CDS indices, since the mechanics on each were so different. The indices were the first to use the auction protocol, the first to have fixed coupons, and they were also the first to eliminate restructuring as a credit event. However, after the SNAC protocol from early 2009, there are few mechanical or contractual differences between the two instruments, and thus we combine them for most sections of this primer. First however, we need to examine exactly how the indices work.

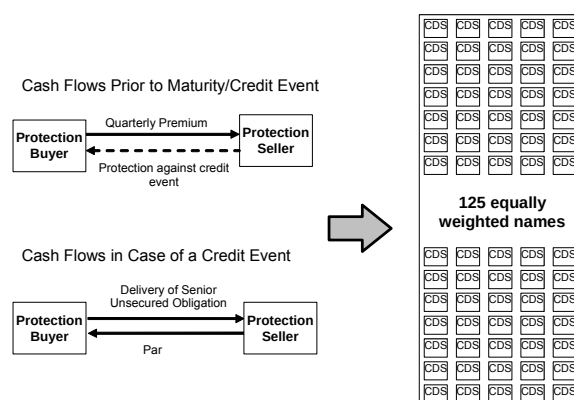
Credit default swap indices are portfolios of single-name default swaps, serving both as investment vehicles and as

barometers of market activity. By buying protection on an index, an investor is protected against defaults in the underlying portfolio. In return, the buyer makes quarterly premium payments to the protection seller, just like a single name CDS. If there is a default, the protection seller pays par less recovery determined in the auction settlement. Exhibit 5 shows the cash flows in an index.

Standardized credit derivatives indices were launched in the corporate credit markets in 2002, with the early products being agreements among a small number of dealers (Synthetic TRACERS was the first such index). Over time, the dealer list grew larger and several “standardizing” forces got involved to make the process more independent. Today the major credit derivative indices covering corporate credit, European sovereign credit, US municipal credit, emerging markets credit, and structured finance are managed by Markit (www.markit.com). Some key characteristics that are unique to the indices are as follows.

exhibit 5

Index Construction



Source: Morgan Stanley

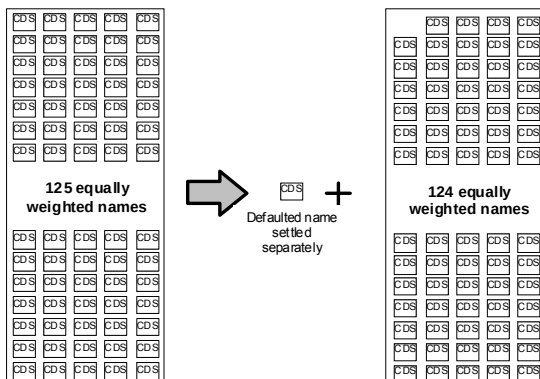
Static underlying portfolio. Once an index composition is determined, it generally remains static, with changes being incorporated in new indices rather than a current index. It is also noteworthy that all names are typically equally weighted by notional, as opposed to market weighted, which is common for benchmark bond indices.

Rolling over of indices. As time passes, the maturity term of indices decreases, making them significantly shorter than the benchmark terms, so new indices are introduced periodically and the latest series of the indices represents the current on-the-run index. When a new index series is launched, typically every 6 months, the maturity is extended and the portfolio constituents are reviewed and can be changed from one series to another. Markets have continued to trade previous series of indices, albeit with somewhat less liquidity.

Quoting convention. One thing to note is that in standard quotation, HY, LCDX, Lev-X and HY indices are quoted on a price basis whereas CDX IG, iTraxx Main and XOver, SovX and MCDX are quoted on a spread basis, even though all trade with the upfront plus coupon format.

exhibit 6

Impact of Default on an Index



Source: Morgan Stanley

Cash flows upon default. When an underlying single name defaults, it is removed from the index and settled separately. For example, if one of the 125 underlying names in CDX IG were to default, the remaining index would have 124 names and the same deal spread. The 1/125th of the notional would be separated and the protection seller would pay par to the protection buyer in exchange for a deliverable obligation. After a default, the premium payments for the index would be $(124/125) \times \text{index coupon} \times \text{original notional}$. It is important to note that an equal-weighted portfolio of underlying names could now have a different spread, given that each of the underlying names has its own spread level and that, depending on which of the 125 names defaults, the average spread for the remaining 124 names could be different from 124/125 of the original spread.

For the remainder of the primer we will continue to reference single name CDS; however, the mechanics and execution are the same for both the indices and single names.

CONTRACT DEFINITION: CREDIT EVENT TRIGGERS

The ultimate value of a CDS contract is derived from the credit events that can trigger it, the bonds and loans that are deliverable upon this event, and their expected recovery. Market sentiment and technicals can greatly influence CDS spreads, but it is important to consider contract language to determine value. CDS contracts have to trigger under a specific type of credit event. Broadly speaking, the most common credit events are bankruptcy, failure-to-pay and restructuring. Other triggers include repudiation/moratorium and obligation acceleration, but these are much less frequent and highly specific to certain types of credit risk.

Based on asset type and geography, a CDS contract may have a combination of the above mentioned types (see Exhibit 7 for the various triggers and classes of debt required to trigger a credit event). The legal and regulatory backdrop in each region, along with differences in the investor base can greatly influence contract language. The vast majority of credit events in the corporate world are bankruptcies, and the CDS contract is now well tested for such events, though the absence of a well-oiled bankruptcy framework makes restructurings more common in Europe and Asia than in the US. One thing to note is that banks do not achieve full capital relief when restructuring is not included in the CDS as a credit event (though this is being discussed with regulators currently and may change). As bank investors are a significant investor base in Europe and Asia, CDS includes restructuring as a credit event in those regions, whereas in North America, corporate CDS contracts do not include restructuring.

Aside from those credit events listed, there are also a few rare instances that can trigger a CDS contract, such as Fannie Mae and Freddie Mac, which triggered the CDS in 2008 after being put into a conservatorship. Though rare, this is relevant for corporates with strong governmental linkages.

For sovereign CDS, bankruptcy does not apply, and instead the credit events are failure to pay, restructuring and repudiation / moratorium, though restructuring is the primary trigger event that predominantly drives valuation. Russia and Ukraine in 1998 would probably have triggered under the repudiation / moratorium clause. There are also several important differences between European EM and Western European sovereigns. First, credit event tests are applicable only to bonds for European EM sovereigns, whereas they are applicable to any borrowed money (including bi-lateral lending agreements) for Western European Sovereigns. Additionally, sovereign lender debt is not applicable in Asia but it is applicable in Europe. Finally, only non-domestic law and non-domestic currency bonds are applicable for European EM sovereigns whereas domestic bonds are also relevant for Western European Sovereigns. See Exhibit 7.

In the municipal market failure-to-pay and restructuring are the only credit events, with the notable omission of bankruptcy. Though some municipalities can file for bankruptcy under U.S. law, they can only file for Chapter 9, and not all municipalities are able to do this according to their individual state constitutions. While the details of this are better left to another report, the important thing to note here is that most bankruptcies would likely include either a failure to pay, a restructuring, or both, and thus would eventually trigger the CDS contract anyway. Municipal restructuring is "old style", which is different from the restructuring type used in the corporate world (see the following section).

CONTRACT DEFINITIONS: CANCELLABILITY IN LCDS

There have been some contract changes in the LCDS market in the US recently, bringing some much needed clarity to the market. The first change is that the single name and index markets will now trade with a standard 250bps coupon, with 100bp and 500bp available if needed. The second change is regarding cancellation risk in LCDS. The key concern with the prior LCDS contract was pricing and valuation uncertainty, given the cancellability or early termination feature of underlying loans, which is akin to a credit defaulting with a recovery of 100%.

The old LCDS contract stated that a simple loan refinancing (with another loan) would not lead to a cancellation/early termination event and the contract would still remain outstanding. However, when a company chooses to replace its existing secured debt by bonds, the loan actually gets paid down, which can happen when a company has access to cheaper unsecured financing, often due to a ratings upgrade. In this case, there is no appropriate reference obligation outstanding in the market to be delivered into the LCDS contract, and under the old contract, the LCDS contract could be terminated. Now, the new LCDS contract does not include optional early termination rights, and in the absence of a credit event, if a valid deliverable obligation does not exist, the “orphaned” LCDS will continue to remain outstanding, similar to the unsecured CDS market.

Additionally, there was some confusion in the LCDS contract regarding succession and refinancing events. The concern was that it was sometimes unclear whether new entities first assumed or became liable for the original loans prior to them being repaid, which could cause some LCDS contracts to be orphaned. To avoid this, the definition of succession and refinancing events have been expanded to include the following, according to Markit (www.markit.com):

- Repayment of Relevant Obligations from the proceeds of new loans or bonds from a new entity
- Repayment of Relevant Obligations where the assets securing them have been acquired by proceeds of new loans or bonds from a new entity
- Repayment of Relevant Obligations where the assets securing them subsequently secure new loans or bonds of a new entity
- Amendment or Restructuring where the Relevant Obligations cease to be obligations of the original Reference Entity and another entity becomes a borrower or provides a qualifying affiliate guarantee
- Any other event that has substantially the same effect as the above.

Morgan Stanley is currently acting as financial advisor to Thomson Reuters Corporation ("Thomson Reuters") with respect to its plans to divest its healthcare business as announced on June 6, 2011. Thomson Reuters has agreed to pay fees to Morgan Stanley for its financial advisory services which are contingent upon the consummation of a transaction. Please refer to the notes at the end of the report.

CONTRACT DEFINITION: RESTRUCTURING

The most common source of credit event ambiguity stems from restructuring credit events. The ISDA 2003 credit derivatives definitions characterize restructuring credit events as a reduction in coupon or principal, a deferral of an interest or principal payment, a change in the ranking of priority of payments, or changes to the currency of the interest and principal payments. The currency criteria state that a redenomination of debt into G7 currencies is permitted, OR redenomination into any OECD currency as long as the local-currency long-term debt rating is AAA or higher by S&P, Moody's, or Fitch. This latter point is important for EMU members with sub-AAA ratings as an event where they leave the EMU and re-denominate their debt into a new local currency would trigger a CDS credit event via restructuring.

There are also additional burdens to prove a restructuring event. Restructuring must be coercive to trigger the CDS – i.e., it must be agreed upon by a sufficient number of holders before it becomes binding on all the holders. Unless Multiple Holder Obligation is specified as not applicable, the obligation has to be held by at least two-thirds of the investors. Furthermore, restructuring is not applicable if any of the above events has not been the result of deterioration in credit quality. Finally, there are variations to deliverable obligations depending on the type of restructuring clauses. In the case of the corporate restructuring, there are limitations on the maturity of the bond that can be delivered into the contract.

As a result of all these requirements, examples of triggering CDS under the restructuring clause have been few and far between. Xerox was a notable example, and Thomson recently experienced a restructuring, which turned out to be a painful learning experience for market participants. Another recent restructuring example is Aiful, which triggered a restructuring credit event (old R) leading to the first CDS auction in Japan.

Post SNAC, restructuring credit events will now be much rarer, as restructuring was eliminated as a credit event in new corporate contracts in the US. Nonetheless, it is important to understand the various restructuring types and the implications therein for the large legacy market, as well as the specific corners of the market where restructuring is still a credit event. It is conceivable for a CDS credit event to trigger from a restructuring where all the bonds of the reference entity remain outstanding and continue trading based on their interest rate risk.

Old Restructuring. This was the original form of restructuring. Old style restructuring implies that, in the event of a restructuring, bonds of any maturity less than 30 years can be delivered, which can introduce a significant amount of interest rate risk into a CDS contract upon a restructuring credit event (i.e., a 30 year fixed-rate bond could trade at a significant discount to par in a high interest rate environment). This is particularly important given that the municipal market tends to be long in duration.

Modified Restructuring and ModMod Restructuring. In 2002 the restructuring of a Xerox obligation brought to light long duration interest rate risk, driving the introduction of modR for the US and modmodR for Europe to the 2003 ISDA definitions for corporate CDS. As a result of this, many CDS

contracts introduced maturity limitations for restructuring credit events.

We have tried to provide a brief overview of the various elements of the contract, but there is a wealth of material on ISDA and Mark-it that we recommend.

exhibit 7

Overview of Standard Credit Events for Various CDS Contracts

	Bankruptcy	Failure to Pay	Failure to Pay (Grace Period Appl.)	Obligation Acceleration	Repudiation/ Moratorium	Restructuring (Old R)	Restructuring Maturity Limitation and Fully Transferable Obligation (Mod R)	Modified Restructuring Maturity Limitation and Conditionally Transferable Obligation (Mod Mod R)	Restructuring (Multiple Holder Obligation Required?)
Asia Sovereign	N	Y	N	N	Y	Y	N	N	Y
Singapore Sovereign	N	Y	N	N	Y	Y	N	N	Y
Latin America Sovereign	N	N	Y	Y	Y	Y	N	N	N
Emerging European & Middle Eastern Sovereign	N	N	Y	Y	Y	Y	N	N	N
Western European Sovereign	N	Y	N	N	Y	Y	N	N	Y
Japan Sovereign	N	Y	N	N	Y	Y	N	N	N
Australia Sovereign	N	Y	N	N	Y	N	Y	N	Y
New Zealand Sovereign	N	Y	N	N	Y	N	Y	N	Y
U.S. Municipal Full Faith And Credit	N	Y	N	N	N	Y	N	N	N
U.S. Municipal General Fund	N	Y	N	N	N	Y	N	N	N
U.S. Municipal Revenue	N	Y	N	N	N	Y	N	N	N
North American Corporate	Y	Y	N	N	N	N	N*	N	Y
European Corporate	Y	Y	N	N	N	N	N	Y	Y
Australia Corporate	Y	Y	N	N	N	N	Y	N	Y
New Zealand Corporate	Y	Y	N	N	N	N	Y	N	Y
Japan Corporate	Y	Y	N	N	N	Y	N	N	N
Singapore Corporate	Y	Y	N	N	N	Y	N	N	Y
Asia Corporate	Y	Y	N	N	N	Y	N	N	Y
Subordinated European Insurance Corporate	Y	Y	N	N	N	Y	N	N	Y
Emerging European Corporate LPN	Y	N	Y	Y	Y	Y	N	N	Y/N
Emerging European Corporate	Y	N	Y	Y	Y	Y	N	N	Y/N
Latin America Corporate B	Y	N	Y	Y	Y	Y	N	N	N
Latin America Corporate BL	Y	N	Y	Y	Y	Y	N	N	Y
Sukuk Corporate	Y	N	Y	Y	Y	Y	N	N	Y/N
Sukuk Sovereign	N	N	Y	Y	Y	Y	N	N	N

Overview of Obligation Characteristics to Trigger a Credit Event

	Obligation Category	Not Subord- inated	Not Sovereign Lender	Not Domestic Currency	Not Domestic Law	Not Domestic Issuance	Specified Curr. & Domest Currency	Full Faith & Credit Oblig. Liab.	General Fund Oblig. Liab.	Revenue Oblig. Liab.
Asia Sovereign	Bond /Loan	Y	Y	Y	Y	Y	N	N	N	N
Singapore Sovereign	Bond /Loan	Y	Y	N	N	N	Y	N	N	N
Latin America Sovereign	Bond	Y	N	Y	Y	Y	N	N	N	N
Emerging European & Middle Eastern Sov.	Bond	Y	N	Y	Y	Y	N	N	N	N
Western European Sovereign	BM	N	N	N	N	N	N	N	N	N
Japan Sovereign	BM	N	N	N	N	N	N	N	N	N
Australia Sovereign	BM	N	N	N	N	N	N	N	N	N
New Zealand Sovereign	BM	N	N	N	N	N	N	N	N	N
U.S. Municipal Full Faith And Credit	BM	Y	N	N	N	N	N	Y	N	N
U.S. Municipal General Fund	BM	Y	N	N	N	N	N	N	Y	N
U.S. Municipal Revenue	BM	Y	N	N	N	N	N	N	N	Y
North American Corporate	BM	N	N	N	N	N	N	N	N	N
European Corporate	BM	N	N	N	N	N	N	N	N	N
Australia Corporate	BM	N	N	N	N	N	N	N	N	N
New Zealand Corporate	BM	N	N	N	N	N	N	N	N	N
Japan Corporate	BM	Y	N	N	N	N	N	N	N	N
Singapore Corporate	Bond /Loan	Y	Y	N	N	N	Y	N	N	N
Asia Corporate	Bond /Loan	Y	Y	Y	Y	Y	N	N	N	N
Subordinated European Insurance Corp.	BM	N	N	N	N	N	N	N	N	N
Emerging European Corporate LPN	Bond /Loan	Y	N	Y	Y	Y	N	N	N	N
Emerging European Corporate	Bond /Loan	Y	N	Y	Y	Y	N	N	N	N
Latin America Corporate B	Bond	Y	N	Y	Y	Y	N	N	N	N
Latin America Corporate BL	Bond /Loan	Y	Y	Y	Y	Y	N	N	N	N
Sukuk Corporate	Bond /Loan	Y	N	Y	Y	Y	N	N	N	N
Sukuk Sovereign	Bond	Y	N	Y	Y	Y	N	N	N	N

Source: ISDA

*Restructuring if specified as applicable in the relevant Confirmation

exhibit 8

Overview of Deliverable Obligations for Various CDS Contracts

	Obligation Category	Not Subordinated	Not Sovereign Lender	Not Domestic Law	Not Domestic Issuance	Maximum Maturity: 30 years	Assignable Loan	Consent Required Loan	Specified Currency	Standard Specified Currencies & Domestic Currency	Specified Currencies & Domestic Currency	Standard Specified Currencies
Latin America Sovereign	Bond	Y	N	Y	Y	N	N	N	Y	N	N	N
Emerging European & Middle Eastern Sov.	Bond	Y	N	Y	Y	N	N	N	Y	N	N	N
Western European Sovereign	Bond /Loan	N	N	N	N	Y	Y	Y	Y	N	N	N
Japan Sovereign	Bond /Loan	N	N	N	N	Y	Y	Y	Y	N	N	N
Australia Sovereign	Bond /Loan	Y	N	N	N	Y	Y	Y	N	Y	N	N
New Zealand Sovereign	Bond /Loan	Y	N	N	N	Y	Y	Y	N	Y	N	N
Asia Sovereign	Bond /Loan	Y	Y	Y	Y	Y	Y	N	Y	N	N	N
Singapore Sovereign	Bond /Loan	Y	Y	N	N	Y	Y	N	N	N	Y	N
U.S. Municipal Full Faith And Credit*	Bond /Loan	Y	N	N	N	Y	Y	Y	N	N	N	Y
U.S. Municipal General Fund**	Bond /Loan	Y	N	N	N	Y	Y	Y	N	N	N	Y
U.S. Municipal Revenue***	Bond /Loan	Y	N	N	N	Y	Y	Y	N	N	N	Y
North American Corporate	Bond /Loan	Y	N	N	N	Y	Y	Y	Y	N	N	N
European Corporate	Bond /Loan	Y	N	N	N	Y	Y	Y	Y	N	N	N
Australia Corporate	Bond /Loan	Y	N	N	N	Y	Y	Y	N	Y	N	N
New Zealand Corporate	Bond /Loan	Y	N	N	N	Y	Y	Y	N	Y	N	N
Japan Corporate	Bond /Loan	Y	N	N	N	Y	Y	Y	Y	N	N	N
Singapore Corporate	Bond /Loan	Y	Y	N	N	Y	Y	N	N	Y	N	N
Asia Corporate	Bond /Loan	Y	Y	Y	Y	Y	Y	N	Y	N	N	N
Subordinated European Insurance Corp.	Bond /Loan	Y	N	N	N	Y	Y	Y	Y	N	N	N
Emerging European Corporate LPN	Bond /Loan	Y	N	Y	Y	N	Y	Y	Y	N	N	N
Emerging European Corporate	Bond /Loan	Y	N	Y	Y	N	Y	Y	Y	N	N	N
Latin America Corporate B	Bond	Y	N	Y	Y	N	N	N	Y	N	N	N
Latin America Corporate BL	Bond /Loan	Y	Y	Y	Y	N	Y	Y	Y	N	N	N
Sukuk Corporate	Bond /Loan	Y	N	Y	Y	N	Y	Y	Y	N	N	N
Sukuk Sovereign	Bond	Y	N	Y	Y	N	N	N	Y	N	N	N

Source: ISDA

In addition: * Full Faith and Credit Obligation Liability, **General Fund Obligation Liability, ***Revenue Obligation Liability

CONTRACT DEFINITION: DELIVERABLE OBLIGATIONS

Another important factor in determining CDS valuation is deliverable obligations. Common considerations for determining deliverability are the currency, maturity, subordination, governing law and market of issuance.

First we address maturity. Corporate CDS in Japan and Asia, most OECD sovereigns, and municipal CDS (as we mentioned above) in the US use “Old R” restructuring, under which maturities of up to 30 years are deliverable if a restructuring credit event occurs, allowing interest rate levels to determine the cheapest-to-deliver bond if bonds do not accelerate (i.e., become due immediately as they would in a corporate bankruptcy).

Type of debt is another interesting point to note when discussing deliverability. For instance, in Europe, only non-domestic debt is deliverable for EM sovereigns unlike Western European sovereigns. Additionally, only bonds are deliverable in European EM and Latin American sovereigns, whereas both bonds and loans are deliverable for Western European and Asian sovereigns. These are all things to keep in mind when evaluating the cheapest to deliver option, which ultimately impacts the value of the CDS.

An important point to consider is the language of the underlying bonds. Bonds issued over different periods tend to

have variation in the clauses as well – this is particularly true in the sovereign space. For example the absence of cross-default provisions in the bond language makes a significant difference to the way bonds across the curve trade relative to the CDS in times of distress.

TRADE INITIATION: FIXED COUPONS REDUCE RISKS

Prior to SNAC in 2009, most CDS contracts traded on an all running basis, with the exception of a few very distressed names. Thus if an investor were to buy protection on a name that was quoted at 110bp, the trade would simply have a coupon of 110bp. During times of credit distress, however, this became an issue as unwinding a CDS contract is complicated when the strike price is very far out of the money, a theme we have highlighted in numerous pieces (see *Credit Volatility – the Unintended Consequences* from April 1, 2005 and *LCDS, After the Trade* from June 1, 2007). Investors who wanted to unwind a trade done without a fixed coupon often found large penalties associated with the unwind due to the mismatched coupon streams. With a fixed coupon this legacy annuity risk is significantly reduced, as any difference in risk premium is paid in the form of an upfront payment. To illustrate this concept, we offer the following example.

Consider an example, where an investor purchased 5-year protection on credit “X” at 100bp in January 2008 and then

subsequently elected to monetize the trade in March 2009 when spreads to the same maturity widened to 1,000bp. To monetize this widening, the investor could do one of two things. The first would be to sell protection at a strike of 1,000bp, and then hold two offsetting trades, one that is paying 100bps and another that is receiving 1,000bps. In this scenario, the investor would be hedged from a default risk perspective, however, now has IO risk in this credit in the event of default.

Why? The key here is the uncertainty of the total cash flows and P&L for the investor depending on the whether or not a credit event is triggered. The investor now receives 900bp (the difference between these two coupon streams) until maturity or a credit event. If a credit event were to occur, both contracts would trigger, the default loss settlement of the two contracts would offset each other, and *both* coupon streams would stop. Thus the 900bps the investor had been receiving on a default “riskless” basis is now gone. Thus while the default settlement is hedged, the coupon stream is still exposed to default risk. In this example, if the default occurs one year later, the investor would have received \$0.9mm on a \$10mm position (900bp * 10mm). If the default is in four years, the investor would have received \$3.6mm in total cash flow from the coupons.

To eliminate this risk, the investor can instead unwind the original 100bp swap and receive the PV of the 900bps difference between the 100bps coupon and the 1,000bps of credit risk the swap is now worth to monetize the P&L immediately. The investor unwinds the original swap struck at 100bp and receives an upfront amount of \$2.9mm, which is the PV of the 900bp stream assuming a remaining duration of 3.2. In addition to having hedged the default risk on the reference entity, the investor has now eliminated the credit risk inherent in the risky coupon (as well as counterparty exposure, which is a separate issue). From the investor’s perspective, the trade is completely closed and monetized.

However the uncertainty in cash flows is not eliminated, but instead transferred to the dealer. Why? Because the dealer now has to hedge the position with the market standard contract. In this example, where floating coupons are market standard, the dealer would have paid out \$2.9mm today to buy protection at 100bp running from the investor, but can only sell protection to another investor on the now current running coupon – in this case 1,000bp. If there is an immediate default, the dealer loses the \$2.9mm paid to the investor but has not received any of the 1,000bp coupon from the protection sold. The dealer has effectively become exposed to the jump risk that the investor just got rid of. In the past, the dealer could buy short dated protection (say 1-year protection) to hedge these early default scenarios, but would most likely pass this additional cost, or at least some fraction of it, on to the investor who was unwinding the trade. This resulted in different spreads shown depending on

whether the investor was unwinding an off-market coupon or putting on a new trade.

Thus prior to the advent of standardized fixed coupons the dealer incurred additional risk and costs when using a non-standard fixed strike CDS arising from a) the cost of funding the upfront payment to be paid to the investor and b) hedging the annuity risk from an early default by buying jump protection. With the fixed strike contracts, each of these legs becomes more fungible and can be hedged immediately with offsetting contracts that have the same coupon and maturity and would involve the transfer of the same upfront amount. This is also important in light of the development of a central clearinghouse, as a clearinghouse would face the same issues when presented with different floating strike contracts from different counterparties.

TRADE INITIATION: FIXED COUPON SENSITIVITY

Following SNAC standardization in 2009, most CDS contracts now trade with a fixed coupon, with the exception of municipal CDS in the US, though that may change in the future. The implication of using fixed coupons is that instead of paying the full CDS premium each year, an upfront payment will be exchanged based on the difference between the coupon and the ‘spread’ or ‘premium’.

The most common coupon strikes are 100bps and 500bps for most CDS contracts around the world. The exact coupon for a name will be a function of the current spread level, the historic spread range and legacy exposures. Three exceptions are a) some European corporates which also have 25bp and 1,000bp available, although the latter is unlikely to be used much; b) Japan corporate and sovereign CDS which have 25bp also available; c) Western European sovereign CDS which has only 25 and 100bp as a choice (and not 500bp) and d) US LCDS contracts that as of 2010 will trade with a standard fixed coupon of 250bp (though 100bp and 500bp will also be available for tight and wide credits, respectively).

exhibit 9

Coupon Standards around the World

CDS	25	100	250	500	1000
Asia Sovereign & Corporate		Y		Y	
Latin America Corporate & Sovereign		Y		Y	
Emerging European & Middle Eastern Sovereign		Y		Y	
Western European Sovereign	Y	Y			
Japan Corporate & Sovereign	Y	Y		Y	
Australia, New Zealand Corporate & Sovereign		Y		Y	
North America LCDS			Y		
North American Corporate & Sovereign CDS		Y		Y	
European Corporate	Y	Y		Y	Y

Source: Morgan Stanley

As we highlighted before, the coupons were standardized to reduce jump-to-default risk inherent in having extremely out-of-the-money coupons when trying to unwind that resulted in legacy annuity risk. However, other considerations remain,

even in light of making all coupons conform to one of 4 strikes. One of the bigger issues we are seeing today even with standardized coupons is the impact of CDS curve structure on upfront pricing at different coupon strike levels.

To illustrate, say we have a 5-year CDS with a par spread of 250bps. The convention is to use Bloomberg CDSW with a flat curve. With coupon of 100bp, the upfront payment on this will be 6.7%. However, since many market participants do not use a flat curve in their internal risk systems, the par spread will be slightly different when used with a full maturity curve to achieve the same upfront levels. In our example the equivalent level is 244bp. If we then change the coupon to, for instance, 500bps using the same upward sloping curve shape as before, we get an upfront level of 11.9%. Finally, we use this upfront to get an equivalent flat curve par spread, and we see that to get this same 11.9% using a flat curve, we have a par spread of 237bps, a difference of 13bp from where we began.

How is this possible? We have several different par spreads for what should be equivalent risk. The answer lies in the default risk implied by the curve shape. It is intuitive that a positively sloped curve is less risky than a flat curve with the same 5-year point, so an upward sloping curve structure will imply lower default probabilities and thus a higher probability of the fixed coupon actually being paid. When we change the coupon to be higher, i.e., from 100 to 500bp, more of the premium paid to the seller of protection is at risk. So to have equivalently risky contracts, a lower “par spread” is used to compensate for the higher running coupon. Consequently, we are seeing that different strike contracts and different curve shapes produce different par spreads, even when the fixed coupon and upfront remain the same.

Another way to look at this is instead to look at the way upfronts change when curve shape changes. In Exhibit 10, we show that given a CDS with a par spread of 250 basis points, the upfront amounts will change pretty significantly based on the type of curve shape used – even though all are using the same 5-year spread of 250bps.

exhibit 10

Curve Shape Affects Upfront Pricing

Curve Shape	5-year CDS Spread	Fixed Coupon of 100bps	Fixed Coupon of 500bp
Upward Sloping	250bp	7.0%	-11.7%
Flat	250bp	6.7%	-11.2%
Inverted	250bp	6.6%	-11.0%

Source: Morgan Stanley

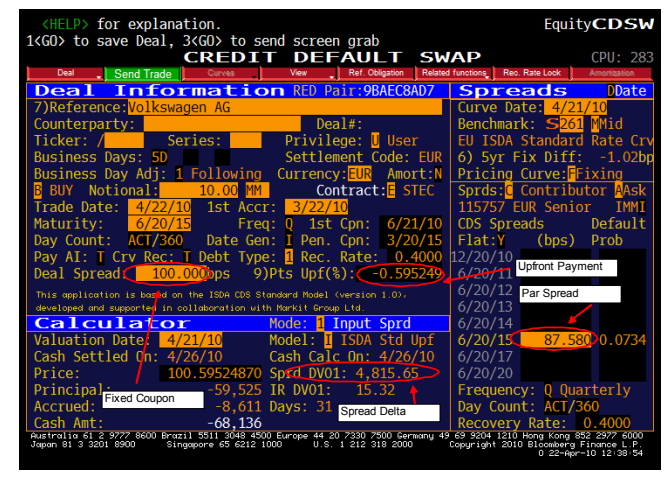
TRADE INITIATION: COMMON METRICS AND MODELS

When trading CDS, an investor goes long risk (sells protection) and earns a “spread” to compensate for the probability of default of the underlying entity in a risk neutral framework. As a result, the present value of all these

incremental cash flows should equal the present value of expected losses. The probability of default of an entity is determined using market CDS spreads across the maturity spectrum.

exhibit 11

Bloomberg CDSW



Source: Bloomberg, Morgan Stanley

The PV01 of a CDS contract (referred to as risky duration) is simply the present value of 1bp of the cash flow at every coupon payment date. It can also be interpreted as the mark-to-market sensitivity of expected losses in a CDS contract to a 1bp parallel shift in the spread term structure of the underlying entity. In contrast to the calculation for the duration of a bond or risk-free interest rate swap, the discount factors for CDS duration are higher, as they have a survival probability term associated with them. This additional discount factor reduces the present value of cash flows, and consequently the risky duration for similar cash flows in CDS vs. less risky assets.

Spread delta or DV01 is a more pertinent metric to look at today, given the advent of fixed coupons. Unlike duration, which is the sensitivity of the contract to a change in expected loss assumptions, the spread delta captures *just* the mark-to-market sensitivity of the contract to a 1bp parallel shift in the credit curve. Thus we get this equation:

$$\text{Upfront payment} = (\text{Par spread} - \text{Fixed Coupon}) * \text{PV01}$$

Morgan Stanley Bank AG is currently acting as financial advisor to the family shareholders of Porsche Holding SE in connection with the potential combination of Porsche Automobil Holding SE and Volkswagen AG.

This report and the information herein are not intended to serve as an endorsement of the proposed transaction.

This report was prepared solely upon information generally available to the public. No representation is made that it is accurate and complete. This report is not a recommendation or an offer to buy or sell the securities mentioned. Please refer to the notes at the end of this report.

DV01 is essentially the first derivative of the upfront payment with respect to par spread. The above equation shows that when a contract trades at par, the spread delta is the same as duration. However, as the coupon differs from the par spread, the change in upfront payment also depends on the change in PV01 with respect to spread (i.e., convexity).

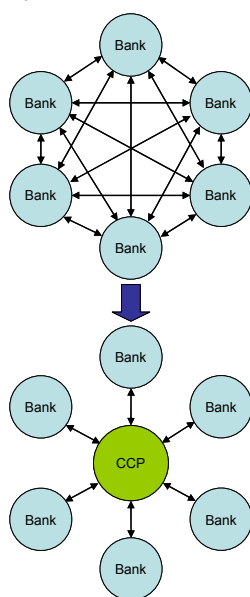
TRADE INITIATION: CLEARINGHOUSES

Mandatory clearing might be a major component of the regulatory reform, at least as far as the credit markets are concerned, with a primary objective of maximizing clearing of derivatives trades in the market. Today, the clearinghouse is legging in on certain types of transactions, starting with dealer to dealer trades on the most liquid products and expanding from there. These trades will eventually include all liquid index products and most single names. As of publication we are already clearing indices and some single names, and the list of eligible trades for the clearinghouse continues to grow.

While the counterparty type will be an important factor in whether or not a trade needs to be cleared or exchange-traded, the trade type may also be considered. For instance, some highly bespoke trades may not be required (or able) to go through a clearing house or derivative exchange. If these trades, or others, do not go through a clearing house, there will likely be reporting requirements imposed. Furthermore, who determines what is required to be cleared is being discussed and right now the CFTC (and SEC when appropriate) are most likely to have the authority to determine what is clearable.

exhibit 12

Central Counterparty Mechanics



Source: Morgan Stanley

AFTER THE TRADE: DEALING WITH SUCCESSION EVENTS

A number of corporate and financing structure decisions also have an influence on CDS. Event risks to keep in mind are corporate refinancings, business restructurings, asset sales, divestitures, M&A and LBOs. While most of what we focus on in this section deals with the 2003 ISDA definitions, we highlight that succession events were one of the many topics covered by SNAC provisions of 2009, specifically regarding look-back dates for determining credit events and succession events. Prior to SNAC, the effective and succession event dates were fixed dates (at the time of trade date plus one), meaning that, in theory, a credit event could be declared anytime between the effective date and the maturity of the CDS contract. One of the issues surrounding the prior standard is that offsetting trades or hedges could have different windows for succession event determination, introducing a certain amount of incremental basis risk. Under the new contract, this long period of potential succession event declaration is reduced to no more than 90 days prior.

While there are numerous corporate situations where the timing of debt exchanges and the ultimate par value of debt that moves between entities determines successor behavior in CDS, we endeavor to provide some basic rules of thumb regarding the impact of succession events (or the lack thereof) on CDS contracts, based on the 2003 ISDA definitions. Here are some of the broader themes:

- When there are corporate successions, CDS contracts follow the debt of a company, rather than equity value, revenues or corporate structure. A corporate succession must result in a “Succession Event,” under the 2003 ISDA definitions, for CDS to change, although CDS can be implicitly affected without such an event.
- The key difference between bonds and CDS in the event of succession is that CDS can be formulaically split, while bonds, by definition, have to go one direction or another (or get taken out).
- It is rare for CDS to be terminated as a result of corporate succession. The only situation where it can happen is where the party to the corporate action is also the protection seller, in which case a “Termination Event” occurs. Even then, it results in a mark-to-market unwind at the option of the protection buyer.
- A debt exchange that is not in connection with a merger (or other terms of a “Succession Event”) will not qualify as a “Succession Event.” As such, there could be situations where no obligations are left to be deliverable into the CDS, although debt issued in the future could be.
- One company guaranteeing the debt of another company (say, after buying its stock) does not qualify as a “Succession Event.” If the debt is assumed by the parent

company and released by the original obligor, then it is a “Succession Event.”

For bond purposes, succession really has more to do with making the credit risk of the instruments of one issuer economically similar to the instruments of another issuer. For CDS contracts, succession is a legal term with a very specific definition (from 2003 ISDA Credit Derivatives Definitions):

“Succession Event” means an event such as a merger, consolidation, amalgamation, transfer of assets or liabilities, demerger, spin-off or other similar event in which one entity succeeds to the obligations of another entity, whether by operation of law or pursuant to any agreement. Notwithstanding the foregoing, “Succession Event” shall not include an event in which the holders of obligations of the Reference Entity exchange such obligations for the obligations of another entity, unless such exchange occurs in connection with a merger, consolidation, amalgamation, transfer of assets or liabilities, demerger, spin-off or other similar event.

The definition is intended to capture most merger and acquisition activity, but because of the exchange exclusion, the actual timing of events can affect whether they qualify as successor events. Once we have defined whether an event is a successor event, we next need to consider how various instruments react. For cash bonds and loans, generally, the corporation will specify the intention to either buy back debt and have it assumed by the new entity, or any other action, which may or may not be subject to approval by bondholders.

exhibit 13

Summary of Succession Events

% of Relevant Obligations That Succeed Original Reference Entity	Impact on CDS	Notes
S1 ≥ 75%	S1 is Sole Successor	
75% > S1 > 25%, REO ≤ 25%	S1 is Sole Successor	
75% > S1 > 25%, 75% > S2 > 25%, REO ≤ 25%	S1, S2 Equal Successors	Can apply to 1 or more successors
S1 > 25%, S2 > 25%, REO > 25%	S1, S2, REO Equal Successors	
All Successors ≤ 25%, REO still exists	REO is the Sole Successor	
All Successors ≤ 25%, REO does not exist	Largest Successor is Sole Successor	Tie breaker is % of all obligations
Seller of Protection “merges” with the reference entity	CDS Terminates with MTM	Merger or other succession event

Source: Morgan Stanley, Thomson SDC

Thus the fate of any individual bond is fairly clear, but the CDS is not so clear. CDS contracts are generally intended to follow the fate of the debt of a company in aggregate. CDS contracts can follow one of three paths: (1) either continue to refer to the original entity, (2) succeed to a new entity or (3)

be divided into contracts that refer to two or more entities depending on what exactly happens to the total debt of the original reference entity.

CDS will move to a sole successor under the following circumstances: When 75% of “Relevant Obligations” move to that successor; when between 25% and 75% move and the original reference entity keeps less than or equal to 25%; or when the original reference entity does not exist and all successors account for less than 25% of relevant obligations (the largest of these will be the sole successor). In this last case, if the original reference entity still exists, then it will be the sole successor. When more than one successor (including the original reference entity) represents greater than 25% and less than 75% of the relevant obligations, they will be equal successors.

Based on 2003 definitions, there is only one way a CDS contract can terminate related to succession events: when the merger is between a reference entity and a CDS counterparty (seller of protection). While we cannot cite a general example of such a situation, it most likely has occurred numerous times historically, for example when credit derivatives dealers (say a large bank) merges with other institutions where CDS activity is common (say a smaller bank). Any investor who bought protection on the smaller bank from the larger bank would have the option to unwind the contract at a mark-to-market.

One thing to note is that corporate management teams are less concerned with the consequences of how a deal is executed for CDS users than they are about the strategic, operational and tax consequences of how they execute a restructuring. These exogenous considerations, which drive decision-making at the corporate level, can introduce a fair amount of risk in the execution of derivatives and cash strategies surrounding restructurings. The key point here is that how a company executes a given transaction can be completely arbitrary from a CDS user’s perspective. While the true motivation can be strategic or cost based, the results for default swap users can be big profits or big losses for transactions that are economically equivalent to bondholders.

AFTER THE TRADE: SPIN-OFFS

In a divestiture or spin-off, the natural question is what happens to existing debt, and the direct corollary of that is what happens to CDS contracts. The details are incredibly important in any discussions of how debt and CDS will be treated in spin offs; as such, making broad statements is difficult to do.

To illustrate the point, consider the following: Whether a parent exchanges debt or assumes the debt of a newly acquired subsidiary, and even the timing of those actions, matters in the potential handling of CDS contracts. Even though the economic consequences for bond holders can be virtually equivalent in either case, the fate of CDS users is

tied to the details. Another example that is critical in spin offs is the differing treatment of guarantees from parent companies to their subsidiaries and guarantees from subsidiaries to their parent companies. Failure to perform on the former would likely result in a credit event for the parent under CDS contract language 2003 definitions, while failure to perform on the latter would likely not result in a credit event for the subsidiary (it is likely that only downstream guarantees matter; we note that this behavior could be different in Europe).

THE AUCTION PROCESS: AN EXAMPLE

To illustrate the auction methodology, we take the example of Masonite from December 2008 for a relatively simple bankruptcy auction. The auction involves a two-stage process.

Stage 1. The dealers establish indicative markets by quoting both a bid and offer for the bonds in a pre-defined size and pre-defined bid / offer, which can vary by credit. In addition, dealers will submit physical settlement requests – that is, requests to buy or sell bonds through the auction. In the case of Masonite, dealers submitted 3pt bid-offers in 2.5MM size and had a net demand to buy bonds through the auction (18MM notional of bonds).

exhibit 2

Stage 1: Dealers Provide Markets

Dealer	Inside Markets		Physical Settlement Requests	
	Bid	Offer	Bid/Offer	Size MM
Dealer A	47.5	50.5	Offer	0.0
Dealer B	48.0	51.0	Bid	0.0
Dealer C	48.5	51.5	Offer	10.0
Dealer D	48.6	51.6	Offer	2.5
Dealer E	49.5	52.5	Offer	0.0
Dealer F	50.0	53.0	Bid	5.5
Dealer G	50.0	53.0	Offer	0.0
Dealer H	51.0	54.0	Offer	0.0
Dealer I	52.0	55.0	Bid	5.0
Dealer J	52.0	55.0	Bid	20.0
Total				18.0

Source: Creditfixings.com

From the bid-offers we arrive at the market mid-points. In Exhibit 3 we sort the bids in descending order and offers in ascending order separately and eliminate tradable markets.

exhibit 3

Stage 1: Arriving at Market Midpoint

Dealer	Bid	Offer	Dealer	
Dealer J	52.000	50.500	Dealer A	Crossed Markets
Dealer I	52.000	51.000	Dealer B	
Dealer H	51.000	51.500	Dealer C	Average top half of the remaining bids and offers
Dealer G	50.000	51.625	Dealer D	
Dealer F	50.000	52.500	Dealer E	
Dealer E	49.500	53.000	Dealer F	
Dealer D	48.625	53.000	Dealer G	
Dealer C	48.500	54.000	Dealer H	
Dealer B	48.000	55.000	Dealer I	
Dealer A	47.500	55.000	Dealer J	
Average	50.125	52.156		

Source: Creditfixings.com

Of the remaining 8 bids/offers, we take an average of the best half (rounded up) of the bids and offers, to get 50.125 as our inside bid and 52.156 as our inside offer. The inside market mid-point (IMM) is the average of these two rounded to the nearest 1/8th, ending up at 51.125 in this case.

Penalties. There is a penalty amount that dealers have to pay for being offside on the market – in this case, Dealer A and Dealer B submitted offers below the IMM, considering there is open interest of 18MM to buy bonds. The dealers have to pay the quotation times the difference between the IMM and the offer, so in this particular case this would be \$15,625 and \$3,125 (2.5MM * (offer – IMM)), as penalty, respectively.

exhibit 4

Stage 2: Establishing Auction Recovery

	Dealer	Offer	Size
17.5MM Limit Orders fully filled	Dealer C	50.125*	5
	Dealer A **	51.125*	2.5
	Dealer B **	51.125*	2.5
	Dealer C **	51.5*	2.5
	Dealer D **	51.625*	2.5
Limit orders partially filled on a pro-rata basis (0.5MM of 7.5MM)	Dealer B	52*	2.5
	Dealer E**	52.5^	2.5
	Dealer F	52.5^	5
	Dealer E	52.625	4
	Dealer F **	53	2.5
	Dealer G **	53	2.5
	Dealer H	53.5	2.5
	Dealer H**	54	2.5
	Dealer H	54	2.5
	Dealer F	54	4
	Dealer A	54.75	2.5
	Dealer I **	55	2.5
	Dealer J **	55	2.5
	Dealer F	55	3.5
	Dealer H	55	2.5
	Dealer E	55.5	5
	Dealer E	56	5

Source: Creditfixings.com

Note: * Limit orders that were filled. **Limit Orders that were derived from inside markets. ^ Limit Orders that were partially filled.

Stage 2. After stage 1 of the auction, market participants get a few hours to evaluate the information from stage 1 and submit bids or offers to close the open interest. Here, investors submitted offers on the bonds to close the net 18MM bid. The bids/offers for the pre-defined size (in this case the offers) from Stage 1 are carried forward to Stage 2. In the case of Masonite, the dealers are deemed to have offered 2.5MM of bonds at the offer levels in Stage 1. Next, the administrator will determine the final price from the auction by matching market orders until the open interest is cleared. In the case of Masonite, 52.5 was the clearing price (see Exhibit 4).

Restructuring. In the case of a mod R or mod mod R restructuring event, the auction is conducted by collapsing bonds and outstanding CDS into several buckets ensuring that the maturity limitation restriction we highlighted earlier is not violated. From a protection buyer perspective, the "Limitation Date" means the first of March 20, June 20, September 20 or December 20 in any year to occur on or immediately following the date that is one of the following numbers of years after the restructuring date: 2.5 years (the "2.5-year Limitation Date"), 5 years (the "5-year Limitation Date"), 7.5 years, 10 years, 12.5 years, 15 years, 20 years or 30 years, as applicable. For each bucket, the restructuring maturity limitation we discussed earlier applies and an auction is conducted for buckets with sufficient CDS outstanding.

The Credit Volatility Culture – An Options Primer

May 7, 2010

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From an investment and portfolio management perspective, credit as an asset class has evolved quite a bit over the past few years. Prior to the financial crisis, much of the risk management effort in credit was focused on default risk, but not necessarily market risk, which is why the CDS market developed quite rapidly while a corresponding options market did not. Since about 2003, there has been some demand for hedging credit market risk as well, and this demand accelerated at the peak of the credit crisis, leaving us a market today that is broader and deeper with much room for growth.

Many core credit investors are able to hold healthy credit instruments on an accrual basis, and as such are more concerned about default risk than market risk. But the investor base and credit culture have changed dramatically over the past few years, as many non-financial institutions have entered the market as buyers of credit risk. For this community, credit is no different than equity in that market volatility is something to be concerned about. Furthermore, CDS usage among both financial and non-financial institutions is generally on a mark-to-market basis, so investors who may use CDS to hedge default risk in corporate bonds and loans (held on an accrual basis) may still introduce credit volatility into their portfolios. As such, credit volatility has become important for a large group of investors, and the credit crisis underscored that premise.

A credit options (also known as credit swaptions) market started as early as 2003, when options on the then-standard US corporate CDS indices began. Pre-crisis, popularity was larger in Europe where liquidity focused on the iTraxx XOver index first (high yield) and eventually iTraxx Main (investment grade) as the crisis intensified. But the market's largest growth began in 2008, in the depths of the credit and financial crisis. Today, the credit options market is fairly active on index products, with options trading on iTraxx XOver, iTraxx Main, CDX IG and HY indices with good liquidity, and SOVX, iTraxx Senior financials and options on the Asian indices also trading in periods of stress.

We have a few years of experience now in credit options, encompassing credit markets before, during and after the credit crisis. Although credit has normalized relative to the crisis, we expect to see continued development of an options culture in credit as the lessons learned from the crisis are secular in our view. Typical investment strategies can be bullish or bearish directionally, and long or short market volatility as well.

Much like equities, we expect to see credit investors lean toward two classes of strategies: downside hedging and income generation. As such, we think the equity market options culture is a good framework for credit investors to understand, keeping in mind that the underlying return potential of credit can differ significantly from that of equities. Credit, especially investment grade credit, is more skewed to the downside at normal spread levels. As such, we would expect the option skew to follow a similar pattern in that there should be more demand for OTM puts or payers than OTM calls or receivers. But despite a more normal distribution of returns in equities, we still see this style of skew in option pricing, owing to the market's demand for hedging and overwriting.

In contrast to considerable hedging flows, a culture of overwriting and generating yield through credit options strategies is less developed – mainly because most credit investors are naturally long risk assets anyway and standalone returns had been attractive coming out of the crisis. This may change if we see risk premiums compress further and indices start trading in a range-bound environment. Many investors look at different ways to monetize a lower volatility environment, and we remind investors that the culture of callable bonds is quite well established in credit; hence, there is precedent for option-related income generation strategies.

In this primer, we trace the evolution of the credit options market through the crisis, examine lessons learned and discuss how this market might develop. We also include a basic primer, describe options nomenclature and common option strategies, and conclude with a discussion of option sensitivities (i.e., the Greeks) and pricing models. We note that all options discussed in this primer trade OTC.

CREDIT VOLATILITY IN CONTEXT

Volatility is a measure of market uncertainty, and data and analysis of the path of volatility in the equity markets demonstrates that the level of uncertainty has a strong cyclical component, rising during periods of weak growth or recession while staying low during periods of normal growth. This is of course intuitive: uncertainty should be higher when we are “outside” the normal market periods and lower when “inside.” And since periods of positive economic growth generally occur more frequently than do recessions, we can justify higher levels of uncertainty and market volatility in cyclically weak periods (see [Volatility in a Recovery – Views for a Turning Cycle](#), September 16, 2009).

Given this cyclical link, volatility across markets has a tendency to be well correlated over long periods of time, especially when considering broad market measures of such

volatility (see Exhibit 1). And given that this volatility can be negatively correlated to the returns of risky assets, many market participants consider volatility to be an asset class unto itself, with important diversification properties.

Indeed through the crisis, we have seen equity investors fishing for credit hedges and credit investors chasing equity hedges, in an attempt to source cheaper or more effective hedges. There are certainly many things the credit options market can learn from the far more mature equity options market. However, there are also many elements that are quite unique to credit options.

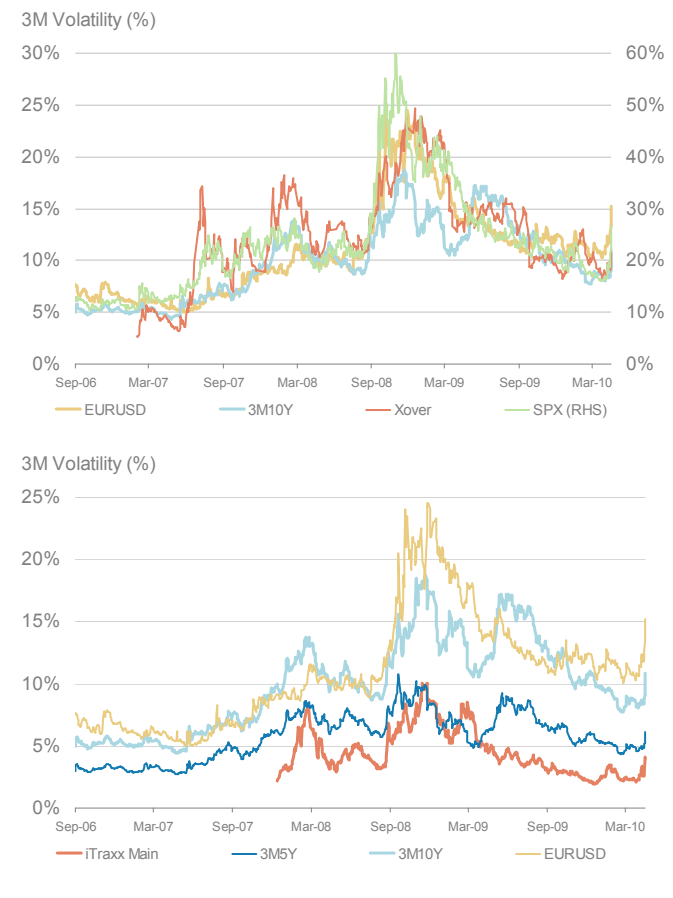
Given that much valuation in credit is done with spreads instead of prices, credit options for IG and iTraxx Main are quoted as spread options, which is similar to the rates markets. Spread options are calculated as the annualized average daily log changes in spreads. Price options, on the other hand, are used for CDX HY, and this type of option quotation is more analogous to the equities and FX markets. For most of our purposes here, we have used price volatility for ease of understanding and comparison.

Most of the activity in the credit options market is relatively short-dated compared to the other markets. Credit index options have standard expiries that correspond to the quarterly CDS payment months, with other expiries gaining liquidity intermittently. For instance, in May of a particular year, the June, September, and potentially even December expiries should trade, with potential liquidity in some May and July expiries as well, and the expiry date is the third Wednesday of the month for credit options. The options that are traded in the market today tend to have expiries that are generally less than one year (3m and 6m options tend to be the most liquid). Credit options investors and dealers prefer options on the OTR index for better liquidity in the underlying and ease of risk management, though occasionally the legacy series will trade, particularly around roll dates.

In Exhibit 1, we illustrate credit volatility compared to FX, and highlight that XOver and HY volatility has traded in ranges that are similar to FX volatility (EURUSD or JPYUSD), though the range of volatility in HY has been wider than FX. Part of the reason for the more frequent and higher spikes in HY vol is the asymmetric nature of the credit asset class.

exhibit 1

Price Volatility Across Asset Classes



Source: Morgan Stanley Research, Bloomberg

For IG vol (we use iTraxx Main for the longer history on the European side), the closest comparable volatility metric is 5yr swaption vol. As seen in Exhibit 1, during the crisis IG vol reached a peak similar to that of swaption vol (around 10%), but also healed quite dramatically relative to swaption vol.

The other aspect that is different for credit is the carry, which is quite high for credit relative to the yield. HY credit, with a 5% coupon, has the largest “dividend” among the assets, compared to dividend yields of 2% or less for US equities and less than 4% for European equities or carry differentials in currency. Much of the return at tighter spread levels for credit is generated through carry and hence overwriting strategies make a lot of sense in credit.

exhibit 2

State of the Credit Options Market

	Coupon/ Dividend	Current	3M Implied Vol		3M Realized Vol			3m Vol	6m Implied Vol
			High	Jun-07	Current	High	Jun-07	Carry	Current
Credit									
iTraxx Main	1.0%	4.9%	10.1%	0.6%	2.7%	6.3%	0.4%	2.2%	4.5%
CDX IG	1.0%	4.6%	11.2%	n/a	2.8%	8.9%	0.6%	1.8%	4.5%
iTraxx XOver	5.0%	14.6%	24.6%	2.6%	9.0%	16.5%	4.6%	5.6%	n/a
CDX HY	5.0%	18.1%	n/a	n/a	12.0%	28.4%	6.3%	6.1%	n/a
Equities									
S&P 500	2.1%	26.7%	63.1%	12.8%	14.2%	72.3%	9.8%	12.5%	26.8%
Russell 2000	1.3%	32.8%	68.9%	18.0%	17.3%	79.3%	13.6%	15.5%	33.6%
Eurostoxx	4.2%	30.3%	61.8%	15.0%	18.4%	65.9%	13.7%	11.9%	29.2%
DAX	3.4%	25.2%	61.1%	16.2%	15.8%	63.7%	14.3%	9.4%	25.2%
FTSE	3.9%	23.4%	60.7%	13.0%	13.0%	62.9%	11.1%	10.4%	23.8%
FX									
EURUSD	0.3%	15.2%	15.2%	24.5%	10.0%	10.0%	22.2%	5.2%	15.1%
GBPUSD	0.3%	14.8%	14.8%	25.5%	10.9%	10.9%	22.7%	3.9%	14.5%
USDJPY	0.3%	15.3%	15.3%	33.5%	13.8%	13.8%	20.4%	1.5%	15.3%
Rates									
USD 5yr Swap	2.4%	6.1%	10.8%	2.7%	4.3%	10.8%	2.9%	1.8%	6.2%
USD 10yr Swap	3.4%	10.8%	18.7%	4.6%	7.8%	18.6%	4.1%	3.0%	11.0%

Note: All Vols are Price Vols. For credit and rates, it is the spread vol * spot * duration.

Source: Morgan Stanley Research, Bloomberg.

CREDIT VOLATILITY INTUITION

As we discussed above, credit spread volatility (implied by index option pricing) is a much different concept than the price volatility that most investors are used to observing in the equity and FX markets, and is instead similar to the lognormal volatility in the rates market. Here, to compare volatility over time or across different markets, investors look at normalized volatility, which is calculated as the lognormal vol multiplied by the level of rates. We must also consider duration, an additional factor in the credit market used to convert normalized volatility into price volatility.

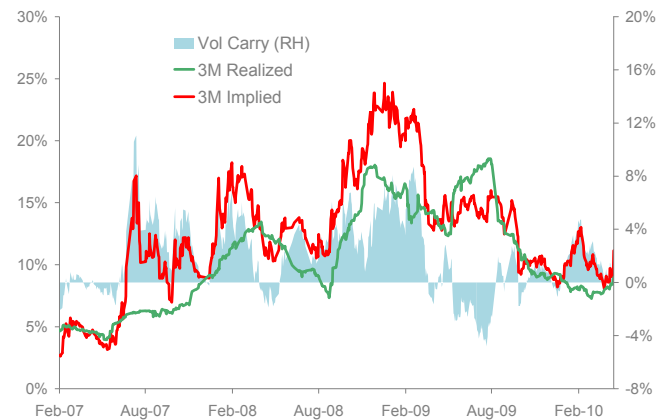
To illustrate, we offer the following example. Say implied volatility on a particular index with a spread of 200bp is 60%. This means that annualized volatility on the index is 60%, which would be equivalent to a daily implied volatility of about 3.72% (60% divided by square root of 260 business days in a year). This is equivalent to a daily spread move of 7.4bp (3.72% times the index spread of 200bp). This, in turn, translates into a standard deviation of daily price moves of about 0.33% (7.4bp times an approximate duration of 4.5). This is equivalent to an annualized price volatility of 5.3% (0.33% times the square root of 260 business days). In Exhibit 2, we show price volatility of a variety of asset classes.

Credit is unique among asset classes in that the upside/downside is very skewed: there is a very high likelihood of a small upside and a very small likelihood of a large downside. This has implications for the volatility carry, skew and the term structure of credit. The out-of-the-money (OTM) and asymmetric nature of credit compared to the equity or FX markets makes it prone to volatility spikes in periods of stress, superimposed over long periods of exceptionally low volatility. Credit spreads have been quite volatile and fat tailed, with IG realized spread volatility ranging from 10–100% over the last several years.

Due to this asymmetric nature, the volatility carry (i.e., the gap between implied and realized volatility) can be substantial in credit. Implied volatility has been consistently higher than realized volatility for the credit indices historically, as investors are willing to pay a premium, keeping in mind these volatility spikes. This is particularly true for IG options – the low absolute cost attracts options buyers. Conversely, sellers of volatility generally need a higher premium over realized in IG options. HY volatility is likely to appeal to sellers of vol to a larger extent due to better absolute premiums.

exhibit 3

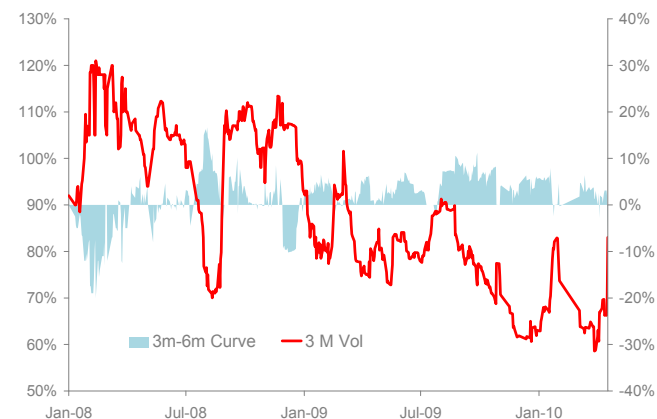
Volatility Carry



Source: Morgan Stanley

exhibit 4

iTraxx Term Structure



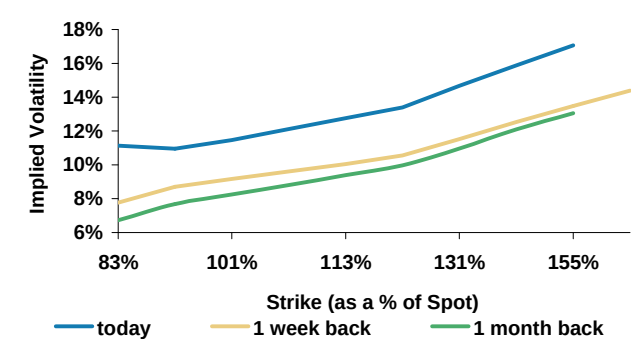
Source: Morgan Stanley

Another implication of the asymmetric nature of credit is that shorter-dated volatility tends to price in a premium to longer-dated volatility in calmer markets, reflecting the risk of periods of very high volatility. The gap between one-month and three-month volatility tends to widen significantly in periods of stress. Whenever this term discount for volatility disappears, it represents a period of relatively low risk premiums embedded in the options market, and in these markets we think that a strategy of buying short-dated options could perform well.

Not only do shorter-dated options trade at a volatility premium, there is also a significant element of skew embedded in OTM options. In particular, there is a strong technical demand for options to buy protection at the OTM strikes. Volatility tends to spike substantially when spreads widen, because even investors who are long credit risk use these OTM options as disaster insurance. Option skews have generally ranged between 10–15% for various options.

exhibit 5

Option Skew for XOver Options



Source: Morgan Stanley

HEDGING STRATEGIES WITH CREDIT OPTIONS

Even before the crisis, hedging was the dominant source of flows in the options market. Loan books, counterparty desks, and structured credit books were the predominant users of options for hedging. The most common strategy was buying OTM payers (puts). Post crisis, setting up a systematic hedging strategy has been one of the most common topics of discussion among investors. The ideal hedging strategies vary based on the scenario being hedged for, the level of volatility and skew, the term structure and the hedging budget.

One strategy is to hedge for smaller tail scenarios, which we define as 30bp wider on IG credit and 150bp wider on HY. In an environment in which realized volatility has been declining, term structures and skew are both steep. One common strategy to hedge for this type of scenario is to buy out-of-the-money puts outright. Put spreads are another effective way to position for such a move, and we have seen investors implement this strategy as well.

To hedge for larger tail scenarios, in which IG credit is 60bp+ wider and HY credit 300bp+ wider, longer-dated strategies might offer better value. Buying long-dated OTM puts is an effective strategy when the indices are trading at tight levels and volatility is relatively low.

To hedge for a market widening (either small or large tail) in periods when implied volatility is fairly high, investors try to reduce the cost of buying puts or put spreads by selling a call to monetize the high implied volatility. The resulting strategies – risk reversals (buy OTM put, sell OTM call) or put spread collars (buy put spread and sell OTM call) – have relatively low exposure to changes in implied volatility (or even short volatility in some scenarios) while retaining a directionally bearish exposure.

To compare the different type of strategies, we calculate the max downside and upside for each strategy based on the bear and bull case for the index. See an illustrative example in

showing the risk/reward of various strategies to hedge a small-tail scenario. Historically a reward to risk ratio higher than 2.5x is desirable and a ratio in 4-5x context for a realistic tail scenario could be considered very good.

YIELD GENERATION STRATEGIES WITH OPTIONS

The use of the options market for yield generation has been a relatively recent development. In environments where carry takes on a stronger role in driving credit returns, but volatility remains elevated, overwriting – i.e., selling call options against long positions as a way to enhance yield – should appeal to investors. Such strategies are common in equities, but less common in credit. In Exhibit 6, we show how much of an impact overwriting can have on realized yields in HY options as an illustration. For example, say an investor can sell a 4-month OTM call on CDX HY that earns an annualized premium of 270bp when CDX HY is at 550bp. Thus the premium is almost half the carry of the CDX HY in this particular example. This is effectively like buying a

callable bond with a “call price” of 100.8 (the breakeven level) which is nearly a 3-point premium to the price of 97.9.

exhibit 6

Overwriting in CDX HY and XOver

CDX HY		Call @ 99		Call @ 100		Call @ 101	
Expiry	Spread	Yield	Breakeven	Yield	Breakeven	Yield	Breakeven
1m	5.6%	8.8%	99.4	3.4%	100.1		
2m	5.6%	6.1%	99.8	3.8%	100.5	2.1%	101.3
4m	5.6%	3.9%	100.2	2.7%	100.8	1.7%	101.5
6m	5.6%	2.5%	100.4	1.8%	101.0	1.3%	101.7

Xover		Call @ 102.9		Call @ 104.0		Call @ 105.0	
Expiry	Spread	Yield	Breakeven	Yield	Breakeven	Yield	Breakeven
1m	4.4%	7.8%	103.2	1.7%	104.0		
4m	4.4%	3.9%	104.0	2.7%	104.8	1.7%	105.5

Assuming CDX HY Index Price is 97.9. XOver Index Price is 102.3. Breakeven level is effectively the “call price” of the package. Prices are indicative.

Source: Morgan Stanley Research

exhibit 7

Bloomberg CDS Option Pricer

GRAB
1<GO> to save Deal

Corp CDSO CPU: 0

CREDIT DEFAULT SWAPTION

Deal	Index Details	Straddles/Strangles
Counterparty:	Deal#:	Market Data
Ticker: /	Series: Privilege: User	Curve Date: 4/29/10
Underlying Index		Benchmark: S261 Mid
CDS Index: MARKIT ITRX EUR XOVER 06/15		CDS Spreads
Ticker: ITRX CDS Series: 13X5 Deal#: SPB84UNI		CDS SPB84UNI
Deal Spread: 500.000 bps Maturity Date: 6/20/15		Spread (bps): 455.747
Notional: 10.00 MM Factor: 1		Recovery Rate: 0.4000
Principal: -182,031.13 Loss: 0.0000		
Accrued: -54,166.67 Currency: EUR		
Cash Amt: -236,197.80		
Option		
Buy/Sell Protection: ☐ Payer Swaption		
Exercise Type: European Knock Out: N		
Strike Sprd: 460.000 bps		
Start Date: 4/29/10 Cash Settlement: 5/ 4/10		
Expiration: 6/16/10 Exercise Settlement: 6/17/10		
Calculator		
Valuation: 4/29/10	Sprd DV01: 2,453.68	Delta: 0.5864
Sprd Vol: 40.000%	IR DV01: -17.25	Gamma (+10 bps): 0.0574
Premium: 1.32545%	ATM Fwd: 471.260 bps	Vega (1%): 2,667.66
MTM: 132,545.16		Theta: -848.40

Australia 61 2 9777 8600 Brazil 5511 3048 4500 Europe 44 20 7330 7500 Germany 49 69 9204 1210 Hong Kong 852 2977 6000
Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2010 Bloomberg Finance L.P.
0675-955-0 29-Apr-10 12:19:55

Source: Morgan Stanley

PRICING MODELS AND INDEX OPTIONS

CDS options can in theory be on any CDS instrument, although in practice the focus in the market is on CDS indices. A simple model based on Black’s formula typically suffices for the pricing of a European option on a single-name CDS. However, index options and single-name options have a key difference in the case of credit events. The option buyer in an index contract continues to retain the option to buy or sell protection on all the remaining names in the event of a default of one of the names. The convention for single-name CDS options is to trade with knock-out, i.e., the single-name option

CDS option specifications include:

- The underlying CDS contract, usually 5-year maturity.
- Strike level specified for the spread (typical range 90–130% of current spread).
- Typical option expiry (liquid maturities extend from one month up to six months).
- The options are European and exercisable only on expiry date and into underlying CDS (physical or cash settlement).
- Options are quoted in bp upfront, i.e., the price of the option is paid upfront. No other cash flows are exchanged until the expiry.

extinguishes upon a credit event. Thus, adjustments may be necessary when applying the same model to index options. The market generally uses various adaptations of the Black model. A simple model is publicly available on Bloomberg (type <CDSO>) for basic pricing and analyzing sensitivities. See Exhibit 7 for more on CDSO.

The convention in the market is for CDS index options to trade “without knock-out” which means that even if a credit does default, the performance of the rest of the index constituents is also important when determining whether or not to exercise

the option. To illustrate, assume an investor has bought deep OTM payers with a strike of 325bp on a 100-name index trading at 200bp. In a scenario where a default occurs with a 40% recovery, the defaulted name is worth 0.6% in P&L (or approximately 15bp in spread terms) for the index, and spreads on the remaining names widen 100bp on average. In this scenario, the net P&L is not enough to make the exercise attractive. Post exercise, the buyer of protection can monetize the default by settling the credit event.

The spread dispersion and number of credits within the index also matter. The pricing of single-name vs. index options has a significant difference, particularly for the OTM payers. In the case of a XOver credit (which has 45 names), single-name OTM payers can be 10-20% more valuable than an identical index option, depending on the strike and the volatility levels due to the default P&L. However, beyond a point, increasing the size of the index does not have a large impact on the pricing of options — for example, increasing the size of the XOver index beyond 50 names has little effect on the price.

Bottom line, the buyer of an index payer option benefits from the implicit protection that is in the option. As such, the cost of a payer option will include the cost of protection for the term of the option. This cost of protection will actually vary depending on the strike, as the protection is less valuable the more out-of-the-money the strike is. Receiver options will not include this implicit cost of protection as recoveries are generally expected to be below par, so there will always be a difference between the full cost of a receiver option and a payer option (over and above what put-call parity would imply).

OPTION CHARACTERISTICS AND GREEKS

Some of the common metrics used to measure and manage risk of an option position are:

Delta: Delta is the ratio of the change in the price of the option to that of the market value of the underlying index. This varies from 60% for in-the-money payers to 30% for the more OTM payers, although it depends on the spread level of the underlying index. Delta (of OTM options) is higher for longer maturities compared to shorter maturities.

Gamma: Gamma, or spread convexity, is the change in delta for a 1bp change in the CDS curve. In a higher volatility environment, option gamma falls and eventually becomes unattractive vis-à-vis simple delta exposures. Shorter-dated at-the-money options have the highest gamma.

Vega: Vega is the change in the value of the option for a 1% change in volatility. The vega of an option increases as volatility levels rise. The vega of longer-dated options is higher than shorter-dated options. The vega of the option can contribute significantly to the P&L of the option.

Theta: Theta, or time decay, is the change in value of the option one day closer to expiry. An option with higher

convexity and sensitivity characteristics tends to have higher theta. While the theta of the longer-dated ATM and OTM options is largely similar, that of shorter-dated options drops dramatically as the option goes out of the money.

A typical trader run specifies the maturity, strike, vol and index level used for exchange of deltas. In Exhibit 8, we show the sensitivities for a typical HY run on 3-month and 6-month options.

exhibit 8

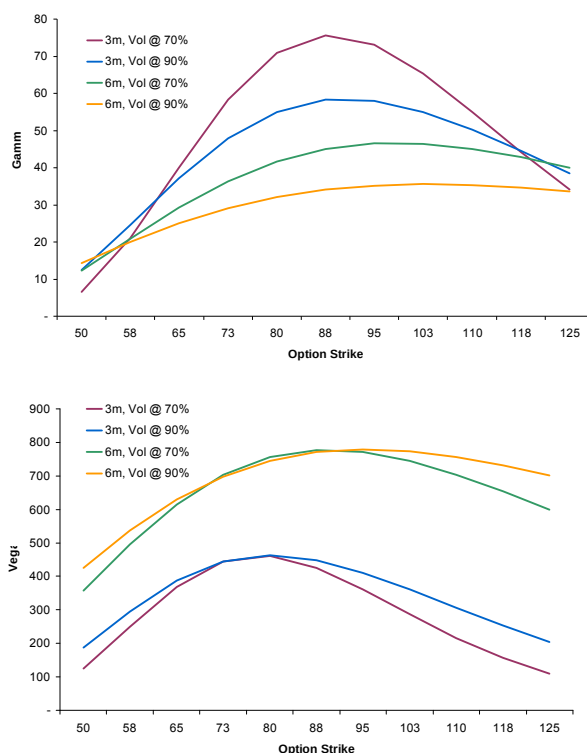
Payer Option Greeks

Expiry	Strike	(%)	Vol	Mid	Delta	Gamma	Theta	Vega
3m	375	85%	69%	343	2,995	7	(2,030)	1,678
3m	400	91%	71%	284	2,710	9	(2,245)	2,041
3m	450	102%	75%	192	2,113	11	(2,448)	2,429
3m	500	113%	79%	130	1,590	11	(2,385)	2,426
3m	550	125%	83%	91	1,185	9	(2,177)	2,210
3m	600	136%	85%	61	865	8	(1,852)	1,895
6m	375	85%	62%	468	2,928	5	(1,442)	2,808
6m	400	91%	62%	411	2,755	6	(1,485)	3,206
6m	450	102%	64%	321	2,375	7	(1,538)	3,795
6m	500	113%	66%	252	2,011	7	(1,527)	4,072
6m	550	125%	66%	192	1,673	7	(1,437)	4,098
6m	600	136%	68%	153	1,396	7	(1,350)	3,954

Note: Greeks calculated on 10MM notional position in the option.
Source: Morgan Stanley

exhibit 9

Option Greeks



Source: Morgan Stanley

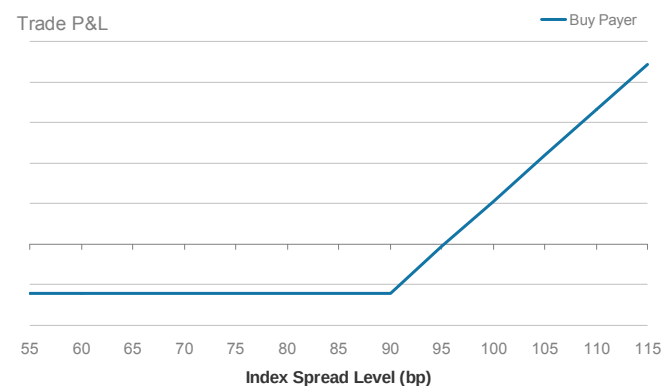
OPTION BUILDING BLOCKS

Credit options are fairly simple instruments, and even though credit and equity volatility have stronger, more direct fundamental linkages, the lingo is probably more analogous to rates options. At the most fundamental level, credit options give the holder the right to enter into a certain credit default swap at a certain time in the future. We discuss the basic option types and strategies in specific detail in this section (all exhibits show P&L of option strategies for changes in underlying spread, assuming a 10MM notional). As in equities, two basic option types form the building blocks for more complex strategies:

Payer (Put): An investor can buy (or sell) a ‘payer’ option, which is simply an option to buy protection at a fixed spread level on a certain notional at a fixed point in the future. Credit payer options are similar to payer options in interest rates wherein the investor hopes to benefit from wider spreads. They are analogous to equity put options in that the investor is buying (or selling) the right to short risk in the market, in this case by buying (or selling) the right to buy protection.

exhibit 10

Payer Option



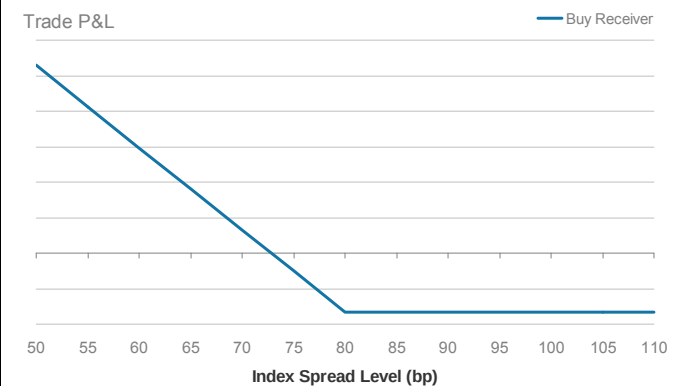
Source: Morgan Stanley

A payer option whose strike level is wider than the index level is referred to as an out-of-the-money (OTM) payer. A payer option whose strike level is the same as the index level is referred to as an at-the-money (ATM) payer. A payer option whose strike level is lower than the index spread level is referred to as an in-the-money (ITM) payer.

Receiver (Call): An investor can also buy (or sell) a ‘receiver’ option, which is an option to sell protection at a fixed spread level on a certain notional at a fixed point in the future. Credit receiver options are similar to receiver options in interest rates wherein the investor hopes to benefit from tighter spreads. They are analogous to equity call options in that the investor is buying (or selling) the right to go long risk in the market, in this case by buying (or selling) the right to sell protection.

exhibit 11

Receiver Option



Source: Morgan Stanley

A receiver option whose strike level is lower than the index level is referred to as an out-of-the-money (OTM) payer. A receiver option whose strike level is the same as the index level is referred to as an at-the-money (ATM) payer. A receiver option whose strike level is wider than the index spread level is referred to as an in-the-money (ITM) payer.

Using these two basic option types, we review some of the standard option combinations used to express different views on directionality and volatility of spreads. We have classified these strategies into:

- expressing directionally bullish views
- expressing directionally bearish views
- expressing long volatility views (typically negative carry trades)
- expressing short volatility views (typically income generating)

exhibit 12

Directional Option Strategies

	Market View (Delta)	Volatility (Vega)	Convexity (Gamma)	Time Decay (Theta)	Upside	Downside
Buy Payer	Bearish	Long	Long	Hurts Position	No cap	Capped
Payer Spread	Bearish	Neutral	Neutral	Neutral	Capped	Capped
Sell Receiver	Bearish	Short	Short	Helps Position	Capped	No cap
Covered Short	Bearish	Short	Short	Helps Position	Capped	No cap
Bear Risk Reversal	Bearish	Neutral	Neutral	Neutral	No cap	No cap
Buy Receiver	Bullish	Long	Long	Hurts Position	No cap	Capped
Bull Spread	Bullish	Neutral	Neutral	Neutral	Capped	Capped
Buy Write	Bullish	Short	Short	Helps Position	Capped	No cap
Sell Payer	Bullish	Short	Short	Helps Position	Capped	No cap
Bull Risk Reversal	Bullish	Neutral	Neutral	Neutral	No cap	No cap

Volatility Strategies

	Market View (Delta)	Volatility (Vega)	Convexity (Gamma)	Time Decay (Theta)	Upside	Downside
Buy ATM Straddle	Neutral	Long	Long	Hurts Position	No cap	Capped
Buy Strangle	Neutral	Long	Long	Hurts Position	No cap	Capped
Short Butterfly	Neutral	Long	Long	Hurts Position	Capped	Capped
Short Condor	Neutral	Long	Long	Hurts Position	Capped	Capped
Cost-neutral Spreads	Neutral/Bearish	Long	Long	Hurts Position	No cap	Capped
Long Butterfly	Neutral	Short	Short	Helps Position	Capped	Capped
Sell ATM Straddle	Neutral	Short	Short	Helps Position	Capped	No cap
Sell Strangle	Neutral	Short	Short	Helps Position	Capped	No cap
Long Condor	Neutral	Short	Short	Helps Position	Capped	Capped
Ladder	Neutral/Bearish	Short	Short	Helps Position	Capped	No cap

Source: Morgan Stanley

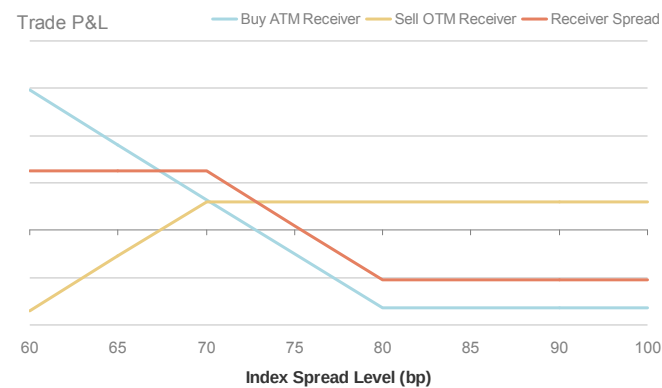
DIRECTIONALLY BULLISH STRATEGIES

The primary issue that options veterans have with buying receivers outright to express a bullish view is that volatility generally declines when markets rally. As a result, a buyer of receivers does not get to monetize the full impact of making the right call on the direction of the index (gamma and vega working at cross-purposes). Thus investors look for strategies that take the sting out of a potential decline in volatility or even aim to benefit from a drop. Common strategies investors use to express a directionally bullish view include (1) buying receivers (calls) outright; (2) buying receiver spreads (call spreads); (3) buying 1x2 receiver spreads (buy 1 ATM receiver and sell two further OTM receiver spreads); and (4) bullish risk reversals (buy OTM call and sell OTM put).

Bull Spreads (Receiver Spreads): Spread trades are constructed by buying and selling the receivers but at different strike levels. A receiver spread is constructed by buying a receiver and selling a more OTM receiver. Receiver spreads (RS) express a bullish directional view but cap both the upside and downside relative to being long the index. Receiver spreads have relatively small exposure to changes in implied volatility, as the option positions are offsetting. Consequently, RS can be considered a more defensive trade compared to buying receivers outright.

exhibit 13

Receiver Spread



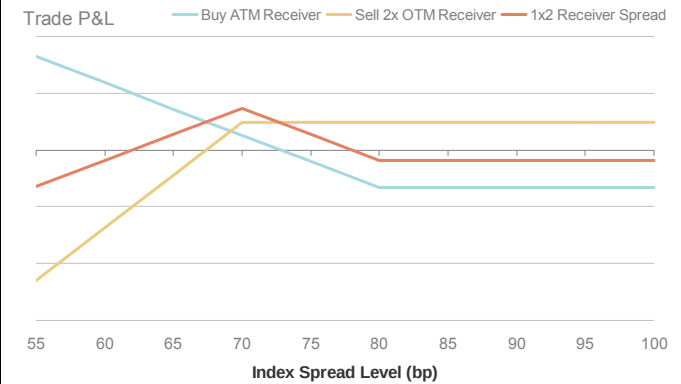
Source: Morgan Stanley

Buy a 1x2 Receiver Spread: 1x2 receiver spread trades involve buying an ATM receiver and selling twice the notional on an OTM receiver. In this trade, the investor is betting on a modest tightening of spreads but is exposed if indices tighten too far. The benefit to this trade is that the downside is reduced compared to the receivers or receiver spreads if spreads actually end up widening. 1x2 receiver spreads are short volatility and short gamma if there is considerable time to maturity. So the value on the 1x2 RS accretes over time and benefits from time decay. As a tighter spread environment in credit is generally accompanied by a

lower volatility environment, some of the short gamma position is offset by positive P&L from a decline in volatility.

exhibit 14

Buy 1x2 Receiver Spread

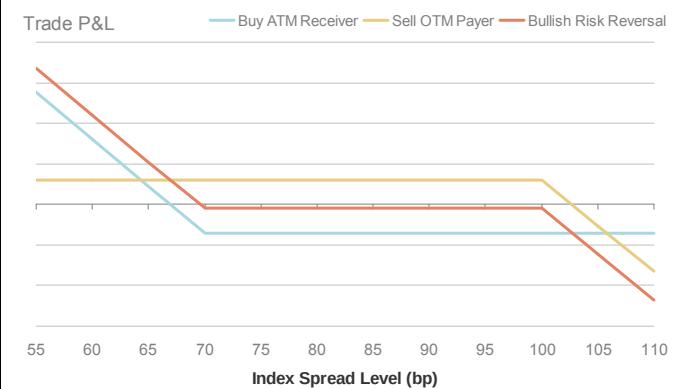


Source: Morgan Stanley

Buy Bullish Risk Reversals: Bullish risk reversals involve buying an OTM receiver and selling a deep OTM payer to finance. Although the opposite trade (bearish risk reversal) is the more popular counterpart, the bullish RR could be one way of getting around the problem of declining volatility in a rallying market. This trade is better suited for an environment with elevated spreads, limited downside for credit but an uncertain upside.

exhibit 15

Bullish Risk Reversal



Source: Morgan Stanley

DIRECTIONALLY BEARISH STRATEGIES

Due to the tendency of vol to spike in periods of market stress, the combined impact of the index move and the increase in volatility makes payers very popular, as gamma and vega work in tandem. Nevertheless, other strategies are useful in reducing volatility exposure and/or reducing cost, especially if the timing of sell-off is uncertain and the intensity of a sell-off is expected to be modest. Also in some

cases, volatility spikes may be expected to be unsustainable. Some of the common strategies investors use to express a directionally bearish view include (1) buying payers (puts) outright both with or without delta; (2) buying payer spreads (put spreads); (3) selling 1x2 receiver spreads (sell 1 ATM payer and buy two further OTM payer spreads); (4) bearish risk reversals (sell OTM call and buy OTM put); and (5) put spread collars (sell OTM call and buy OTM put spread).

Bear Spreads (payer spreads): Spread trades are constructed by buying and selling the same type of option but at different strike levels. A payer spread (PS) is constructed by buying a payer and selling a more OTM payer. PS express a bearish directional view but cap both the upside and downside relative to being outright short and are amongst the most common trades in the credit options world. They are particularly useful when volatility levels are high and outright payers are expensive. They are also popular if there is uncertainty regarding the timing of a widening or the expectation of a more modest widening.

exhibit 16

Payer Spread

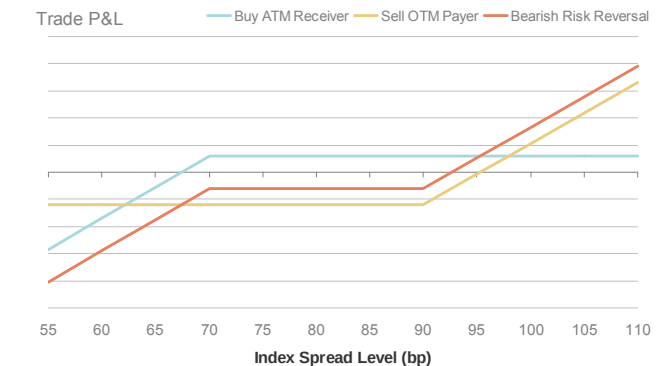


Source: Morgan Stanley

Bearish risk reversal: Bearish risk reversals involve buying an OTM payer and selling an OTM receiver to finance it. Bearish RR are also quite popular as they play to the theme of declining volatility in a rallying market and vice versa, which cushions the MTM if spreads rally. Bearish RR are long gamma in a sell-off and short gamma in a rally and have relatively muted volatility exposure.

exhibit 17

Bearish Risk Reversal



Source: Morgan Stanley

Payer spread collar: A payer spread collar (PSC) is constructed by buying a payer spread and selling an OTM receiver to further reduce the cost of playing a sell-off in spreads. PSC are less expensive than outright payer spreads but are short gamma and volatility if spreads tighten substantially. If spreads remain range-bound, this is a much cheaper trade than buying payers or payer spreads outright.

exhibit 18

Payer Spread Collar

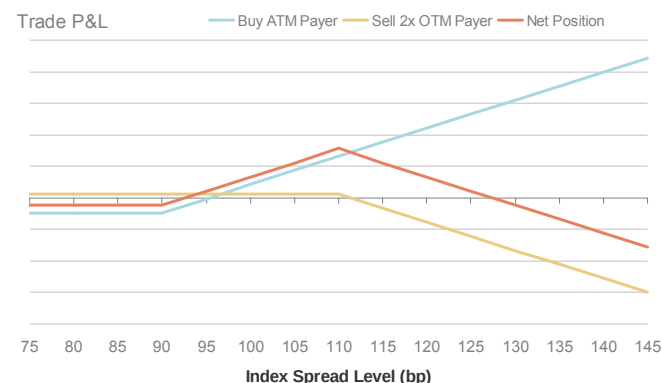


Source: Morgan Stanley

Buy 1x2 payer spreads: Buying 1x2 payer spreads involves buying an ATM payer and selling twice the notional on an OTM payer. In this trade, the investor is betting on a modest widening of spreads, but exposed to sharp moves wider. 1x2 payer spreads are short volatility and short gamma if there is considerable time to maturity.

exhibit 19

Buy 1x2 Payer Spread



Source: Morgan Stanley

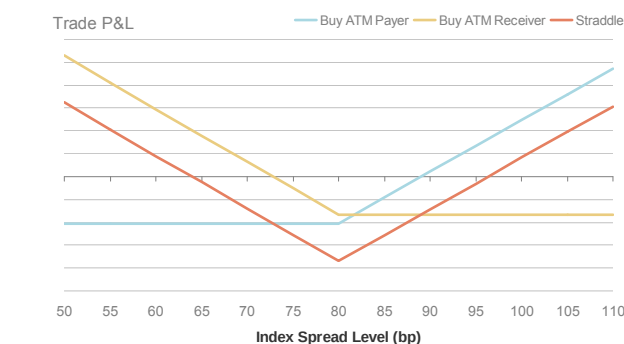
LONG VOLATILITY STRATEGIES

Long volatility strategies are typically negative carry trades – as implied volatility typically trades at a premium to realized volatility. These strategies therefore suffer with the passage of time to varying degree. Some common strategies used to express a bullish view on volatility (or big out of the range moves) include (1) buying a payer or receiver with delta; (2) buying straddles (buy put and call with the same strike); (3) buying strangles (buy OTM payers and OTM receivers); (4) selling 1x2 payer spreads (sell 1 ATM payer and buy 2x OTM payers); and (5) bearish risk reversals (sell an OTM receiver and buy an OTM payer).

Straddle: A straddle payoff is created by buying a payer and a receiver with the same strike price and expiry. Depending on the strike levels, straddles can be a play on volatility or bets on both the direction of any future price change and its magnitude. While at-the-money straddles struck around the futures price have low deltas, making those more of a volatility play, far-from-the-money straddles have high deltas and represent bets on the level of the underlying spread as well.

exhibit 20

Straddle

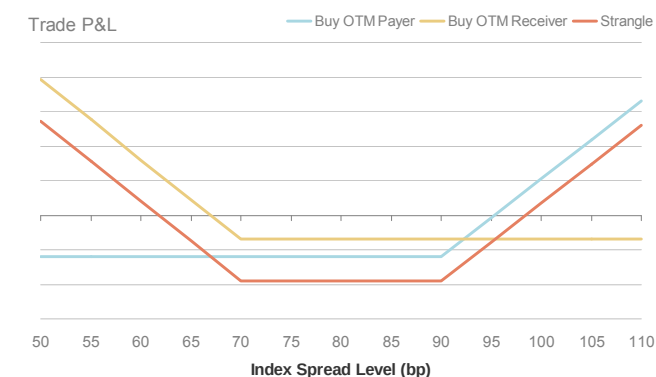


Source: Morgan Stanley

Strangle: A strangle payoff is similar in construction to the straddle, except that the payer option is struck at a wider spread than the receiver option with the same expiry. Again, at-the-money strangles struck around the futures price have low deltas, making them pure volatility plays; far-from-the-money straddles and strangles represent bets on the level of the underlying spreads as well.

exhibit 21

Buy Strangle

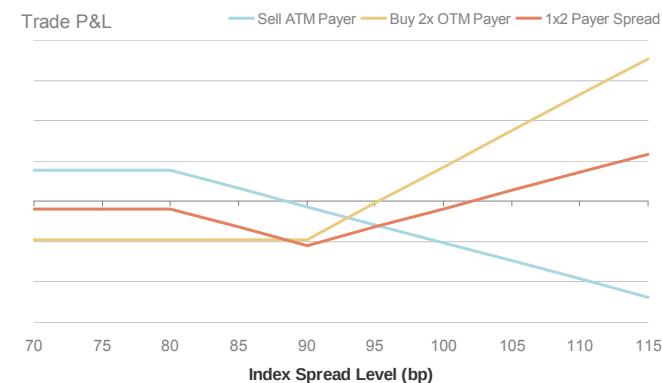


Source: Morgan Stanley

Sell 1x2 payer spreads: 1x2 payer spreads are a popular way of expressing a long volatility view. Unlike the straddles or strangles which benefit from a large move wider in either direction, the 1x2 PS benefit only from a large move wider in spreads and volatility. The worst case scenario for the seller of the 1x2 PS is a modest widening into the option expiry.

exhibit 22

Sell 1x2 Payer Spread



Source: Morgan Stanley

SHORT VOLATILITY STRATEGIES

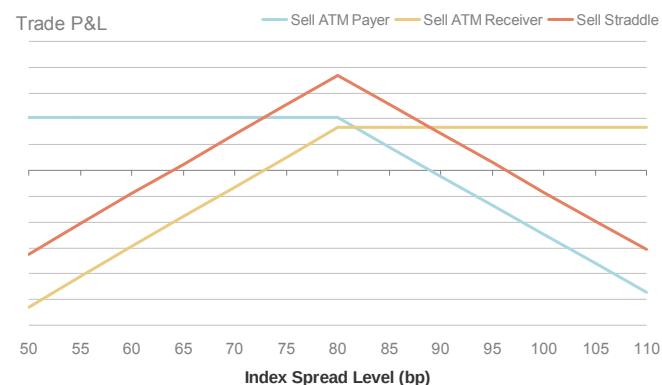
Some of the common strategies investors use to express a bearish view on volatility and/or range-bound markets include (1) selling a payer or a receiver with delta; (2) selling straddles i.e., sell put and call with the same strike; in addition investors may hedge the widening tail with a further OTM put; (3) selling strangles, i.e., selling OTM payer and OTM receiver; (4) buying 1x2 receiver spreads (buy 1 ATM receiver and sell two further OTM receivers); and (5) payer ladder (buy 1 ATM payer, sell 1x OTM payer and 1x deep OTM payer).

We differentiate between two types of trades: one which express a view that the implied volatility is too high (irrespective of views on the underlying) and the second that the underlying will be range-bound. The challenge with short volatility trades is that they are also short gamma – some of the trades below seek to achieve short vega exposure but remain relatively less short/neutral on gamma.

Sell Straddles: Selling straddles is a strategy for monetizing the high level of ATM implied volatility. The trade is short volatility and significantly short gamma.

exhibit 23

Sell Straddles

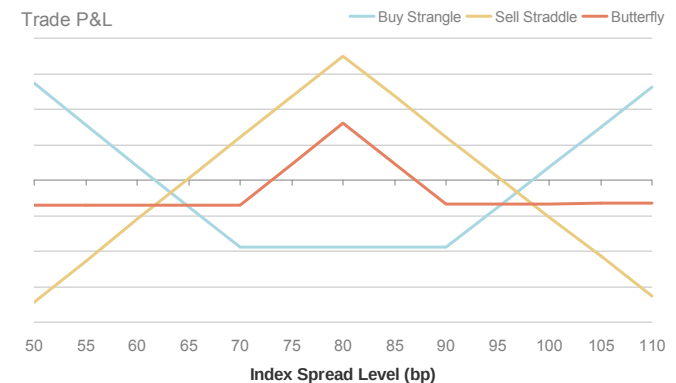


Source: Morgan Stanley

Butterfly: A long butterfly position is similar to a short straddle, but involves an additional hedging of the extremes by buying an OTM payer and OTM receiver. Butterflies express a view that spreads are likely to remain in a range, although they imply a willingness to pay some premiums to hedge the extreme outcomes. Butterflies are often used to bet the index will stay in a specific range rather than a bet on volatility.

exhibit 24

Butterfly

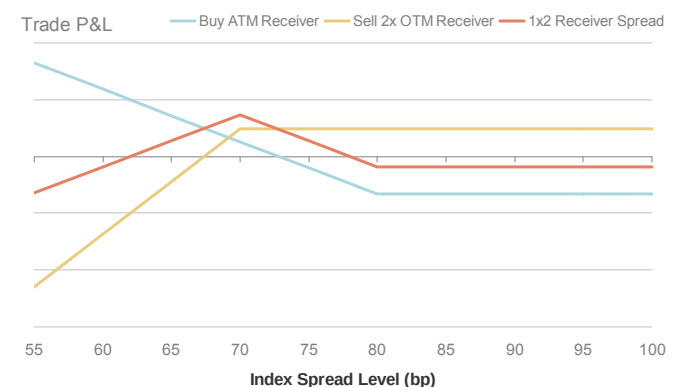


Source: Morgan Stanley

Buy 1x2 Receiver Spread: 1x2 receiver spread trades involve buying an ATM receiver and selling 2x of an OTM receiver. In this trade the investor is betting on a modest tightening of spreads but is exposed if indices tighten too far. 1x2 receiver spreads are short volatility and short gamma if there is considerable time to maturity. So the value on the 1x2 RS accretes over time and benefits from time decay. As a tighter spread environment in credit is generally accompanied by a lower volatility environment, some of the short gamma position is offset by positive P&L from a decline in volatility.

exhibit 25

Buy 1x2 Receiver Spread

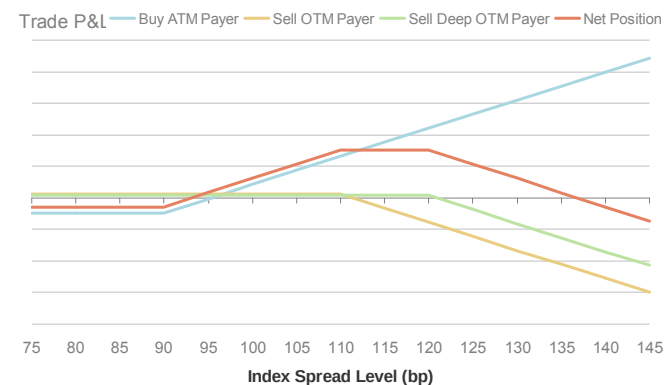


Source: Morgan Stanley

Buy Payer Ladders: Payer ladders (buy ATM payer, sell 1x each of an OTM payer and 1x deep OTM payer) are a more popular way of expressing a short volatility view than 1x2 receiver spreads due to lower short-gamma exposure thanks to spreading one of the strikes further out. Similar to buying 1x2 receiver spreads, the value in this trade accretes over time. The risk is a sharp and extreme widening in spreads rather than a gradual drift wider.

exhibit 26

Buy Payer Ladder



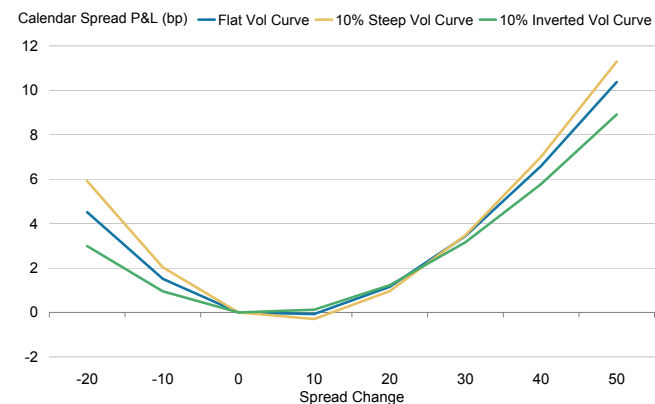
Source: Morgan Stanley

Calendar Spreads: Another popular way to express a view on volatility declining but not taking short gamma exposure is through calendar spreads. This would typically involve selling longer-expiry options (typically payers) and buying

excess shorter-dated options (again payers) to be relatively flat on convexity and short volatility.

exhibit 27

Calendar Spreads MTM



Source: Morgan Stanley

Tranche Primer – A Tale of Two Markets

July 22, 2010

Sivan Mahadevan
Ashley Musfeldt
Phanikiran Naraparaju

The credit crisis will have a lasting impact on the corporate credit markets, from the experience of hitting the historical wides in spreads and highs in credit volatility, to the lessons learned from housing and consumer credit contagion, to the significant changes we will continue to see with respect to market trading standards and financial regulation. While derivatives in general should continue to see much change, the corner of the corporate credit markets most affected by all of this has been tranches.

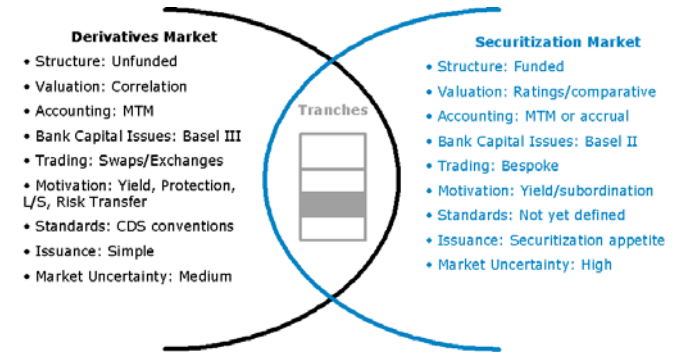
The tranche market is known by many names, ranging from the term itself (which means “slice” in French, as in *tranche de pain*, or slice of bread), to CSOs, synthetic CDOs, bespokes, or simply structured credit. Part of the reason for this complexity is that tranches live in two markets, namely the world of derivatives, where they act like long-dated options on defaults, and the world of securitization, where they represent an investment in a pool of credit with enhancement in the form of subordination to protect against defaults. To understand tranches, we must follow both of these perspectives, and in this updated market primer, we attempt to juxtapose both markets.

Before we dig into the tranche market, it is worth asking what direction this market may take given the significant fallout from the credit crisis. The future of tranches is in some sense a tale of these two markets, derivatives and securitization. Unlike Dickens’ novel, we do not necessarily see a stark contrast in the two sides of the market, as both face headwinds and tailwinds.

The securitized form of tranches (bespokes) was a large and influential market pre-credit-crisis, while the derivatives form was important (but probably less so). These roles reversed during the credit crisis, as market volatility introduced many trading opportunities, but depressed valuations on the largely mezzanine tranches on the securitization side. Today, that theme continues with much more activity on the derivatives side, and a growing stressed opportunity on the securitization side. We do not expect to see the securitization side of the tranche market become as big as it once was; however, more balance between both markets could make the whole space big enough to matter, depending a variety of outside factors, including the rules behind derivatives regulation and the general health of the securitization market away from corporate credit (mortgages, commercial real estate, consumer credit).

exhibit 1

Tranches and a Tale of Two Markets



Source: Morgan Stanley Research

In the remainder of this primer, we provide a market primer on tranches, taking both a derivatives approach and a securitization approach.

Derivatives Approach. In this approach the main sources of uncertainty are the impact of financial regulation and Basel III, although the latter is turning positive. Topics relating to this approach include:

- Index tranches
- Valuation: risk-neutral models/correlation
- Pricing types and variations
- Tranche sensitivity

Securitization Approach. Here, the main source of uncertainty is whether the appetite for securitization will be there going forward. Topics included under this approach are:

- A history of the bespoke new issue market
- Ratings models
- The legacy CSO market
- Basel and CPM themes

DERIVATIVES APPROACH: RATIONALE AND UNCERTAINTY

In the derivatives world, the rationale for using tranches is quite simple. They are long-dated options on defaults in an asset class (corporate credit) that has historically had significant risk premium embedded in credit spreads relative to actual defaults in the market. Interest is clearly two-way as investors both lean against market implied defaults (to earn yield), or position for defaults with potentially large payoffs

on thinner subordinate tranches. Bespoke activity is quite common, especially within subordinate tranches, and this is one part of the “new issue” tranche market that has been somewhat active since the credit crisis.

The index tranche market provides the most liquid point in tranches, and it is here where much relative value and correlation trading takes place. Much standardization progress has been made in the index tranche market in 2009, coinciding with the CDS trading standards that emerged at the same time. These standards have made netting of risks much easier, and the eventual clearing and exchange trading of such trades much more feasible.

DERIVATIVES APPROACH: STANDARDIZED INDEX TRANCHES

The emergence of standardized, tradable, and relatively diverse default swap indices in 2003 provided the opportunity for an on-the-run market for tranches to develop shortly thereafter. Today, the tranching index instruments are by far the largest liquidity point in the structured credit market and serve as an entry point for many investors. We covered the underlying indices on which the tranches are based more thoroughly in our single-name primer in Chapter 1. Some key characteristics of both the indices and index tranches are as follows:

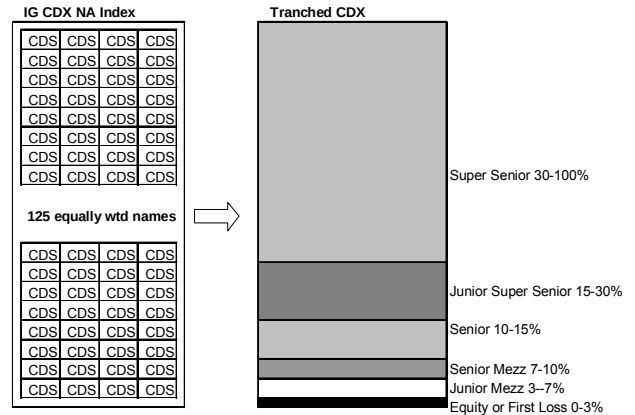
Static underlying portfolio. Once an index composition is fixed, no names can be added or deleted from the portfolio and all names are equally weighted, unlike common benchmark bond indices.

Standardized payment and maturity dates. Just like the single name CDS, the cash flow dates of indices and tranches are also standardized – the 20th of March, June, September, and December of every year with standardized maturity dates of the indices and tranches on the June and December dates.

Payment of accrued premiums. If an investor enters an index or tranche transaction in between the payment dates, the protection seller would make a payment of accrued premium to the protection buyer, to reflect the fact that the protection buyer would pay premium for the full quarter on the next payment date but the protection is in effect only for part of the quarter.

exhibit 2

CDX IG Tranches



Source: Morgan Stanley Research

Standard coupon. The indices have a predetermined “Deal Spread”, which is paid on a quarterly basis. Consequently, if the index is currently trading away from the deal spread, an upfront payment is required to reflect the difference between the current market spread level and the deal spread, equal to the present value of the difference between the two, adjusted for default probabilities. Tranches also have a standard coupon. Prior to the implementation of SNAC (Standard North American Contract, including small bang in Europe and big bang in the US), only the equity tranches and occasionally the junior mezzanine tranches traded with an upfront plus running coupon structure. Post-SNAC however, all tranches now trade with a standard coupon. See Exhibit 3 for a list of standard coupons on the various tranches.

exhibit 3

Standard Tranches and Coupons Across Indices

CDX IG		CDX HY		iTraxx Main	
Tranche	Coupon	Tranche	Coupon	Tranche	Coupon
0-3%	500	0-10%	All Upfront	0-3%	500
3-7%	500	10-15%	All Upfront	3-6%	500
7-10%	500	15-25%	500	6-9%	300
10-15%	100	25-35%	500	9-12%	100
15-30%	100	35-100%	500	12-22%	100
30-100%	100			22-100%	25

Source: Morgan Stanley

Index rolls vs. tranche rolls. New indices are introduced every six months and the latest series of the index represents the current on-the-run index. Initially, tranches referencing the indices were rolled as well, up until IG9 for CDX IG, S9 for iTraxx Main, and HY10 for CDX HY. While tranches were introduced on new series of these indices, for one reason or another specific to each index, the new tranches never gained much liquidity, and thus the main liquidity point at the time of this publication is in those series mentioned above. Liquidity in outright index trading as well as trading on options continues to be on the on-the-run index

series in each of these markets, however. Going forward, tranches will only roll once a year on odd numbered indices in the US.

Attachment and detachment points. The attachment point determines the subordination of a tranche. The first tranche, also referred to as the “equity” tranche, takes the first X% of the losses on the portfolio. When the portfolio has accumulated large enough losses to exceed X% of notional, the next tranche, X-Y%, will incur losses from any additional defaults up to Y% of losses, and so on. The detachment point determines the point beyond which the tranche has lost its complete notional. It is important to note that we are discussing total losses, not defaults – losses incorporate recovery value. The difference between the attachment and detachment points is referred to as “thickness” of tranche.

Restructuring definitions. US indices and thus tranches trade with no restructuring, whereas European indices and tranches trade with Mod Mod R.

Default settlement. Indices are settled via the standardized auction process, which we discuss in great detail in Chapter 1.

These are the standard terms regarding the mechanics of the index tranches. Pricing and valuation terminology are covered in the next two sections.

WHAT IS CORRELATION?

Before we address what correlation is, it is worth a few words to address why correlation is of interest to credit investors. Portfolio structured credit products are generally based on portfolios of assets whose performance is interrelated. For example, default rates are economically cyclical and entire sectors can suffer from downturns, resulting in higher-than-expected default experiences. These baskets of correlated assets behave differently from identical baskets of independent assets in several significant ways. Markets have long recognized these differences and prices have reflected the differing risks. Correlation is the missing link necessary to go from today’s standard derivative pricing models to the prices we observe in credit markets, or vice versa.

In financial markets, correlation is best understood using a statistical definition. Most often, correlation in financial markets describes the relationship of the price changes (or returns) of two different assets. This is the type of correlation many investors and asset allocators are most accustomed to, and it has long been discussed in financial literature.

Another example of this type of correlation is in the world of equity index options where we can examine the correlation between securities, rather than the correlation between a security and the market. Consider the market for options on the S&P 500 and all of the components. Traded option markets tell us that the weighted average 6 month implied

volatility for the portfolio is roughly 32% while the observed implied volatility for options on the actual index was roughly 26%. The difference is explained by the correlation among the components.

exhibit 4

Portfolio Volatility Affected by Correlation

Correlation	Index Volatility
0%	3%
50%	23%
100%	32%
68% (Index Option Implied)	26%

Source: Morgan Stanley

In Exhibit 4, we summarize the implied index volatility as calculated under several correlation assumptions, as well as the observed index volatility from the options market. If the credits are 100% correlated then there are no diversification benefits and the volatility of the portfolio is simply the weighted average of the components, which is 32%. If the components are independent of one another, there are significant diversification benefits, with an implied index volatility of 3%. The reality is somewhere in between, and in our example, the correlation implied by market pricing was approximately 68%.

In order to simplify the analysis above, we have used one common correlation variable for each pair of components and varied that number. In reality, this assumption of a single correlation between these assets is probably inadequate and the market pricing likely reflects the varying levels of correlation among the assets in the portfolio. While historical correlations among the returns of the index components are observable (because we have active equity markets for all the components), these are backward-looking estimates, like realized volatility.

CORRELATION IN STRUCTURED MARKETS

Correlation in tranches has a different meaning than in most financial markets. We highlight the key differences below:

Correlation as a model output. First, the correlation quoted in structured credit markets is generally implied from the market price of a tranche, under a certain set of assumptions using a given model. Therefore it is usually an output, rather than an input, and it is generally forward-looking rather than historical.

Timing of defaults, not mark to market. Second, and probably most important, the correlation we are actually measuring is not the correlation of the change in prices (or spreads) but rather it is most closely related to the correlation of the timing of default events within the portfolio. In contrast to equity derivative models, structured credit models generate values based on default probabilities derived from today’s market prices of the underlying instruments rather than the expected value based on the future market prices of

the underlying instruments. This is a fundamental difference in the way credit derivatives are valued as compared to derivatives in other asset classes.

Correlation between names. Third, the quoted correlation is a single number that represents the relationship between all the pairs of names in a given portfolio. Correlation is generally a concept that measures the relationship between any two assets. In standard credit correlation models, the same number is used to represent the relationship between every pair in the portfolio. When we say the price of the tranche has an implied correlation of 30%, we are essentially saying that the correlation for each distinct credit pair is implied by the market price to be 30%. While this definition does not allow us to vary correlation for each pair, it does provide a measure of the aggregate correlation in the portfolio.

For all but the most junior tranches, the aggregate correlation of a portfolio is a more important driver of pricing than the correlation between any two credits. For the most subordinate tranches, the correlation between a single pair of credits can be an important driver as well, given their exposure to idiosyncratic risk. For example, if the two widest trading names in a basket announced a merger for which the financing plan indicated increased leverage (and by implication, higher levels of default risk) for the joint entity, this would be doubly bad for the most junior tranches, while having only a marginal impact on the senior tranches due to the idiosyncratic nature of the risk. Default propensity increased only for the two names and not for the broader universe. Correlation jumped between only these two entities, but their correlation to the rest of the portfolio likely dropped.

Consider how muted the benefit would be to the same tranche if two of the most high-quality credits announced a credit-enhancing merger. Situations like this help, in part, explain the difference in implied correlations for actively traded tranches of the same portfolio.

VALUATION IN STRUCTURED MARKETS

Valuation models for tranches can vary, but they generally follow an approach to what we describe here. As with single-name credit derivatives, pricing for tranches or N-th to default baskets is a function of the single-name par spread levels, the spread at which the contract is struck, the risk-free interest rate and a recovery assumption, but we need to also add the following variables for tranches: attachment and detachment points (either in losses or in number of defaults) and the correlation of default events.

As with the single-name models, the spreads are converted into risk-neutral default probabilities, which is effectively the same as generating a loss distribution for the single-name credit. We can think of the single-name default probabilities as being represented by a very simple Bernoulli probability

distribution, where default for a given single name is distributed in the following way:

1 if default with probability p
0 if no default with probability $1-p$

Therefore, losses are distributed as follows, using a fixed recovery assumption:

$(1-\text{recovery rate}) \times \text{notional amount}$ if default with probability p
0 if no default with probability $1-p$

The key to valuing tranches of a given portfolio is the ability to generate a distribution for the number of defaults in the entire portfolio over a given time horizon. This, in turn, gives us the ability to calculate the probability that losses in the portfolio exceed given thresholds (like the attachment and detachment points of tranches). Once we have these probabilities of various levels of default for the portfolio, we can generate expected losses, cash flows and value tranches accordingly.

CORRELATION AS A RELATIVE VALUE METRIC

The language of implied correlation has evolved to provide a useful relative value metric in the form of the often quoted base correlation. The first time we heard the term “implied correlation,” it was used to refer to what we now call “compound correlation” – that is, the correlation implied by the price of a tranche, given all the other market inputs. The problem with compound correlation is that it fails to give meaningful results in some of the mezzanine parts of the capital structure. Additionally, the idea of using compound correlation as a relative value metric is even further complicated by the fact that rising correlation is good for some tranches and bad for others.

Base correlation, which uses the implied correlations from a series of first loss tranches with detachment points equal to the detachment points of the actual tranches, does not suffer from either of the two issues above. It does not have tranches for which there are multiple solutions because the correlations are implied from first loss tranches. Base correlation provides a good relative value benchmark because it is fairly uniform in its meaning (i.e., rising base correlation means tighter spreads on that tranche, all else being equal), and it provides a unique price for a given base correlation. Additionally, it provides price sensitivities that are more in line with what we have observed in the market when compared to compound correlation, which can imply unreasonable price sensitivity values for very high or very low implied correlation values.

exhibit 5

Tranche Equivalence – Long / Short Combinations

Benchmark Tranche	First Loss Tranche	Tranche Position	Implied Benchmark Tranche Delta
3-7%	0-3%	Short	8.8x
	0-7%	Long	
7-10%	0-7%	Short	5.2x
	0-10%	Long	
10-15%	0-10%	Short	1.9x
	0-15%	Long	

Source: Morgan Stanley Research

WHAT IS CORRELATION MISSING?

In structured credit markets, it has never been the case (at least to our knowledge) that all the tranches of a given index/portfolio traded to the same correlation. There are several likely explanations for this (aside from liquidity), but most relate to assumptions in today's standard models. The use of a single correlation number fails to capture the fact that there are stronger relationships between specific companies and even entire sectors. These subtleties are more important for the more subordinate parts of capital structures and can move prices in ways that force our single implied correlation metric to move away from the correlation in other parts of the capital structure.

Today's models are risk-neutral in nature and therefore assume that investors are indifferent when faced with a choice between risky assets that can be hedged and risk-free assets. Lower correlation for more subordinate portions of the capital structure can reflect the increasingly levered exposure to (and subsequent risk premium charged for) jumps in single-name prices or exposure to idiosyncratic risk. Another factor in the relative pricing of tranches simply represents the different supply and demand dynamics for instruments with differing risk levels. Ratings may add an institutionalized component, exacerbating these supply/demand dynamics and increasing the skew in correlation. Finally, most of today's standard models use the normal distribution (or the Student T) to generate correlated defaults. It is very possible that the assumed distribution introduces error into the models simply because the distribution is not the best approximation of the real world.

What happened during the credit crisis in 2007-2008? Correlation models indeed had problems during the credit crisis, driven mainly by extremely high market implied default rates and extreme underperformance of senior tranches as correlation rose owing to a systemic crisis. At one level, the directional change in market implied correlation was absolutely correct given the nature of systemic crises, and such market data was very telling at the time. However, many standard models broke as implied correlation levels rose above levels that made intuitive sense.

Have there been improvements in models to address these issues? First, there is no guarantee that any solution will work in all market environments, as market technicals can have an

upper hand. For example, in the weeks following the Lehman bankruptcy in 2008, put-call parity, perhaps one of the most basic properties in options pricing models, failed in the US equity options markets given a ban on short-selling of financial shares.

In credit correlation models, we see development along two paths, one much simpler than the other. On the simpler side, we note that market-implied prices for credit can give us expected losses, but most models must then break that down into a default rate, and a recovery rate, given that losses are not continuous but triggered by a credit event. But both of these rates are actually assumptions, as many different default and recovery rate combinations can lead to the same expected loss. Furthermore, history tells us that recovery rates tend to fall in high default rate environments, and vice versa, owing to shifts in actual asset values the bondholders have claim to. More advanced correlation pricing models are able to address these shifts in default and recovery rates and come up with better solutions in extreme environments such as systemic crises or severe recessions.

On the more complex side, we note that risk-neutral correlation pricing models tend to assume that expected losses are fixed based on the current underlying CDS market spreads. But we know that such spreads change continuously in the market, so modeling the "volatility" of these credit spreads could also be an improvement in credit correlation pricing, addressing some of the issues we brought up in the beginning of this section. Computational complexity increases dramatically in this context, but the rapid development of the credit options markets recently can provide the market with some valuable volatility information for correlation models that attempt to go down this path.

DERIVATIVES APPROACH: TRANCHE PRICING TYPES

There are 4 primary formats in which to trade tranches – all running, all upfront, PO, IO, and a fifth, upfront plus running, is a combination of two. While one should, in theory, be indifferent between them when priced accordingly, they are affected differently by default timing, and thus investors might have a preference for one over the other given default expectations.

To illustrate, we consider a mezzanine tranche trading at 400 bp, with a maturity of 7 years, and a corresponding duration of 6 and assume an investor goes long risk, or sells protection, on \$10mm notional. Additionally, we assume there are a certain number of defaults such that the tranche is written down by 10% of its original principal amount, a loss of \$1mm on the trade.

All running. This is the most simplistic form. To sell protection, the investor receives the fair value spread on a running basis, generally paid in quarterly installments. So in our example, the 400 bp is paid in 100 bp increments 4 times throughout the year, and no upfront costs are exchanged at

trade inception. Timing of default is very important here – the seller of protection would prefer defaults to be as late as possible, as the seller of protection loses income from the coupon stream with each default and also has to pay out losses to the buyer of protection each time there is a default.

All upfront. Also simplistic, the seller of protection receives the present value of the coupon stream all at trade inception. The calculation for the upfront amount is simply the fair value running spread times the duration, in this case $400 \text{ bp} * 6 = 24\text{pts}$. This calculation is less sensitive to timing of defaults than its all running counterpart, though the seller of protection would still prefer that they are later, given the loss amount that needs to be paid out. No coupon amount is at risk however, as this is exchanged entirely at trade inception.

Upfront plus running coupon. This pricing type is a hybrid of all upfront and all running in that the seller of protection receives a fixed coupon (100 bp and 500 bp are common) and the PV of the difference is exchanged at trade inception. The counterparty who pays the upfront amount is determined by whether the coupon is struck above or below the fair value spread. In our example where the credit is trading at 400 bp, the seller of protection would receive the upfront premium if the coupon were struck at 100 bp, and the seller of protection would pay the upfront if the coupon were struck at 500 bp. This type varies in sensitivity to timing of default between all upfront and all running, depending on what percentage of the risk is paid upfront.

PO value. Here, there is also an upfront payment; however, it is the seller of protection who pays, and then receives par minus losses at maturity. Upfront value should equal par minus the all upfront value minus the PV of the interest rate component, so in this case we have $100 - 24\text{pts} = 76\text{pts}$. Assuming the interest rate component is worth 4pts, the seller of protection would pay 72pts upfront. At maturity, the seller of protection would receive par minus losses, in our example losses are 10%, so the seller of protection would receive 90% of par. The PO investor is entirely indifferent to timing of default, as the timing of the payments are fixed: one at trade inception, the other at maturity.

IO value. Somewhat more complicated, in this type of pricing, the seller of protection pays par upfront and receives a very large coupon each year equivalent to the future value of the par premium plus the running fair value spread. So in our example, the investor would pay \$10mm upfront, and receive an annual coupon of 22%, which would compensate the investor for the original upfront in addition to the 400 bp of risky spread. This type of pricing is highly sensitive to the timing of default because the coupon is so large and the seller of protection has made an upfront payment in the amount of the total potential losses on the trade.

DERIVATIVES APPROACH: TRANCHE SENSITIVITY

We now attempt to put some math behind the intuition of how different tranches should behave in various scenarios and review some of the measurable parameters for this. Some of the basic concepts should be fairly familiar for investors from the world of equity/FX/rate derivatives, where option sensitivities are well-documented and understood. So concepts like delta (sensitivity to market move) and theta (impact of passage of time) are very similar. However, the tranches of credit portfolios have one additional dimension in terms of sensitivity measures – distribution of spreads in the index. In credit, it matters whether a 10% move in the index is achieved because of a uniform 10% widening of all names or, say, a 50% widening of select names or just a single name going to default. A fattening of tails in the portfolio will have a different impact on the various tranches. Exhibit 6 shows the basic definitions of various sensitivities for structured credit.

Using these metrics, we demonstrate how one could separate default risk from spread risk, and to take credit positions that become long when spreads are falling and short when spreads are rising. This performance emanates directly from differences in sensitivities of tranches. Also, the structure of the tranche, i.e., how much of the tranche value is paid as a coupon stream and how much is paid as an upfront payment (which we discussed previously in this primer), influences the tranche sensitivity. This is particularly relevant for junior tranches and we will address this separately.

exhibit 6

Tranche Sensitivity Metrics

Year (Index)	Key Events
Delta (δ)	Tranche price sensitivity to changes in underlying portfolio spreads, measured as a ratio of tranche PV01 to index PV01 (PV10% is also used sometimes)
M-Gamma ($m-\gamma$)	Tranche price sensitivity of a delta-neutral position to parallel shifts in spreads of underlying names. It represents a form of convexity (M = Market)
I-Gamma ($i-\gamma$)	Tranche price sensitivity of a delta-neutral position to jump-to-default risk or changes in spread distribution of the underlying portfolio. It represents a form of convexity to moves in a single credit while all others remain constant (I = Idiosyncratic risk)
Rho (ρ)	Change in tranche value due to changes in default correlation
Theta (θ)	Change in tranche value due to the passage of time

Source: Morgan Stanley Research

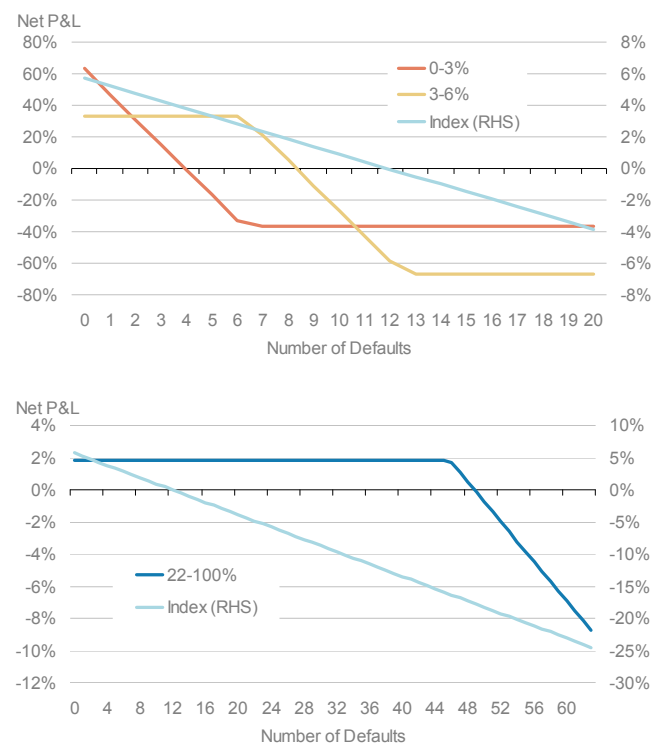
One way of thinking about tranches is to view them as options on losses from default in the underlying portfolio, which helps us develop a feel for the sensitivity of tranches to different variables, such as time decay, spread changes, default exposure, etc. We can easily estimate expected losses for a portfolio from its market spread level by simply taking the present value of the running spread level. So if a 5-year index is trading at 125 bp and the duration is 4.5, the all upfront equivalent is 5.6%. This can also be interpreted as the market implied expected loss on the index in a risk-neutral sense.

The tranche attachment and detachment relative to the index expected losses is a measure of the “moneyness” of the tranche and indicator of the tranche behavior. For example, 0-3% can be viewed as a deeply “in-the-money” tranche with respect to losses, as it detaches well below the index expected loss of 5.6% in our example. The 3-6% tranche would be considered “at-the-money” since it attaches below and detaches above the market implied expected loss. The more senior tranches would be considered “out-of-the-money” as they attach above the expected loss point.

As the index spread widens, index expected losses increase and senior tranches could expect greater risk of notional losses. Consequently, the subordination level that separates in-the-money and out-of-the-money tranches can and does change. Thus, junior tranches that trade near the expected loss can change character – behaving like senior tranches at low spread levels and equity tranches as spreads widen significantly. Additionally, different indices and even maturities within one index series can have different expected loss levels, causing the subordination level that separates the in-the-money and out-of-the-money tranches to be different in different maturities of the same index.

exhibit 7

Tranches Are Options on Default



Source: Morgan Stanley Research

For example, 10-15% tranches in CDX HY detach below the expected loss and are thus in-the-money, implying that they behave more like equity tranches, while the 15-25% tranche

behaves more like the investment grade mezzanine tranche. Similarly, in 5-year CDX IG, the 7-10% tranche is out-of-the-money, while it is in-the-money in the 10-year maturity, causing markedly distinct sensitivities. For this reason, as time passes and spreads roll down the credit curve, tranches can move from in-the-money to out-of-the-money and their sensitivities would also change.

SENSITIVITY TO SPREAD CHANGES OR “DELTA”

As the spread of the underlying index changes, the impact on different tranches varies. Overall, it is consistent with the index directionally, i.e., as index spreads widen, so do tranche spreads. For small spread movements, one can estimate the impact using tranche delta, which is defined as the ratio of change in the tranche value to change in the index for a given spread change. While it is usually calculated as the ratio of PV01s, it is increasingly common to use PV10%. (PV01 is the change in the mark-to-market for a 1 bp move in each of the underlying credits in the portfolio. PV10% is the change in mark-to-market for a 10% increase in spreads). However, for bigger moves in spreads, the delta-based calculation is only approximate, as the impact of tranche convexity becomes more meaningful, which we will discuss later.

Broadly speaking, junior tranches have higher deltas than senior tranches which makes sense, given that junior tranches take losses before senior tranches, and thus wider spreads should make them proportionally more likely to be affected by the increased likelihood of defaults implied by wider spreads. In other words, the “default PV” (present value of expected losses from default) rises faster than the index for junior tranches, but slower than the index for senior tranches, bringing the average default PV for tranches in-line with the index default PV. Junior tranches with their low cushions against losses are riskier tranches and would generally move more than the index or the senior tranches. Thus, junior tranches would typically have a delta >1, while the senior-most tranches have a delta less than 1.

The exception to the phenomenon of the tranche delta decreasing with seniority arises when spreads widen so much that the junior-most tranches become risk saturated and start pricing in a high likelihood of being fully written down. Therefore, the next tranche in line starts to absorb the impact of any incremental widening in spreads and has higher delta than the tranche below it.

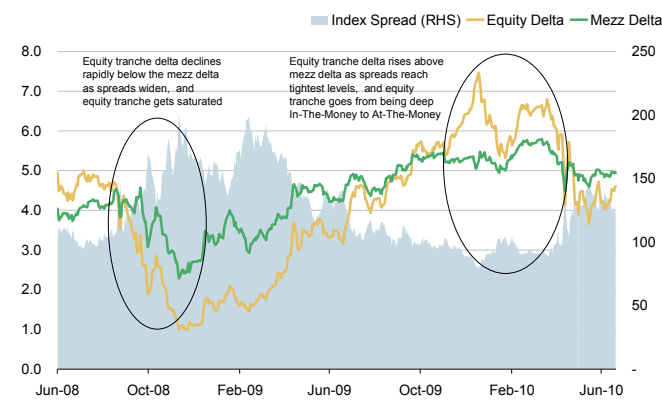
A corollary to this phenomenon of risk-saturated junior tranches in wider spread environments is that this can happen as we move out in maturity in an index as well. Due to the longer duration and often the higher spread of the index as we move out in maturity, the expected loss point is higher in the longer maturities, causing the equity and junior mezz tranches to become saturated as we move out the maturity spectrum. Consequently, we often see the delta of the equity tranches fall as we move out in maturity while the deltas of the senior tranches rise. Expressed differently, the delta for tranches that

are in-the-money rises as time passes, whereas deltas for out-of-the-money tranches decline over time. A delta-neutral position on an equity tranche may not be neutral a year later, as the equity tranche delta could have risen during that time and vice versa for senior tranches.

The same logic can help explain delta behavior in different spread environments. When spreads widen, the delta on junior tranches can fall as they become more in-the-money. The equity tranche gets saturated and the tranche delta eventually drops below that of junior mezzanine. Similarly, more senior tranches start becoming at-the-money as the level of spread increases and the tranche delta keeps rising as a result.

exhibit 8

Tranche Delta in Different Spread Environments



Source: Morgan Stanley Research

Understanding these sensitivities can help us fine-tune investment strategies to achieve desired deltas under different spread scenarios. For example, it is possible to construct trades that become delta positive when spreads tighten and delta negative when spreads widen, resulting in a more tailored payoff. Suppose an investor believes that spreads are currently too tight and will widen in the near future, but start tightening down the road, say next year. It is possible to implement a tranche combination that starts out delta negative but becomes more delta positive as time passes or spreads widen.

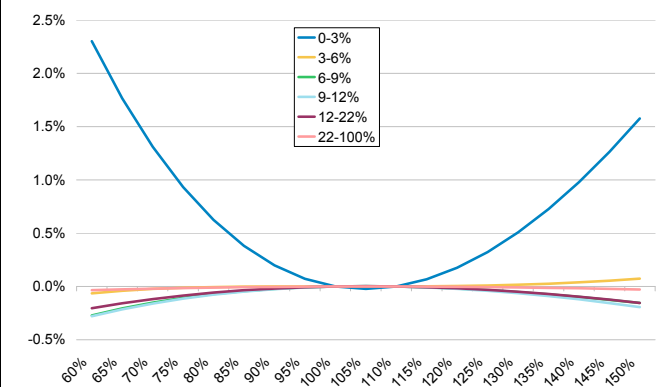
“M-GAMMA” OR CONVEXITY

As we discussed earlier, for wide spread moves, the relationship between tranche value and index value does not remain linear. This difference between linear approximation (using deltas) and the actual movement in market value is a tranche’s convexity or gamma. We typically measure it as a ratio of PV100 to $100 \times \text{PV01}$, i.e., the ratio of tranche mark-to-market for a 100 bp move in underlying spreads to 100 times PV01 of the tranche.

The easiest way to observe convexity is to plot the P&L of a delta-neutral transaction of tranches, as we have done for the 5-year iTraxx S9 tranches in Exhibit 9. Tranches that are in-the-money are positively convex, while out-of-the-money tranches are typically negatively convex (from the perspective of the protection seller). In other words, a delta neutral seller of protection in five-year equity tranche risk would have a positive mark-to-market for large changes in spreads, while a delta neutral 6-9% or 9-12% tranche of the same index would have negative mark-to-market.

exhibit 9

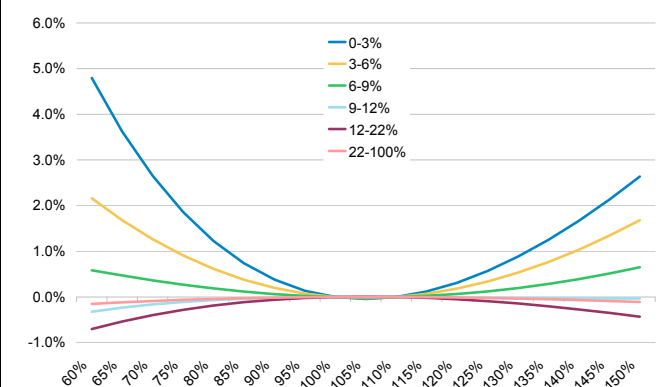
iTraxx S9 5-Year Tranche Convexity (Delta Neutral)



Source: Morgan Stanley Research

exhibit 10

iTraxx S9 10-Year Tranche Convexity (Delta Neutral)

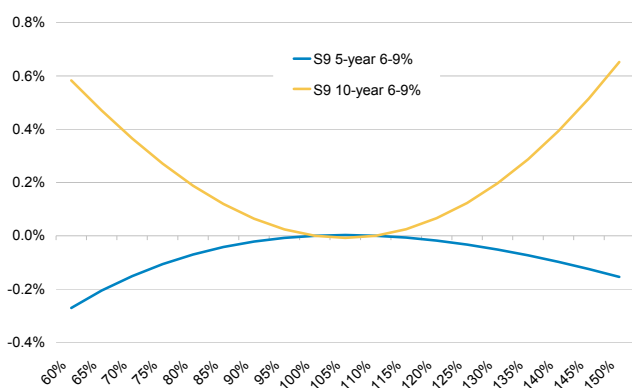


Source: Morgan Stanley Research

Exhibit 10 shows the convexity characteristics of 10-year IG tranches, where 0-3%, 3-6% and 6-9% tranches are all positively convex, as they are already underwater. To further illustrate this point, in Exhibit 11, we show the convexity of the 6-9% tranche in both the 5-year and 10-year maturities. Clearly, the 6-9% tranche is negatively convex (and out-of-the-money in this example) for 5-year, but it is positively convex (and in-the-money) for the 10-year maturity.

exhibit 11

Mezzanine Changes Character with Maturity (Delta-Neutral P&L)



Source: Morgan Stanley Research

While trading tranches of different underlying indices, delta-neutral convexity can be harder to measure, as the underlying indices are different and have convexity differences themselves. In addition, curve shape changes are also important to consider; a steepening of the 5s10s credit curve would affect 10-year tranches but not the five-year tranches.

“I-GAMMA” OR SENSITIVITY TO SPREAD DISTRIBUTION

While the impact of overall spread changes on tranches is more or less obvious, the effect of changes in the distribution of the underlying spreads, especially when the overall portfolio average remains unchanged, is subtle and worth exploring further. Wide or even average credits moving significantly wider affect junior mezzanine and first-loss tranches, depending on the size of the move. We have a number of historical examples of single name spread moves that involved a major reshaping of portfolio distribution.

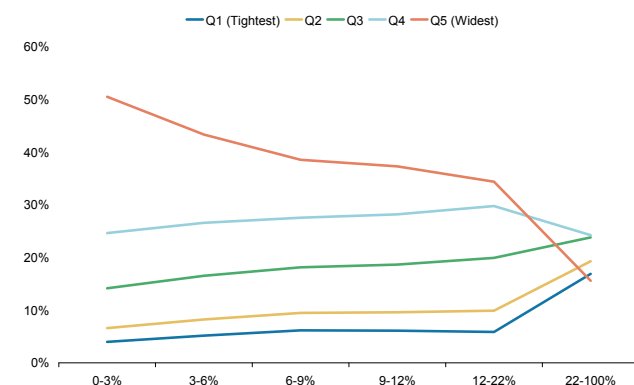
Each tranche has varying sensitivity to different spread buckets and each is affected differently by these moves. In Exhibit 12, we show the changing sensitivity (as measured by PV10%) of the different tranches to spread quintiles (Q1 being the tightest quintile and Q5 being the widest quintile). As illustrated, equity tranches can get 50% or more of their PV10% from the widest quintile (Q5). This share of the widest quintile drops as we move higher up in capital structure. For example, the 22-100% only gets about 18% of its PV10% from the widest quintile.

Thus, the shape of the risk distribution within the portfolio influences the pricing of tranches. It should be fairly intuitive to recognize that the length and thickness of the right tail influences the pricing of subordinate tranches, meaning that the bigger the tail, the riskier the equity and mezzanine tranches. Similarly, the shape of the middle to left side of the distribution should influence the more-senior notes. As credit

risk increases (i.e., shifts from left to right within the tail), senior tranches should become riskier.

exhibit 12

Sensitivity to Different Spread Buckets



Source: Morgan Stanley Research

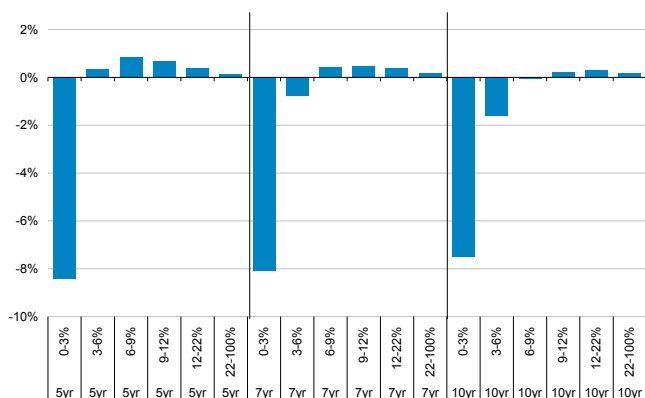
JUMP-TO-DEFAULT SENSITIVITY

While higher default sensitivity for junior tranches and lower for senior tranches is intuitive, given their relative positions in the capital structure, comparing sensitivities across different maturities and indices provides valuable insight into how defaults affect different tranches. For example, one 40% default on a 125 name index translates into a 0.48% loss (i.e., $(1-40\%) / 125$) for the index, which in turn is a 16% loss (i.e., $0.48\% / 3\%$) for the equity tranche. Because equity tranches are very in-the-money today, the default impact is very similar across maturities. The impact of default on the mezzanine tranche is different across the maturities – a default adds to the value of a 5-year investment grade mezzanine tranche but reduces the value of 10-year mezzanine tranche.

In Exhibit 13, we have summarized the average impact of one default on various tranches, assuming a 40% recovery on a delta neutral basis, i.e., assuming that we bought protection on the index to offset the tranche’s spread sensitivity. Tranche thickness, expected number of defaults for the index, and presence of an upfront payment are important variables to watch in assessing default sensitivity. A thicker tranche generally has a lower sensitivity to a default. Similarly, if the index is expected to have a large number of defaults, a single default has a smaller impact. Additionally, a tranche with an upfront payment has lower default sensitivity, compared to the same tranche without an upfront payment because with an upfront payment the protection seller does not lose premium due to a default, while in the case of a the running premium, the amount of premium paid would decline proportionally to the amount of tranche notional lost due to default.

exhibit 13

P&L Across the Capital Structure from One Default (Delta-Adjusted)



Note: Assuming 40% recovery for IG.

Source: Morgan Stanley Research

How do we determine if a tranche has more default risk or spread risk? Delta-neutral default sensitivity, as shown in Exhibit 13, helps us determine the subordination level in the capital structure where tranches transition from being net default sensitive to net spread sensitive. For example, the mezzanine tranche (3-6%) of five-year investment grade is net spread sensitive (i.e., the delta-neutral protection writer will have a positive mark-to-market in case of a default), while the 10-year mezzanine tranche is net default sensitive.

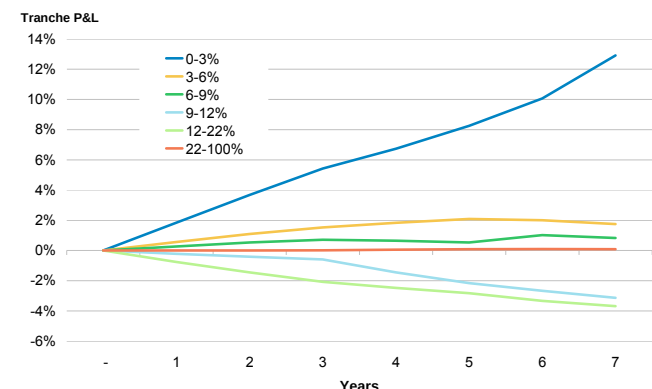
Using the distinction between net default and spread sensitivities of tranches, we can construct trades that are more efficient in expressing our credit views. For example, a combination of long equity and short mezzanine in a delta neutral ratio effectively expresses a credit view that is constructive on default risk but not on spread levels.

“THETA” OR TIME DECAY

As a credit default swap approaches maturity, the spread is bound to converge to zero, since credit protection provided by a CDS eventually becomes worthless if a default does not occur. As with any credit/rate product, the passage of time generates return or “time decay”. The “theta” or time decay in tranches is a function of three elements: a) the slope of the credit curve b) reduced time to maturity and therefore reduced riskiness as measured by duration, and c) changes in risk allocation in the index capital structure in different maturities.

exhibit 14

Time Decay for 5yr Tranches

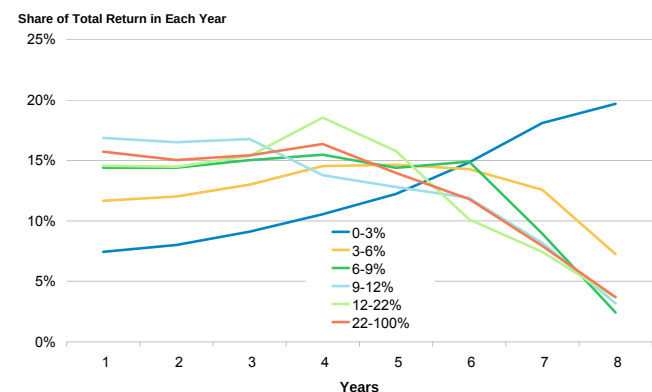


Source: Morgan Stanley Research

The first two factors are fairly simple to understand. In general credit curves are upward sloping, although not always, and even if curves are inverted, the portfolio with the shorter maturity will have a lower duration, which means the index will generate a positive return in a year’s time (assuming no defaults, of course, and that spreads remain unchanged).

exhibit 15

Time Decay as % of Total Return Over 8 Years



Source: Morgan Stanley Research

The last factor – risk allocation change due to reduced maturity – is slightly more complicated. Since junior tranches are levered investments on defaults, their value changes faster than the index. Another way to think about this is that as the index gets closer and closer to maturity, the risk tends to flow down the capital structure into the more junior tranches. Exhibit 14 shows the total return of iTraxx S9 five-year tranches for different time horizons, from the perspective of the protection seller. The value of protection declines precipitously initially and decelerates toward the maturity. On an absolute basis, tranches with higher than 1x delta lose value faster than the index, while senior tranches with deltas of less than one lose value more slowly than the index.

However, when we analyze a tranche's time decay relative to itself over time (i.e., how much of the total value is realized every year), we find that equity tranche value decays slower than the index while all other tranches decay faster than the index, in the case of investment grade tranches. For example, equity realizes less than 10% of its cumulative value in each of the early years. Senior tranches such as 12-22% get a higher than average share of the value in early years (see Exhibit 15).

"RHO" OR SENSITIVITY TO CORRELATION

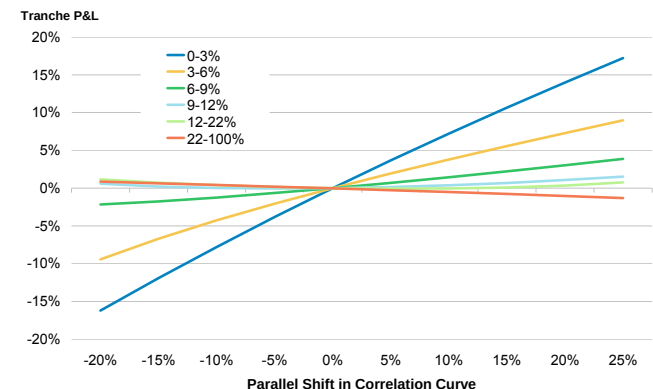
As we discussed earlier, junior tranches are long correlation, senior tranches are short correlation and mezzanine tranches are relatively insensitive to correlation. In other words, higher correlation is better for junior tranches and worse for senior tranches (from the perspective of protection sellers). Exhibit 16 shows the sensitivity of 5-year investment grade tranches to parallel shifts in the base correlation curve. For example, for the +5% scenario, we assumed that the equity tranche's correlation increases 5% and the base correlation skew curve remains unchanged. Thus, for all tranches, the attachment and detachment point correlations change by the same amount. As seen in the Exhibit, the junior tranches generate a positive P&L as correlations rise, whereas super senior tranches are short correlation.

To isolate the impact of correlation changes, we have assumed that the base correlation skew remains unchanged. In practice, skew can and does change. Since base correlation considers all tranches as a portfolio of two equity tranches, as the skew (i.e., the difference between the implied correlations for attachment and detachment points) increases, the par spread decreases, just like the equity tranche. It is important to note that correlation sensitivity of tranches changes as the underlying index spread moves. The more in-the-money the tranche is with respect to index expected losses, the more equity-like the tranche is and therefore the longer the correlation. So as seen in Exhibit 17, the 3-7% tranche will be less sensitive to correlation if spreads tighten a lot as it becomes more at-the-money.

In Exhibit 17, we analyze the mark-to-market impact of a 1% increase in correlation for 5-year investment grade tranches at different spread levels for the index. As shown in the exhibit, senior tranches are relatively correlation insensitive for small changes in spread but become significantly more sensitive as the index widens materially. The converse is true for the equity tranche, i.e., it is rather sensitive for small changes to the index, but beyond a certain point of widening, correlation sensitivity starts to decline.

exhibit 16

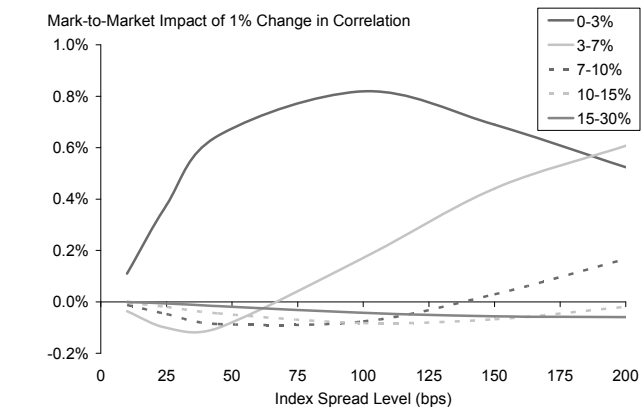
Correlation Sensitivity of Tranches (iTraxx Sg 7-Year)



Source: Morgan Stanley Research

exhibit 17

Correlation Sensitivity vs. Spreads – CDX IG 5-Year



Source: Morgan Stanley Research

IMPACT OF TRANCHE STRUCTURE ON SENSITIVITY

Last but not the least, the pricing structure of the tranche influences the sensitivities meaningfully – a tranche with an upfront payment tends to have a lower delta compared to the same tranche without an upfront payment. Why does an upfront payment lower the delta of a tranche? The answer lies in understanding how rising spreads affect the present value of expected premiums (premium PV) and default PV. An upfront payment with no running coupon implies that as spreads widen, the tranche value is affected only by higher expected defaults, and not lower expected premiums, since the protection seller has already collected the entire premium. On the other hand, an all-running contract will have both negative impacts – higher expected defaults and lower expected premiums due to more defaults making it more sensitive.

SECURITIZATION APPROACH: RATIONALE AND UNCERTAINTY

In the securitization world, the rationale for using tranches is two-fold. At one level, it allows investors to make portfolio investments in corporate credit with credit enhancement (i.e., protection from some amount of defaults). The rationale here is very similar to what we see in the broader securitization world where the underlying credit risk is a bit more esoteric, including non-agency mortgages, consumer credit like autos and credit cards, and commercial real estate. The second is to be part of the so-called “shadow” banking system itself. Regulators are clearly supportive of the existence of a shadow banking system, where credit is provided through capital market solutions that include securitization.

SECURITIZATION APPROACH: BESPOKE NEW ISSUE

Synthetic CDOs, or CSOs, first started appearing in the previous cycle during the rapid growth in popularity and usage of single-name CDS in the default-intensive cycle of 2001-2002. The development of index tranches and correlation technology led to a move away from the fully distributed approach favored by the cash CDO market and more toward the single-tranche CSO model, as dealers were able to more efficiently hedge individual tranche exposure using single-name deltas and correlation hedges in the liquid tranche market. Relative to their cash CDO counterparts in the securitization world, CSOs provided increased flexibility, as synthetic technology made it possible to customize risk exposures with respect to currency, cash flow, tenor, portfolio, maturity, subordination and size of exposure. Since there is no need to identify and acquire a specific asset, the maturity of the transaction is not constrained by the maturity of the pool of assets and furthermore, there is a significantly lower ramp-up period and therefore lower carry costs. Finally, bespoke structures allowed investors to customize an asset according to rating preference and desired yield, something neither the cash CDO market nor the single-name market had been able to do with so much flexibility.

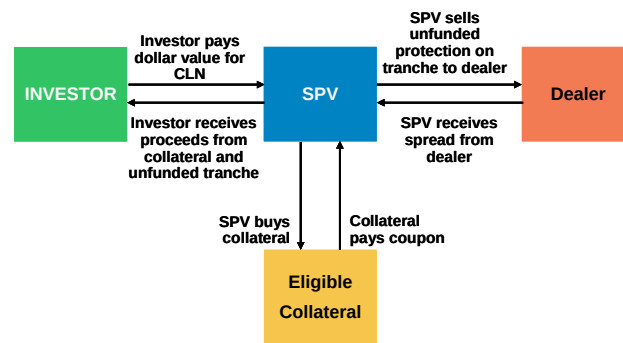
From a mechanics perspective, bespoke mezzanine tranches look less like fully funded cash CDOs and instead better resemble index tranches on the swap side, with a funding component. An SPV is set up, and the investor pays par to the SPV, which in turn purchases eligible collateral to earn a LIBOR based coupon, and then enters into a swap agreement with a dealer (see Exhibit 18). We have covered the various types of collateral elsewhere (see Chapter 31).

Early transactions had capital structures that were fully placed, usually with a 5-year maturity, mezzanine attachment points ranging from 2% to 10% and thicker tranches on portfolios that were predominantly IG, though it wasn't until the ratings model changes of late 2005 when HY credits were almost completely eradicated from structures. The bulk of mezzanine issuance that fueled the structured credit machine occurred between 2004 and 2007. The portfolios for these transactions were bespoke in nature, with thin mezzanine tranches and collateral that was either managed by third-party managers,

managed by the investors themselves, or simply static. These transactions were more varied from a performance perspective than their early-cycle counterparts, and we cover the distressed opportunity that arose from these mid-to-later cycle deals later on in this primer.

exhibit 18

Typical Bespoke Mezzanine Funded Structure



Source: Morgan Stanley

One question we hear repeatedly is whether structured credit activity will return to the corporate credit market. Whether or not one intends to use synthetic forms of structured credit, flows here had a tremendous impact on investment grade markets in the last cycle, so the question is certainly an important one. From a derivatives perspective, structured credit activity never really died in the cycle as both increased volatility and pronounced amounts of credit risk created unprecedented trading opportunities, mainly in the liquid index tranche markets, and more recently in unfunded bespoke transactions (i.e., in swap form).

From a securitization perspective, however, new issue bespoke volumes came to a standstill, with many investors focused on restructuring their existing, or legacy, bespoke positions to meet ratings and capital guidelines. With the healing of the credit markets, securitization has been deemed vital to the credit system, mainly because it offers lending institutions an efficient alternative to unsecured borrowings (hence the motivation for the government programs). The source of this efficiency comes from the secured nature of the borrowing, the economies of scale associated with pooling assets into portfolios, and the cheaper costs of taking first loss exposure and borrowing against tail risks. Given that securitization is most prevalent in cash securities, issuers (mainly banks) benefit most directly.

SECURITIZATION APPROACH: DISTRESSED MEZZANINE

In early 2009 we first identified the enormous opportunity that the legacy bespoke mezzanine market was offering in the wake of the downgrade and default of many corporate issuers (see Chapter 20), causing traditional bespoke mezzanine investors to liquidate their holdings for a variety of reasons, including ratings downgrades, changes in internal investing rules from risk managers, and others who were spooked by the markets in

general and were fleeing to higher quality assets. Most of these assets were bespoke transactions from the 2004-2007 vintages and were trading at highly distressed levels. For anyone willing to put into place the necessary tools to evaluate these structures, there were plenty of opportunities to be had. We categorized the distressed assets as follows:

IO-Value. These are the low dollar price instruments (below \$10), which are priced like IOs – typically equal to about 2-3 years of coupon. It is expected that the principal value will be little to nothing at maturity; the valuation is based on the expectations for how many coupons the structure will pay out before the notional goes to zero. The yield to maturity of these structures could be in the 60%+ range.

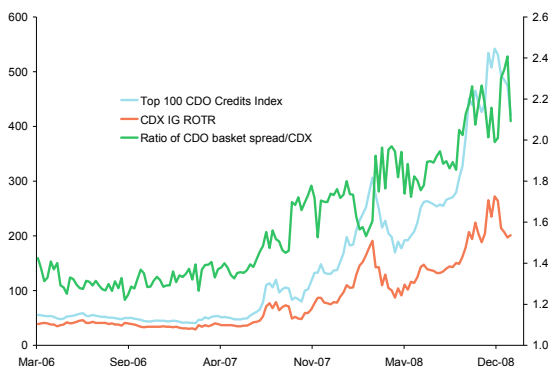
Option-Value. These are mid-priced structures (\$5–30) that capture the interest component plus the small potential likelihood of principal recovery as well. Yields for these deals are likely in the 30-60% range.

Potentially Money-Good. These structures can have any or all of the following: shorter maturities, cleaner portfolios, higher subordination, greater thickness – and consequently they have higher prices associated with them (\$20–80). These are categorized as potentially money-good because while nowhere near a certainty, the likelihood that there will be at least a partial principal repayment is higher. Yields here are much lower (25–50%) but still fairly attractive compared to the high single-digit and low-to-mid-teen risk-adjusted returns for plain vanilla IG/loans or HY credit.

The primary driver of value in the legacy mezzanine bespoke market was tail risk. Exhibit 19 illustrates just how much wider the tail names that were common constituents in many CSOs were than the CDX IG index at the peak of 2008. The distressed mezzanine opportunity is and was primarily a levered play on spread compression and healing in this part of the portfolio.

exhibit 19

Top 100 CSO Names vs. CDX



Source: Morgan Stanley Research

Taking advantage of this dislocation required some background knowledge of the market, though we highlighted when we first identified this opportunity that correlation models aren't as necessary here as they were earlier. Instead, we argued that investors should focus on the following characteristics to determine value:

Subordination. Many of the financials defaults were common names in bespoke CLNs and thus many of these assets had at least some loss of subordination.

Thickness. Thin tranches were the dominant theme when these assets were originally structured, to increase structural leverage and thus spread. This matters because if the tranche were touched at all, it could potentially require only one additional default for full write-down. This is very different from the index tranches, which are all at least 3% thick and can withstand a greater number of defaults once breached.

Portfolio composition. This is clearly important, particularly if there is little subordination left in the structure. Tail risk in particular is a primary concern. To combat some of the extreme dispersion present in these portfolios, we indicated a preference for paying for risk management and recommended using some amount of the coupon income to pay for a manager and/or buying protection on tail names.

Recovery. The majority of these structures are floating recovery, and recovery can be much lower than expected at times. Some, however, are fixed recovery, generally 40%, and this added certainty regarding number of defaults needed to impact the tranche can be very valuable.

Managed or static. Some CLNs are managed synthetics, and the added presence of a manager can be valuable, depending on the manager.

Maturity. Earlier vintage deals had much shorter maturities while transactions done in the 2006-2007 time period were typically 7-10yrs. We argue that maturity is far less important than the first five components listed above, though still worth noting, particularly in the very short end.

Coupon. While we argue that the deeply discounted dollar price is the more eye-catching price component, for those bespoke we consider to fall under the category of "IO opportunity", coupon is important.

SECURITIZATION APPROACH: RATINGS METHODOLOGY

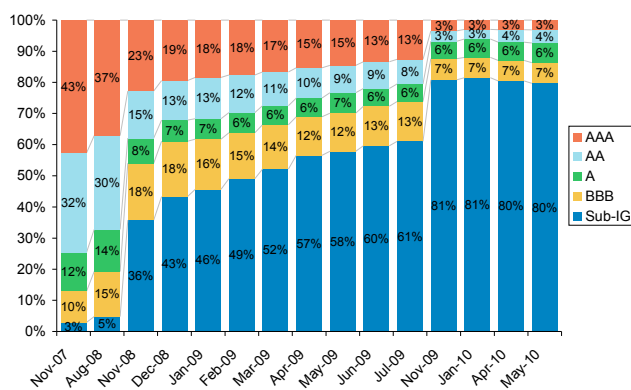
Rating agencies have been an important part of the tranche market since inception. Because ratings are closely linked to the capital held against CSO tranches, we expect the agencies to wield considerable influence on the trajectory of the CSO market in the future, especially in the post-Basel II environment. While the synthetic CSO market has evolved considerably since its introduction ten years ago, the core of

the rating agency approach has remained relatively constant. (See “CDO Evaluator Applies Correlation and Monte Carlo Simulation to Determine Portfolio Quality” by S&P, November 13, 2001; “Moody’s Approach to Rating Corporate Collateralized Synthetic Obligations” by Moody’s, September 14, 2009; and “Update to Global Methodologies and Assumptions for Corporate Cash Flow and Synthetic CDOs” by S&P, September 17, 2009). Both S&P and Moody’s use a Gaussian copula model as the basis for inferring portfolio default distributions from a set of individual default assumptions and correlations amongst various assets in the portfolio. Both S&P and Moody’s methodologies specify for each credit a) some kind of default rate trajectory based on ratings of the underlying portfolio names and b) recovery assumptions based on the domicile, sector and nature of the credit in the portfolio.

Although the methodology remains same, the threshold levels for the default rates, recoveries and asset correlations have undergone various changes over time. One of the important shifts was the change in default rate inputs by S&P in 2005, which benefitted IG corporates and penalized HY corporates. A positive outcome of this was that much of the CSO portfolios from 2006 did not have much high yield exposure, though they had heavy concentrations of financials and insurers instead. The heavy presence of banks (including the Icelandic banks, Washington Mutual, Lehman) and the insurers proved to be the weakest link for CSOs in the 2008-09 default cycle and distressed financials continue to dominate the tails of legacy CSOs.

exhibit 20

Ratings Distribution of Sample CSO Tranches



Source: S&P, Morgan Stanley Research

Post-crisis, the rating agencies have once again altered the CSO landscape by making ratings criteria more stringent. The experience with sub-prime CDOs caused rating agencies to scale up default assumptions and correlations while also

lowering assumed recoveries and introducing additional stress tests and safeguards. These include, among others, applying additional stresses to base default rates and introducing concentration tests for sector and geographic location and are particularly stringent for tranches that will be rated AAA. All these changes have made achieving AAA/AA ratings with a viable yield difficult to achieve for any new potential CSOs.

The changes to methodologies have also contributed to turning the ratings distributions of legacy CSOs upside down – from just 3% of the universe below investment grade in 2007, to more than 80% today. As seen in Exhibit 20, the CSO universe suffered two shocks – one in late 2008 due to the default of the financials and the second in the second half of 2009 due to the ratings criteria change.

As we mentioned earlier, both S&P and Moody’s have broadly similar approaches in terms of arriving at portfolio default rates. The key difference between the two agencies is that while S&P focuses on the probability of default of the tranche, Moody’s focuses on the expected losses from the tranche. The result is that Moody’s assigns different ratings to tranches of same attachment but different thicknesses, whereas S&P would give the same rating for tranches of same attachment point, regardless of tranche thickness. Both agencies emphasize additional overrides to the model suggested ratings in changes to methodologies made after the crisis.

Both agencies have a similar approach to specifying correlations. Moody’s correlation assumptions are based on a) the inter-industry correlation, which applies to two corporates in different sectors, and b) the intra-industry correlation, which applies to two corporates within a same sector. Moody’s also classifies industries as “global”, “semi-local”, or “local” to describe the extent to which credits within a single industry, but in different regions, are correlated. In a similar vein, S&P specifies correlation parameters for a) firms in the same corporate industry and b) firms in different corporate industries. In addition, S&P assigns a correlation factor between assets from different industries in different geographic regions.

With respect to recovery assumptions, both agencies now use recovery assumptions that are correlated to the default levels in the simulations. Both agencies link the mean and volatility of the recovery for each credit in the portfolio based on the type of credit (bond vs. loan) or priority/seniority (secured or unsecured) and jurisdiction of the entity.

exhibit 21

Hypothetical Ratings of Index Tranches

S9	3Y	5Y	7Y	10Y
3-6%	BB	B	CCC+	CC
6-9%	A+	BBB	BB+	B+
9-12%	AA+	A+	BBB+	BB+
12-22%	AAA	AA+	A+	BBB+
22-100%	AAA	AAA	AAA	AA+

IG9	3Y	5Y	7Y	10Y
3-7%	B-	CCC-	CC	CC
7-10%	A	BB+	B+	CCC+
10-15%	AA	A-	BBB-	BB-
15-30%	AAA	AA+	A+	BBB+
30-100%	AAA	AAA	AAA	AAA

Source: Morgan Stanley Research

As a part of the criteria changes after the crisis, agencies have increased the correlation parameters within different parts of the portfolio, thereby raising the probability of tail scenarios and correlated defaults. To illustrate these changes we show in Exhibit 21 hypothetical ratings on index tranches. The 9-12% and 10-15% tranches, which would have been AAA previously, are now only able to achieve BBB ratings. Among the index tranches with 5 years or more to maturity), only super senior tranches would qualify for a AAA rating by S&P.

SECURITIZATION APPROACH: BASEL AND CPM THEMES

Ultimately, the future of tranche markets rests on the securitization part of the market, which has the potential to be the dominant force. Undoubtedly, the derivatives part of the market is the more dynamic, agile and tactical, but the rebuilding of the securitization market is critical for the long-term health of this market. And this recovery, in turn, has been hamstrung in part by the uncertainty around ratings methodologies and capital treatment regime in 2009.

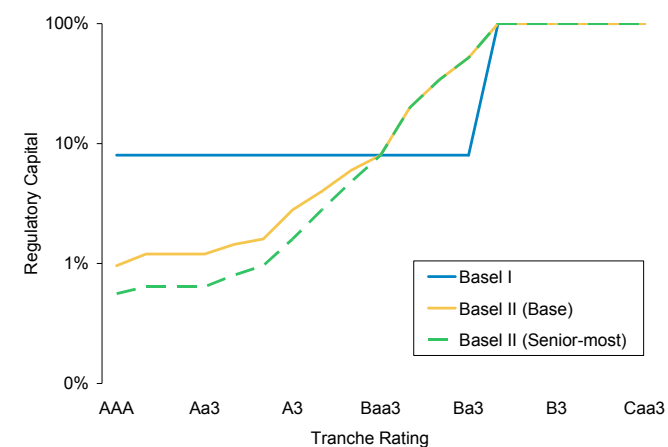
The ratings uncertainty is removed now as the framework from Moody's and S&P seems to have stabilized, although criteria are fairly conservative now. On the regulatory side, there has been much focus in the market on the implications of new financial regulation on structured credit flows, in particular at banking institutions who use structured credit to both invest in credit and hedge credit exposures. Most banks outside of the US adhere to Basel II, and much of the rhetoric around changes to Basel II (i.e., Basel III) have not really addressed securitization or structured credit within banking books, as Basel II is reasonably comprehensive with respect to securitizations. Much uncertainty remains in the market regarding potential paths of regulation, but much of the uncertainty centers on the liability side rather than the banking book asset side at this point. As it stands now, the Basel II approach for securitization would govern most banking books from a structured credit perspective.

The bottom line for corporate credit risk held in structured/securitized form is that some form of rating (either internal or external depending on the institution) will determine the risk weights, which, when combined with market pricing will determine regulatory capital efficiency. The good news is that this approach is more aligned with economic capital than Basel I, and with the rating agencies' more conservative approach to ratings methodologies in the last two years, there is much less opportunity for any form of ratings arbitrage. We provide a brief summary of key regulatory capital themes and encourage readers to consult Chapters 25-29, which address capital efficiency quite extensively.

We provide a quick regulatory capital refresher here for investors completely new to Basel. Basel I (introduced in 1988) was an extremely simple first attempt by the Basel Committee on Banking Supervision (BCBS) to strengthen the international banking system in a consistent way. Basel I introduced 'risk-weightings', whereby assets on a bank's balance sheet are separated into different classes, with prescribed risk-weightings. Basel I differentiated risk weights by asset class but without much regard for credit quality – for example all corporate credit got a 100% risk weighting regardless of whether it was AA or BBB. Basel II is a significant change wherein the amount of regulatory capital is a function of the credit quality, bringing alignment in regulatory and economic capital.

exhibit 22

Basel II Tranche Capital Requirements: Senior Tranches Benefit



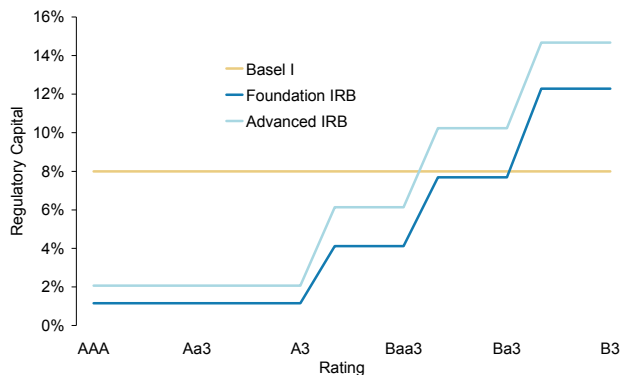
Source: BIS, Morgan Stanley Research

While Basel II is new to the US, European banks have already adhered to Basel II since 2008. As is clear from the discussion above, banks with a greater proportion of IG assets may actually end up reducing their risk-weighted assets. In a world where banks err towards more liquid and higher quality credit and have greater amount of buffers, Basel II could end up being more favorable from an asset perspective. If the European Banks’ experience is any indicator, most large banks will ultimately seek approval from regulators to use the Advanced Internal Ratings Based Approach (AIRB). AIRB specifies methods to calculate the risk weighting based on internal estimates of the Probability of Default (PD), Exposure at Default (EAD) and Loss Given Default (LGD).

Because Basel II regulatory capital is closely aligned with economic capital, changes in market pricing can influence strategies to a greater extent than did Basel I. While senior tranches generally benefit relative to junior tranches, relative value perceptions – for example, between junior mezzanine and senior mezzanine or between senior and super senior – can change based on the pricing.

exhibit 23

Basel II Corporate Risk Capital Requirements: High Quality Credit Benefits



Note: IRB – Internal Ratings Based Approach.
Source: BIS, Morgan Stanley Research

Impact on CSO market. As far as bespoke CSOs from the last cycle are concerned, they are deeply underwater from a ratings perspective, as we have highlighted previously. Clearly, these mostly mezzanine CSOs look unfavorable under Basel II. Despite the ratings issue that has existed for some time now, unappetizingly low valuations were a key impediment for sellers early in 2009, as were lacks of both an organized buyer base and adequate depth. However, a maturing ratings-insensitive buyer base has developed in 2010 and has largely been able to absorb these sales as they occur. Orderly CSO BWICs were a regular occurrence in 1H 2010 and appetite for secondary CSO paper remained fairly strong.

Securitization and Capital Relief Strategies. Last but not least, securitization and capital relief strategies have experienced a complete U-turn from Basel I to Basel II. The most common Basel I strategy involved retaining the yieldiest part of the capital structure, i.e., the equity tranche, and buying protection on 3-100% part of the capital structure. Under Basel II, the optimal strategies for Europe proved to be the exact opposite – the focus is on what part of the capital structure is most efficient to retain, and as we discussed above, it is now the senior, higher quality tranches that are the most attractive risk to retain. Investors have a number of types of hedging choices (see Exhibit 24).

exhibit 24

Thinking About Structured Hedging Alternatives

Long Index			
Equity tranche (0-5%) hedged	5-100% tranche risk retained		2T – H-L
Equity tranche (0-5%) hedged	5-9% retained	9-100% tranche risk hedged	3T – H-L-H
Equity tranche (0-5%) hedged	5-9% retained	9-12% hedged	12-100% tranche risk retained
Equity tranche (0-5%) No position	Sell Protection on 5-100% tranche risk		2T – N-L

Source: Morgan Stanley Research

Baskets Go Back to the Future

February 26, 2010

Sivan Mahadevan

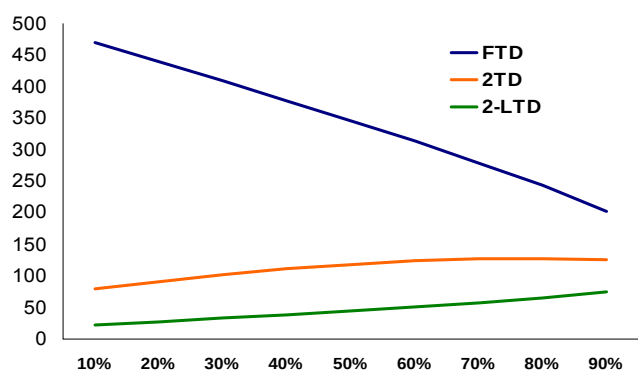
Ashley Musfeldt

Phanikiran Naraparaju

First-to-default, or more generally, nth-to-default, baskets were one of the credit derivatives market's earliest instruments, dating back to the beginnings of the CDS contract itself in the late 1990s. Such baskets (which typically have exposures to less than 10 corporate or sovereign credits) have much in common with the broader world of tranches on larger portfolios in that they have "correlation" risk. However, the smaller basket market tends to fall in the middle of the large worlds of single-name CDS and portfolio tranches, and it has even been ignored by market participants for extended periods as investors instead focused on the larger flows in CDS and portfolio tranches.

exhibit 1

Basket Pricing Depends on Correlation



Source: Morgan Stanley Research

As structured credit heals from the fall-out of the credit crisis, there is certainly a "back to simplicity" bid in the market, and as such, we see increased flows in simple baskets from investors, both as buyers and sellers of protection. The motivation includes still wide credit spreads in many corporate sectors, reasonably high correlation, and stress in the sovereign and US municipal sectors, both of which have burgeoning single-name CDS markets. Given our fair value to constructive views on corporate credit along with our belief that default correlation has room to continue to fall, we in general like buying FTD protection and selling 2nd-to-Nth to default protection in today's market environment, but details certainly matter. In the remainder of this chapter, we provide some recent history and updates on the FTD/NTD basket market, some examples of investment and hedging strategies that we favor today, and an updated primer on the market.

NOTIONAL BASKET HISTORY

FTDs began trading, as we mentioned earlier, very early on in the CDS market. Initially they were simple tools to get additional structural leverage on a basket of names. While they were and are a correlation product, they behave differently from tranches in that when a name in the basket is triggered, they behave more like single-name CDS contracts than tranches because the "loss" doesn't erode subordination, but rather the entire name is removed and settled separately like a single-name contract. In the case of FTDs, after this one default, the contract is over then, regardless of recovery. This structural simplicity makes them easier to deal with than tranches, and this is why many investors then found them to be a good introduction to the correlation market. As the CDS and structured credit market matured in 2003-2006, FTDs fell in popularity as bespoke tranches on much larger "baskets" (more appropriately deemed portfolios) of 100+ names became the bread-and-butter product of most structured credit market participants.

exhibit 2

Investor Motivation for Basket Trading

Basket	Protection Seller	Protection Buyer
FTD	- Receive higher yield for same notional by taking more levered exposure to basket - Express credit views in levered form - Position for rising correlation	- Avoid idiosyncratic risk on a basket - Buy cheaper protection vs. individual swaps on all underlying names, but retain tail risk - Express views of falling correlation between individual names or industries
2nd-through-Nth	- Express positive views on a basket, while avoiding idiosyncratic risk, but taking tail risk - Receive possibly higher yield than some of the low yielding names in the underlying basket - Position for falling correlation	- Buy cheaper protection against the tail risk vs. buying protection on all underlying names - Manage risk for extreme scenarios

Source: Morgan Stanley, Moody's

In late 2006 we saw a brief resurgence of interest in FTDs, though bespoke tranches still ruled the day, but this time we saw greater two-way flows and more interest in other default baskets, such as second-to-defaults (2TDs), second-through-last-to-defaults (2-LTDs), higher Nth-to defaults (NTDs) and even some Nth-through-last-to-defaults (N-LTDs). Investors this time around were interested in using these products for sector plays, to get either long or short 5-10 names in a sector, as de-levered hedges with low carry,

such as 2-LTDs, or as a levered way to play a small basket of high conviction longs.

Now, after another dry spell brought on understandably by the credit crisis, we are again seeing a resurgent interest in these products. Again we are seeing interest in both FTDs and their unlevered senior cousins 2-LTDs. We are seeing interest in these in both swap and CLN form based on investor type. Some new additions this time around, however, are the constituent base; namely European sovereign and US muni CDS are now being put into baskets

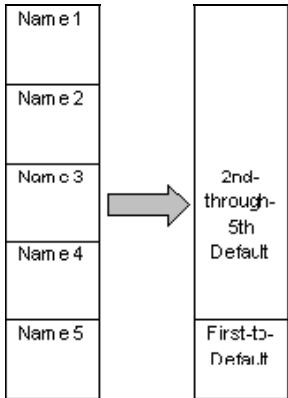
STRUCTURAL OVERVIEW

An Nth-to-default basket is a product in which the investor gains either long or short exposure to a relatively small basket of credits. The typical size of baskets ranges from 4 to 10 credits. The investor is either selling or buying protection on the Nth credit to default in the basket, as with a plain vanilla credit default swap, except that the reference credit is a basket instead of a single name. Five-name baskets are most common, and protection is typically bought or sold on either First-to-Default (FTD) or 2nd-through-last-to-default (2-LTD).

Suppose an investor sells FTD protection on a five-name basket of reference credits. The investor will receive a periodic payment (the “premium”) in exchange for taking on the credit risk of the basket. If no credit events occur during the term of the basket default swap, the swap expires. If a credit event occurs during the term of the basket, the swap is terminated and the seller of protection in the basket loses par minus the recovery value of the defaulted asset, and the single-name contract is settled via the ISDA auction process. This process is analogous to a single-name credit default swap. Exhibit A1 shows an FTD swap with an underlying basket of five names:

exhibit A1

Nth-to-Default Structure



Source: Morgan Stanley Research

We also note that FTD and NTD baskets have “cousins” in other markets, most notably equities and currencies, where option flows in “best of” and “worst of” baskets are quite common. Taking equities as an example, it is common for investors to, for example, buy a call option on the “worst performing” name in a small basket of stocks, as a way of cheapening the premium vs. the alternatives (buying call options on each of the individual stocks, or buying a call option on the whole basket). Such a strategy is long correlation.

exhibit A2

FTDs’ Cousins in the Equity Markets

Credit Strategy	Equity Strategy	Correlation Exposure	Rationale
Buy FTD Protection	Buy Worst of Put	Short Correlation	Buy put or default protection on “worst” name, so you have first-loss hedging without buying puts or protection on full portfolio (smaller notional exposure)
Sell FTD Protection	Sell Worst of Put	Long Correlation	Take a bullish view on defaults or equity price performance and position for the scenario that all assets do well or avoid default.

Source: Morgan Stanley

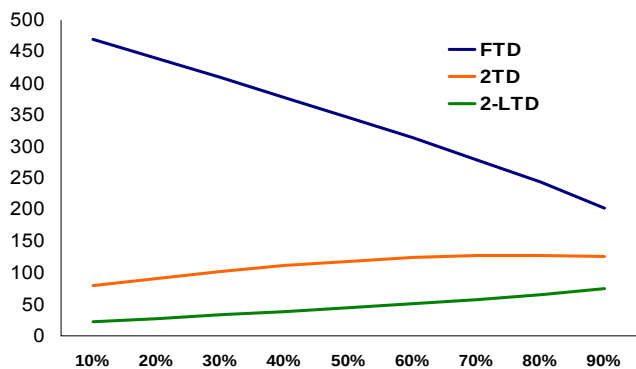
In comparing these equity strategies to FTD baskets, it is probably best to focus on puts. In Exhibit A2 we highlight common strategies. Buying a worst of put in equities would be similar to buying first-to-default protection, in that one has “protection” from the worst name (but not the others), and it is a short correlation strategy.

FTD MECHANICS AND CORRELATION

Although basket default swaps resemble single-name credit default swaps mechanically, particularly in the event of default, their risk/return and valuation profiles are more akin to bespoke tranches. FTD baskets are thus analogous to equity tranches, 2TDs would be mezz-like, and 2-LTDs are similar to senior or super senior tranches. Even if fully funded, the default swap basket can be thought of as a levered investment on a basket of credits, with non-recourse leverage provided by the counterparty. The seller of protection in an FTD, however, is only exposed to the first credit event, and the principal loss is directly related to the recovery value of the defaulted asset. This aspect is unlike equity tranches, where the trade can still remain after the first loss and will remain until it matures or experiences enough losses to wipe out the tranche.

exhibit A3

Basket Pricing Depends on Correlation



Source: Morgan Stanley Research

Like their tranche counterparts, Nth-T-D baskets are priced using correlation. As an example, we consider a first-to-default swap on a basket of five credits ranging from 50bp to 150bp, with an average spread of 100bp. In the case of 0% correlation, a first-to-default swap would carry a premium of approximately 500 bp, the sum of the individual default swap premiums. In the case of 100% correlation, the first-to-default swap would have a premium of 150bp, the maximum of the individual default swap premiums. A graph of the premiums on the first-, second-, and second-through-fifth-to-default swaps as a function of correlation is shown in Exhibit A3.

INVESTOR MOTIVATION

Why would an investor either sell or buy first-to-default protection on a basket of credits? Say an investor has a positive view on a basket of five credits with an average spread of 100bps each in the CDS market. By entering into five individual default swaps, the investor could gain \$2 million notional exposure to each credit and effectively receive nothing upfront and receive 100bp of premium per annum on \$10 million notional for five years if no credit events occur. If one or more credit events occur, then there will be a principal loss in each individual default swap (equal to par minus the recovery value of the default asset).

Alternatively, an investor can express a positive view on the same basket of credits by selling first-to-default protection on the basket with a notional amount of \$10 million. From the previous example, if we assume a correlation of 40%, the premium earned will be approximately 380bp (76% of the portfolio spread); therefore, implementing this view via an FTD basket results in a higher-yielding investment for a given notional exposure. To get an equivalent yield from a single-name, an investor would have to write protection on nearly four times the notional. Therefore, an FTD is a much more leveraged way of expressing a positive view on credits, which is obviously a double-edged sword.

Why would an investor buy protection on a basket of credits? It is effectively a way of shorting a basket of credits with less premium outlay than buying protection individually on each underlying credit. However, the swap gets terminated when the first credit event occurs, so the upside for the protection buyer is equal to par minus the recovery value of the defaulted credit only. This makes the basket a better hedge for default risk than spread risk.

Buying protection is also a way of hedging existing long exposure to credits, particularly for investors who want to reduce concentration risk but do not necessarily have an outright negative view on credits. For example, rather than buying protection individually on a portfolio of names, buying protection on the basket will be cheaper and more useful if the investor has a generally positive view on the credits but would like to reduce concentration risk. The buyer of protection is effectively short correlation.

REPLACEMENT LANGUAGE AND MERGERS

What happens if two of the names in a basket merge? The mechanics behind merger treatment in baskets are important to understand. Initially the market offered two ways of handling merging entities in baskets.

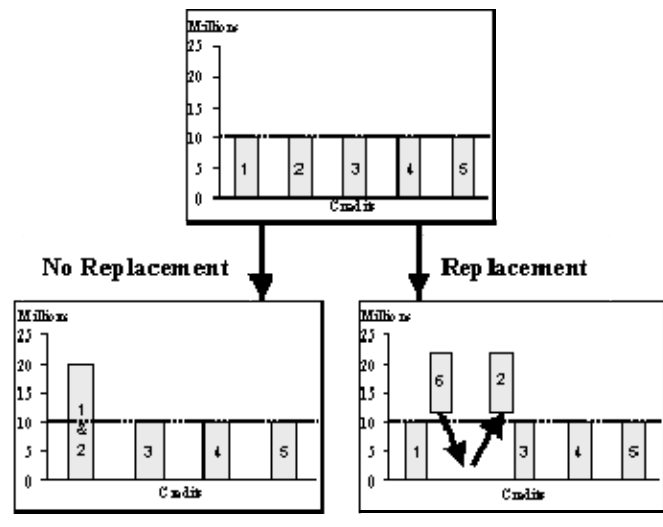
With replacement. This is the market standard today for handling merger events in a basket. In this instance, when a merger occurs, the buyer and seller of protection must mutually agree on a replacement credit, and often this replacement credit is one that trades at a similar spread level. Clearly, there is significant correlation risk in this exercise, and the buyer and seller of protection have opposite correlation views, however standard documentation offers specific parameters for selecting the replaced name and a process by which both the buyer and seller of protection come to an agreement on which name to choose. Still, the operational aspects of going through this process for each basket that experiences a merger can be arduous.

Without replacement. This language was used sporadically in early trades in this market and today is very rare. However, since some dealers are willing to trade with this language, we will highlight some of the mechanics and rationale for this type of language here. If a basket trades without replacement a five-name first-to-default basket where two reference entities merge effectively becomes a four-name basket once the merger closes. The “contractual” notional amount of the merged entity in the basket doubles to reflect that it was once two reference entities. Yet, this change in exposure size of the merged entity does not matter to the seller or buyer of first-to-default protection. A credit event in any of the four names (including the merged entity) would still trigger the contract. A credit event experienced by the merged entity, however, has a larger notional impact. Therefore, it would simultaneously trigger both a first-to-default and a second-to-default contract. The seller of first-to-default protection sees a reduction in risk when there is a

merger, so the seller of second-to-default protection must see an increase in risk, for which he or she must be compensated. We would argue that the merger option is one factor that keeps implied correlation in small baskets higher than in large baskets.

exhibit A4

Replacement Mechanics



Source: Morgan Stanley Research

SECOND-THROUGH-NTH-TO-DEFAULTS

A second-through-Nth default basket is similar to a senior tranche, and any credit events after the first default trigger the contract. Unlike FTD baskets, the swap remains in effect until maturity or until all the covered credits default. The risk/return profile is similar to that of a senior tranche in that a protection seller is protected from idiosyncratic risk, but exposed to severe loss scenarios where multiple credits default (also known as “tail risk”). However, a key

difference is that the return profile of the protection seller in a second-through-fifth default basket is independent of the severity of the first loss.

Why would an investor sell protection on a second-through-fifth default basis? As with senior versus equity tranches, selling protection in a second-through-fifth default basket is potentially a safer or less volatile way to gain long exposure to a basket of credits. In exchange for a smaller premium, the protection seller has direct exposure to any defaults after the first credit event, but is protected from the first credit event. An investor with a positive view on the underlying credits can sell protection in this swap and potentially earn more premium than selling protection on some of the individual names. In contrast to first-to-default baskets, the seller of protection is short correlation (i.e., an increase in correlation among the underlying names would have a negative impact).

Why would an investor buy protection on a second-through-fifth default basket? In general, it can be a significantly cheaper way of hedging existing long credit exposure when the investor is willing to take the risk of some losses on the first credit event, but wants to hedge against more severe default scenarios.

FUNDED BASKETS

Most basket default swaps are structured and executed in “unfunded” form and are not assigned credit ratings. However, it is possible to combine an FTD basket with a deposit to create a funded product. In that case, the protection writer would buy a note with spread over Libor coming from an FTD position. In case of a default in the underlying portfolio, the note holder would incur a loss of principal on the note, reflecting the payment of par minus recovery on the FTD basket.

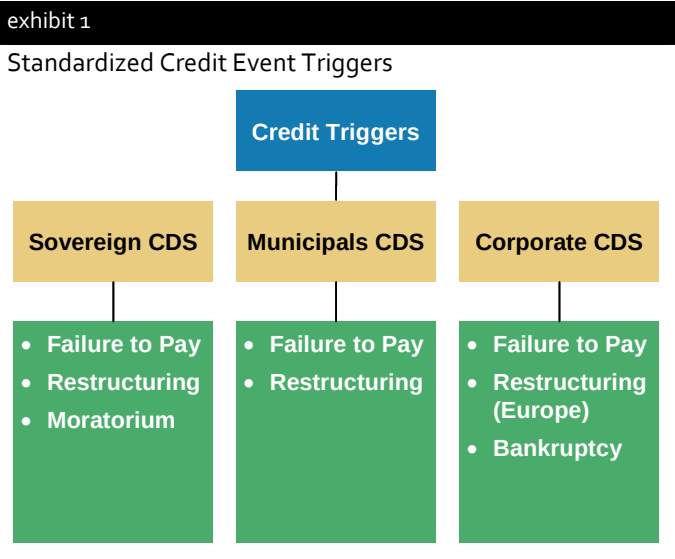
What Happens in a Default?

February 23, 2012

Ashley Musfeldt
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In a risk neutral framework, the concept of default and resultant loss is a primary source of value for credit derivatives and index based products, and determining an efficient and standardized value from default has been a key focus of the market since its beginnings over a decade ago.

Over the past several years we've seen numerous developments in the CDS market as it pertains to determining the loss from default after a credit event is declared. There has been significant evolution on everything from what constitutes a credit event, to what is deliverable in the contract, to finally how that deliverable is assigned a value in a timely manner post default.



Source: Morgan Stanley Research

Initially, given that all contracts were bilateral, market participants used physical settlement and the buyer of protection would actually deliver the bond and get par in return. Additionally, the buyer of protection had to trigger the CDS with each individual counterparty, rather than go through a centralized determination committee.

While there have been numerous changes to the contract to make it more fungible and standardized, the credit event triggers are still the primary drivers of valuation, and a contract with more potential triggers (such as restructuring) is generally more valuable or riskier than one without (see Appendix A for exact triggers and deliverables).

With regard to settlement, the market has come quite a long way from the first credit events, and today a standard auction process takes place shortly after a credit event is determined. The auction determines one fixed recovery level at which all CDS contracts are settled, including single name CDS, indices, baskets, tranches, options, and bespoke trades. Please see Appendix B for a sample list of CDS credit events that have been settled via the auction process.

While many in the market understand the process for single-name CDS, there is less appreciation for credit event settlement on portfolio products including indices, index options, baskets and tranches. This chapter endeavors to dig into various credit events and most importantly, how they impact various credit derivatives products. An overview of the impact on each product is as follows:

Single name CDS and FTD baskets. Upon default, a single name CDS contract (or first-to-default basket) will have an immediate payout of principal (following the auction), and the contract will terminate, with no future coupon streams.

Indices. An index will have a principal payout commensurate with the weighting of the name and recovery. The contract does not terminate, and the coupon is instead reduced through a smaller notional amount.

Options. In standard credit index options (where the option includes protection to expiry), a principal payment is not immediate post-default, and indeed only takes place if and after the option is exercised. The breakeven strike at which the option buyer would exercise will shift however.

Tranches. The impact on a tranche trade can include principal loss, reduction in subordination, loss of coupon or a combination of these, depending on the tranche and part of the capital structure.

WHAT CONSTITUTES A CREDIT EVENT?

Before we dig into the mechanics of a default and impact on various products, we must first determine what, exactly, constitutes a credit event. We divide credit events into two main buckets: “hard” credit events which include failure to pay and bankruptcy, and “soft” credit events, most notably restructuring. Currently US corporate CDS contracts do not include any type of restructuring after the SNAC protocol from 2009. European corporates, sovereigns, and municipal CDS all do, but the deliverables can vary.

Hard Credit Events: Bankruptcy and Failure-to-Pay. In a hard credit event, the CDS is automatically triggered once the ISDA Determination Committee rules a credit event has occurred and a list of valid deliverables is produced by ISDA. A single auction is conducted, and all CDS positions are settled based on the one recovery.

Soft Credit Events: Restructuring. In a restructuring, the CDS is not automatically triggered, and both the buyer and the seller have the right, but not the obligation, to trigger the CDS. For sovereign and municipal CDS, there is no maturity limitation, and anything with less than 30yrs to maturity is deliverable. European corporates have a different type of restructuring, known as ModModR, and deliverability of bonds depends on the maturity of the bond relative to CDS maturity.

A CREDIT EVENT IS DECLARED

In the past, credit events on single name CDS were settled either entirely in cash or entirely physically (with bonds or loans), and the standardized index tranches had a hybrid cash/physical settlement mechanism. However, both the sheer volume of outstanding CDS contracts on the indices and the demand for index tranches to be fungible after a credit event created huge demand in the marketplace for a standardized settlement process. Beginning in 2005 with the Collins & Aikman default, numerous investors have participated in standardized industry-wide settlements. The benefits of an auction are manifold, but the primary benefit is that it is operationally efficient to use a common recovery for all trades to eliminate basis risk between products.

The International Swaps and Derivatives Association (ISDA) publishes CDS protocols for defaulted entities, and the resulting auctions (including both senior and subordinated debt) are administered by MarkIt Partners and CreditEx. The protocols are available on the ISDA website, www.isda.org, and details of the auctions, including past results, are available on www.creditfixings.com. Both websites have credit event auction primers and presentations as well.

The objective of the auction is to concentrate trading in the bond(s) during a short window of time to arrive at a recovery. To get to an auction, there is a three step process:

1. Market participants submit credit event. To initiate this process, market participants submit a request to the appropriate regional Credit Derivatives Determination Committee (DC), a group comprised of both credit derivatives dealers and non-dealers. After the SNAC Big Bang protocol in the US, the request must be within two months of the credit event, as now all contracts have a fixed 60-day lookback period in which to declare a credit event.

2. DC decides if credit event has taken place: The DC then decides whether the credit event has occurred and what type.

3. The DC determines whether there will be an auction. In case the outstanding CDS in a name is very small, no auction may be conducted.

The process determines one recovery rate (or one for each maturity bucket in the case of a restructuring event), which is used to cash settle the credit event in all single name CDS contracts, index transactions and determine losses (for equity tranches) and subordination levels (for non-equity tranches) for tranches in all covered indices. We encourage readers to visit the ISDA website (www.isda.org) to get current information on the methodology, as our summary should in no way be considered a complete or accurate description.

A side note on restructuring. There are some additional provisions that are specific to the “Small Bang” protocol in Europe:

- The DC publishes an auction protocol which outlines the maturity buckets for which an auction will be conducted, and the deliverables for the same.
- Unlike other Credit Events, when a DC resolves that a Restructuring Credit Event has occurred, this does not automatically trigger the CDS.
- Both the buyer and the seller have the option, but not the obligation to trigger the CDS. When the final list of deliverable obligations is published, depending on the notifying party, the seller of protection has 2 business days and the buyer of protection has 5 business days to trigger the transaction.

THE AUCTION PROCESS: AN EXAMPLE

To illustrate the auction methodology, we take the example of Masonite from December 2008 for a relatively simple bankruptcy auction. The auction involves a two-stage process.

Stage 1. The dealers establish indicative markets by quoting both a bid and offer for the bonds in a pre-defined size and pre-defined bid / offer, which can vary by credit. In addition, dealers will submit physical settlement requests – that is, requests to buy or sell bonds through the auction. In the case of Masonite, dealers submitted 3pt bid-offers in 2.5MM size and had a net demand to buy bonds through the auction (18MM notional of bonds).

exhibit 2

Stage 1: Dealers Provide Markets

Dealer	Inside Markets		Physical Settlement Requests	
	Bid	Offer	Bid/Offer	Size MM
Dealer A	47.5	50.5	Offer	0.0
Dealer B	48.0	51.0	Bid	0.0
Dealer C	48.5	51.5	Offer	10.0
Dealer D	48.6	51.6	Offer	2.5
Dealer E	49.5	52.5	Offer	0.0
Dealer F	50.0	53.0	Bid	5.5
Dealer G	50.0	53.0	Offer	0.0
Dealer H	51.0	54.0	Offer	0.0
Dealer I	52.0	55.0	Bid	5.0
Dealer J	52.0	55.0	Bid	20.0
Total				18.0

Source: Creditfixings.com

From the bid-offers we arrive at the market mid-points. In Exhibit 3 we sort the bids in descending order and offers in ascending order separately and eliminate tradable markets.

exhibit 3

Stage 1: Arriving at Market Midpoint

Dealer	Bid	Offer	Dealer	
Dealer J	52.000	50.500	Dealer A	Crossed Markets
Dealer I	52.000	51.000	Dealer B	
Dealer H	51.000	51.500	Dealer C	Average top half of the remaining bids and offers
Dealer G	50.000	51.625	Dealer D	
Dealer F	50.000	52.500	Dealer E	
Dealer E	49.500	53.000	Dealer F	
Dealer D	48.625	53.000	Dealer G	
Dealer C	48.500	54.000	Dealer H	
Dealer B	48.000	55.000	Dealer I	
Dealer A	47.500	55.000	Dealer J	
Average	50.125	52.156		

Source: Creditfixings.com

Of the remaining 8 bids/offers, we take an average of the best half (rounded up) of the bids and offers, to get 50.125 as our inside bid and 52.156 as our inside offer. The inside market mid-point (IMM) is the average of these two rounded to the nearest 1/8th, ending up at 51.125 in this case.

Penalties. There is a penalty amount that dealers have to pay for being offside on the market – in this case, Dealer A and Dealer B submitted offers below the IMM, considering there is open interest of 18MM to buy bonds. The dealers have to pay the quotation times the difference between the IMM and the offer, so in this particular case this would be \$15,625 and \$3,125 (2.5MM * (offer – IMM)), as penalty, respectively.

exhibit 4

Stage 2: Establishing Auction Recovery

	Dealer	Offer	Size
17.5MM Limit Orders fully filled	Dealer C	50.125*	5
	Dealer A **	51.125*	2.5
	Dealer B **	51.125*	2.5
	Dealer C **	51.5*	2.5
	Dealer D **	51.625*	2.5
Limit orders partially filled on a pro-rata basis (0.5MM of 7.5MM)	Dealer B	52*	2.5
	Dealer E**	52.5^	2.5
	Dealer F	52.5^	5
	Dealer E	52.625	4
	Dealer F **	53	2.5
	Dealer G **	53	2.5
	Dealer H	53.5	2.5
	Dealer H**	54	2.5
	Dealer H	54	2.5
	Dealer F	54	4
	Dealer A	54.75	2.5
	Dealer I **	55	2.5
	Dealer J **	55	2.5
	Dealer F	55	3.5
	Dealer H	55	2.5
	Dealer E	55.5	5
	Dealer E	56	5

Source: Creditfixings.com

Note: * Limit orders that were filled. **Limit Orders that were derived from inside markets. ^ Limit Orders that were partially filled.

Stage 2. After stage 1 of the auction, market participants get a few hours to evaluate the information from stage 1 and submit bids or offers to close the open interest. Here, investors submitted offers on the bonds to close the net 18MM bid. The bids/offers for the pre-defined size (in this case the offers) from Stage 1 are carried forward to Stage 2. In the case of Masonite, the dealers are deemed to have offered 2.5MM of bonds at the offer levels in Stage 1. Next, the administrator will determine the final price from the auction by matching market orders until the open interest is cleared. In the case of Masonite, 52.5 was the clearing price (see Exhibit 4).

Restructuring. In the case of a mod R or mod mod R restructuring event, the auction is conducted by collapsing bonds and outstanding CDS into several buckets ensuring that the maturity limitation restriction we highlighted earlier is not violated. From a protection buyer perspective, the "Limitation Date" means the first of March 20, June 20, September 20 or December 20 in any year to occur on or immediately following the date that is one of the following numbers of years after the restructuring date: 2.5 years (the "2.5-year Limitation Date"), 5 years (the "5-year Limitation

Date"), 7.5 years, 10 years, 12.5 years, 15 years, 20 years or 30 years, as applicable. For each bucket, the restructuring maturity limitation we discussed earlier applies and an auction is conducted for buckets with sufficient CDS outstanding.

A NOTE ON PHYSICAL SETTLEMENT WITHIN THE AUCTION

One concern for investors who hold cash bonds and use CDS to hedge their positions is that the auction recovery may be different from the price of the specific bond the investor holds. To eliminate this basis risk, the investor can also physically settle the cash bond at the auction price as well.

As we note in this example, using the physical settlement requests, it is possible to settle the cash position at the CDS settlement price. Say an investor buys 100MM bonds and buys 100MM 5yr CDS protection. If a credit event were declared, the investor would have delivered the cash bond and received par from the investor before the auction process was standardized.

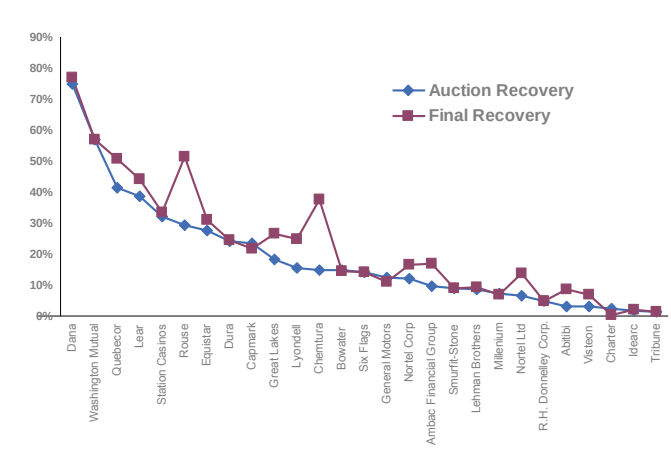
How can the same outcome be realized in the auction? First, the investor would have to submit a Physical Settlement Request to sell bonds in stage 1 of the auction process (Note that PSR represents a firm commitment to sell bonds at the auction final price). When the final recovery is determined (say 40%), the proceeds from selling this bond would be 40MM (100MM * 40%). The 100MM CDS protection is also cash settled and the investor receives 100MM * (1 - 40%) = 60MM. Thus, across the two positions the investor receives 40MM + 60MM, for a total of 100MM, or par of the original position. Note that this amount would add up regardless of the Auction Final Price.

AUCTION RECOVERY VS. FINAL RECOVERY

How predictive is the auction in terms of final recoveries? It should be noted that CDS credit events settle fairly early in the typical lifecycle of a corporate bankruptcy workout, where such workouts can take a few months at a minimum and several years as a maximum. So there is a frequent misconception in the market that CDS settlement represents the true "recovery" for a corporate entity when in fact it can take significant time for the workout to materialize, and investors in this workout process demand a return during the workout process.

exhibit 5

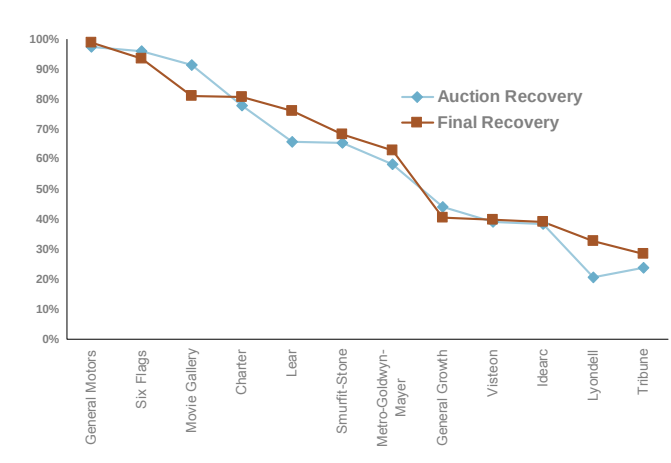
Auction Recoveries vs. Final Recoveries for Sample Set of Senior Unsecured Bonds



Source: ISDA, CreditFixings, Moody's

exhibit 6

Auction Recoveries vs. Final Recoveries for Sample Set of Senior Secured Bonds



Source: ISDA, CreditFixings, Moody's

Based on 27 corporations that have been through a full bankruptcy restructuring process and also had CDS settlement (for unsecured debt), we find that the average CDS recovery was 20.6% and the average final recovery in the bankruptcy process was 24.6%. The difference of 4 points represents a return of about 19%, which is in line with what many investors demand from a workout process (time period dependent). For secured debt, the average CDS recovery was 59.9% (on 12 names) and the average final recovery was 61.9%, representing an average 2 point return during the bankruptcy process.

In Exhibits 5 and 6, we look at a sample of US names for which we could get both auction and final recovery data to compare.

CREDIT EVENT SETTLEMENT MECHANICS: SINGLE NAME CDS

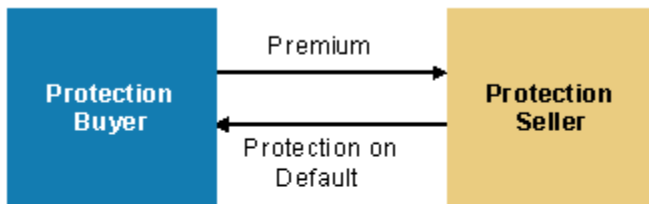
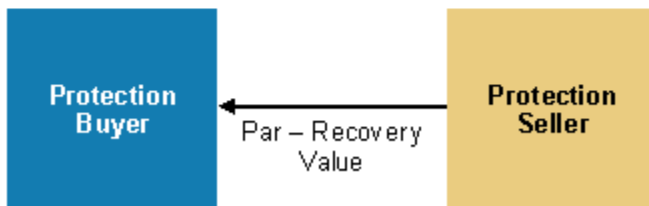
Prior to a credit event, a single name CDS protection buyer pays a running coupon (plus upfront at trade inception). If a credit event is declared, the auction process will determine recovery and the protection seller will make a payment of the notional amount of the contract, minus the recovery determined at auction.

Example: Investor buys 10MM CDS protection on Company XYZ, which defaults and a 40% recovery is determined at auction.

The investor receives 6MM from the seller of protection
 $10\text{MM} * (1 - 40\%) = 6\text{MM}$

Upon default, the contract terminates and no further coupon payments are made.

exhibit 7

Default Cashflows for Single Name CDS**CDS Cash Flows before Maturity / Default****Cash Settlement After Auction**

Source: Morgan Stanley Research

CREDIT EVENT SETTLEMENT MECHANICS: INDICES

When a default occurs in an index, the name is removed and settled at auction. The index then trades without the underlying name (typically with the title v2 to differentiate it) and on a smaller notional. The settlement of the defaulted credit occurs based on the scaled notional of that credit (thus, 1/125 in the case of CDX IG and iTraxx Main).

Example: Investor buys protection on 100MM of a 125 name index and Company A defaults. A recovery of 40% is determined at auction.

The investor receives 480k
 $100\text{MM} / 125 = 800\text{k per name. } 800\text{k} * (1 - 40\%) = 480\text{k}$

The investor now owns protection (and pays coupon) on 99.2MM index (v2) with 124 names
 $100\text{MM} - 800\text{k} = 99.2\text{MM}$

In Exhibit 8, we highlight historical indices and series and number of historical credit events per index.

exhibit 8

Number of Defaults Across Indices

Series	CDX IG	CDX HY	iTraxx Main	iTraxx Xover
4	7	16	1	3
5	5	18	1	3
6	6	16	1	3
7	5	16	1	4
8	4	16	0	5
9	4	17	0	5
10	4	16	0	4
11	1	16	0	2
12	1	9	0	1
13	0	4	0	1
14	0	3	0	1
15	0	3	0	0
16	0	3	0	0
17	0	3		

Source: Morgan Stanley Research

CREDIT EVENT SETTLEMENT MECHANICS: OPTIONS

In an index options trade (where protection is included through the expiry of the option), payout of principal will only occur if and after an option is exercised – nothing happens until the option expiry. If the credit event auction settlement has not occurred prior to expiry, the option exercises normally, and the trade becomes a standard index trade.

If, however, the auction takes place prior to expiry, the buyer of protection will not only receive protection on the remaining index (known as version 2), but also a payout of par minus recovery (determined at auction) on the defaulted name. Because the option is on the version 1 index, at expiry, the option buyer will base the exercise decision on the equivalent version 1 index spread, which consists of the version 2 index spread plus loss payout from the defaulted name.

Payer Option Example: An investor buys a 100MM notional 130 strike payer option with an expiry in December. Credit ABC defaults in October, and settles (prior to expiry) via the auction process at 40% recovery.

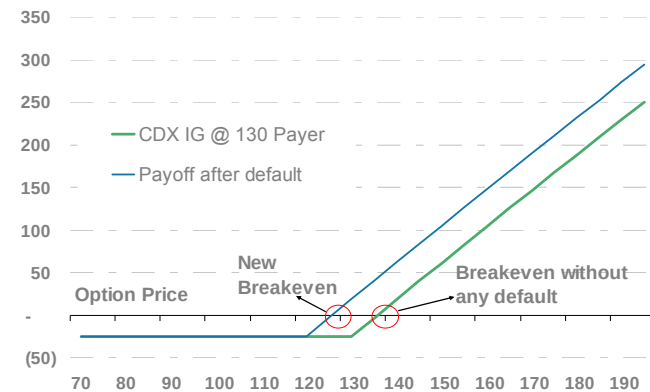
If the option is exercised at expiry, the investor will enter into a trade to buy protection on the 124 name version 2 index at 130bps and will also receive 480k from the protection on the defaulted credit
 $100\text{MM} / 125 = 800\text{k} * (1 - 40\% \text{ recovery}) = 480\text{k}$

To calculate the new breakeven, we divide the additional payout of 0.48% by the duration of the index to get the

version 2 equivalent strike price ($0.48\% / 4.8 = 10\text{bps}$). Thus, it would still be economical to exercise the 130 strike payer if the version 2 index were trading wider than 120bps. See Exhibit 9 for an illustration.

exhibit 9

Payer Option Breakeven With and Without Default



Source: Morgan Stanley Research

Receivers. For an investor who bought a receiver in the previous example, the exercise is similar. At expiry, the investor would have the right to sell protection on the 124 name version 2 index, but would owe the 480k payout to the buyer of protection (receiver seller in this case). Thus the decision to exercise the option would be based on whether the spread at expiry was *tighter* than 120, rather than the strike of 130, because he or she would still owe 480k to the buyer of protection.

CREDIT EVENT SETTLEMENT MECHANICS: TRANCHES

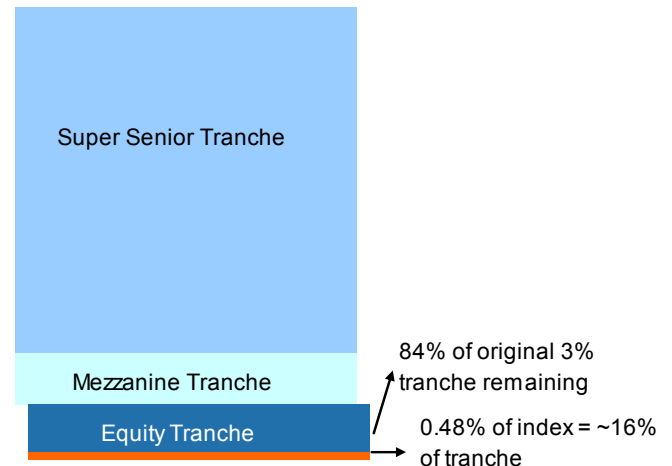
In tranches, a loss in the underlying index will have a different impact depending on the part of the capital structure represented by the tranche. A default can result in principal loss (first loss pieces), reduction in subordination (all *except* first loss pieces), and / or reduction in coupon (first loss and super seniors only).

Unlike options, tranches settle immediately (loss of subordination, principal, or coupon) following the auction, like single name CDS and indices. From a pricing standpoint, a default will impact various tranches differently but will depend highly on how expected the default was. For equity tranches, it can be the case that the remaining tranche actually trades *tighter* after the default, if the name in question was priced in and trading at a very wide level whereas the impact on super senior can be more mixed. We describe the mechanics of each in greater detail as follows:

First Loss Tranche: Principal Loss. In the case of an equity tranche, a loss on the index would equate to a principal loss on the tranche, and a reduction in coupon due to the reduced notional.

exhibit 10

Equity Tranche: Principal Loss



Source: Morgan Stanley Research

Example: The investor buys 30MM protection on a 0-3% tranche in an index with 125 names. The tranche exposure per name is 8MM.

$30\text{MM} / 3\% \text{ thickness} = 1\text{BN Index equivalent exposure}$
 $/ 125 \text{ names} = 8\text{MM exposure per name}$

One name defaults and recovers at 40% at auction. The loss to the tranche is 4.8MM of the original 30MM, or 16% of the original tranche notional

$8\text{MM} * (1-40\%) = 4.8\text{MM} / 30\text{MM tranche notional} = 16\%$

The tranche continues to receive coupon on the remaining 25.2MM notional.

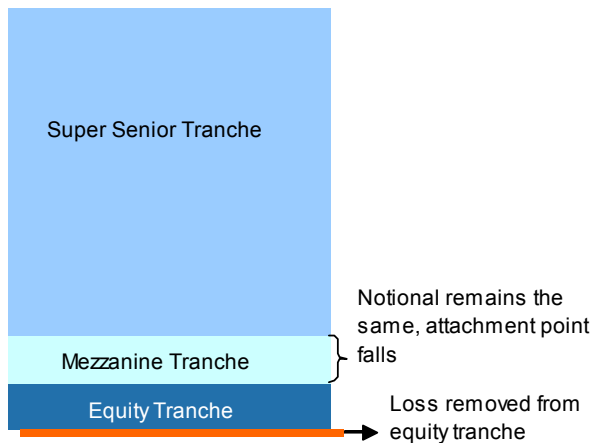
$30\text{MM} - 4.8\text{MM} = 25.2\text{MM}$

Note that in the example, the tranche would now have 25.2MM notional remaining, and this would constitute 2.5403% of the remaining capital structure (25.2MM remaining tranche notional / 992MM remaining index notional). When a tranche experiences a principal loss, new trades will be quoted on a factor, or percent of the original tranche notional. In our example above, the tranche would now be quoted on a factor of 84%.

Mezzanine Tranche: Loss of Subordination. For a more senior tranche, such as 7-10%, a loss of less than 7% in the index would NOT translate to a loss on the tranche itself, but rather a loss of subordination.

exhibit 11

Mezzanine Tranche



Source: Morgan Stanley Research

Example: The investor buys 30MM protection on a 7-10% tranche in an index with 125 names. The tranche exposure per name is 8MM.

30MM / 3% thickness = 1BN Index equivalent exposure
/ 125 names = 8MM exposure per name

The amount of losses in the underlying index required to impact the tranche is 70MM, and after 100MM losses in the index, the tranche will be wiped out

1BN index equivalent * 7% attachment = 70MM

1BN index equivalent * 10% detachment = 100MM

One name defaults and recovers at 40% at auction. While there is no reduction in principal, the subordination of the tranche will change. There will be a 0.48% writedown in the original index on the subordination of the tranche, and the amount of losses needed for the tranche to get hit will be reduced. In our example, instead of 70MM subordination, the tranche will now have 65.2MM subordination.

70MM original attachment – 4.8MM losses = 65.2MM remaining attachment

Thus, the new tranche will be 6.573% - 9.597%

65.2MM new attachment / 992MM reduced index notional = 6.573%

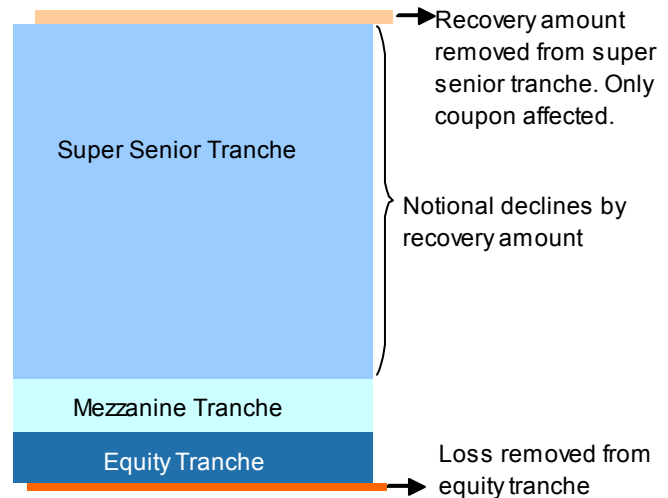
The tranche continues to receive coupon on the 30MM original notional.

Super Senior Tranche: Writedown from Above. In a super senior tranche that attaches at 100%, the same reduction in subordination will apply as in the mezzanine tranche, and additionally, there will be what is known as write down from above, in which the recovery value is removed from the super senior tranche, and the coupon paid on the tranche is reduced, though no principal payment is

exchanged. The reason for this is that the index notional is reduced by the full notional exposure of the defaulted name, whereas the equity tranche is reduced by only the loss amount (par minus recovery). Thus, to make the sum of the tranche notionals add up to the index, the recovery value is removed from the top.

exhibit 12

Super Senior Tranche: Decline in Coupon



Source: Morgan Stanley Research

Example: The investor buys 700MM protection on a 30-100% tranche in an index with 125 names. Exposure per name is still 8MM

700MM / 70% thickness = 1BN Index equivalent exposure
/ 125 names = 8MM exposure per name

The amount of losses in the underlying index required to impact the tranche is 300MM

1BN index equivalent * 30% attachment = 300MM

One name defaults and recovers at 40% at auction. Again, there will be a 0.48% writedown in the original index on the subordination of the tranche, and the amount of losses needed for the tranche to experience losses will be reduced. Thus, instead of 300MM subordination, the tranche will now have 295.2MM subordination.

300MM original attachment – 4.8MM losses = 295.2MM remaining attachment

Because this is a super senior tranche, the recovery value of the defaulted name will be taken from the tranche principal. Thus, the 700MM notional will be reduced to 696.8MM and the coupon will be paid on this reduced notional.

700MM – (40% * 8MM = 3.2MM) = 696.8MM

Thus, the remaining tranche would be 29.758% - 100%, and would now constitute 70.242% of the remaining index.

295.2MM / 992MM = 29.758% attachment
696.8MM / 992MM = 70.242% thickness

CREDIT EVENT SETTLEMENT MECHANICS: BASKETS

For first-to-default basket trades, the process is identical to a single name CDS trade. The contract settles as though it were a single name contract on just the defaulted credit.

Example: Investor buys 10MM first-to-default protection on a basket of 5 names. One of those names defaults, and a 40% recovery is determined at auction.

The investor receives 6MM from the seller of protection
10MM * (1 - 40%) = 6MM

Upon default, the contract terminates and no further coupon payments are made.

A second-to-default basket would be similar mechanically, but only if an additional name in the basket were to default. So upon default of only one name, nothing happens, and the coupon would remain the same. After the second default in the original basket, that name would be settled and the contract would terminate.

A second-through-Nth-to default is only slightly different. In a basket where “Nth-to-default” is more than one name, the mechanics would work something like this:

Example: Investor buys 40MM second-through-Nth-to-default protection on a basket of 5 names. One of those names defaults, and nothing happens. The second of those

names defaults, and a 40% recovery is determined at auction.

The investor receives 6MM from the seller of protection
40MM * (1/4 names) * (1 - 40%) = 6MM

Upon default, the contract remains outstanding and coupon is paid on the notional amount of 30MM
40MM – 10MM notional of defaulted name = 30MM

In a basket trade, merger language can play a role in terms of what happens in a default. There are two ways to handle mergers in a basket, each with its own implications in terms of defaults:

With replacement. This is the market standard today for handling merger events in a basket. In this instance, when a merger occurs, the buyer and seller of protection must mutually agree on a replacement credit, and often this replacement credit is one that trades at a similar spread level.

Without replacement. If a basket trades without replacement (which is rare) a five-name basket where two reference entities merge effectively becomes a four-name basket once the merger closes. The “contractual” notional amount of the merged entity in the basket doubles to reflect that it was once two reference entities. This change in exposure size of the merged entity does not matter for an FTD trade. A credit event in any of the four names (including the merged entity) would still trigger the contract. A credit event experienced by the merged entity, however, has a larger notional impact and would simultaneously trigger both a first-to-default and a second-to-default contract.

Appendix A

exhibit A.1

Overview of Standard Credit Events for Various CDS Contracts

	Bankruptcy	Failure to Pay	Failure to Pay (Grace Period Appl.)	Obligation Acceleration	Repudiation/ Moratorium	Restructuring (Old R)	Restructuring Maturity Limitation and Fully Transferable Obligation (Mod R)	Modified Restructuring Maturity Limitation and Conditionally Transferable Obligation (Mod Mod R)	Restructuring (Multiple Holder Obligation Required?)
Asia Sovereign	N	Y	N	N	Y	Y	N	N	Y
Singapore Sovereign	N	Y	N	N	Y	Y	N	N	Y
Latin America Sovereign	N	N	Y	Y	Y	Y	N	N	N
Emerging European & Middle Eastern Sovereign	N	N	Y	Y	Y	Y	N	N	N
Western European Sovereign	N	Y	N	N	Y	Y	N	N	Y
Japan Sovereign	N	Y	N	N	Y	Y	N	N	N
Australia Sovereign	N	Y	N	N	Y	N	Y	N	Y
New Zealand Sovereign	N	Y	N	N	Y	N	Y	N	Y
U.S. Municipal Full Faith And Credit	N	Y	N	N	N	Y	N	N	N
U.S. Municipal General Fund	N	Y	N	N	N	Y	N	N	N
U.S. Municipal Revenue	N	Y	N	N	N	Y	N	N	N
North American Corporate	Y	Y	N	N	N	N	N*	N	Y
European Corporate	Y	Y	N	N	N	N	N	Y	Y
Australia Corporate	Y	Y	N	N	N	N	Y	N	Y
New Zealand Corporate	Y	Y	N	N	N	N	Y	N	Y
Japan Corporate	Y	Y	N	N	N	Y	N	N	N
Singapore Corporate	Y	Y	N	N	N	Y	N	N	Y
Asia Corporate	Y	Y	N	N	N	Y	N	N	Y
Subordinated European Insurance Corporate	Y	Y	N	N	N	Y	N	N	Y
Emerging European Corporate LPN	Y	N	Y	Y	Y	Y	N	N	Y/N
Emerging European Corporate	Y	N	Y	Y	Y	Y	N	N	Y/N
"Latin America Corporate B"	Y	N	Y	Y	Y	Y	N	N	N
"Latin America Corporate BL"	Y	N	Y	Y	Y	Y	N	N	Y
Sukuk Corporate	Y	N	Y	Y	Y	Y	N	N	Y/N
Sukuk Sovereign	N	N	Y	Y	Y	Y	N	N	N

Source: ISDA

*Restructuring if specified as applicable in the relevant Confirmation

exhibit A.2

Overview of Obligation Characteristics to Trigger a Credit Event

	Obligation Category	Not Subordinated	Not Sovereign Lender	Not Domestic Currency	Not Domestic Law	Not Domestic Issuance	Specified Curr. & Domestic Currency	Full Faith & Credit Oblig. Liab.	General Fund Oblig. Liab.	Revenue Oblig. Liab.
Asia Sovereign	Bond /Loan	Y	Y	Y	Y	Y	N	N	N	N
Singapore Sovereign	Bond /Loan	Y	Y	N	N	N	Y	N	N	N
Latin America Sovereign	Bond	Y	N	Y	Y	Y	N	N	N	N
Emerging European & Middle Eastern Sov.	Bond	Y	N	Y	Y	Y	N	N	N	N
Western European Sovereign	BM	N	N	N	N	N	N	N	N	N
Japan Sovereign	BM	N	N	N	N	N	N	N	N	N
Australia Sovereign	BM	N	N	N	N	N	N	N	N	N
New Zealand Sovereign	BM	N	N	N	N	N	N	N	N	N
U.S. Municipal Full Faith And Credit	BM	Y	N	N	N	N	N	Y	N	N
U.S. Municipal General Fund	BM	Y	N	N	N	N	N	N	Y	N
U.S. Municipal Revenue	BM	Y	N	N	N	N	N	N	N	Y
North American Corporate	BM	N	N	N	N	N	N	N	N	N
European Corporate	BM	N	N	N	N	N	N	N	N	N
Australia Corporate	BM	N	N	N	N	N	N	N	N	N
New Zealand Corporate	BM	N	N	N	N	N	N	N	N	N
Japan Corporate	BM	Y	N	N	N	N	N	N	N	N
Singapore Corporate	Bond /Loan	Y	Y	N	N	N	Y	N	N	N
Asia Corporate	Bond /Loan	Y	Y	Y	Y	Y	N	N	N	N
Subordinated European Insurance Corp.	BM	N	N	N	N	N	N	N	N	N
Emerging European Corporate LPN	Bond /Loan	Y	N	Y	Y	Y	N	N	N	N
Emerging European Corporate	Bond /Loan	Y	N	Y	Y	Y	N	N	N	N
"Latin America Corporate B"	Bond	Y	N	Y	Y	Y	N	N	N	N
"Latin America Corporate BL"	Bond /Loan	Y	Y	Y	Y	Y	N	N	N	N
Sukuk Corporate	Bond /Loan	Y	N	Y	Y	Y	N	N	N	N
Sukuk Sovereign	Bond	Y	N	Y	Y	Y	N	N	N	N

Source: ISDA

exhibit A.3

Overview of Deliverable Obligations for Various CDS Contracts

	Obligation Category	Not Subordinated	Not Sovereign Lender	Not Domestic Law	Not Domestic Issuance	Maximum Maturity: 30 years	Assignable Loan	Consent Required Loan	Specified Currency	Standard Specified Currencies & Domestic Currency	Specified Currencies & Domestic Currency	Standard Specified Currencies
Latin America Sovereign	Bond	Y	N	Y	Y	N	N	N	Y	N	N	N
Emerging European & Middle Eastern Sov.	Bond	Y	N	Y	Y	N	N	N	Y	N	N	N
Western European Sovereign	Bond /Loan	N	N	N	N	Y	Y	Y	Y	N	N	N
Japan Sovereign	Bond /Loan	N	N	N	N	Y	Y	Y	Y	N	N	N
Australia Sovereign	Bond /Loan	Y	N	N	N	Y	Y	Y	N	Y	N	N
New Zealand Sovereign	Bond /Loan	Y	N	N	N	Y	Y	Y	N	Y	N	N
Asia Sovereign	Bond /Loan	Y	Y	Y	Y	Y	Y	N	Y	N	N	N
Singapore Sovereign	Bond /Loan	Y	Y	N	N	Y	Y	N	N	N	Y	N
U.S. Municipal Full Faith And Credit*	Bond /Loan	Y	N	N	N	Y	Y	Y	N	N	N	Y
U.S. Municipal General Fund**	Bond /Loan	Y	N	N	N	Y	Y	Y	N	N	N	Y
U.S. Municipal Revenue***	Bond /Loan	Y	N	N	N	Y	Y	Y	N	N	N	Y
North American Corporate	Bond /Loan	Y	N	N	N	Y	Y	Y	Y	N	N	N
European Corporate	Bond /Loan	Y	N	N	N	Y	Y	Y	Y	N	N	N
Australia Corporate	Bond /Loan	Y	N	N	N	Y	Y	Y	N	Y	N	N
New Zealand Corporate	Bond /Loan	Y	N	N	N	Y	Y	Y	N	Y	N	N
Japan Corporate	Bond /Loan	Y	N	N	N	Y	Y	Y	Y	N	N	N
Singapore Corporate	Bond /Loan	Y	Y	N	N	Y	Y	N	N	Y	N	N
Asia Corporate	Bond /Loan	Y	Y	Y	Y	Y	Y	N	Y	N	N	N
Subordinated European Insurance Corp.	Bond /Loan	Y	N	N	N	Y	Y	Y	Y	N	N	N
Emerging European Corporate LPN	Bond /Loan	Y	N	Y	Y	N	Y	Y	Y	N	N	N
Emerging European Corporate	Bond /Loan	Y	N	Y	Y	N	Y	Y	Y	N	N	N
"Latin America Corporate B"	Bond	Y	N	Y	Y	N	N	N	Y	N	N	N
"Latin America Corporate BL"	Bond /Loan	Y	Y	Y	Y	N	Y	Y	Y	N	N	N
Sukuk Corporate	Bond /Loan	Y	N	Y	Y	N	Y	Y	Y	N	N	N
Sukuk Sovereign	Bond	Y	N	Y	Y	N	N	N	Y	N	N	N

Source: ISDA

In addition: * Full Faith and Credit Obligation Liability, **General Fund Obligation Liability, ***Revenue Obligation Liability

Appendix B

exhibit B.2

Overview of Corporate Credit Event Auctions and Recoveries

Credit Event and Auction Protocol Summary
As of February 2012

Credit Event	CE Date	Auction Date	Recovery			Credit Event	CE Date	Auction Date	Recovery		
			Snr Sec	Sen Unsec	Sub				Snr Sec	Sen Unsec	Sub
Collins and Aikman	5/17/2005	6/14/2005		43.625%	6.375%	Idearc	3/31/2009	4/23/2009	38.500%	1.750%	
Northwest Airlines	9/14/2005	10/11/2005		28.000%		Bowater	4/16/2009	5/12/2009		15.000%	
Delta Airlines	9/14/2005	10/11/2005		18.000%		General Growth	4/16/2009	5/13/2009	44.250%		
Dana	3/3/2006	3/31/2006		75.000%		Syncora	5/1/2009	5/27/2009		15.000%	
Dura	10/30/2006	11/28/2006		24.125%	3.500%	JSC BTA Bank	5/1/2009	6/10/2009		10.250%	
Movie Gallery	9/15/2007	10/23/2007	91.500%			HLI Operating Co.	5/13/2009	6/9/2009	9.500%		
Quebecor	1/21/2008	2/19/2008		41.250%		R.H. Donnelley Corp.	5/18/2009	6/11/2009		4.875%	
Tembec	9/4/2008	10/2/2008		83.000%		Dex Media East	5/18/2009	6/24/2009	77.250%		
Fannie Mae	9/7/2008	10/6/2008		91.510%	99.900%	R.H. Donnelley Inc.	5/18/2009	6/24/2009	78.125%		
Freddie Mac	9/7/2008	10/6/2008		94.000%	98.000%	Dex Media West	5/18/2009	6/24/2009	82.875%		
Lehman Brothers	9/15/2008	10/10/2008		8.625%		Georgia Gulf	5/19/2009	6/10/2009	83.000%		
Washington Mutual	9/27/2008	10/23/2008		57.000%		JSC Alliance Bank	5/19/2009	6/18/2009		16.750%	
Landsbanki	10/8/2008	11/4/2008		1.250%	0.125%	Visteon	5/28/2009	6/23/2009	39.000%	3.000%	
Glitnir	10/8/2008	11/5/2008		3.000%	0.125%	General Motors	6/1/2009	6/12/2009	97.500%	12.500%	
Kaupthing Bank	10/9/2008	11/6/2008		6.625%	2.375%	Six Flags	6/13/2009	7/9/2009	96.125%	14.000%	
Masonite	11/14/2008	12/9/2008	52.500%			Lear	7/6/2009	7/21/2009	66.000%	38.500%	
Hawaiian Telecom	12/1/2008	12/17/2008	40.125%			Bradford & Bingley	7/9/2009	7/30/2009		94.625%	5.000%
Tribune	12/8/2008	1/6/2009	23.750%	1.500%		Thomson	8/12/2009	10/22/2009		65.125%	
Ecuador	12/18/2008	1/14/2009		31.375%		NJSC Naftogaz	10/14/2009	12/16/2009		83.500%	
Lyondell	1/7/2009	2/3/2009	20.750%	15.500%		Metro-Goldwyn-Mayer	10/19/2009	11/10/2009	58.500%		
Millenium	1/7/2009	2/3/2009		7.125%		Hellas II	11/19/2009	12/15/2009		1.375%	
Equistar	1/7/2009	2/3/2009		27.500%		Thomson (Bankruptcy)	12/1/2009	12/10/2009		77.750%	
Nortel Ltd	1/14/2009	2/10/2009		6.500%		FGIC	12/3/2009	1/7/2010		26.000%	
Nortel Corp	1/14/2009	2/10/2009		12.000%		Cemex	12/14/2009	2/18/2010		97.000%	
Smurfit-Stone	1/26/2009	2/19/2009	65.375%	8.875%		Aiful	12/30/2009	3/2/2010		33.875%	
Station Casinos	3/4/2009	3/31/2009		32.000%		Ambac Assurance Corp	3/26/2010	6/4/2010		20.000%	
Chemtura	3/19/2009	4/14/2009		15.000%		Boston Generating	8/18/2010	9/21/2010	96.750%		
Great Lakes	3/20/2009	4/14/2009		18.250%		Ambac Financial Group	11/8/2010	12/10/2010		9.500%	
Rouse	3/20/2009	4/15/2009		29.250%		Dynegy Holdings	11/8/2011	11/29/2011		71.250%	
LyondellBasell	3/23/2009	4/16/2009		2.000%		PMI Group	11/23/2011	12/13/2011		16.500%	
Abitibi	3/27/2009	4/17/2009		3.250%		AMR Corporation	11/29/2011	12/15/2011		23.500%	
Charter	3/27/2009	4/21/2009	78.000%	2.375%		Eastman Kodak	1/19/2012	2/22/2012		23.875%	
Capmark	3/30/2009	4/22/2009		23.375%							

Source: MarkIt Partners, CreditFixings

Basis Basics in a Normalized World

March 19, 2010

Sivan Mahadevan
Ashley Musfeldt
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From the beginnings of the corporate CDS markets, one of the most basic valuation measures has involved comparing a corporate bond with a CDS contract. While it is simple in concept, this basis relationship is quite complicated, as one instrument is not necessarily an easy substitute for the other. That complexity led many to treat basis trades as an investment strategy with an opportunity to arbitrage pricing and earn carry for taking on specific “basis” risks. Consequently, large basis books were built up across the financial markets.

However, the basis opportunity was perhaps oversimplified, and many basis trades suffered from the actual differences between the instruments during the financial crisis. Differences range from “bond math” type things like coupons, duration, price, yield, and convexity to much broader market themes like liquidity, deliverability/fungibility and of course funding costs. In fact the last point is one that is lost on most pricing models as the dearth of capital at the peak of the crisis pushed the basis to levels that were more negative (cash trading wider than CDS) than the market had ever seen in the history of the CDS market.

The negative basis trade post-Lehman was a significant market opportunity (along with many others), and one that we argued required a long holding period to realize value (see [The Basis Moves from a Trade to an Asset](#), October 31, 2008). Today, the corporate basis relationship has nearly completely healed from the systemic woes that began with the Lehman bankruptcy, as liquidity has returned to the market. While the basis remains negative in the US, it is and has actually been positive in Europe for some time.

So in a more normalized world of funding, liquidity, and risk premiums, we see a return to more normalized basis relationships, and market themes can push the basis to one extreme or the other. Trading the basis is a notion that feels like it is back in markets today; however, as we highlighted above, basis trades do have a funding/liquidity component to them that we are not willing to ignore even if it remains difficult to model. In a sense, the market should demand a premium for taking on this risk, and can look to funding markets as a guide. In this chapter, we use our pre-crisis, crisis, and post-crisis experience with the corporate basis relationship to provide our thoughts on basis relationships generally. Summaries of the themes are below.

Over a decade of history. From the early days of CDS, the corporate basis relationship has moved through several cycles.

It started out as positive owing to the “youth” of the then CDS contract and high default risk in the 2001/2002 credit cycle. The basis then moved to being consistently negative in the midst of the demand for synthetic credit during the structured credit bid during 2004-2007. Extreme negative basis levels appeared during the peak of financial crisis as funding became dear, but today we have moved to a more normalized environment.

Basis metrics: Over the years, we have developed three approaches to measuring the relative value of CDS vs. corporate bonds. We review our simple raw basis measure, more elaborate adjusted basis measure, and our fair value CAPS basis model to gain some insight into valuation.

Interest rate impact. Rates too can have an impact on basis pricing, as rates impact both bond pricing and discounts on CDS upfronts, as well as play a role in determining the steepness of CDS curves. We examine the impact of potentially rising rates on the basis.

High yield as a special case. The high yield basis has always been a different animal than the investment grade one, owing to a variety of factors including higher default probabilities, dollar price differences, the structure of cash flows and of course the callable nature of much of the market (a meaningful portion of the HY universe is callable). The high yield basis still trades at more negative levels than investment grade in both the US and Europe.

A BIT OF HISTORY OF CORPORATE CDS FLOWS

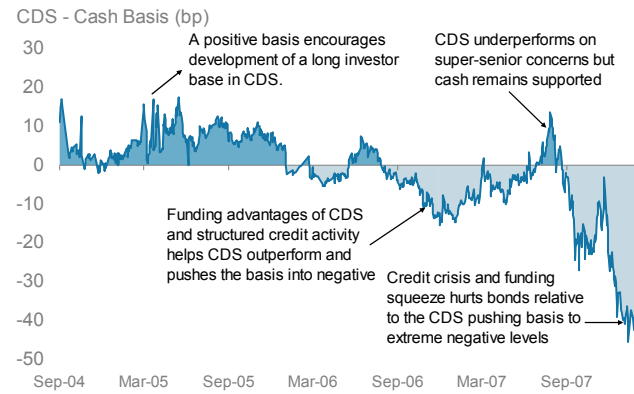
Corporate CDS markets were established in the mid-1990s, mainly as a way for banks to hedge their loan exposures, and much of the initial shape of the contract (restructuring risk, shorter maturities) was influenced by this community. But a one-way trade never makes markets, so it was critically important for the market to embrace CDS from an investment perspective as well. As such, in the late 1990s and early 2000s, CDS took a form that was more “bond-like” and subsequently the contract became more investor friendly.

As the CDS market development progressed, we began to see large investment flows through the sale of protection in corporate CDS contracts, mainly in portfolio form through structured credit and market indices. This natural seller of protection was an important factor in market growth by providing a balance in the market. Eventually this flow became a dominant force for CDS market performance, at least until the systemic risk period of the recent crisis, although even during the crisis cash products suffered more from their inherent funding disadvantage. The most recent changes to the corporate CDS market – the big and small bang protocols – have made the contract more fungible by

standardizing coupons, requiring adherence with ISDA protocols, and addressing the concerns surrounding restructuring risk.

exhibit 1

Evolution of Corporate CDS-Cash Basis (2004-07)

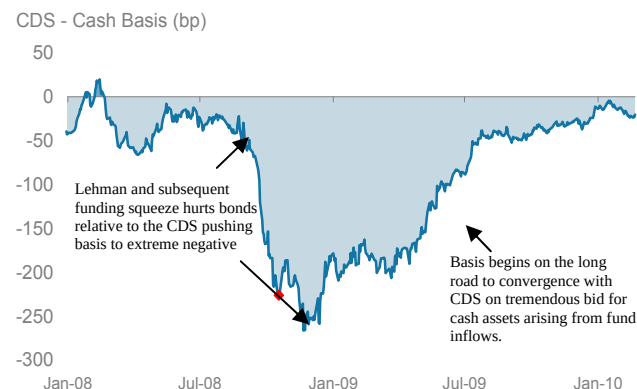


Source: Morgan Stanley Research

Throughout the history of the development of the CDS market, the basis between CDS and cash bonds has been an important relationship to monitor. Cash bonds once helped price CDS contracts, as cash bond spreads over Libor were the starting point to evaluate credit risk in CDS, after which they were adjusted for restructuring credit events and cheapest-to-deliver optionality. In the early days, CDS generally traded wide to cash instruments as the newness and nuances of the contract motivated protection sellers to demand a premium. Indeed the positive basis was a marketing point for market-makers wishing to promote liquidity in the derivatives contracts. As CDS markets matured, differences became more granular, and market participants built more sophisticated models. In these early days funding costs of cash instruments were largely ignored and liquidity on both sides of the basis remained robust.

exhibit 2

Evolution of Corporate CDS-Cash Basis (2007-10)



Source: Morgan Stanley Research

The corporate negative basis trade – positive carry long bond vs. buy protection through CDS – that emerged circa 2004 was essentially a balance sheet usage allowance for those investors who were able to fund at attractive levels and use this balance sheet to earn a small carry for taking on some amount of counterparty risk, but no actual credit default risk. Regulatory capital charges (for banks) for perfectly matched bond and CDS positions were and continue to be minimal for basis trades. There was also the added benefit of optionality through covenanted basis packages. Investors bought bonds with covenants (poison puts, rating step-ups) which outperform in LBO/M&A scenarios and with ratings deterioration, and buy CDS protection. The basis trade played a part in absorbing protection sold by investors in structured credit and at the same time provided the investment community with a way to retain illiquid secondary market cash positions.

Around 2005 the cash-CDS basis turned negative, and has remained largely negative ever since, save for a short period when super senior concerns drove it into positive territory (see Exhibit 2). From 2005 to 2007, the basis traded primarily in the negative 15 to 20bp range. In an environment where AAA/AA assets traded under 10bp, 15-20bp seemed reasonable compensation for basis trades at that time.

In 2008 the basis started dipping to new lows in the negative 50bps range and hovered there until the Lehman bankruptcy triggered a larger credit squeeze. Conventional wisdom would suggest that CDS should underperform cash bonds in a high default environment, given the fact that CDS is a pure play on credit risk, whereas cash bonds are a combination of interest rate and credit risk and are short cheapest-to-deliver optionality. However, during the credit crisis, liquidity issues trumped default spreads, with balance sheet usage becoming a bigger concern for investors and the market than default risk. The basis re-priced to unseen levels, reflecting the additional premium charged by the market on fully funded assets. Unsurprisingly, financials were the worst performers during this time period. With the basis approaching negative 300 bp at the height of the crisis, we opined in our report from October 31, 2008 [The Basis Moves from a Trade to an Asset](#) that the basis had become an asset unto itself, which could offer fully funded investors high unlevered returns for no credit risk and little counterparty risk. Abundant liquidity, accommodative policy and deleveraging meant that the basis began on the long road to recovery in the second half of 2009.

MEASURING THE BASIS: FACTORS TO CONSIDER

The simplest measure of the basis (raw basis) is the difference between the 5 yr CDS spread and the actual cash spread of the bond. This basis can be considered as an “actionable” metric based on bond Z spreads and CDS spreads quoted widely in the market. For bonds trading close to par and roughly 5 years to maturity, the raw basis is a pretty good estimate of the yield pickup an investor can gain by exploiting the arbitrage opportunities between the cash

and CDS spread. However, apart from the potential maturity mismatch between the bond and CDS, this metric does not take into account the effect of credit curve shape and bond dollar price, which greatly affect the fair value of the basis.

Credit curve shape. Extremely negative basis packages are often driven by flat to inverted credit curves as seen during the credit crisis of 2008-09. Inverted credit curves in theory imply a lot of near-term risk to bond coupons, which in turn makes bonds less attractive in a fair value sense. The raw basis calculation uses a simple Z spread measure, which assumes a flat credit curve and does not assign survival probabilities to bond coupon cash flows, even if it is implied in the CDS market.

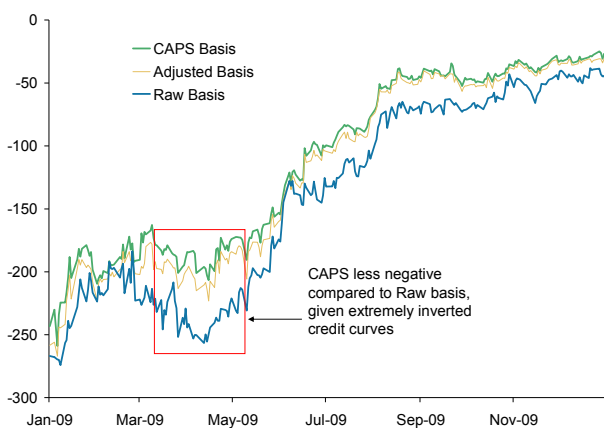
Bond dollar price. For bonds trading at a discount (or premium), buying CDS protection on par notional could mean that an investor is buying too much (or too little) protection. For bonds trading at a significant discount or premium, this tends to distort the fair value of the basis.

BASIS METRICS: ADJUSTED AND CAPS BASIS

The adjusted basis uses the simple raw basis as a building block, and then makes an adjustment based on the dollar price of the cash bond. To compensate for the deviation of bond dollar price from par, we purchase (or sell) additional protection for a premium (discount) bond, based on the average forward price of the bond for each year until maturity and calculate an adjusted Z spread metric. The maturity mismatch of the cash bond and CDS contract is also adjusted for by interpolating the CDS contract along the CDS curve.

exhibit 3

US IG Fair Value versus Raw Basis (2009)



Source: Morgan Stanley Research

The curve adjusted par spread (CAPS) basis uses the adjusted basis and takes it one step further by adjusting for the risk implied by the CDS curve. CDS curves can be interpreted as indicators of default risk over time, given that they are par

instruments, unlike corporate bonds. The fair value measure (alternatively the curve adjusted par spread or CAPS) is based on a curve adjusted Z spread, which is more accurate than the simple Z spread, given that it takes into account how default risk varies over time. It also takes into account that losses from default are a function of dollar price and assumes a certain recovery, whereas the conventional Z spread assumes zero recovery. Inverted credit curves mean that near term bond coupons are more risky than implied by flat or steep curves.

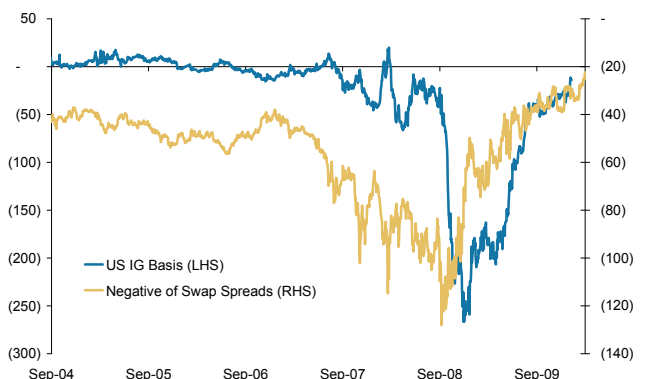
Exhibit 3 compares the three metrics (raw, adjusted and CAPS) through 2009. The first half of the year was characterized by inverted credit curves and low dollar prices never seen before. The difference between the CAPS and raw basis was nearly 70 bp around April 09, when spreads were at their widest and curves at their most inverted. Normalization of credit markets was accompanied by curve steepening, which meant that the basis began tightening. Naturally, we saw a convergence between the raw basis and the fair value basis through the second half of the year.

WHAT OUR BASIS METRICS DO NOT CAPTURE

The most important shortcoming of the basis metrics we discussed above is the inability to capture shifts in funding environment. Being long the basis is ultimately a funding trade – something investors took for granted pre-crisis. As the cost of funding rises and falls, it is natural for the basis to move as well. A rise in funding costs was the key driver of a negative basis in 2008-09. With hindsight, swap spreads (or AA cash spreads in Europe) were a good leading indicator in terms of signaling a regime shift in the basis both going into and coming out of the crisis (Exhibit 4). With swap spreads at historic lows post crisis, it seems that the basis should be driven by the factors and technicals we discussed earlier and funding should be a less important driver of basis.

exhibit 4

US Swap Spreads Highlight Basis Funding Risks



Source: Morgan Stanley Research

Another omission is differentiation of funding costs – for example, of IG (which get better funding) vs. HY bonds, or

as has come under focus recently, bonds of different sovereigns within Europe. Many investors look at the spread to government bonds as a measure of value and the significant divergence between bonds of core and peripheral Europe made it difficult to assess value. Because the basis metrics we discuss above are based on value relative to the swap curve, it totally circumvents the choice of benchmark issue. As long as the swap curve is not differentiated across investors across sovereigns and largely representative of the cost of funding for high quality counterparties, the basis is unaffected by this divergence in government bond performances.

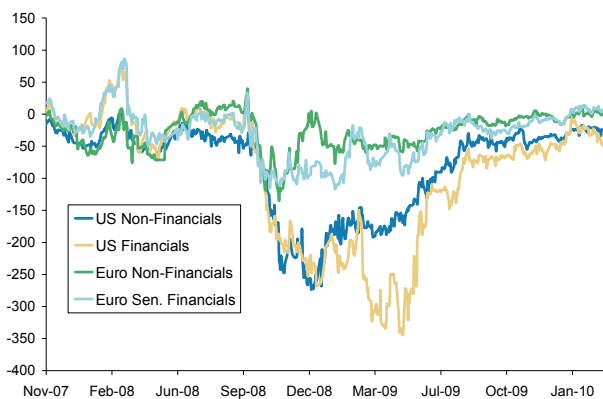
Finally, we advocate using greater caution when analyzing basis trades in the HY space, particularly for those with a high probability of default, as these credits can trade with a significant difference in the structure and timing of cash flows between the bond and the CDS. We discuss this in greater detail in the HY basis section later in the chapter.

INVESTMENT GRADE BASIS: EUROPEAN VS. US

The US basis hit more extreme negative levels relative to Europe given credit events in actual investment grade names (Lehman, FRE, FNM, WaMu), the significant presence of non-bank financials, and a generally lower quality and wider trading universe of credits in US. Unsurprisingly, financial cash bonds were the worst hit and the sector basis touched an all-time low of -350 bp and at the time traded almost 150 bp tighter than the rest of the universe. The financials basis remained negative through the first half of 2009, even as non-financial cash credit began the long road to recovery. In fact, the basis in financials credits touched all-time wides in April 2009, when the non-financials basis had already tightened more than 50 bp from its wides seen in Oct 08 following the Lehman bankruptcy.

exhibit 5

Financials' Basis Was Much More Stressed



Note: Our IG universe includes over 250 issuers, with 700+ bonds across maturities. The European universe includes over 100 issuers, with 400+ bonds across maturities. The aggregate cash spreads, CDS and basis numbers are calculated as median values for the set of bonds belonging to a specific maturity bucket.

Source: Morgan Stanley Research

Post crisis, the European basis has normalized and sectors are trading with a positive basis today. In contrast the US basis is modestly negative and negative basis trades are still available albeit in the riskier end of the credit spectrum. Overall, there is more two-way flow in the basis now. We see positive basis ideas in Europe, where taking risk through CDS in credit linked note format can generate more yield than in corporate bonds with potential for lowering dollar prices. We also see negative basis opportunities in HY and US IG and the first signs of a return of M&A/Buyout risk, which should encourage buying of covenanted basis packages.

RATES AND THE BASIS

We first addressed the impact of rising rates on the cash bond – CDS basis in Chapter 15. In this publication, we advocated selling the basis, or more accurately, unwinding negative basis positions that were close to flat or already positive, because of the advantages of selling protection vs. buying bonds. We also advocated going long floating rate CLNs as opposed to fixed rate bonds to take advantage of potentially rising rates, something that led to several trades marketed as “positive basis” trades, whereby an investor would sell a bond to the dealer and buy a CLN in one package, thus swapping out the fixed rate cash exposure for floating rate funded CDS exposure.

On the bond side, a key factor for the IG corporate bond universe is dollar prices. Higher dollar prices are likely to be an impediment to the performance of bonds and also make the basis look less attractive, as the higher initial cash outlay would require more CDS hedges, all things being equal.

On the CDS side, curve steepness can be affected by rates as well. Shorter-maturity CDS generally trade with a negative upfront (pay an upfront amount and then receive 100bp when selling protection), whereas the longer-maturity CDS contract is likely to trade with a positive upfront, making it more attractive for protection sellers from a funding perspective. Bottom line, from both a CDS and bond perspective, the longer-maturity bond holders are likely to find selling CDS interesting.

Furthermore, the structure of the upfront contract after the SNAC protocol was implemented has rates implications as well. If the upfront value is positive, which we define as a situation in which the seller of protection receives the upfront payment in addition to the running coupon, the benefit is to the seller of protection, or the investor going long the credit risk. The opposite is true of the buyer of protection, who should prefer to receive the upfront as well.

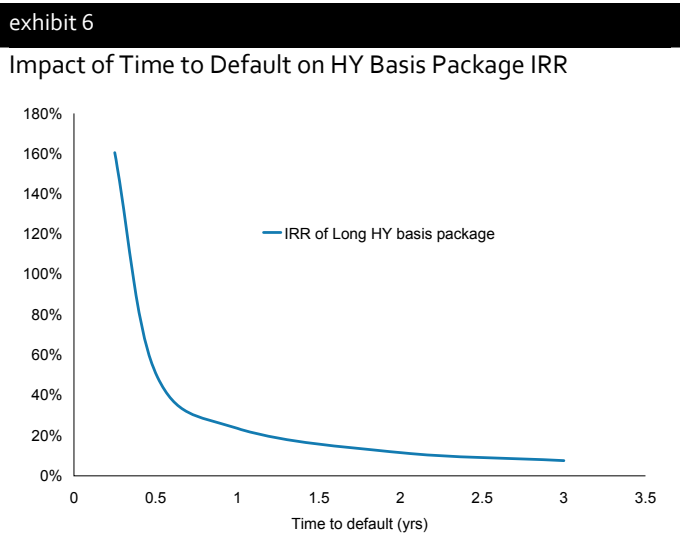
In addition to the standard funding reasons we’ve discussed in the past, the benefit of receiving an upfront prior to a rise in rates is that as rates rise, the discount factor applied to PV the running coupon increases as well, thus *lowering* the present value of the remaining coupon. To illustrate, if a CDS were trading at 600bp and had a 500bp coupon and a duration

of 4, the upfront would be 4% (100bp x 4). Even if credit risk on the single name stayed the same, the 500bp coupon would be subject to MTM fluctuations as rates changed. Alternatively, if the coupon had been 100bp, the upfront would be much greater and the coupon at risk would be much lower.

THE HIGH YIELD BASIS: A DIFFERENT ANIMAL

The high yield basis has always been a much harder relationship to analyze for a plethora of reasons as we discuss below.

High default probabilities: The basis metrics we capture are a probability weighted measure of fair value vs. CDS. For IG names, the probability of survival to maturity is quite high and is the more likely outcome. In the case of HY many outcomes are more probable than survival to maturity (default in year 1 or year 2, etc.), and the payoffs for each of these scenarios are different.



Source: Morgan Stanley Research

Dollar prices and structure of cashflows: HY bonds have a greater tendency for \$ prices to trade away from par due to higher coupons and volatility of spreads. HY CDS also traded on an upfront basis even before the shift to fixed coupon convention creating substantial mismatch in timing of cashflows in the CDS vs. bonds. High default probabilities and mismatch in cash flows result in very different P&Ls depending on timing of a default. Consider a hypothetical bond trading at a dollar price of 50 and coupon of 500 bp, and assume CDS protection on the name trades at 30pts upfront plus coupon of 500 bp. The basis package is now similar to a zero coupon bond (price of 80pts), *except* that the entire principal may be redeemed before bond maturity, if the name defaults. Depending on when the credit defaults we end up with different IRRs (See Exhibit 6).

Call features: Since many HY bonds have issuer calls, unlike CDS which is a bullet instrument, calculating the basis becomes more complicated, as these features introduce significant duration mismatches into the equation. We adjust for this by calculating a basis to the worst date of the bond – i.e., buy the bond and buy protection to the worst date rather than the maturity as is the case with IG bonds. While this approach is also not perfect as the worst date can move around and there is still duration mismatch risk, this is one way of assessing relative value between HY bonds and CDS.

Trading Recovery Risk – The Missing Link

September 16, 2005

Sivan Mahadevan

The advent of credit default swap instruments has helped both seasoned and new credit market participants more rigorously analyze default risk over the past several years. We now have very important tools to determine default probabilities, which helps in the pricing of both single-name and structured credit instruments. As credits become stressed, these default probabilities are particularly meaningful in many contexts, from forming fundamental views to debt capital structure arbitrage and even the pricing of CDO tranches.

However, this process works best only when we have a good sense of what recovery value might be, in the event of a default or a bankruptcy filing. Some of our early work in the airline space (see *The Airline Triangle* from November 7, 2003) was indeed feasible because we could make an assumption about the recovery value of unsecured airline debt (single digits %) without much debate, which, in turn, made the pricing of many other relationships in the debt capital structure much easier. Yet, in most cases, the process of determining recovery values is itself complicated; thus, recovery values are the missing link in any type of stressed credit analysis, with the market providing us little information.

There is a tiny and perhaps budding market for trading recovery risk through conceptually simple instruments like recovery locks, a specific form of a recovery swap, which are actually the net position of a more complicated trade. Most of the trading activity we have seen in the recovery space is in stressed fallen angel credits. Trading recovery risk adds a whole new dimension to the credit puzzle, and it may take a turn in the credit cycle to become more mainstream in usage. Yet, there are indeed motivating factors today, including the rise of idiosyncratic risk in select sectors and the proliferation of synthetic bespoke tranches issued with some form of fixed recovery protection.

exhibit 1

Very Volatile – Senior Unsecured Bond Recovery Rates

Year	Mean	Median	Min	Max	StDev	Observations
2003	41.2%	34.0%	0.1%	99.5%	24.7%	34
2004	50.1%	47.0%	15.0%	95.8%	22.3%	33

Source: Morgan Stanley, Moody's

HOW RISKY IS RECOVERY?

Before we delve into both the motivation behind trading recovery risk and the instruments used, it is worth at least mentioning how large the topic of recovery analysis really is. Ironically, recovery valuation is not always about

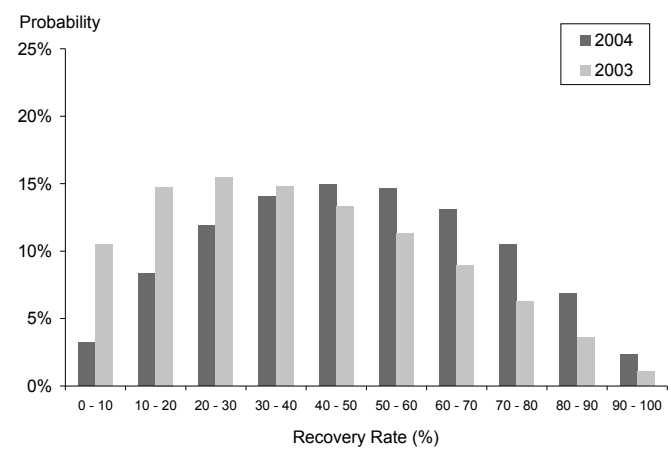
determining the value of a firm's unencumbered assets. Most US companies that file for bankruptcy protection do so under Chapter 11 with the idea that they will restructure instead of liquidate (which would be Chapter 7). In these cases, recovery value is something that is negotiated through bankruptcy court, which is easy to lose sight of.

One can and should think about recovery risk almost independently of default risk, and then put the two together to make valuation decisions. Some recent examples can highlight this point. Northwest Airlines' decision to file for bankruptcy protection was a bit of a surprise, and one could argue that most of the movement in unsecured instruments was related to the probability of this event, not recovery once it happened, as unsecured debt recoveries in the airline sector have generally been low. Also, much further away from the mainstream, an Australian court recently ruled that unsecured creditors' claims on an Australian-domiciled entity would be *pari passu* to those of equity shareholders, which demonstrates the independence of recovery risk with respect to default risk, at least in this example.

The question of how uncertain recovery risk is remains difficult to answer, but, anecdotally, we can use aggregate rating agency data to get a sense for the distribution of recovery rates on defaulted issuers (derived from data in "Moody's Default and Recovery Rates of Corporate Bond Issuers, 1920-2004"). The results are summarized in Exhibit 1 and offer some interesting insight. There is a significant amount of uncertainty around recovery rates, which is not surprising (the standard deviations quoted by Moody's are 25% and 22% for 2003 and 2004, respectively).

exhibit 2

Fitted Recovery Rate Distributions



Source: Morgan Stanley

Based on this data for senior unsecured debt, we fitted beta distributions to get a more robust sense of how uncertain recovery actually was in 2003 and 2004. The resulting distributions are shown in Exhibit 2 and illustrate that, in addition to being very volatile, recovery rates are likely skewed to the downside and appear even more skewed the lower the average recovery (at least based on this limited dataset).

MOTIVATION FOR TRADING RECOVERY RISK – STRESSED NAMES AND SPECIALIZED CDOs

Although any fundamentally oriented credit investor ought to be interested in isolating and trading recovery risk, it is still a new concept and the real motivation for the limited activity we see in the market is the result of two phenomena, in our view. First, the fact that we remain in a fairly benign credit environment from a default perspective reduces both the interest of end investors in trading recovery and the dollar value of doing so successfully. In other words, given today's spreads, the dollar value of a recovery point is much less than for an environment in which spreads, on average, trade wider. This phenomenon helps explain the concentration of recovery trading activity in the wide names in the market like Delphi and Calpine.

Second, investor demand for synthetic CDOs where underlying default swaps have fixed recoveries creates supply of fixed recovery protection in the market without a natural other side. These types of CDOs have gained in popularity largely because fixed recoveries reduce the uncertainty of tranche losses, making them appealing to end investors.

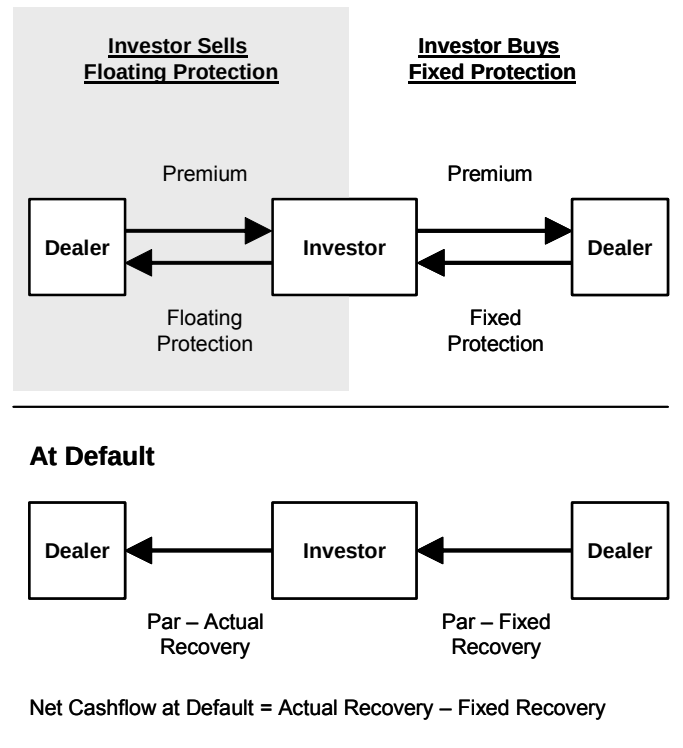
TRADING RECOVERY RISK – WHAT ARE THE INSTRUMENTS?

The standard credit default swap was crafted partly with the motivation of being as bond-like as possible. While this one point was critical for credit derivatives to gain acceptance in the corporate bond community, it did one disservice to investors: it did not allow for the dis-aggregation of default risk from recovery risk. Early variants of credit defaults swaps had fixed recoveries, which made it easier to think of default risk independent of recovery risk, but made bond versus credit default swap comparisons much more difficult.

Today, a form of recovery swaps that isolates recovery risk is termed a “recovery lock,” which is simply a pairing of a standard (floating recovery) default swap with a fixed recovery default swap. A simple example serves best to explain the structure.

exhibit 3

What Is a Recovery Lock?



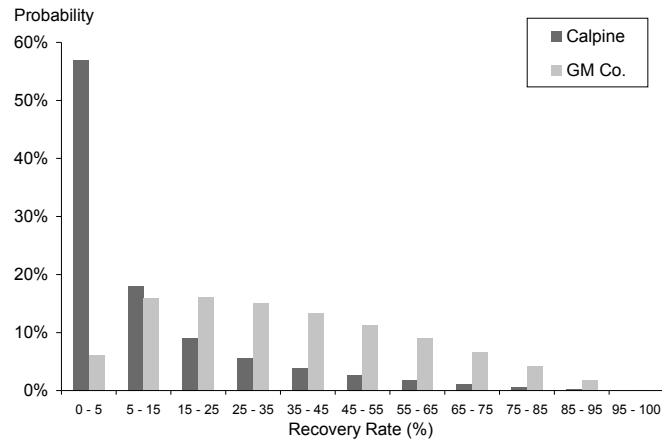
Source: Morgan Stanley

If one sells protection on a credit using standard CDS (where recoveries are not fixed but floating) and then buys protection on the same credit and term using fixed recovery (say 40%), then the net position the investor has is one that is long recovery risk, i.e., the investor would want the recovery on the issuer at default to be as high as possible. When there is a default, the investor would pay par and get delivered a bond (based on the terms of selling regular CDS protection) and at the same time receive par and pay the fixed recovery of 40% (based on the terms of buying fixed-recovery protection). The par payments cancel out and the net position is that the investor paid 40% for the defaulted bond. Clearly, the investor would want the actual recovery on the bond to be as high as possible, so therefore he or she is long recovery risk.

In the above example, it is common practice for the premiums on both legs of the default swap trades to be the same. In this case, the instrument (or actually net position) is termed a recovery lock. The fixed recovery required to make the two premiums the same is the market-implied recovery value, and it is this value that is quoted in the market for recovery swaps.

exhibit 4

Fitted Recovery Distribution (Calpine and GM Co.)



Source: Morgan Stanley

THINKING INSIDE THE RECOVERY BOX

While the ability to trade and hedge recovery risk is indeed one step in the direction of solving the missing link in credit analysis, recovery risk ultimately needs to be modeled just like default risk. If we combine the notion that recovery values are volatile (Exhibits 1 and 2) with recovery values that we observe in recovery swaps, we can measure the uncertainty of recovery values in a given credit.

When we consider fixed versus floating instruments, one concept that jumps out at us is the idea that taking default risk with fixed recovery should generate less premium than

taking the same risk with uncertain recovery, assuming the fixed recovery is set at the correct level. What the current market for recovery locks gives us is that correct level, since the strike spread and maturity on both the fixed and floating leg are generally the same.

This happenstance gives us the ability to compare the risk associated with floating CDS (with their uncertain cashflows in default) and fixed CDS (with their certain cashflows in default). Conceptually, if we can come up a way to weight the various floating recovery scenarios, then we can isolate the default component of the risk and gain some insight into how investors think about recovery risk. In Exhibit 4, we demonstrate this uncertainty associated with recovery for two credits, combining market pricing on standard CDS, recovery locks and our results for the distribution of Moody's recovery rates.

RECOVERY ANALYSIS IS THE MISSING LINK

While the current overall credit environment is not necessarily ripe for the development of a robust recovery swap market, there is potential for this space to expand in the more stressed corners of the credit markets, as we are beginning to witness now. Despite the simplicity of recovery locks, recovery risk itself is a whole dimension in the credit analysis puzzle that today is meshed with default risk in market standard instruments, like bonds and traditional default swaps. Recovery instruments will help to isolate these risks, but the market is still in need of development.

Floating a New Idea – CMCDS

January 21, 2005

Sivan Mahadevan

Much of the development that resulted in today's standard credit default swap contract was driven by definitions of credit events, sparked, in turn, by the many bankruptcies, defaults and restructurings that the investment grade market experienced during the past credit cycle.

The development of and liquidity injection into the credit curves is still a relatively recent phenomenon (over the past 15 months or so). Active credit curves are important inputs into pricing instruments like credit options and constant maturity credit default swaps. Additionally, from an investor's perspective, there is little dispersion in the market place today, particularly in investment grade, where spread compression is even higher than it was in 1997 (see "Couch Potato Prognosticating," January 14, 2005). Tight spreads and a lack of differentiation create a natural reach for yield phenomenon, but also causes concern among those who must be fully invested and don't feel great about the upside potential.

We rarely discuss very new credit derivatives instruments in our research, partly because until we see some indication of liquidity potential, it is difficult to determine strategic opportunities. However, many market participants have recently asked about one new variant in particular – constant maturity credit default swaps (CMCDS).

As a result, we focus this chapter on ideas involving CMCDS, an instrument that provides investors with a convenient way to string together a series of forward credit curve trades. We feel that varying risk premiums along the credit curve, combined with the potential for spread regime shifts, can result in impractical forward spreads. We can therefore think of CMCDS as a convenient (and positive carry) means to lean against the forwards.

CMCDS MECHANICS

A constant maturity credit default swap is a default swap where the premium is reset (on a quarterly basis) to equal a fixed percentage (called the participation rate) of the then-prevailing premium of a plain-vanilla default swap for a certain term. While this is very much a developing market, a typical CMCDS trade today has a 5-year term and references a fresh 5-year default swap every quarter during that 5-year term. Assuming a 50% participation rate, the seller of CMCDS protection would receive 50% of the prevailing premium on a 5-year default swap every quarter, until the

CMCDS expires (in five years) or until a credit event occurs (see Exhibit 1). Consequently, if spreads widen, the quarterly payment would also increase and the concomitant mark-to-market impact could be significantly lower than a regular default swap. The premium on a 5-year default swap is inferred from the market, generally by some type of a fixing process on the reset date by a calculation agent. There can also be a cap on the premium, usually at stressed premium levels.

exhibit 1

CMCDS – Sample Quarterly Premium Calculation

Notional	\$10,000,000		
Participation Rate	50%		
Quarter	5 Yr CDS Spread (bp)	CMCDS Spread (bp)	Quarterly Payment (\$)
1	100	50	12,500
2	125	62.5	15,625
3	150	75	18,750
4	120	60	15,000
5	100	50	12,500

Source: Morgan Stanley

PRICING – DETERMINING THE PARTICIPATION RATE

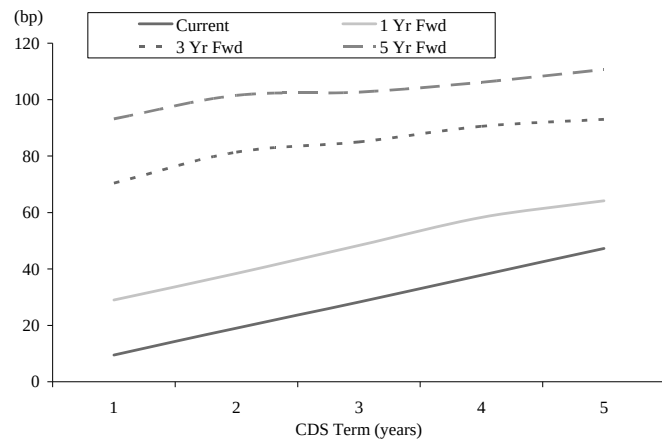
Since the protection provided by a CDS and a CMCDS is essentially identical in case of a default, the pricing of the two instruments should be directly linked, as well. Said differently, buyers of protection in either instrument should expect to spend the same amount for the protection at the inception of the contracts. This linkage is enforced through the concept of a participation rate.

We start by using an analogy from the world of interest rate swaps. The fair fixed rate on a swap is the one that equates the present value of floating leg cash flows to the present value of fixed leg cash flows. Employing the same heuristic, the fair participation rate is the rate that equates the present value of payments of a regular CDS to the present value of CMCDS payments.

To determine the expected payments of a CMCDS, we need the implied forward CDS rates, just as we need forward Libor rates to calculate the fixed rate in the case of interest rate swaps. Once we have forward rates, we can determine the participation rate that generates cash flows with a present value matching a plain-vanilla CDS.

exhibit 2

IG CDX Spot and Forward Curves



Source: Morgan Stanley

INTUITIVE FEEL

There are effectively two ways one can think of CMCDs. First, as we mentioned above, CMCDs is a convenient way to string together a series of forwards. If the curve shape and spread levels implied by forwards are realized over the term, the CMCDs and CDS should have the same return at maturity, and this is the basis for pricing. Thus, a position in CMCDs (versus one in CDS) is a way of expressing the view that the forwards will not be realized. Second, ignoring forwards for the moment, CMCDs is really just a floating rate instrument, but the credit premium is what actually floats, as there is no interest rate. A floating premium can have more muted mark-to-market volatility than a fixed premium instrument.

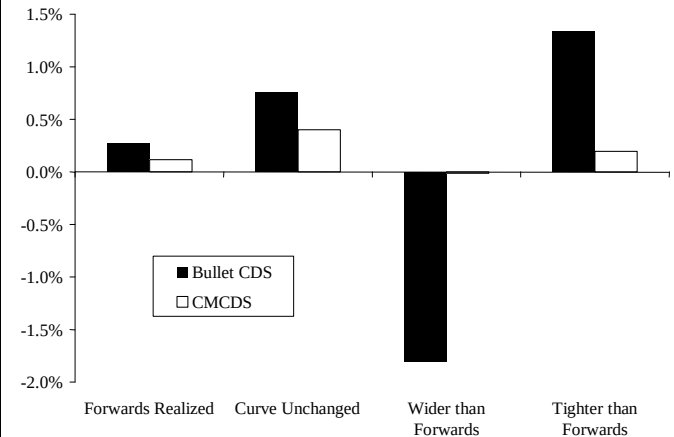
STRATEGIES AND HORIZON ANALYSIS

From a strategic perspective, the most interesting aspect of CMCDs is that it separates default risk from spread risk; these are packaged together in more traditional instruments, such as bullet bonds and CDS. This characteristic allows takers of risk to be paid a premium for exposure to default risk, while avoiding exposure to the market risk associated with spread movements. Given the pricing techniques we discussed above, we thought it would be worthwhile to examine how a CMCDs would perform against a bullet CDS under a variety of scenarios.

While the CMCDs does not have any direct exposure to spreads, the realized and implied spreads in the future can have an impact on the price of the swap. We highlight the performance of both a bullet 5-year CDS and a 5-year CMCDs for several scenarios 1 year forward (see Exhibit 3, where pricing is model-based). The first point to note is that price movements are less muted in CMCDs (compared to CDS) across all of the scenarios, but there is still some price volatility. Second, of the scenarios we chose, CMCDs outperforms CDS only in the case where spreads move wider than implied by the forwards. Part of the underperformance

can be attributed to the fact that the floating premium on the CMCDs starts out lower than the CDS level. It should be noted that the floating premium is also expected to increase well above the initial CDS level as time passes.

exhibit 3

Muted Price Movements – CMCDs vs. CDS
(One Year Forward)

Source: Morgan Stanley

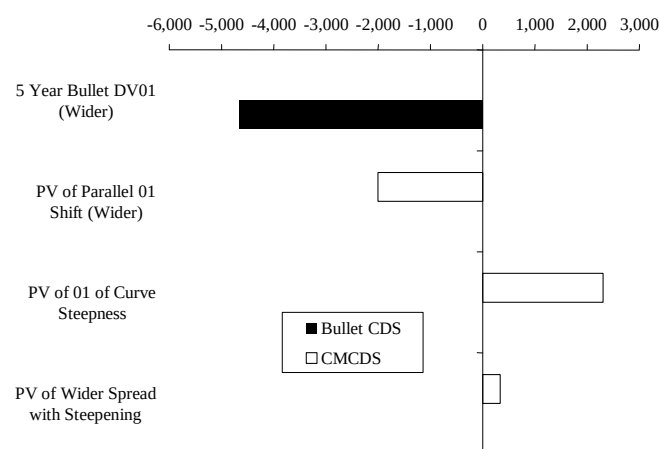
To explain this sensitivity, consider the simple example of a world with perfectly flat credit curves. In this world, the participation rate on a CMCDs would be 100%. As the curve gets steeper, the “fair” participation rate will decline. A CMCDs struck in a flat curve environment should thus appreciate slightly if a steeper curve environment ensues.

PLAYING CREDIT CURVE SHAPE

This sensitivity to credit curve shape leads to interesting implications for price sensitivity. Given a credit curve, we can distinguish between a parallel shift and a steepening or flattening to the same 10-year level. In Exhibit 4, we show the price sensitivity of a 5-year CDS to spread moves and a 5-year CMCDs to parallel shifts in the credit curve, as well as to flattening/steepening. Under some scenarios, these moves can offset one another to a large extent. For example, the negative impact of increased spread levels, which imply increased default risk, makes the CMCDs marginally less valuable. This can be offset if the shift is accompanied by a steeper curve, implying larger floating coupon payments as the instrument reaches maturity, which increases the value. Which one dominates will depend largely on the magnitude of the relative changes.

exhibit 4

CDS and CMCDs Spread and Curve Exposures (Price Sensitivity of \$10MM 5 Year Positions)



Source: Morgan Stanley

WHO ARE THE NATURAL BUYERS AND SELLERS?

Right now, there are not many of either, and there may never be. Yet, as we saw above, the risk and return implications of CMCDs are significantly different from CDS in many scenarios (except when the forwards are realized) so it is indeed worth considering who might benefit from buying or selling CMCDs protection. In the current credit curve environment, the seller of protection gives up quite a bit of current premium, but could have less mark-to-market volatility and will outperform CDS when spreads go wider than implied by the forwards. For those who must be invested, but have a negative view on credit (over and above what is implied by forwards), selling CMCDs protection would be beneficial. It is also a simple way of implementing the view that default risk is benign, but spread levels are not very exciting (which fits our view, see “Breaking Down the Barriers,” December 17, 2004). Long/short strategies in tranches can also implement this view, but can be more complicated in nature.

The natural buyer of CMCDs protection is perhaps more straightforward, given that lower current premiums might seem appealing. For any protection buyer who feels that the forwards overstate spread widening over the term, buying CMCDs protection on its own would be more attractive than CDS protection itself. In either case (buyer or seller), lower potential mark-to-market volatility of CMCDs may be appealing to investors who value the lower volatility, or who must treat cash and derivative instruments differently.

SOME PRACTICAL ISSUES

For investors who are excited about the convenience of packaging forwards into a single instrument, and the resulting different risk and return characteristics (compared to traditional CDS), we caution that these instruments are still relatively nascent. Transaction costs and the potential that market prices can drift away from model-based valuations are important practical implications (consider how correlation skew has moved with spread levels). Also, while the cap in CMCDs is generally deep out of the money at inception, the pricing of this cap as stress enters a particular credit can be somewhat subjective, as well.

What little activity we have seen in the market for CMCDs has been centered around the benchmark indices, rather than individual credits. One debate in the market has focused on how to treat the index roll. From a risk-management perspective, we find it simpler to think of a CMCDs on an index to always be based on today’s composition of the index; however, as indices roll and become off-the-run, it may prove difficult to apply a fixing process to levels on an off-the-run index. As a result, some market participants have proposed that, as the index rolls, the CMCDs should refer to the new index, which leaves the CMCDs users exposed to changes in index constituents. In any event, what we find interesting about this instrument is its vastly different risk and return profile, which can encourage investors to think outside of the box.

Corporate Bond Options – From Macro to Micro

June 14, 2011

Phanikiran Naraparaju
Sivan Mahadevan
Ashley Musfeldt

The significant growth in credit index options during and after the credit crisis has been driven by investors wishing to manage the MTM volatility of their corporate credit portfolios. Much of that flow has been in a macro, or top-down manner, using options on the popular CDS indices as the instrument of choice to hedge, position for spread ranges or generate yield. Along this growth path, many investors have asked us about single-name options as well to help manage volatility at an issuer-specific level, akin to the vast market for single-name options within equities. While we will not rule out single-name CDS options as a viable alternative, at this point we believe that the natural market for such options usage is within the world of corporate bonds given the liquidity and volume advantages, and the notion that most investors' single-name risk exists in their corporate bond portfolios.

What is the scope for growth in the more micro world of credit options? At the macro/index level, option skew is very steep today in both equities and credit. In equities, this stands in contrast to option skew in single-stocks, where there is more balance and the fall in volatility has been greater than at the macro level. In a sense, the equity options markets tell us that there is more macro uncertainty than micro uncertainty, and in fact the style of investment in single-name equity options is quite different (and more bullish in nature) than at the index level (see [Macro-Micro Disconnect](#), May 19, 2011).

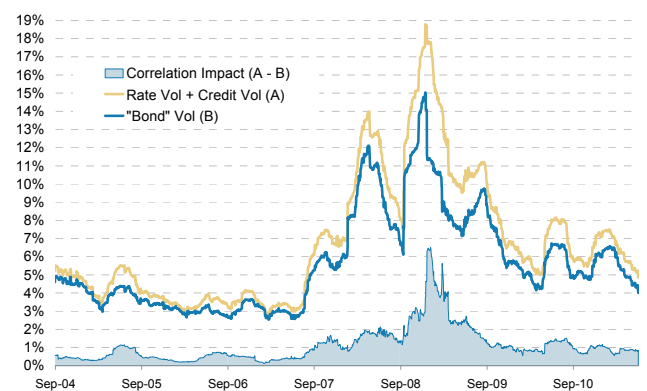
As a corporate bond options market develops, do we expect to see the same macro-micro disconnect as we see in equities? We believe that a similar theme will develop, in particular because we expect investors to be a bit less focused on hedging (a negative credit view on a particular issuer might best be implemented by simply selling the bonds) and more focused on directional themes, such as selling puts to establish entry points, selling calls to add yield, and buying calls to participate in upside.

Options on corporate bonds, the right to buy or sell a bond at the strike price during a specific period, were among the earliest credit derivative instruments in the credit markets. Most outstanding corporate bond options are embedded in corporate bonds and are thus implicitly held by either bond issuers (for call options) or bond investors (for put options). Given the widespread prevalence of embedded call options especially in the HY space and in bank sub-debt paper, it is somewhat surprising that options on corporate bonds are not more popular. The closest comparable market for corporate bond options would be government bond options. However, options on Treasuries are also considered not very liquid by

the standards of the rates markets. At the short end, liquidity tends to be concentrated in the bond futures options market. Beyond six months expiry, the swaptions market offers the best volatility cues. If one really needs to track volatility on Treasuries instead of swaps (e.g., if one were concerned about spreads), then one could also trade options on constant maturity Treasury yield, again sacrificing some liquidity.

exhibit 1

Credit and Rates: The Whole Is Less than the Sum of the Parts



Source: Morgan Stanley Research, Bloomberg

The picture in Europe is similar with bund futures and options dominating the short-end and swaptions dominating the long-end market for volatility. The diverging fortunes of different sovereigns makes sovereign bond options more appealing but liquidity is not inspiring from a rate investor's point of view. Nevertheless, by the standards of corporate bond markets (most are quoted in 5MM clips), these rates volatility products would count as 'liquid' (trading 100MM to 500MM sizes).

We have written previously, many years ago, in detail about the theory and the different analytical approaches to valuing and using corporate bond options. In this note we emphasize the practical aspects, review how the volatility dynamics have changed since the crisis and ideas that are appropriate in the context of today's pricing. We focus on two 'sectors' in particular, a) senior financial bonds in Europe, and b) long-end non-financial bonds in US and Europe.

TWO PRICING APPROACHES

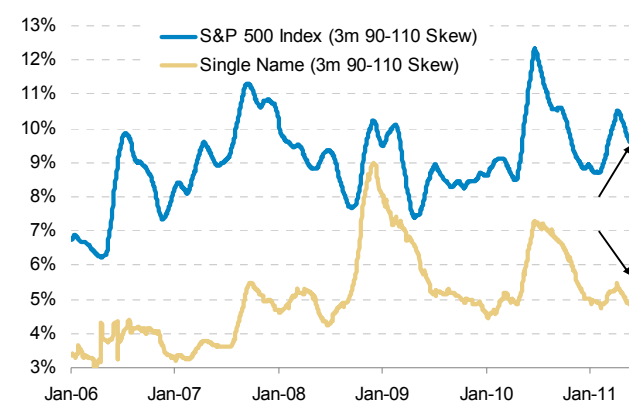
There are two potential approaches to pricing bond options: a) the first approach is to build up the bond volatility from the components, i.e., rates and credit volatility as well as the correlation between the asset classes, and b) a second simpler approach is to ignore the components and just price the

option as one would on any other asset class, using the standard suite of pricing models.

Certainly, there are merits to the first approach: we have good pricing data for credit and rates volatility from the respective options markets. But there is still the question of estimating the correlations between credit and rates which has varied over time. There is also the issue of the basis risks in applying the index volatility on a CDS index to single-name volatility of a cash bond. The data on the equity options side show a major disconnect in the micro (single-name options) and macro (index options) (see Exhibit 2). While skew is generally flatter (less differentiation between OTM puts and OTM calls) in single-name options compared to indices during most time periods, the gap between the S&P 500 index and single-name skew is actually quite high today, and moving in different directions. Increasingly flatter skew at the single-name level sets a more bullish tone while steeper skew (OTM puts richer than calls) at the index level paints a more bearish picture.

exhibit 2

Skew Much Flatter for Single-Names than Index



Note: Both skews are 2m rolling averages. Source: Morgan Stanley Research, Quantitative and Derivative Strategies, Bloomberg

In essence the potential uncertainties in the estimation of some parameters might make this approach less worthwhile relative to just pricing this as an option on any asset class in general, keeping in mind realized volatility of the bond as a whole. Nevertheless, it is instructive to analyze the underlying components to determine overarching trends.

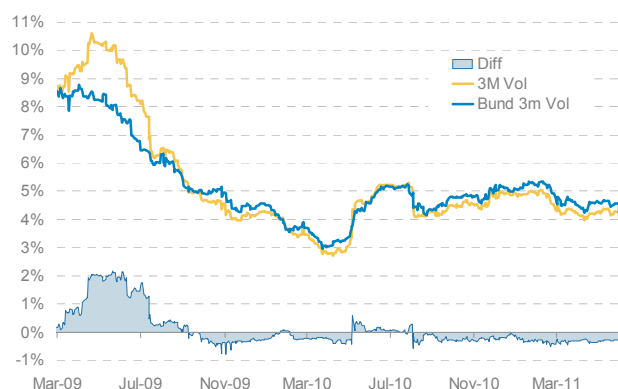
CORRELATION DAMPENING BOND VOLATILITY

Although corporate bonds have been more volatile during the crisis than government bonds, the gap has since collapsed (see Exhibit 3). A steady decline in credit volatility even as rates (and FX) volatility stay elevated has driven this

convergence. Another key element is the correlation between credit and rates. Historically, this correlation has been flat or modestly negative (see Exhibit 4) as rates fall when spreads widen and vice versa. In fact, many a time, the correlation offset more than compensates for the credit volatility. This is neither just a European nor a financials phenomenon.

exhibit 3

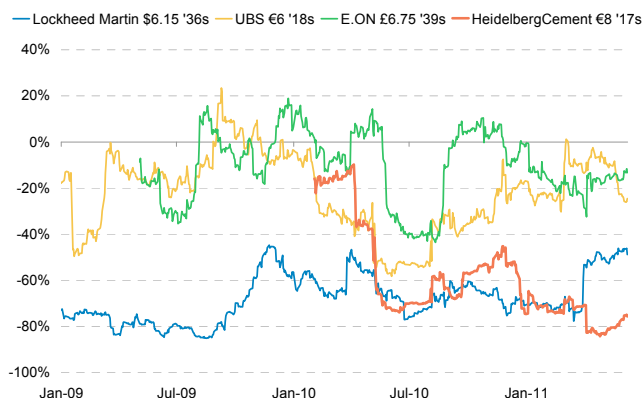
UBS Senior Bond Volatility vs. Bunds Vol



Source: Morgan Stanley Research, iBoxx, Bloomberg

exhibit 4

Correlation between Rate and Credit Components

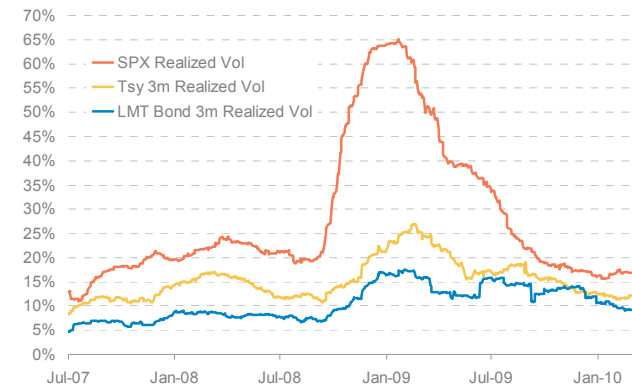


Source: Morgan Stanley Research, iBoxx, Bloomberg

For example, long-end bonds' correlation with similar maturity Treasuries is extremely negative, leading to corporate bonds volatility to trade inside of Treasury volatility (see Exhibit 5 overleaf). Also interestingly, the gap between equity volatility and long-end bond volatility has converged considerably. Given the macro environment we are in – with fears of a growth slowdown, there is a case to be made for selling bond options to fund downside protection on equities.

exhibit 5

Lockheed Martin Long-End Bonds, Long-End Treasuries and Equity Vol



Source: Morgan Stanley Research, iBoxx, Bloomberg

APPLICATIONS AND THE USER BASE

In our view, the primary use of bond options will, at this stage, be to better express a directional view on the bond rather than to trade volatility. Anecdotal activity and discussions suggest bond options are likely to appeal most to credit investors as a way to establish entry and exit points in specific bonds.

A very topical concern today is the historically high dollar prices (and low absolute yields) which makes corporate bond investors more cautious about the entry and exit points for their bond holdings. Investors can use bond options to set an exit price for existing bond holdings through overwriting (i.e., sell call options against existing bond holdings), which is equivalent to taking profits if the bond hits an exit price. If the bond does not reach the exit price, the investor earns yield over and above that from just holding the bond. This is a bread-and-butter strategy for funds in the equity options world and is also increasingly well established in the credit index options market.

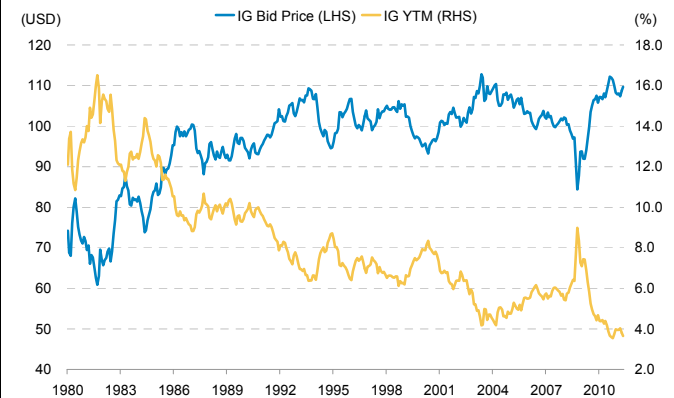
Overwriting on long-end bonds (10y+) in particular should be attractive for pension funds, insurers, etc. Demand by pension funds causing long-end bonds in the sterling market to become difficult to source and trade rich is a well known phenomenon. These long-end portfolios tend to be more passive and are characterized by longer-holding periods. Overwriting is a good way to add yield on a tactical basis to such portfolios.

From a fundamental point of view, the attractiveness of overwriting in corporate bonds today is a combination of several different factors: i) dollar prices are at historical highs and yields at historical lows (see Exhibit 6), ii) rate implied volatility remains elevated relative to other asset classes and relative to what is being realized in the market, notwithstanding recent moves (see Exhibit 7), and iii) the

pick-up through overwriting is considerable both from the rates carry and the negative correlation component. As a result, our overall bias is towards selling options with the most prominent theme being overwriting.

exhibit 6

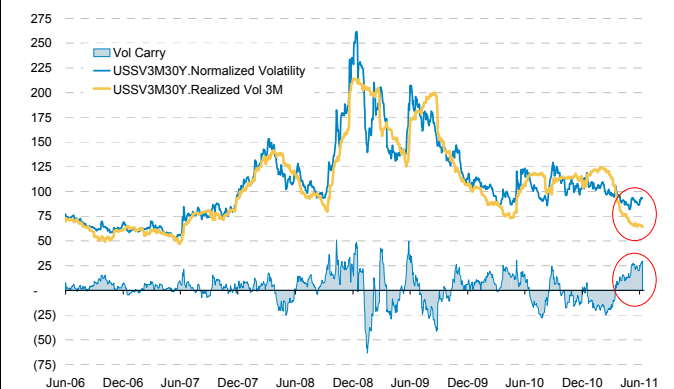
High Dollar Prices, Low Yields Favor Overwriting



Source: Morgan Stanley Research, Bloomberg

exhibit 7

Rates Volatility Carry Is Attractive



Source: Morgan Stanley Research, Bloomberg

Alternatively, bond options are also useful to set entry points. Investors who like a specific bond but are uncomfortable about the valuation levels may set an entry point by selling put options at a price level they would be comfortable buying the bond. If the bond never hits the buy level, the investor keeps the option premium. Thus the investor is better off than earning nothing, as they would have if they passively waited for the market to reach their buy target levels. A particularly apt situation to implement a strategy of selling puts is in the context of under-allocation in a strong new issue bond. Note though that options on new issue bonds trade after the issue is seasoned.

We show some key volatility metrics for a cross-section of liquid corporate bonds in US and Europe in Exhibit 8.

Interestingly, corporate bond realized volatility is very similar to the rate volatility, i.e., the credit volatility is either largely counteracted by the correlation component or the credit volatility is small relative to the rates volatility. The correlation is negative for all the bonds in this list. The yields generated on overwriting are quite significant for long-end bonds, often a multiple of the spread of the underlying bond. Overwriting yields on shorter maturity bonds are inline with the underlying bond spreads

CONCLUSION

As we highlighted above, overwriting in long-end bonds looks like a fairly compelling trade from a fundamental and valuation standpoint. Overwriting also makes sense in a

sector that is a source of intense valuation debate: European financials. European financials is a strong case for a negative correlation between rates and credit spreads. There is a clear flight-to-quality to Bunds when sovereigns are in stress and credit spreads widen and vice versa. Moreover, the credit volatility for the 5y-10y bucket bonds is much smaller than the rate volatility itself. For all these reasons, the realized volatility of financials bonds can be relatively low. The level of vol and hence premium is shorter for the shorter-maturity bonds, but the pick-up relative to the spread is still meaningful enough. From a macro point of view, selling calls on long-end corporate bonds can be used to buy downside hedges elsewhere, for example in equities or FX.

exhibit 8

Corporate Bond Options Overview

	Bond Price	Yield	Asset Swap Spread	Realized Volatility Breakdown						Overwriting Yield	
				Rate Volatility	Credit Volatility	Correlation	Bond Volatility	Implied Volatility	Volatility Carry	ATM Call	101% Call
IG Long-end (10yr+)											
Lockheed Martin \$6.15 '36s	111.7	5.3%	154	8.8%	13.9%	-49%	12.3%	12.5%	0.2%	7.7%	6.2%
EDF €4.625 '24s	102.6	4.3%	77	6.6%	3.0%	-30%	6.4%	6.8%	0.4%	3.8%	2.6%
E.ON £6.75 '39s	120.2	5.2%	148	6.4%	3.6%	-13%	6.9%	7.3%	0.4%	4.1%	2.8%
Financials (5yr-10yr)											
UBS €6 '18s	113.0	3.8%	90	4.2%	1.2%	-25%	4.0%	4.5%	0.5%	2.0%	1.0%
Credit Agricole €3 '15s	99.1	3.2%	68	3.1%	1.2%	-28%	2.9%	3.2%	0.3%	1.7%	0.7%
Cyclicals (5yr-10yr)											
HeidelbergCement €8 '17s	111.0	5.7%	302	3.4%	6.4%	-76%	4.4%	4.8%	0.4%	1.6%	0.8%
Lafarge €5.375 '18s	101.5	6.2%	317	4.2%	4.4%	-36%	4.9%	5.4%	0.5%	2.5%	1.4%
Xstrata €5.25 '17s	107.5	3.8%	99	3.9%	1.4%	-24%	3.8%	4.2%	0.4%	1.9%	0.9%

Note: All prices are indicative only. Implied volatility is based on the BBG OV. Overwriting yield is annualized yield.

Source: Bloomberg, Morgan Stanley.

Appendix

BUILDING BLOCKS

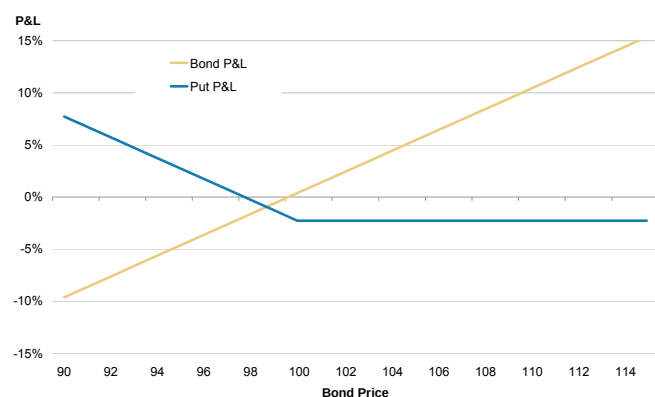
Bond options language is similar to equities, i.e., options are quoted on a price. The basic option types are Call and Put Options

Put option: A put option gives the right to sell the bond at a fixed price on the expiry date. Buying a put expresses bearish view on bond prices.

Call option: A call option gives the right to buy the bond at a fixed price on the expiry date. Buying a call expresses bullish view on bond prices.

exhibit 9

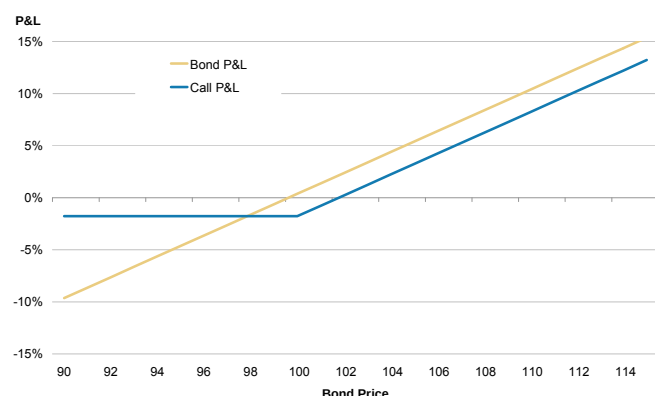
Buying a Put Option



Source: Morgan Stanley Research

exhibit 10

Buying a Call Option



Source: Morgan Stanley Research

TRADING AND SETTLEMENT CONVENTIONS

We review some of the key trading and settlement conventions for corporate bond options, many of which are similar to those for credit index options.

Expiries: The quoted expiries typically range from one month to one year but expiries under three months should be the most popular.

Underlying: Option references a specific bond.

Exercise type: Options are European style, exercisable only on the maturity date.

Strike Levels: The strike level is specified as a price level on the bond. Note that the strike is on the clean price.

Delta exchange: Prices are quoted with delta exchange as the standard.

Type of settlement: Physical settlement is the standard, i.e., upon the exercise of an option; bonds get exchanged at the strike price to settle the contract.

Exercise date and window: Option expiry for a month is on the third Wednesday of the month, similar to the CDS index options. Exercise window is between 9AM and 11AM EST for options on US corporate bonds and between 9AM and 4PM BST for options on European corporate bonds.

Option valuation: Options Valuation (OV) on Bloomberg is the closest we have to a public pricing model (see Exhibit 12). It has the standard suite of option pricing models: Black-Derman-Toy [B], lognormal [L], and normal mean reverting [N] models and Price based [P] model.

We acknowledge the contributions of Vishwas Patkar to this chapter.

exhibit 11

Bloomberg OV

GRAB
Change values, or for quick use, see OV <Help> for help on tails.

Corp OV

OPTION VALUATION
Option **LMT 6.15 09/01/36** → Reference bond

TRADE 6/ 6/11 Mon 0 trad 3 PRICE 112.133 5.2740 Conv. Worst
EXPIRY 9/21/11 Wed 107 days exercise 5 STRIKE 112.000 P 5.2790 Conv. 9/ 1/36
9/26/11 exercise settle @ 100.000

Save ? N

Option Strike can be specified either on the bond price or the yield

A/E EUROPEAN

	CALL		PUT	
	dec.	32nds	dec.	32nds
Option Price	2.830	2-26+	4.491	4-15 ³ / ₄
Price I. Vol	15.000		15.000	
Yield Vol(%)	21.166		21.166	
dPdY	15.07			
Delta (Price)	.443		-.556	
Gamma (Price)	.044		.044	
Vega (Price)	.236	0-07+	.236	0-07+
7-day decay	.118	0-03 ³ / ₄	.118	0-03 ³ / ₄
Time Value	2.830	2-26+	2.830	2-26+

Model P → Model Specification

B - Black-Derman-Toy
L - Lognormal
N - Normal Mean rev
P - Price based → Price of put option (% of notional)

Model properties

Price based
1 factor: bond price → Price Volatility used for pricing put
mean reversion: no
bond price lognormal → Yield Volatility used for pricing put
vol: const, relative

Override? N
fwd px 110.339 → Forward price, implied by repo rate
repo .100 → Repo rate

Australia 61 2 3777 8600 Brazil 5511 3048 4500 Europe 44 20 7330 7500 Germany 49 69 3204 1210 Hong Kong 852 2377 6000
Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2011 Bloomberg Finance L.P.
SN 884795 G675-955-3 06-Jun-11 16:46:57 GMT+1:00

Source: Bloomberg, Morgan Stanley

Section B

Market Themes

Tracking the Credit Cycle – A History of Tranche Returns

January 13, 2011

Sivan Mahadevan
Ashley Musfeldt
Phanikiran Naraparaju

Structured credit markets comprise many different types of instruments that touch various underlying credit assets. Within corporate credit, structured credit has focused on high yield, loans and investment grade over the past dozen years through two economic cycles. Beginning in 2003, the market for liquid standard tranches tied to the various corporate CDS indices has provided an enormous amount of valuable market data on performance through a historic credit cycle including the financial crisis of 2008. During this period, we witnessed the extremes of a full cycle, including the historic wide in credit spreads, a default cycle matching the peaks of previous cycles, and significant systemic risks. Every part of the tranche capital structure has been tested, providing both important lessons learned and investment insight.

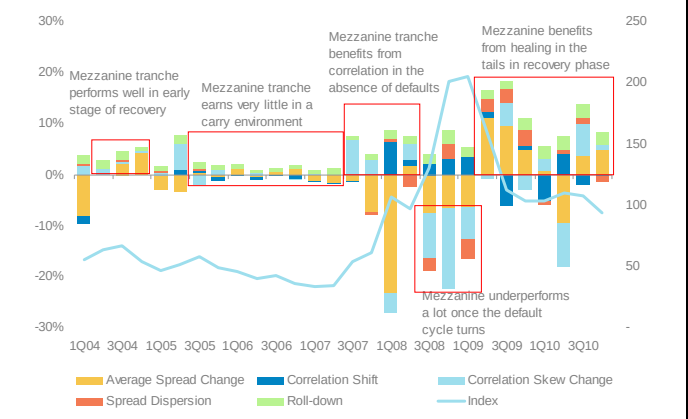
In short, it has been an interesting period in which to examine credit performance, and we believe that the tranche market has highlighted, sometimes in extreme ways, the performance characteristics and credit fundamentals of the underlying market, as it should. Ultimately, the rationale for the existence of the tranche market is to be able to manage the different credit environments, and the brief history of tranche trading has many lessons to offer. Furthermore, the long/short nature of both the index and the bespoke market means that investors are not forced to make do with just the best long opportunity, but instead can focus on the attributes of long or short positions that are most appealing for a particular period of the credit cycle, from carry themes to systemic risks and default concerns. The derivative nature of tranches implies that their performance should be “derived” from the underlying asset, and that is the experience we had through the cycle.

The reason the tranche market has witnessed “rational” performance has a lot to do with the nature and simplicity of the structures. Given that losses (from defaults) are assigned to the various tranches using a simple bottom-up system, the derivatives approach used to value tranches provides fairly good tools in translating underlying credit performance to tranche performance. We can use these tools to ascertain how much performance a given tranche has experienced owing to various market factors like changes in spread, portfolio dispersion, defaults, and the passage of time (with no defaults). The plug factor is correlation, or the relative

change in tranche pricing that cannot be explained by the other measures. This is similar to implied volatility in an options pricing model. We know that concepts like implied volatility and skew move with markets, and we see this in tranche implied correlation too; moving significantly higher in systemic/macro dominated environments, and falling substantially in idiosyncratic environments.

exhibit 1

CDX Junior Mezzanine Performance 2004-2010



Source: Morgan Stanley Research

In this chapter we use market data over the past nearly eight years to detail the performance characteristics of standardized tranches with respect to the underlying corporate credit markets. For example, equity tranches underperformed during the auto-sector woes in 2005, while super senior tranches underperformed during the beginnings of the systemic crisis in the summer of 2007. As the credit crisis gained momentum in 2008, systemic risks pushed senior trades wider initially while significant amounts of corporate default risk pushed mezzanine tranches wider, resulting in these tranches underperforming the market later in the year. As recovery started in 2009, super senior tranches initially outperformed as systemic risks fell, and then mezzanine tranches outperformed as idiosyncratic risks healed as well.

We organize the remainder of this chapter by focusing on each part of the capital structure, providing a detailed look at the performance characteristics of the underlying credit market, and how particular tranches reacted to them.

exhibit 2

Key Milestones in the Index Tranche Market

Year (Index)	Key Events
2003	Standardized index tranches begin trading on the US and European CDS indices (then called Trac-X)
2003	Parmalat is the first default for the newly launched CDS indices and tranche market. A low recovery (20%) and isolated nature of the default hurt equity tranches and to a lesser extent 3-6% tranches. In a sense this set the stage for the correlation crisis in 2005.
2005	Auto sector got downgraded to sub-investment grade sparking fear of losses and considerable dislocation for equity tranches.
2005	Auto names removed from the on-the-run IG indices and IG5 is a considerably cleaner portfolio than IG4. Both IG4 and IG5 tranches continue to trade due to significant portfolio differences.
2006	Perception of lack of default risk and flows from structured credit risk-taking helped credit spreads reach cycle lows. Levered super senior flows help compress super senior spreads to single digits.
2007	Super senior sells off in late 2007, owing to early stages of subprime crisis and rising correlation. Junior tranches remain supported due to lack of defaults.
2008	The financial crisis causes several financial credit events globally with mezzanine tranches being the biggest underperformers. Market focuses on legacy mezzanine risk and tranche rolls stop.
2009	Recovery starts, but defaults remain high in the early part of the year. Tail credits still an issue in the first quarter.
2010	Energy sector moves (Anadarko and Transocean) along with widening in the distressed financials in US and cyclical fears in Europe cause junior mezzanine tranches to underperform.

Source: Morgan Stanley Research

LOOKING FORWARD – INVESTMENT IMPLICATIONS

In the US, we see a lower probability of severe systemic risks given that the growth debate is less about growth and perhaps more about inflation. We do not see much default risk in either investment grade or high yield markets, as is typical in early- to mid-cycle recoveries, particularly those where the weaker companies have already defaulted (during the recession). We do see some rising idiosyncratic risks that are more related to corporate actions (LBOs and the like) that are more equity friendly than bondholder friendly. As such, in the US, we are constructive both on equity and mezzanine forms of risk generally, although rising LBO risks bias us to be long near-dated equity tranches (where default risk is low). With mezzanine risk we are happy to move further out the

maturity spectrum (time decay is good). We prefer short positions as hedges in more senior parts of the capital structure owing to their low cost. We caution that these are large tail risk hedges, for the most part.

For Europe, we believe the fate of credit is linked to the trajectory of the European sovereigns. So far the strong liquidity and balance sheet trends for corporates have kept the index and tranche spreads relatively reined in. Most of the risk premium has been absorbed by the senior tranches as the market perceives the risk of corporate defaults to be relatively low. A few surprise credit events, mainly in the financials sector, would hurt junior tranches. In fact, short-dated junior tranches make for good hedges against a very bearish tail scenario involving groups of credits moving wider. Otherwise, for a systemic crisis, we have been recommending senior tranches as low-cost hedges to insure against generic IG spread widening.

OUR PERFORMANCE ATTRIBUTION APPROACH

Many factors ultimately influence the performance of a credit portfolio. In Exhibit 3, we highlight the nine factors we use in our performance attribution model for the standardized index tranches in the US and Europe. Taken together, we think these factors are relatively comprehensive when evaluating tranche performance.

exhibit 3

Factors Used in Tranche Performance Attribution

Factor	Description
Average 5yr spread change	Spreads at all maturities for the individual names change by same percentage as the index 5yr spread change
Average credit curve change	Spreads at each maturity for the individual names change by same percentage as the index spreads change at the respective maturity
Correlation curve shift	Entire correlation curve shifts by the same amount as change in equity correlation
Correlation skew	Correlation skew changes for all maturities
Spread dispersion	Spread changes that deviate/not explained by index moves alone
Time decay	Impact of reduced time to maturity
Rates shifts	Impact of shift in interest rate changes
Carry	Coupon paid during the period
Defaults	Losses paid out due to defaults in the underlying portfolio

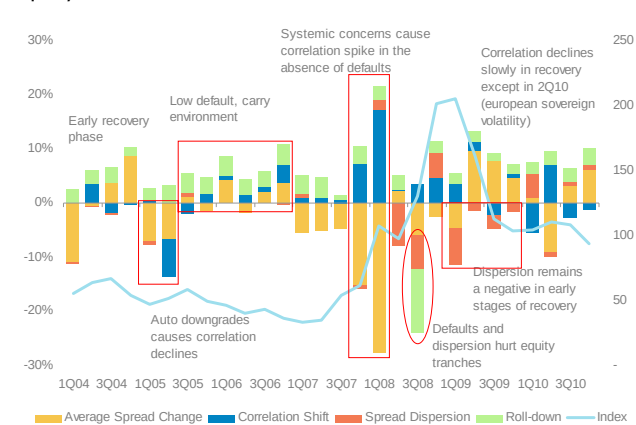
Source: Morgan Stanley Research

EQUITY TRANCHE PERFORMANCE THEMES

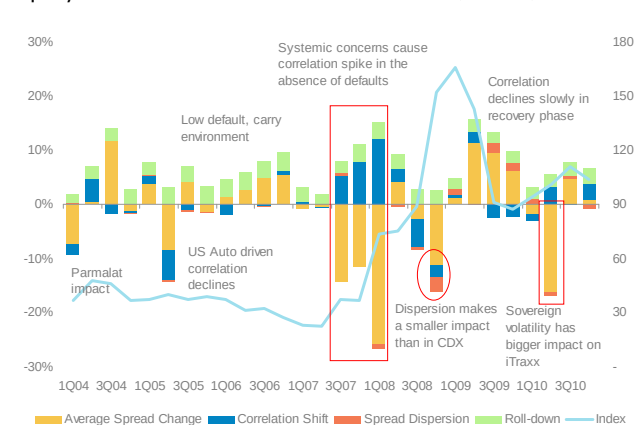
Equity tranches, or first-loss tranches, are levered investments on the index and are most sensitive to defaults, by definition. Because equity tranches are high delta instruments (deltas range from 3x to as high as 25x in tight spread environments), a big component of the absolute return of equity tranches will be the sensitivity to index moves. Indeed, that levered exposure is one of the primary attractions of equity tranches. Correlation and dispersion are driven by expectations of defaults, and time decay and carry represent the compensation for this risk. In general, time decay and carry are quite substantial for equity tranches and junior tranches. We review some key historical performance themes.

exhibit 4

Equity Tranche Performance Attribution (CDX)



Equity Tranche Performance Attribution (iTraxx)



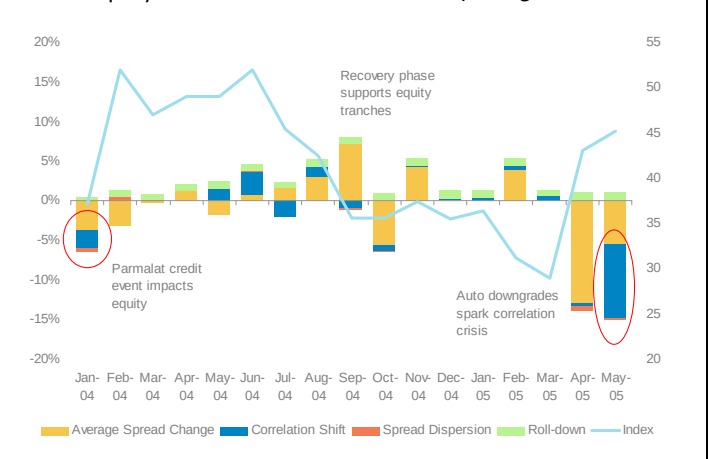
Source: Morgan Stanley Research

Parmalat default (late 2003). Parmalat was the first default for the newly launched CDS indices and the tranche market (see *Leverage and Long Lasting Milk*, January 16, 2004). The Parmalat default is probably the closest example the tranche market has of a purely idiosyncratic event (a European credit event arising out of accounting issues), and

one that was a “jump-to-default” event as well. A low recovery (20%) and the isolated nature of the default hurt equity tranches, mainly in Europe and to a slightly lesser extent 3-6% tranches as well. The 0-3% tranche posted an overall loss of 28.7pts, much of it from the default payout for Parmalat. The 3-6% tranche widened out 100bp, i.e., about 4.3pts, whereas the 9-12% tranche widened just 6bp. In a sense, the Parmalat default provided some important lessons on the risk to junior tranches. Unsurprisingly, the idiosyncratic nature of the event meant that it did not have a big enough impact on market psychology, and market participants were still caught off-sides by the tail risks created by the auto crisis in 2005.

exhibit 5

iTraxx Equity Tranche Performance (2004-1H05)



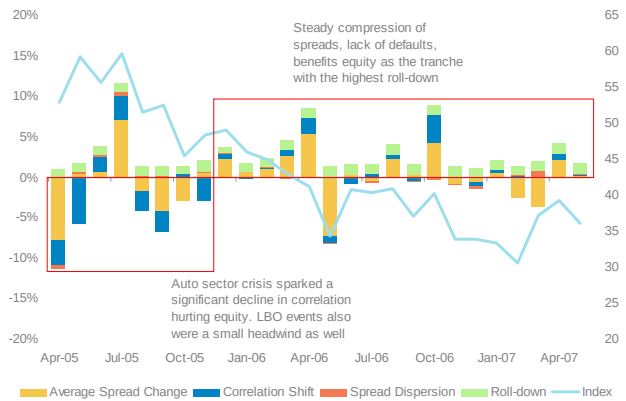
Source: Morgan Stanley Research

US auto sector crisis (2005). Even though Parmalat was the first credit event for the tranche market, the auto sector driven correlation crisis was the first major test of the index tranche market and specifically equity tranches (see *Correlation – Don’t Throw Opportunity Out with the Model*, May 13, 2005). Despite Parmalat and the auto repricing and eventual downgrades that triggered the sell-off, the overall spread, default and volatility environment was fairly benign as the economic recovery took hold. Equity tranches performed extremely well in 2004 and Q1 of 2005. Perhaps this engendered complacency, and many investors were drawn into variations of a long equity tranche risk trade, leveraging on the low default environment. In particular, long equity, short mezzanine was a popular trade in early 2005. Another was holding long equity tranches and using the carry to buy protection on single names and managing single-name default risk.

Stress in the US auto sector served as a trigger for many equity tranche investors who under-hedged spread and/or default risk in the autos. In addition, single-name credit markets and convertible arbitrage communities may have forced levered investors to reallocate risk and rebalance portfolios, resulting in the unwinding of equity tranche trades.

exhibit 6

CDX Equity Performance During Auto Crisis and the Subsequent Low Default Environment (2005-07)

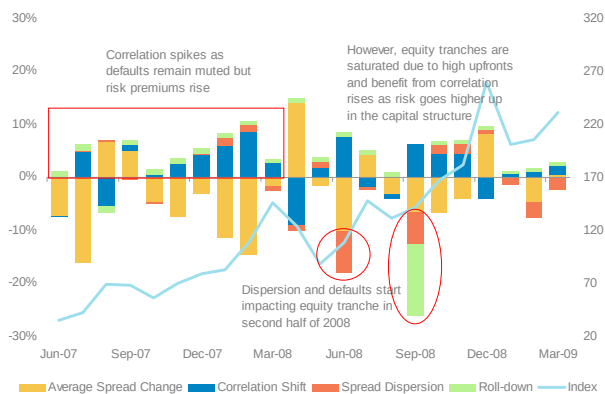


Source: Morgan Stanley Research

Low default, carry environment (2H05 to 1H07). For almost two years following the auto sector upheaval, credit enjoyed a low default, low volatility environment with credit spreads declining to cycle tight. Tranche performance was dominated by tighter spreads and spread compression – generally positive for equity tranches due to their levered exposure. Roll-down (time decay and carry) was a strong and consistent positive every month, as there were no IG defaults. Structured credit activity was an important driver – partly helping push risk out of mezzanine tranches into equity and sometimes senior tranches to a small extent. We see this in the small negative impact of correlation on equity tranche returns in this period.

exhibit 7

CDX Equity Tranche During the Credit Crisis



Source: Morgan Stanley Research

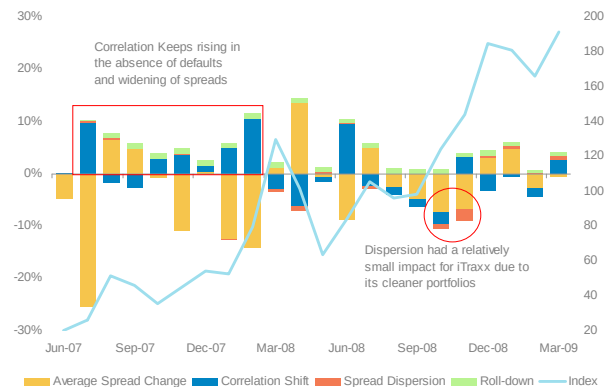
Rising risk premiums, systemic crisis (2H07-1H08).

Although May 2007 marked the tightens for spread levels, it took almost a year for defaults to pick up. And even then the on-the-run indices had only a couple of defaults (FNM, FRE,

WAMU) in the US and none in Europe. So the early stage of the credit crisis was characterized by rising risk premiums associated with large correlation rises which mitigated the impact of the spread widening. In absolute terms, wider spreads did push equity lower, but the relatively strong performance on a delta hedged basis proved to be justified in the absence of defaults.

exhibit 8

iTraxx Equity Tranches During the Credit Crisis



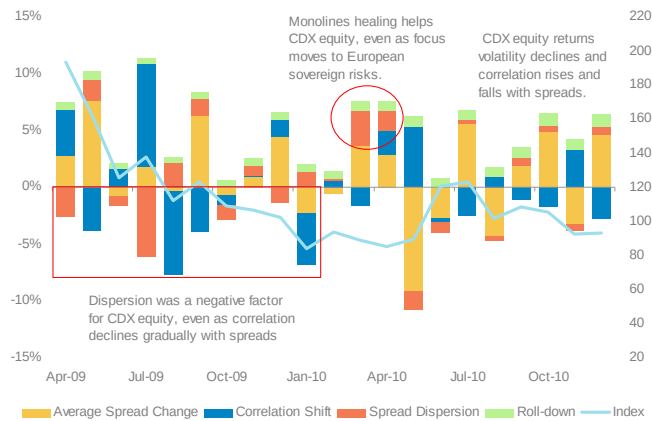
Source: Morgan Stanley Research

Default cycle turns, hurting equity (2H08-1Q09). In the second phase of the credit crisis, correlation continued to rise modestly, but default concerns rose and the associated rise in dispersion as well as actual defaults dominated performance. Time decay was only a small contributor to returns. There was a big difference between the performance of equity tranches in CDX and iTraxx due to the differences in the portfolio quality. While CDX suffered actual defaults and a tail of distressed financials, iTraxx performed relatively well.

Recovery and beyond (2Q09 to today). In the post-crisis period, we have seen all the different environments on a small scale: a considerable spread rally and normalization of correlation, healing in tails and reduced dispersion, shades of a systemic crisis from European sovereigns and even a potential repeat of a correlation crisis in the energy sector following the Gulf of Mexico events. Time decay is becoming more important in terms of contribution as volatility declines. In Europe, the magnitude of spread moves has dominated equity tranche performance, although correlation is rising to offset some of the spread impact. Dispersion has been only a modest negative in iTraxx so far. In CDX, healing in the tails is causing a positive contribution from dispersion.

exhibit 9

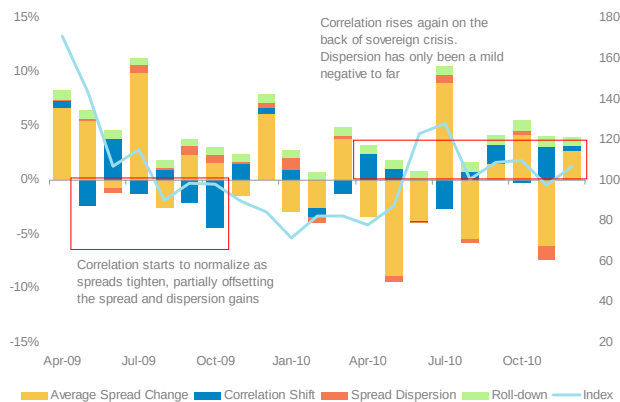
CDX Equity Tranche in the Recovery Phase



Source: Morgan Stanley Research

exhibit 10

iTraxx Equity Tranche in the Recovery Phase



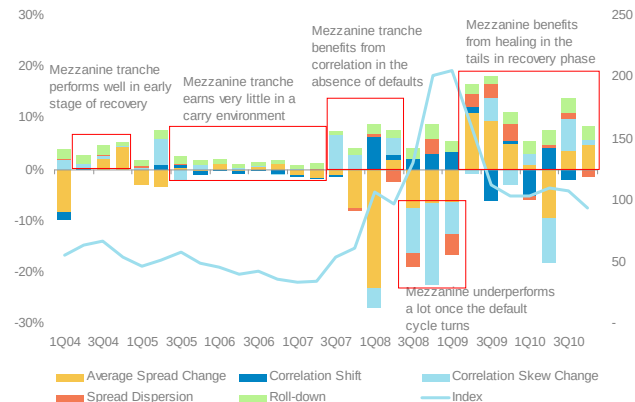
Source: Morgan Stanley Research

Junior Mezzanine Tranche Performance Themes

Mezzanine tranches have had a special place in the structured credit markets over the years. They are junior enough in the capital structure to add, in some cases, significant amounts of yield over vanilla portfolios of credit, but they tend to have enough subordination to protect investors through most default cycles, and they have historically had investment grade ratings as well. But they are by definition pro-cyclical investments, with long positions performing well through growth periods, and short positions being among the best market hedges per unit cost during recessions. These index tranches also are important proxies for the large market of mezzanine risk distributed via bespoke during the previous cycle. Typically these were 1% to 3% thick tranches, and, due to the quality of the portfolio, these usually mapped to the CDX IG 3-7% and iTraxx 3-6% tranches.

exhibit 11

CDX Junior Mezzanine Performance 2004-2010

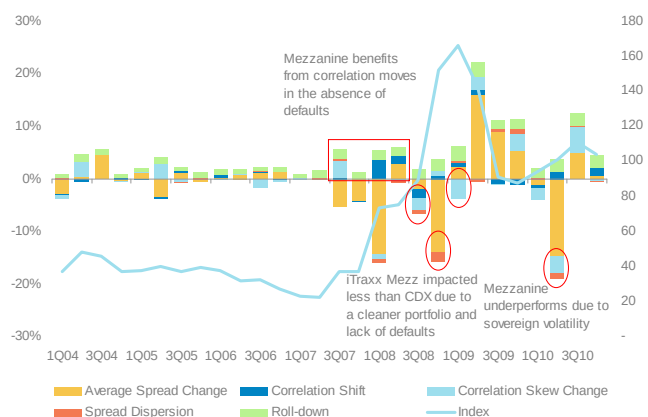


Source: Morgan Stanley Research

Overall, as we observe junior mezzanine performance from 2004 through 2010, we find that positive performance in 2006-2007 was driven by spread compression, time decay and relative pricing in favor junior mezzanine (correlation skew). At the outset of the credit crisis, corporate credit super senior tranches initially underperformed on rising correlation/systemic risks, and mezzanine was a net beneficiary. But as the credit crisis developed into a broader financial crisis and ultimately a global recession, mezzanine tranches underperformed substantially, owing primarily to skew (as markets favored other tranches) and spread dispersion. The healing market since then has been positive for junior mezzanine tranches with both spread compression and risk allocation benefitting the middle of the capital structure.

exhibit 12

iTraxx Junior Mezzanine Performance 2004-2010

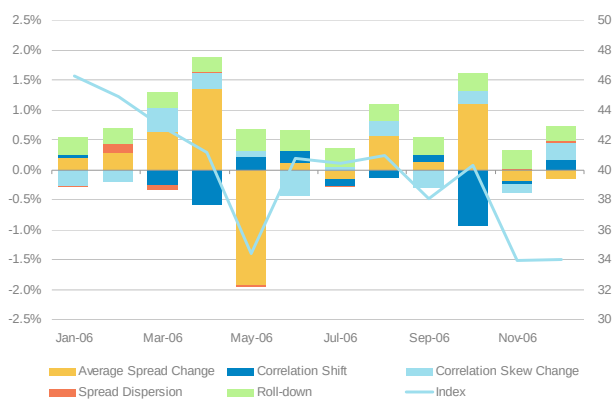


Source: Morgan Stanley Research

Time decay has generally been a positive factor as defaults have been lower than what is implied by single-name spreads. We note that skew and dispersion seem to work together, so when dispersion is rising, investors tend to avoid junior mezzanine risk, and such tranches are driven lower both by market spreads and mezzanine risk aversion. We also note that skew is directional and tends to fall when markets are selling off, which exhibits the beta nature of junior mezzanine tranches. We look below at more detailed performance of junior mezzanine risk per calendar year, highlighting key credit market themes.

exhibit 13

2006: CDX Junior Mezzanine Returns Are Low



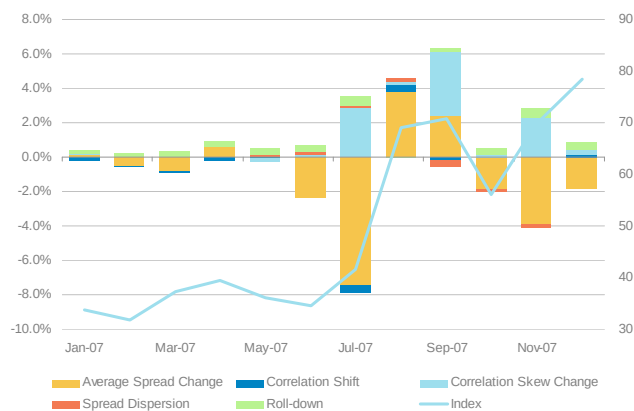
Source: Morgan Stanley Research

2006. Credit risk was stable during the year, with bespoke mezzanine issuance driving spreads to extremely tight levels. LBO activity was the primary driver of idiosyncratic risk through the year. Regarding junior mezzanine risk specifically, skew performance was mixed as bespoke mezzanine flows were quite substantial in the market, but not necessarily evenly distributed. Time decay was a steady positive in a zero default environment, and compression was a mild positive, but LBO risk at times hurt mezzanine tranche performance.

2007. The first half of the year was very supportive for corporate credit risk. After the first whispers of trouble from the collapsing Bear Stearns funds, the summer brought the initial contagion from the subprime crisis and along with it, the first significant re-pricings in corporate credit. Much of the returns in junior mezzanine came from time decay and dispersion in the first half of the year. The outperformance owing to skew in the second half of the year was related to correlation rising and strong underperformance of super senior and senior tranches. Dispersion started to be negative in the second half of the year.

exhibit 14

2007: CDX Junior Mezzanine Outperforms Somewhat on Rising Correlation

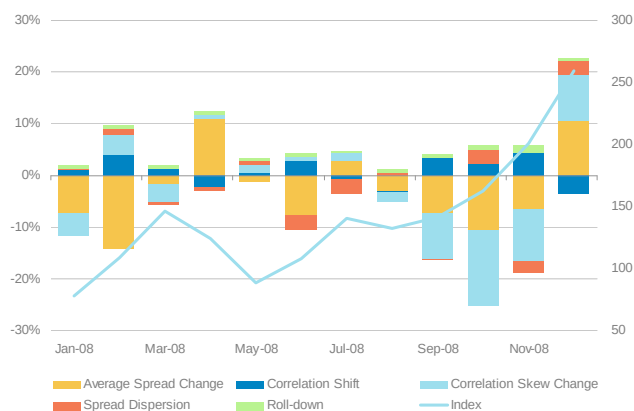


Source: Morgan Stanley Research

2008. Volatility in the credit markets was elevated throughout the year, with the Bear Stearns' rescue in March serving as an introduction for the extreme events that would ensue in September. Credit spreads reached historic highs in the last several weeks of the year. As credit risk grew more negative, dispersion began to rise, weighing negatively on junior mezzanine performance for most of the year. This was partially offset by skew in the first 8 months of the year; however, even skew turned significantly negative in the risk-off environment following the Lehman bankruptcy. Time decay returns fell significantly, rising only in December.

exhibit 15

2008: CDX Junior Mezzanine Underperforms Substantially on Dispersion and Skew Changes

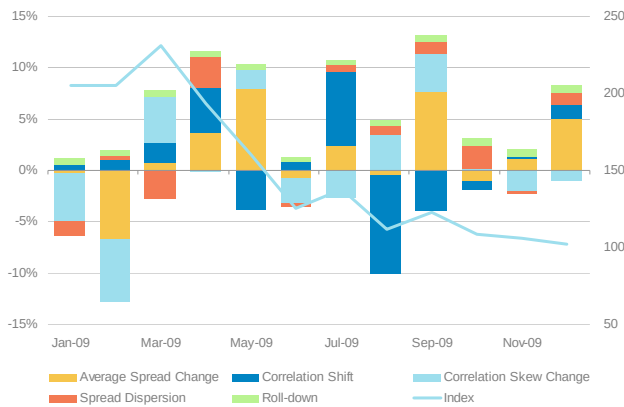


Source: Morgan Stanley Research

2009. The market got off to a weak start in the first quarter, but many dislocated relationships healed. Credit had a strong rally in the remainder of the year, with more dislocation healing, and most new risk taking was in corporate bonds rather than CDS, healing the basis. Skew was a strong positive for junior mezzanine for most of the year as 2008 dislocations began to heal, causing mezzanine to outperform. Dispersion was high early in the year as default levels remained elevated, but as recovery gained momentum and defaults fell, spread compression generated positive performance for junior mezzanine. Time decay was positive throughout the year.

exhibit 16

2009: CDX Junior Mezzanine Heals Eventually with Spread Compression Aiding

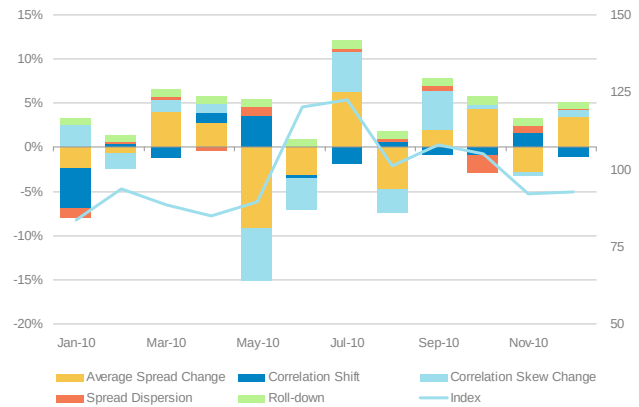


Source: Morgan Stanley Research

2010. Credit markets rallied early in the year, which was ultimately tempered by significant European sovereign volatility in May/June. The summer brought on a slow healing, followed by a strong year-end rally. Spread compression was a positive for junior mezzanine performance throughout the year. Skew was driven by a risk-on / risk-off environment given huge moves in underlying spreads. Time decay was a solid positive through the year as default risk was low.

exhibit 17

2010: CDX Junior Mezzanine Performs Well Outside of Sovereign Crisis on Compression and Skew



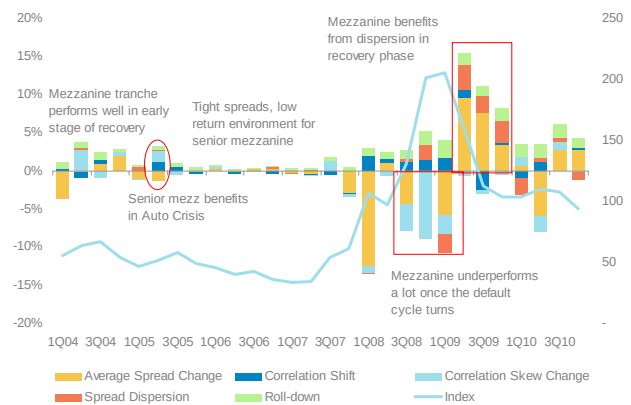
Source: Morgan Stanley Research

Senior Mezzanine Tranche Performance Themes

Senior mezzanine tranches during much of the last credit cycle were considered “orphaned” credit risk. Too senior to generate enough yield for those with strong appetites, yet too junior for those investors who required extremely remote risk, these tranches (CDX IG 7-10% and iTraxx 6-9%) were often an afterthought, at least until the credit crisis drove risk allocations up the capital structure and ratings models and increased defaults served to differentiate them from their senior tranche brethren. In this section we examine the ways this more subordinated yet still “mezzanine” risk behaved throughout the cycle.

exhibit 18

CDX Senior Mezzanine Performance 2004-2010

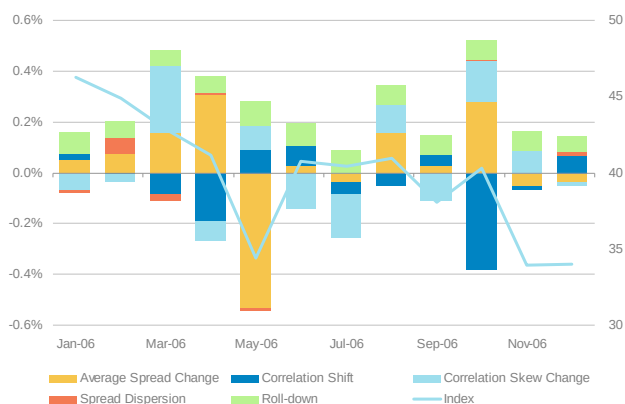


Source: Morgan Stanley Research

2006. This was a rather uneventful year for senior mezzanine risk, as market flows and attention were dedicated to the enormous wave of bespoke flows which had an outsized impact on the tranches just below senior mezzanine. Time decay provided a steady, if small, positive performance, while dispersion was flat for the entire year. Skew was positive at the beginning of the year, with a brief, small negative period in the third quarter, ending the year relatively flat.

exhibit 19

2006: CDX Senior Mezzanine Largely Uneventful

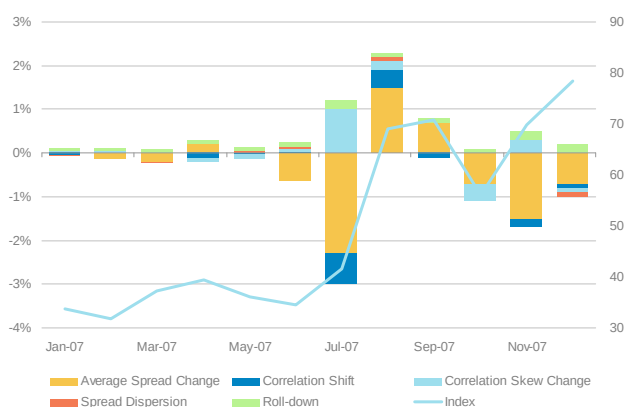


Source: Morgan Stanley Research

2007. While the bottom of the capital structure was digesting the first hints of the impending crisis, senior mezzanine risk performance began to get more interesting. Time decay picked up steadily throughout the year, while dispersion began to factor into tranche performance, though still small. Skew was overall flat the first half of the year, and became a much bigger contributor to positive returns over the summer given the underperformance of more senior tranches.

exhibit 20

2007: CDX Senior Mezzanine Benefits from Super Senior Issues, But Turns Weaker Later

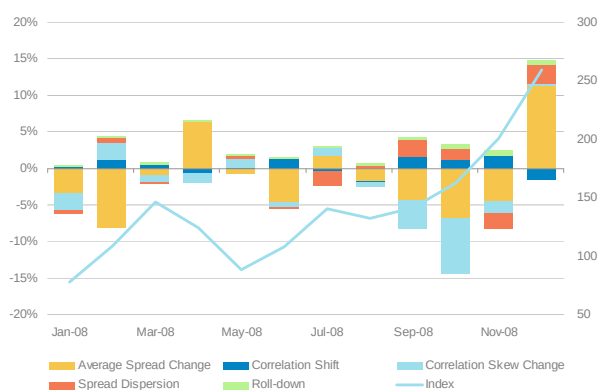


Source: Morgan Stanley Research

2008. The impact of rolldown (time decay) became more pronounced as the expected loss on the underlying index rose substantially, up through 3-7% and into the senior mezzanine tranche, substantially changing the fundamental nature of this risk. Whereas before this tranche tended to behave more like its more senior relatives, elevated overall credit risk made this now solid mezzanine risk. Dispersion started to have a positive impact in the second quarter and really drove performance during the peak of the crisis. Skew, however, had increasingly negative momentum and more than offset any gains made by dispersion and time decay.

exhibit 21

2008: CDX Senior Mezzanine Suffers with Other Mezzanine Risk



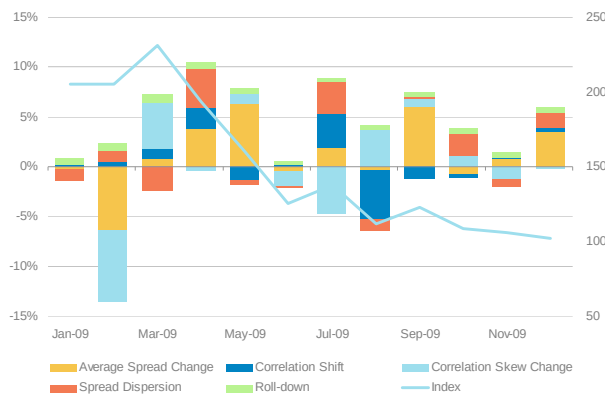
Source: Morgan Stanley Research

2009. The market got off to a weak start, but credit had a strong rally in the back half of the year. In senior mezzanine risk, rolldown generated positive performance, peaking in the first quarter of 2009 and remaining elevated throughout the rest of the year. Performance due to dispersion was volatile; in the beginning of the year it contributed negatively, like the junior mezzanine tranches, but soon turned positive in the spring as spreads came down slightly, and remained positive for the rest of the year. Skew remained negative, though to a lesser degree than 2008.

2010. As credit markets continued to heal, time decay remained a major positive factor in senior mezzanine risk. In the first quarter, this was offset by spread dispersion, though between spring and fall spread dispersion was a positive contributor to returns. Skew performance was mixed and somewhat volatile, ending the year flat.

exhibit 22

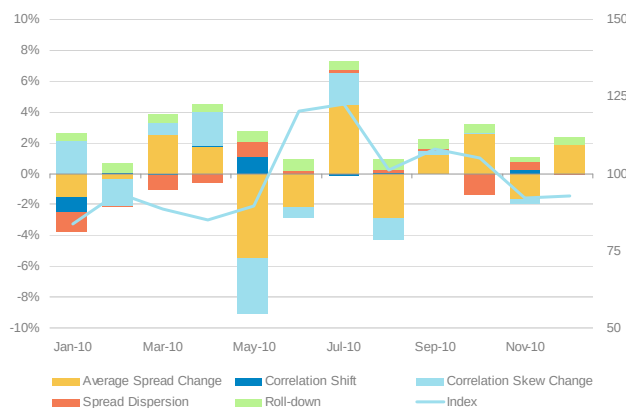
2009: CDX Senior Mezzanine Recovers More Quickly than Other Mezzanine



Source: Morgan Stanley Research

exhibit 23

2010: CDX Senior Mezzanine More Mixed with Sovereign Volatility



Source: Morgan Stanley Research

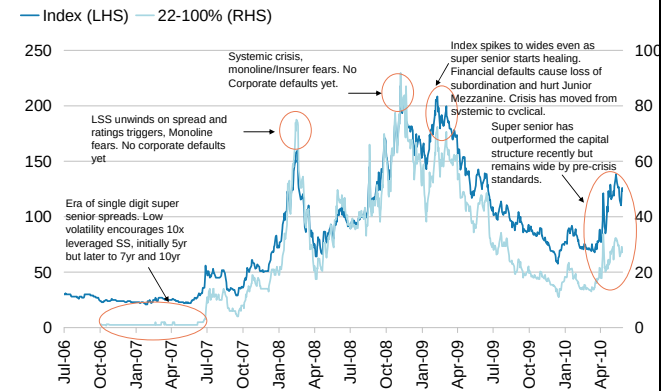
Senior Tranche Performance Themes

As equity and junior mezzanine tranches jostled for the spotlight in the 2005, super senior tranches with spreads in the teens that rarely moved were the less interesting part of the market, until the advent of levered super senior (LSS) flows. In an environment where central banks were considered to have mastered the control of tail scenarios, super senior was considered dead money and any level of super senior spread would be considered worth selling into. LSS flow pushed super senior spreads into single digits. The combination of the extremely low volatility environment, high leverage, long duration, the presence of unwind triggers (both spread and ratings driven), and in hindsight, poor entry points proved to be the undoing of super senior structures.

Many of these structures were being unwound quite early in the crisis (late 2007 and early 2008).

exhibit 24

Super Senior vs Index Spread (iTraxx)



Source: Morgan Stanley Research

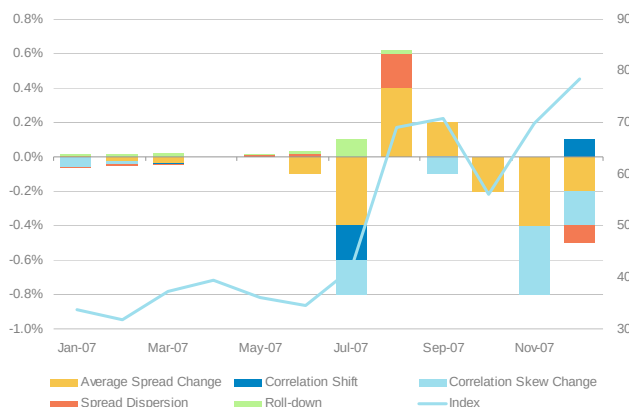
Junior super senior risk, represented by the 15-30% tranche in CDX IG and the 12-22% in iTraxx, has been a relatively less active part of the tranche market, with pricing primarily driven by equity and mezzanine below and levered super senior flows above. A key difference in analyzing the importance of underlying factors in driving tranche performance for senior risk is that with their lower deltas and subordination, roll-down has a smaller role to play relative to the higher delta junior tranches. Relative to the index, (delta-adjusted), these tranches have a “negative” roll-down.

2H07 – 1H08. Unlike equity or mezzanine, the impact of correlation (parallel or skew) is not as straightforward to analyze. In general we have found that correlation moves tend to have a largely directional impact for junior tranches, i.e., large moves wider in spreads tend to be accompanied by a move higher in correlation for equity (lower in skew for mezzanine), which then has an offsetting effect for equity and exacerbates the impact of spreads for mezzanine.

In Exhibit 25 we show that correlation skew is a cyclical indicator rather than just a reflection of spread directionality for the more senior tranches. Through the summer of 2007 and early 2008, senior tranches underperformed junior tranches as developing subprime concerns sparked fears of a credit crisis. This period also witnessed leveraged super senior unwinds driven by spread and rating fears. Through this period, risk flowed into senior tranches, which can be seen by the significant negative contribution provided by the skew component.

exhibit 25

2007: Senior Underperforms in the Rising Systemic Risk Environment

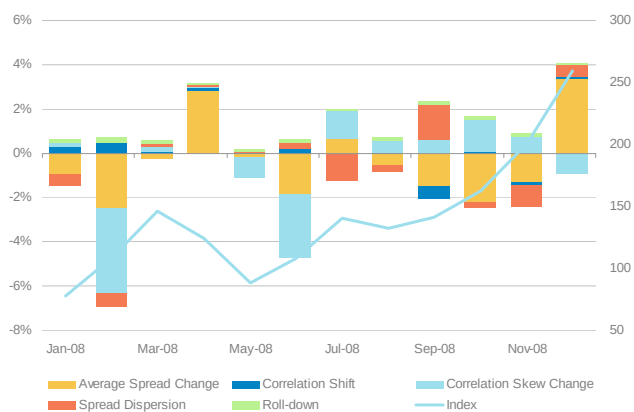


Source: Morgan Stanley Research

2H08. As the credit crisis morphed into a full-blown economic recession accompanied by significant corporate defaults, senior risk outperformed equity and mezzanine risk through the second half of 2008. A combination of policy action aimed at reducing systemic stresses and idiosyncratic default concerns on the back of defaults in the IG index led to risk flowing down the capital structure. Exhibit 26 captures the positive impact of moves in correlation skew for the 15-30% tranche even as spreads continued to widen through the second half of 2008. This was in stark contrast to the market moves seen from July 2007 – May 2008, when spread changes and correlation skew both contributed negatively to tranche performance.

exhibit 26

2008: Senior Underperforms Early on Systemic Risk, but Outperforms During the Recession



Source: Morgan Stanley Research

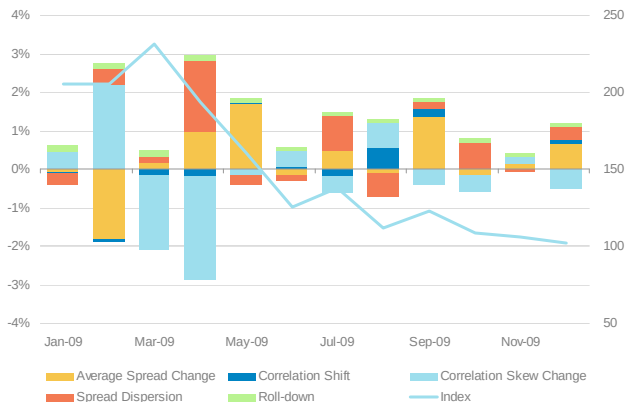
2009. Dispersion played a more significant role in senior tranche performance in early 2009. Even as credit spreads

started tightening from elevated levels, tails risks persisted (largely in the form of distressed financials), as did defaults, which meant that idiosyncratic risk still existed. This dispersion worked favorably for senior risk, which was now increasingly OTM from a default breakeven perspective, because fat portfolio tail risks were being disproportionately allocated to the junior tranches. This trend subsided towards the end of the year as the economic backdrop improved and defaults started subsiding.

2010. Spread compression occurred at a portfolio level in 2010 and as such tail names tightened significantly, leading to outperformance of junior tranches. Exhibit 28 shows the negative impact that spread dispersion had on 15-30% performance through most of the year.

exhibit 27

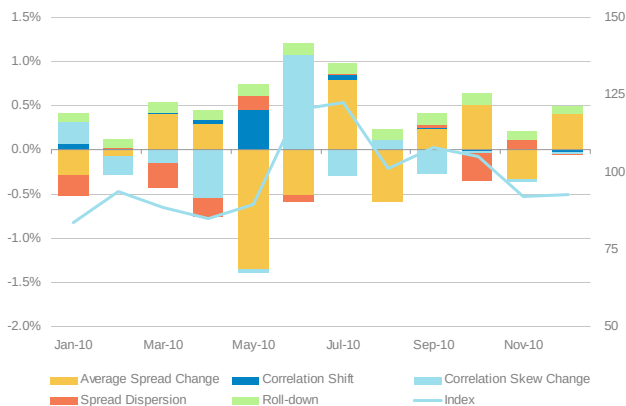
2009: Senior Benefits from Dispersion



Source: Morgan Stanley Research

exhibit 28

2010: Senior Underperforms on Sovereign Risks and Improving Credit Quality



Source: Morgan Stanley Research

The Credit Hedger's Manual

October 28, 2011

Sivan Mahadevan
Phanikiran Naraparaju
Ashley Musfeldt
Vishwas Patkar

Corporate credit markets have struggled to find their proper footing since the 2008 financial crisis, as increased volatility has dominated what has otherwise been a relatively healthy credit fundamental backdrop including both significant deleveraging and debt refinancing. The source of this volatility has been both macro and systemic, and numerous events have served to bolster this volatility – including European sovereign solvency, slow global growth and record low levels of developed market interest rates.

If frequent bouts of volatility are part of the fabric of the next phase of credit markets, then we argue that strategies that dampen this volatility should be part of the playbook to create a more efficient investment landscape. The solid growth of credit options activity since 2008 is certainly testament to this theme, as a significant number of market participants have moved their credit investment mandates to a more equity-style model where both positioning for and monetizing market volatility is common. Hedging has increasingly become a part of the credit portfolio management process today for many investors, even those with a “long only” mindset and those who come from outside credit (equities, rates, currencies) but are focused on attractive hedging alternatives. The asymmetric nature of credit (limited upside but large downside) and the ability to implement truly long-dated hedges in a liquid way is certainly part of the hedging appeal.

Of course, hedging is nothing new to credit investors, as the CDS market dates back to the mid-1990s and many investors have incorporated hedges through structured credit solutions for well over a decade. While some investors have seasoned hedging programs in place, we believe many face a learning curve in terms of getting the best out of their hedging buck.

Although many investors are comfortable with the “hockey sticks” that are associated with standard options and tranche strategies, credit hedging is more than a buy-and-hold strategy, as it requires some amount management to find efficient solutions, monetize gains, and choose between instruments that offer varying relative value and payoffs. In short, there is a systematic side to credit hedging as well.

We have published extensively on credit hedging instruments. The point of this chapter is to focus on the systematic investment process behind credit hedging, including

determination of budgets, selection of instruments and strategies, and performance measurement and monetization. Over the time periods studied, we make the following conclusions:

Systematic Option Strategies: In investment grade, ATM payer strategies have had the greatest positive impact on volatility reduction, max drawdowns and overall P&L, while in high yield, collared strategies and those that had a short volatility bias performed the best.

Systematic Tranche Strategies: In tranches, senior, super senior and X-100% tranches have had the greatest impact on reducing volatility in a portfolio, have the lowest cost, and require less attention to entry point for systemic hedging. Junior tranches, while highly expensive, can have the greatest convexity if idiosyncratic risk rises. However, given their extreme cost, we do not recommend these as systematic hedges, and much attention must be paid to entry points.

We cover the themes outlined below and provide details in the remainder of this chapter.

Credit Hedging Landscape: We classify credit hedgers into 5 broad categories based on objectives and styles including both small and large tail hedgers and even those who lean against hedging flows (yield generators).

Hedging Budgets: We have not seen standard sized hedging budgets among credit investors, but offer some suggestions based on current yield levels and expected payoffs.

Overview of Hedging Instruments: There are three classes of instruments in credit: 1) Delta-one instruments like CDS indices and single-names baskets; 2) Options on CDS indices; 3) and Tranches that focus on specific default scenarios. We compare and contrast these instruments.

First Step – Defining the Hedge Scenarios: We divide our scenarios into large tail and small tail hedges, and in tranches divide further into idiosyncratic vs systemic risk hedges.

Second Step – Choosing Hedging Strategies: We review the standard option and tranche strategies, comparing their objectives and structures.

Third Step – Monitoring Hedge Performance: While we argue that results can be very time period specific, the performance mentioned in the above summary comprises what we learned from our historical backtests of systematic strategies.

CREDIT HEDGERS – WHO ARE THEY?

The tremendous growth in credit hedging activity has been broad-based since the financial crisis. We estimate average monthly options volumes in credit to be around \$150-200 billion notional, based on anecdotal evidence. We estimate average monthly tranche volume to be on the order of magnitude of \$70-90 billion. These are not delta adjusted, so the delta adjusted levels for options would be smaller, while tranches would be larger.

The flows in options and tranches, however, have certainly not been evenly balanced, thereby introducing equity-market like option skew and higher levels of implied vs. realized volatility as discussed earlier. The flows in options are skewed towards hedgers, although a class of “yield generators” does exist and is important for the market. The flows in tranches are generally skewed toward credit risk takers, so it stands in contrast to options. This imbalance is related to the nature of the participants in credit hedging and their very different motivations and styles. The overlap between options and tranche users, until recently, has actually been quite small, as they are somewhat different skill sets and can come from different credit backgrounds as well. Nevertheless, we highlight below five communities of market participants that define the technical landscape we see among hedgers.

Small Tail Focused Hedgers (Size = Large): The community of investors who focus on small tails in investment grade and high yield credit in both the US and Europe is perhaps the largest segment of the market. This community is opportunistic in the sense that they will hedge when they feel necessary or when they see value. Some take more of a buy-and-hold approach, while others can manage their options portfolios very actively as we highlighted earlier. We have seen more focus of late on tranches in this community, especially at times when credit volatility becomes high and investors look for further convexity opportunities through rises in market correlation or actual idiosyncratic events.

Natural and Systematic Hedgers (Size = Medium): Natural hedgers generally have business models that demand regular hedging, including the management of risks from loan books and counterparty exposures over both short and long-term periods, and they tend to take very systematic approaches as well. There is also a medium-sized community of banks who engage in first-loss protection hedging that can provide important regulatory capital relief to lending portfolios.

Long-Term and Large Tail Hedgers (Size = Medium): Many investors in this category are looking for a lock-in-the-drawer type long-term hedge that costs little and doesn't require much day-to-day management. This is also a community that is more likely to invest in third-party tail risk funds, which will in turn engage in such activities in the

market. Long-term hedgers do use credit options, but the lack of a well-developed longer maturity market is a deterrent, as is a generally high volatility environment, which has impacted their equity hedges as well. Within credit options, this community will focus on far OTM strikes thereby owning skew. The longer-duration and far OTM nature of tranche hedges are appealing as well, and are in many ways a more natural fit, and activity here has increased.

Yield Generators (Size = Medium): This community of participants are not hedgers and are generally not inclined to spend premium on option or tranche strategies, as they tend to focus on relative return versus benchmarks and their peer groups. Instead, they look for opportunities to monetize excessive market volatility through the sale of options and have effectively translated their capabilities in rates, FX and equities into credit. Flows from this community are important to balance the large community of hedgers, and the market would welcome more yield generators for this reason.

Cross-Asset Hedgers (Size = Small): This growing community of investors is increasingly active in options and to some degree tranches to move between asset classes relatively rapidly, or to simply identify the cheapest hedge or yield idea across equities, credit, rates, currencies, and commodities. They can play an important role in balancing the technicals between markets, and have often engaged in credit vs. equity hedging themes (i.e., selling options in one market to fund the purchase of options in another).

BUDGETING – HOW MUCH TO PAY FOR HEDGES?

Before we delve deeper into the dynamics of a robust credit hedging program, we first highlight the necessity of having a hedging budget, which we see as helpful when narrowing down the range of hedging strategies. Deciding a hedging budget is an iterative process – a function of the portfolio being hedged and its yield, how much of a contribution is desired from the hedges and the current cost of sourcing hedges.

We generally target payout ratios of 3.5x or higher for hedging strategies in normal environments. Thus for every \$1 spent on hedging, we would expect a payoff of about \$3-6 if the hedged scenario materializes. During market dislocations; however, we drop this target payout to 2.0x-2.5x. With IG portfolios yielding between 4-5% today and HY credit yielding 10% today, we believe an IG hedging budget of 0.5%-0.75% of AUM and a HY hedging budget of 1.5-2.5% of AUM should be typical.

Increasing the budget can facilitate the purchase of hedges with unlimited upside, whereas with a more limited budget investors may have to settle for strategies with capped upside. Assuming the hedges are successful and the hedging budget generates 3.0x leverage, that can translate into 2% in hedge P&L for IG portfolio and 4.5-7.5% for HY portfolios. We

emphasize setting guidelines on constructing a hedging budget, and remaining adaptable as the market changes.

THE HEDGING INSTRUMENTS

The three things that drive the choice of instrument for hedging are: liquidity, correlation to investor assets and cost vs. convexity. Inevitably, the choice of hedge instrument involves trading off one of the above for the other. Recently it seems to us that liquidity is the aspect hedgers are least willing to compromise on, with 2008 still a fresh memory for many investors. Thus, indices and products that derive their liquidity from the index (such as index options and index tranches) are the most popular. However, we have also seen the use thematic baskets (baskets of consumer cyclicals, corporates with exposure to peripherals, etc) as well as hedges from other asset classes. Within credit, we divide the universe of potential instruments as follows:

Credit indices (delta-one, no leverage): CDS indices, of which there are over 20 globally, remain the most liquid way to express a portfolio view in credit. Much like using futures in rates and equities markets, credit indices become an easy and liquid way to temporarily “take off” risk without selling actual cash assets, which can be less liquid and harder to “re-source” when an investor wants to put risk back on. The drawback is that the exposure is linear, meaning that the investor can lose money if the market improves, and hedging can be expensive if risk premium is already in the price.

Index options (buying an option on protection): Liquidity in options has been growing steadily over the last 2 years or so, and these are now more liquid than individual CDS. This, along with the asymmetric payoffs, the ability to customize payouts, and the ability to define maximum costs at trade inception has attracted a variety of investors to options for hedging. One thing to note is that these are still very short dated instruments, with expiries typically in the 3m–6m range.

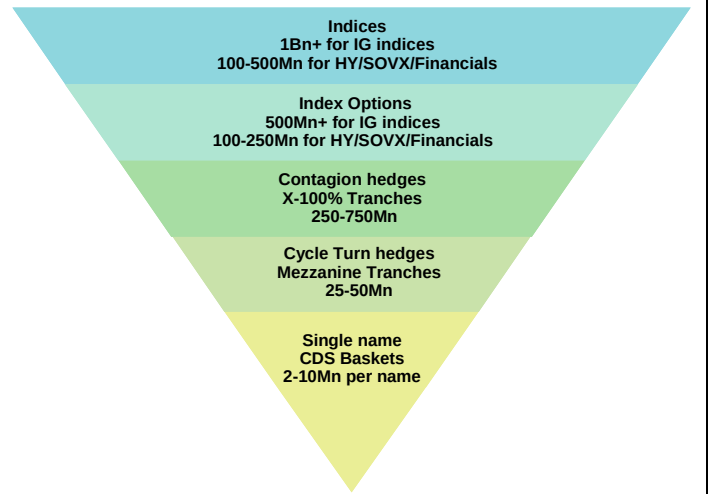
Credit tranches (buying protection on a certain amount of portfolio losses from default): By definition, tranches are designed to express a view on defaults and risk premiums separately. Their long-dated nature makes them ideal for hedging when one is uncertain about time-frame of events. These are also relatively liquid (more so than individual single names, less so than the index), and can vary in cost depending on the strategy. Like the index and unlike options however, these do have more than capped downside if the market improves.

Single-name CDS and baskets (buying customized credit default protection): Individual single name CDS (or a basket thereon) are less liquid than the indices, but more “customized” in that the investor can pick and choose which names to use. We like using this strategy to hedge exposure to highly specific exposures or risks (for example, an

investor could buy protection on core sovereigns or banks to hedge contagion or costs of a rescue of the peripherals).

exhibit 1

Hedger's Liquidity Pyramid – Typical Trade Sizes



Source: Morgan Stanley Research

Given that owning convexity is a key part of the hedging motivation, many investors look beyond the “delta-one” instruments and reach for products with greater convexity – namely tranches and options. These instruments both have their merits and drawbacks, including the impact of time decay, the timing of the scenario hedged, the cost involved, the delta to the underlying index if the hedged scenario does materialize, and how much negative MTM an investor is willing to tolerate should the market actually improve. We breakdown the primary differences between the two strategies as follows:

Options as a hedge: An investor who is long a credit payer or put is long the “option” to buy protection at a pre-determined price (strike). The probability of this option being in the money at expiry is somewhere between 0% and 100%, which is roughly equivalent to the option’s delta. If the option expires in the money, then the investor also owns protection from any defaults that occurred up through the option expiry. Time decay is a big factor in an option hedge, as the non-linear nature of the time decay means an OTM option loses value at an accelerating rate as expiry approaches.

Tranche as a hedge: An investor who is long tranche protection is already short the market with 100% certainty much like simply owning CDX protection or selling equity futures. The tranching defines the payoff for scenarios where there are actual losses from default, and the delta of the tranche estimates how much the tranche’s price will move with respect to small price changes in the underlying index/portfolio. This delta is driven by how sensitive the tranche price is to changes in the market-implied level of

defaults in the underlying portfolio/index. A tranche that is ATM in terms of expected defaults will have the highest delta. The biggest concern for tranches is the ongoing carry involved, unlike an option strategy where the payment is set upfront and occurs only once. In tranches, while the time decay is less severe (unless maturity is less than a year or so), the longer the investor waits for the market widening, the more expensive the hedge.

If we think of an option in the context of a tranche, then the put buyer's best case scenario is being long 0-100% protection at an attractive price (at expiry) relative to the then market price of that protection. There is no additional leverage to the market in the form of default exposure.

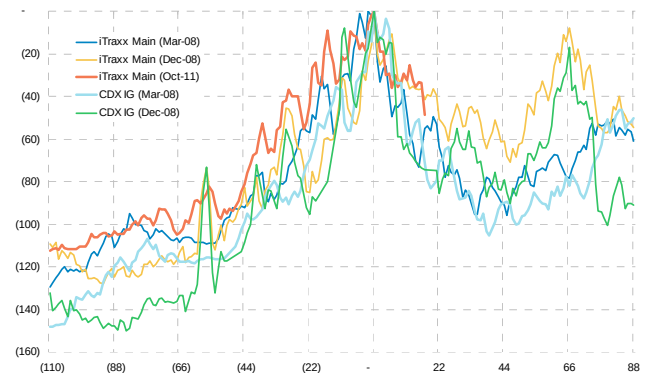
THE FIRST STEP: DEFINING HEDGE SCENARIOS

Before choosing the hedge instrument, the investor must first determine exactly the scenario being hedged (how much credit widening and over what period). We find it useful to look at past credit sell-offs to assess the extent and pace of bear market credit spread deterioration. For the moment, we ignore the basis between cash and CDS. For CDS indices specifically, we have a more limited history than the broader credit market, but the examples we have show roughly the same intensity and trajectory as the underlying credit market for various market declines. On average, a far tail scenario for CDS indices in IG seems to be about 130bp of widening over five months from trough to peak, with the bulk of this in the two months preceding the peak (see Exhibits 2 and 3). The equivalent move for high yield indices is around 400-600bp over a five month period with nearly 300bp of that in the 4-8 weeks before the peak.

Smaller tail scenarios are more frequent and involve 30-65bp move in a 1-3 month timeframe with the bulk of this occurring in the 4-6 weeks prior to the peak. In HY, a small tail is a move of about 150-300bp with a similar timeframe. The easiest way to hedge a small tail is by using options, which would involve buying a rather expensive ATM or close to ATM payer, coupled with a short call or OTM put to cheapen the cost. Hedging for small tail scenarios is more challenging in tranches, as spread impact on individual tranches is less clear in smaller spread widening scenarios, and the cheapest hedges can have little convexity in a moderate sell-off.

exhibit 2

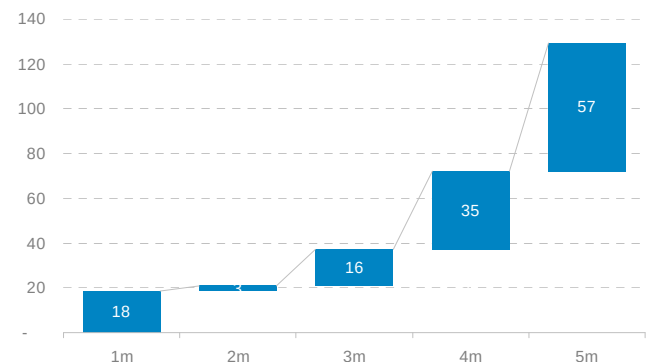
Many Examples of IG Large Tail Moves



Source: Morgan Stanley Research

exhibit 3

An Average Large Tail Widening in IG



Source: Morgan Stanley Research

Another factor to consider when choosing a hedging instrument is timing. Here, tranches have the upper hand, as the maturity is usually considerably beyond the time horizon for the hedged scenario. In options, however, choosing a time horizon is a considerably more involved exercise. For instance, an investor buying deep OTM options to an expiry of 3 months or less may not see much in the way of a payout, particularly for a small move, but even for a large move.

In fact, in many instances the investor may have been better off using a 3m payer spread to hedge the first leg of widening, and switching strategies to a longer expiry bearish risk reversal to hedge the last leg wider to the peak. For investors who believe the spread widening peak is near, covered shorts (buying index protection and sell OTM puts) can be a strategy to consider. Finally, 3m bearish butterflies, with strikes set such that the peak P&L occurs around 30bp wider than current spreads can be another interesting alternative for an investor who wants both limited downside and limited cost, but wants attractive upside in a moderate drift wider.

exhibit 4

iTraxx Main Small Tail Moves

Start	End	Time Months	Spread Move
3/7/2005	5/17/2005	2.37	32
6/4/2007	7/30/2007	1.87	46
10/11/2007	11/21/2007	1.37	35
5/19/2008	7/8/2008	1.67	45
2/9/2009	3/9/2009	0.93	63
1/11/2010	2/8/2010	0.93	29
3/17/2010	6/8/2010	2.77	66
1.70			45

CDX IG Small Tail Moves

Start	End	Time Months	Spread Move
3/8/2005	5/17/2005	2.33	36
6/5/2007	8/3/2007	1.97	47
4/15/2010	6/9/2010	1.83	50
7/4/2011	10/3/2011	3.03	61
2.29			49

Source: Morgan Stanley Research

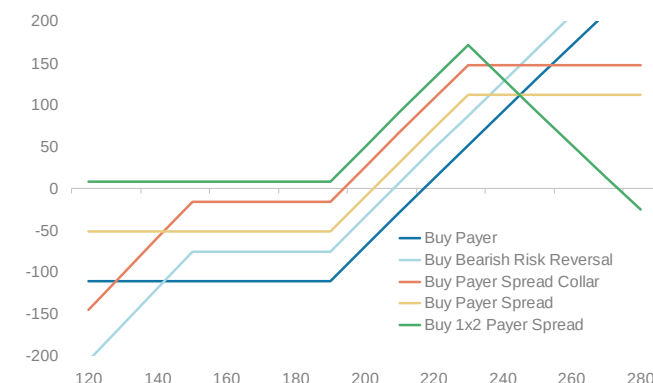
THE SECOND STEP: CHOOSING A STRATEGY

Once the investor has determined what type of scenario is being hedged, the next step is choosing the instrument and strategy. In options, the most common trades are outright payers, payer spreads, payer spread collars, risk reversals, and 1x2 payer spreads, each with its own benefits and risk factors. The simplest strategy is the outright payer, a trade with capped downside and unlimited upside if spreads widen dramatically. The trade-off here is that this strategy is also usually the most expensive, depending on how far out of the money the strike is. The next question therefore is how to cheapen the purchase of a payer outright?

One way to cheapen the cost of an outright payer is by capping upside in a widening (via selling a further OTM payer) to make a payer spread. While the payer spread is cheaper than the payer, it does not have the unlimited upside of a payer. A second strategy is selling away a spread tightening scenario by selling an OTM receiver in a bearish risk reversal. Bearish risk reversals have the advantage of unlimited upside when spreads widen, but also have unlimited downside in a rally. Payer spread collars are the cheapest in a range-bound market, capping the upside if spreads widen but also taking the risk of a significant tightening of spreads.

exhibit 5

The Playbook of Option Hedging Strategies



Source: Morgan Stanley Research

For tranches, the investor can think of hedges as falling into one of two broad categories. First, the investor can put on a hedge that is more systemic in nature, which should protect the underlying portfolio when the market widens rapidly due to systemic risk concerns but not individual corporate issues. Buying protection on tranches that detach at 100% would fall into this category, whether it's the standard super senior tranche, or the more recently popular X-100% tranches which combine the super senior with any number of the tranches beneath it. These all performed well going into the credit crisis, when systemic risk was rising. These trades are generally done outright (no-delta) with an annualized cost of under 100bps, depending on the attachment used and when the trade was implemented (super seniors traded in the single digits prior to the crisis for 5yr risk!). These tranches usually have a delta of less than one, which rises as markets widen.

The second broad category of tranche hedge is cyclical, which should protect the portfolio from more idiosyncratic risk, or actual corporate defaults. These would include buying protection on equity tranches, which can be prohibitively expensive. For that reason, most investors prefer mezzanine tranches, which can widen considerably with a default, given the loss of subordination. Typically these types of hedges are paired with a light delta long on the index to reduce cost.

exhibit 6

Credit Hedging Menu: Characteristics of Options vs. Tranches

	ATM Put	OTM Put	ATM Put Spread	Put Spread Collar*	Bearish Risk Reversal**
Absolute Cost	High	Medium	Medium	Low	Medium
Downside	Limited	Limited	Limited	Unlimited	Unlimited
Upside	Unlimited	Unlimited	Limited	Limited	Unlimited
MTM Impact of..					
Spread Direction (Delta)	High	Medium	Medium	High	High
Magnitude of Spread Change (Gamma)	High	Medium	Medium	Low	High
Implied Volatility (Vega)	High	High	Low	Medium	Low
Volatility Skew	Low	Medium	High	High	High
Time Decay (Theta)	High	Medium	Medium	Low	High
	Equity Tranche	Junior Mezzanine	Senior	Super Senior	X-100
Absolute Cost	High	High	Medium	Low	Medium
Downside	High	Medium	Medium	Low	Low
Upside	Full Writedown	Full Writedown	Medium	Low	Medium
MTM Impact of..					
Increase in Systemic Risk	Low	Low	Medium	High	High
Increase in Tail Risk	High	Medium	Low	Low	Medium
Implied Correlation	High	n/a	n/a	n/a	High
Correlation Skew	n/a	High	High	High	n/a
Time Decay (Theta)	High	High	Medium	Medium	Low

Source: Morgan Stanley Research. Note: *Buy ATM Payer, Sell OTM Payer and OTM Receiver, **Buy OTM Payer, Sell OTM Receiver. n/a implies that the correlation impact on tranche performance depends on other factors.

These trades work best when there is an expectation of actual defaults and the underlying index has some risky tail names.

STEP THREE: MONITORING THE HEDGE (OPTIONS)

Next, we take a look at how some of these hedges and instruments performed in actual tail scenarios, starting with options. What we find is that systematically hedging with options has been cost effective, volatility reducing, and loss protecting, on the whole. In Exhibit 7, we analyze the returns on a portfolio, assuming the investor rolls the option hedges a month before expiry, rather than holding them to expiry. In absolute terms, the cost of implementing the systematic hedge program has been less than 15bp over the course of two years, yet without costing much in performance (over a full up and down cycle), the use of any one of these options hedging strategies helped significantly reduce the volatility of the underlying portfolio by as much as 15% to 60%. Furthermore, the hedged portfolio experienced a similar 15% to 60% reduction in the max drawdown.

We can further parse the results and conclude that for investment grade, "expensive" strategies such as ATM payers look better than OTM payers both in terms of both volatility reduction and drawdown reduction, which seems to suggest that the skew has perhaps been far too steep. However, ATM options experience much higher impact from theta in the 4-6 weeks prior to option expiry, relative to OTM puts, put spreads and put spread collars. In strategies such as put spreads and put spread collars, the value is really monetized in the last few months.

exhibit 7

Systematic Hedging with Options: ATM and OTM Payers Consistently Reduce Volatility

iTraxx Main Jan-08 to Dec-09	P&L Impact	Volatility Reduction	Max Monthly Drawdown Reduction
ATM Payer	-15	-43%	-35%
OTM Payer	-15	-32%	-25%
OTM RR	-18	-46%	-41%
PS	-13	-15%	-6%
PSC	-16	-34%	-24%
1 X 2 payer	-8	-19%	-17%

iTraxx Main Jan-10 to Sep-11	P&L Impact	Volatility Reduction	Max Monthly Drawdown Reduction
ATM Payer	3	-66%	-65%
OTM Payer	-1	-46%	-54%
OTM RR	1	-55%	-59%
PS	-3	-17%	-15%
PSC	-2	-26%	-23%
1 X 2 payer	-1	-27%	-39%

CDX IG Options Jan-10 to Sep-11	P&L Impact	Volatility Reduction	Max Monthly Drawdown Reduction
ATM Payer	-4	-27%	-47%
OTM Payer	-6	-23%	-35%
OTM RR	-8	-21%	-33%
PS	-9	-15%	-12%
PSC	-11	-14%	-14%
1 X 2 payer	-2	-11%	-23%

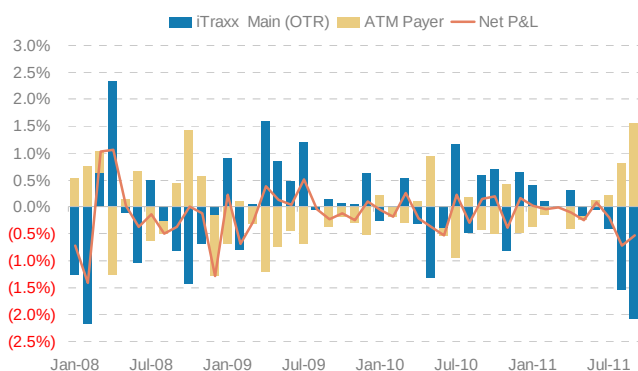
Source: Morgan Stanley Research

Note: i) P&L Impact: shows the average monthly return of systematically using the hedging strategy. E.g. a negative number - 5bp shows that the hedging overlay cost 5bp of P&L per month during the period. Conversely, a positive number implies, the hedging strategy added to the P&L on average over the period. ii) Volatility reduction: shows the % decline in volatility of the hedged portfolio over the unhedged portfolio – e.g., If the unhedged portfolio has an annualized volatility of 10% and the hedged portfolio has an annualized volatility of 5%, that is a 50% reduction in the portfolio vol. iii) Drawdown reduction: shows the % reduction in the performance of worst month of the hedged portfolio over the worst month of the unhedged portfolio – e.g., If the worst month for the unhedged portfolio was March 2009 with a -3% return and the worst month for the hedged portfolio was June 2009 with -1.5% return, the drawdown reduction is 50%.

Option strategies have low deltas (<1), with ATM payers having a higher delta and payer spread collars having a lower delta. Although the delta rises with a widening in spreads, it does not rise above 1. The implication is that a 1 to 1 option strategy cannot fully negate the downside of a long portfolio. Thus even if a strategy is very cost effective and leverage (upside/cost) is very good, it may be necessary to scale up the notional amount. For example, in a month with a major spread decline, an investor would need to buy 2x the notional of a put spread collar with a 25% delta to compare the performance (and cost) against an ATM payer with 50% delta.

exhibit 8

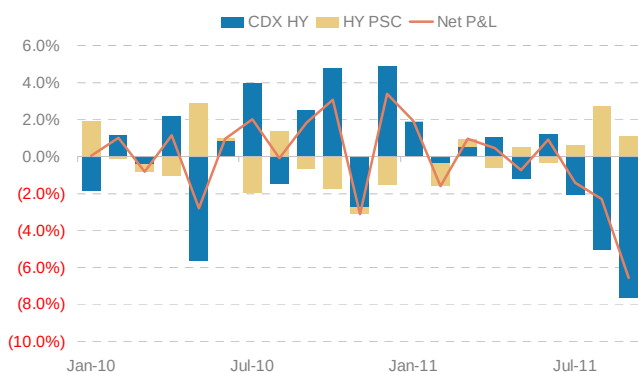
Systematic ATM Payers Strategies in Investment Grade Have Worked Well (Monthly P&L)



Source: Morgan Stanley Research

exhibit 9

Systematic Collared Strategies in US High Yield Have Worked Well (Monthly P&L)



Source: Morgan Stanley Research

We do note a few cautions in hedging with options. First, management of the hedging program and entry point can be important, given the propensity of investors to want to put on the same hedge at the same time. Indeed just earlier this summer, we saw a near inelastic demand for tail hedges, regardless of volatility and skew levels, until the negative headlines globally started to decrease. Furthermore, basis

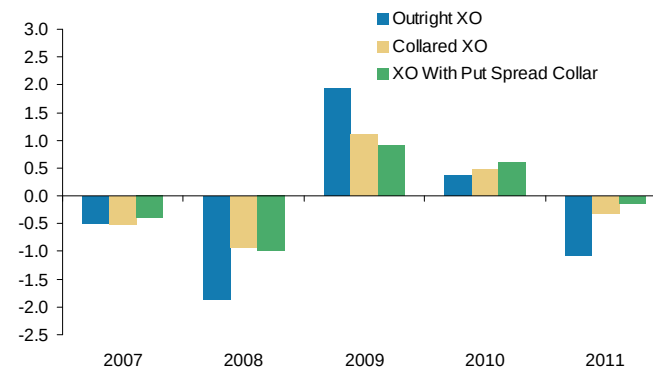
risks between the liquid indices and illiquid cash portfolios can exist, and be exacerbated in times of market dislocations. Additionally, there is a greater risk of a big sell-off compared to small moves, and investors may be less likely to trim positions early on pre-emptively. Finally, implied volatility in credit options generally trades at a premium to realized volatility, which can make hedging expensive by and large.

Turning our attention to high yield, we find that high yield credit behaves differently compared to investment grade, particularly from a volatility perspective. In our view, the HY market has arguably made a transition from being “credit-like” (~5% volatility) in the pre-2007 bull market to more equity-like (10-20%) today, as reinforced by some of the daily spread moves in recent months. This is a key differentiating factor when thinking about optimal hedging strategies from a systematic as well as macro standpoint.

Our analysis shows that for HY credit, unlike IG, strategies that work best from a risk adjusted standpoint are those that either have a short volatility bias or sell away extreme upside. Part of the reason for this divergence is that most of this analysis is based on a post credit crisis period, in which valuations and volatility have been somewhat dislocated. With implied volatility generally trading at a significant premium to realized, ATM payers and OTM payers have not been the best hedges even though they offer unlimited upside.

exhibit 10

Collared Strategies in European High Yield Resulted in Superior Sharpe Ratios



Source: Morgan Stanley Research, Note: Sharpe Ratio = Total Strategy Return/Return Volatility

Put spread collars and risk reversals that do not take an outright long volatility view benefit from a pure cost standpoint. These strategies provide reasonably high delta exposure to moves wider in the index and have generally outperformed over the past two years. From a MTM perspective, risk reversals also benefit from steeper downside skew when spreads widen, while put spread collars benefit from lower volatility and time decay as expiry approaches. Collared strategies are popular as systematic hedges in equity

markets for the same reasons. Given HY market pricing today this congruence in the two markets makes sense to us.

STEP THREE: MONITORING THE HEDGE (TRANCHES)

There are several differences in the economics of hedging using tranches and options, including amount paid upfront, time decay, correlation versus volatility sensitivity, etc. While options are a good medium to express broad index level hedges, tranches can be used to fine tune views to account for default correlation, portfolio dispersion and tail risk. Index tranches are an important liquidity point in the corporate credit landscape in part due to their potential for outsized returns at different points in the cycle, but also because they are among the few “lock-in-the-drawer” longer duration instruments available in the credit markets from a hedging perspective.

exhibit 11

Systematic Hedging with CDX Tranches: Senior and X-100% Strategies Consistently Reduce Volatility

CDX Tranches Jan-06 to Sep-07	P&L Impact	Volatility Reduction	Max Monthly Drawdown Reduction
7-100% (No Delta)	1	-39%	-55%
30-100% (No Delta)	0	-33%	-37%
15-30% (No Delta)	2	-49%	-79%
3-7% (Delta Scaled)	1	-59%	-73%
7-10% (Delta Scaled)	3	-39%	-49%
10-15% (Delta Scaled)	-3	38%	34%
15-30% (Delta Scaled)	9	49%	-70%

CDX Tranches Jan-08 to Dec-09	P&L Impact	Volatility Reduction	Max Monthly Drawdown Reduction
7-100% (No Delta)	-3	-37%	-45%
30-100% (No Delta)	-2	-27%	-41%
15-30% (No Delta)	-8	-33%	-44%
3-7% (Delta Scaled)	12	77%	75%
7-10% (Delta Scaled)	14	27%	21%
10-15% (Delta Scaled)	16	22%	-20%
15-30% (Delta Scaled)	1	-24%	-44%

CDX Tranches Jan-10 to Sep-11	P&L Impact	Volatility Reduction	Max Monthly Drawdown Reduction
7-100% (No Delta)	-1	-39%	-45%
30-100% (No Delta)	-1	-16%	-16%
15-30% (No Delta)	3	-51%	-61%
3-7% (Delta Scaled)	-4	3%	-37%
7-10% (Delta Scaled)	0	-29%	-63%
10-15% (Delta Scaled)	-2	-30%	-51%
15-30% (Delta Scaled)	-1	-45%	-49%

Source: Morgan Stanley Research

Note: i) P&L Impact: shows the average monthly return of systematically using the hedging strategy – e.g., a negative number -5bp shows that the hedging overlay cost 5bp of P&L per month during the period. Conversely, a positive number implies, the hedging strategy added to the P&L on average over the period. ii) Volatility reduction: shows the % decline in volatility of the hedged portfolio over the unhedged portfolio – e.g., If the unhedged portfolio has an annualized volatility of 10% and the hedged portfolio has an annualized volatility of 5%, that is a 50% reduction in the portfolio vol. iii) Drawdown reduction: shows the % reduction in the performance of worst month of the hedged portfolio over the worst month of the unhedged portfolio – e.g., If the worst month for the unhedged portfolio was March 2009 with a -3% return and the worst month for the hedged portfolio was June 2009 with -1.5% return, the drawdown reduction is 50%.

We took a deep dive into drivers of tranche performance in Chapter 10 and isolated tranche performance in varying credit environments, including both systemic and idiosyncratic risk environments. Today, we endeavor to reverse our framework, and look at tranches purely from a

hedging perspective to examine how different parts of the capital structure would have performed as shorts during the last several years.

Using the same framework we used to assess systematic hedging in the options market, we look at tranche hedge performance history. What we find is that for systematic hedges, very senior tranches, as well as some X-100% tranches with mid-level attachment points can be cost effective, portfolio volatility reducing and loss protecting hedges. Intuitively this makes a great deal of sense to us, because these tranches are very much akin to puts (OTM would be 30-100% or 22-100%, ATM would be the lower attaching X-100% tranches) and put spreads (15-30%, 12-22%, etc.). Looking at Exhibits 11 and 12, we can see that across both iTraxx and CDX in 2008 through the present these tranches performed very well.

exhibit 12

Systematic Hedging with iTraxx Tranches – Senior and X-100% Consistently Reduce Volatility

iTraxx Tranches Jan-06 to Sep-07	P&L Impact	Volatility Reduction	Max Monthly Drawdown Reduction
6-100% (No Delta)	1	-32%	-43%
22-100% (No Delta)	0	-22%	-27%
12-22% (No Delta)	2	-59%	-79%
3-6% (Delta Scaled)	-3	-54%	-58%
6-9% (Delta Scaled)	1	-55%	-80%
9-12% (Delta Scaled)	4	-37%	-87%
12-22% (Delta Scaled)	7	19%	-90%

iTraxx Main Jan-08 to Dec-09	P&L Impact	Volatility Reduction	Max Monthly Drawdown Reduction
6-100% (No Delta)	0	-55%	-61%
22-100% (No Delta)	-1	-41%	-50%
12-22% (No Delta)	0	-44%	-41%
3-6% (Delta Scaled)	3	-19%	-35%
6-9% (Delta Scaled)	8	-50%	-64%
9-12% (Delta Scaled)	3	-50%	-72%
12-22% (Delta Scaled)	-2	-36%	-19%

iTraxx Tranches Jan-10 to Sep-11	P&L Impact	Volatility Reduction	Max Monthly Drawdown Reduction
6-100% (No Delta)	10	-57%	-56%
22-100% (No Delta)	4	-34%	-30%
12-22% (No Delta)	25	-50%	-76%
3-6% (Delta Scaled)	17	-60%	-69%
6-9% (Delta Scaled)	17	-63%	-76%
9-12% (Delta Scaled)	18	-69%	-77%
12-22% (Delta Scaled)	18	-61%	-74%

Source: Morgan Stanley Research

Note: i) P&L Impact: shows the average monthly return of systematically using the hedging strategy – e.g., a negative number -5bp shows that the hedging overlay cost 5bp of P&L per month during the period. Conversely, a positive number implies, the hedging strategy added to the P&L on average over the period. ii) Volatility reduction: shows the % decline in volatility of the hedged portfolio over the unhedged portfolio – e.g., If the unhedged portfolio has an annualized volatility of 10% and the hedged portfolio has an annualized volatility of 5%, that is a 50% reduction in the portfolio vol. iii) Drawdown reduction: shows the % reduction in the performance of worst month of the hedged portfolio over the worst month of the unhedged portfolio – e.g., If the worst month for the unhedged portfolio was March 2009 with a -3% return and the worst month for the hedged portfolio was June 2009 with -1.5% return, the drawdown reduction is 50%.

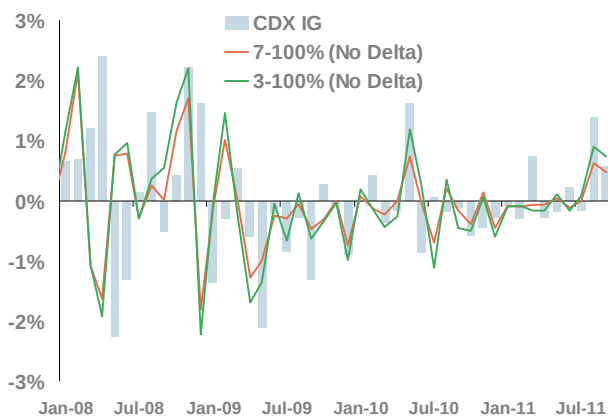
Conversely, the more junior tranches ranged substantially in terms of cost, portfolio volatility reduction and maximum drawdown reduction. For instance, in 2008-2009, the junior mezzanine tranche hedge actually *increased* volatility.. To continue the options analogy, these would be more similar to an ITM to ATM put spread – which explains the cost and portfolio volatility. To us, this signals that these tranches should remain more tactical in nature. Given this history, we classify our tranche hedges into two broad categories, systemic and cyclical.

Systemic: Senior tranche shorts. When hedging for systemic or contagion risk, senior and super senior tranches are ideal, given the cost and convexity profile. These tranches are more “option like” in their return profile, can be thought of as deep OTM puts, but on defaults rather than spreads. On an annualized basis, a super senior tranche usually costs much less than most OTM puts, yet can offer decent payoffs if systemic fears manage to push risk up the capital structure, as we saw in late 2008. Both IG and HY senior and super senior tranches offered attractive payouts relative to their initial annualized cost. If the systemic scenario does not materialize, these usually have limited downside, both in terms of carry and MTM, given that they can at most fall to 0.

One variation on the super senior tranche we’ve seen in the last several months is the “thicker” version, made by combining the super senior with several tranches below it, resulting in 3-100%, 6-100% and 9-100% in iTraxx (7-100% and 10-100% in CDX IG). The appeal of these is that they compare favorably in cost terms to hedging with the index, yet offer decent convexity if and when systemic risk actually does arrive, as they benefit from both rising correlation and higher deltas as spreads widen.

exhibit 13

X-100% Credit Hedging Has Good Performance

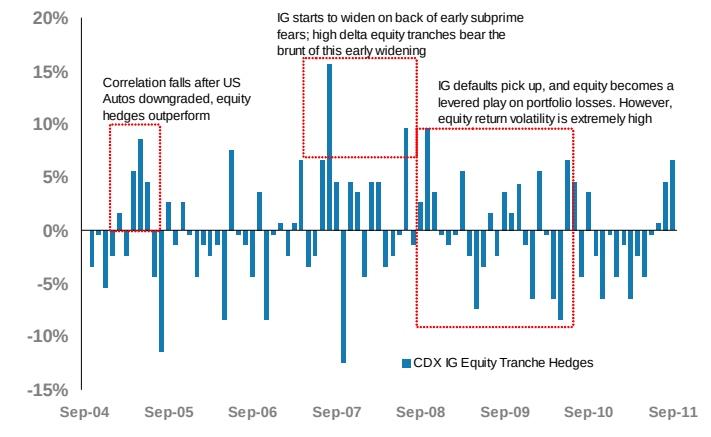


Source: Morgan Stanley Research

Furthermore, as the risk rises up the capital structure out of equity, these tranches capture all (or almost all, depending on attachment point) of this move, unlike the pure super senior, which may only capture the last leg of the sell off. This hedge worked particularly well this past summer, especially given the attractive entry points pre-August. In Exhibit 13 we show the comparison of using the index versus the X-100% tranches for hedges, and given the fact that they traded at a significant discount to the index, performed quite well.

exhibit 14

Equity Tranche Hedge Performance



Source: Morgan Stanley Research

Cyclical or idiosyncratic: Junior tranche shorts. For environments where systemic risk is low, spreads on the whole are generally low, yet there is risk of a few names widening dramatically or even defaulting, junior tranches are appealing tools for hedging. To illustrate, we look at tranche performance in mid-2005, when the US autos were downgraded, causing fears of potential defaults in names from that sector, despite an overall robust credit market.

Given the low spread backdrop and lack of tail names in the portfolio, the equity tranche was not yet saturated with risk prior to the downgrade. Thus risk flowed here first, while more senior tranches remained flat or even tightened (see Exhibit 14). Junior mezzanine tranches too can offer this type of return profile as a hedge if the equity tranche is saturated already, as we see in the series 9 portfolios today, particularly IG9. They key is determining the ATM tranche, and then shorting it on a delta adjusted basis to reduce the cost, since the cost of being wrong for too long can be a drag on eventual returns, even if the hedged scenario materializes. We also like these shorts for turns in the cycle. When it appears that credit may be headed for a downturn, but before defaults happen, the junior mezzanine tranche can be an effective hedge. Earlier this year, we liked this trade, paired with an equity tranche *long* for a delta hedge rather than the index, to capture the convexity of the first move wider and up the capital structure.

Understanding Credit in a Low Yield World

November 4, 2011

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“Yields are low, but spreads are wide, so is credit rich or cheap?” This is a common question in credit markets today, and the answer depends largely on who is asking. The credit investment community comprises both those who make credit investment decisions based on all-in yields and on credit spreads or CDS premiums. And we argue that there are important structural forces in a low yield world that impact credit spreads and can distort the view that spreads represent default risk premium alone.

Today’s low yields in the US are the result of a 30-year bull market for fixed income. Many investors might have preferred a 30-year equity bull market instead, but the reality is that we are now witnessing a regime representing the lowest level of yields since the 1920s in the US. Furthermore, students of the Great Depression have often cited early monetary tightening as the reason for the follow-on recession that occurred in the late 1930s, so we may be in a low yield world until very strong and sustainable growth returns.

In the corporate bond world, we have been used to the notion that a bond’s spread above Treasuries represents its default risk premium, and that this premium can be viewed independently of the actual level of yields. But with yields outside a “normal” range, we cannot ignore the impact that low yields have on spreads.

How investors view and trade credit differs quite substantially as well. Within broad fixed income mandates, credit allocation decisions are often made on an “excess return” basis with duration management done by rates portfolio managers. Insurance companies are an important exception, and often think of corporate bonds in an asset/liability framework where yield is a key measure. Many derivatives and structured products users will focus on CDS, which is more of a pure play on credit without the “yield” component. Many outside of the credit world watch both the CDS indices (pure spread) as well as the popular bond ETFs (pure corporate bonds), which are not perfectly comparable. However, wide bond spreads can drive CDS premiums higher simply in sympathy. In the high yield world, separate mandates are more common (mutual funds and the like), so total yield becomes a more important measure than spread.

In this chapter, we provide our views on how to value credit in a low yield world, focusing both on structural forces and

strategic market opportunities. We summarize below our key themes, and provide detail in the remainder of this chapter.

Low Yields Tend to Coincide with Higher Spreads and Default Rates: While low yields are often associated with slow growth, thereby justifying wider credit spreads, other factors can keep spreads wide, including the trouble companies have in inflating away their nominal debt (higher default risk). History tells us that if growth eventually picks up while yields stay low (1930s-1950s), spreads can indeed normalize.

Improving Duration Management: Treasuries can be an imperfect interest rate hedge for corporate bonds in a low yield world. We demonstrate that systematically overlaying traditional Treasury hedges with OTM rate receivers can improve hedging efficiency.

Penalizing Premium-Priced Bonds: Our fair-value models indicate typical premium-priced investment grade bonds require 10-70 bp of additional spread to compensate for above-par prices. We find little consistency in actual pricing of long-dated corporate bonds relative to this fair-value metric, and suggest investors consider this framework.

High Yield Hurt More than Investment Grade: Owing to several factors, including callability, economic sensitivity, duration and default risk, we find that high yield tends to perform worse than investment grade in falling rate and low yield environments.

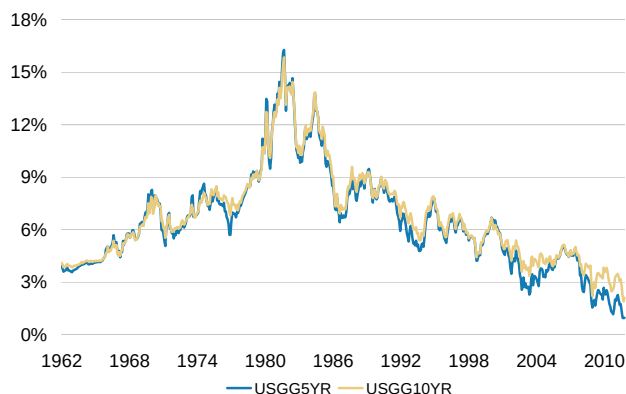
CDS as a Pure Play on Default Risk: CDS premiums should avoid the yield-induced spread issues with corporate bonds, but CDS can move wider in sympathy. With adequate compensation for worst-case default risk, we favor selling protection on names and structured portfolios to capitalize on this market dynamic.

Insurance Companies and Replication: We recommend that insurance companies consider replication strategies with CDS overlays to diversify corporate credit risk at attractive prices. We also feel that senior and thick tranches in the structured credit market can be good alternatives to single-names and indices.

Dividend Yield vs. Corporate Bond Yield: With corporate bond and dividend yields narrowing, we highlight a screen of stocks where our equity strategy team sees stable 3%-6% dividend yields that are higher than corporate bond yields. Investors can also consider collaring such strategies to create more bond-like payoff profiles.

exhibit 1

30-Year Bull Market for Fixed Income



Source: Morgan Stanley, Bloomberg

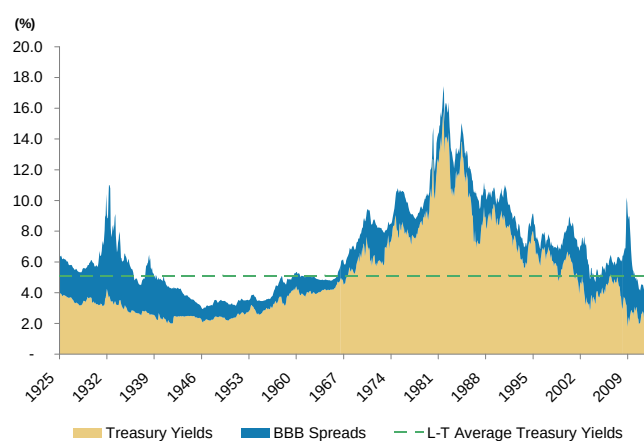
LOW YIELDS TEND TO COINCIDE WITH WIDER SPREADS AND HIGHER DEFAULT RATES

Going back to 1925, we can make two general observations about the relationship between spreads/default rates and the level of yields. In a low and falling rate environment, spreads tend to be wider and default rates higher on average than in periods when yields are high and rates are rising. Yet details matter, and we explore those below.

Granted, there are not many periods since 1925 when Treasury yields were as low as they have been over the last two years, and only a handful when they were as low as they are today, making meaningful comparisons difficult. For example, the only period when Treasury yields stayed below 2.5% was between 1939 and 1951. During that time, BBB credit spreads rallied broadly, tightening from around 250bp in 1939 to around 75bp in 1951, as the U.S. rebounded from the Great Depression, entered World War II and reached new heights of economic prosperity by the early 1950s. In the 1930s, when spreads were wide, it is worth noting that while Treasury yields were low relative to those observed from 1960 onwards, they were not necessarily low by historical standards, when considered in the context of the rate environment prior to 1925.

exhibit 2

Historically, Wider Spreads Correlated with Lower Yields



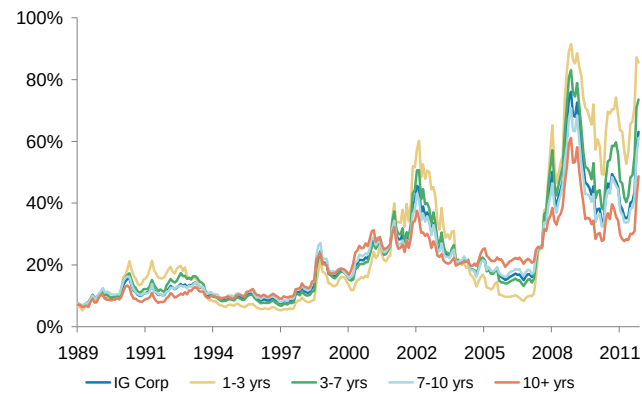
Source: Morgan Stanley, Moody's, the Yield Book, Bloomberg

But the key focus is on the tails, as peaks in spreads tend to occur in low yield environments, including those with high inflation, while the tights tend to be in higher yield environments. That said, spreads certainly can spike when yields are high, as we witnessed during the choppy economic environment of the 70s and early 80s. However, even then, the spike in spreads usually came as risk-free rates were at least temporarily falling. A key reason for this spread/yield relationship is fairly straightforward. Low yields are often associated with slow growth or even recessions, so wider credit spreads can certainly make sense in that environment. However, again, this is not always the case. For example, yields remained low for much of the 1940s and 1950s, when they were not accompanied by slow growth, and spreads steadily tightened over that period. However, that was after many years of low yields. So while we have only one period of extended low rates (albeit a very long one), history tells us that spreads can tighten in a low yield world, but it can take years as markets and corporations adjust.

Spreads also become more important when yields are low, as they represent a much larger percentage of the overall yield, especially in the front end of the credit market, as we show in Exhibit 3. In part, this explains why if yields stay low for a long-enough period of time (for example the story in Japan over the last decade), the demand for credit can increase, as the incremental yield pickup moving from risk-free rates to credit becomes attractive in percentage terms. Today, spreads make up nearly the largest percentage of overall yields looking across the last twenty years.

exhibit 3

Spread Percentage Contribution to Yield Rises in Low Yield Environments

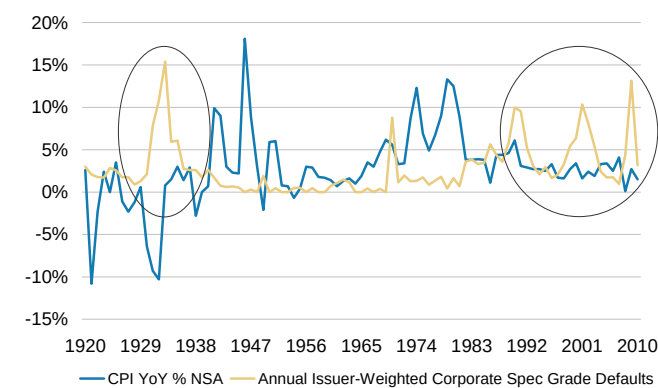


Source: Morgan Stanley, the Yield Book

While one could argue that default rates could be high during times of higher yields owing to higher debt service cost, the opposite is actually true. High inflationary environments allow corporations to inflate away their nominal debt as their assets (and revenues) grow with inflation, leading to lower default rates. Low inflation environments, like the one we've had for the past 25 years, tend to be ones where defaults can spike.

exhibit 4

Defaults Rates Are Often Higher in Low Inflation Environments



Source: Morgan Stanley, Moody's, the Yield Book, Bloomberg

DURATION MANAGEMENT AND FOCUSING ON THE TAILS

While managing duration/interest rate risk by hedging with Treasuries is a key part of credit portfolio management for many market participants, investors quite often bear the brunt of being short/underweight Treasuries in times when rates are headed lower if spreads do not follow suit. US Treasuries have long been the safe haven asset class across the globe and periods of risk aversion tend to see a flight to quality

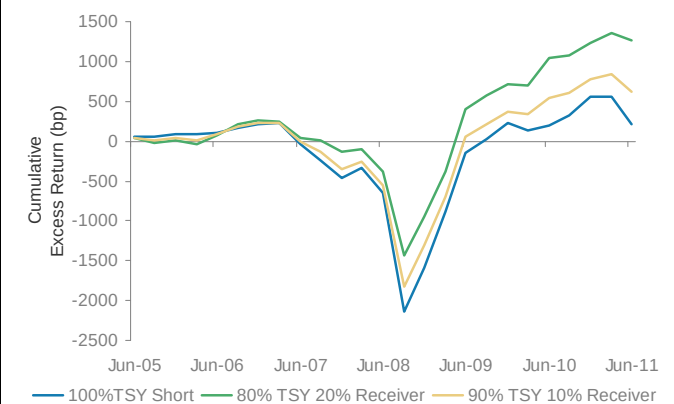
towards these bonds, which often leads to a sharp move lower in rates (as opposed to a grind lower).

Most corporate credit investors and traders tend to get frustrated with Treasury hedges when rates move sharply lower. Treasury hedges are clearly imperfect given what we have described in the previous section respect to spread and yield relationships. We conducted a study of a systematic strategy of overlaying traditional Treasury hedges with out-of-the-money (OTM) receivers (think call options on lower yields) as a way of monetizing a big move lower in rates and found that they worked better than full notional Treasury-hedged approach.

Our analysis shows that there is significant value in owning the "tail risk" of lower rates by buying OTM receivers at the expense of some of these long duration hedges (90%-80% notional of Treasury hedges with 10%-20% of OTM receivers). This does create a duration mismatch between the risky portfolio and the risk-free hedge, which understandably does not perform as well when rates move higher. Nevertheless, given the tendency of rates to "gap" lower rather than grind, we find that owning this tail could have actually proved quite valuable for an investor looking to maximize excess return.

exhibit 5

Buying Rate Receivers Systematically Would Have Led to Better Excess Returns



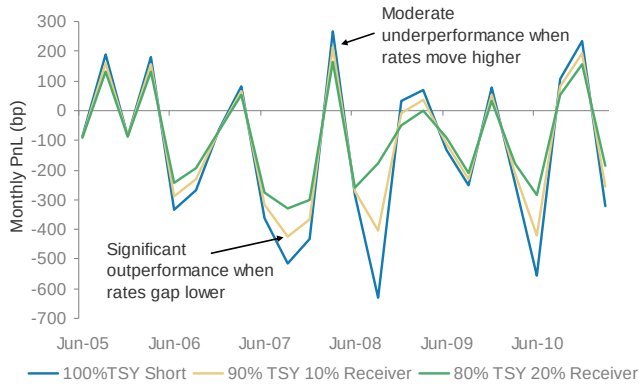
Source: Morgan Stanley Research

Our framework assumes that an investor reduces a certain notional of his or her Treasury hedge to invest in an equally weighted basket of 5Y and 10Y 25 bp OTM rate receivers (thus roughly mirroring the duration of the corporate index). Exhibit 5 shows using such an approach systematically (using quarterly rebalancing) would have resulted in superior excess returns since 2005. In months where spreads widened dramatically, the OTM receiver would have mitigated significant negative MTM at the same time costing very little in months where rates move higher. In effect, buying receivers monetizes the convexity of rate moves by buying relatively cheap optionality on lower rates (as well as long

rates volatility exposure) at the expense of full duration hedging.

exhibit 6

Overlaying Treasury Hedges with Rate Receivers Benefits from Convexity

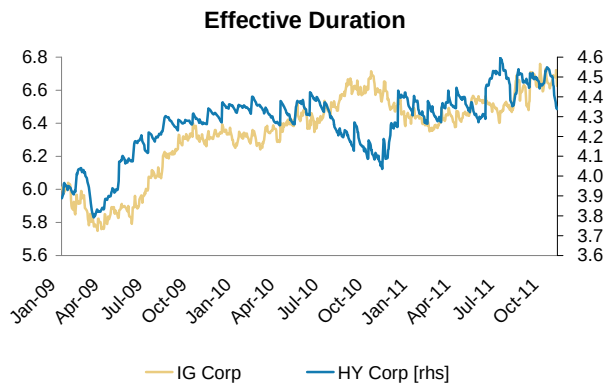


Source: Morgan Stanley

Furthermore, the importance of managing interest rate risk in corporate bond portfolios only increases in a low yield world. Among the consequences of lower overall yields is higher sensitivity to a change in yields. The effective duration of the IG corporate index today is about 6.6 years, an increase of almost 1.2 years since the onset of the credit crisis. Given today's market value of about \$3.1 trillion for the US investment grade corporate bond market (based on IG Corp cash index values), a 100bp move in yields would result in a loss in market value of approximately \$200 billion.

exhibit 7

Lower Yields Lead to Higher Rates Sensitivity



Source: Morgan Stanley, Moody's, the Yield Book

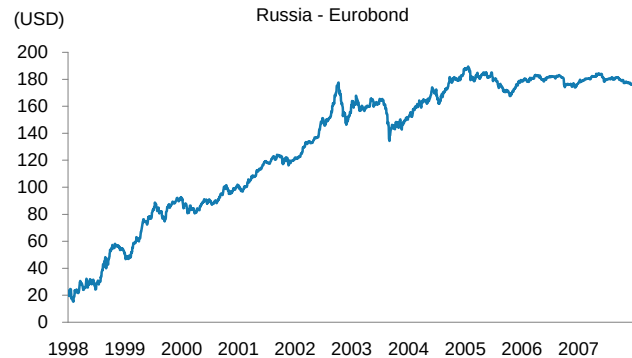
PENALIZING PREMIUM-PRICED BONDS – WHAT IS IT WORTH?

Corporate credit markets are always concerned about paying too much in dollar price terms for a bond, given the par claim at bankruptcy. When markets move away from credit to other fixed income risk, dollar price anxiety can disappear, with

Russia's famous 30-year Eurobond issue from 1998 being perhaps the best example (it trades in the \$180 context today). However, we do not typically see such a lack of focus on dollar prices within the corporate world given the potential for more specific idiosyncratic risk.

exhibit 8

A Dollar Price Story: Russia 12.75 30-Year USD Eurobond



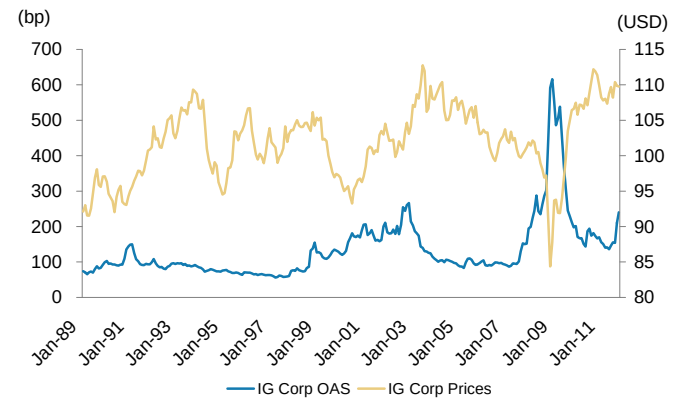
Source: Morgan Stanley, Bloomberg

How bad is the dollar price problem in US investment grade credit?

The average dollar price for the investment grade corporate index is above \$110, near its recent peak and well above most periods since 1989 (in 2003 the average price was at similar levels for a brief period). The 10+ year maturity bucket is close to an average \$120 dollar price.

exhibit 9

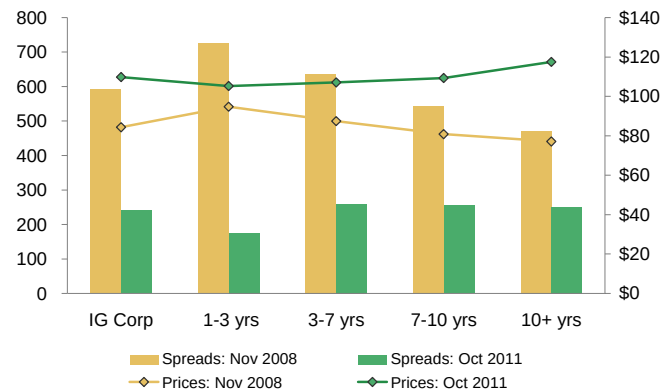
IG Dollar Prices Near 20-Year Highs



Source: Morgan Stanley, the Yield Book

exhibit 10

IG Dollar Prices in 105-120 Range on Average

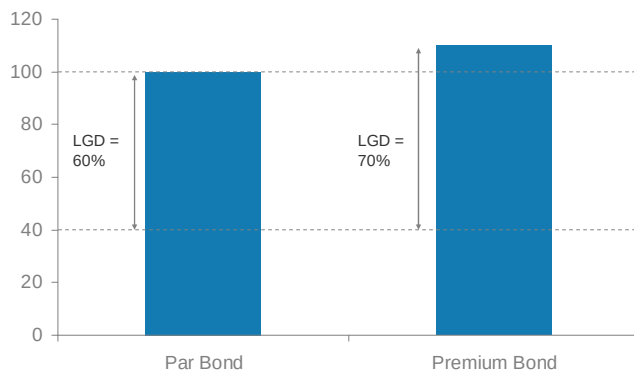


Source: Morgan Stanley, the Yield Book

How much should premium bonds be penalized in today's markets? We quantify this penalty by estimating the additional cost of protecting the “premium” part of the bond above cost of protecting par. We then express this additional cost as an annualized spread by accounting for the duration of the bond. Before discussing our approach in detail, there are three underlying factors that clearly affect this calculation.

exhibit 11

Par vs. Premium Bonds: Same Probability of Default, Different Loss Given Default



Source: Morgan Stanley Research, LGD = Loss Given Default
Assuming Recovery = 40%

Amount of premium above par: Bonds with a higher dollar price face a higher loss in the event of default and investors need to be compensated accordingly.

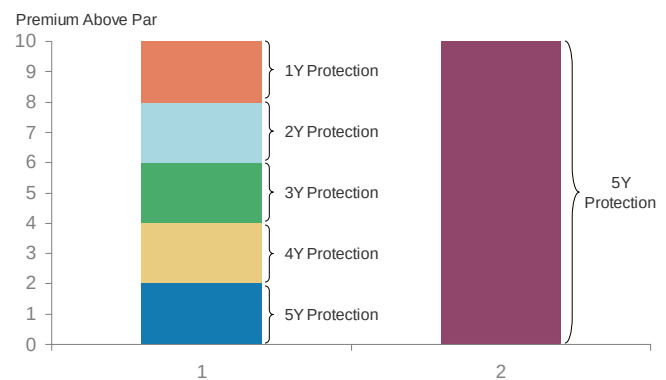
Probability of default / CDS spread of the bond: We take a fair-value approach by assuming market-implied probability of default through CDS curves for an issuer. As such, in the case of bonds of different issuers with the same dollar price, the one with the wider CDS spread needs to compensate more because of the higher probability of default/loss of

principal. Another way to interpret this is that the cost of protecting the premium portion of such a bond is higher.

Recovery assumption: In most cases of corporate defaults/credit events, all bonds have an equal claim on the company's residual assets and hence trade at the same dollar price (i.e., recovery) post default, irrespective of maturity and coupon. In theory, to remain perfectly default hedged, an investor needs to buy “excess” protection (scaled up by $1/(1-R)$) on the premium portion of the bond to account for the fact that recovery on this part of the principal is zero. The recovery assumption should determine the scaling factor.

exhibit 12

Compensating for Premium-Priced Bonds



Source: Morgan Stanley

To illustrate the calculation of this incremental compensation, assume that a company with a CDS spread of 150 bp has a 5-year bond trading at 110 and another at par. In order to remain perfectly default-hedged, an investor owning the premium bond will buy additional protection on the 10 pts above par.

We highlight two approaches that can be used. The first is to buy 5Y protection on the entire “premium” portion scaled by a factor $(1/(1-40\%))$ to account for zero recovery in case of default. The second approach is to estimate forward bond prices based on the risk-free curve assuming constant yield and buying piecewise maturity-matched protection to account for the natural decay in bond price to par (see Exhibit 12). For example, if the bond price is expected to decay from 110 to 108 in one year, an investor needs to buy just one year's protection on these 2 pts (as opposed to five years' in the previous case). This approach takes into account the actual credit curve shape of the reference entity of the CDS as well.

We acknowledge that there are some shortcomings in both these approaches such as the constant forward yield assumption and no room for change in rates over time. However, we do get an interesting starting point to estimate the fair value of protection. Bottom line, based on these two

approaches, we believe that for every 10 pts of premium above par, a premium bond should offer a spread pick up of 15-20 bp relative to a maturity-matched par bond of the same issuer. Exhibit 13 illustrates indicative levels for this premium for bonds in different maturity buckets, which can range from 7-10 bp in the front end to 65-70 bp in the long end for a relatively healthy investment grade credit.

exhibit 13

Required Additional Spread for Premium-Priced Bonds

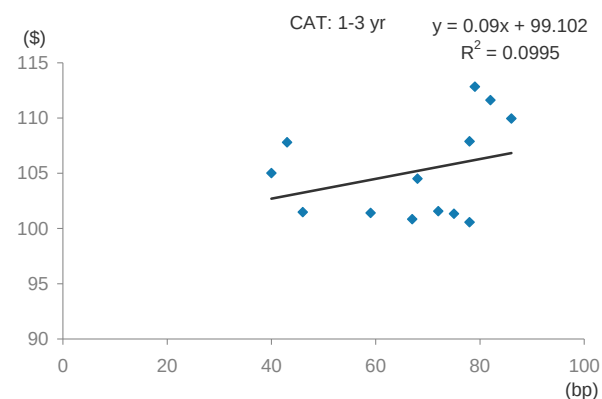
Maturity (yrs)	Average Bond Price	CDS	Compensation (bp per year)
3	105	150	7 - 10bp
5	110	150	15 - 20 bp
7	115	150	25 - 30bp
10	120	150	35 - 40bp
30	130	150	65 - 75bp

Source: Morgan Stanley

Does the market “price-in” this premium-priced penalty? The short answer is not consistently. We show in Exhibits 14-15 dollar prices vs. spreads for two industrial issuers where this is no consistent spread premium for higher dollar price bonds, and we find that in general this is true across many issuers in the investment grade corporate universe. Clearly issues of liquidity of older (and presumably higher coupon) bonds compared to more recent issuance is a driving factor as well, but nevertheless, we believe there are opportunities in the market to identify relative value among high dollar price issues, and we suggest the above approach to find the fair compensation for premium bonds.

exhibit 14

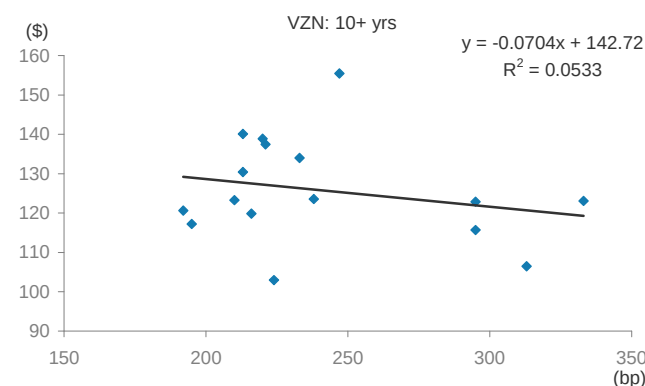
Front-End Example: Spread vs. Price Relationship Is Not Consistent



Source: Morgan Stanley, the Yield Book

exhibit 15

Back-End Example: Spread vs. Price Relationship Is Also Not Consistent



Source: Morgan Stanley, the Yield Book

HIGH YIELD VS. INVESTMENT GRADE IN A LOW YIELD WORLD

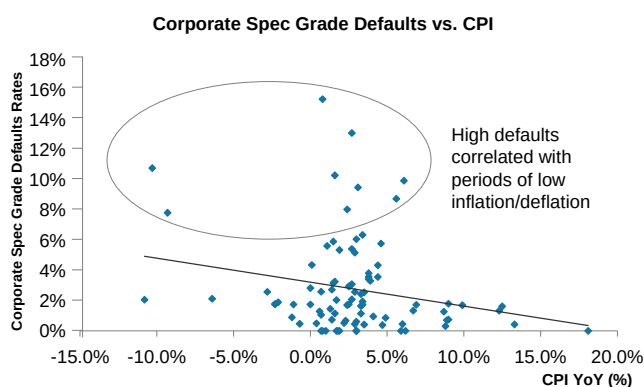
The level and change in Treasury yields should have a very different effect on high yield vs. investment grade credit. Putting aside the fact that HY investors are typically much more focused on total returns compared to IG investors who often care about excess returns, we think there are four key reasons why low and falling risk-free rates are more harmful for HY than IG spreads.

- 1) **Callability:** About 70% of the HY market is made up of callable bonds versus a very small percent of the IG market. When prices are low, this is a non-issue, but earlier this year with the HY index trading at \$104, near the average call price, the market becomes very negatively convex. At these high prices, as rates drop, the potential for spread tightening also drops as bonds get priced to shorter call dates.
- 2) **Economic Sensitivity:** High yield companies are generally much more sensitive to changes in economic growth expectations than investment grade corporates. Usually when rates are dropping (at least over the past few years), it is due to, and/or correlated with, weak economic data. As a result, when rates are dropping, excess returns for HY credit typically fall more than for IG, and the opposite is true when rates are rising.
- 3) **Duration:** HY bonds trade at a duration of 4.4 versus 6.8 for the average IG bond. When rates are falling (or expected to fall), IG bonds will experience greater price appreciation (due to the rate move), and hence will be more attractive to total return investors. Also, often when rates are falling, the Treasury curve is steep, as the Fed is cutting. A steep Treasury curve benefits longer duration credit (investment grade), all else equal, given a bigger yield pickup from moving out the curve.

- 4) **Default Risk:** Given defaults are a much bigger problem for high yield investors, high dollar prices (and hence low yields) should require a bigger spread premium in the high yield market than in investment grade. The other side of the argument is that high yields are a burden on companies, given greater interest costs, which is clearly a bigger issue in high yield. That said, as mentioned earlier, we have usually found that periods of deflation (low rates) are much more likely to coincide with high defaults than periods of inflation (high rates), as it is much easier to service a fixed debt burden when pricing power, the value of assets, and earnings are rising.

exhibit 16

High Defaults Rates Correlated with Low Inflation/Deflationary Environments



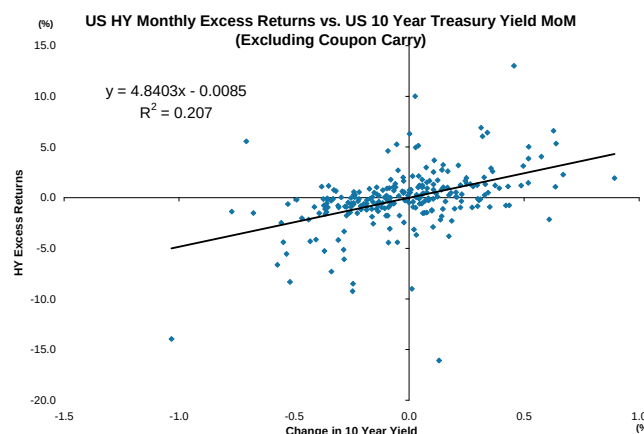
Source: Morgan Stanley, Moody's, the Yield Book, Bloomberg

In the following exhibits we plot monthly excess returns (excluding the carry – so the price return due to the change in spread), given various changes in the 10-yr Treasury yield. Our data points date back to 1990.

First, as we show in Exhibit 17, as one would expect, falling Treasury yields are historically correlated with positive excess returns for high yield, though the relationship is not a very strong one.

exhibit 17

Falling Rates Historically Correlated with Wider Spreads in HY...

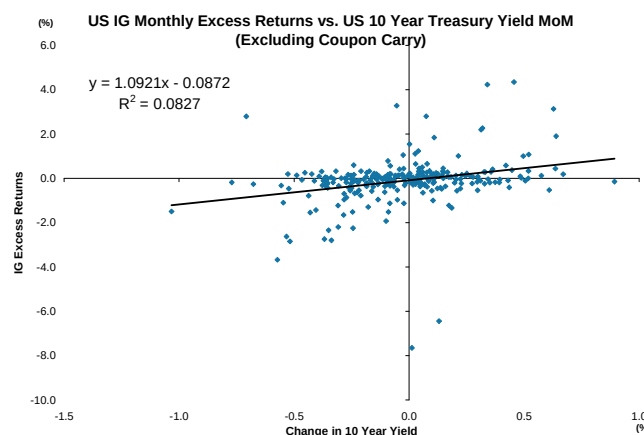


Source: Morgan Stanley, the Yield Book, Bloomberg

For investment grade, the story is the same, as Treasury yields drop, credit spreads widen, though for the reasons mentioned above, the relationship is an even weaker one than for high yield.

exhibit 18

...We Find a Similar but Weaker Relationship in IG

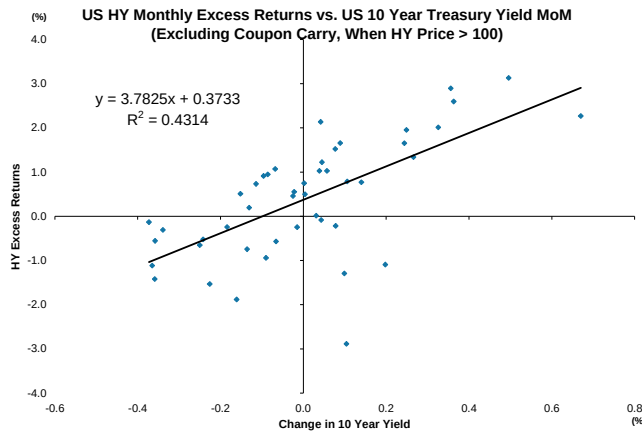


Source: Morgan Stanley, the Yield Book, Bloomberg

However, if we only look at periods when the high yield index is above par, the correlation between the change in Treasury yield and excess return increases. This makes sense to us, as at high dollar prices, high yield is very negatively convex (as loss given default rises), and returns become very asymmetric (due to default risk). As a result, falling yields become more problematic for high yield at high dollar prices than they do at low prices.

exhibit 19

Falling Rates a Bigger Problem for HY Spreads When Prices Are High...

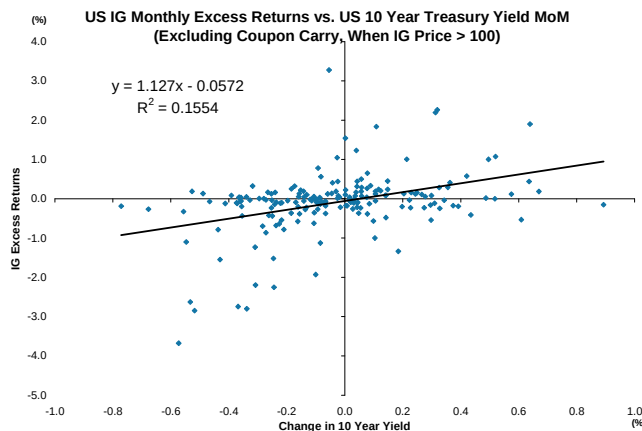


Source: Morgan Stanley, the Yield Book, Bloomberg

For IG, even at high dollar prices, the relationship between falling yields and falling excess returns is a weak one. Once again, this makes sense given default risk and callability are much smaller issues for IG investors, and hence high dollar prices are not as big a problem.

exhibit 20

... Even When Prices Are High the Spread/Yield Relationship Is Weak in IG



Source: Morgan Stanley, the Yield Book, Bloomberg

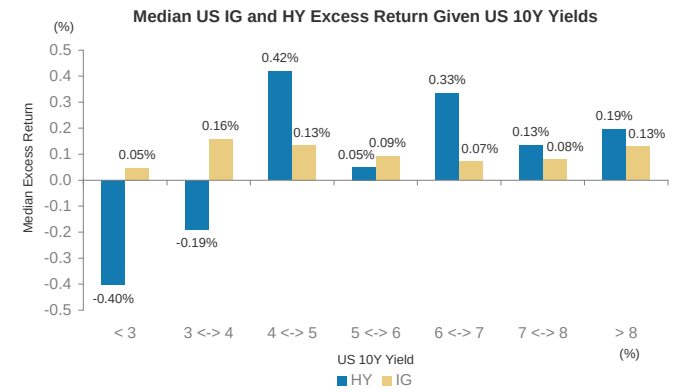
Finally, most of the analysis above shows excess returns given various *changes* in yields. In Exhibit 21 we show excess returns given the *absolute level* of Treasury yields. What we find is that for high yield, low yields are bad for excess returns, as expected. But very high yields are also not ideal. This is likely due to the fact that in high rate environments, not only do interest costs become a burden for companies, but the incremental yield pickup in percent terms also gets unattractive (a 500bp spread when risk-free rates are 10% is 33% of the all-in yield vs 50% when risk-free

rates are 5%). All else equal, 10-yr yields in the 4-5% range have historically been ideal for high yield.

For IG, we find less variation in returns given various levels of rates, though a similar overall trend. Very low Treasury yields are the worst for IG, high yields are fine though 3-5% Treasury yields have historically been ideal.

exhibit 21

Risk-Free Rates That Are 'Not Too High' and 'Not Too Low' Are Historically Ideal



Source: Morgan Stanley, the Yield Book, Bloomberg

CDS AS A PURE PLAY ON DEFAULT RISK

If corporate bond spreads are wide owing to the low yield issues discussed in the previous sections, then all else being equal CDS premiums should be much lower and the basis far more negative. The reason is that investors should not demand the additional compensation in CDS premiums (given that it is unfunded) as they would in corporate bonds given yield boogies. Simply put, if we find that the basis (CDS premium minus bond asset swap) is not negative enough during low yield environments, selling protection is a good alternative to buying bonds for those who have a choice. And for those who still require fixed rate exposures, CDS can be used in replication strategies (in insurance company parlance; see next section) or credit-linked notes with fixed rate collateral.

How much compensation does one require in CDS relative to corporate bonds? Our answer is the same as in the previous section on determining fair compensation for premium-priced bonds. Our basis model performs these adjustments.

Often CDS and structured credit users are less concerned about interest rate risk and the issues of corporate bonds, focusing more specifically on compensation for default risk. So if low yields are pushing spreads wider, investors need to make sure that they are being adequately compensated for the additional default risk we have seen historically in low rate environments. Today's wide spreads give us the luxury of focusing on worst-case defaults, and in Exhibit 22 we show

that spread compensation today is more than adequate for this risk based on the CDX IG index.

exhibit 22

Corporate CDS Provides More than Adequate Compensation for Default Risk

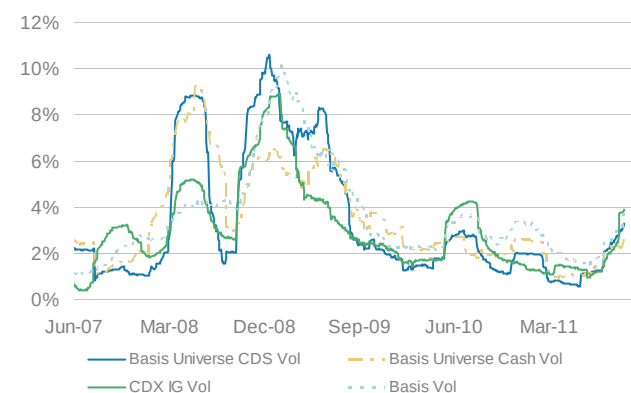


Source: Morgan Stanley, Moody's

One risk when corporate bond spreads and CDS premiums are driven by different factors is increased “spread” volatility. A negative basis in theory argues in favor of owning cash bonds over synthetic credit from a pure valuation standpoint. However, given differing technicals associated with the cash and CDS markets, they do not always move in tandem making the basis a volatile relationship. We looked at this relationship in detail in Chapter 19 and found that the basis is actually quite volatile for the yield it offers today. Indeed, in a relatively benign default environment with a narrow negative basis, we argued in favor of selling protection outright in CDS form as a better risk-adjusted bet. While credit volatility has picked up significantly, the basis has remained choppy as well, as seen in Exhibit 23. As such, we still advocate selling protection outright to monetize excessive default premiums embedded in credit spreads today.

exhibit 23

Credit Is More Volatile but So Is the Basis



Source: Morgan Stanley Research

INSURANCE COMPANIES – CONSIDER REPLICATION

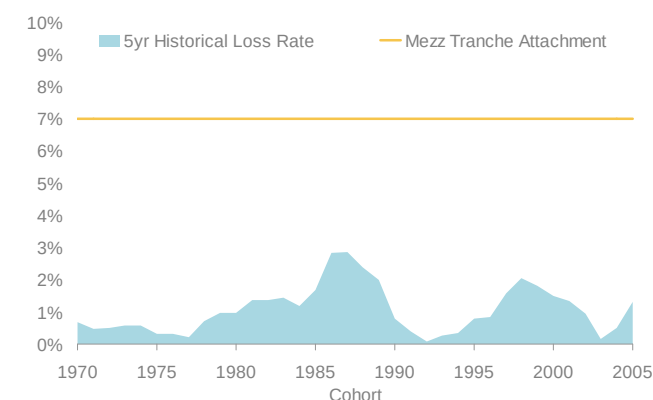
Wide levels of CDS premiums relative to required compensation for default risk can be captured by many investors through the sale of protection in various structures. For insurance companies, we believe that replication strategies, where CDS can act as a credit overlay to a fixed rate Treasury (or other high quality fixed income) portfolio, become an additional source of credit risk exposure that is attractively priced today.

While single-name and index CDS exposures can be adequate, we find that higher quality tranches in the structured credit market can provide even better additional compensation for default losses and with relatively good capital treatment.

As an example, if we focus on a 7% attachment point for a CDX IG portfolio, we find that such a tranche has more than double the necessary subordination to protect from worst-case default losses over five years (see Exhibit 24). The issue with such strategies in the past has been the negative convexity of thin tranches, but we now find that significantly thicker tranches (for example 7-15%) can still provide good compensation (275bp, which is more than double the 127bp spread of the CDX IG index) with much less convexity issues.

exhibit 24

7-15% Tranches: Yield Pickup with Subordination



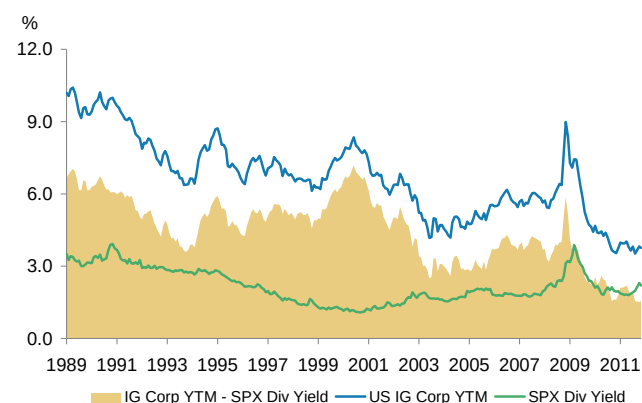
Source: Morgan Stanley Research, Moody's

WHEN ARE DIVIDEND YIELDS MORE ATTRACTIVE THAN CORPORATE BOND YIELDS?

The differential between high grade corporate bond yields and the dividend yield on the S&P 500 is now as low as it has been since our data series started in 1989, as we highlight in Exhibit 25. While the dividend yield remains low by historical standards, corporate bond yields reaching the lows not seen since the 1950s has caused the differential between the two to compress to the lowest level in several decades.

exhibit 25

Compressing Differential between Bond Yields and Dividend Yields



Source: Morgan Stanley, Bloomberg, the Yield Book

As our US equity strategy team recently noted, the equity market dividend yield, currently at about 2.2%, is well below its long-term average of 2.8% (based on Top 500 dividend yields since 1964), despite US corporates reporting decent earnings and holding nearly \$1.5 trillion in cash on their balance sheets. The dividend contribution to the S&P 500 total return is also at very low levels, around 13.4% in 2010 compared to over 40% since 1930. However, our equity strategists believe dividend yields will rise over the next year or two, driven by increased payout ratios and factors outlined in their recent publication¹. The team also highlights a list of large-cap stock ideas of companies with 3% to 6% dividend yields that appear moderate, or sustainable (i.e., companies with evidence of ample cash and dividend payout ratios of less than 50%), in a strategy they believe will outperform.

¹ "US Equity Strategy: Dividend Yield Will Rise: One Way or Another", Adam Parker et al, September 18, 2011

exhibit 26

Large Cap Stocks with Dividend Yield Between 3% and 6% that Appears Sustainable

Ticker	Company	Sector	Median Bond YTM	Dividend Yield	YTM - Dividend Yield
CPB	Campbell Soup Co	Consumer Staples	1.47%	3.53%	-2.06%
PG	Procter & Gamble Co/The	Consumer Staples	1.44%	3.35%	-1.91%
GD	General Dynamics Corp	Industrials	1.37%	3.03%	-1.66%
CVX	Chevron Corp	Energy	1.49%	3.10%	-1.61%
DPS	Dr Pepper Snapple Group Inc	Consumer Staples	1.98%	3.44%	-1.46%
LMT	Lockheed Martin Corp	Industrials	4.10%	5.36%	-1.26%
ETR	Entergy Corp	Utilities	3.72%	4.81%	-1.08%
RTN	Raytheon Co	Industrials	3.12%	3.97%	-0.85%
K	Kellogg Co	Consumer Staples	2.40%	3.20%	-0.80%
INTC	Intel Corp	Information Technology	3.08%	3.52%	-0.44%
DRI	Darden Restaurants Inc	Consumer Discretionary	3.24%	3.62%	-0.38%
BLK	BlackRock Inc	Financials	3.34%	3.60%	-0.26%
COP	ConocoPhillips	Energy	3.62%	3.81%	-0.18%
NOC	Northrop Grumman Corp	Industrials	3.54%	3.59%	-0.06%
SRE	Sempra Energy	Utilities	3.62%	3.61%	0.02%
NTRS	Northern Trust Corp	Financials	3.17%	2.83%	0.34%
HAS	Hasbro Inc	Consumer Discretionary	3.54%	3.15%	0.40%
SWY	Safeway Inc	Consumer Staples	3.53%	3.03%	0.50%
TRV	Travelers Cos Inc/The	Financials	3.42%	2.86%	0.56%
CAG	ConAgra Foods Inc	Consumer Staples	4.67%	3.81%	0.86%
TWX	Time Warner Inc	Consumer Discretionary	4.52%	2.82%	1.70%
AFL	Aflac Inc	Financials	5.96%	2.98%	2.98%

Source: Morgan Stanley, Moody's, the Yield Book, Bloomberg

As we recently noted², with companies acknowledging a slower organic growth environment post-crisis, and equity investors expecting to be compensated increasingly through income, corporates will be motivated to return capital to shareholders and bolster equity returns through some form of financial leverage and other tweaks favoring equity investors over bondholders. Finally, for credit investors who are looking for a more bond-like payoff profile in holding high dividend-yielding stocks, we suggest collaring stocks (buying OTM puts or put-spreads and selling OTM calls), which can add additional yield in most equity volatility regimes (see [Turning a Stock into a Bond](#), October 14, 2009).

² "Credit Basis Report: The Unintended Consequences of Low Yields on Credit", Riz Hussain et al, October 28, 2011

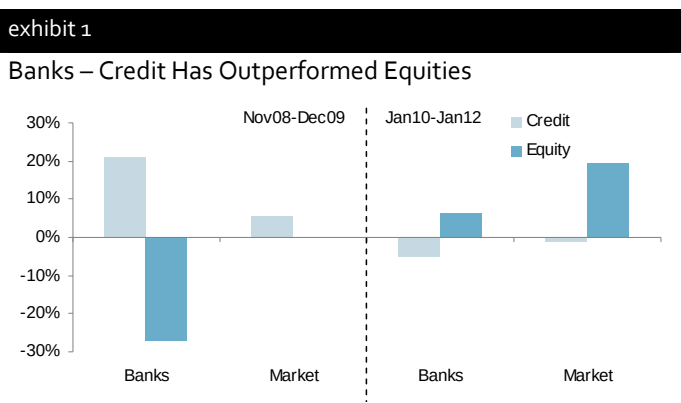
US Bank Credit – Regulatory and Market Dynamics

February 1, 2012

Sivan Mahadevan
Rizwan Hussain
Peter Mallik
Vishwas Patkar

Financials in general, and banks in particular, have been the biggest performance outliers in both credit and equity markets since the financial crisis and this should be clear to any and all investors in either market. In fact, it is quite amazing that corporate credit and equity markets have recovered as much as they have since the financial crisis, with financials punching well below their mighty weight. The positive price performance of banks in the early part of 2012 has certainly not gone unnoticed, helping the broader equity and credit markets significantly and reminding us how important the sector (and positioning within it) is to the total return story.

With equity valuations still at a fraction of tangible book value and credit spreads still materially wider than pre- or post-crisis tight, despite an enormous positive change in balance sheets and funding sources, we and most investors we speak with can agree on one thing: US bank credit is not trading to fundamental value alone. As such, much of the discussion quickly turns to the issues that do drive valuations, including regulatory challenges, corporate bond and CDS market dynamics, and the inherent sector beta.



Notes: Credit bars are excess returns, while equities are price returns ex-divs. Banks / market for credit are Banks sector / IG Corp, while for equities it's BKW index / S&P 500.

Source: Morgan Stanley Research, Yieldbook, Bloomberg.

In this chapter, we take a deep dive into the regulatory and market dynamics that are impacting US bank credit valuations. We focus on 8 broad topics, summarizing our views below and providing much detail in the remainder of this chapter. In our view, these issues result in significant market beta to the US banks credit sector that will likely not subside for some time. However, we continue to have a

favorable view of US bank credit, as we feel that many of the challenges facing bank credit can be overcome, even if they remain a headwind to equity market valuations based on an ROE or ROC framework. But it is by no means a straight-line performance story, and as such it is important to follow the technical structure of the market.

1) Dodd-Frank and Regulation: The Volcker rule, regulation of swaps trading and new risk-based capital guidelines are the key areas we are focused on that will both impact bank business models and the markets in which bank credit trades. We see the swaps regulation as being a net positive for bank CDS, as counterparty hedging needs fall while two-way flows increase, but the risk-based capital guidelines run the risk of decreasing lending efficiency relative to Basel, and uncertainty remains on Volcker rule implementation given the industry response to it.

2) Financials & the Corporate Bond Market Structure: 27% of IG trading volume is concentrated in just 7 issuers, all of which are financials. As a result, financials, and banks in particular, remain the dominant avenue for 'risk on' or 'risk off' expressions and market beta.

3) Financials & the CDS Market Structure: Banks also dominate CDS market volumes given appetite for counterparty hedging. However, the absence of banks in CDX make the CDS flows less balanced, especially compared to Europe, where their inclusion in iTraxx supports more two-way flows. The drop in bespoke structuring activity has presented another lack of natural sellers of protection on banks. The development of Swap Execution Facilities (SEF) will help relieve these imbalances in the US, in our view.

4) Comparisons to Europe's Banks: Despite sovereign concerns and a slower macro environment, European senior financials trade tighter than their US counterparts. This is partly technical, in our view, as a better developed sub-debt culture in European banks attracts more capital structure plays, while US investors focus more on senior debt for such views (and sub-debt has been reduced through exchanges).

5) Comparisons to Equities: While bank credit and equity performance is well correlated, credit has clearly outperformed equities on a risk-adjusted basis since the depths of the crises, and we believe some stability in equity price/book multiples could support further healing in bank credit. Furthermore, equity volatility markets suggest a more favorable stance on credit outright.

6) Rating Agency Methodology: Much change has taken place in the methodology used by credit rating agencies for

financial issuers. Pre-crisis, US banks benefitted from favorable ratings, which meant that they were candidates for inclusion in many structured vehicles. The change in the rating agencies' framework for analyzing bank credit, which now includes gauging the probability of sovereign support as well as more stringent economic and industry risk factors, is likely to be a key valuation driver going forward.

7) The State of Mortgage Put-backs: Agency put-back claims have been quantified for some banks, but there is still more to be done. Non-agency claims, ongoing state AG litigation on servicing and foreclosure practices, as well as monoline insurer claims remain a key hurdle for further repair in both equity and credit pricing.

8) Looking to the Savings & Loan Crisis as A Guide: The S&L crisis of the early 1990s serves as a pertinent and relevant roadmap for the performance of bank spreads post-crisis. However, the US sovereign downgrade in August did materially reset the basis between banks and industrials (also driven by an escalation of the European banking and sovereign crisis). We note that this measure in this cycle had been tracking quite closely to the compression (in both pace and magnitude) of the same measure during the early 1990s.

We provide details on each of these 8 areas in the remainder of this chapter.

1) DODD-FRANK AND REGULATION

While implementation of the 800+ page Dodd-Frank act in many ways will transform the banking sector, we find much focus among credit market participants on three specific sections of the reform, namely: a) Section 619, otherwise known as the 'Volcker rule', b) regulation of the swaps markets, and c) improvements in the regulation of securities focused specifically on credit ratings and securitization. These sections of Dodd-Frank collectively impact bank credit in two important ways. First, they impact the business models within banks that can drive profitability and the capital efficiency of holding corporate debt, and second, they impact the trading activity of corporate bonds and CDS, of which banks credit are a major part.

1A) THE VOLCKER RULE

Section 619 of Dodd-Frank, otherwise known as the Volcker Rule, prohibits and restricts financial institutions supervised by the Board of Governors of the Federal Reserve System from engaging in proprietary trading and having interests or relationships with hedge funds and private equity funds. The ramifications of the Volcker rule include potentially changing market-making activities in securities where dealers act as a principal, which impacts secondary market activity in corporate bonds generally.

The OCC, Board of Governors of the Federal Reserve System, the FDIC and the SEC have issued a Notice of Public Rule Making ("NPR" – November 7, 2011) to

implement the Volcker rule. There has been much market reaction to the impact on market-making activities in the corporate bond market as a result of this rule and NPR, including a recent congressional hearing. In a nutshell, the concern is that significant restrictions on holding inventory in a market-making function will ultimately have a negative impact on liquidity, thereby reducing the value of corporate credit (including bank credit), raising the cost of financing for corporations (including financial institutions), and increasing the cost of transacting in the market – thereby potentially reducing the efficacy of investment management activity.

While uncertainty remains as to the implementation details of the Volcker rule, the good news is that we expect significant progress in the first half of 2012 on rule making. While a severe interpretation can change the character of the market, clarity can also be helpful to all participants who need to plan for it as well. Either way, the Volcker rule is an important area for the market to gain clarity on, affecting bank credit from both valuation and market-trading perspectives.

1B) REGULATION OF SWAPS

Over the past 15 years, the credit default swap market (for corporates and financials) has gone from being insignificant in terms of volumes and impact to being a market with larger volume than cash bonds (just prior to the financial crisis), before falling back in importance substantially just after the crisis. While CDS was a center of blame for many aspects of the financial crisis, the market has moved forward with significant amounts of standardization (even before Dodd-Frank was legislated), and over the next 1–2 years, we expect CDS markets to benefit (in terms of liquidity and transparency) substantially from Dodd-Frank implementation that will mandate central clearing and electronic trading (through swap execution facilities or SEFs). Assuming current market participants (who trade OTC largely via the telephone) would continue to trade at similar volumes in a world where execution would be via SEFs, we believe there is upside potential for investor participation from those who are more comfortable with electronic trading platforms, including equity market participants.

The downside is that netting of collateral both for initial and variation margin could be an issue that reduces the efficiency of certain trading strategies and risk books, owing to central clearing of some but not all instruments. This might be a problem that disappears in the long run, as more instruments become centrally cleared, but there are transitional risks.

We see three potential impacts for bank credit related to the development of swaps (CDS) regulation:

1. As more CDS transactions move to central clearing and central counterparties, we see less demand for bank counterparty hedging, although there will still be plenty of hedging demand for non-standardized exposures.

- 2. We expect central clearing, standardization and the implementation of SEFs to improve market liquidity and transparency, mainly through the increase in the number of market participants. Given that banks are a big part of the CDS complex, bank credit should benefit and also become more “two-way” in flow as investors seek relative value (also related to less hedging as mentioned above).
- 3. We also expect higher CDS trading volumes to be incrementally positive to bank revenues.

1C) RISK-BASED CAPITAL

The third area of regulation that we focus on is one of the biggest sources of uncertainty for banks and the broader credit market, namely an inconsistency between the Basel Committee on Banking Supervision (BCBS), which relies on the use of credit ratings, and Dodd-Frank, which prohibits it. A recent NPR (*Risk-Based Capital Guidelines: Market Risk; Alternatives to Credit Ratings for Debt and Securitization Positions*, December 21, 2011) proposes rules for alternatives to credit ratings that are “generally consistent” with Basel’s “market risk framework,” which relies on the use of credit ratings.”

We focus here on the impact on both financial and non-financial corporate credit. Our securitized products research group has written extensively on risk-based capital guidelines for the securitization market within this NPR (see the report [Proposed Alternatives to the Credit Ratings for Risk-Based Capital](#), January 9, 2012).

NPR for Corporate Credit: For those unfamiliar with it, the Basel Accord assigns risk weights ranging from 0.25% to 12% for corporate debt positions within banks (0.25% to 1.6% for investment grade), based on term to maturity and ratings, as we show in Exhibit 2.

exhibit 2	
Basel Risk Weightings for Investment Grade Corporate Debt	
Remaining Contractual Maturity	Specific Risk Weighting
Residual term to maturity less than 6 months	0.25%
Residual term to maturity less than 6 months and less than 24 months	1.00%
Residual term to maturity exceeding 24 months	1.60%

Source: FRB, FDIC, OCC

The NPR from the OCC, Fed and FDIC (‘federal agencies’) cited above proposes an alternative methodology that is based on some fundamental metrics, namely EBITDA to Assets and Debt to Assets ratios, and stock return volatility to determine risk weights that would range from 0.25% to 12%.

exhibit 3

Proposed Risk Weightings for Non Financial Corporate Credit				
EBITDA-to-asset ratio	Stock Return Volatility*	Debt-to-asset ratio		
		<0.2	Between 0.2 and 0.5	>0.5
Greater than 0				
	Less than 10%	0.25%-1.6%	8%	8%
	Between 10% and 15%	8%	8%	8%
	Greater than 15%	8%	8%	12%
Less than 0				
	Less than 10%	8%	8%	8%
	Between 10% and 15%	8%	8%	12%
	Greater than 15%	8%	12%	12%

Source: FRB, FDIC, OCC, * - Defined as standard deviation of monthly stock returns over the past year, without annualizing

Given that much of the above fundamental information is less relevant for financials credit, the federal agencies are taking an alternative approach for financials credit. For corporate credit issued by financial institutions, the federal agencies are currently proposing a risk weight of 8.0%, but acknowledge that they are working on developing other methodologies.

The federal agencies are also considering an alternative approach based purely on relative bond spreads (bond spreads relative to popular CDX indices), with details in Exhibit 4. While the spread based measure is less complicated relative to the cyclical measure, there is less clarity with regards to the exact type of spread measure to be used (e.g., Z spread, asset swap spread, yields spread, etc.). In addition, the approach does not define the tenor of the underlying index to be used for the risk characterization in the exhibit below.

exhibit 4

Alternative Proposed Risk Weights Based on Market Spreads

	Risk characterization	Possible specific risk-weighting factor
Average spread < CDX.NA.IG.FIN	"low risk"	4.0%
CDX IG $\leq$ average spread < CDX.NA.HY.B	"medium risk"	8.0%
Average spread $\geq$ CDX.NA.HY.B	"high risk"	12.0%

Source: FRB, FDIC, OCC

NPR for Non-Financial Credit: We analyzed a large universe comprising over 350 US IG corporate issuers to get a sense of the possible impact of the NPR on non-financial corporate credit holdings for banks. As noted earlier, the new guidelines propose the use of three fundamental and market metrics to calculate capital requirements,

- Debt to asset ratio
- Stock market volatility based on monthly returns over a 12-month horizon
- 12M trailing EBITDA to asset ratio

One analysis reveals that general market uncertainty (i.e., stock return volatility) is a key defining factor for investment grade corporates in this new proposed framework, and this can make capital requirements quite onerous even for relatively healthy credits and short dated liabilities. Based on the universe we looked at, while 75% of issuers would fall under the "<10%" equity volatility band, the other 25% of the IG universe would have a capital charge of 8% or higher (see Exhibit 3). This implies that a material portion of the investment grade market would get much more onerous risk weights relative to the BCBS approach.

Our interpretation of proposed RBC treatment of corporate credit: If the OCC/Fed/FDIC go down a path similar to what is proposed in the NPR of Dec 21, 2011, we believe it potentially increases the volatility of risk weights (and hence the volatility of credit assets overall) given both the pro-cyclical nature of the metrics and the use of market based information (equity market capitalization and equity market volatility).

Macro uncertainty reflected in asset price fluctuations is likely to weigh heavily on regulatory capital requirements for banks. IG credits that could be characterized as 'tail' credits could realize 8% or 12% risk weights versus the far more capital friendly 1.6% under Basel. While, any new proposed approach is widely expected to be more onerous towards unhealthy tail credits, the sheer size of the tail (greater than 20%) could be onerous.

The impact on investment grade is far greater than on high yield, as a market accustomed to 1.6% Basel risk weights would move to many examples of 8% and 12% risk weights, including 8% on financials' debt. However, this would largely apply to bank lending books, which tend not to own lots of corporate bonds but instead have loan exposures. Trading books would have other capital measures apply that could be more problematic.

Nonetheless, these proposed changes make lending to IG companies potentially less capital efficient for banks, reducing ROE/ROC unless efficiency is made up in other lines of business, or credit spreads move materially wider. As it currently stands, we would qualify this as an 'unintended negative consequence' of rule making on the real economy.

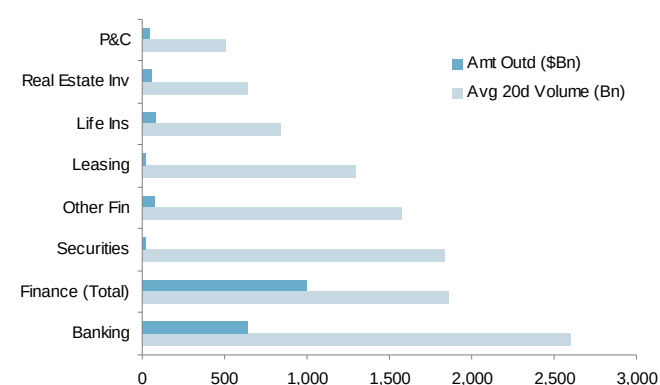
2) FINANCIALS AND THE CORPORATE BOND MARKET STRUCTURE

Financials, and specifically large banks, are of particular importance since they are the most frequently traded, make up the largest segment within the IG cash universe, and historically have had the least macro-driven price action (prior to the financial crisis). In terms of size, Financials issues make up 34% of a \$2.8Tn (in amount outstanding) market, or 26% of a 4,000 IG bond universe, while Financials volume is nearly double non-financials volume (using a 20d average). This is notable even as Financials shrink as deleveraging continues in the space.

Moreover, they are important from a positioning perspective as their volatility of returns is significantly higher than non-financials issues as they are a high beta sector today. Therefore, they can be (and we argue, have been) the asset of choice for portfolio managers looking to rebalance or change overall risk profiles. We expect Financials to remain prominent along each of these dimensions, despite their declining notional size overall.

exhibit 5

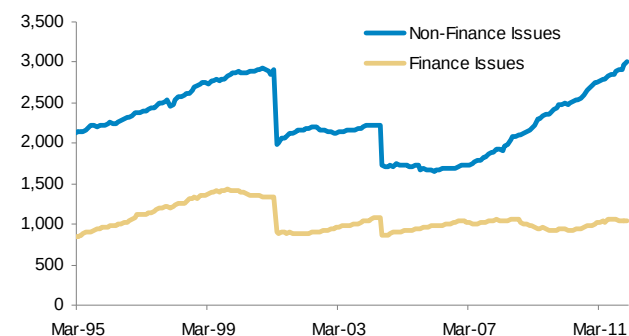
Banking Remains the Largest and Most Liquid Subsector in the IG Universe...



Source: Morgan Stanley, FINRA Trace, Yieldbook.

exhibit 6

...Despite Financials' Smaller Share of the Market



Source: Morgan Stanley, Yieldbook.

Within Financials, large-cap banks hold an even more prominent position as size and liquidity are concentrated in their bonds. The banking subsector itself makes up 70% of the notional outstanding in Financials issues and twice the traded volumes of the average of the remaining subsectors. But in fact the size and liquidity is even more concentrated. Of the top 20 most frequently traded issuers, the top 7 make up 27% of annual trading volume and all are Financials (and 6 are banks). This trend is true historically both pre- and post-crisis as we see evidence in December of 2007 and 2009 that Financials, banks, and financial benchmark bond volumes were higher than the mean volume for the average non-financial IG index bond regardless of an increasing (pre-crisis) or decreasing (post-crisis) number of Financials issues.

exhibit 7

The Most Liquid and Largest Issuers Are Financials

No	Issuer	% Volume	Cumulative % Volume	Actual Volume (Bn)	Sector
1-7	6 Largest Banks + GE	-	27%	835	Financial
8	AT&T INC	1.4%	28%	45	Telecom
9	WAL-MART STORES INC	1.4%	30%	42	Cyclical
10	VERIZON COMMUNICATIONS INC	1.2%	31%	36	Telecom
11	HSBC	1.1%	32%	33	Financial
12	BARCLAYS BANK PLC	0.9%	33%	29	Financial
13	BERKSHIRE HATHAWAY INC	0.9%	34%	28	Financial
14	CREDIT SUISSE GROUP-SPON ADR	0.9%	35%	27	Financial
15	KRAFT FOODS INC-CLASS A	0.8%	36%	26	Non-Cyclical
16	ANADARKO PETROLEUM CORP	0.8%	36%	26	Energy
17	AMERICAN EXPRESS CO	0.8%	37%	25	Financial
18	MATERIAL TECHNOLOGIES INC	0.8%	38%	24	Industrial
19	ANHEUSER BUSCH INBEV NV	0.8%	39%	23	Industrial
20	AMERICAN INTERNATIONAL GROUP	0.8%	39%	23	Financial

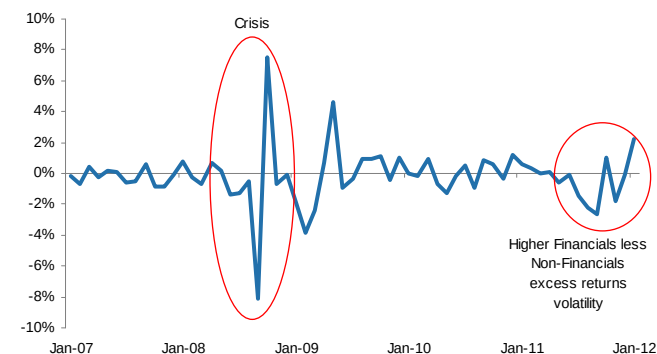
Notes: Actual Volume traded is of total issues traded during 2011.

Source: FINRA Trace, Bloomberg.

Particularly during times of market stress post-crisis, Financials' excess returns have been more volatile, driving index return volatility higher as in 2H08 and 2011. As such, increasing or reducing Financials exposure remains the cleanest way to rebalance risk, adding to the market beta theme within the sector.

exhibit 8

Financials Less Non-Financials Excess Returns Has Once Again Become More Volatile Post-Crisis



Source: Morgan Stanley Research, Yieldbook.

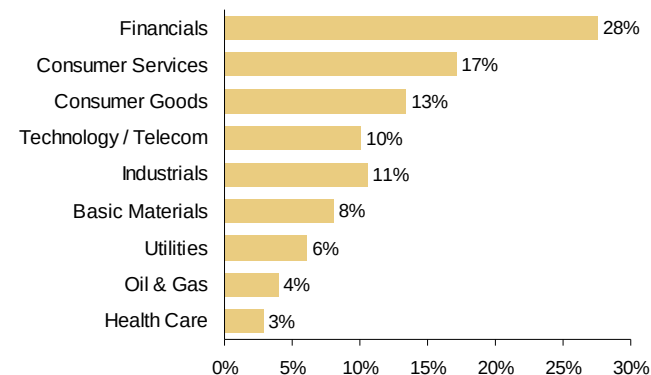
Going forward, we expect Financials and benchmark bank issues to remain a market driver despite our belief that US banks will be negative net issuers as deleveraging and less reliance on debt financing continues in 2012. However, this shrinkage may be a boon to Financials, as less supply should be supportive in the face of lower yields. Meanwhile, with any reduction in near-term jump risk views, investors may favor holding Financials issues to maturity instead of selling them as a way to mitigate risk, as they are shorter maturity than the broader market.

3) FINANCIALS AND THE CDS MARKET STRUCTURE

On the CDS front, financials are the biggest sector at a single-name level as per DTCC data (see Exhibit 9). Indeed, US and European banks figure at the top of the list of most widely traded single-name corporate reference entities (see Exhibit 10). However, there is one key aspect which differentiates US banks from their European counterparts, namely their presence in the benchmark indices.

exhibit 9

Financials Remain the Largest Sector in Single Name CDS by Gross Notional Outstanding



Source: DTCC

Over the past decade, the development of standardized index products has been a key motivator for risk taking in synthetic form, and the absence of banks from these indices has been a key technical headwind for credit spreads. Since the development of CDS indices in the US and Europe followed different paths (with different parties involved), their standards toward the inclusion of banks (who were active CDS dealers) differed in an important way: they were excluded in the US but were included in Europe. Exhibit 10 shows that, while all of the most liquid credits in Europe are part of one (or more) iTraxx index, 6 out of the 8 most liquid US financials are banks and not part of any standardized index.

With most of the single-name CDS focus arguably oriented towards hedging (counterparty exposure, prime brokerage exposure, etc.), we believe that the market does lack sufficient natural sellers of protection in US banks to balance this flow. We argue that this difference between US and European banks is important to understanding why the latter trade tighter in many cases, despite more complex capital structures as well as a weaker macro backdrop.

CDS flows in European financials tend to be more balanced because many large issuers are included in the iTraxx Main index and the senior financials portion of this index actively trades as well. Given the enormous liquidity gap between the indices and single-names, we argue that more balanced flows

in iTraxx Main can lead to a more fair outcome to the names included within the index because there are both natural buyers and sellers of index protection. In the single-name market, while we can list numerous natural buyers of protection on banks, there has not been a large base of natural sellers of protection since bespoke flows slowed after the financial crisis. Thus, diminished bespoke flows, as well as exclusion from the standardized indices, can explain some of the underperformance of large cap US banks relative to their European peers.

exhibit 10

US Banks Underrepresented in the Standardized Indices Relative to Their European Counterparts

Reference Entity	Region	CDX constituent	iTraxx constituent
GENERAL ELECTRIC CAPITAL CORPORATION	US	X	
MBIA INSURANCE CORPORATION	US	X	
JPMORGAN CHASE & CO	US		
BANK OF AMERICAN CORPORATION	US		
THE GOLDMAN SACHS GROUP	US		
WELLS FARGO AND COMPANY*	US		
CITIGROUP INC	US		
DEUTSCHE BANK AKTIENGESSELLSCHAFT	EU		X
BANCO SANTANDER S.A	EU		X
INTESA SANPAOLO SPA	EU		X
MERRILL LYNCH AND CO INC	US		
RADIAN GROUP	US	X	
BBANCO BILBAO VIZCAYA ARGENTARIA, SOCIEDAD	EU		X
AIG	US	X	
BARCLAYS BANK PLC	EU		X
THE ROYAL BANK OF SCOTLAND PUBLIC COMPANY	EU		X
SWISS REINSURANCE	EU		X
COMMERZBANK AKTIENGESSELLSCHAFT	EU		X
AXA	EU		X

Source: DTCC, *-Wells Fargo was a constituent in older CDX IG indices, but is not part of the newer indices,

4) COMPARISONS TO EUROPE'S BANKS

Given the stresses that European financial institutions are under from a slower economy, bank deleveraging, sovereign exposures, and potential tail events derived from Euro-zone stresses, it would seem odd that European senior financials often trade tighter than senior US bank debt, both in CDS form and relative to Bunds/Treasuries. What bridges the gap?

European financials have more complicated debt capital structures, with larger amounts of subordinated debt, which tend to attract the underweight or otherwise negative positioning in the financials space. Also, senior debt holders

have generally been treated well in bank failures, as they remain untouched in the Euro-zone and the UK so far in this crisis, even in those countries which already have resolution regimes in place which ostensibly allow for write-downs. According to our European Banks' analyst Jackie Ineke, an EU-wide resolution regime will be formally proposed later this year, but this is likely to only introduce the potential for senior debt write-downs from 2016. Furthermore, much European senior debt is issued at the operating company level, while in the US it tends to be more at the holding company level (which itself introduces a degree of subordination).

exhibit 11

US Banks in Comparison, Have Very Little Sub Debt

	Senior Bank Debt	Subordinated Bank Debt
Average spread < CDX.NA.IG.FIN	~1600	135
European Banks (\$Bn equivalent)	2400*	680*

Source: Deal Logic, FDIC, Company Data

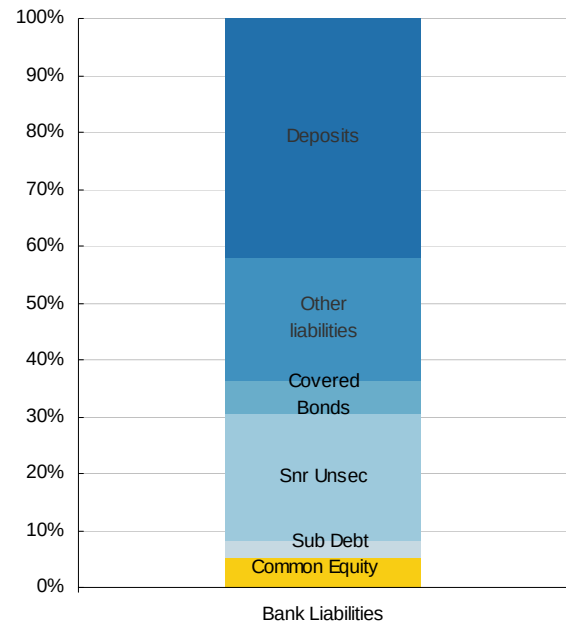
Note: * - Assuming EURUSD exchange rate of 1.3

Additionally, in the US, there tends to be a less subordinated debt for banks, so there is more attention on expressing a view on banks through the senior bonds, and there have also been bank failures that have impaired senior debt (Washington Mutual and Lehman Brothers). According to data from DealLogic, the size of European bank senior and sub debt market is €1.85 Tn and €522 Bn, respectively. In contrast, the size of the US banks subordinated debt market is a mere \$135 Bn.

While these numbers could be slightly skewed by the different composition and weightings of the underlying constituents in the two indices, our European colleagues published a report in November 2011 (see [The Most Profitable Bank in Europe](#)), where they looked at an average "liability mix" for European banks which were part of the iTraxx Senior Financials index (see Exhibit 12). The exhibit reaffirms that subordinated debt even for the biggest and most liquidly traded banking entities in Europe is a significant part of the capital structure. US banks have TruPS, but sub debt is a shrinking market given shifts in their usage for regulatory capital requirements.

exhibit 12

Euro Banks Have a Relatively Complex Capital Structure



Source: Morgan Stanley Research, SNL Financials, Company Data.

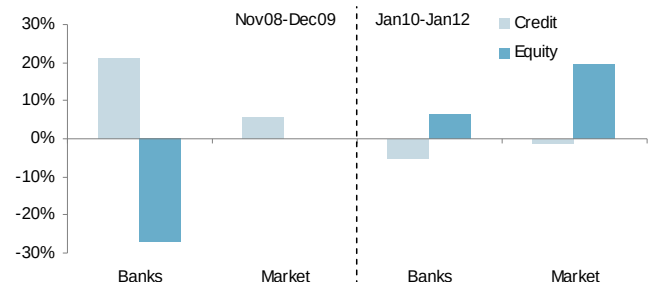
Note: Liability "mix" is the average mix of the 16 banks in the iTraxx Senior Financials index as of 2Q11.

5) COMPARISON TO EQUITIES

With the second largest sector weight in the S&P 500 and one of the highest betas, Financials remain a key sector in the equity markets as well. However, performance for Financials equity and credit has diverged post-crisis as increased regulation expectations, deleveraging, capital structure uncertainty and lower rates have driven equities to underperform the broader market more so than in credit. Going forward, do we expect this trend to continue and, more importantly, what do the equity markets tell us about the Financials sector compared to credit markets?

exhibit 13

US Financials and Bank Credit Have Cumulatively Outperformed Their Equity Counterparts Through the Cycle



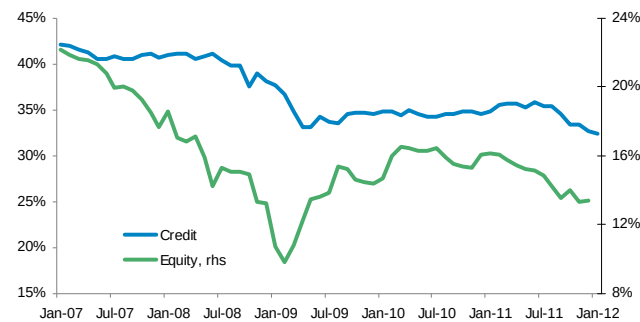
Notes: Credit bars are excess returns, while equities are price returns ex-divs. Banks / market for credit are Banks sector / IG Corp, while for equities it's BKW index / S&P 500.

Source: Morgan Stanley Research, Yieldbook, Bloomberg.

For this analysis, we compared performance of the S&P 500 Financials sector and the KBW Banks index to their counterparts in the IG cash credit universe. We should note that composition-wise the IG credit index is more heavily weighted towards Financials and Banks than in equities. Proportionally the credit and equity sectors shrunk by similar amounts since 2007; however, the maximum decline was much more pronounced in equities, mirroring the price action.

exhibit 14

The Relative Size of Financials in Equity and Credit Indices Has Fallen Since 2007

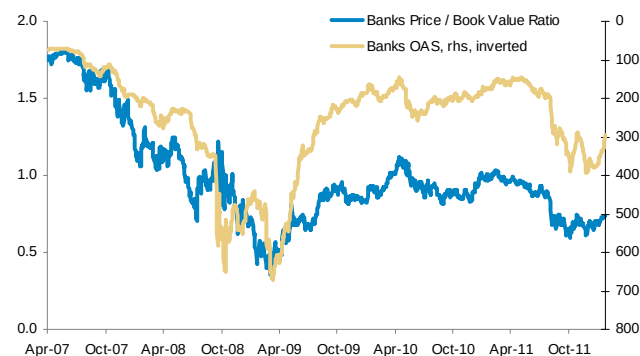


Notes: Credit sector is ratio of Financials to overall IG corporate index market value, while Equity is ratio of Financials to S&P market caps. Source: Morgan Stanley Research, Yieldbook.

In the healing period post-crisis (Nov08 – Dec09), financials and bank stocks underperformed the S&P 500, while in contrast financial and bank credit outperformed the IG index significantly. This trend has continued, albeit to a lesser degree, since 2009. With European deleveraging contributing to slowing growth, the potential for QE3 to flatten NIM, and regulatory uncertainty still omnipresent, equity performance may continue to decouple from credit.

exhibit 15

Some Stability in Price/Book Ratios for Banks Could Serve to Further Drive Credit Spreads...



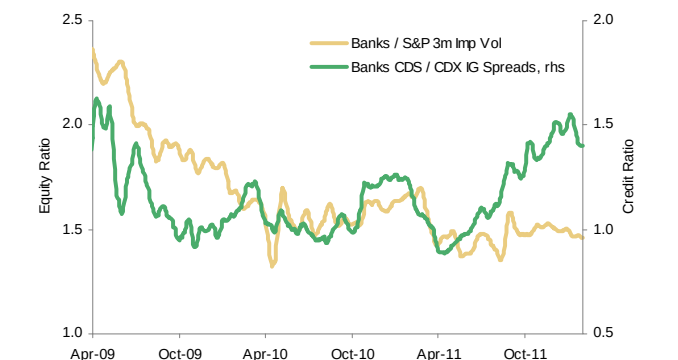
Notes: Banks for credit is IG corp bank sector, while for equities it is the KBW BKX index. Source: Morgan Stanley, Bloomberg.

With the general underperformance of bank equity and credit (even after YTD performance), what could stem the tide, even without a meaningful pop in top-line fundamentals? Continued European bank deleveraging may cause higher-quality borrowers to migrate to US banks in the short term. Lighter regulation, litigation settlement, or movement towards a pro-business Congressional agenda in 2013 may push equities higher; although each of these may be a low probability event (see [Large Cap Banks: 2012 Outlook: In-Line as Deleveraging Accelerates](#), Nov 28, 2011). As such, spreads may tighten if any of these catalysts materialize, even if bank fundamentals do not improve. This has happened before, as post-crisis in mid-2009, spreads rallied considerably as price-to-book value ratios remained below 1.0x, while pre-crisis the two moved roughly inline.

In contrast to cash equity markets, equity volatility markets support a more constructive credit outlook for banks. While total returns have not kept pace, Financials and Bank volatility at the index and single-name levels have followed the broader market lower since the peak in late 2008, diverging from comparably higher relative banks credit spreads, indicating that banks credit upside could be considerable. Financials and Banks 90-110% skew and term structures concur, as they have improved relative to the S&P recently.

exhibit 16

...And Bank Equity Volatility Markets Indicate Credit Upside Is Considerable



Notes: Banks is the 10d moving avg of the 3m implied vol of the KBW BKX index for equities and the average CDS for names in the KBW BKX index for credit. Source: Morgan Stanley Research, Bloomberg.

Today, both equity and credit investors appear to share our cautiously bullish outlook. According to our US Equity Banks research team's Dec 2011 Financials survey, 43% of respondents are underweight the sector even though they picked Financials as the top performing sector for 2012 (see [North American Financials: 2012 Outlook: Own Diamonds in the Rough](#), Nov 28, 2011). Similarly in credit, canvassing of US investors in our most recent Global Credit Investor Survey yielded senior financials rated as the most attractive

subsector in US credit, with lower than normal issuance expected, yet very little re-risking having been reported.

6) THE IMPACT OF RATINGS AGENCY METHODOLOGY CHANGE

Moody's and S&P downgraded several US banks last year in September and November, respectively, as part of the implementation of their new ratings methodology for financials. While we do not go into the details here of all the changes they outlined, the stand-out criteria in both cases related to the likelihood of a sovereign backstop (or lack thereof) for banks going forward. Moody's virtual wholesale downgrade of US banks in 2011 was specifically based on the possibility of Dodd Frank legislation in 2012 effectively precluding any kind of government support for banks.

S&P has placed an increased emphasis on economic risk and industry risk in setting the starting point for rating financial institutions (see Exhibit 17), which is then adjusted for bank-specific factors. These specific factors are implemented to rate hybrid bank capital as well as preferred stock of banking institutions. For rating senior unsecured issues, the revised framework also considers potential for direct support from the sovereign or the parent company. As banks continue to delever and improve capital ratios, bank-specific factors should change favorably. However, the additional emphasis on economic and industry risk is likely to remain a headwind on bank credit ratings going forward. In addition, the 'willingness' (as well as 'ability') of government support remains a topic of wide debate, but strict interpretation of Dodd-Frank and other regulation could outright prevent future bank bailouts.

exhibit 17

Summary of S&P's Bank Rating Methodology

Macro Factors	
	Economic Risk Score
	Industry Risk Score
Bank-Specific Factors	
	Business Position
	Capital and Earnings
	Risk Position
	Funding and Liquidity
External Support	
	Group Support
	Government Support

Source: S&P

So why are bank credit ratings important? Aside from the obvious funding advantages for higher quality institutions, credit ratings continue to flow through the investment landscape in an important way. Banks in particular benefitted from favorable ratings in an environment pre-crisis, where

ratings efficiency was one of the primary drivers of structured investment flows. While the notional of such portfolios outstanding is much lower than it was pre-crisis, we have seen some interest of late in credit risk in customized form (see *Chapter 35*). Even though focus on ratings optimization does not seem quite as important in these deals, the recent change in the rating agencies' methodology is likely to resurface as a negative for bank credit.

7) THE STATE OF MORTGAGE PUT-BACKS

Bank securities litigation provides a powerful fundamental headwind that we expect to persist as uncertainty in the settlement size and litigation term cast a shadow over performance. On the agency ABS side, we have some precedent of settlements already, but there is more to be done. In 2010, BAC reached settlements with the GSEs on agency put-backs, leading our Securitized Products Research team to estimate system-wide put-back losses to be up to 2% of unpaid principal balances or up to \$78 billion system-wide (see [Morgan Stanley ABS Market Insights](#), Jan. 10, 2011). And while the amounts are large, the fact that settlements are in place and some put-backs have already occurred provides the market a sense of certainty.

On the other hand, the non-agency MBS put-back story is considerably cloudier. First of all, the litigation success threshold is higher with a breach in rep/warranty and materiality needed to be proven for non-agency MBS legal claims. There is an emerging precedent as BAC has agreed to an \$8.5Bn settlement with 22 large institutional investors to cover past and expected losses on select Countrywide RMBS. But the settlement still must be approved by the court and each investor, which may not happen without substantive changes, as we do not expect resolution until mid-2012 at the earliest ([ABS Market Insights - Bank of America Settlement: Implications to Non-Agency RMBS](#), Jul 7, 2011). We note that there are also MBS related claims from monoline insurers that can continue to be a headwind to banks.

For individual banks, calculating aggregate legal liability is challenging since in addition to the put-back claims, they face litigation from both the FHFA and various state attorneys general. On the FHFA case, our US large cap banks analyst Betsy Graseck estimates losses for the top 4 covered US banks to be \$16Bn and losses for the top 7 covered European banks to be \$13Bn (see [We Estimate FHFA Securities Losses in US Bank Suit](#), Sep 5, 2011). Simultaneously, large mortgage services are negotiating with state AG's to reach an industry settlement on servicing and foreclosure practices, which we expect shortly. Although a settlement may have limited impact on equity prices, since most banks already have reserves to cover immediate fees and penalties (\$5Bn estimate) and future "soft" losses (\$20Bn estimate) (see [State AG Settlement Likely Gains Traction Post Tomorrow's SOTU](#), Jan 23, 2012), we do

believe some clarity can serve as a catalyst to firm both equity and credit's recent moves higher/tighter.

8) LOOKING TO THE SAVINGS & LOAN CRISIS AS A GUIDE

At the opening of this note, we suggested that much of the credit underperformance in the US banking sector was better explained by future, but yet to be fully developed, regulatory changes, as well as bond and CDS market dynamics (not to mention systemic issues of concern as well, primarily emanating from Europe). As a result, we've avoided focusing on the secular improvement in fundamentally based measures such as Tier 1 capital ratios, funding and liquidity positions, as well as supportive changes to business models generally. However, is there any precedent we can note from prior US banking system crises to guide our view on bank spreads now?

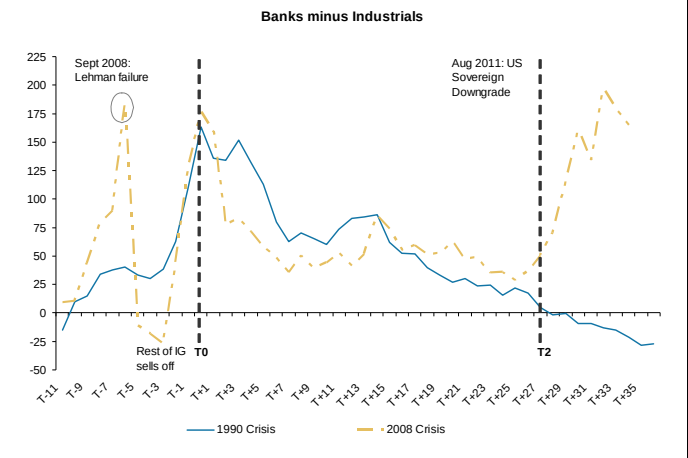
Perhaps our best modern-day example we can point to in US credit markets comes from the early 1990s and the related S&L crisis. What are some of the elements in common post both crises?

- Deregulation of the banking sector leading to faulty over-extension of credit and over-investment in particular sectors/asset classes;
- Real estate at the core, but not the only source of impairments to be ultimately realized;
- Government support to eventually backstop the financial system as a credit crisis followed; and
- An ensuing economic recovery, characterized as 'subpar' and 'jobless'.

Admittedly every crisis and/or recession has unique identifying characteristics, drivers and challenges. However, the comparison of today's environment versus the early 1990s seems applicable nonetheless. In Exhibit 18, we have tracked the absolute spread basis between the banking and industrial sectors in this cycle as well as the early 1990s, with T0 defined as the point of maximum differential in each cycle. We would characterize both of those starting points as a 'point of maximum uncertainty' – March 2009, preceding the release of the Fed's banking system stress test results, and November 1990, shortly preceding a bottoming of the economy in 1Q91 as well as shortly after the Iraq invasion of Kuwait, and resulting US military response,

exhibit 18

This Cycle's Repair in Banks Mapped Well with the Recovery After the S&L Crisis, Until the US Downgrade



Source: Morgan Stanley Research, Yieldbook

Note that the narrowing of the banks-industrial basis we had seen for 2 ½ years in this cycle was marching almost hand in hand with the recovery the banking sector enjoyed following the S&L crisis. However, we've had another point of 'maximum uncertainty' in this cycle with the downgrade of the US sovereign rating last summer, resetting the basis once again (and again, almost consistent with levels seen at prior peaks).

BOTTOM LINE – BANK CREDIT IS ABOUT BETA

Regulatory forces and market dynamics are playing a large role in the price of bank credit. They were responsible for the considerable underperformance in 2011 and for the outperformance to date in the early part of 2012. In many ways, bank credit is about beta, especially as financial regulation is a key driver of markets. We expect this theme to remain for some time, and as such it is important to understand the roles Dodd-Frank, cash market liquidity, the CDS complex, debt capital structures and rating agency methodologies play in setting the price for bank credit.

We continue to have a favorable view of US bank credit in this environment, although we acknowledge that this view has an embedded market call in it given both the high beta nature of the cohort as well as the size of the group relative to the credit market overall. We feel that many of the challenges facing bank credit can be overcome, even if they remain a headwind to equity market valuations based on an ROE or ROC framework. But it is by no means a straight-line performance story, even though the last few months have shown otherwise.

Section C

Valuation

Libor Metrics

June 27, 2003

Sivan Mahadevan
Rizwan Hussain

A combination of low spreads, high dollar prices and a steep yield curve has made relative value analyses difficult today, both within the universe of corporate bonds and between bonds and default swaps. Looking at bonds on a Libor basis is supposed to be the common denominator approach, but confusion persists. Using Libor spreads is a bit like the Metric System: everyone agrees that it is better, but it is hard to develop intuition for the measures when the market trades on Treasury spread and dollar price.

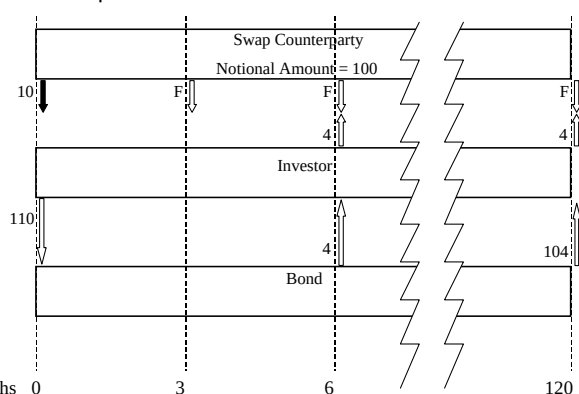
We focus on describing the various Libor spread measures in simple terms, recommend using the intuitive Z-Spread approach for relative value purposes (for both cash investors and derivatives users), and show some practical examples to illustrate relative value in an environment where a handful of basis points really matter.

FOUR LIBOR MEASURES

There are four Libor spread measures commonly used by market participants. Par and market value asset swaps are meant to be used by those doing real asset swaps (i.e., converting a fixed rate bond to a floating rate instrument), while interpolated swap curve spreads and Z-Spreads are relative value measures used by those who are focusing on fixed rate assets.

exhibit 1

Par Asset Swap – 8% 10-Year Bond Priced at \$110



Note: The floating payment F equals $\text{Notional Amount} \times (\text{LIBOR} + \text{Spread}) / 4$.

Source: Morgan Stanley

ASSET SWAPS

A par asset swap is the most common type of fixed-for-floating swap used by credit investors. If a bond trades at par, then the swap simply involves an exchange of coupon

payments for floating rate Libor plus a fixed spread. When a bond trades at a premium, the swap becomes off-market, and there is typically an upfront payment from the swap counterparty to the investor to make up for the premium (see Exhibit 1).

A market value asset swap is less common and involves converting the fixed-rate bond into a floating-rate note with par equal to the original bond's dollar price.

MAKING UP FOR UNACCOUNTED CREDIT RISK

While asset swaps are practical vehicles for converting fixed-rate bonds to floating-rate, the "spread" over Libor paid out by the swap counterparty is not necessarily an accurate measure of the bond's credit risk, particularly when there are upfront or residual payments.

In a par asset swap, the present value of all the periodic cash flows is equal to the premium or discount on the bond. This present value is calculated using the Libor term structure, while the premium on the bond is a result of cash flows, which are discounted using a credit-risky rate. As a result, asset swap spreads reflect the shape of the Libor yield curve but fail to fully incorporate the impact of the credit-risky nature of the bond cash flows. This mismatch in discount rates introduces a bias in the asset swap spreads, which is particularly acute for bonds with significant premiums or discounts and bonds with wide credit spreads. One point we want to make clear is that asset swaps are not incorrectly measuring credit risk. Indeed, the swap itself has no credit exposure (to the bond issuer) because all of the payments on the swap are due whether or not the bond defaults prior to maturity.

GOING BACK TO BASICS

For investors who are simply using Libor measures to make relative value decisions between bonds and/or default swaps, we consider two other measures more relevant. Both measures involve a common and intuitive practice, namely comparing a bond's yield or cash flows to a benchmark. An interpolated swap spread is one measure and is simply the yield to maturity of a bond minus the interpolated yield on the swap curve. This spread is termed I-Spread (or yield-on-yield spread by asset swappers).

I-Spread ignores the shape of both the Libor yield curve and the credit curve, and thus does not reflect any impact for the actual timing of payments. Two bonds with the same maturity and yield but different coupons (and thus different duration) would get the same I-Spread. The second measure involves an OAS model. If we take a step back and think about bond basics, what investors require is a method to compare a series of risky cash flows to a risk-free yield curve that is not biased by dollar prices and coupons. An OAS

model, using a zero Libor curve can solve this problem. However, OAS models build a tree of paths, adding unnecessary complexity to a relatively simple problem.

CONVERGING ON Z-SPREAD

An OAS model with a zero volatility input can reduce the tree to a simple yield curve, which results in an intuitive price/yield type of calculation. Under this method, we calculate a fixed spread over a series of zero-coupon Libor rates that equates the price of the bond with the present value of the cash flows. We call this spread the Z-Spread, which one can think of as zero-volatility Libor OAS. Z-Spread may be easier to explain as an equation than in words. The general price/yield relationship of a credit-risky bond is as follows:

$$P = \sum_{i=1}^n \frac{\text{BondPayments}}{(1 + \text{Yield}_i)^i}$$

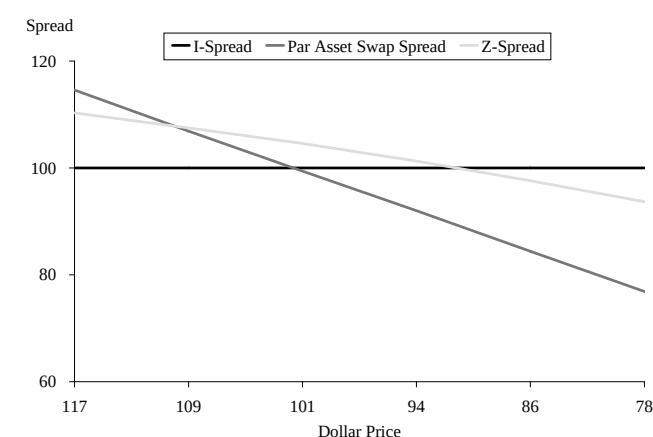
We can then decompose the yield into a Libor component and a spread component:

$$\text{Yield}_i = \text{ZeroLibor}_i + \text{ZSpread}$$

Solving for Z-Spread in the previous equation will give us the desired measure. This method has the advantage of not being biased by premium or discount bonds. It takes the shape of the Libor curve into consideration, but it assumes a fixed spread over all the Libor zero rates. More elaborate models that consider credit spread curves can be used, but they can confuse the issue when an investor's goal is relative value.

exhibit 2

Measuring Spread Sensitivity to Price Changes



Source: Morgan Stanley

MEASURING SENSITIVITY

In Exhibit 2, we use a hypothetical example of a 10-year bond with a fixed yield of 10-year Libor + 100 bp. As we vary the coupon of the bond, we can observe the sensitivity of the Libor measures to changes in the bond's price. The Libor curve does not change in this example, so the I-Spread curve is (by definition) flat. The par asset swap spread curve

is steep, implying that it is very sensitive to bond prices, while the Z-Spread is flatter (but not completely flat), demonstrating a more muted sensitivity to price changes.

We highlight three points. First, at a par price, the asset swap spread and I-Spread are roughly equivalent. Second, the Z-Spread has an intuitive slope; however, it has "shifted" higher because of an upward sloping yield curve. The basis point "shift" increases with Libor curve steepness, absolute credit spread and maturity. Third, in today's high dollar price and steep yield curve environment, the spread measure with the largest "error" is the I-Spread metric.

REAL RELATIVE VALUE

A practical example of a relative value trade where the attractiveness of a transaction depends on the spread metric used comes from a Morgan Stanley Fixed Income Research June 17, 2003, report "Consuming Ideas." The report suggests selling May Department Stores' 2011 issue and swapping into Federated Department Stores' 2011 bonds. As rationale for the trade, the report cites Federated's lower leverage (Debt/LTM EBITDA of 1.8x versus 2.3x for May), as well as the fact that it has paid down debt and generates four times as much cashflow as May. This is a classic "up-in-quality" trade of the type we suggest investors should pursue in an environment of relatively undifferentiated, tight valuations.

exhibit 3

Comparing Valuations in a Trade Idea

		Treasury Spread	Asset Swap Spread	I-Spread	Z-Spread
Buy	FD 6.625 2011	T+80	96	84	93
Sell	MAY 7.45 2011	T+105	118	100	111
Difference		-25	-22	-16	-18

Source: Morgan Stanley

The pricing details of the trade are shown in Exhibit 3. The trade is a 25 bp give-up on a spread to Treasuries basis, but a less onerous 18 bp give-up on a Z-Spread basis (the par asset swap give-up is 22 bp). While the numbers may not be that striking on the surface, we believe that, in a world where "nips for blips" is making a comeback, market participants should take note.

CONCLUSION

We recommend using the Z-Spread measure as a relative value tool, given its simplicity, intuitive feel and accuracy. A flat credit spread curve assumption is a shortcoming, but solving this problem may introduce more complexity than value. Finally, there are approaches to valuing bonds that are built on a risk-neutral framework, where premium bonds would suffer when recovery rate assumptions are fixed. While such an approach may be useful in an absolute return framework (such as in a synthetic CDO), they are less applicable to the day-to-day relative value world that many credit investors live in.

Rates and the Credit Culture

November 13, 2009

Sivan Mahadevan
Phanikiran Naraparaju
Ashley Musfeldt

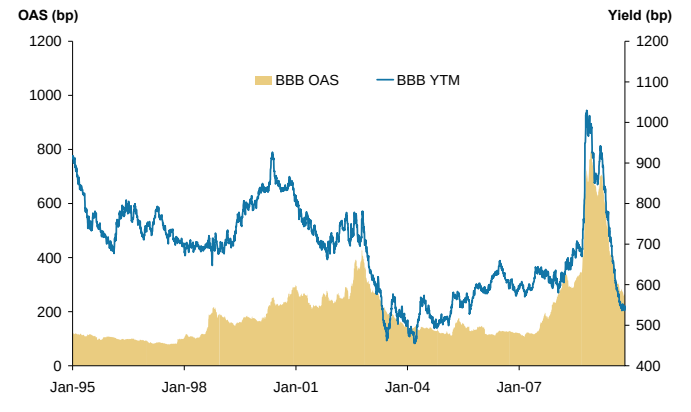
For much of 2009, investment grade corporate bonds were the clear credit investment of choice. Spreads were wide, yields were high, and the fallout of the financial crisis favored an investing style of simplicity over complexity. While credit is still attractive as an excess-return play over Treasuries, the risk of rising rates has changed the total-return equation dramatically. Our credit and equities teams have addressed rising rate risk on risky assets, and we believe it is prudent for credit investors to seek some shelter from the rising-rate storm. Within the equity world, we recently highlighted both hedging and upside plays based on interest rate scenarios (see [Equities in the Bond Bear Camp](#), November 11, 2009). Our investment grade strategy team discussed a variety of ways to play a rising-rate environment in credit in the Credit Basis Report from October 23, 2009, [Rate Relief](#).

Among other things, they highlighted the likelihood of floating-rate issuance and recommended investors create similar products using asset swaps on fixed-rate corporates as well as hedging interest rates generally through interest rate cap structures. From a bond perspective, converting fixed-rate products to floating-rate makes sense. However, the pool of floating-rate notes available to investment grade investors is smaller than fixed-rate, and asset swapping bond by bond can be difficult to do at the portfolio level. Furthermore, there are contingent risks in an environment where single-name credits can widen and/or default while interest rates rally. Here, the value of the rate swap would fall while the riskiness of the fixed coupon from the corporate bond would rise dramatically.

Another consideration in a rising-rate environment is the impact on the cash / CDS basis. While the large negative basis (cash bonds trading wider than CDS on a Libor basis) was an obvious healing trade a year ago, what was not so obvious was whether markets were actually going to heal. Today, the negative basis in non-financials continues to compress, and we think that flat-to-positive basis situations are worth looking at more closely. Furthermore, unfunded CDS as well as funded notes can address the interest rate risk problem quite elegantly. In the derivatives world, we see three important investment themes related to rising rates, which we summarize below. We provide more detail in the remainder of this chapter.

exhibit 1

Yields Below Historical Average; Spreads Are Not



Source: Morgan Stanley Research

Sell protection. First, selling protection has many advantages over buying fixed-rate bonds in a rising-rate environment, owing to a lack of interest rate duration, the benefits of receiving upfront payments when short rates can rise (vs. dollar prices on bonds that will fall with rising rates), and importantly the ease of execution relative to the alternatives in the cash world (buying floaters or asset swapping bonds).

Sell the basis. Second, because of the above advantages of selling protection vs. buying bonds, we advocate selling the basis, or more accurately, unwinding negative basis positions that are close to flat or already positive, as we do not see reasons for CDS to trade significantly wider than bonds in a positive credit environment.

CLN exposure. Third, as the basis normalizes (and it has in Europe and select sectors of the US market), credit-linked notes (where investors sell protection in a funded format akin to a floating-rate bond) become easy methods to gain exposure to credit without interest rate risk and to take advantage of CDS trading wider than bonds. We extend our CLN view to portfolios of credit, and of course tranches, which are the most popular instruments within CLN structures.

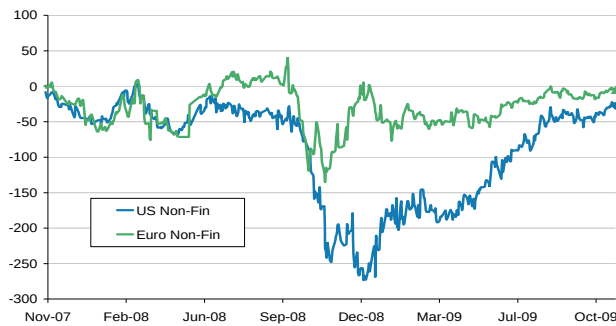
BASIS VALUATION AND DRIVERS

The negative basis observed in 2008 was clearly a function of higher premiums attached to funded cash instruments, fueled by global liquidity concerns. With credit unavailable and banks unwilling to lend to each other, the yield for assets requiring large cash outlays went up. We have previously outlined the basis tightening (less negative) as a systemic healing trade, which would play out as entire fiscal and monetary stimulus measures kicked in. Indeed this has

progressed in the last year or so, and consequently we think it is time to take the negative basis trade off in some parts of the market.

exhibit 2

Cash/CDS Basis: Finding Some Stability



Source: Morgan Stanley Research

What has changed? Valuations for one are expensive – especially on the European side. The European IG basis is now close to flat (see Exhibit 2). In North America, the negative basis opportunities are more attractive towards the shorter end of the curve and in HY/XOver credits. In both Europe and the US, we actually see credits where the basis is now *positive*. A year ago, these were very rare, but today there are quite a few, as we highlight in Exhibit 3. Financials are the one exception where bonds are still substantially wider than CDS due to the overhang of legacy cash positions.

exhibit 3

European Positive Basis Situations

Company	Cpn	Maturity	Price	CDS Spread	Z-Sprd	Adj. Basis	CAPS Basis
Bertelsmann	4.75	Sep-16	100	227	187	37	24
Carrefour	4.375	Nov-16	103	86	69	19	9
Tesco Plc	5.875	Sep-16	113	101	84	25	9
Siemens	5.625	Jun-18	112	82	63	21	2
Telecom Italia	4.75	May-14	105	132	110	16	20
France Telecom	3.625	Oct-15	101	54	54	4	0
Enel	4.75	Jun-18	105	120	65	55	39
Iberdrola	7.5	Nov-15	120	96	74	32	15

Source: Morgan Stanley Research

On the bond side, a key factor for the IG corporate bond universe compared to last year is dollar prices. With many of the bonds issued at much higher absolute yield levels, the average IG bond today is trading above par (average prices of 106) compared to an average price of 93 just at the start of the year. Today's higher dollar prices are likely to be an impediment to the performance of bonds and also make the basis look less attractive, as the higher initial cash outlay would require more CDS hedges, all things being equal. Much of the decline in rates occurred in the last quarter of 2008. As

we see in Exhibit 4, a 100bp rate hike makes a significant difference to the dollar prices on the longer-maturity bonds.

exhibit 4

Bond Prices – Rate Decline Driving Dollar Prices

Ticker	Cpn	Maturity	Jan-09	Today	Rate +100bp
TITIM	4.5	Jan-11	96.5	103.0	101.8
TITIM	7.5	Apr-11	101.8	107.5	106.0
TITIM	6.25	Feb-12	99.6	107.4	105.2
TITIM	7.25	Apr-12	100.9	110.1	107.6
TITIM	6.875	Jan-13	99.4	110.8	107.7
TITIM	4.75	May-14	88.6	104.4	100.3
TITIM	5.375	Jan-19	82.0	105.6	98.3

Source: Morgan Stanley Research, iBoxx, Bloomberg

On the CDS side, two important factors are at play. First, CDS curves are much steeper than they were at the start of the year, which is clearly a contributing factor to the basis being a bit more negative at the shorter maturities than at the longer maturities. Second, CDS structure has changed following the adoption of the fixed-coupon contracts. Shorter-maturity CDS generally trade with a negative upfront (pay an upfront amount and then receive 100bp when selling protection), whereas the longer-maturity CDS contract is likely to trade with a positive upfront, making it more attractive for protection sellers from a funding perspective. Bottom line, from both a CDS and bond perspective, the longer-maturity bondholders are likely to find selling CDS interesting.

exhibit 5

CDS Structure Has Changed (TITIM CDS Curve)

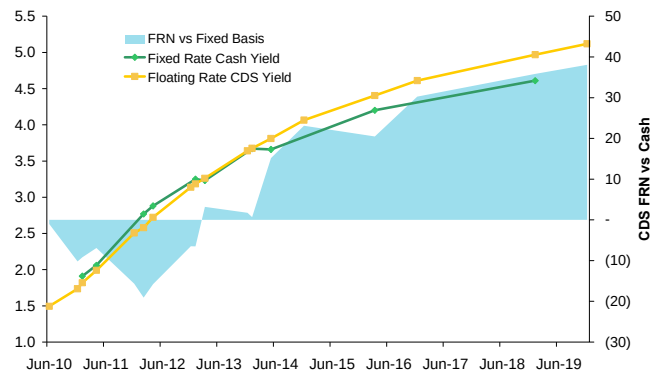
	Spread Levels			Today	
	Jan-09	Today	Chg	Coupon	Upfront
6M	415	45	-89%	100	-0.3%
1Y	415	45	-89%	100	-0.6%
2Y	415	70	-83%	100	-0.6%
3Y	410	90	-78%	100	-0.3%
4Y	400	108	-73%	100	0.3%
5Y	390	125	-68%	100	1.2%
7Y	378	140	-63%	100	2.5%
10Y	365	154	-58%	100	4.3%

Source: Morgan Stanley Research

To illustrate the positive basis, we take the specific example of Telecom Italia (Exhibit 6), which is characteristic of the rest of the universe generally. The CDS yield here is represented by CDS spread + swap rate to the particular maturity. The basis at the short end is more negative (short-end cash is more attractive), whereas the CDS is more attractive at longer maturities. This is partly a consequence of an easing of default concerns leading to short-dated protection finding limited takers. A positive basis of 30-40bp on a 7-10yr bond can add as much as 3-4 points in upfront terms.

exhibit 6

Telecom Italia – Basis Across the Curve



Source: Morgan Stanley Research

THE CREDIT-LINKED NOTE CULTURE

Many investors like today's spread levels but are grappling with the problem of low absolute yields. It is tempting to lock in to historically wide credit spreads and add a lot of credit duration, but this comes with rate exposure as well. A possible solution would be to diversify credit portfolios away from fixed-rate bonds into longer-dated FRNs (see [Rate Relief](#), October 23, 2009). However, historically, the opportunity set for FRN investors has been rather limited, either in terms of availability of specific credits or the maturities. Moreover, corporates will be highly motivated to lock in yields at today's historically low levels and are more likely to issue fixed-rate debt. We think the answer lies in CDS 101 – take synthetic exposure to credit via vanilla credit-linked notes, in names and maturities customized to an individual investor's appetite. We've recently started hearing more discussion of single-name CLN products being issued.

A credit-linked note is simply a synthetic way of creating exposure to credit, but in funded form much like a bond. However, the culture of CLNs involves creating exposures that are difficult to find in the cash bond market, and the vast majority of CLNs issued in the corporate credit space are linked to tranches of CDS portfolios, motivated both by scale of a portfolio approach and the leverage of mezzanine-type exposures in the structured credit capital structure. But the market did not start this way, as early in this decade, it was common to create CLNs that referenced single issuers. The rationale was to gain exposure to a credit that had limited bond availability, attractive pricing in CDS relative to cash bonds, or a different interest rate exposure (floating rate instead of fixed). There is, however, an additional risk factor as funding collateral generally has credit risk as well, so both the risk of that collateral and the additional yield of it must be taken into consideration.

In today's market, we do see the rationale for the creation of single-name CLNs motivated by all three factors above, although the positive basis and the floating-rate nature of a

CLN might be the most attractive elements, given the potential interest rate environment. Indeed, in a return to an old culture, we have seen a reasonable flow in single-name CLNs this year, particularly in Europe, where the flat-to-positive basis opportunity is bigger than in the US.

Away from single names, the rationale for investing in CLNs of CDS portfolios makes sense, although it will be hard to source a credit portfolio of names where the basis on average is flat to positive, so the real motivation needs to be scale of portfolio construction and potential floating-rate exposure in an environment of rising rates. The bottom line is that we believe that the CLN culture makes a lot of sense in today's market environment of a more compressed basis and floating-rate exposures, not to mention the relative value of tranching in the key mezzanine part of the structured credit capital structure.

CDS UPFRONTS AND RATE BENEFITS

Earlier this year, the CDS market made dramatic changes to trading conventions with the SNAC protocol. We've written extensively about these changes and their effect (see [CDS Sea Change](#) and [CDS SNAC – A Large Market Reinventing Itself](#), March 6, 2009 and April 17, 2009). In these pieces along with Chapter 1, we've highlighted the impact of changing the running spread on single-name CDS to an upfront-plus-standard-coupon structure.

In the past, an all-running coupon was the standard CDS structure, so if an investor were to go long risk in a single-name credit by selling protection, there would be no upfront payment. Post-SNAC, there is still a running coupon of either 100 or 500bp, but there is an upfront component as well. In the US market today, about half the names trade where the seller pays the upfront, and the other half where the buyer pays the upfront.

We have not yet discussed the implications of these changes in a rising-rate environment. If the upfront value is positive, which we define as a situation in which the seller of protection receives the upfront payment in addition to the running coupon, the benefit is to the seller of protection, or the investor going long the credit risk. The opposite is true of the buyer of protection, who should prefer to receive the upfront as well.

In addition to the standard funding reasons we've discussed in the past, the benefit of receiving an upfront prior to a rise in rates is that as rates rise, the discount factor applied to PV the running coupon increases as well, thus *lowering* the present value of the remaining coupon. To illustrate, if a CDS were trading at 600bp and had a 500bp coupon and a duration of 4, the upfront would be 4% (100bp x 4). Even if credit risk on the single name stayed the same, the 500bp coupon would be subject to MTM fluctuations as rates changed. Alternatively, if the coupon had been 100bp, the upfront would be much greater and the coupon at risk would be much lower.

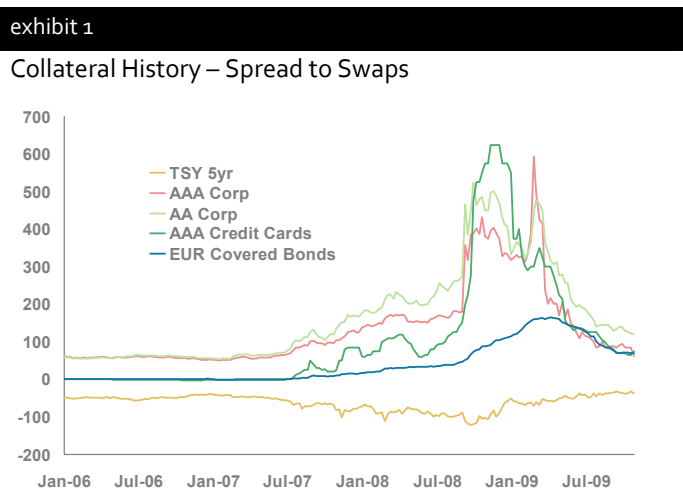
Collateral – Full Faith or Credit?

November 20, 2009

Ashley Musfeldt
Sivan Mahadevan
Phanikiran Naraparaju

Of the many unexpected side effects of the credit crisis, one was the impact of both mark-to-market volatility and correlation of synthetic credit assets with the funding collateral used to create credit-linked notes (CLNs). The hybrid nature of CLNs, or more broadly structured notes, exposes investors to both the underlying market risk of the instrument being securitized and the funding collateral used to provide the “Libor” component of the yield. The increased correlation and MTM volatility of the funding collateral with respect to the underlying risk was felt not just in credit markets, but equities and commodities as well.

If we focus specifically on credit-linked notes in the corporate credit markets, much of what was issued was bespoke mezzanine tranches, in funded form. In the 2005-2007 heyday of bespoke mezzanine issuance, much attention was paid to tranche thickness, structural leverage, rating agency model inputs, portfolio selection and tranche maturity. The primary goal was to achieve a reasonable yield for underlying market risk, and with much of CLN funding assets trading at Libor flat and little to no historical volatility, much more attention was paid to the credit risk in the synthetic portion of the CLN than to the collateral.



Source: Morgan Stanley Research

The credit crisis brought about much change to these previously staid assets, and consequently many types of collateral experienced MTM stress during this period, though certainly some more than others. In Exhibit 1, we illustrate many common sources of CLN funding and their relative performance throughout the credit crisis. Immediately several themes emerge. First, bonds with less or no credit risk fared

the best, including US treasuries, and covered bonds in Europe. On the other end of the spectrum, securitized products fared the worst. While some of the MTM experience in these assets was and is attributable to a genuine repricing of fundamental risk in such assets, many of these suffered from contagion as the market as a whole suffered.

We devote the first part of this chapter to discussing several types of collateral in existing CLNs. These include:

AAA credit cards. These were popular in the 2005-2007 issuance period, and while they widened considerably during the peak of the crisis, we think some of this was due more to regulatory uncertainty than an actual increase in default risk. This asset class is more important for existing CLN holders, as we think other types of collateral will be more likely in new issue CLNs.

Covered bonds. These were popular in European CLNs, and as far as collateral types go generally, these have held up pretty well throughout the crisis.

Corporate bonds. As these were required to have a rating equal to or higher than the synthetic tranche in the CLN, AAA corporate bonds were highly coveted.

GICs. These guaranteed investment contracts were also used to fund many CLNs. With recent monoline stress, many of these have been unwound in the last year or so, though some remain.

The good news is, in today’s market, there are options for funding CLNs that retain the positive qualities of these earlier funding vehicles while further mitigating some of the risk factors that came up during the crisis, such as concentration risk, correlation to the credit swap in the CLN, and risk to the issuing dealer. In the remainder of the chapter we discuss our favored approaches to funding collateral, including the following.

MMFs and Treasuries. Depending on the geographic region, we think these lower-risk, higher-liquidity instruments would mitigate some of the funding concerns that arose in 2008-2009, as they have dramatically less MTM volatility than their corporate and securitized counterparts. Specifically, Treasuries offer greater relative safety and money market funds (MMFs) offer greater market diversification.

Repos. While these were used sporadically before the crisis, we expect repos to be even more common, as they are more effective at mitigating default risk of the issuing dealer.

CURRENT COLLATERAL – CORRELATION AND CONTAGION?

The recent focus on SPV collateral has shined a spotlight on the funding assets used during the structured credit issuance heyday. There were several assets that were used, though the majority were one of four types, depending on the geographic region of the SPV and the issuing dealer.

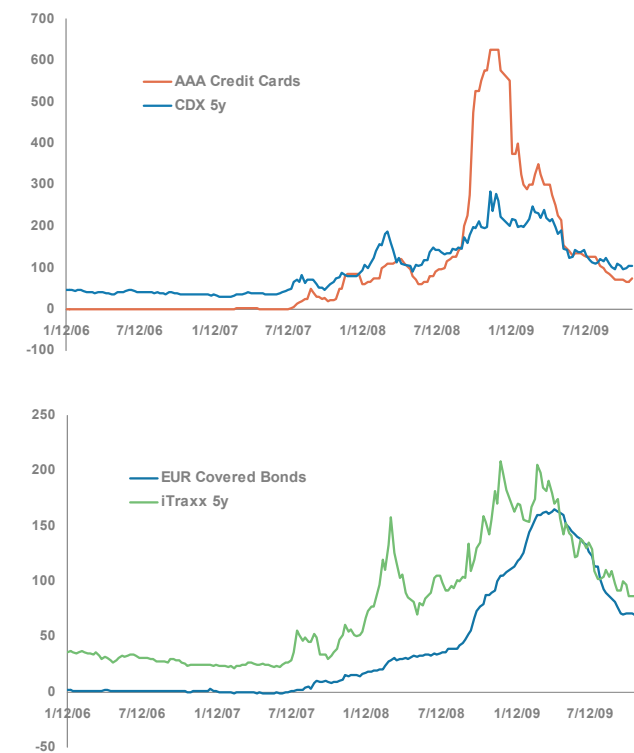
Credit card securitizations were one popular asset used as collateral, and they have been much discussed recently, owing primarily to the regulatory ambiguity surrounding credit card securitizations throughout 2009. Traditionally, these assets were protected in the event of a bankruptcy by the credit card issuer via the FDIC safe harbor provision – see Moody's report: *Safe Harbor Uncertainty Leads to Uncharted Waters for Card ABS*, September 25, 2009 for a full explanation. Originally, there were concerns that Moody's might downgrade some of these AAA assets (as virtually all of the credit card securitizations used to collateralize structured credit CLNs were AAA) based on the senior unsecured rating of the issuing bank.

A second popular funding source was covered bonds, particularly in Europe. These had additional credit protection for the holder, as they were secured by the on balance sheet assets of the issuer and effectively have second-to-default risk to both the secured assets and the issuing institution.

We use these two first examples to highlight an important point when considering and comparing collateral types – that of correlation to the underlying swap. In Exhibit 2, we illustrate this point by comparing credit cards to the CDX index, and covered bonds to iTraxx. While both of the funding assets widened in the systemic crisis, covered bond spread movement was much more muted, and only briefly broke wider than the general market. Credit cards, on the other hand, were not only much more correlated with CDX, but also, at the peak, traded multiples above the credit index.

exhibit 2

Collateral Correlation



Source: Morgan Stanley Research

Corporate bonds were also used as a relatively simple and straightforward funding vehicle, and the primary concern was that the bonds had a higher rating than the tranche in the swap portion of the CLN. This made GECC and Berkshire Hathaway bonds two very popular corporate bonds used in SPVs. During the credit crisis however, these widened dramatically, and in Exhibit 1, we show that AAA corporate bonds reached wides around 600bps.

Finally, guaranteed investment contracts, or GICs, were used frequently as collateral in CLNs. We highlighted these earlier in the year during our distressed bespoke mezzanine discussions (see Chapter 31). These have the benefit of additional margin requirements when the monoline or other issuer breaches certain ratings thresholds, but they are somewhat opaque and more difficult to price. Many of these have already been unwound, though there are still a fair number of these in existence with the more stable monolines.

THE FUTURE OF COLLATERAL

While many of these collateral types in existing CLNs have tightened in from the wilds at the peak of the credit crisis, we still expect that different types of collateral will be more popular than some of those used in the past.

First, we think that the safe havens of Treasuries and money market funds (MMFs) will be used with more regularity. Both have been used in CLNs in the past, but due to their sub-Libor funding and less focus on collateral MTM issues prior to the credit crisis, they weren't as popular as some of their securitized and corporate bond brethren. Today we're seeing existing SPVs as well as new SPVs taking a closer look at these safer havens, for varying reasons. Both are stable and exhibited remarkably low volatility throughout the crisis. Additionally they are far less correlated to the swap portion of the CLN, whether that is a tranche or a single name. Finally, MMFs offer a greater degree of diversification than using a corporate bond referencing a single entity, or a securitized product referencing a specific set of cash flows.

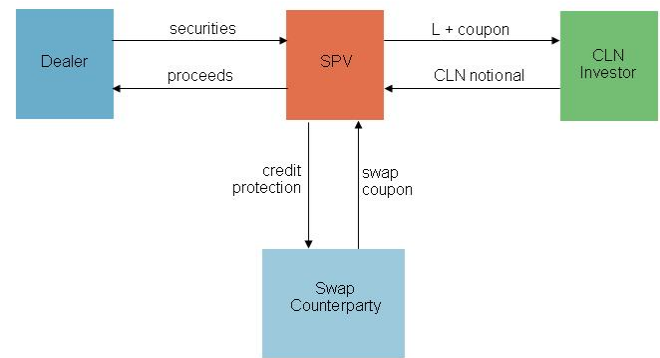
Repurchase agreements present another alternative investors are using to fund CLNs. In repos, the SPV that will issue the CLN will buy securities from a sponsor, usually a dealer, and in return will have the right to sell them back on a certain date at a pre-set price. In the specific instance of CLN collateral, these repos will be sold back at par a few days before the CLN matures. These securities used to fund the repo can be anything from Treasuries to corporate bonds to securitized products, but the difference here is that the dealer is on the hook for any MTM fluctuations and credit risk in the securities, instead of the CLN noteholder.

If there are MTM changes on the securities, the dealer is required to post additional collateral to the SPV as margin. Additionally, if the securities experience any losses, the dealer is responsible for those as well. At end of trade, the SPV sells back the securities, receives the predetermined price, and pays off the CLN notes with the proceeds.

Besides MTM protection, another major reason for using this type of agreement is that the CLN holder has additional credit protection against the dealer who issued the notes. If the sponsoring dealer were to go into bankruptcy, the repurchase agreement would terminate; however, the SPV would own the securities, which would not be available to the creditors of the issuing bank. Furthermore, theoretically any current MTM fluctuations in the price of the securities should have had margin posted against them, and thus the SPV could theoretically be unwound with collateral at par.

exhibit 3

Repurchase Agreement Mechanics at Inception



Source: Morgan Stanley Research

While the possibility of bankruptcy would have seemed extremely remote just two years ago, the stress and failure of many large financial institutions have shown this is not impossible. Thus the motivation for using this type of agreement as collateral is very clear: in effect, the investor has “second-to-default” risk to the collateral AND the dealer – both have to experience losses before the CLN holder would.

A Credit-Rates Booster Shot

September 10, 2010

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What happens in the interest rate markets is inherently important to credit investors and today's low-rate environment is of particular importance. Traditionally, most investors in investment grade corporate bonds have not been directly responsible for managing interest rate risk, even though much of the time the interest rate risk of a high-grade bond can dominate the credit risk. In funds managed to fixed income market benchmarks, interest rate duration is generally set and adjusted by the interest rate teams, and in insurance companies, liabilities are effectively the hedge for interest rate risk. The corporate credit risk is essentially an overlay strategy that adds yield and an alternative source of alpha (credit selection).

However, over the past decade, a new type of investment grade investor has entered the market, one not tied to fixed income indices but instead looking for absolute returns (53% of our recent global survey participants cited absolute returns as their target). Prior to the credit crisis, this investor primarily used CDS contracts and related structured instruments, such as indices and tranches, owing to their better liquidity and ability to isolate credit risk. Higher levels of short-term rates made Libor-funded structures attractive from an all-in yield perspective. Just after the credit crisis, this community turned away from CDS and focused simply on corporate bonds, as both the credit and funding crises made capital dear and bonds significantly cheaper than CDS.

Using cash instruments worked like a charm in 2009 as the credit component of the total return dominated the interest rate component, despite the fact that 10-year Treasury yields rose over 160bp. In 2010 to date, the total-return party has continued, but for exactly the opposite reasons. Credit spreads have been volatile yet largely range-bound while the 10-year Treasury yield has returned to within 40bp of early 2009 levels.

The recent strong contribution of rates poses challenges for future performance of corporate bonds. If rates were to rise, negative total returns driven by higher rates could well create a negative overhang for the absolute-return investor in credit. The sky-high dollar prices in the cash market (average prices of IG non-financials bond in EUR, USD and GBP are 112.4, 115.2 and 116.6 respectively) are also likely to prove a deterrent.

While potential deflation risks, monetary policy and double-dip debates, combined with strong fixed income fund flows can keep interest rates low for a while, seeking absolute

returns in corporate bonds at such high dollar prices requires having at least a neutral if not positive view on Treasuries and higher quality Euro government bonds.

For absolute-return investors, the question is how one can position a positive view on credit without a strong view on benchmark government bonds. The obvious answer is to take credit risk without interest rate risk, and both floating rate corporates and CDS are the easiest solutions. However, simply buying floaters today is a strategy that would underperform fixed rate bonds in a scenario of benign moves in interest rates, as the extremely steep government yield curves provide a significant yield boost. We devote the remainder of this chapter to credit and interest rate hybrid solutions that can use the current level of rates, forward expectations for rates, and rates volatility to position absolute-return-oriented credit investments in today's low-rate world. Our key themes are as follows:

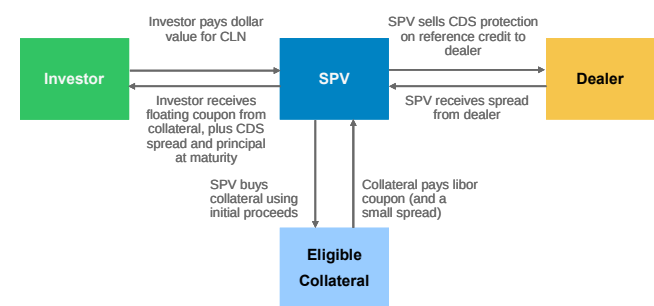
- Steep Libor curves imply higher short-term rates over time. However, the rise is gradual and buying vanilla FRNs is only part of the answer.
- To counteract low short-term rates, we favor CLN structures where yields are boosted through a simple combination of a rate floor and cap strategy that would effectively position investors for the short-term rates that are implied by forwards today, but avoid most interest rate duration exposure unless short-term rates rise above cap levels (using only a floor would avoid this risk).
- The cost of such a "collared" interest rate strategy today ranges from 30 to 100 bp per annum, and we believe that credit selection through single names or structured solutions can more than pay for this premium quite easily.
- We provide a variety of single-name CLN solutions today where a positive cash-CDS basis is a motivator and which can be used to fund the interest rate yield boost. These structures' expected yield is at least as much as the equivalent fixed-rate bond yield of the same issuer today.

CREDIT LINKED NOTES VS. CASH BONDS

Credit linked notes (CLNs) form the backbone of most credit and interest rate hybrid structures, and we briefly summarize them here. CLNs involve a CDS and a funding component. An SPV is set up, and the investor pays par to the SPV, which in turn purchases eligible collateral to earn a LIBOR-based coupon. The SPV enters into a swap agreement with a dealer. The collateral can also be substituted by an issuer (i.e., the funding amount is retained by the issuer) and the investor is exposed to the risk of the issuer as well as the underlying CDS in this case.

exhibit 1

Typical (SPV) Credit Linked Note Structure



Source: Morgan Stanley

These SPV structures can be used to convert synthetic forms of credit into funded structures, and are exposed to the credit risk of both the funding collateral (or issuer) and the CDS. Yield on such structures can be boosted through structural leverage as well, for example by fixing recovery assumptions on the CDS (at 0% for example). This is similar to a corporate hybrid security (where issuance is high today). These hybrids are subordinated to traditional senior unsecured debt and are effectively ways to improve yield by accepting the risk of a lower recovery in a default.

Apart from the fundamental case for managing rate risk, a positive basis (CDS trading wider than bonds) supports taking risk through CDS-based structures. The basis is near flat now in the US and positive on average in Europe, driven by strong bond flows. The premium part of the bond price is not factored into a recovery, and thus this portion of the price assumes “zero recovery,” which can further positively skew the basis. Hedging flows on the CDS side and liquidity support on the cash side have resulted in CDS trading wider than cash bonds in several cases. Notably, the sector with the best positive basis opportunities is Western European sovereigns, followed by financials and peripheral utilities.

exhibit 2

Basis Is Positive in Europe, Near Flat in the US



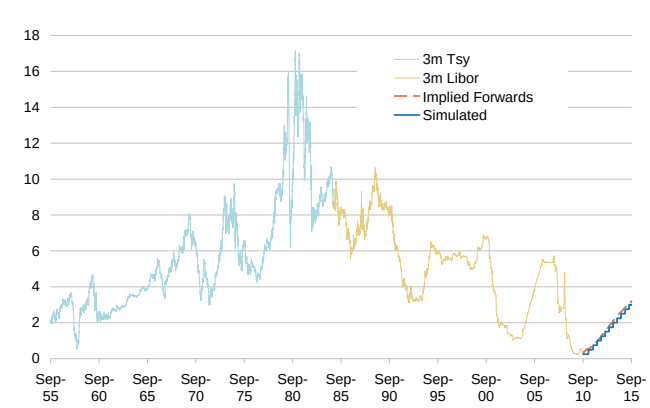
Source: Mark-It, Bloomberg, Morgan Stanley Research

WHAT ARE INTEREST RATE MARKETS TELLING US?

Before exploring different types of rate exposure that the CLNs can be used to achieve, we put current rates and expectations in context. In Exhibit 3, we show the implied forward rates and a simulated scenario of rate hikes relative to historical rates. The red line shows the forwards implied by the market (USD 3-month rates), which are similar in path to the simulated rate hikes. It is pricing in just about 300bp cumulative rate hikes by 2015 (about a 25bp hike every six months). That is conservative, both in terms of pace and magnitude compared to previous rate cycles, in which we saw rates rise as much as 400bp in a shorter span than is being priced into the forwards today. In Exhibit 4, we show the expected forward rates in EUR, USD and GBP. This context is important in terms of evaluating the coupon caps and floors in rate hybrids as we discuss later.

exhibit 3

Historical Context 3M Rates



Source: Federal Reserve, Bloomberg, Morgan Stanley Research

exhibit 4

Implied 3M Forwards – A Gradual Rise

	Forward Rates (3m)		
	USD	EUR	GBP
Sep-10	0.3%	0.9%	0.7%
Dec-10	0.5%	1.0%	0.8%
Dec-11	0.8%	1.2%	1.6%
Dec-12	1.5%	2.2%	2.1%
Dec-13	2.2%	2.2%	2.7%
Dec-14	2.8%	2.6%	3.2%
Dec-15	3.3%	2.9%	3.6%
Dec-17	3.8%	3.3%	4.3%
Mar-20	3.9%	3.5%	4.5%

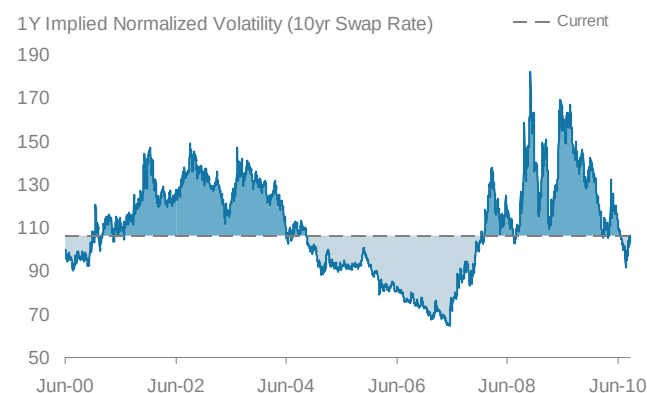
Source: Federal Reserve, Bloomberg, Morgan Stanley Research

The other relevant factor in looking at hybrids is rate volatility, which is currently in the middle of historical ranges (Exhibit 5). That said, realized volatility in interest rates is as high as implied volatility and is currently less expensive than credit and equities, where options-implied volatility is well above realized volatility.

Also noteworthy is that volatility has tended to be the highest when rates are relatively low (see 2002 and 2009) and vice versa, which implies that when buying options on rates, buying receiver options generally work because volatility and rate directionality work in tandem. Conversely, in a rising-rate environment, implied volatility tends to ease, reducing MTM pain for the option seller.

exhibit 5

Rate Volatility in the Middle of Historical Ranges

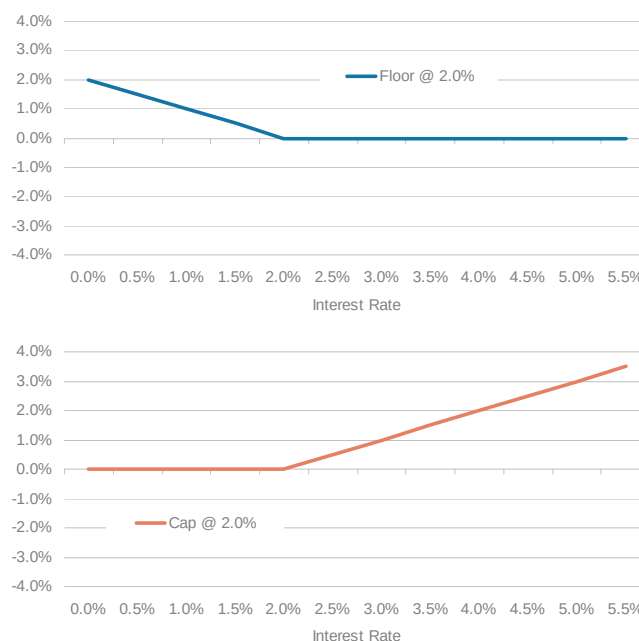


Source: Bloomberg, Morgan Stanley

If we combine both forward expectations for short-term rates and volatility levels, we can see what the costs are for introducing Libor floors and caps in various markets. Exhibit 7 summarizes indicative spreads for floors and caps set at different coupon levels and expressed as an annual spread. For example, if we were to create a USD Libor floater in which 3M Libor was capped at 2.5% over five years, we would earn 34 bp annually for selling this option. Conversely, if we were to floor Libor at 2.0% (compared to the current level of 0.30%) for five years, it would cost 89 bp annually. If we combined both, we could create a collared Libor floater (with coupon ranging from 2% to 2.5%) for a net cost of 55 bp annually.

exhibit 6

Interest Rate Floors and Caps



Source: Morgan Stanley

exhibit 7

Running Cost (bp) of Floors and Caps on 3M Libor

Strike	Floors			Caps		
	EUR	GBP	USD	EUR	GBP	USD
1.0%	16	-	29	94	-	109
1.5%	42	-	53	66	-	78
2.0%	80	75	89	46	65	51
2.5%	122	112	127	33	49	34
3.0%	168	155	169	25	37	22
3.5%	217	200	215	19	29	13
4.0%	268	248	264	15	24	8
4.5%	320	298	316	12	19	4
5.0%	373	349	368	10	16	2

Note: Illustrative only.

Source: Bloomberg, Morgan Stanley

CREDIT-RATE HYBRIDS REFERENCING SHORT-END RATES

So this quick tour through interest rate forwards and volatility provides a bit of background to help us solve our original problem, namely the risk that rising rates can negatively skew corporate bond total returns substantially. As we mentioned in the outset, we favor CLN structures that limit interest rate risk. The issues are that pure floaters will yield less today than fixed-rate assets if rates remain low or even rise modestly. So what is the solution? To counteract the low level of short-term rates today, we go back to our previous example in which we introduce a Libor floor and a cap into a CLN, with a combined cost of 55 bp. What we have created is essentially a CLN in which the interest rate portion of the structure would fluctuate between 2% and 2.5% over the life of the structure, for a net cost of about 55 bp.

So now the question is, can we find credit opportunities that are cheap enough (by 55 bp) to the bonds to fund such an interest rate structure? In some sense, we are using CDS cheapness relative to bonds to pay for interest rate hedges, thereby locking in rates in a range that often surpasses longer-term yields on bonds, but with less interest rate duration risk.

CONCLUSION

We note that such CLN structures that limit interest rate risk are similar to some of the floating rate opportunities we see in the corporate markets, though these are less common for core investment grade and sovereign credits. For example, leveraged loans often incorporate Libor floors, a feature that paid investors well as monetary policy pushed rates to current levels. In the financial sector, there is a considerable universe of floating rate instruments, including banks senior FRNs and LT2 FRNs. There is also a large Tier 1 asset class with fixed-to-floating coupons. In addition, there is a smaller set of bonds linked to the 10-year swap rate (CMS) or government bond yields, as well as bonds linked to the shape of the swap curve. While not very common, some bonds also embed floor and cap features. The bottom line is that there is precedent for such interest rate structures in vanilla bonds and loans, so we find structured solutions with the same goal to be appealing.

Learning Curves

November 2, 2010

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The yield compression on the back of strong global liquidity is contributing to a resurgence of a variety of carry trades, curve trades being one of them. While we have been through a tumultuous credit cycle and have some good CDS curve experience under our belt, credit curve relationships are arguably less mature than in the rates world. The existence of liquid curves where investors can go long and short to different dates implies that investors can position for “forward” credit risk, a concept that is still relatively nascent compared to the interest rate world, where curves are in fact monitored closely as indicators of market activity and trends.

Fortunately, we can borrow quite a bit of math and market experience from the interest rate world in determining forward credit spreads, but there are key differences as well. Most importantly, credit instruments are “risky” assets, and thus, any calculation of implied forward rates must take into account the probability of default. As credits become more risky from a default perspective, the “pari passu” claim of all bonds (regardless of maturity) in default is a powerful force that will push credit curves flatter and eventually inverted, a trend that lasted for over two years during the credit crisis. Furthermore, while the credit markets have matured in the last several years, we still have nowhere near the granularity and liquidity of the rates markets regarding term structures.

Given this background, it is clear that the drivers of credit term structures are significantly different from what we observe in interest rate markets, even though some of the math is similar. In early- to mid-cycle periods, like we are experiencing now in both the US and Europe, it is common for credit curves to be steep and remain steep, as early deleveraging and the “cleaning up” of weaker companies from a default risk perspective tend to point to low levels of near-term default risk while investors still demand default-risk premium and term premium for longer maturities. As such, it is common for investors to focus on curve steepener trades, given their appeal of being both positive carry and long near-term default risk. Later in the cycle, a reach-for-yield phenomenon can cause a bull flattening of the credit curve as investors look to longer maturities to meet yield boogies.

In this chapter, we review credit fundamental and technical drivers of credit curves, incorporate what we have learned from the credit crisis, take into account the economics and motivation of curve trades and highlight current market ideas. We see the following themes:

Investment grade credit. Historically the incremental default risk of the last five years of a 10-year investment grade credit is about 59% higher than the default risk in the first five years. This implies a naturally steep curve in these less risky credits. IG curves reflect this today, with the 3s5s curve in US already quite steep, although 5s10s steepeners in both US and Europe still make sense.

High yield curves. Refinancing and liquidity risk are important for HY curves, and are somewhat different in the US and Europe. While both cash and free-cashflow levels are close to cycle highs in both regions, the European HY universe faces a refinancing profile that is front-ended relative to the US. So far, a strong liquidity profile is trumping refinancing concerns and driving the steepening trend. We expect curves to remain steep across both regions but have greater fundamental conviction in the US than in Europe.

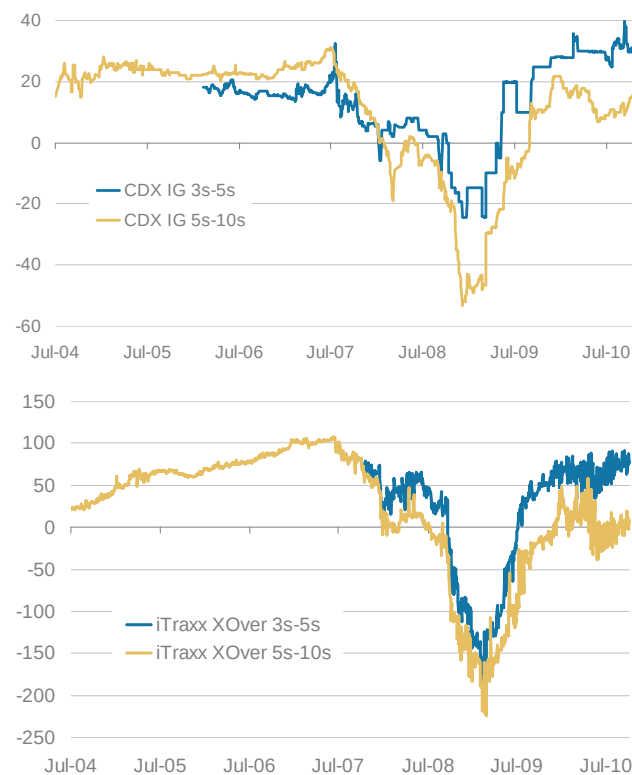
Technicals. Demand for risk managing notional exposure from loan syndicates and trading books keeps short-dated spreads elevated when there is anticipated issuance. Structured credit flows also have important implications for curve shape, particularly in the IG market, and given the continued importance of credit ratings, we expect to see a steepening bias here as well.

Regulation and risk management. Independent of the default arguments, investors may demand more risk premium for longer duration assets versus shorter duration assets to account for greater mark-to-market volatility, a trend we see in new regulatory frameworks (e.g., Solvency II).

Economics of curve trading. Carry and rolldown on curve trades can actually drive curve relationships as much as some of the debt distribution and historical default risk relationships. We show different styles of curve trading. Although curves have steepened recently, attractive steepener economics kept curves consistently much steeper between 2003 and 2007.

exhibit 1

CDX IG and iTraxx XOver 3s5s and 5s10s History



Source: Morgan Stanley Research

WHAT CAN WE LEARN FROM RATES?

In a nutshell, a forward interest rate is simply the break-even rate that makes all investments on the curve equally rewarding. If the forwards are realized, an investor should be indifferent about which point to invest in on the curve. As such, forward curves are important inputs into risk-neutral interest rate derivatives pricing models, which assume, among other things, that there is no relative value among various opportunities, given market pricing. The following equation shows the calculation of one-year implied forward rate starting at the end of year 1, $F_{1,2}$, given the one-year spot rate S_1 and the two-year spot rate S_2 :

$$F_{1,2} = (1 + S_2)^2 / (1 + S_1) - 1$$

The first step toward calculating implied forward rates is to calculate default probabilities for each payment period. To simplify, let us assume that we have two default swap contracts, CDS_1 and CDS_2 , maturing at the end of year 1 and year 2, respectively, with annual spread payments. Now we can determine the implied probability of default at the end of year 1 from CDS_1 , given a recovery rate. Similarly, given the probability of default in year 1 and CDS_2 spread level, we can calculate the probability of default in year 2, given the reference entity does not default in year 1. Thus, we can impute default probabilities for each period from a whole

credit curve. The combination of CDS_1 and a forward default swap, which starts at the end of year 1, replicates CDS_2 . Therefore, by equating the two cash flow streams, we can determine the implied forward default swap level.

Although calculating implied forwards from a spot curve involves little more than basic algebra, it is relevant because it represents a ‘no arbitrage’ relationship between instruments that are tradable in the market. We would argue that if the implied forward rates from a swap curve deviate widely from realized rates, one should not be surprised if credit spreads also ultimately diverge from what ‘forwards’ have implied, although we have less experience with the latter. After all, rate forwards have a number of inherent advantages that should make them a ‘cleaner’ calculation. A lack of default risk is a big one, but better liquidity throughout the swap curve, especially at 1Y and 2Y points, should not be overlooked. However, the unique nature of a credit forward relative to these other assets is important to consider. The underlying idea of a forward is the same in credit, FX, or rates: what is the level one is willing to receive in the future that makes one indifferent to what can be locked in today. But while the ideas are the same, credit forwards contain unique properties that need to be considered.

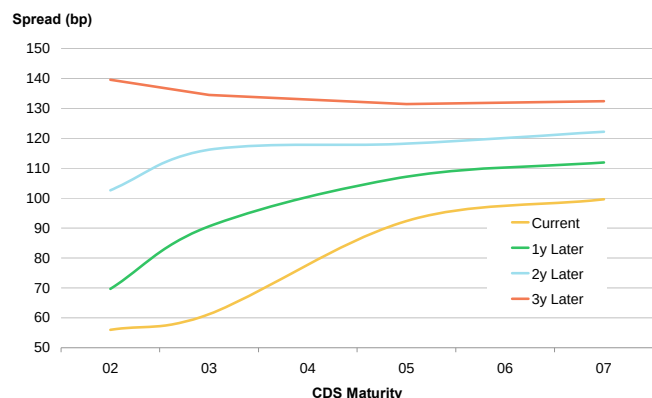
Default risk. A credit forward trade involves a risk of default that would cancel the forward, a risk that does not exist in FX or rates. Let’s consider two ways an investor can take 10-year credit risk: Selling 10-year protection today, or selling 5-year protection today and 5-year protection, 5 years forward. If the curve is upward-sloping, in a default the investor was better off doing the former (selling the 10-year protection), as they pocketed more carry for the same loss. The greater the default risk on the first leg (5-year spot) of the trade, the higher the spread that should be demanded for the second leg (5-year, 5 years forward) in order to keep an investor indifferent, assuming an upward sloping curve.

In this way, we caution that “forwards” in the credit market are less expectations of future spreads than they are a measure of indifference between the two types of trades mentioned above. It is the spread an investor is indifferent to receiving assuming that the credit survives to that point and, as such, it must compensate an investor for all the scenarios in which it might not survive. It will therefore overstate the spread we would expect to see if we, say, held an option to invest 5 years forward, and could therefore check for survival. This overstatement will be very small at tighter spreads, but it increases with default risk and curve steepness.

Liquidity. Another factor affecting the ‘efficiency’ of forward trades is that liquidity issues can make the forwards harder to monetize. Going long 7-year CDX 15 and short 5-year CDX 15 is a long 2-year position, 5 years forward, constructed from two of the most liquid points on today’s curve. But 5 years forward, the market will trade CDX 25, and the market for 2-year CDX 15 risk could be diminished.

exhibit 2

CDX IG Forward Curves Are Wider and Flatter



Source: Bloomberg, Morgan Stanley Research

Ratings migration risk. When thinking about forward spreads in corporate credit, we note that any company can look different several years down the road. A steep curve in an investment grade name could simply represent an expectation that the entity will be lower rated in five years time, rather than a mispricing.

DEFAULT TERM STRUCTURE AND CURVES

In theory, credit curve shapes ought to be dictated by fundamental drivers of default risk along the curve. Credit migration and evolution of default risk over time are an important factor. We only have history as a guide, but when we look at incremental default risk from a ratings perspective, we find a striking difference. On average, considering 40 years of default history, the incremental default risk for an IG credit is higher in the last five years rather than the first five years. In fact, five-year default risk five years forward is 59% higher than default risk of the first five years (see Exhibit 3). From AAAs to CCCs, the forward increment falls steadily, so AAAs are the riskiest by this measure (should have the steepest relative curves) and CCCs are the least risky (should have the most inverted curves). The BB point is flat (0%), while BBBs are very positive (45%). But averages can be misleading, and given the cyclical nature of default risk, if we look over time (cohorts), we find that BBs have periods of negative forward default risk (see Exhibit 4).

exhibit 3

Incremental Default Risk in IG and HY

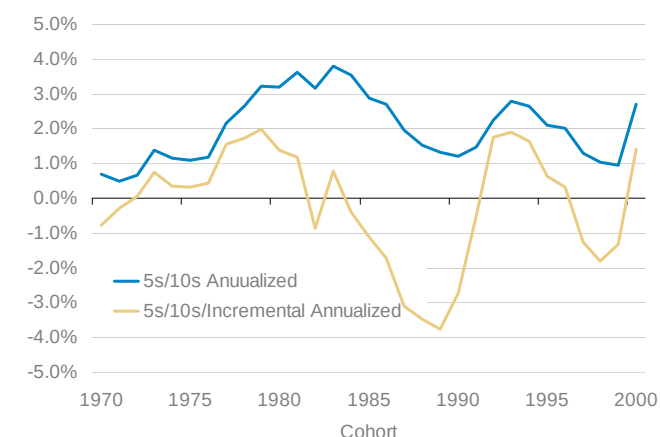
	Annualized Default Rate					3s/5s to 3yr	5s/10s to 5yr
	3-Yr	5-Yr	10-Yr	3s/5s	5s/10s		
IG	0.14%	0.19%	0.24%	0.27%	0.30%	96%	59%
Aaa	0.00%	0.02%	0.06%	0.05%	0.09%	n/a	314%
Aa	0.02%	0.05%	0.08%	0.09%	0.11%	277%	113%
A	0.09%	0.12%	0.17%	0.18%	0.22%	110%	93%
Baa	0.30%	0.40%	0.50%	0.54%	0.58%	84%	45%
HY	3.7%	3.6%	3.3%	3.2%	2.5%	-15%	-31%
Ba	1.7%	2.0%	2.2%	2.3%	2.0%	34%	0%
B	5.3%	5.1%	5.1%	4.1%	3.6%	-23%	-30%
Caa-C	15.3%	12.5%	12.4%	4.6%	5.5%	-70%	-56%

Source: Morgan Stanley, Moody's

In HY, the incremental default risk between years five and ten is lower than years zero to five for high yield credits (by 31%), suggesting that 5s10s credit curves ought to be inverted. This makes intuitive sense, given that higher levels of leverage imply near-term default risk for most speculative grade companies.

exhibit 4

5s10s Incremental Default Risk – Very Cyclical



Source: Morgan Stanley, Moody's

REFINANCING AND LIQUIDITY PROFILES

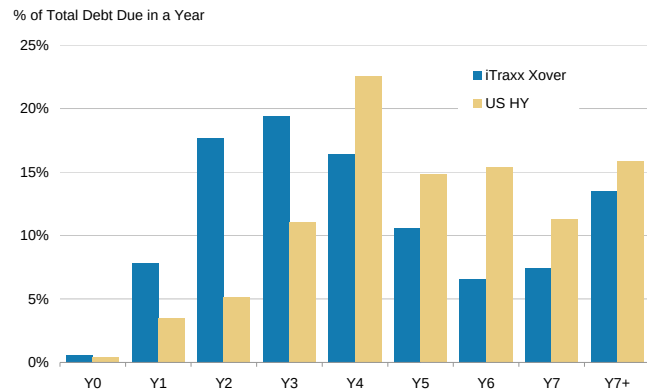
Near-term default risk for high yield companies is driven by refinancing cycles and maturity walls, and we have seen numerous examples of credits where large and concentrated debt maturities influence curve shape. Today, we see a meaningfully different profile between the US and Europe.

In Europe, the rash of LBO financings pre-crisis meant that the maturity wall was back-ended, with peak maturities at the 5- and 7-year points (see Exhibit 5). With the passage of time, that maturity wall is now closer with the peak falling between 2012 and 2014. The average iTraxx XOver credit today has more than 70% of its total debt due by year five. The US maturity profile, on the other hand, is more back-ended, with the peak maturities in years 4 and 5. This argues

for a more benign view on near-term default risk and thus steeper curves in the US relative to Europe.

exhibit 5

CDX HY and iTraxx Over Maturity Profiles



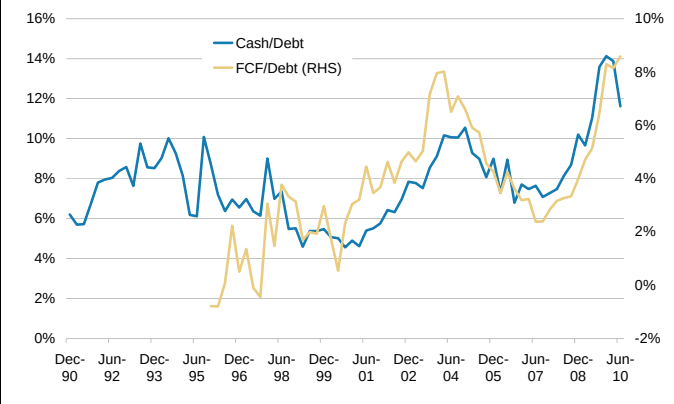
Source: Bloomberg, Morgan Stanley Research

The mixed refinancing picture in Europe should be examined in conjunction with the liquidity profile, including cash flows and liquidity buffers. In contrast to the different maturity profiles in the US and Europe, the liquidity picture is fairly strong in both regions. Names with negative net cash flows (often due to CAPEX requirements) or poor liquidity buffers tend to have flatter curves, which reflect the increased near-term default risk. In 2010, corporate liquidity and free cash flow trends have been extremely strong. In the HY space, a cash to total debt ratio of 14% is well above the peaks we saw in the recovery phase of the previous cycle. In addition, steep CAPEX cuts have also helped keep FCF close to the peaks of the last ten years (see Exhibit 6). The bottom line is, despite refinancing risks, the strong liquidity profile has pushed curves steeper in the US and Europe.

A clear expression of this tension between liquidity and leverage was the use of curve steepeners in names with LBO risk. Putting on IG shorts in anticipation of an LBO was a challenging trade, as the risk of not picking the right name is quite high. And even then, if the investor did pick the right name, the timing uncertainty regarding when the LBO would materialize made these tough negative carry shorts in an environment where structured credit flows steadily pushed IG spreads to tight levels. Instead, investors often used curve steepeners under the rationale that freshly LBO'd capital structures typically had stronger near-term liquidity but higher long-term leverage after sponsors refinanced all existing debt with 5-7yr maturities and often held or were willing to inject cash to support the buyout in the near term. We have a number of examples of curve steepeners that were successful in names where an LBO did materialize.

exhibit 6

Free Cashflow and Cash/Total Debt for US HY



Source: Bloomberg, Morgan Stanley Research

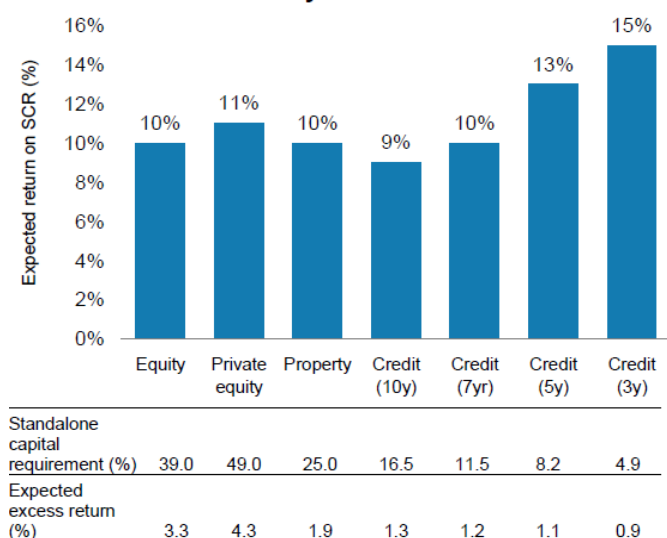
REGULATORY BACKDROP AND CREDIT CURVES

Another important factor that comes into play over the medium term is the regulatory backdrop. The credit crisis has had a profound impact on attitudes toward risk management and capital buffers. There is a clear shift in the mindset of market participants and regulators – from a ratings-focused, hold-to-maturity (HTM) centric world to a volatility-focused, mark-to-market (MTM) centric world. One expression of this can be found in Solvency II proposals.

The incorporation of spread shocks implies that longer-maturity bonds suffer bigger P&L moves and require greater capital than shorter-maturity bonds, whereas credit curves are relatively flat. Credit curves may be compensating for incremental default risk with time, but may not be paying a lot for the increase in volatility that comes with extension. As per our insurance equity research team, the expected return on SCR for a short-maturity credit (3yr bucket) can be 50% higher than that on a longer-maturity credit (10yr bucket – see Exhibit 7). Even if the volatility and duration focus is not enshrined in all regulatory regimes, there is a clear focus from management on MTM risk and implications for economic capital held. All this should favor shorter-maturity bonds.

exhibit 7

Solvency II Incentivizes Buying Shorter-Dated Credit on a Risk-Adjusted Return on Capital Basis



Assumes that liabilities are cash flow-matched using swaps and other interest rate derivatives.

Source: QIS5 technical specifications, Bloomberg, Morgan Stanley Research, Oliver Wyman

All of these fundamental factors are important, but over the last several years, ‘technical’ have also influenced the performance of curves as we discuss below.

DEFAULT AND JUMP RISK HEDGING

Hedging flows generated from anticipated or underwritten issuance is also a factor from time to time in curve shapes. Syndicates and underwriters have tended to use short-dated CDS to hedge execution risk in loan deals. In essence these underwriters have a curve flattener on, as they are long 5yr risk on the underlying loan and hedged with 1yr CDS. The advantage (or disadvantage depending on one’s perspective) of using short-dated CDS is that it has little duration risk, and consequently is often used to hedge jump-to-default risk.

At the peak of the credit crisis, pressure on trading books to reduce notional exposures (i.e., default exposure) caused the purchase of short-end protection, contributing to both the curve inversion as well as a perceived increase in default risk.

Today, we are in a completely opposite environment – central banks’ aggressive provision of liquidity, record levels of cash and high free cashflow in corporates, and the disintermediation of banks by loan reduction in favor of bond issuance are all factors making jump hedging less popular today. These are all factors in our conviction that short-end curves can remain steep in this environment.

STRUCTURED CREDIT FLOWS AND IMPACT ON CURVES

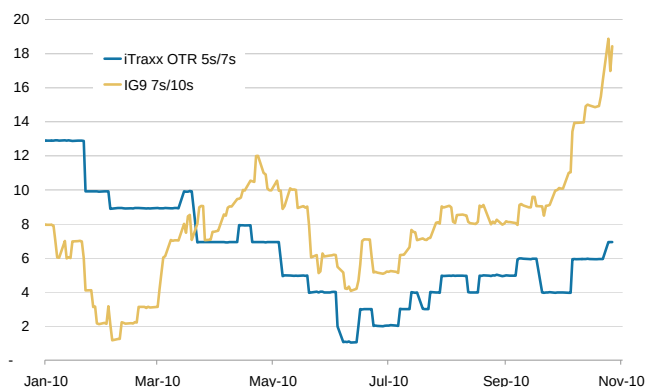
Structured credit activity, which was one of the important factors driving credit spreads between 2005 and 2007, was

also influential in terms of the shape of the credit curves. In the early stages of the cycle, much of the issuance was at the 5yr point due to the visibility on default cycles and still attractive yields. Thus, the early flows in structured credit activity caused 5s/10s curves to steepen. As 5yr spreads tightened under the weight of structured CDS BWICs, investors increasingly extended maturities to the 7yr point in a hunt for yield. As a result, the later phase of structured credit activity had a curve flattening effect, particularly the 5s7s curve.

Today, structured credit activity is but a fraction of peak volumes. Most activity is concentrated in the shorter dated 3yr and 5yr maturities as we are still very early in the product cycle. This is also reflected in the index tranche market where the 3yr and 5yr points in IG15 are most active.

exhibit 8

IG9 Curves Impacted by Structured Credit Activity



Source: Bloomberg, Morgan Stanley Research

MOTIVATIONS FOR CURVE TRADES

The inherent appeal of curve trades is the ability to trade one type of risk (default risk) for another (spread risk). Curve positions can minimize spread risk and retain default risk or vice versa, whereas outright long or short positions in a credit carry both default and spread risk. Short-dated CDS has the same default risk as long-end CDS but carries less spread duration.

Carry was a primary motivation for curve trades in the heyday of curve trading, and curve steepeners were popular for this reason. The carry in the curve steepener was compensation for taking purely default risk, and in 2006 and 2007, faith in central banks’ ability to provide liquidity and stave off default cycles was strong. However, like many carry trades, curve steepeners did not work well during the credit crisis. Most obviously, the sharp increase in defaults and concurrent rise in systemic risk caused curves across many assets to flatten and invert. In credit, this was exacerbated by the fact that curve steepeners had been a crowded trade going into the crisis. Intense pressure on trading desks to reduce notional exposures and counterparty

risk also contributed to curve trade unwinding, similar to the unwinding of many other notional intensive trades such as index arbitrage.

While we are in an environment where demand for carry and yield remains strong – an argument for steeper curves today – lessons from the crisis suggest that this strategy will be highly correlated to the broader liquidity environment as well as performance of traditional carry trades.

THE ECONOMICS OF TRADING CURVES

One point we would like to stress is that curve economics (i.e., carry and rolldown on curve trades) can actually drive curve relationships as much as some of the debt distribution and historical default risk relationships that we described.

In Exhibit 9, we show three different ways to execute 5s/10s steepener trade in CDX IG: (1) Notional-weighted steepener, (2) Duration-weighted steepener; and (3) Duration-weighted steepener with jump risk hedge.

(1) Equal notional curve trades – take spread risk instead of default risk. A notional-weighted steepener (in this case sell 5-year protection, buy 10-year protection of same notional in CDX IG) has no default risk until the shorter-leg (5yr) matures. But it does carry duration risk as the 10yr leg has a higher duration than the 5yr; in this case the steepener is net short duration (of about 3.6) and is exposed to market direction – it loses money for a parallel tightening of spreads and vice versa. The trade has no running coupon (as both legs have the same coupon) but has net positive rolldown (0.28%) as the 5yr point benefits from disproportionately steep short-end curves. Despite some directional exposure in the trade, notional-weighted steepeners are popular due to easy execution and liquidity.

(2) Duration-weighted curve trades – take default risk instead of spread risk. To further reduce the directional exposure, an investor can buy the steepener on a duration-

weighted basis (sell protection on 10MM of 5yr CDS, and buy protection on only 5.8MM of 10yr CDS, given the higher duration). This makes the position much less sensitive to parallel shifts in spreads, but there is exposure to curve reshaping. The trade-off is default risk during the life of the trade. The duration weighted steepener pays better (0.81%) than the notional weighted steepener (0.28%) as the investor is buying less protection on the long end and taking on default risk in the process. The duration-neutral steepener is popular in the market today because of the higher carry and rolldown and investors' comfort taking near-term default risk.

(3) Duration-weighted curve trades with hedging/monetization of jump. One way to manage this default exposure in duration-weighted trades is to buy/sell short-dated protection to the residual notional (for example, buy 4.2MM 1-year protection for the duration-weighted steepener). Buying short-dated protection helps reduce the default risk in the duration weighted steepener at the cost of reduced yield (of 0.67% compared to 0.81% on the unhedged steepener).

The use of 1yr CDS to manage jump exposures in curve trades is more prevalent in flattener trades. Duration-weighted flatteners are generally negative carry trades that benefit in a default scenario. One way to “monetize” this positive jump-to-default exposure is by selling 1-year protection and earning some premium to offset the negative carry in flatteners.

In all these cases, we are assuming that the curves remain unchanged to calculate carry and rolldown. We remind investors that durations can clearly drift over time. The steepener is duration-neutral initially, but gradually gets short the market because the 5-year duration drops at a faster rate than 10-year duration. Conversely, the duration-neutral flattener gradually becomes long the market.

exhibit 9

Different Ways to Play a Steepener (CDX IG 5s10s)

Maturity	Notional Sold (MM)	Coupon	Spread	Current Duration	Upfront	Spread	1yr Later Duration	Upfront	Carry	Rolldown	Total P&L	Default P&L
(1) Notional Weighted 5s/10s Steepener												
5y	10.0	100	92	4.9	-0.39%	76	4.0	-0.94%	1.00%	0.56%	1.56%	(6.0)
10y	-10.0	100	108	8.4	0.67%	105	7.8	0.39%	1.00%	0.28%	1.28%	6.0
Net	0	-	(16)	(3.6)	-1.06%		(3.8)	-1.33%	0.00%	0.28%	0.28%	-
(2) Duration Weighted 5s/10s Steepener												
5y	10.0	100	92	4.9	-0.39%	76	4.0	-0.94%	1.00%	0.56%	1.56%	(6.0)
10y	-5.8	100	108	8.4	0.67%	105	7.8	0.39%	1.00%	0.28%	1.28%	3.5
Net	4.2	42	30	(0.0)	-0.78%		(0.5)	-1.17%	0.42%	0.39%	0.81%	(2.5)
(3) Duration Weighted 5s/10s Steepener (With hedging of jump exposure)												
5y	10.0	100	92	4.9	-0.39%	76	4.0	-0.94%	1.00%	0.56%	1.56%	(6.0)
10y	-5.8	100	108	8.4	0.67%	105	7.8	0.39%	1.00%	0.28%	1.28%	3.5
1y	-4.2	100	42	1.2	-0.67%				1.00%	-0.67%	0.33%	2.5
Net	0.00	-	12	(0.5)	-0.50%		(0.5)	-1.17%	0.00%	0.67%	0.67%	-

Source: Morgan Stanley

One important factor to consider when putting on steepeners is the impact of fixed coupons, especially in the case of shorter-dated steepeners across IG indices. At a fixed coupon of 100 bp, the indices can be compared to bonds trading at a premium. For such bonds, as the time to maturity decreases, the price can be impacted significantly by the duration change (i.e., pull to par). This applies specifically to the 3-

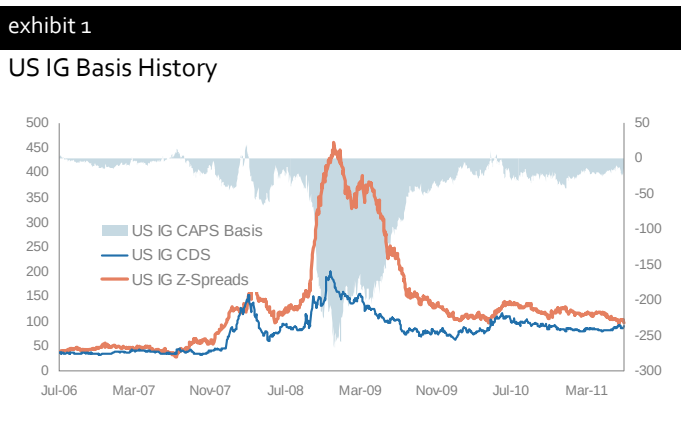
year point across IG indices, where the scope for absolute rolldown is much lower, given how tight they are trading. Overall, the impact of the 100bp fixed coupon on the trade is to make it more expensive from an initial outlay perspective, but also to increase potential gains from the higher coupon carry.

Trading the CDS Basis Today

July 19, 2011

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Ashley Musfeldt
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The basis between corporate bonds and CDS is a much watched and well-trafficked corner of the credit markets, with an interesting history. The negative basis trade, in which the CDS premium trades tighter than the equivalent corporate bond spread (allowing one to buy a bond and buy equivalent maturity credit protection resulting in a positive spread) remains a popular trade in the market, despite the volatility experienced post-crisis. The rationale for this trade can range from simply hedging corporate bond exposure to creating a default-risk-free positive carry trade, but the motivation will focus on the carry earned, the MTM volatility, the trading opportunity and, of course, capital efficiency.



Source: Morgan Stanley Research

At the peak of the crisis, funding issues drove the investment grade basis into the -200 bp range in the US, and around the -100bp range in Europe, a level that made it almost an asset unto itself (see [The Basis Moves from a Trade to an Asset](#), October 31, 2008). Despite the fact that many credit relationships have yet to fully heal from the crisis, the basis in US IG corporates is about -20 bp on average today, near pre-crisis levels (when spreads on IG CDS were about 50 bp tighter). In Europe, the IG basis is +20bp on average, though driven by a different set of factors. The basis relationship in both regions remains volatile, driven by uncorrelated moves in CDS and bond spreads, which are, in turn, related to the significant rise in government bond volatility today.

In this chapter, we explore the new post-crisis basis environment focusing on valuation, volatility, and market dynamics, and we highlight the reasons for the divergent landscape we see in the US and Europe. We also offer thoughts on how the basis could evolve given the credit cycle, Dodd-Frank implementation and the potential behavior of

banks and insurance companies with respect to capital and regulation. Our key points are as follows:

The US basis today: We believe that the continued negative basis in the US is related to lower levels of loan growth (so less bank loan hedging activity), the move to standardized NoR language in CDS contracts (banks are penalized for not having restructuring as a credit event in protection under Basel), and a credit environment where fundamentals remain strong and debt maturities have been pushed out through a significant refinancing process since the crisis.

The European basis today: In Europe, the basis is positive on average, due to several factors. First, there has been loan growth post-crisis, and the overall level of loans outstanding remains at around 3x that of the US, resulting in more potential bank loan hedging activity. Additionally, CDS standards in Europe include restructuring as a credit event, resulting in better capital treatment for banks hedging under Basel. Finally, sovereign credit concerns have led to more corporate credit hedging, pushing CDS premiums higher.

Negative basis as a trade today: The relatively low carry combined with higher levels of volatility makes the basis more of a trade than an investment in our view. The higher levels of volatility are driven by less correlated CDS and bond spread moves along with higher levels of interest rate volatility and CDS index-based hedging. The opportunity, in our view, is executing at attractive entry and exit points created by higher levels of volatility. In high yield, negative basis situations are prevalent as well, but we view this as more of a micro story given bond callability and the potential for large capital structure changes.

The basis tomorrow: How might some of these relationships change? We see potential US insurance hedging flows, given recent changes in the treatment of CDS protection within total capital, and little of this flow has been realized yet. More clarity on Dodd-Frank implementation could grow CDS volumes generally, which would smooth the basis volatility. Any increase in traditional structured credit issuance through bespoke tranches could drive tighter CDS levels, although we feel that this would occur in a tighter spread environment. Finally, the credit cycle thus far has been dominated by deleveraging, so any re-leveraging might contribute to a less negative or even positive basis in the US.

A BRIEF LOOK BACK

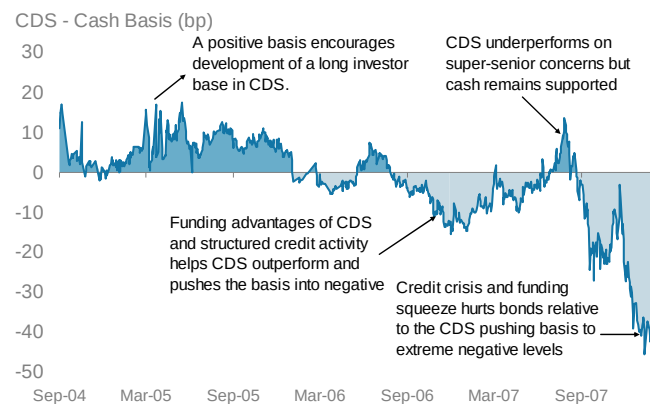
Corporate CDS markets were established in the mid-1990s, but it wasn't until nearly a decade later that the contract and product came to look like it does today. Sometime during the nascent stages of the CDS market, the basis between CDS and cash bonds became an important relationship to monitor. Cash bonds once helped price CDS contracts, as cash bond

spreads over Libor were the starting point to evaluate credit risk in CDS, after which they were adjusted for restructuring credit events and cheapest-to-deliver optionality.

In the early days, CDS generally traded wide to cash instruments as the newness and nuances of the contract motivated protection sellers to demand a premium, and this positive basis was a marketing point for those wishing to promote liquidity in the derivatives contracts. As CDS markets matured, differences became more granular, and market participants built more sophisticated models. In these early days funding costs of cash instruments were largely ignored and liquidity on both sides of the basis remained robust.

exhibit 2

Evolution of Corporate CDS-Cash Basis (2004-07)



Source: Morgan Stanley Research

The corporate negative basis trade – long the bond vs. buy protection on the CDS – that emerged circa 2004 was essentially a balance sheet usage allowance for those investors who were able to fund at attractive levels and use this balance sheet to earn a small carry for taking on some amount of counterparty risk, but no actual credit default risk. There was also the added benefit of optionality through covenanted basis packages, in which investors bought bonds with covenants (poison puts, rating step-ups) which outperform in LBO/M&A and ratings deterioration scenarios and bought CDS protection. Additionally, the basis trade helped absorb protection sold by structured credit investors and also provided the investment community with a way to retain illiquid secondary market cash positions.

Around 2005 the cash-CDS basis turned negative and has remained so ever since (at least in the US), save for a short period when super senior concerns drove it into positive territory (see Exhibit 2). From 2005 to 2007, the basis traded primarily in the negative 15 to 20bp range. In an environment where AAA/AA assets traded under 10bp, 15-20bp seemed reasonable compensation for basis trades at that time. In 2008, the basis started dipping to new lows, in the negative 50bp range and hovered there until the Lehman bankruptcy triggered a larger credit squeeze.

Conventional wisdom would suggest that CDS should underperform cash bonds in a high default environment, given that CDS is a pure play on credit risk, whereas cash bonds are a combination of interest rate and credit risk and are short cheapest-to-deliver optionality. However, during the credit crisis, liquidity issues trumped default spreads, with balance sheet usage becoming a bigger concern for investors than default risk. The basis re-priced to unseen levels, reflecting the additional premium charged by the market on fully funded assets. Unsurprisingly, financials were the worst performers during this time period. Abundant liquidity, accommodative policy and deleveraging set the basis on the long road to recovery in the second half of 2009.

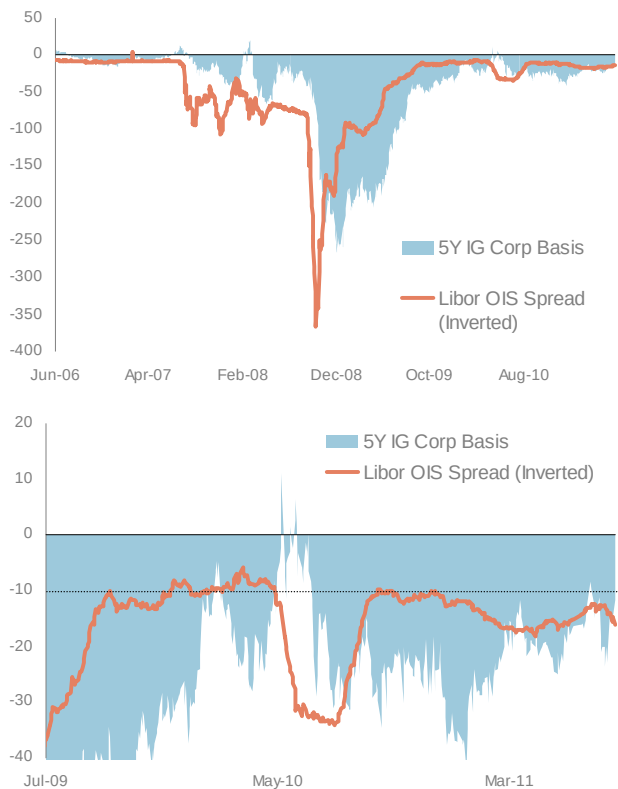
THE BASIS AS A FUNDING TRADE

Today, the basis has returned to a period of greater stability in terms of ranges, and actually looks much like it did in the pre-crisis timeframe. We caution however, that optics can be deceiving, and that there are many different factors at play in today's market environment. Primarily, funding remains of paramount importance, something our basis metrics do not capture (see Appendix). In the pre-crisis period, given the easy access to capital, funding was not a major factor in basis trading. Looking at historical Libor-OIS as a proxy (though admittedly an imperfect one) for funding costs, we see that they were minimal during this period. In 2008, however, capital became dear and the Libor-OIS spread reflected this gap in funding premium. Not surprisingly, the basis spiked to new lows just months after the Libor-OIS spread reached its peak.

One could argue, of course, that in an extreme crisis period, correlation between asset classes goes to 1 and thus the levels of Libor-OIS and the basis were both casualties of a larger market meltdown. To mitigate this effect, we isolate a more recent period, spring 2010, and can see (in Exhibit 3) that again, just after a sharp decline in LIBOR-OIS, the basis falls, and then peaks at its local negative amount. Additionally, on average, Libor-OIS has remained generally higher than local ranges pre-crisis, reflecting a new normal in persistent higher funding costs. While not the only factor, clearly funding costs can and will play a role going forward.

exhibit 3

The Funding Connection: Libor-OIS vs. US IG Basis



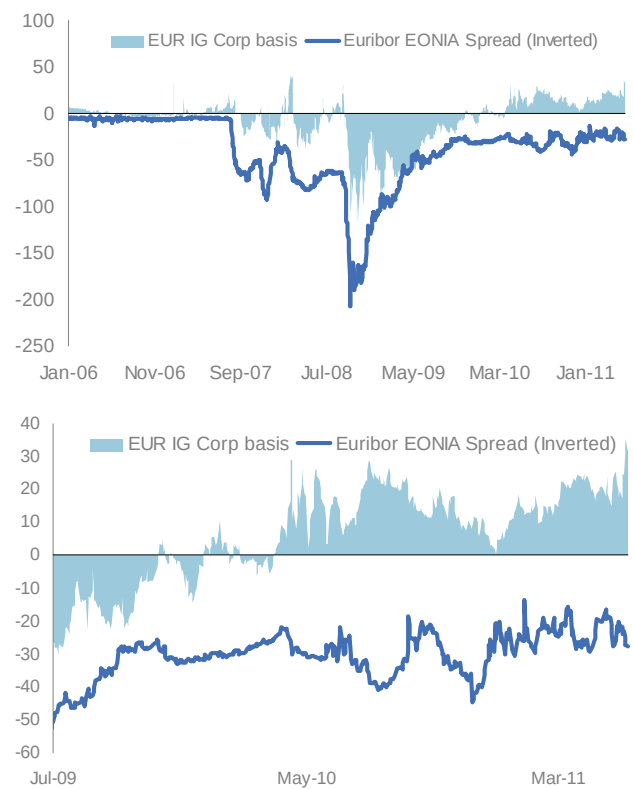
Source: Morgan Stanley Research, Bloomberg

In Europe, we look at Euribor-EONIA as a proxy for funding costs, and see largely the same story as the US, leading up to and during the crisis. Pre-crisis we see relative stability, coupled with a basis that hovers near zero. As the Euribor-EONIA spread moves, so does the European basis, with a slight lag, which makes sense given the relative liquidity levels in credit and the velocity of basis trades. During the crisis, a familiar theme emerges and from the summer of 2008 until early 2009 the US and EU funding and basis look virtually identical, save for differences in magnitude.

Post-crisis, however, the EU and US basis and funding stories begin to diverge. While the respective proxies for funding both normalized, albeit without quite returning to their pre-crisis levels, the EU basis shot straight through a low negative level into positive territory, where it has remained for over a year. We will explore the positive basis story in Europe a bit later, but clearly funding environment alone is only part of the story.

exhibit 4

The Funding Connection: Euribor-EONIA vs. EU IG Basis



Source: Morgan Stanley Research, Bloomberg

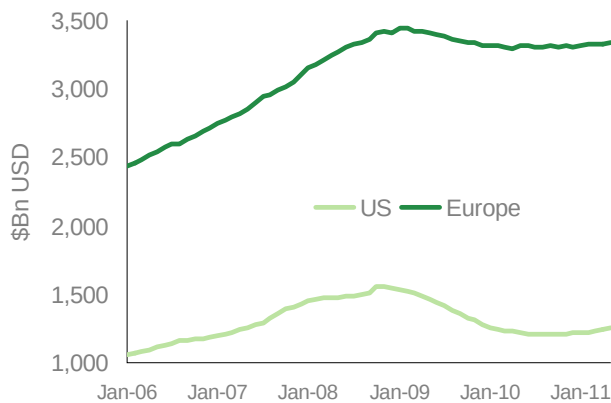
THE BASIS AS A HEDGE

Aside from funding, another factor driving the basis today is the fact that all standardized CDS now trade without restructuring as a credit event in the US. This is a significant change from the pre-crisis markets, in that a CDS is now truly a contract that only hedges the default expectation on the underlying bond, while before, when CDS traded ModR, the contract was more in line with what the hedging side of the market was trying to protect against. Though it is difficult to quantify exactly how much this “contract basis” should be worth, it is worth highlighting that banks do not get as good regulatory treatment for using NoR CDS to hedge their underlying portfolios, as regulatory rules do not give full capital relief when hedging with mismatched instruments. We think the lack of this particular flow is one reason it will be difficult to see a positive basis in the US for awhile.

Additionally, one of the primary sources of hedging flows in the market pre-crisis were bank loan books, using CDS as hedges for their broader loan portfolios. In addition to worse regulatory capital treatment today with NoR as the standard language on restructuring, bank lending activity in the US is significantly below pre-crisis levels, consistent with overall balance sheet shrinkage in US banks. In Exhibit 5, we illustrate the sharp decline in bank lending volumes, and thus reduced demand for CDS hedges in the US.

exhibit 5

Bank Loans Outstanding – US Lower and Much Smaller than Europe



Source: Federal Reserve, ECB, Datastream

Europe, on the other hand, has historically been dominated by bank loan issuance, and continues to be so. While it is difficult to parse out CDS flows and assign them to particular types of hedging activities, it's a relatively safe assumption to say this could be a possible driver of CDS protection buying activity in Europe.

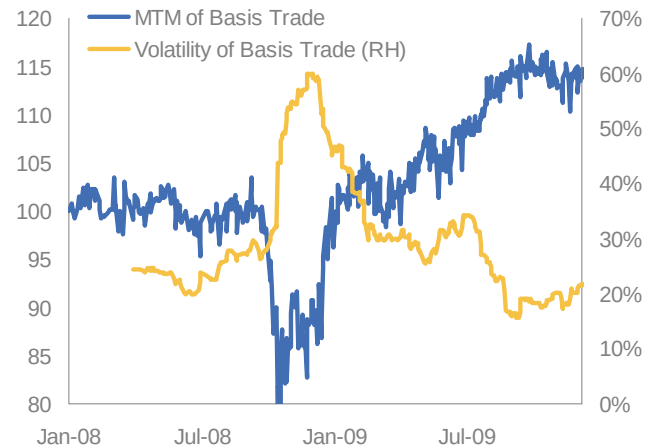
Finally, one potential source of hedging activity going forward could be the NAIC rules on CDS hedging for insurance companies. Under the new rules, insurance companies get better capital treatment for using CDS (even NoR CDS) as hedges for bonds and bond portfolios. We haven't seen much in the way of this type of hedging flow yet, but this could be due to the elevated level of spreads in today's market.

IS CDS A GOOD HEDGING VEHICLE FOR CASH?

Given the funding environment and uncertainty surrounding both hedging and long flows in the corporate markets, our next question is, given the basis history, whether CDS is a good hedge for cash bonds. To address this, we look at the MTM of a representative cash bond / CDS pair in two different market environments. First, we start our period of analysis in the beginning of 2008, when markets were weaker, but the extreme negative basis hadn't yet developed. In Exhibit 6, we can see that the MTM of such a trade got significantly negative in the sell-off, indicating that, given all of the externalities that CDS does not capture (namely in this case, the funding environment), CDS may not be a great hedge for a bond portfolio in an extreme market environment.

exhibit 6

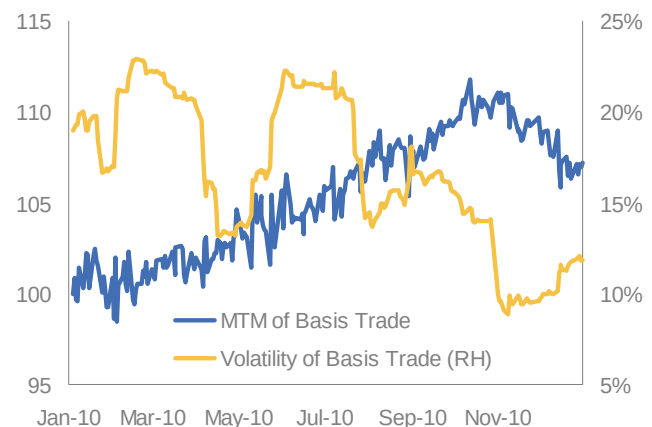
Basis Trade MTM: Through the Crisis



Source: Morgan Stanley Research

exhibit 7

Basis Trade MTM: Post-Crisis



Source: Morgan Stanley Research

For a major spread widening such as this, a better hedge may have been a credit instrument more levered to the underlying market. We have long advocated the use of super senior tranches for major spread widening hedges, and in this scenario, such a trade would probably have been a better choice. Options hedges, particularly those with no capped PnL in a downside environment would have worked as well.

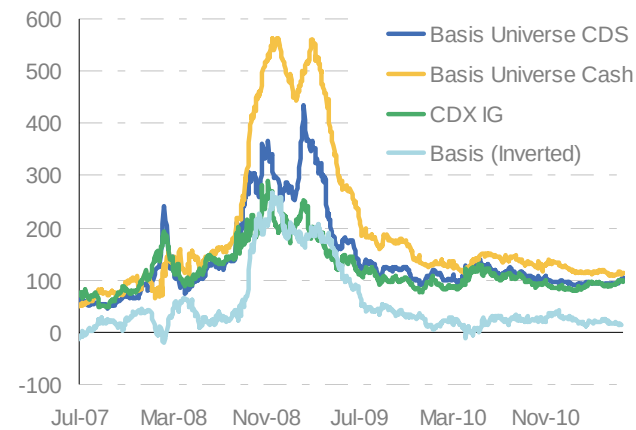
However, when we look at our single name cash / CDS pair in FY2010, we see that, despite some noise (indicated by the volatility of the MTM of the trade), the CDS hedge actually performed as it should have throughout this period. The MTM of the basis trade climbed as the market healed and rates remained low. Thus, for smaller moves in quieter environments, where rates volatility remains low, rates themselves remain low, and funding does not become a driving force, CDS can be a good hedging instrument.

TODAY'S NEGATIVE BASIS VALUATION

So where then do we see value when looking at the corporate cash and derivatives markets, given high funding costs, reduced hedging flows and the potential for rising rates? To answer this, we look at four different unlevered ways to take exposure to the corporate credit markets: cash bonds, CDS, CDX IG, and the basis (shown in Exhibit 8, with the basis inverted).

exhibit 8

Negative Basis Is Low Today Relative to Spreads

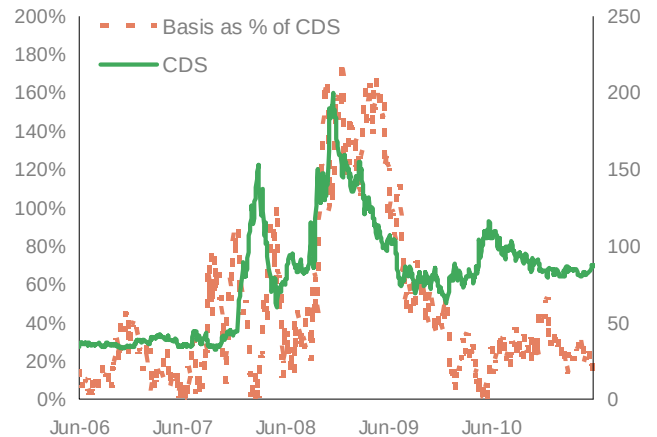


Source: Morgan Stanley Research

As the markets have healed, spreads have fallen, and concurrently, the basis, representing their difference, is at local lows (in terms of spread generated by putting on a trade). To put this in context, we can think about the basis not just on an absolute level, but in relation to the spread environment. In Exhibit 9, we show the historical US IG basis as a percentage of the spread of the underlying CDS, thus taking into account the fact that while a basis package at the peak of the crisis may have generated 300bp+, an outright CDS long could have yielded around 200bp, putting in context the risk reward of executing a package trade. What we see in this graph is that the basis today is actually relatively low, given the still somewhat wide overall spread environment.

exhibit 9

Basis as Percentage of CDS Spreads Lower Today



Source: Morgan Stanley Research

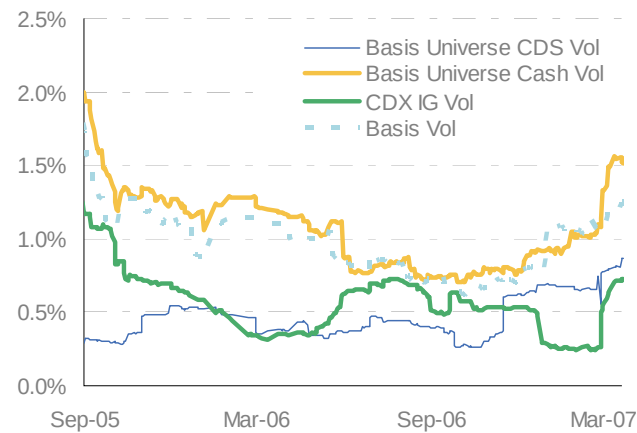
BASIS VOLATILITY

Next, we look at the volatility environment of direct credit investments and basis trades. In theory, a duration matched basis package should have little exposure to the actual credit spread of the underlying entity. However, the basis itself does move around independently at times, and monetizing the basis can involve MTM volatility. Although somewhat counterintuitive, the basis has not been a low volatility entity historically. In fact, even though cash spreads and CDS spreads largely move in the same direction in the long run, short term moves can be uncorrelated for many reasons, among them the difference in investor base, varying exposures to rates (Treasury yields have been more volatile than corporate bond yields of late), funding environments and differing uses.

One of the primary motivations for executing basis trades is that in purely funded unlevered form, they can generate a (credit) riskless premium for taking on some amount of counterparty and mark-to-market risk, and by using financial leverage, the yield pick-up on this trade can be enhanced significantly. However, as in any trade, the MTM volatility of the trade is important.

exhibit 10

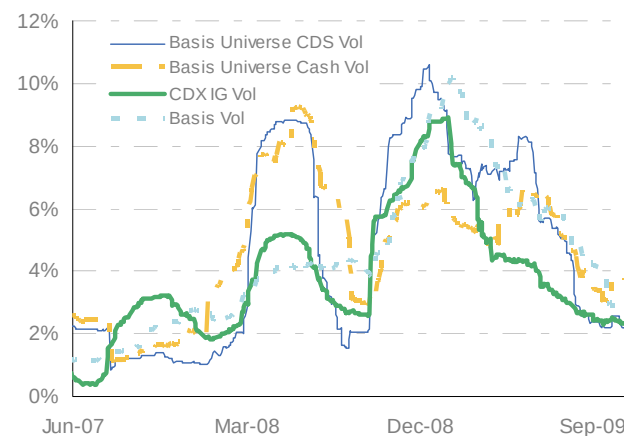
Pre-Crisis: US IG Basis Volatility Lower Than Bonds



Source: Morgan Stanley Research

exhibit 11

In-Crisis: US IG Basis Volatility Relatively Higher

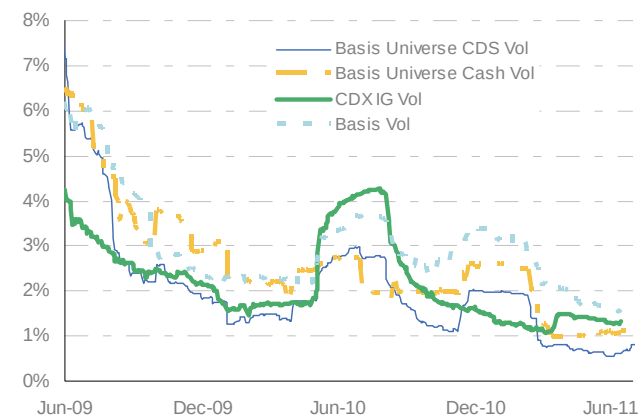


Source: Morgan Stanley Research

However, valuations on average are not as attractive anymore, primarily because outright CDS spreads are still so wide while being relatively attractive from a fundamental standpoint, while the relative carry from a basis trade compared to actual CDS levels is close to its lowest point. In our view the increased volatility owing to the relatively low correlation between bond spreads and CDS premiums is a big part of the negative basis trade today, which can lead to interesting opportunities, if investors pay attention to entry and exit points.

exhibit 12

Post-Crisis: US IG Basis Volatility Higher than Bonds and CDS



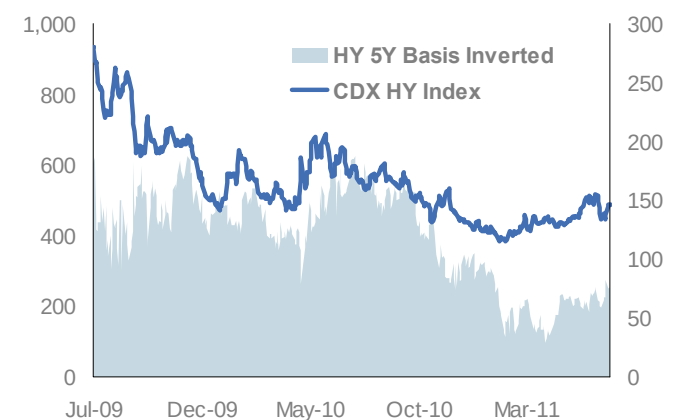
Source: Morgan Stanley Research

THE HIGH YIELD BASIS: A MICRO STORY

The high yield basis is a more complicated relationship to analyze, given the higher probability of actual default and the prevalent callability feature. A negative basis trade on a high yield issuer can be a cheap play on jump to default risk, but given healthy corporate fundamentals generally, the current environment is not necessarily one that justifies basis trades on wide spread names just for the jump-to-default convexity.

exhibit 13

US HY Basis Is More of a Micro Story



Source: Morgan Stanley Research

Additionally, the percentage of callable bonds and the probability of bond prices moving away from par is larger in high yield. Bond prices are high today driven by lower interest rates, which increases the possibility of bonds trading close to their call price, and could impose a cap on the dollar price. In fact, based on the current high yield cash index composition, more than 80% of the bonds maturing in the 4-6 year time frame have a dollar price above par. These

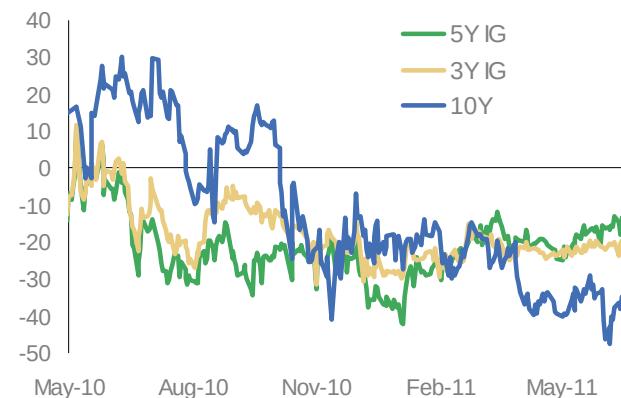
features make it difficult to quantify real relative value between HY cash bonds and CDS. One possible remedy would be to buy protection to the “worst date” of the bond rather than the maturity, though this approach is imperfect given that the worst date can move around. This can lead to duration mismatches between the two legs of the basis trade and is likely to make the MTM even more volatile. As such, we believe that the HY basis is more of a micro story today than it was during the crisis, and needs to be examined on a more granular level.

THE US IG BASIS: SHORTER DATED OPPORTUNITIES

While we think that given other opportunities in the market the basis may not be the most attractive trade in corporate credit, it does represent an arbitrage rather than an outright long, and thus we do acknowledge that there are pockets of opportunity for investors with the right funding mix and leverage capability. In particular, given steep curves in the CDS market (see [Credit Curve Conundrum](#) from July 12, 2011) we think the best opportunities are going to be in the shorter-dated maturities because of the relatively benign MTM impact of rolldown on the CDS leg, and reduced long-end interest rate volatility for the cash bond leg. Exhibit 14 shows that the 3yr basis trades in the same range as other maturities and is actually more negative than the 5yr basis.

exhibit 14

US IG Basis Trades – Shorter Dated Attractive



Source: Morgan Stanley Research

Apart from monetizing the basis at the short end, basis trades can also be used as micro hedges at current levels to hedge against select single name credit risk. As we have already highlighted, the negative basis trade overall may not be the best systemic hedge when funding and liquidity concerns dominate the MTM impact on an otherwise risk free trade. But given that CDS represents pure credit risk relative to a cash bond, a negative basis trade on a credit where the investor is fundamentally bearish may make sense.

Of course, as always, details matter, and we highlight that investors need to be tactical in taking advantage of opportunities as they arise. In Exhibit 15, we highlight some sample trades that, given the risk factors we’ve cited, we think still offer a compelling spread in today’s market.

exhibit 15

US IG Basis Screen

Name	Ticker	Maturity	CDS	Z Spread	Adjusted Basis*	CAPS Basis*
ALTRIA GROUP INC	MO	Nov-13	57	205	-162	-150
PITNEY BOWES INC	PBI	Mar-15	77	168	-76	-69
SOUTHWEST AIRLINES CO	LUV	Oct-14	80	137	-52	-49
DUKE ENERGY INDIANA INC	DUK	Sep-13	22	60	-44	-41
ANHEUSER-BUSCH COS INC	ABIBB	Jan-14	21	53	-36	-34
SAFEWAY INC	SWY	Aug-14	62	95	-32	-30
HOSPIRA INC	HSP	Jun-14	61	89	-29	-27
BUNGE LTD FINANCE CORP	BG	Apr-14	103	126	-29	-25
MEDCO HEALTH SOLUTIONS	MHS	Aug-13	89	100	-22	-20
KRAFT FOODS INC	KFT	Oct-13	39	49	-21	-19

* - Please refer to Appendix for the definition of our basis measures.
Source: Morgan Stanley Research

THE EUROPEAN BASIS: A MORE COMPLEX STORY

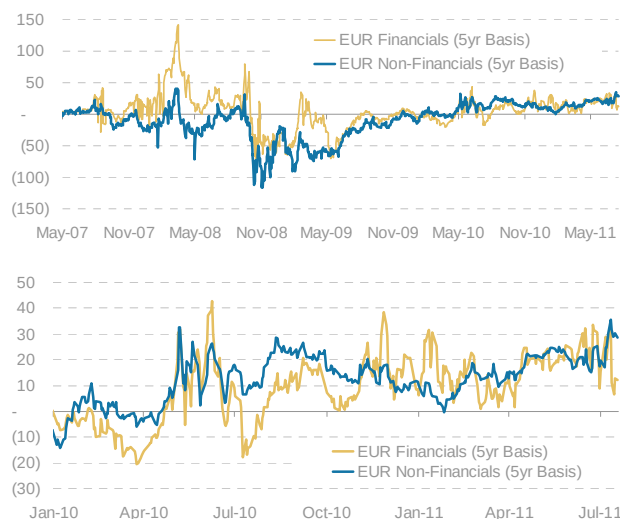
Turning our attention to Europe, we see a far more complex story in light of the macro backdrop. It is striking that the performance of the basis has followed two very different paths: first, during the credit crisis and then second, during the sovereign crisis today. At a broad market level, the median basis in Europe is positive (about +15-30bp) for both financials and non-financials as opposed to during the crisis when it was extremely negative (as low as -100bp). The funding component, which, as we mentioned earlier, was the primary driver of the basis during the credit crisis, is clearly not an issue for bond investors today, thanks to central bank liquidity provisions.

The problem in Europe is primarily perceived to be an issue of sovereign balance sheets and less of corporate or investor balance sheets. Investors have resorted to hedging with CDS on a tactical basis to mitigate MTM volatility rather than liquidate corporate bond positions altogether. The net result is a positive basis across Europe (see Exhibit 16).

While sovereign driven hedging is a major contributing factor for the positive basis recently, in general, we believe the appetite for CDS hedging is more robust in Europe due to the greater involvement of banks in the European credit system, relative to the US. Additionally, contrary to the US CDS market, corporate CDS contracts in Europe still have restructuring as a credit event. Under Basel II rules, banks get significant capital relief when they hedge loan portfolios with CDS contracts that incorporate restructuring as a credit event. The European credit system is moving towards the US in terms of institutional and retail investors stepping in to provide corporate financing, as banks retrench. Still, in the near-term bank hedging will be an important driver in supporting a positive basis.

exhibit 16

European Basis – Financials vs. Corporates



Source: Morgan Stanley Research, iBoxx, Mark-It

We expect today's hedging flows to abate even as the benign funding conditions created by central banks is eventually withdrawn, which would cause the European basis to trend negative. With Euribor-EONIA swap spreads and currency basis showing the first signs of stress, it makes sense to switch out of vanilla bonds into CDS or credit linked notes referencing CDS in many European corporates for better yield and lesser funding sensitivity. Additionally, although credit spreads are above average, yields are at historical lows, and CLNs with floored Libor coupons or CMS coupons can be a good way to overcome the "good spread, bad yield" problem faced by credit investors. In Exhibit 17 we show names where this cash for synthetic switch trade may make sense.

exhibit 17

Switch Out of Bonds and Into CDS

Ticker	Cpn	Mty	Bond Price	CDS	Z-Spread	CAPS Basis
LGFP	4.25	Mar-16	101.6	298	146	146
GLEINT	5.25	Mar-17	101.5	311	235	96
UU	4.25	Jan-20	104.6	145	64	78
TKAGR	4.375	Mar-15	104.6	135	81	76
ENELIM	5.25	Jan-15	104.8	216	158	75
ENIIM	4.75	Nov-17	106.4	162	89	75
TITIM	8.25	Mar-16	110.0	378	335	74
ENELIM	4	Sep-16	98.8	235	174	72
FNCIM	5.75	Dec-18	105.9	250	195	71
EADFP	5.5	Sep-18	113.1	123	60	68
VINCI	5.875	Oct-16	113.4	117	54	66
VW	5.375	May-18	112.2	119	58	63

Source: Morgan Stanley Research

However, there is an important exception to the above. While the median basis in Europe is positive, there is a difference between the basis performance of the peripheral corporates and the rest of Europe. Given the volatility in the periphery, there is a genuine desire among peripheral corporate bond holders to de-risk and liquidate those holdings. The market has absorbed the supply, but at a price causing the basis in peripheral financials and corporates to trade at relatively wide levels. As a result, the basis on several peripheral corporates and financials can be very negative, as the CDS hedging appetite, while meaningful, is still less powerful than cash bond liquidations so far. The peripheral negative basis levels today are consistent with what we saw during the credit crisis.

We acknowledge the contributions of Vishwas Patkar to this chapter.

APPENDIX: MEASURING THE BASIS

The simplest measure of the basis is the difference between the 5yr CDS spread and the actual cash spread of the bond, also called the *raw basis*. For bonds trading close to par and roughly 5 years to maturity, the raw basis is a decent estimate of the yield pickup an investor can gain by exploiting the arbitrage opportunities between the cash and CDS spread. However, apart from the potential maturity mismatch between the bond and CDS, this metric does not take into account the effect of credit curve shape and bond dollar price, which greatly affect the fair value of the basis.

Curve shape can affect the basis as inverted credit curves imply a lot of near-term risk to bond coupons, which in turn makes bonds less attractive in a fair value sense. The raw basis calculation uses a simple Z spread measure, which assumes a flat credit curve and does not assign survival probabilities to bond coupon cash flows, even if they are implied in the CDS market. Additionally, bond price affects the basis calculations, because for bonds trading at a discount (or premium), buying CDS protection on par notional could mean that an investor is buying too much (or too little) protection. For bonds trading at a significant discount or premium, this tends to distort the fair value of the basis.

To compensate for these two factors, we can also look at two additional basis measures, the adjusted basis and the CAPS basis. The *adjusted basis* takes the simple raw basis and makes an adjustment for the price of the cash bond. To compensate for the deviation of bond dollar price from par, we purchase (or sell) additional protection for a premium (discount) bond, based on the average forward price of the bond for each year until maturity and calculate an adjusted Z spread metric. The maturity mismatch of the cash bond and CDS contract is also adjusted for by interpolating the CDS contract along the CDS curve.

The *curve adjusted par spread (CAPS)* basis uses the adjusted basis and takes it one step further by adjusting for the risk implied by the CDS curve. CDS curves can be interpreted as indicators of default risk over time, given that they are par instruments, unlike corporate bonds. The CAPS basis is based on a curve adjusted Z spread, which is more accurate than the simple Z spread, given that it takes into account how default risk varies over time. It also takes into account that losses from default are a function of dollar price

and assumes a certain recovery, whereas the conventional Z spread assumes zero recovery. Inverted credit curves mean that near term bond coupons are more risky than implied by flat or steep curves.

Of course, there are several shortcomings in our basis calculations, even those that take bond price and curve shape into account. Most notably is the inability to capture shifts in funding environment, as the cash component of the basis is ultimately a funding trade. Thus changes to the cost of funding can be a primary driver of basis relationships, which was illustrated by the negative basis in 2008-09. Additionally, the choice of benchmark funding costs and differentiation therein is not captured in our metrics, which matters because the basis metrics we discuss above are based on value relative to the swap curve.

Finally, we advocate using greater caution when analyzing basis trades in the HY space, particularly for those with a high probability of default, as these credits can trade with a significant difference in the structure and timing of cash flows between the bond and the CDS.

Another factor to consider in measuring the basis is the impact of the rates environment, which we first addressed in Chapter 15. On the bond side, a key factor is dollar prices, as higher dollar prices can make the basis look less attractive, as the higher initial cash outlay would require more CDS hedges. On the CDS side, rates can impact curve steepness. Shorter-maturity CDS are more likely to trade with a negative upfront (pay an upfront amount and then receive 100bp when selling protection), whereas the longer-maturity CDS contract is likely to trade with a positive upfront, making it more attractive for protection sellers from a funding perspective.

Finally, the structure of the upfront contract after the SNAC protocol was implemented has rates implications as well. The buyer or seller of protection who receives the upfront stands to benefit the most, because in addition to the standard funding reasons we've discussed in the past, the benefit of receiving an upfront prior to a rise in rates is that as rates rise, the discount factor applied to PV the running coupon increases, thus *lowering* the present value of the remaining coupon.

Introducing the CDX IG Macro Credit Model

November 29, 2011

Sivan Mahadevan
 Peter Mallik
 Rizwan Hussain
 Adam Richmond
 Phanikiran Naraparaju

In any economic or market environment, macro factors are an important driver of credit market prices, and the investment process requires a good balance of those macro forces with credit- and company-specific factors. The real macro-micro mix in the market can often get quite skewed, as we have seen since the financial crisis where cyclical, systemic, and technical factors have played an important role in setting the price for credit. Fundamentals will always matter, but they may not always be the dominant factor in driving credit pricing, particularly in investment grade if fundamentals are in a reasonable range of safety.

Whether macro or micro factors should drive credit markets has been part of the debate for a long time, and the reality is that both matter a lot, although not always at the same time. While the macro influence on credit can be cyclical (almost by definition since credit becomes far more sensitive to it in slow-growth periods), we believe macro will always matter. It is only a question of balance with micro. And of course, macro can quickly become micro as well, as systemic shocks and significant changes in economic growth impact sector-specific supply / demand quite quickly.

We intend to develop a series of macro credit models that use historical relationships to help predict short-term and longer-term price movements in credit markets, both for corporate bonds and CDS indices. In this chapter, we introduce a short-term macro credit model that attempts to predict spreads on a 3-month forward basis for the on-the-run 5-year CDX IG index.

How do we intend to use such a model? While the model's results are not intended to be traded, they do provide insight into the direction and magnitude of future spread changes and the underlying drivers of those changes. The models are also tools that can help gauge the sensitivity of predicted spreads to changes in the factors, which can be helpful in scenario testing. Over time, the model's independent variables may change as regimes change, providing further insight into market dynamics.

A summary of our CDX IG macro credit model appears below, and in the remainder of the chapter, we provide much more detail.

CDX IG Model Factors: Our historical analysis has resulted in selecting the following six factors, given their stability and

strength in predicting CDX IG spreads. For cyclical variables, we chose PPI, ISM PMI and ISM New Orders. As a fundamental factor, we chose S&P 500 EPS long-term growth based on consensus bottom-up estimates, and as a systemic factor, we selected S&P 500 3-month variance (a measure of equity volatility). Finally, for a technical measure, we chose Trace investment grade trading volume.

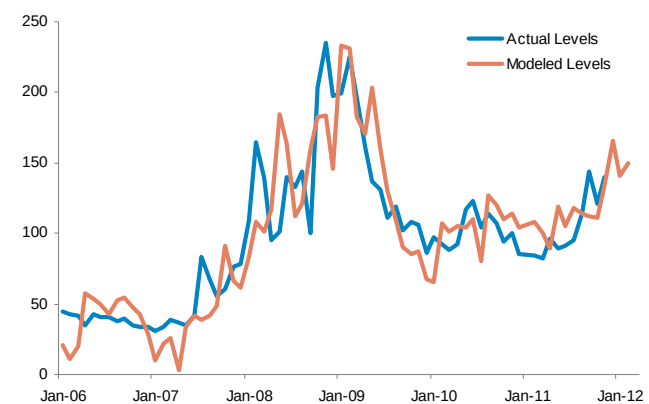
CDX IG Model Results: The six-factor model was able to predict incremental CDX IG spreads 3 months forward with an adjusted R-squared of 61% (and a spread level R-squared of 75%) since January 2006 (covering three regimes) and spreads ranging from a tight of 31bp to a wide of 235bp.

Current Model Predictions: Based on November 2011 expectations for the cyclical variables, our model predicts CDX IG 5-year spreads 3 months forward to be 10bp wider than current levels of 140bp for a prediction of 150bp. Our bear case is 168bp or nearly one standard deviation wider, while the model's bull case is 131bp or one standard deviation tighter than the prediction.

Model Sensitivity: The model is most sensitive to changes in the cyclical and fundamental factors. The model projects higher spreads for higher PPI, and equity variance, while it projects lower spreads for higher trading volume and higher manufacturing activity (as measured by the ISM indexes) and projected S&P EPS long-term growth.

exhibit 1

CDX IG Macro Credit Model Performance



Source: Morgan Stanley Research estimates

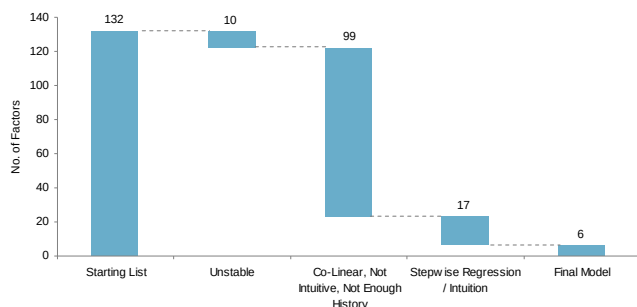
BUILDING THE CDX IG FACTOR MODEL

For the purposes of this model, we focused strictly on calculating CDX IG spreads 3 months forward based on distinct types of factors. We used our equity volatility factor model methodology ([The Volatility Debate](#), June 21, 2011) as a blueprint given that our equity and credit models are similar. But in comparison, the credit model incorporates a wider range of variables, including fundamental and technical factors (in addition to cyclical and systemic) to account for credit's position higher up the capital structure and (potentially) lower liquidity.

Choosing the Factors: We began by thinking about factors that could impact credit spreads going forward and from there assembled a list of 132 potential variables. We adjusted each factor's data to ensure stationarity and dropped names that could not be made stationary through differencing. We further trimmed the list by dropping factors that were too closely correlated with a more intuitive factor, did not have a long enough history, or did not exhibit enough variability, bringing the number of relevant factors to 23. Next we cleaned the potentials dataset by eliminating outliers. We then performed a stepwise regression on spread differences over the pre-crisis (Jan06 – Mar08), crisis (Apr08 – Jul09), post-crisis (Aug09-Oct11), and entire (Jan06 – Oct11) time periods to determine our final variables to include in the model (as determined by their p-values). We also calculated factor correlations to spread differences for illustrative purposes (the absolute value is shown below). We chose the final 6 factors based on the stepwise regression results and the factor's intuitiveness. In an effort to build a comprehensive model, we made sure to include representatives from each of our 4 categories described below.

exhibit 2

Factor Selection Process



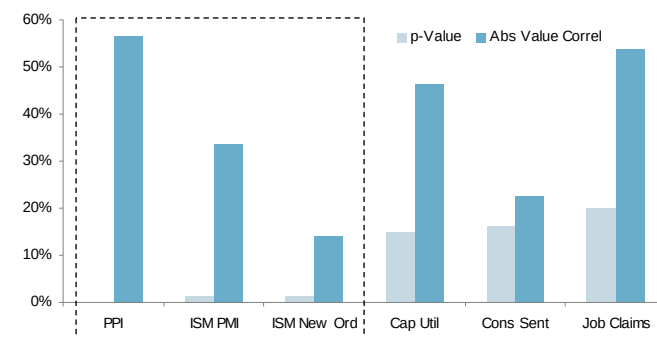
Source: Morgan Stanley Research

Cyclical Factors: Based on the stepwise regression, the candidate pool included: (1) PPI, (2) ISM PMI, (3) ISM New Orders, (4) Capacity Utilization, (5) US Initial Jobless Claims, and (6) Consumer Sentiment. From the set, PPI tested as the most significant overall, and the ISM indexes were most impactful in the post-crisis periods. These three were selected for the model. The chart below exhibits both

the measure of confidence in the independent variable being significant (p-values fell below the significance threshold) and the factor's correlation to spread differences.

exhibit 3

Cyclical Factor Selection Pool



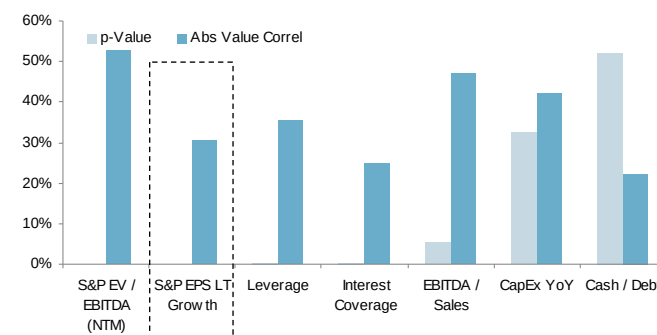
Source: Morgan Stanley Research

Note: P-value is for entire time period (Apr06-Oct11) and significance cutoff is 10%. Correlations are calculated over the pre (Apr06-Mar08) and post-crisis (Aug09-Oct11) periods.

Fundamental Factors: The candidates included: (7) S&P EPS long-term growth based on consensus estimates and (8) S&P EV to EBITDA (NTM), and non-financial average (9) cash / debt ratio, (10) CapEx YoY, (11) EBITDA / sales ratio, (12) leverage, and (13) interest coverage. Unfortunately, the accounting-based fundamental metrics are published with a quarterly lag, which led us to favor the S&P-related factors. Between those two factors, we selected EPS growth since the relationship between EV to EBITDA and credit spreads is less intuitive – namely, pre-crisis, a higher EV to EBITDA ratio (due to higher EV) correlated with lower spreads, whereas post-crisis, a higher ratio (due to lower EBITDA) correlated to lower spreads.

exhibit 4

Fundamental Factor Selection Pool



Source: Morgan Stanley Research

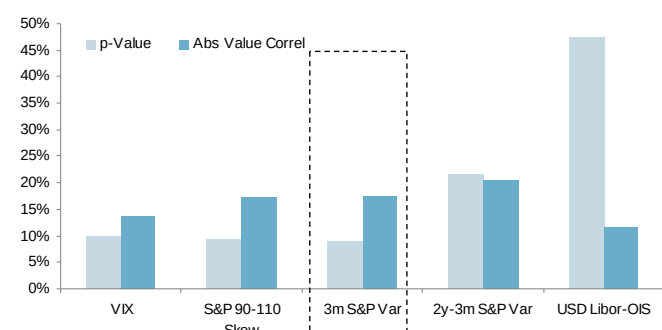
Note: P-value is for entire time period (Apr06-Oct11) and significance cutoff is 10%. Correlations are calculated over the pre (Apr06-Mar08) and post-crisis (Aug09-Oct11) periods.

Systemic Factors: The candidate pool included: (14) VIX, (15) S&P 500 90%-110% volatility skew, (16) 3m S&P 500

variance, (17) 2y-3m S&P 500 variance term structure, and (18) USD Libor-OIS spread. As expected, systemic factors had less correlation and lower impact on CDX IG spread differences than the cyclical factors, and therefore the p-values were generally higher, with VIX, S&P skew and 3m variance the most significant of the bunch. Much like the cyclical factors, these factors had the highest correlations in the post-crisis period, highlighting investor concerns about tail risk scenarios after 2008 and the fact that these variables have a cyclical component to them. We ended up selecting S&P 500 variance for the model due to its familiarity and 3-month term.

exhibit 5

Systemic Factor Selection Pool



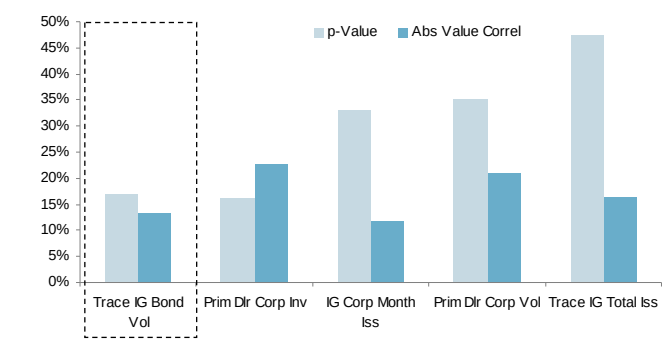
Source: Morgan Stanley Research

Note: P-value is for entire time period (Apr06-Oct11) and significance cutoff is 10%. Correlations are calculated over the crisis (Apr08 – Jul09) period.

Technical Factors: The candidates comprised of: (19) FINRA Trace IG bond dollar volume and (20) total issues traded, (21) primary dealer corporate securities trading volume and (22) inventory, and (23) IG corporate monthly issuance. These factors had the lowest impact on the model, regardless of time period. We selected Trace IG bond dollar volume for the model.

exhibit 6

Technical Factor Selection Pool



Source: Morgan Stanley Research

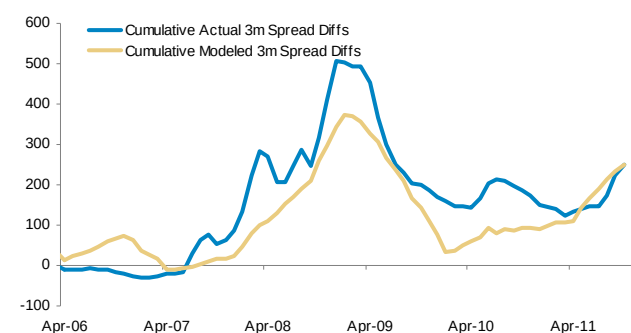
Note: P-value is for entire time period (Apr06-Oct11). Correlations are calculated over the pre- (Apr06-Mar08) and post-crisis (Aug09-Oct11) periods.

CURRENT MODEL COMPOSITION

The factor model includes the following 6 independent variables: (1) ISM PMI (levels), (2) ISM New Orders (levels), (3) PPI (YoY returns), (4) FINRA Trace IG Bond Dollar Volume (MoM returns), (5) S&P LT EPS growth (MoM differences), and (6) 3m S&P variance (MoM differences). We combined these factors to create a factor model with an adjusted R-squared of 61% for 3-month spread differences (and an R-squared of 75% for spread levels), while the highest individual factor correlation is 32% (PPI).

exhibit 7

Cumulative Model 3m Spread Changes vs. Actuals



Source: Morgan Stanley Research, MS Quotient

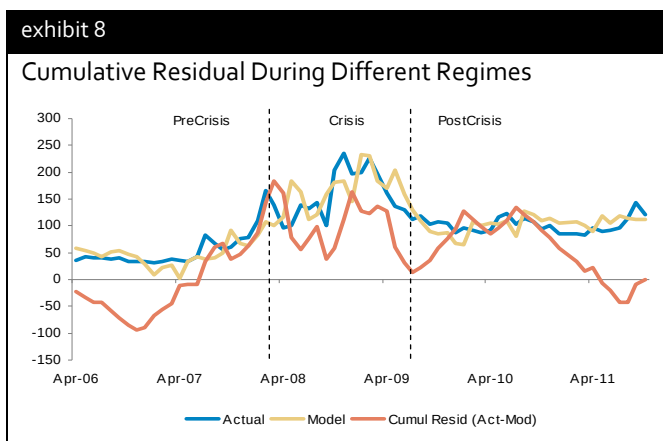
Model Sensitivity: The model is most sensitive to changes in the cyclical factors followed by the fundamental factors, and least sensitive to the systemic and technical factors over the entire time period. The model projects higher spreads for higher PPI and S&P 3-month variance, while it projects lower spreads for higher trading volume, manufacturing activity, and S&P EPS growth. The model does project higher spreads for higher (ISM) manufacturing new orders, but this impact is generally offset by the (ISM) PMI factor assuming similar changes in both month-over-month.

Historical Performance: The model performance varies over time, reflecting macro trends during regimes. Pre-crisis (Apr06 – Mar08) the model underestimates credit spreads due to lower inflation and higher EPS growth compared to the post-crisis period. Much of the underestimation occurred from June 2007 to March 2008, when spreads rose from 42bp to 139bp culminating in the Bear Stearns collapse. At the time, perception was that the crisis was limited to the subprime sector and therefore macro indicators did not turn down until late 2008, despite the rise in credit risk.

Conversely, the model predictions overshot actual spreads during the post-crisis period (Aug09-Oct11), particularly after July 2010. Three months previous, the Federal Reserve halted asset purchases (QE1) and subsequently macro indicators fell, causing the model to predict incrementally higher spreads. After the Federal Reserve resumed monetary stimulus (QE2), commodity prices (and PPI) rose, accounting

for the recent overshoot up till Sep 2011, when actual spreads spiked due to renewed European debt concerns.

During the crisis period (Apr08–Jul09), the model predicts marginally higher spreads than the market due mainly to the frequent spikes of market spreads. Since we are modeling spread differences (instead of levels), the predicted spreads generally overshoot market spreads 3 months after a spike despite factor and credit spread trends largely mirroring each other.

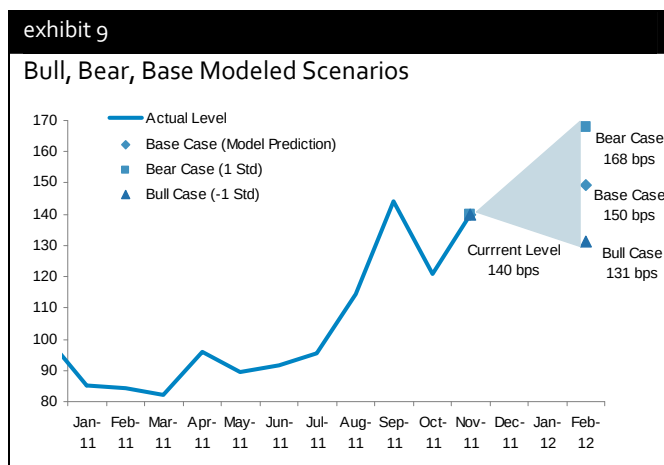


Model Calibration: Our current model is calibrated over the entire time period and will be recalibrated every quarter. We considered using a shorter calibration period (2006-2010), but the results for the two models were nearly identical.

Model Issues: As mentioned above, the model predicts 3-month spread differences instead of spread levels as levels are non-stationary. Therefore 3-month forward predictions become somewhat dependent on the current spread level. Generally this is not a problem, but when spread levels spike, the model generates incrementally higher predictions given high current levels, even though spreads usually decline 3 months after the spike.

INTERPRETING THE RESULTS

The model predicts a 10bp rise from current spread levels of 140bp for a 3-months-forward predicted spread level of 150bp. For this projection, we use estimates for the November cyclical variables and October values for the non-cyclical ones. The main drivers of the rise are the slowdown in long-term EPS growth (0.1% MoM rise in Nov 2011 versus 0.3% in Jul 2011) and the rise in S&P 3m variance (7 var points), which outweighed a projected fall in PPI (of 0.8% YoY from Oct to Nov estimates). The 10bp rise is approximately one half standard deviation above the average spread change (0bp) for the post-crisis period. In fact, when we model a one standard deviation deterioration or strengthening in the factors, we see the resulting predictions rise or fall by nearly 20bp (168 and 131bp bear and bull cases).



Introducing the iTraxx Main Macro Credit Model

March 9, 2012

Phanikiran Naraparaju
 Andrew Sheets
 Serena Tang
 Sivan Mahadevan
 Peter Mallik

Late last year we explored the influence of various cyclical, systemic and technical factors in forecasting credit spreads (see Chapter 20). We received a lot of investor interest in that model, which forecasted CDX IG credit spreads based on six factors: (1) ISM PMI (levels), (2) ISM New Orders (levels), (3) PPI (YoY returns), (4) FINRA Trace IG Bond Dollar Volume (MoM returns), (5) S&P LT EPS growth (MoM differences), and (6) 3m S&P variance (MoM differences). In this report, we have tried to create a similar model in Europe, i.e., a medium-term macro credit model that attempts to predict spreads for the on-the-run 5-year iTraxx Main index.

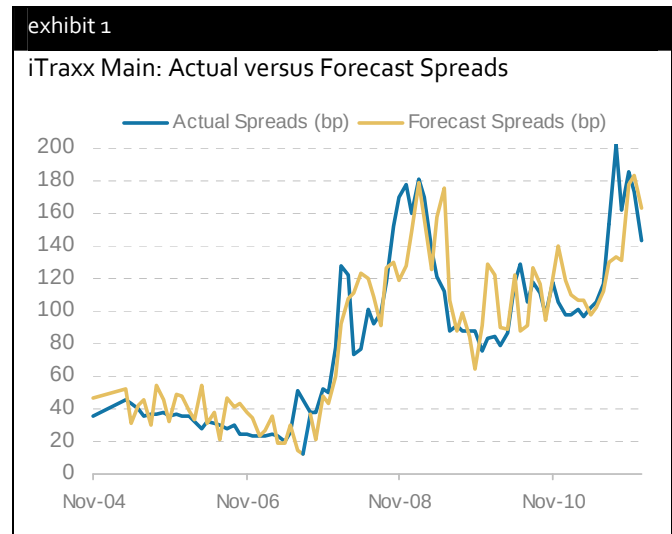
Building a robust forecasting model for iTraxx Main has been even more challenging than in the US due to the regime shifts we have had over the last several years. Given enough variables, we could potentially fit any series, but we limited ourselves to five variables. In doing so, the best model for the period 2006 to today came up with a spread level R^2 of 75%.

How do we intend to use such a model? As with the CDX IG model, we think of it more as a medium-term fair value model rather than a trading model, and a good indicator of the factors and their relative significance in driving the future spread movements. Beyond the model itself, we found it quite instructive to study the factors which proved to be the most significant drivers of credit in different macroeconomic environments. We intend to recalibrate the model at regular intervals, and changes in model should provide further insights into changing market dynamics. A summary of our iTraxx Main macro credit model follows:

iTraxx Main model factors: Based on the statistical analysis of iTraxx index data from 2006 to date, we finally selected five factors from our factor pool which did best in explaining the historical movements of the iTraxx index. From model results, we identified UK PPI (YoY) and EU CPI (YoY) as cyclical variables. As a fundamental variable, we identified EU growth expectations as a significant factor. For technical/systemic factors, we have identified EURUSD 30-day realized volatility and iTraxx index 90-day realized volatility.

iTraxx Main model results: The model was selected through a rigorous selection of five factors with maximum adjusted R^2 . The model has a spread level-adjusted R^2 of 75% for data from January 2006, with spreads varying from a

tight of 21bp to a wide of 216bp. We were able to achieve even better fits for individual regime models.



Source: Bloomberg, Morgan Stanley Research forecasts

Model sensitivity: Cyclical factors (EU growth expectations, EU CPI index) were among the most important variables on a consistent basis. EURUSD vol is the systemic risk metric for Europe (akin to SPX variance in the CDX IG model). One somewhat surprising factor was the strong mean-reversion element in credit spreads on a three-month timeframe. A spike in iTraxx Main realized vol (3m) forecasts tighter spreads three months later.

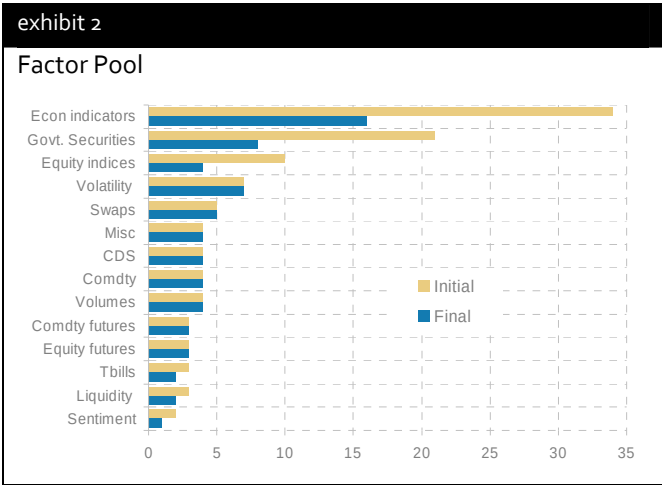
Current model predictions: The three-month model forecasts a 17bp tightening in iTraxx Main spreads from the February levels of 129bp to 112bp. The bulk of this move is driven by improved EU growth expectations in February (-20bp), partially offset by the sharp decline in iTraxx vol which portends wider spreads (+7bp).

BUILDING THE ITRAXX MAIN MODEL

As with our CDX IG Model (see again Chapter 20), our main focus was to predict iTraxx Main spreads three months forward based on different macro factors. We used the index spread history from 2006 to date, and the factors were chosen from a wide pool to account for the varying regimes European credit has been through.

Factor selection process: We started with a pool of factors which we thought could potentially indicate a changing market overview of credit spreads going forward. European markets have gone through different regime shifts in the last 15 years, historically taking cues from US markets, but the recent prolonged European sovereign crisis has dramatically

changed the market dynamics. Against this backdrop, we selected factors representing significant market and economic activity, not just in the European markets but also in global markets.



Source: Morgan Stanley Research

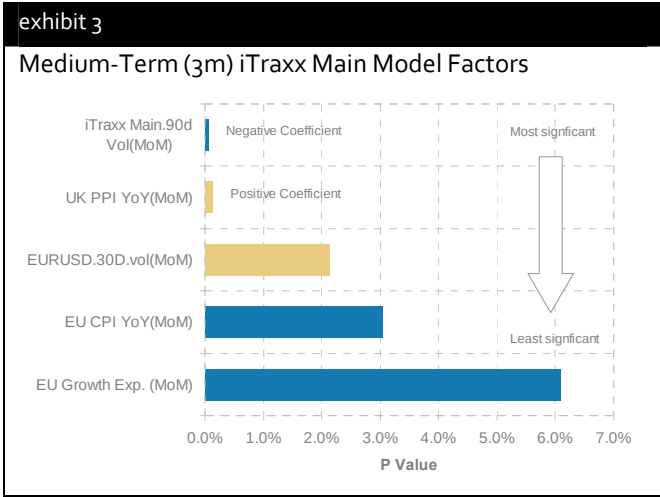
Beginning with a list of 107 variables, we classified them primarily into market and economic indicators. The variables came from a vast pool of major global equity indices, short and long-term gilt yields, major commodities, EU and global economic health indicators, equity and commodity futures, realized and implied market volatilities, CDS markets, swaps, futures trading volumes and sentiment indicators.

Acknowledging the problem of non-stationarity in most of the financial time series variables during the period in question (2006 to date), we corrected them by first and second-order differencing and dropped variables which could not be made stationary by this method. Next, we removed fields which were highly correlated and kept the more intuitive and appealing variables, ending up with a shortlist of 67. The final selection for a regression for a period was based on a best fit as measured by statistical analysis subjected to statistical significance.

Model selection process: The non-stationary behaviour of iTraxx index spreads forced us to model incremental change in spreads rather than the original level. With an objective of getting a robust, statistically significant model, we used stepwise leap regression procedure with a constraint of maximizing adjusted R^2 for a given number of variables. We capped the number of independent variables we can have in a given model at five. We also checked for multicollinearity (redundancy of variables) in the final model.

MODEL SENSITIVITIES

For the period from mid-2006 to date, we ended up with a five-factor model to predict the three-month change in iTraxx Main spreads. Exhibit 3 lists the independent factors from the most significant to least significant (increasing order of p-values), i.e., iTraxx Main 90-day realized volatility is the most significant factor, whereas EU growth expectations was the least significant among the chosen variables. The factors with negative coefficients (marked blue) indicate a tightening in spreads with increase in the factor values, whereas factors with positive coefficients (marked yellow) indicate a widening in spreads with increase in the factor values. For example, improved growth expectations help in healing of credit spreads, as shown by the negative coefficient for the EU CPI and EU growth expectations. In the CDX IG model, we had ISM PMI and new orders as the equivalent cyclical variables.



Source: Morgan Stanley Research

In contrast, an increase in EURUSD 30-day realized volatility, a sign of systemic risk in European region, is a predictor of wider spreads for iTraxx Main. In the CDX IG model, we had SPX variance as the equivalent systemic risk factor. An increase in UK PPI is also a predictor of widening in spreads, which is similar to the sensitivity of the CDX IG model to the US PPI.

All the above parameters are largely intuitive. The parameter that was the most surprising in the model to us was the iTraxx index current 90-day volatility for the extent of its (high) significance in the model as well the direction of the sensitivity (negative). Essentially, a sharp increase in three-month vol is a contra-indicator, signalling mean-reverting behaviour in the index. Despite our considerable efforts to replace this variable in the model, we could not find an equally strong replacement.

In Exhibit 4, we provide the coefficients and the sensitivity of the model (partial deltas) to each of the individual factors based on the largest moves in either direction and a one standard deviation move in a period for that factor.

exhibit 4				
Model Deltas to Different Factors				
	Coefficient	Forecast Spread Move (bp)		
		Max	1 σ	Min
iTraxx_90D_Vol	-1.0	33	(12)	(50)
UK.PPI.YOY	31.2	37	18	(53)
EURUSD.30Drvol	5.2	28	11	(34)
EU.CPI.YOY	-32.9	36	(10)	(20)
EU.Growth.Exp	-0.8	26	(9)	(20)
iTraxx Main		96	32	(59)

Source: Morgan Stanley Research

MODEL CALIBRATION AND HISTORICAL PERFORMANCE

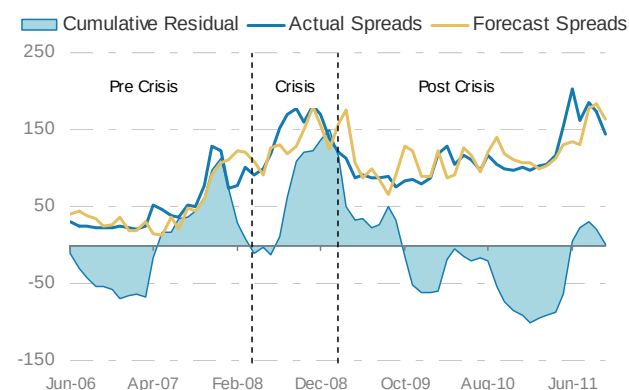
Our current model is calibrated over the time period starting from mid-2006 to date and will be recalibrated every quarter. Given the dynamic nature of the European market, it is possible that other factors may become significant going forward. Exhibit 5 plots the actual and forecast spreads and the cumulative three-month residual.

As mentioned before, Europe has been through different regimes in last six years. We found it a challenging task to fit a statistically significant model with a limited number of factors and a reasonable adjusted R^2 . Acknowledging the different regimes Europe has been through in the last six years, we also investigated if different models are more successful in shorter horizons and offer a better understanding of the factors that are significant in different environments rather than a single model across all regimes (see Appendix). We believe this can be of great value once European sovereigns normalize.

What did not work? The most obvious omission in the full-period model is clearly equity market prices and vols, both US and European. The sovereign crisis has caused European credit to disconnect from stocks and risk assets globally. Even equity volatility, which we have historically perceived to be strongly correlated to credit spreads, proved to be less useful. In the individual regime studies in the Appendix, we find that systemic risk/technical indicators such as IG bond trace volumes, basis swap levels, gilt levels and even stock returns scored relatively high. The other major omission is Bunds, which are a strong risk-aversion indicator in recent times but did not prove to be significant factor in the model due to the coincident nature of moves with credit spreads recently and the weak correlation to credit spreads before the credit crisis.

exhibit 5

Cumulative Residual Over Time



Source: Bloomberg, Morgan Stanley Research

We avoided using fundamental factors such as earnings and leverage for the purposes of this model due to the large time lag for the data – as much as six months – and the mismatch in frequency of the data – quarterly as opposed to monthly for most other parameters.

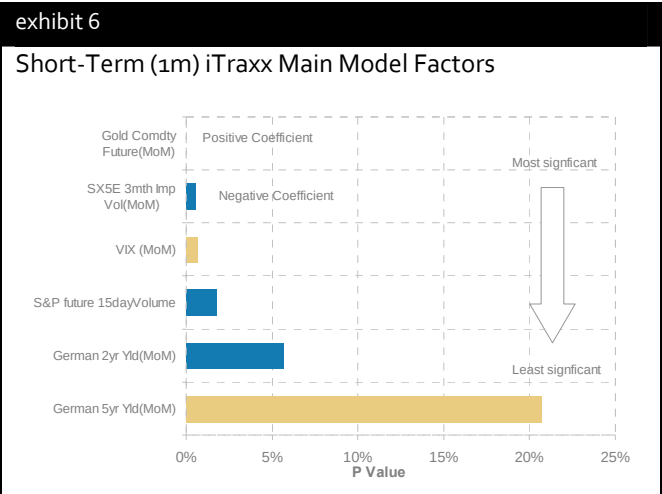
We also explored the role of momentum and mean-reversion in European credit and find that the recent changes in iTraxx Main spreads (1-3m) are not very useful in forecasting future spread changes (1m or 3m). Instead, what seems to work is changes in realized volatility, which show up in the model above. The second-order moves (i.e., vol of change or acceleration) in the index are useful rather than the first-order moves (change of spreads or velocity).

Although we did not use the 2004-06 data for calibration, we used the model for forecasts in that period (a small 'out of sample' test if you will) and find that the forecast spread moves, while more volatile than actual spread changes, were of the same order of magnitude during that period (see Exhibit 1).

SHORT-TERM SPREAD MOVEMENTS (ONE MONTH FORWARD)

We also tried to build a short-term one-month macro factor model to look at indicators of credit movement in a shorter timeframe. As seen in Exhibit 6, model results indicate that iTraxx spreads are sensitive to curve changes in equity vol (VIX versus 3m vol). One-month vol as measured by VIX is a momentum factor, whereas three-month volatility is a contrarian indicator.

Gold commodity futures price increases forecast wider spreads in the short term. Increase in S&P trading volumes tend to cause spreads to heal. On a less significant level, the model also hints at spread widening with an increasing German 5-year yield and decreasing 2-year yields, indicating an accompanied yield curve twist, with risk rising in 5-year Bunds and iTraxx 5-year index spreads following.



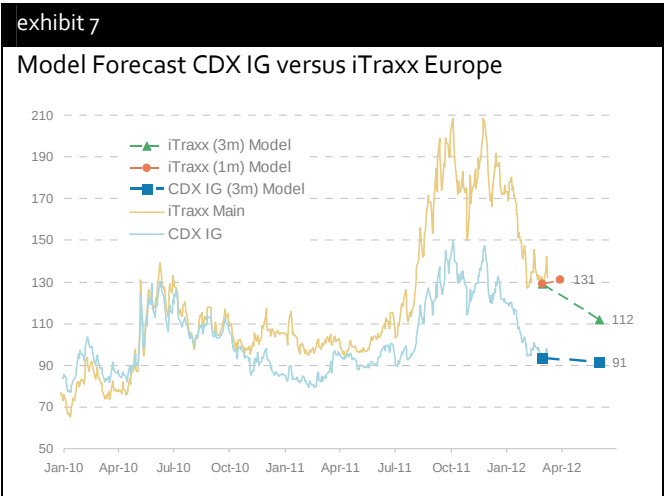
Source: Morgan Stanley Research

MODEL FORECASTS

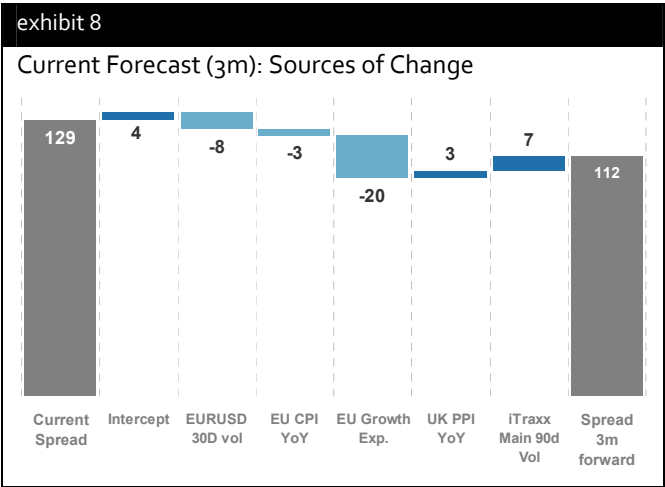
The medium-term (3m) model forecasts a three-month change in the iTraxx index of about 17bp from end of February spread levels (see Exhibit 7), which is a considerable move in the index. The short-term (1m) iTraxx Main model, in contrast, is forecasting a 2bp widening in the index levels in the month of March. Interestingly, the CDX IG model is forecasting just a 2bp tightening in credit spread from end-February levels.

The biggest source of change for the iTraxx model (3m) is a 20bp decline forecast due to rise in EU growth expectations in February (see Exhibit 8). Improved cyclical parameters – EU CPI and decline in EURUSD volatility – also suggest

that spreads should tighten. The key factor arguing for wider spreads is iTraxx 90-day realized volatility, where the decline in volatility is cause for caution on spread levels, in our view.



Source: Bloomberg, Morgan Stanley Research

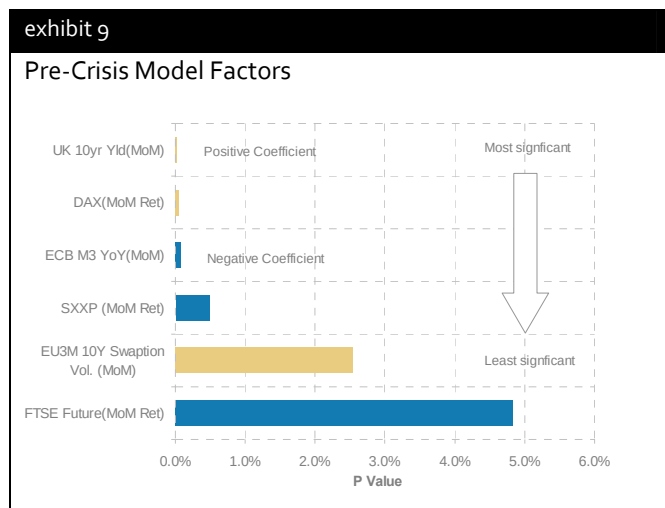


Source: Bloomberg, Morgan Stanley Research

Appendix: Regime Study

As mentioned earlier, we investigated how the models change in different regimes for iTraxx Main. We have divided the history of iTraxx Main into three regimes: i) June 2004 to September 2007; ii) October 2007 to September 2009; and iii) January 2009 to date.

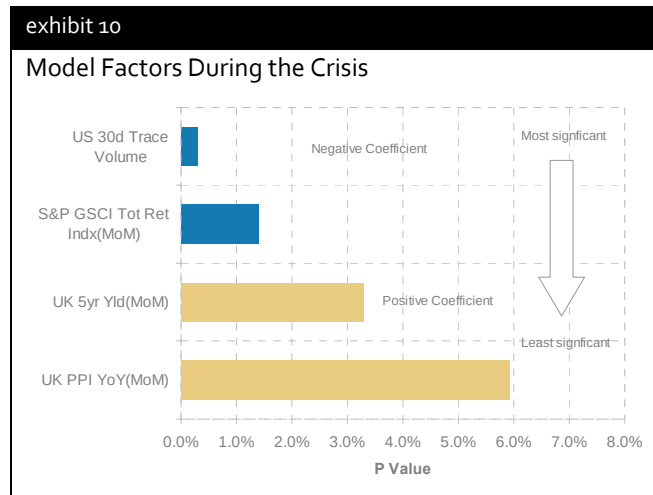
Pre-crisis model: Using the same methodology from earlier, we developed a six-variable model for predicting the iTraxx index level three months forward for the period from mid-2004 to late 2007 with a spread level R^2 of 60%. Exhibit 9 shows the significant independent variables in increasing order of p-values (decreasing significance). During this period, spreads moved in a very tight range of 30bp. The model suggests that iTraxx Main spread moves were dominated by a combination of growth indicators and risk-inversion activity. The model predicts credit tightening with lower UK 10-year yields (the most significant factor). The model also predicts credit tightening for an increase in SXXP and FTSE futures returns but a fall in DAX returns. A fall in Euro 3m10y volatility and rise in ECB M3 growth are also indicators of credit tightening over the next three months.



Source: Morgan Stanley Research

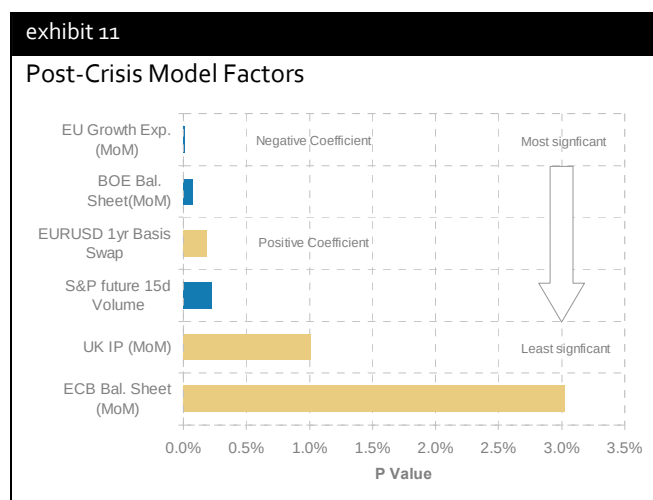
Crisis model: For the credit crisis period, we limited our variables to four, keeping in mind the relatively few data points. The model had a spread level R^2 of 74%. Exhibit 10 shows the four significant variables in the increasing order of p-values (decreasing significance). Improving liquidity, as measured by the increase in 30-day US trace volume

(corporate bond trading volumes), and better stock market performance, as measured by S&P GSCI total return, forecast tighter spreads. An increase in UK 5-year yields forecasts wider spreads. As with the overall model, an increase in the UK PPI again correlated well with a widening in iTraxx Main spreads, although with less significance than in other periods.



Source: Morgan Stanley Research

Post-crisis model: Finally, in the post-crisis period, a combination of growth (EU growth), central bank balance sheet increases (BoE) and improved equity futures volumes presaged a tightening in iTraxx Main spreads. A rise in systemic risk indicators (EURUSD basis swaps) forecast wider spreads, as does the UK CPI.



Source: Morgan Stanley Research

Falling Forward

June 2, 2011

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 Peter Mallik
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 Viktor Hjort

Portions of this report regarding non-US options are not intended for US clients, other than Institutional Clients. Investing in options is not suitable for all investors. Please see the disclosures at the end of this report and discuss whether this or any particular options strategy is suitable with your Morgan Stanley representative. Please direct all market-specific questions to the coverage analyst and all options-specific questions to Sivan Mahadevan, Derivative Research Strategist.

As investors use today's relative low volatility environment to debate bullish positioning and market hedges, we find that one common theme across both credit and equity markets globally can have a meaningful impact on the price and attractiveness of certain option strategies.

The forward level of any market instrument is an important concept in finance: it represents the market's expectation of the price of that instrument, absent any new market information. This forward price can be derived from an arbitrage-free view of the world, or the concept or risk-neutral pricing, i.e., the notion that all market instruments (independent of their yield or risk) will provide the same total return over time. Forwards in a sense provide a fair market price for an instrument on some future date, and are therefore very important inputs in the pricing of options across most markets including equities and credit. We find that fixed income investors focus on forwards quite often (given the importance of term structures), but equity investors do less so.

If you lean against the forwards, how likely are you to be right? Or, to frame the question in another way, how 'correct' are the forwards to begin with. The answer, generally speaking, is not often, which in itself is quite interesting. Not realizing forwards does not necessarily mean that the market is wrong, or inefficient. It is a reflection of other factors that influence pricing in the future including any new information that becomes available and technical factors including carry, which have huge implications on the performance of certain strategies (see [Leaning Against the Forwards](#), January 26, 2007 for a study we conducted in interest rate markets).

Some of the skew we see in option pricing can in effect be a function of markets leaning against the forwards, especially when they are extreme. With low levels of front-end implied volatility, steep interest rate and credit term structures, and a disconnect between stock dividends and short-term interest-

rates, forwards today are an important driver of option prices. As such, we see some attractive opportunities in leaning against such forwards using options strategies, and we focus on that theme in this chapter, covering both credit and equity markets. A summary of the ideas are below and in Exhibit 1, with details in the remainder of this chapter.

Global Equity Options: The relationship between dividends and short-term interest rates pushes forward equity index levels higher than spot levels in higher inflation markets like Brazil and India, and we find further support for hedging through OTM puts in these markets as a result. In the US and Taiwan, the opposite is true and OTM calls price favorably and provide support for positioning for upside moves.

VIX Opportunities: The steep VIX futures curve makes OTM VIX put options attractive from a forward perspective, and this is a trade that we now reinstate given recent moves.

US and European Dividends: In the S&P 500, we favor buying 2013 dividends and selling 2016 dividends, while in Europe, we like buying SX5E 2012 / 2013 dividend futures, in both cases driven partly by forward relationships.

US Credit: Steep credit curves driven by term premium support buying OTM receivers on CDX IG, and even selling far OTM payers as a way to fund the trades.

exhibit 1

Preferred Trades to Lean Against the Forwards

Trades	Price (% of Spot)	Discount Due to Fwd (% of Price)
Equity		
SPX 3m 105% Call	1.0%	9%
DAX 3m 95% Put	2.0%	1%
NIFTY 3m 95% Put	1.6%	6%
TWSE 6m 110% Call	0.7%	55%
Bovespa 3m 95% Put	1.7%	29%
VIX Jul11 18 Put	7.1%	25%
Buy 2013, Sell 2016 SPX Divs ¹	-5.3 pts	
Credit		
CDX IG Sep11 80 Receiver	10 bps	30%
CDX IG Sep11 80/130 Bull RR	1 bps	

Note: Dividend price is the difference between dividend swap levels in 2013 and 2016. Pricing is indicative.

Source: Morgan Stanley Research

WHAT'S IN AN EQUITY FORWARD PRICE?

Forwards in equity markets are very different from forwards in interest rate markets. While the latter embeds the consensus market view of future short-rates (plus liquidity, term, and other offsets), equity forwards do not provide information on the market's expectation of future equity

prices. Equity forwards are constrained by cash-and-carry arbitrage and thus reduce to the product of the current spot price, and the difference between interest rates and expected dividends.

The true measure of an option's "moneyness" is the distance between the strike and the forward price – not the spot price. So with an inverted forward curve, a put option with a certain strike is actually closer to the money than it would be if the forward curve was flat, making the option more expensive. Likewise inverted forward curves make calls inexpensive. On the flip side, upward sloping forwards make calls rich and puts cheap.

Is it valuable to lean against equity forwards? Stock prices should move along the forward line in the long-run – equity investors should be compensated for the time value of money, and dividends are part of the total return as well (so should be excluded from stock price appreciation lest equity holders get over-compensated). But with this forward containing no other information on the balance of risks in the market, or the equity risk premium, they are arguable safe to lean against in the short-term.

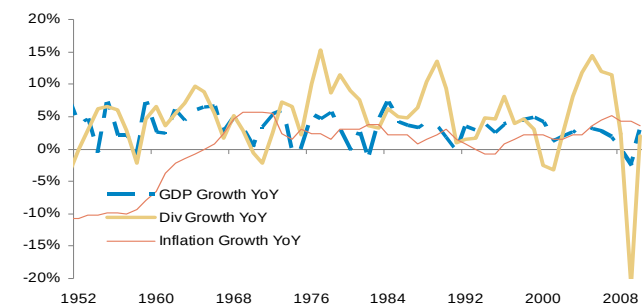
The difference between forward prices and spot prices is typically slow to change in the equity market. Interest rates are generally low, and changes modest versus the level of dividends (i.e., you need eight 25 bps Fed hikes to equal the current 2% dividend yield on the S&P 500). But there are differences across markets in both the level and shape of interest rate and dividend curves that make leaning against the forwards more attractive in some places than others.

Both growth and inflation influence equity forwards through their impact on interest rates and dividends, and today inflation is a big driver of the differences in forwards we see across markets globally. While theoretically the impact of both growth and inflation should cancel out (higher levels of both should raise rates and dividends in lock-step), in practice this is not the case as different markets place varying emphasis on either growth or inflation, leading to very different forward curves. Typically stronger growth leads to higher real rates, and dividends as well – history bears this out as correlation between GDP and dividend growth is nearly 40%. Expectations for higher inflation also drive increases in nominal rates, and should increase both earnings and dividends as well, although history shows a weaker relationship here.

In most developed markets, the current environment is one of inverting forward curves as interest rates are low, with dividends also somewhat low as payments have not yet fully recovered from the crisis. Looking ahead, markets are priced for increases in both, but interest rate curves are much steeper than dividends, so the discount of forward to spot falls as we extend to further maturities.

exhibit 2

GDP and S&P 500 Dividend Growth Are Correlated, Inflation Less So

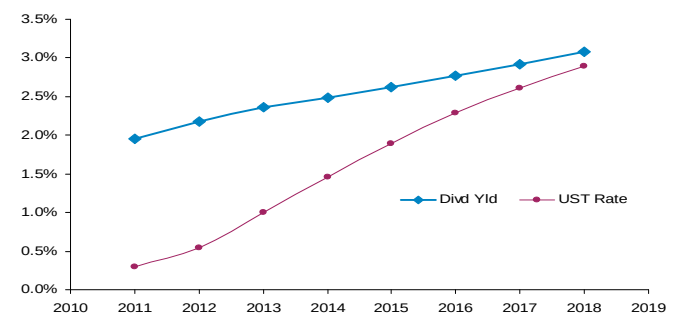


Source: Morgan Stanley Research, Bloomberg

We see that very strongly in the US, where the level of interest rates should remain low for some time as the Fed continues a loose monetary policy even as inflation moves higher. In Europe the level of rates should also remain low, settling at 2% by next spring on our forecasts after two more hikes. But dividend yields are much higher in Europe, driving forwards even lower versus spot than in the US. The forward curve flattens more quickly in Europe though as we go out in time, as the ECB is being more aggressive than the Fed, and the EuroStoxx dividend market prices much less year-over-year dividend growth than the S&P 500, and is actually inverted from 2012 to 2016.

exhibit 3

US Interest Rates Priced to Remain Under Dividend Yields, Inverting Forward Curves

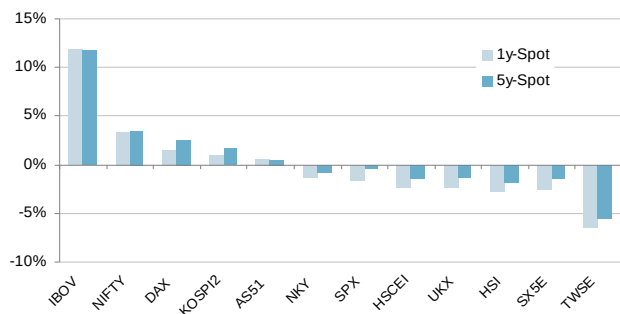


Source: Morgan Stanley Research, Bloomberg

In Asia and Emerging Markets, forwards show a range of shapes. In Hong Kong, China, and Japan, the differential between interest rates and dividends is similar to that in the US and Europe, leading to inverted forward curves. Meanwhile economies with the highest inflation rates – India, Australia, Korea, and Brazil – have equity index forwards that are well above spot prices. With inflation and likely rate hikes to contain it a priority for authorities, these forward curves should remain upward sloping.

exhibit 4

Equity Forwards vs. Spot across the Globe



Note: Bovespa and DAX are total return indices, so the forward curve for the tradable instrument only reflects the level of interest rates
 Source: Morgan Stanley Research, Quantitative and Derivative Strategies, Bloomberg

How can we take advantage of these curve shapes?

Forwards are not the principal driver of whether we like buying or selling options on a given index, but they can help pricing on strategies that are also attractive for other reasons. And for some strategies like risk reversals, they are a significant pricing driver (although we focus on simple puts and calls here, risk reversals are similar to synthetic forwards, and thus have a high sensitivity to forward levels). Here we highlight interesting trades in our view, where pricing can also benefit from the shape of equity forward curves.

In the US, low volatility, a steep upside skew, and potential reacceleration in growth in 2Q motivate buying 3m 105% S&P 500 calls for ~100 bps. The inverted forward curve reduces the cost of this trade by ~10 bps compared to if the curve was flat. In Europe, we think a potential sovereign restructuring could be a downside catalyst and with vol and skew flat, we like buying puts on DAX and SX5E. The inverted SX5E forward curve works against our view. But for the DAX dividends are reinvested, driving an upward sloping forward curve. We prefer SX5E as a hedge given flatter skew and higher exposure to financials, but the price benefit of moving to DAX is interesting – 3m 95% puts cost 2.0%, versus 2.3% for the SX5E.

Meanwhile in EM we are bearish on India due to inflation concerns and like buying OTM puts that are cheaper due to the steep forward curve. Our underweight fundamental view, falling spot and vol, and falling skew make outright protection attractive. We like buying 3m 95% puts for 1.6%. Taiwan has the most inverted forward curve globally, driven by a high dividend yield (~4% annually) and negative rates implied by the offshore market, although our house view is constructive. We like buying 6m 110% calls on TWSE for 70bps. Brazil has the steepest forward curves globally, driven by policy rates in excess of 12% and reinvested dividends (IBOV is a total return index). After recent underperformance, our LatAm and EM strategy teams have both turned more bullish on Bovespa due to valuations and

moderating inflation. With high forwards, we like 3m 110% calls costing 1.2% of spot on the USD denominated MSCI Brazil to play for upside, and local index puts for those looking for an EM hedge as Brazil is a high-beta index. 3m 95% puts on IBOV cost just 1.7% of spot, ~70 bps less than if the forward curve was flat.

VOLATILITY CURVES ARE STEEP

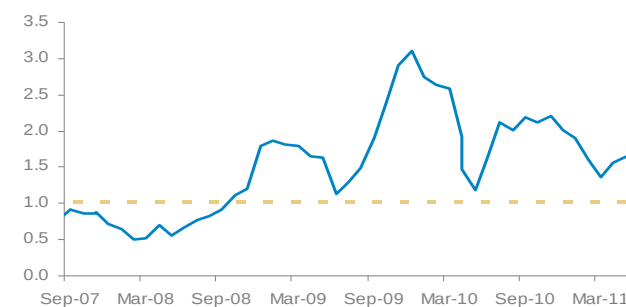
Much like interest rate curves, volatility curves are steeply upward sloping. Some of the drivers are the same – with QE2 the Fed reduced downside risk by expanding its balance sheet and keeping rates low (*Swapping S&P Puts for Bernanke Puts*, Nov 9, 2010). But this support only buys time for the economy to heal, and with the unresolved structural issues longer-dated volatility remains elevated.

The steep volatility term structure is perhaps most visible in the VIX futures curve, which has received much focus over the past two years with the explosion of VIX-based products. Currently, the gap between the 3m VIX future and VIX spot is over 2 volatility points. For perspective, this spread was as high as 7 volatility points as recently as April. But there is another important similarity between interest rate and volatility forwards – both contain information on what the market's expectation for the price of that asset will be in the future. For VIX this is possible because investors cannot buy the VIX, finance it, and deliver it into a forward contract.

How accurate is the information content in VIX? We don't have a deep enough history to say for sure, but over the past two years we've found that VIX futures typically roll more to spot, than spot VIX rolls to the futures level.

exhibit 5

VIX Futures Tend to Roll to Spot More than Spot Rolls to the Futures



Notes: Ratio above is the absolute value of the difference of the 1m VIX future 1m before expiry and spot at expiry over VIX spot 1m before expiry less spot at expiry. $[(F(-1m) - Spot(Exp)) / (S(-1m) - Spot(Exp))]$

Source: Morgan Stanley Research, Quantitative and Derivative Strategies, Bloomberg

Combined with a view on the fair level of volatility, the current steepness of the VIX creates attractive trading opportunities. Today we reinstate our view of buying 2 month VIX puts. We believe we are entering a lower

volatility part of the cycle, but there are still risks in the near-term that could drive spikes higher in volatility. Upside VIX options price this jump risk as skew is steep, but this makes VIX puts relatively cheap. With the futures curve upward sloping, we can buy strikes that are OTM versus the futures, but ATM versus spot – the July 18 put costs \$1.30, at a 32 delta, despite the strike being very close to current VIX (July forward is 19.5). As futures roll down to spot, this put will become closer to ATM and appreciate.

On a further flattening of the futures curve we'd put on more direct curve trades like diagonal put spreads (calendar put spreads with different strikes). While still limiting exposure to upside moves, this spread gains value if the gap between the front and further dated futures expands ([Trading the VIX – Which Way to Lean?](#), Mar 29, 2011).

FOCUS ON DIVIDENDS

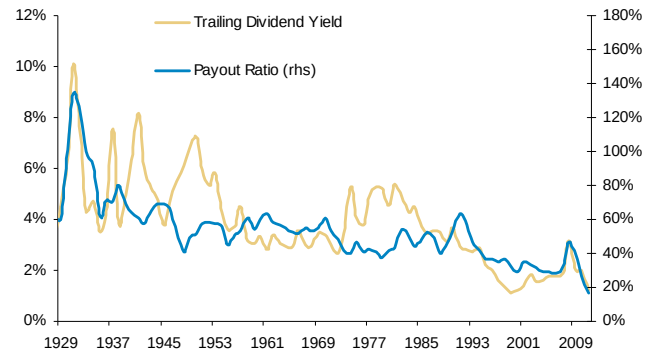
With dividends such a big driver of equity forwards – particularly in the US and Europe given low interest rates today – it is useful to dig into the story a bit more. We find there are interesting opportunities to trade the dividends that are implied by forward prices.

Since July 1, 2010 (the 1-year S&P low), near-dated dividends have underperformed the S&P 500 as corporate profits have risen, but companies are still keeping much of that cash on the balance sheet or using it for M&A and share buybacks. Meanwhile longer-term dividends have outperformed as the curve has steepened. Going forward, we expect shorter-term dividends to outperform, as implied dividend growth from 2011 to 2012 of 11% is converging to consensus EPS growth rate of 13%, while longer-dated dividends may face headwinds as a +5% year-over-year growth looks too optimistic.

Payout ratios are historically low, and as recovery progresses and uncertainty falls, payout ratios may increase. In fact, a modest increase in payout ratios to 30% from the 2010 level of 27%, still well below the historical average of 46%, would drive nearly 3.2 points of upside to 2012 dividend swap levels. We also feel there is less downside risk to near-term dividends – if companies pay dividends commensurate with 2012 swap levels, the payout ratio would be 26% on consensus earnings estimates, which would be the lowest on record.

exhibit 6

Limited Further Downside to Payout Ratios

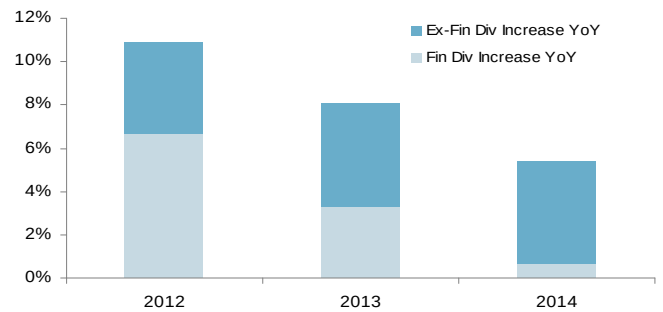


Source: Morgan Stanley Research, Quantitative and Derivative Strategies, Bloomberg

Financials are a big part of the near-term dividend story. If Financials achieve our Banks team's estimates for earnings and dividends, sector payout ratios may more than double from 12% now to 27% in 2013, which would drive an incremental 3.6 dividend points for the S&P 500 – a 14% increase above current 2011 levels *just from this one sector alone*. Currently 2013 is priced for 9% year-on-year growth for the entire index.

exhibit 1

Financials Share of Implied Dividend Growth Rate



Source: Morgan Stanley Research, Modelware, Bloomberg

Meanwhile longer-dated implied dividends are more reflective of longer-dated GDP growth estimates than specific company or sector forecasts. In light of this, we think longer-term dividends growth rates are too high today. The 2014 to 2018 annual growth rate is 5.5%, which is in the 90th percentile since we have data beginning in June 2008. This level of growth is in-line with the median actual dividend growth rate since 1950, so in a normal environment would be reasonable. But given the structural demographic and savings challenges in the US economy, we think the implied growth rate may be too optimistic, and is at least a poorer risk / reward than near-term dividends. To play this

theme, we favor buying 2013 dividends at ~31.2 and selling 2016 dividends at ~36.5 (theoretical mids).

In Europe, we like buying SX5E 2012 / 2013 dividend futures due to their accelerated growth post trough, outperformance of equities during selloffs, and potentially higher contributions from the Energy, Industrials, and Consumer Discretionary sectors. Since we are constructive on earnings growth, the SX5E dividend term structure may normalize absent a double-dip scenario ([SX5E Dividends – Defensive Upside](#), Apr 11, 2011).

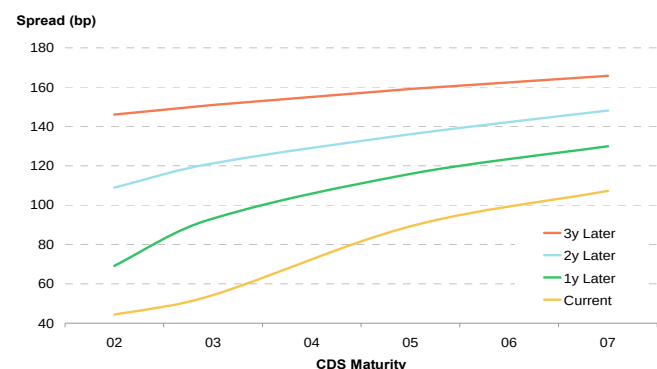
CREDIT FORWARDS

The very same term premium of volatility that we see in many asset classes also manifests itself in the shape of credit curves. Recall that credit spreads are compensation for two factors, namely defaults and volatility. While high yield credit spread is mostly compensation for default risks, investment grade credit spread is largely compensation for volatility. The term premium in credit curves arises from both these factors – i.e., a term premium arising out of increased default risk over time as well as term premium attributable to volatility. Technicals also play a part. As a result of macro/sovereign concerns 5yr spreads have been at elevated levels as investors use the most liquid part of the market as a hedge, whereas greater visibility on short-end default risks makes investors more comfortable about risk-taking in the short-end.

Bottom line – credit curves are extremely steep, both in a historical context as well as in an absolute sense. This is evident in the level of curve forwards as seen in Exhibit 8. The 5yr spread in 2yrs time for CDX IG 5yr is 50% higher than the current spread of 89bp, and nearly 80% higher in 3yrs time. We are firmly of the view that these levels of forwards will not be realized and believe forward credit spreads are attractive at these levels.

exhibit 8

CDX IG Curve Forwards



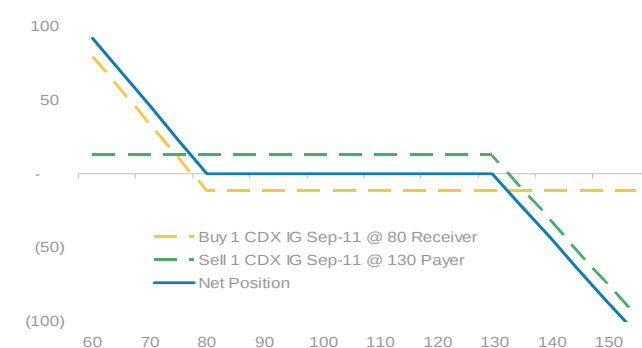
Source: Morgan Stanley Research

CREDIT OPTIONS TRADES

What is a good way to position for credit forwards not being realized? Strategies that involve buying receivers on CDX IG, specifically bullish risk reversals, make sense. Option pricing would typically take into account the forwards, whereas we believe rolling down the curve is a more likely outcome. The current IG16 3s5s curve is as steep as 35bp, implying IG16 tightens by 8-9bp (about 10%) over a 6 month horizon due to spread roll-down (as opposed to forwards being realized which would imply spread was 15% wider). Thus the incremental spread tightening required to monetize the receiver is relatively small. The buyer of receivers is leaning against the high forwards implied by today's steep credit curves.

exhibit 9

Buy CDX IG Bullish Risk Reversals to September



Source: Morgan Stanley Research

We are also comfortable with the sale of OTM strike puts in CDX IG as we are relatively constructive on downside risks for IG credit on the back of strong fundamentals as well as attractive pricing in terms of a steep skew. One could also hedge September expiry bullish risk reversals with shorter-dated (Jun/July expiry) slightly OTM puts on a tactical basis.

STRATEGY RISK FACTORS

Long Call: The maximum loss is the premium paid.

Long Put Spread: The maximum loss is the premium paid

Bullish Risk Reversal: The maximum loss is short put strike and long call premium less the short put premium.

Dividend Futures/Swaps: Investors long dividend futures/swaps participate 1:1 in movements on the underlying dividend levels. If the dividend index rises the position will show a profit and a loss if the index falls below the entry price. Eligible investors can gain long or short synthetic exposure to the return on the basket through a swap. Typically, the party gaining long exposure via swap will be required to make financing payments to the swap provider. Such payments will be based on various factors, including,

credit, liquidity, and borrowability at the time of the transaction. With all swaps, this type of basket swap offers customized exposure based on clients' interest in a given investment strategy and weighting profile. Please contact your Morgan Stanley sales representative for more details. The information contained herein has been prepared solely

for informational purposes and is not a solicitation of any offer to buy or sell any security or other financial instrument or to participate in any trading strategy. Swaps may not be appropriate for every investor. Please consult with your legal and tax advisors before making any investment decision.

Two Tales About Two Tails

February 24, 2011

Sivan Mahadevan

Ashley Musfeldt

Peter Mallik

Phanikiran L. Naraparaju

Many investors look for extreme tail risk hedges, the type that one can buy (at a reasonable cost) and lock in a drawer hoping that it has a reasonable payoff in a 2008 type scenario. While picking the right market and the right instrument is always a challenge, systemic risk plays are something that hedgers are after, and liquidity is a key characteristic, as hedges are not worth much unless monetization is possible in a time of need. Many markets and instruments offer short-term extreme tail risk plays, but given that long duration is a requirement for this type of hedge, the credit and equity markets have only a handful of instruments that would fit this bill.

The equity and credit derivatives markets have some common ground and common investors, but can otherwise be quite separated by a cultural investing divide. Nevertheless, we often find ourselves comparing the challenges and opportunities in both markets, with tail hedging being a common task. One such situation involves two corners of the market that are not necessarily connected on first glance, but share many similar market traits. When the market for long-dated variance swaps on equity indices were trading at historical high levels in the summer of 2010, we found the experience quite similar to one we encountered in super senior tranches (loss-remote tranches with a lot of subordination) in corporate credit three summers prior, during the early stages of the credit crisis. The markets for super senior protection in credit and long-dated variance swaps in equities both represent extreme tail and systemic risks and as such can be used to hedge such scenarios. These trades are not necessarily well correlated as individual market factors and timing have driven their performance over the past few years.

Compared to most valuation levels seen in history, both super senior tranches and long-dated variance trade with a significant premium to their fundamental value. They are reasonably liquid instruments that carry a fair amount of systemic risk in their pricing, but less fundamental risk. In fact, fundamentals are not so much the point, as investors focus on liquidity, the upside/downside risk/return profile, and what other parts of the market (capital structure) imply for their valuations. The argument to sell protection and/or sell variance is to capture the difference between the market premium and the fundamental premium, but such strategies tend to be low carry and very negatively convex in times of stress. This is why buying protection and/or buying variance can become attractive as large tail risk hedge.

However, if fundamental risk is low, how can we gauge value? Systemic risk and market technicals become very important. Compared to a pre-crisis world where many institutions had business models that allowed them to comfortably sell long-dated variance and super senior protection, the short side of this market is much more fragmented today. While sellers of both risks have not disappeared, they tend to be much more opportunistic today, more focused on valuations and trading ranges rather than regulatory-capital based efficiency (which has become far less attractive). Long positions in super senior protection and long-dated equity index variance swaps tend to both be long correlation (although different types of correlation) and long convexity. There has been some academic work done attempting to link the relative pricing of the two markets¹.

exhibit 1

Summary of Costs and Payoffs

Instrument	Current Level	P/L to All-time Wides	P/L to 2006/7 Tights	Risk / Reward
IG 30-100%	16	3.6%	-0.7%	5.3x
IG 15-30%	29	7.5%	-1.2%	6.1x
IG 15-100%	30	3.6%	-1.3%	2.8x
HY 35-100%	145	23.4%	-5.6%	4.2x
5-year SPX Variance	28%	9.7%	-4.8%	2.0x
10-year SPX Variance	30%	4.0%	-5.6%	0.7x

Source: Morgan Stanley Research

In this chapter, we describe the instruments and markets for super senior protection and long-dated variance swaps, and compare them as long-duration tail-risk mitigation strategies in the current market environment, detailing costs and payoff profiles. Our conclusions are as follows:

- While both instruments can serve as long-duration tail risk hedging instruments, in the current market environment, we find certain investment grade and high yield super senior tranche protection as more attractively priced.
- Within investment grade, we find five-year 15-30% junior super senior protection the most attractive, with a price of 29 bp and payoffs of 6.1x if we compare the P&L from today's level to the 2007 tights and 2008 wides.
- Within the high yield world, we find 35-100% solutions the most interesting, with a 4.2x payoff if we compare P&L from today's levels to 2007 tights and 2008 wides. The absolute cost, at about 145bp per annum is higher, but the

¹ See "On the Relative Pricing of long Maturity S&P 500 Index Options and CDX Tranches," P. Collin-Dufresne, R. Goldstein and F. Yang, January 25, 2010

P&L to the wides is approximately 23%, a very equity-like payoff.

- While five-year S&P 500 variance can be an attractive tail risk hedge, we believe that today's levels of approximately 28% are not an attractive entry point, we would favor levels at or below 20%, last reached in 2007.

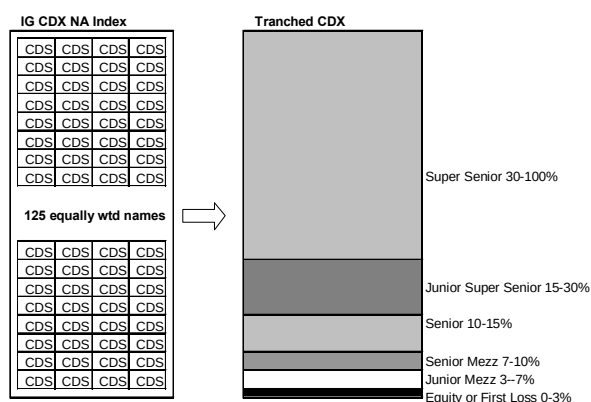
WHAT IS A SUPER SENIOR TRANCHE?

A super senior tranche is, by definition, the senior most tranche of a portfolio, usually with sufficient subordination to withstand a multiple of historical worst-case defaults. On-the-run maturities range from 3 to 10 years. The seller of super senior tranche protection will generally earn a small premium (on a quarterly basis) for taking this very remote default risk. A typical investment grade super senior tranche will have an attachment of 15% to 30%, while worst-case 5-year losses on a typical investment grade portfolio are around 3.0%.

A high yield portfolio will typically attach at about 35%, which while much higher, represents only a small cushion above historically worst-case losses (about 25%) for high yield credit. Index super senior tranches are fungible with the remaining index tranches and the underlying index itself, so such a property can keep pricing in a sane range during distressed periods where market liquidity falls. Given that super-senior protection is a portfolio product, its valuation is influenced by a measure of correlation. The measure most market participants gauge is termed default correlation, which is somewhat different from the correlation implied by index equity options (for more details on tranches and tranche pricing, see Chapter 3).

exhibit 2

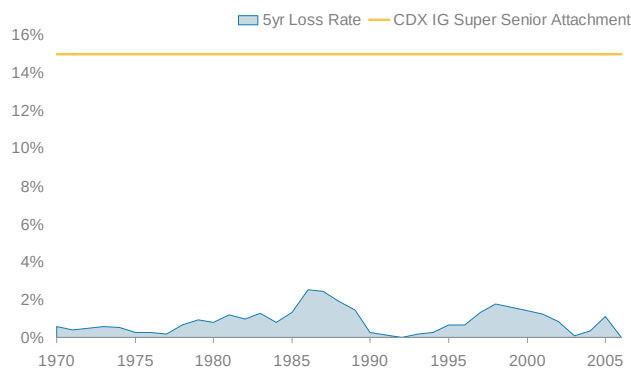
What Is a Super Senior Tranche?



Source: Morgan Stanley Research

exhibit 3

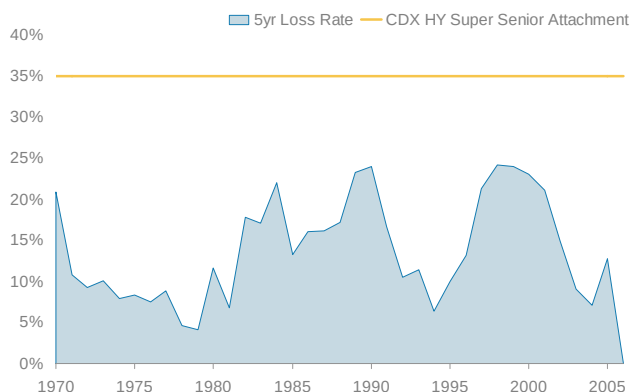
5-Year Cumulative Loss Rates – IG



Source: Morgan Stanley Research, Moody's

exhibit 4

5-Year Cumulative Loss Rates – HY



Source: Morgan Stanley Research, Moody's

THE MARKET FOR SUPER SENIOR RISK

Within the world of credit derivatives, one of the biggest structural changes has been at the top of capital structure, as super senior technicals have changed drastically during the crisis. Prior to the credit crisis, there was a fair amount of super senior risk taking, mainly by institutions that were setup well to monetize what was considered a risk premium trade. Attachments on investment grade portfolios ranged from 10% to 30%, and sometimes they were true super seniors (X-100%) and other times they were junior super senior (i.e., 15-30%). The super senior protection providers were generally financial institutions designed to take very remote AAA-rated risks including monoline insurers and SIVs (structured investment vehicles). With the super senior contagion risk that spread from subprime ABS and commercial real estate, most of these investment strategies performed poorly and are no longer taking on new super senior risk, if they exist at all. Given that the credit crisis separated corporate credit super senior risks (where actual losses were not realized), from subprime ABS and CMBS (where actual losses have been or are likely to be realized in

some cases), a different community of investors has taken interest in selling corporate super senior protection, although not nearly as aggressively as before.

We must also add that to the degree that financial institutions (especially banks) consider this space again, the capital treatment is unlikely to be as attractive as before the crisis given Basel considerations, although this does not apply necessarily non-financial institutions.

So in a nutshell, we do not expect the type of super senior protection selling today as we saw prior to the financial crisis, which drove super senior spreads to single-digit areas. Nevertheless, spreads on super senior can tighten meaningfully from here, owing to overall market tightening, and this risk must be considered in any hedging strategy. We do not expect super senior risk to outperform (delta-adjusted) other tranches as we see core credit investors favoring more junior tranches for risk taking today. The net result is that we expect super senior protection to enjoy favorable technicals as a tail hedge instrument.

SUPER SENIOR TAIL RISK PRICING AND PAYOFFS

We consider four simple super senior strategies across IG and HY indices, comparing their current pricing with the wides experienced in 2008 and 2010 and the tights in 2007.

We find 30-100% tranches appealing owing to their low absolute cost and good risk/return to a 2008 widening scenario (5.3x). But they do suffer from an absolute payoff of only about 3.6% in such a scenario, lower than what many tail risk hedgers are looking for. On that front, 15-30% looks a more suitable hedge, as the payoff (at 7.5% for 2008 scenario) is interesting with a slightly better payoff ratio (6.1x). HY 35-100% has a 23.4% return in a 2008 scenario appealing enough for equity investors, but the cost is about 5.6% if we move to the 2007 tights, resulting in a 4.2x payoff.

exhibit 5

Super Senior Pricing and P&L to 2007, 2008, 2010

	Current	All-time Wides	2010 Wides	2007 Tights	Payoff Tights to Wides
Spread (bps)					
IG 30-100%	16	95	38	1	
IG 15-30%	29	195	83	2	
IG 15-100%	30	110	46	1	
HY 35-100%	145	729	166	6	
PnL					
IG 30-100%		3.6%	1.0%	-0.7%	5.3x
IG 15-30%		7.5%	2.4%	-1.2%	6.1x
IG 15-100%		3.6%	0.7%	-1.3%	2.8x
HY 35-100%		23.4%	0.8%	-5.6%	4.2x

Source: Morgan Stanley Research

Note: 2007 tights are HY7 and IG7 5y. All-time wides are IG9 5Y and HY10 5y. May 2010 wides are HY10 5y and IG9 7y.

EQUITY VARIANCE AS A RISK PREMIUM TRADE

Equity variance has the similar risk-remote property as super senior protection, but in a more intangible form as there is no structural subordination. In an equity variance swap, a seller (risk-taker) receives a fixed amount (at expiration) and agrees to pay the realized variance (volatility squared) over the life of the contract times a notional. In a fair world, the fixed amount received would equal the expectation for volatility over the life of the contract. But given that volatility can rise more than it can fall (it is bounded by zero), and the mechanics of the contract (variance is the square of volatility, so it has natural short convexity for a seller), we find that variance typically trades above what a fair value would be just looking at history, not dissimilar to super senior credit protection.

exhibit 6

Realized Variance Statistics vs. Current Strikes

	1m	3m	1y	5y	10y
Min	1.7%	3.8%	5.2%	8.4%	9.7%
75th %ile	9.3%	9.8%	10.3%	11.6%	13.2%
Median	12.5%	12.8%	13.6%	14.9%	15.0%
Average	15.5%	15.9%	16.4%	16.9%	17.1%
25th %ile	17.7%	18.2%	18.5%	19.9%	18.3%
Max	109.6%	72.4%	56.5%	43.0%	35.4%
Current	21%	22%	25%	28%	30%

Source: Morgan Stanley Research, Bloomberg

As Exhibit 6 shows though, this holds more for longer-dated maturities. And right now, it doesn't hold for any, as implied strikes have fallen below the maximum volatilities realized during the Great Depression (during the summer of 2010 10-year variance reached loss-remote levels).

This does not mean we favor buying long-dated variance as a tail risk hedge here though, just because worst-case historical levels are above where the market prices today. All of the observations of 5-year realized volatility above 22% (11% of total) came after World War II, while we had many observations of 5-year implied volatility above the current 28% mark (30% of the time since 2000).

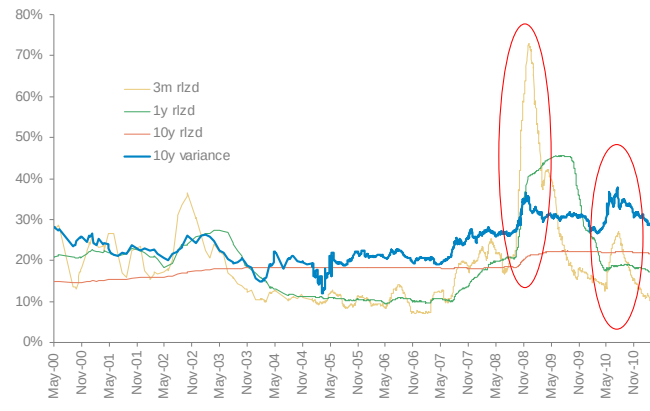
We see four main bridges from realized volatility to long-dated implied variance: 1) the implied-realized spread, analogous to the risk premium over defaults present in credit spreads generally; 2) the implied volatility term structure, analogous to term premiums in other markets; 3) the impact of skew which raises variance versus ATM volatility, which makes the moneyness of equity more similar to credit; and 4) the correlation embedded in equity index products, which has a risk premium of its own and is analogous (but not the same as) the correlation embedded in credit tranches.

The last of these carries most of the systemic risk premium we see in long-dated variance, and while it is present in all equity index products, it is most acute in long-dated variance

given the longer time to maturity and accumulated impact of term structure and skew. Combined, all of these forces mean things have to deteriorate severely, and broadly, across stocks, for long-dated variance to get “touched” by heightened realized volatility and see losses.

exhibit 7

Long-term Variance Overreacted in Spring / Summer 2010



Source: Morgan Stanley Research, Morgan Stanley Quantitative and Derivative Strategies

From a technical perspective, long-dated variance has similar imbalances to super senior as well. The long-dated volatility community has a large group of those who “have to buy,” including insurance companies who are short similar options to their customers (variable annuities), or increasingly institutions seeking long-term tail exposure, whether through variance, vanilla options, or structured products. Post financial crisis, the community of volatility sellers willing to supply this risk has become smaller, as volatility broke through what was previously viewed as the achievable highs. This has combined with long-term economic risks to drive the back-end of the curve up, and as we saw during the summer of 2010, it can spike higher as the fewer players that do supply risk to this market were forced to cover as volatility levels rose.

EQUITY VARIANCE PRICING AND PAYOFFS

The relatively low likelihood of realizing losses on selling long-dated equity variance makes it much like super senior – more of a mark-to-market trade to play perceptions of tail risk and higher realized volatility, rather than a trade on realized volatility itself. In Exhibit 8, we consider the P/L from buying 5-year and 10-year S&P 500 variance swaps assuming moves to the highs experienced in October 2008 and May 2010 and the lows in 2006. We also consider the speed to ramp to those levels, using realized volatility leading into each time period.

exhibit 8

Equity Variance P/L: Risk / Reward

Holding Period	5-year Variance P/L to...				10-year Variance P/L to...			
	Current Strike	Oct 2008 Levels	May 2010 Levels	2006 Lows	Current Strike	Oct 2008 Levels	May 2010 Levels	2006 Lows
1d	27.7%	9.7%	5.8%	-4.8%	30.4%	4.0%	4.3%	-5.6%
1w	27.7%	9.8%	5.8%	-4.8%	30.4%	4.0%	4.3%	-5.6%
1m	27.7%	10.5%	5.7%	-4.8%	30.4%	4.4%	4.3%	-5.6%
3m	27.7%	10.3%	5.3%	-4.9%	30.4%	4.4%	4.0%	-5.7%
6m	27.7%	9.6%	4.8%	-5.0%	30.4%	4.1%	3.8%	-5.7%
1y	27.7%	9.4%	3.7%	-5.4%	30.4%	4.2%	3.3%	-5.9%
2y	27.7%	6.7%	5.0%	-5.9%	30.4%	3.5%	4.0%	-6.3%
3y	27.7%	3.8%	3.1%	-6.1%	30.4%	2.6%	3.0%	-6.5%
4y	27.7%	0.9%	0.8%	-5.1%	30.4%	1.8%	1.7%	-6.3%

Source: Morgan Stanley Research, Morgan Stanley Quantitative and Derivative Strategies

Exhibit 8 illustrates that the risk/reward at current levels (~2x in the best scenario above) is less attractive than in credit senior tranches, as long-dated variance could fall to below 20% in a return to 2006 lows, while rise to 35-40% in a severe sell-off (from the high 20% range today). While we view both of these scenarios as somewhat extreme in the near term, they do illustrate that the risk/reward is not yet as attractive as we’d like it to be. In addition, the table also shows how the holding period is important: a quick move up in implied volatilities is clearly much more advantageous than a slow grind.

THE TWO TAILS – WHERE DO WE COME OUT?

How do we evaluate pricing between these markets? There is no perfect way, as they are different instruments as the following analogy shows: protection on IG super senior (15%-100% tranche) currently costs ~30bps running for 5 years. All upfront, this equals about 1.6%. An equivalent cost S&P 500 5-year option would be one that is about 70% OTM (or a 400 strike). Both are clearly very far OTM, but at expiration they are very different, as the likelihood of being touched is lower in credit than in equity, and the risk/reward is higher clearly making IG super senior much more of a tail risk trade than the equity put.

Despite the linkages, we caution that super senior and long-dated variance are not necessarily well correlated, as recent history has demonstrated different drivers in the market. In 2007 and early 2008, both super senior and equity variance repriced meaningfully wider, but it was credit that took the lead. Systemic tail risk was the key pricing driver, and super senior reacted in a predictable way here. While variance “should have” moved as well on similar risks, the equity markets were more complacent, so the repricing was more muted.

In late 2008, equity variance caught up, as the equity market finally realized it was in trouble too. Long-dated variance did not re-price in excess of other equity market instruments, but suffered at the same hands of higher volatility, higher correlation, and steeper skew as short-dated variance did. In 2009 both markets healed, but the 2010 sovereign crisis

impacted equity variance more. This was largely a result of technicals, as the lack of natural sellers exacerbated a short squeeze as volatility rose overall.

Today, we feel that super senior protection is more attractively priced than long-dated S&P 500 variance for long-duration large tail risk hedges. While we expect pricing on super senior protection to get tighter over time as a base scenario (credit improving), the downside costs are limited and still provide payoffs in the 4x to 6x range as we highlighted earlier. And credit healing from the crisis highs has been more sustained than in equity, which suffered from the summer of 2010 sell-off that still has lingering effects, leaving us with a less attractive risk/reward tradeoff.

When would we favor buying long-dated S&P 500 variance? We would look to establish long variance positions in the ~20% range for 5-year, and the low 20s for 10-year. This would be back near the lows of 2005 and 2006, and would likely coincide with realized volatility of 10% (2004-2006 average), a short-term volatility premium of 4%+ (implied

less realized that is 1% above bull market averages given persistent risk premiums), and a 1m-5y term structure of ~6%. We do expect term structures to remain somewhat more upward sloping than we have seen on average, as the current imbalance will likely persist for some time, and fundamentally structural economic risks remain as well, which deserve some term premium.

STRATEGY RISK FACTORS

Senior Tranche Credit Protection: Held to expiry, the maximum potential loss is the stream of coupon payments over the life of the trade, plus or minus any upfront pair or received. On a mark-to-market basis, buyers of protection take the risk of spreads tightening.

Long Variance Swap: At expiry, if realized volatility over the life of the contract is above the strike price, there will be a gain, and if realized volatility is below the strike price, there will be a loss. The maximum potential loss is the level of the strike squared.

Trading Tails in Credit-Equity

January 24, 2011

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Comparing and trading credit and equity markets has been an age-old opportunity and challenge. The two major asset classes are well correlated simply by virtue of being risky assets that are driven by similar dynamics (macro and company specific factors). But Merton models suggest credit investors are short the proverbial bankruptcy option to equity investors, so comparing credit spreads with equity implied volatility has been common as well. In this stage of the recovery, we see increasing flows and opportunities in trading OTM equity options with CDS at the company level that are driven in part by this Merton relationship. However, at the macro level, there are many other factors that can influence the relationship between credit and equity, not just the short-option relationship implied by such theory.

In this chapter, we take a fresh look at credit vs. equity relationships using a more macro approach, taking advantage of recent developments on both ends of the corporate capital structure. In credit, we now have a robust index options market, having grown quite substantially from pre-crisis levels to a market with fairly large volumes, broad participation, and an equity-like culture of favoring puts over calls. In equities, we continue to have robust activity on equity index options, but we also have had growth in VIX related products, including VIX options, which, as a “volatility of volatility” measure, should have some theoretical link to credit option-implied volatility.

Where do we start? We focus this chapter on two themes: First, we present the results of our historical studies measuring the strength of equity and rates market factors on credit market variables, namely changes in credit spreads, implied volatility and skew. Second, we consider these relationships and our current market views to identify credit-equity opportunities we see in today’s markets.

Our conclusions from the historical study are as follows:

- **Credit Spread Changes:** While many factors in our models serve to explain credit spreads changes, we find changes in equity prices and fixed strike volatility to be among the strongest.
- **Credit Implied Volatility Changes:** In the US, we find changes in spreads and changes in fixed strike equity volatility to be the strongest drivers. In Europe, we have found changes in spreads and equity skew to be strong

drivers, owing to more tail event risk within European credit markets.

- **Credit Skew Changes:** In the US, changes in spreads as well as implied volatility from the VIX options markets are the biggest drivers. In Europe, changes in spreads along with changes in equity skew are significant.

In terms of trading themes, as we mentioned above, we see a credit options culture developing in a manner that is similar to equities where net flows in the market are biased toward long put and short call positions, thereby effectively reducing market risk and creating the familiar steep skew relationship. But there are important disconnects in both markets as well. In particular, we think tails (upside and downside) are perhaps too extreme in credit options, and we favor trading those tails for ones in equities where they are more valuable given valuations, our fundamental views, and the nature of equities in general. We suggest two trade themes:

- **European Tails:** We see much demand to hedge risk in Europe, where credit has underperformed equities over the past year. Within options, we find that skew in iTraxx Main is steep relative to skew in SX5E and iTraxx Main volatility is modestly rich. We suggest selling OTM payors in iTraxx Main to fund the purchase of OTM puts in SX5E for roughly zero cost using our preferred strikes and hedge ratios. For the same theme but less tail exposure, we suggest buying an iTraxx Main payor spread and selling a put spread in SX5E at zero cost. For investors long credit, we suggest selling OTM calls over these strategies to improve costs.
- **US Upside:** Front-end volatility levels have fallen significantly in both CDX IG/HY and the S&P 500. While upside skew has flattened in both credit and equity, we see it as flatter in CDX HY versus the potential upside at this stage in the cycle where HY is closer to fair value while equities can continue to gain on improving earnings, the macro environment and M&A. We advocate selling OTM calls in CDX HY and buying S&P 500 OTM calls (for a small net cost) or call spreads (for a small net credit).

CREDIT-EQUITY: FINDING WHICH VARIABLES MATTER

As a starting point in looking at opportunities in the credit and equity markets, we explored the strength of relationships between benchmark instruments in spot and options markets in both asset classes. In this analysis, we used only market-based information (equity, interest rate, and credit spot and volatility measures) to understand when dislocations arise, rather than fundamental information to try to determine fair value. After several series of linear regressions exploring what best explained credit spread changes, credit implied volatility, and credit implied volatility skew, we found the metrics in the below table to have the best explanatory power for each:

exhibit 1

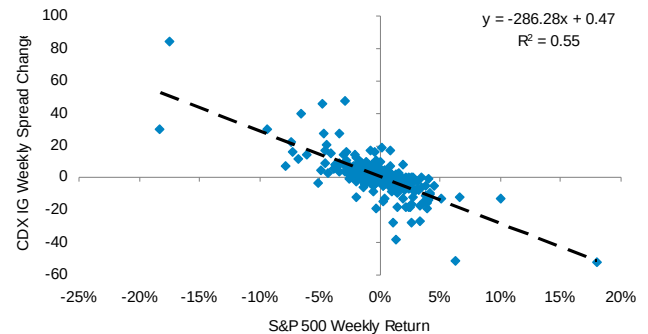
What Explains Credit?

Credit Metric	Explanatory Variables	R ²
CDX IG Spread	S&P: Price, Skew, 3m Fixed Strike Vol UST: 2y Rates, 2y-5y Slope, 5y Vol	70%
3m CDX IG Implied Vol	CDX: Spread S&P: 3m Fixed Strike Vol	85%
3m CDX IG Skew	CDX: Spread VIX: 1m Vol	67%
iTraxx Main Spread	SX5E: Price, 3m Fixed Strike Vol EURIBOR: 2y Rate, 2y-5y Slope, 5y Vol	47%
3m iTraxx Main Vol	iTraxx: Spread, Spread (squared) SX5E: Skew	87%
3m iTraxx Main Skew	iTraxx: Spread, Spread (squared) SX5E: Skew	78%

Note: R² include insignificant variables. R² based on only variables above are ~2% lower.
Source: Morgan Stanley Research

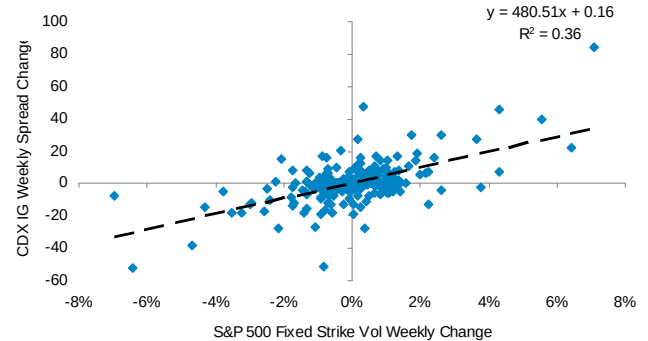
Credit spreads. In a traditional Merton model the level of credit spreads are related to both equity prices (asset values) and equity implied volatility as a forward measure of uncertainty around asset values. And while we find modeling the asset classes as single large companies difficult for day-to-day moves, it is not unsurprising that the variables that have the most explanatory power for credit spread variation are changes in the equity spot prices and fixed strike volatility, in both the US and Europe. We used fixed strike volatility to abstract away from the effect of rolling up and down the skew, and focus on just the shift in uncertainty explained by actual moves up and down in the volatility surface. Implied volatility skew was marginally explanatory, but not as much as a Merton model approach, where credit is a far OTM put, might suggest.

exhibit 2

CDX IG Spread vs. S&P 500 Price Changes

Note: 2005-present. Source: Morgan Stanley Research, Bloomberg

exhibit 3

CDX IG Spread vs. S&P 500 Fixed Strike Vol Changes

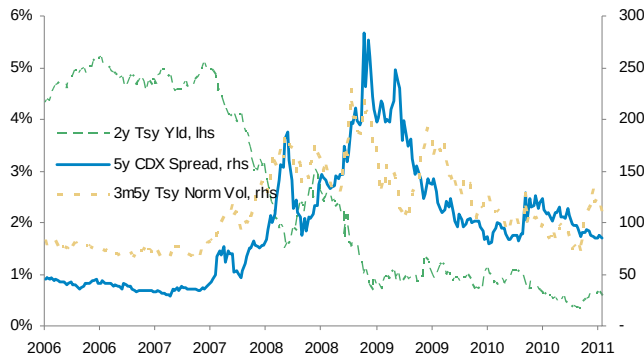
Note: 2005-present

Source: Morgan Stanley Research, Quantitative and Derivative Strategies, Bloomberg

Interest rate market variables mattered in our analysis as well. Pre-financial crisis relationships were somewhat weak. But as the economy fell into crisis the Fed cut rates dramatically, while credit spreads widened reflecting increased systemic risk. As the economy regains its footing correlations have wobbled some, but it seems we are returning to the pre-crisis paradigm of increasing rates and tight spreads. In addition, the total yield matters for the community of credit investors that trade bonds on a price and funded basis, so some degree of negative correlation (spreads / yields) is not surprising. Given these relationships we see a link with volatility in rates and credit as well. But what is most interesting is that interest rate variables were significant in our model, in addition to the equity metrics, indicating that these results are not just because "every market is correlated."

exhibit 4

Credit Spreads Track 5y US Rates Vol, While Moving Inversely to Rates

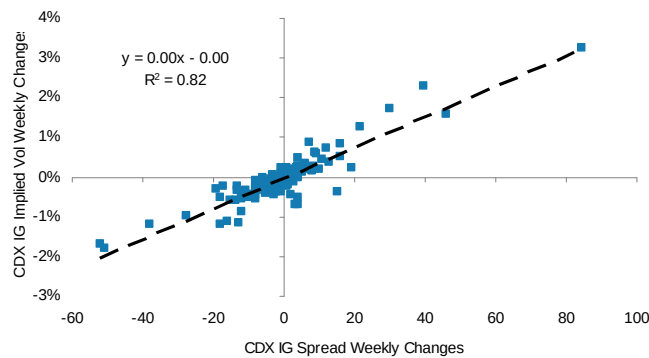


Source: Morgan Stanley Research, Bloomberg

Credit implied volatility. The biggest driver of credit volatility is naturally the level of spreads, and there is a skew to the market similar to equities. While we already knew this intuitively, we included the change in credit spreads as an independent variable in this regression, so we could examine the incremental explanatory power of equity and interest rate variables beyond just the effect of changes in spot prices.

exhibit 5

CDX IG Implied Vol vs. CDX Spread Changes

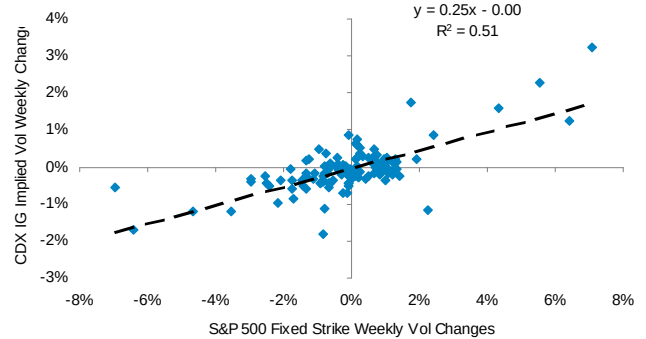


Note: Oct 2008-present

Source: Morgan Stanley Research, Bloomberg

exhibit 6

CDX IG Implied Vol vs. S&P 500 Fixed Strike Vol Changes



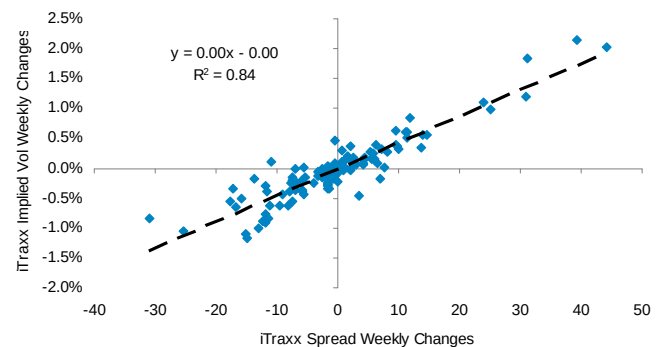
Note: Oct 2008-present

Source: Morgan Stanley Research, Quantitative and Derivative Strategies, Bloomberg

And we did find that equity volatility metrics (S&P fixed strike volatility in the US and EuroStoxx skew in Europe) were significant. Equity skew's significance with credit options may be heightened in Europe where continuing sovereign concerns and tail risk is particularly prevalent. The squared change in credit spreads, a proxy for realized volatility, was significant as well, indicating that credit implied volatility does indeed move with realized, and reacts to big moves in the market regardless of direction. Interestingly the implied volatility of volatility (on the VIX in the US) did not seem to explain credit volatility, despite the theoretical link in a Merton context (it is more important for skew, however, as we highlight next).

exhibit 7

iTraxx Main Implied Vol vs. iTraxx Spread Changes

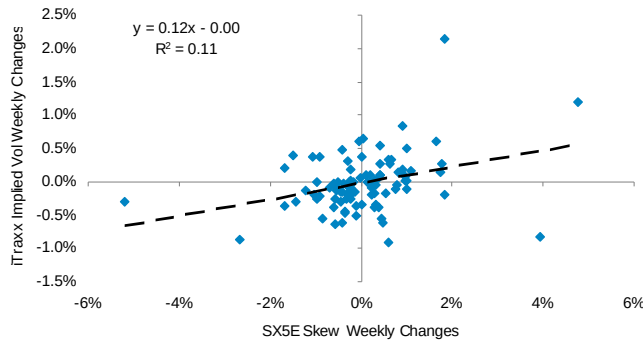


Note: Oct 2008-present

Source: Morgan Stanley Research, Bloomberg

exhibit 8

iTraxx Main Implied Vol vs. EuroStoxx Skew Changes



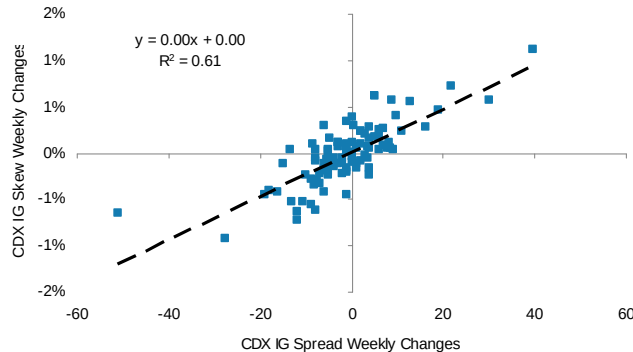
Note: Oct 2008-present

Source: Morgan Stanley Research, Quantitative and Derivative Strategies, Bloomberg

Credit implied volatility skew. Our regression of credit skew shared many of the significant independent variables with the credit implied volatility analysis. Credit spread changes were the main driver of changes in credit skew, but interestingly VIX 1m implied volatility was significant as well in the US. Vol of vol tends to accelerate in market corrections suggesting that it may influence skew rather than volatility in credit markets. In Europe, where we did not have implied vol of vol data, we found that SX5E implied volatility skew was somewhat useful in explaining iTraxx skew. This fits with the link between skew and vol of vol, as well as the relationship in Europe between ATM credit volatility levels and equity skew, and when we remove vol of vol from the US regression, equity skew does become more significant.

exhibit 9

CDX IG Skew vs. CDX IG Spread Changes

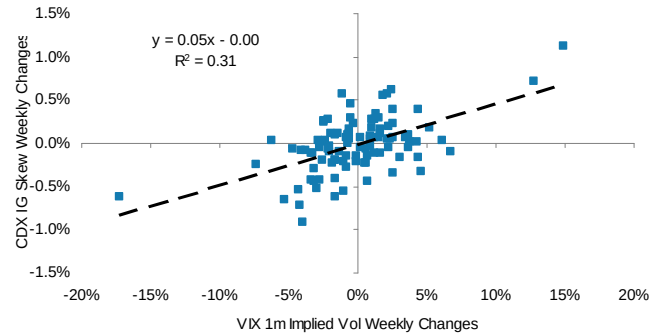


Note: Feb 2009 – present

Source: Morgan Stanley Research, Bloomberg

exhibit 10

CDX IG Skew vs. VIX 1m Implied Vol Changes



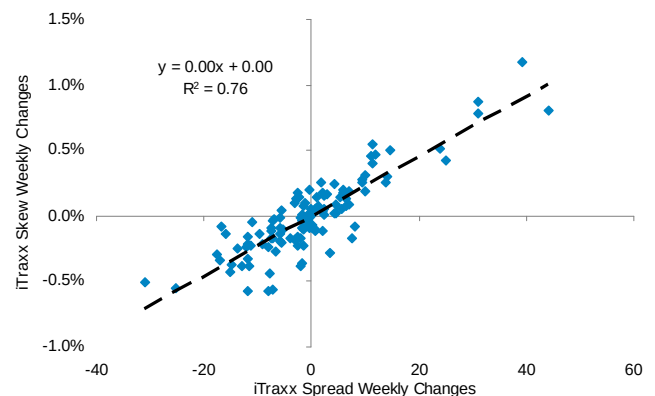
Note: Feb 2009 – present

Source: Morgan Stanley Research, Quantitative and Derivative Strategies, Bloomberg

We note that since credit options markets are relatively new, we have a smaller post-crisis dataset to consider for these regressions. And with macro drivers dominating financial markets over idiosyncratic events during the past few years, many of the independent variables in our analysis are highly correlated, and this work does not necessarily suggest causality. Instead we use these results to help us find descriptive metrics for the variation in credit markets. Only when we pair the regression results with an understanding – and forward looking view – on other markets do we find opportunities.

exhibit 11

iTraxx Main Skew vs. iTraxx Main Spread Changes

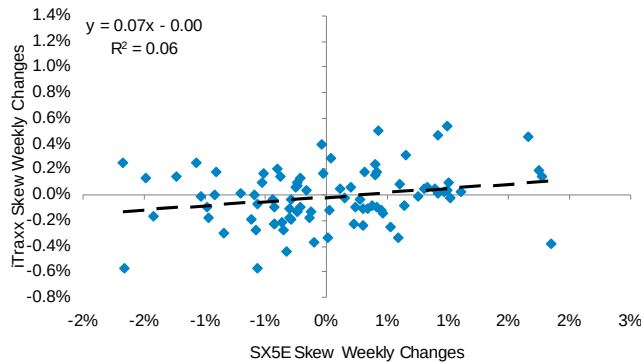


Note: Apr 2009 – present

Source: Morgan Stanley Research, Bloomberg

exhibit 12

iTraxx Main Skew vs. EuroStoxx Skew Changes



Note: Apr 2009 – present

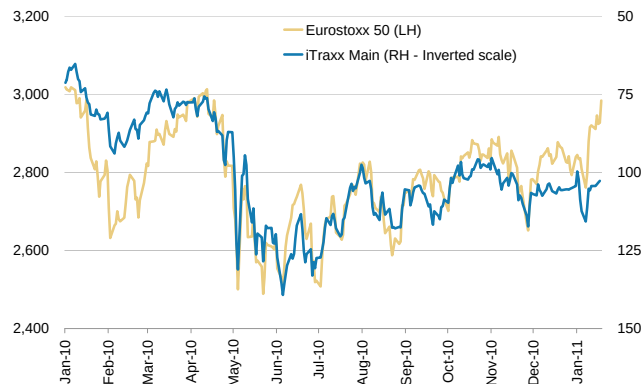
Source: Morgan Stanley Research, Quantitative and Derivative Strategies, Bloomberg

TRADE IDEA – TRADING PUTS & SPREADS IN EUROPE

Uncertainty in Europe over sovereign solvency and the plan for backstopping financials has created some dislocations. Our regression work points to focusing on equity spot prices and implied volatility in explaining credit spreads, and when looking over the past year we see equity as having generally outperformed credit.

exhibit 13

iTraxx Main vs. EuroStoxx Performance

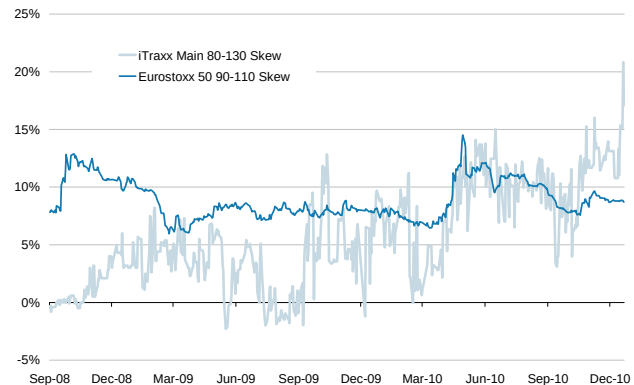


Source: Morgan Stanley Research, Bloomberg

We see this dynamic reflected in volatility markets as well, with credit volatility elevated relative to equity volatility. Further, we see one of the biggest dislocations when comparing skew between European credit and equity markets. SX5E implied volatility and skew are not particularly elevated compared to history, and while implied volatility continues to trade at a premium to realized volatility we see realized volatility at a significant discount to implied in credit markets.

exhibit 14

iTraxx Main vs. EuroStoxx Skew



Source: Morgan Stanley Research, Quantitative and Derivative Strategies, Bloomberg

Much of the underperformance in credit in Europe has come from the financial sector, as highlighted in [European Credit Strategy: A Difficult Consistency](#), Jan 14, 2011. But SX5E has a 29% concentration in Financials, and SX5E options are one of the most liquid proxies for the Financial sector in Europe. Sovereign stress could certainly put more pressure on banks, but the greater risk to credit, and to equity markets, is if it spreads more broadly to impact funding levels and the corporate sector. While credit could sell-off hard on sovereign stress, like we saw in April / May 2010 (iTraxx 50bps wider, or -2.2% price return, versus -15% in SX5E), in more severe dislocations equities can catch up quickly – while equity lagged the credit selloff in 2008, it fell 44% versus a 6.4% return for the iTraxx. With iTraxx volatility high and skew rich, it appears to us that much of the “first-mover” nature of credit could already be priced-in.

To take advantage of the pricing differentials in credit and equity markets, and/or position for potential equity underperformance, we like selling June iTraxx Main payors to fund the purchase of SX5E OTM puts. Looking at historical relationships, we can get a credit:equity hedge ratio of about 6.5x, and while this is not far from the ratio of ATM implied volatilities in both markets, or how the markets behaved in May 2010, the OTM implied volatility ratio looks cheap at 5.5x. If we scale our strikes by the same 6.5x ratio as our notionals, we are selling a June 140bp iTraxx Main payor (1.6% OTM) and buying a June 10.4% OTM EuroStoxx 50 put, for nearly zero cost. This trade is a tail trade, where we position for a relative steepening of equity skew, or outperformance in credit where it falls less than 6.5x the decline in equity. While hedged in theory, given the short credit put, the maximum potential loss is the notional less recovery value.

exhibit 15

iTraxx Main vs. Eurostoxx Put Pair Trades

	Strike / Spread	Distance to Strike (%)	Distance to Strike (€mm)	Option Premium (%) of Notional *	Option Premium (\$mm) *
Equity Notional = €10mm					
Spot	2963				
Short Near Strike	2904	-2.0%	-€ 0.20	5.46%	€ 0.55
Long OTM Strike	2655	-10.4%	-€ 1.04	-2.88%	-€ 0.29
Net		8.4%	€ 0.84	2.58%	€ 0.26
Credit Notional = €65mm					
Spot	103				
Long Near Strike	110	-0.3%	-€ 0.20	-0.82%	-€ 0.53
Short OTM Strike	140	-1.6%	-€ 1.04	0.45%	€ 0.29
Net		1.3%	€ 0.84	-0.37%	-€ 0.24
Net premium - buy SX5E put / sell iTraxx put				0.04%	€ 0.00
Net premium - sell SX5E ps / buy iTraxx ps				0.17%	€ 0.02

* Positive option premium is an inflow (credit), negative is an outflow (cost)

Source: Morgan Stanley Research, Bloomberg

As an alternative trade to position for corporate spreads retracing their outperformance versus financials, we like buying 110-140 bp June iTraxx payor spreads and selling June EuroStoxx 98-89.6% put spreads. Like the put versus put trade, this is close to a zero net cost, and is long equity skew. In the case of a large tail scenario where both of the put spreads are fully in-the-money, at a 6.5x notional weighting the P/L from each cancels out. The maximum potential loss is the width of the equity put spread.

An overlay for those starting with a long credit position that we like for either of these trades is to sell a receiver (call). This is an attractive way to boost carry in our view, and also provide some cushion in case equity does underperform credit. The added carry is particularly attractive on the put spread trade, as it creates a positive P/L even in the large tail scenario. Over longs, the risk is that upside is capped beyond the strike.

TRADE IDEA – TRADING CALLS IN THE US

In the US, we see similar pricing differentials from our models, but think the upside story is better in equities and would prefer to play the trade from the risky-asset positive side. But IG CDX receivers are fairly inexpensive on an absolute basis, so here we look to the high yield market, where volatility is a bit higher, and upside skew flatter.

With HY CDX at a price of 103.5 and spread of ~415, the next leg up will likely be more challenging than the last. Our Leveraged Finance strategy team is cautious near-term as sentiment may be too bullish for a continuation of the 45bps rally (in cash) since mid-December, despite their positive view for 2011. Higher interest rates, while not an outright negative, is at least a headwind given the potential for flows into the asset class to slow down or stop. In a bull market LBOs are always a risk, and higher quality HY should lag lower quality in an improving economy. In a bear market

credit should be under pressure with other assets classes, although falling rates could act as a cushion. With this backdrop carry becomes more important, and overwriting feels attractive to us to boost yields.

On the equity side, more upside is possible in an improving growth environment or if M&A picks up. Equity upside skew has flattened and prices some of this in, but much less on the index level than the single-name level, at least for shorter-dated maturities.

To play for relative upside in equity, we like buying a call option on the S&P 500, funded by selling a CDX HY call. Looking at historical relationships, we get a HY:SPX hedge ratio of about 2x, which fits with the ATMF implied vol ratio of 1.7x (June). Selling a June 104 call on high yield (the forward is 102.5) brings in 85 bps, which at a 2:1 ratio to equity offsets ~80% of the cost of a 105% S&P 500 June call.

exhibit 16

Sell HY CDX Upside to fund S&P 500 Upside

	Strike / Spread	Distance to Strike (%)	Distance to Strike (\$mm)	Option Premium (%) of Notional *	Option Premium (\$mm) *
Equity Notional = \$10mm					
Spot	1283				
Long Outright Call	1348		\$0.50	-2.0%	-\$0.20
Call Spread Near Strike	1325	3.2%	\$0.32	-2.7%	-\$0.27
Call Spread Far Strike	1375	7.1%	\$0.71	1.2%	\$0.12
Net		3.9%	\$0.39	-1.5%	-\$0.15
Credit Notional = \$20mm					
Spot	103.6				
Fwd	102.5				
Short OTM Call	104.0	0.4%	\$0.07	0.85%	\$0.17
Net premium - sell HY call / buy SPX call				-0.32%	-\$0.03
Net premium - sell HY call / buy SPX call spread				0.17%	\$0.02

* Positive option premium is an inflow (credit), negative is an outflow (cost)

Source: Morgan Stanley Research, Bloomberg

While we like the relative volatility story, we acknowledge that the relative skew story between US HY and equities is less interesting. And given the strong rally we've seen in equities in the past few months, further upside could come slower. We think an attractive zero-cost variant on this trade is to use the sale of the HY 104 receiver to fund a 1325-1375 (3.2%-7.1% OTM) SPX call spread. For either trade, shifting the HY strike further OTM to 105 is an alternative for those a bit more bullish on HY. With the premium falling from 85bps to 50bps, for similar cost offsets we would buy a S&P 500 107.5% (1380), or a 1350-1380 call spread. Given the short HY call, the maximum potential loss for any of these trades is unlimited outright, and over longs upside is capped beyond the strike.

STRATEGY RISK FACTORS

Long Puts: Over longs, the position is protected below the strike. The max potential loss in isolation is the premium paid.

Long Calls: The maximum potential loss is the premium paid.

Short Puts: The maximum loss is the strike less the premium.

Short Calls: The maximum loss is unlimited. Over longs, the risk is that the underlier rallies through the short call strike, with the asset called away at the strike price, limiting participation in further upside. Downside exposure is retained.

Put Spreads: Overlaid on a long position, the position is protected between the strikes (but not below) at expiration. The maximum potential loss in isolation is the premium paid.

Call Spreads: Overlaid on a stock position, the position is protected between the strikes (but not above) at expiration. The maximum potential loss in isolation is the premium paid.

Section D

Credit Portfolio Management

Bank Loan Hedging Today – CPM Part I

May 30, 2008

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As much as credit derivatives flows over the past year have been dominated by a wide range of institutions trading, hedging and speculating on credit risk, bank loan hedgers are both the market's original and natural buyers of protection. Corporate CDS contracts were designed for this community more so than any other, and many of the nuances investors deal with in CDS contracts relate to the idiosyncrasies of loan risk that is on the books of banks.

Credit portfolio management (CPM) is the industry term used to describe the practice of managing credit exposures in corporate lending institutions, mainly large banks but also the lending businesses within dealers and insurance institutions to some degree. Every major bank lending institution has a significant effort devoted to it. While this is not a new art by any means, CPM practitioners face both a very interesting and challenging environment going forward, a function of the current credit environment, a significant repricing in corporate and structured credit, market liquidity issues and the secular regulatory changes related to the transition from Basel I to II, accounting shifts and general local jurisdiction oversight.

For those within the CPM world, there is a lot to consider, and in this chapter we attempt to outline ten themes to set the stage for further research and strategies. For the majority of market participants who will watch CPM activity from the sidelines, it is difficult to say whether the resulting credit flows are bullish or bearish, but structurally speaking, as banks move from a Basel I to Basel II framework, economic capital and regulatory capital will become more closely aligned. For many banking institutions, full implementation of Basel II is only 18 months away, with parallel processing going on now, and some amount of Basel II implementation starting at the beginning of 2009.

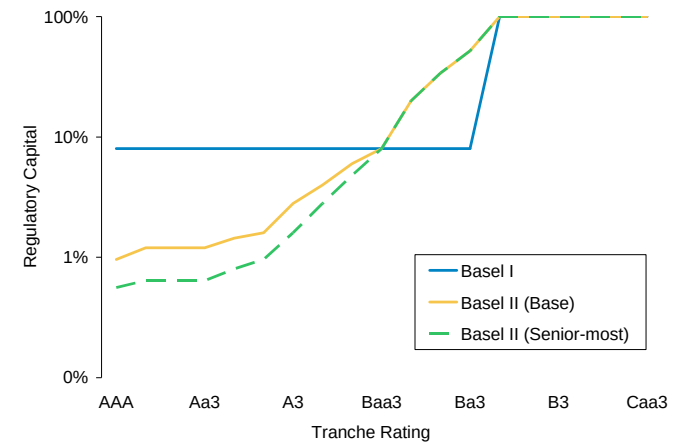
WHY IS CPM IMPORTANT?

For many, bank loan hedging vernacular is a foreign tongue. For those who are willing to learn a bit (and we encourage it), we recommend a few resources. Credit exposure management by banks and other financial institutions involves both long and short credit investments, employs credit derivatives technology, and creates flows that are driven both by economic and regulatory capital needs. The International Association for Credit Portfolio Managers (IACPM) is a professional organization that brings CPM practitioners together globally to address issues and

opportunities. There are currently 88 financial institutions that are members of the IACPM, from 16 countries (for a general description of CPM activities and links to many of the relevant public documents, see www.iacpm.org).

exhibit 1

Capital Requirements: Basel I vs. Basel II



Source: Morgan Stanley

A lot of CPM work involves interpreting Basel I and Basel II regulatory capital regimes, and mixing it with local jurisdiction regulatory environments, accounting practices and other factors to create efficient exposures to retained credit risk while meeting economic capital goals.

CHALLENGES FACING BANK LOAN HEDGERS

We have spent a considerable amount of time recently with the CPM community, both to understand their challenges and to discuss strategies for resolving them. In this market and regulatory environment, the CPM community faces many challenges and has a lot of work to do. The good news is that economic and regulatory capital will be more closely aligned when they are done, and this will make the financial system more sound, as at least some of the problems of this systemic crisis can be blamed on disconnected regulatory and economic capital regimes. We list and briefly describe below some of the challenges and potential solutions for a number of topics that we believe are relevant, and caution that the list is by no means all-encompassing.

1. BASEL I TO BASEL II TRANSITION

Perhaps Basel I/II regulatory capital regimes are the most commonly associated term with CPM, and a lot of what banks need to do over the next several quarters is related the transition from Basel I and II.

In simple terms, one of the big differences between Basel I and II is the regulatory capital treatment for retained credit risk. Owning the junior portion of credit risk versus owning only the senior exposure of credit risk in many cases is not treated differently under Basel I, so bank loan hedgers can efficiently gain regulatory capital relief by buying protection on senior parts of a portfolio (i.e., 3-100%), because senior parts trade at tighter spreads. One does not have to be a market expert to realize that such a regulatory capital approach stands in contrast to economic capital, where junior tranches have more MTM and default risk than senior tranches, for a given notional amount.

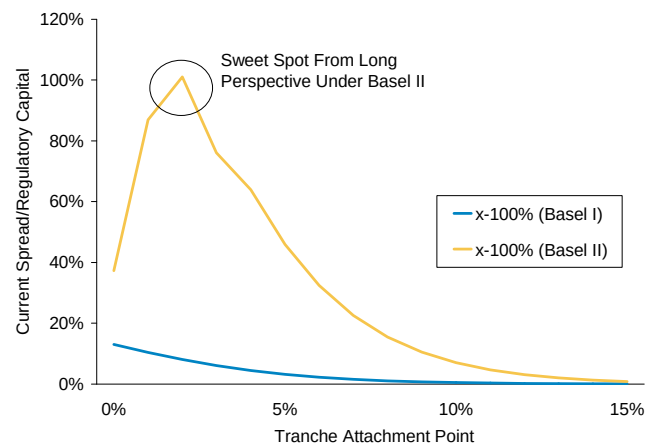
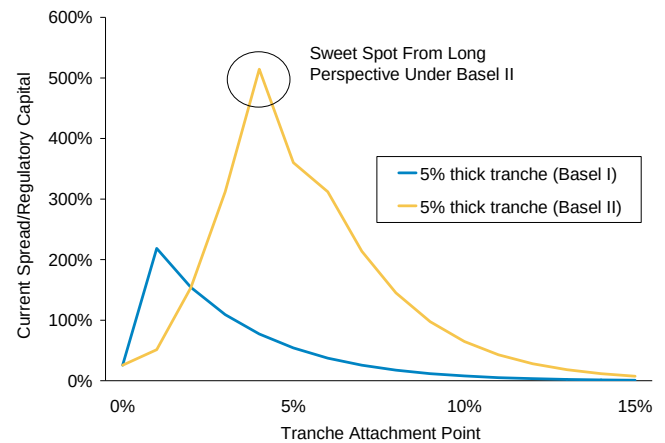
Under Basel II, the risk weightings are high for junior tranches and fall off dramatically as one goes up the capital structure, which is very much in line with economic capital, in our view (see Exhibit 1). This is also true for low rated assets compared to highly rated assets. We offer the caveat that local jurisdictions can treat risk rated below investment grade differently (full deduction in some cases for BB- and below), but the general theme is consistent. For investment grade portfolios, it is possible to release more capital at the portfolio level simply by transitioning from Basel I to Basel II – risk weights move to 50% to 75% for BBB portfolio depending on the approach used relative to a 100% under Basel I. The implication is that, generally speaking, holding securitized risk of investment credit portfolios is more attractive under Basel II, although there are important implications based on the tranche rating itself (see Exhibit 2).

For below investment grade portfolios, the outcome is a bit more complex – depending on the assumptions/approach, the risk weight for such portfolios could be higher under Basel II, although the higher quality securitized tranche benefits under Basel II. This actually creates opportunities for significant differentiation in capital management in the below investment grade space – partially hedging a loan portfolio has a lot of scope under Basel II. We provide a flavor of impact on investment grade securitized risk here and will follow up in detail on the impact on both investment grade and below investment grade securitizations.

In Exhibit 2, we show the efficiency of retaining certain parts of credit risk in a portfolio of corporate credit with a similar ratings and risk profile to the 5-year CDX IG (tranche attachment vs. the spread of this risk divided by the product of risk weights and an 8% capital regime). The top chart illustrates that the sweet spot for retained credit risk moves from a very junior attachment under Basel I to a more senior attachment (say around 4% or 5%) under Basel II. This example assumes tranches that are 5% thick. Very junior tranches have the most to lose under Basel II.

exhibit 2

Retained Credit Efficiency Shifts Radically from Basel I to II



Source: Morgan Stanley

However, since many bank loan hedgers are used to thinking about X-100% hedges given their efficiency under Basel I, in the second chart in Exhibit 2, we show the poor efficiency of such hedges under Basel II (or said differently, they actually can look attractive from a long perspective today given the market repricing).

As a caveat, we note that this is a very simplified example, and the details of any portfolio and hedging scheme can result in different outcomes, although we expect the general themes to be consistent with this analysis.

One of the key challenges for CPM practitioners today is efficiently moving from Basel I type hedges to Basel II. In our simple view of the world, the two capital regimes stand in stark contrast to one another, and as such there is no “magic” way of hedging efficiently under both regimes, although there are partial solutions that, in our view, would involve owning protection on tranches with junior mezzanine type attachment points.

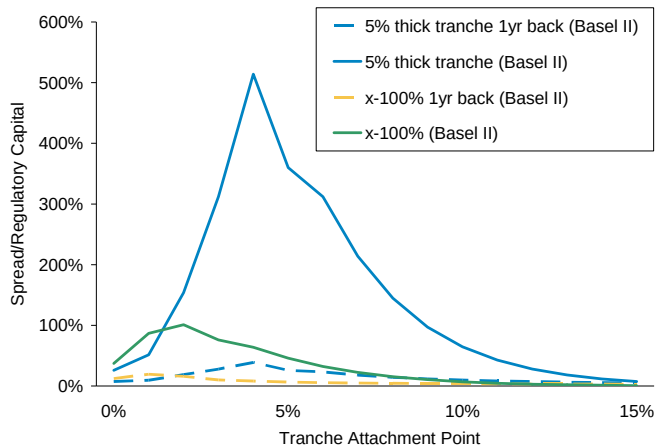
2. DYNAMIC MANAGEMENT UNDER BASEL II

One of the key benefits to operating under a Basel II world, in our view, is that there is scope for dynamically managing portfolio hedges, as market pricing changes. Given the non-constant nature of risk weightings for retained risk, CPM practitioners will have an opportunity to make adjustments to their hedges based on market movements in structured credit pricing, single-name dispersion, etc.

The flat risk weight structure under Basel I implies that ratings migration of the underlying portfolio has little impact. Even cataclysmic shifts in pricing across the structured credit capital structure like the one we recently went through do not alter the hedging framework much – even though economically the risk environment has changed dramatically. Stability of regulatory capital strategies under Basel II is a function of both pricing and ratings. Under Basel II, given the repricing from last year, senior mezzanine risk is significantly more attractive today – and while X-100% hedges with low attachments might have made sense in 2007, there is more incentive to be long senior mezzanine type of risk today rather than using it in a hedge (see Exhibit 3). On the other hand negative ratings migrations in today's environment will compel investors to add subordination to make it regulatory capital efficient under Basel II – consistent with economic logic.

exhibit 3

Basel II Retained Credit Risk – Repricing Matters



Source: Morgan Stanley

3. LOAN OPTIONALITY

Most bank loans have variable maturities, potentially variable coupons, and even their utilization can vary (i.e., untapped or partially tapped revolving). Simply put, loan portfolios have short convexity exposures, given the options that the lenders have sold to the issuers. While the pricing of the loan instruments might compensate banks for some amount of negative convexity, in general most CPM specialists need other tools to hedge convexity exposure.

As credit investors have begun to dominate the momentum of credit derivatives markets over the past many years, CDS contracts have been trending toward bullet type contracts as most investors prefer simple fixed maturity credit risk. Restructuring is not a standard credit event in some forms of LCDS today, and while non-standard CDS is always a possibility, the illiquidity of such instruments is generally an issue.

In the mortgage world, hedgers tend to use swaptions as a way to hedge the negative convexity exposure of being short prepayment options. In many ways, their problems are analogous to bank loan hedgers, and we would argue that credit option instruments might present an interesting solution to convexity issues. Index options are a natural solution to hedge overall portfolio convexity, and we expect activity to increase here. There are solutions at the single-name level that will be necessary as well, either through cancellable forms of CDS/LCDS contracts (more likely) or options on single-names (less likely).

4. ILLIQUID CREDITS

Many CPM specialists we meet cite illiquid credits as being a top challenge for hedging. Illiquidity can have many faces too, including both smaller companies (from a debt market capitalization perspective) and loans that are linked to (or secured by) hard to value assets, cashflow streams and the like. There is no single solution to this problem, but perhaps some creative ones. The growth of loan indices (LevX and LCDX) is certainly improving liquidity for quite a number of secured credits in both the US and Europe, and such recent liquidity boosts might help CPM practitioners in this regard. There are also overlay strategies including the indices themselves or credit proxies. But for more illiquid direct exposures, in this market environment, the dealer community is unlikely to want to take on highly illiquid credit risks, so any transfer of such credit risks by CPM specialists will likely need to involve linking the credit risk to a product that could be offered to an end investor, or employing hedges on the real assets (e.g., commodities) being referenced by the credit, or both.

5. LIQUIDITY/FUNDING

Liquidity risk from a funding perspective (i.e., using short-term funding to finance the purchase of long-term assets in SIV-like structures) was one of the most painful lessons learned in this credit cycle, both for those who lent and borrowed in this framework. Liquidity and funding costs have driven performance of AAA assets and the CDS-bond basis, among other relationships. Funding has also affected product securitization strategies. Banks might also consider more efficient mechanisms for funding, given the current state of their balance sheets, and some of the options might include securitizing credit risk into CDOs whose senior tranches can be posted to the Fed/ECB as collateral, or using more elaborate CPPI type structures to create funding efficiencies.

6. COUNTERPARTY RISK

Clearly a topic with a lot of market focus today, counterparty risks on CDS trades (long or short) are a key focus point for CPM practitioners. It is certainly good news that Basel II considers the impact of counterparty risk in its accord, but the approach might be overly simple. There are generally two methods to account for counterparty risk, the substitution approach and the double-default approach. Double-default provisions are meant to reduce risk weights cognizant of the lower probability of joint default of the exposure and protection bought. However, double-default provisions are not applicable to multiple name credit derivatives or index-based products, synthetic securitization – so substitution is the fallback.

Under substitution, protection purchased on a credit will only reduce risk weightings on the credit to the level of the protection provider (i.e., buying protection on a BBB credit from a AA entity would result in a net risk weighting of AA). Selling protection on a single tranche say 5-7% results in different capital requirement from selling protection on the index and buying protection on 0-5% and 7-100% parts of the capital structure due to the difference in the way counterparty exposures are accounted. For example, 5-100% protection purchased against 0-100% exposure (rated Baa1) would result in a capital requirement of \$6.02MM, which is higher (because of counterparty charges) than from simply being long \$5MM of 0-5% (rated B2) which requires a full deduction, i.e., capital of \$5MM.

But there is one potential problem here. The vast majority of CDS trades today are fully collateralized to their market value, with certain thresholds for daily moves depending on the quality of the counterparty. This effectively reduces counterparty risk in the system to the “gap risk” of not having enough collateral (see [Stress Testing the CDS Market](#), January 18, 2008). We do not believe that the Basel II Accord addresses this collateral posting, which is a shortcoming in our view.

7. RATINGS

One of the criticisms of Basel II is its reliance on ratings within structured forms of credit including securitizations (for the rest, banks will rely on internal credit assessments). Ratings are clearly an issue today with the credibility of rating agencies being questioned in the market. Generally, when a public rating is available, banks must use them under the standard or RBA approaches to calculate risk weightings on retained credit risk. The supervisory formula is available when public ratings are not available, and the calculation is similar in scope to rating agency models in that subordination, tranche thickness, number of credits in the pool, and loss given default of the underlying portfolio is taken into consideration.

Nevertheless, there is a general reliance on ratings under Basel II for structured credit risk, and as such the Financial

Stability Forum (FSF) is addressing rating agency operating issues including the quality of the ratings process, potential conflicts of interest, more information on risk characteristics, data quality, due diligence by ratings users, and the role of ratings by regulators¹. We do not see a lot of direct changes to ratings of retained corporate credit risk as a result of the FSF efforts, except for re-securitizations like CDO-squared transactions. The ratings process might be better vetted as a result of FSF, but whether or not it is successful, we believe that bank regulatory capital will continue to be heavily dependant on external ratings.

8. FAIR VALUE OPTION, ACCOUNTING, CASH VS. CDS

Among the most complicated aspects of the CPM world is related to accounting, especially in the world of fair value. Separation of risks from the banking and trading books is a central issue, and the differing accounting treatment of cash assets and synthetic hedges continues to be problematic for many banking institutions from an income statement perspective. It has been common in the past and will likely be in the future to add in macro forms of long credit exposure (without much idiosyncratic exposure) to trading books offset MTM volatility of credit hedges.

9. MODEL RISK

The systemic nature of the current credit crisis was atypical in the sense that there is a disconnect between mark-to-market losses and investor perception of the real losses (outside the obvious troubled sectors). This disconnect is so extreme as to force rethinking of models including correlation calculations, recovery modeling and the like. While events of such magnitude may not be an annual occurrence, it has stirred up a debate on fair value accounting and the use of models with and without market inputs.

10. INSURANCE COMPANY ISSUES

While much of the focus in CPM is related to banks and their corporate credit assets, insurance companies practice CPM as well, and Solvency II is the set of regulatory requirements that apply to European insurance companies. Similar in theme to Basel II, Solvency II will move regulatory and economic capital closer together.

¹ See *Report of the Financial Stability Forum on Enhancing Market and Institutional Resilience*, April 7, 2008.

Basel II Based Hedging – CPM Part II

June 13, 2008

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We continue with our focus on the secular changes that are afoot in the bank loan hedging community, the market's natural and original buyers of protection. In Chapter 25, we described the landscape today and the challenges and opportunities facing credit portfolio managers (or CPM, the industry term used for bank loan hedging). In this chapter, we focus on a specific theme, namely capital efficiency under Basel II for those institutions that use tranching portfolio protection programs as a means of hedging (which is quite common). The market implications of these hedging programs are quite significant in our view.

Under the Basel I regime, much investment grade rated credit risk is treated equally, and as such, banks who operate under Basel I have not had much motivation to either invest directly or indirectly (through retained credit risk after hedges) in senior forms of credit risk, given their lower yields. In fact, one could argue that the steady trend of the WACC curve moving to the lower rungs of the investment grade world has something to do with the lack of motivation among banks (under Basel I) to finance high quality corporations.

Under Basel I, from the perspective of a bank with exposure to investment grade rated loans and revolvers, deciding which part of the portfolio to retain becomes more a function of spread rather than expected losses, ratings, default-risk remoteness, seniority and the like. This is one of the reasons why buying 3-100% protection (i.e., banks retaining the 0-3% part of the loan portfolio) has been a reasonably good Basel I strategy, given the much lower cost. But as we discussed in Chapter 25, under Basel II, we move to a regime where economic and regulatory capital become more closely aligned, and as such, what is efficient to retain is a function of risk remoteness, expected losses and market prices.

We focus this chapter on assessing the efficiency of retained credit risk for portfolios of investment grade loan exposure under Basel II, using a host of tranching portfolio hedging alternatives. Details and assumptions clearly matter, and we caution that our approach should be considered a general one. We also note that investment grade portfolios benefit from regulatory capital relief simply by moving from Basel I to Basel II, and that a repriced credit and structured credit market will alter portfolio efficiency quite significantly as well.

exhibit 1

Sample of Basel II Efficient Solutions

	CDX 5yr	Bespoke	CDX 10yr
Portfolio Spread (%)	1.1%	1.7%	1.1%
Portfolio Rating	Baa2	Baa1	Baa2
Spread per unit of RC	26%	40%	27%

Efficient Solutions and SRC (Spread per unit Reg Capital)

Type	CDX 5yr	Bespoke	CDX 10yr
Two Tranche	Retain 2.5-100% (SRC 115%)	Retain 2-100% (SRC 192%)	Retain 3.5-100% (SRC 117%)
Two Tranche	Retain 4-100% (SRC 110%)	Retain 3-100% (SRC 189%)	Retain 4.5-100% (SRC 108%)
Three Tranche	Retain 0%-0.5% and 2.5%-100% (SRC 109%)	Retain 0%-0.5% and 1.5%-100% (SRC 173%)	Retain 0%-0.5% and 3.5%-100% (SRC 105%)
Four Tranche	Retain 3%-9% and 10%-100% (SRC 108%)	Retain 2.5%-7% and 8.5%-100% (SRC 190%)	Retain 4.5%-10% and 11%-100% (SRC 109%)

Source: Morgan Stanley

exhibit 2

Tranching Alternatives and Nomenclature

Long Index			
Equity tranche (0-5%) hedged	5-100% tranche risk retained		2T – H-L
Equity tranche (0-5%) hedged	5-9% retained	9-100% tranche risk hedged	3T – H-L-H
Equity tranche (0-5%) hedged	5-9% retained	9-12% hedged	12-100% tranche risk retained
Equity tranche (0-5%) No position	Sell Protection on 5-100% tranche risk		2T – N-L

Source: Morgan Stanley

OUR METHODOLOGY

Basel II does not provide a framework for regulatory capital relief from a hedging perspective specifically. Instead, for the analysis included in this chapter, we focus on a bank's long credit exposure, net of hedges (and the associated counterparty costs of those hedges), using the following framework:

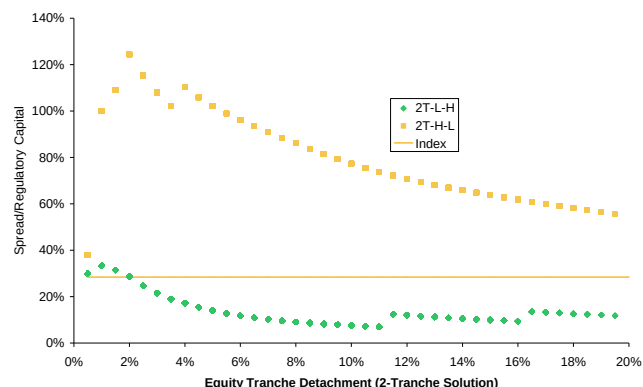
- We assume that a bank is long the credit risk of a loan portfolio and considers tranche protection as a way of isolating and selling off the "expensive" parts of this risk (from a capital perspective) so that this bank can retain the more efficient parts of the credit risk of the portfolio.
- We assume two portfolios, one mirroring CDX and the other a global bespoke portfolio that represents a wider

spread universe of investment grade names, but marginally higher in quality with more sector concentration. Clearly, results are very dependent on the actual portfolios, and any specific solution may differ materially from what we present here. Solutions involving non-investment grade portfolios will be considerably different.

- We calculate the return on capital by translating the retained credit risk into a rating(s), using the Moody's synthetic CDO rating approach for corporate credit, which is closest to Basel II's internal approach, given the focus on expected losses instead of first dollar losses (the latter being the S&P approach).
- We note that banks may use their internal frameworks or the ratings of any agency, so our approach may not exactly match the approach of any one CPM practitioner, but we believe it is generally useful.
- There is a counterparty-related regulatory capital cost for hedging credit risk, based on the actual counterparty rating (internal or external). We assume a counterparty rating of A1 and assume the substitution approach attracts a capital of 1.2%.
- There are an infinite number of ways that a portfolio of loans could be sliced into tranches. For simplicity's sake, we limit our approach to solutions that incorporate two, three and four tranches and make thickness assumptions as well (see Exhibit 2 for the nomenclature we use).
- For each tranche, we estimate an implied rating using the Moody's approach, and we also imply tranche pricing from CDX/iTraxx and bespoke correlation surfaces.
- We calculate spread on capital as the spread earned on the notional amount, divided by the product of the risk weight of the retained credit risk and 8% (representing our assumption for the amount of capital a bank will hold for a non-deducted asset). We subtract the regulatory capital charge for hedging.
- This approach is very simple, and we fully realize that any one bank's approach can be an evolving process given that interpretation of guidelines is still not certain across different jurisdictions.
- The minimum amount of risk transfer and type of risk transfer required to be eligible for regulatory capital relief also varies across different jurisdictions.

exhibit 3

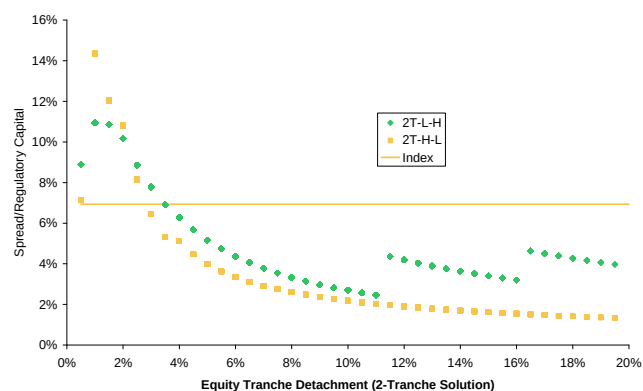
Two Tranches – X-100% Attractive to Retain Today



Source: Morgan Stanley

exhibit 4

Two Tranches – Closer Call a Year Ago



Source: Morgan Stanley

TWO-TRANCHE SOLUTIONS

In Exhibit 3, we show the results of a two tranche solution for the 5-year CDX index, graphing the spread/required capital (which is a measure of return on capital) on the Y-axis. On the x-axis we graph various detachment points between the equity tranche and senior tranche. The flat line represents the return on capital for simply holding the CDX portfolio with no hedges, so it is the return-on-capital bogey to beat. The two curves in the graph represent: a) retaining the equity tranche and hedging with the senior tranche (the lower curve), and b) retaining the senior tranche and hedging with the equity tranche (the higher curve). We make a few observations.

- In general, retaining the senior risk and hedging the equity risk is far more efficient than the other way around, with the optimal points of efficiency achieved by retaining tranches between 2-100% to 5-100%.

- The same analysis using market pricing from one year ago shows quite a different picture. Retained equity solutions could have competed reasonably well with retained X-100% solutions a year ago even under Basel II assumptions, given the very tight pricing of senior and super senior risk. Today retained equity solutions are considerably poorer value (see Exhibit 4).
- Today any retained X-100% type strategy attaching anywhere up to 20% is more efficient than plain vanilla portfolio exposure.
- The moral of course is that big shifts in pricing can alter the efficiency of specific solutions, and strategies are likely to be more dynamic under Basel II.

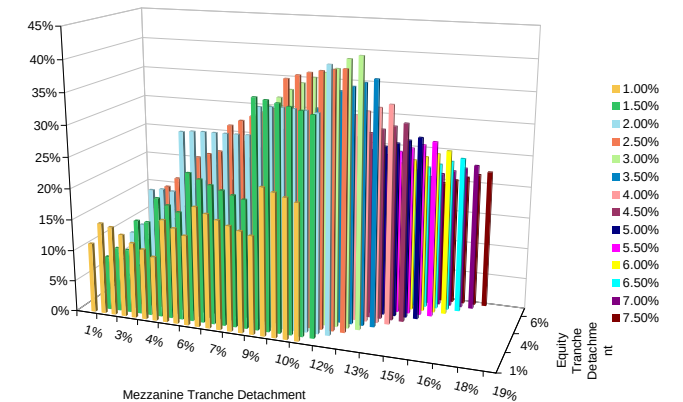
THREE-TRANCHE SOLUTIONS

The world does not necessarily have to be limited to 2-tranche solutions, so we extend the analysis to three tranches, and hence show a 3-dimensional graph (Exhibit 5, where the Z-axis is the detachment of the equity tranche, and the X-axis is the detachment of the middle mezzanine tranche). We add two more solution types: a) Hedging mezzanine risk and retaining equity and super senior risk; b) hedging equity and super senior risk while retaining mezzanine risk.

- Retaining senior mezzanine risk (say 3-9%) is among the most efficient of the pure three-tranche solutions.
- Retaining thin equity tranches plus super senior tranches that include senior mezzanine risk is also efficient (i.e., buying 1-2% protection and retaining 2-100% and 0-1% risk).
- At higher attachments, say greater than 3%, retaining 3-10% risk is more efficient than retaining 0-3% and 10-100% risk. Overall, retaining senior mezzanine still provides the most stable solution, in our opinion.
- However, two-tranche X-100% strategies outperform either of the three-tranche strategies described above, as the cost of buying super senior protection is high.

exhibit 5

Three Tranches – Senior Mezz Attractive to Retain



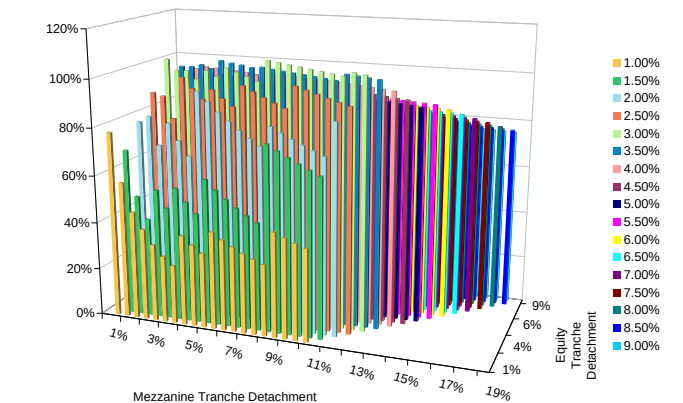
Source: Morgan Stanley

FOUR-TRANCHE SOLUTIONS

Extending the same analysis to four tranches, we find there is additional capital that could be released by hedging out thin tranches. There are senior tranches that rate poorly due to their thin size (and hence provide an opportunity to hedge and achieve capital relief) but still may not cost as much since they are fairly far from the loss thresholds. Solutions that involve retaining X-100% with X ranging from 2.5% to 4% and hedging out a 1% thick tranche attaching around 6.5% to 8% offer the best regulatory capital yield (better than the equivalent retained X-100% strategy).

exhibit 6

Four Tranches – Hedging Thin Middle Tranches is Efficient



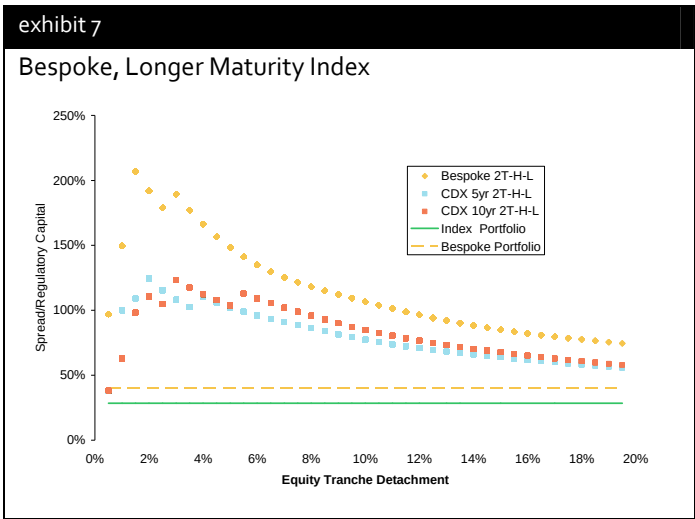
Source: Morgan Stanley

LONGER MATURITIES

Applying the same analysis to the 7-year and 10-year parts of the curve, we find that the broad conclusions do not change (see Exhibit 7). Solutions involving being long senior mezzanine risk remain attractive, although the optimal equity detachments move higher (3% at the 7-year point and 4% at the 10-year point), reflecting the higher expected losses for longer maturities.

BESPOKE PORTFOLIOS

The diversity of portfolios for a bank loan hedging program can be quite significant, so it is difficult to draw conclusions from any one analysis. We extended the work done for CDX to a bespoke portfolio that is riskier in terms of spread, slightly higher quality in terms of rating, and more concentrated than the index. Naturally, pricing for such a portfolio would reflect the higher correlation expectations. Here we find that X-100% type of solutions dominate even further compared to CDX (see Exhibit 7). The dispersion of spreads as well as the correlation differences make going long senior tranches very attractive.



Source: Morgan Stanley

Securitization as a Solution – CPM Part IV

September 5, 2008

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We continue with our focus on the secular trends and market challenges that encompass the bank loan hedging world, owing to a variety of factors including current credit market conditions, the tremendous repricing in credit markets over the past 1+ year, and the shift toward the Basel II regulatory regime. In Chapter 25, we highlighted ten challenges that CPM practitioners face in today's markets as they move toward improving banks' capital positions through investment, hedging and funding activities. We have focused much effort on bank loan portfolio hedging techniques using synthetic CDS and tranche technology in both investment grade and leveraged loan portfolios (see Chapters 25-26).

While such hedging techniques are quite useful to CPM practitioners, they do not meet all of a given bank's hedging and funding needs. Synthetic solutions are part of a broader world of securitization approaches to managing credit risk in the CPM world. Many portfolios are either too illiquid or too complex structurally to fit neatly into simple hedges built from standardized unsecured and secured CDS, and tranching portfolios of such CDS contracts. There are also true sale issues that are supportive of a transfer of actual cash assets.

The traditional solution to the above challenges for many banks has been issuance of balance sheet CDOs, where illiquid assets are pooled together and securitized and structural complexities at the loan level are dealt with using waterfall structures within the securitization structure. The CLO market was originally designed for such transactions, as balance sheet deals dominated the early flow in the 1990s.

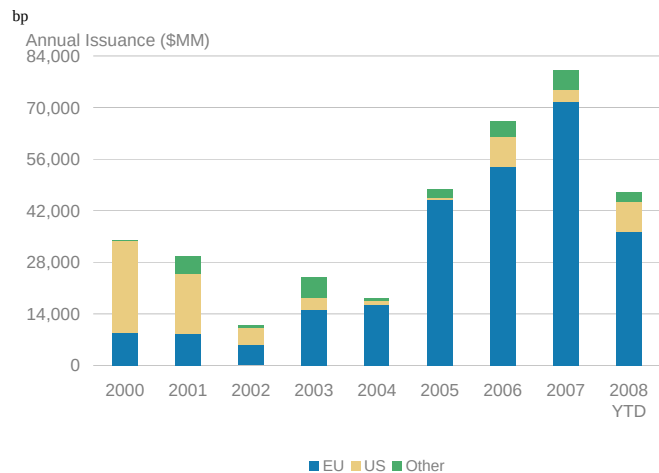
Although the loan securitization process remains difficult today given a variety of factors, we believe that balance sheet style securitization transactions will be an important regulatory and economic capital solution for banks as we move forward, especially in the less liquid corners of the corporate credit markets and in commercial real estate loans. The motivation will range from regulatory capital relief under Basel II (where efficient solutions involve retaining senior risk and selling junior risk) to improving funding for loans and loan commitments (which involves posting super senior risk as collateral in secured central bank borrowing programs such as those sponsored by both the ECB and the Fed).

In a nutshell, we see traditional securitization techniques as being an important tool for CPM practitioners as we move

forward, focused on the less liquid and more structurally complex parts of their loan portfolios. Old ideas are not necessarily irrelevant ideas, and in this case, one of the oldest corners of the structured credit markets (balance sheet transactions) should prove quite useful in the coming quarters. We focus this chapter on highlighting the usefulness of securitization technology for less liquid parts of both corporate and commercial real estate loan portfolios to improve regulatory capital efficiency and funding.

exhibit 1

Balance Sheet Transactions Are a Solution



Source: Morgan Stanley

BALANCE SHEET DEALS REMAIN ROBUST

Since the late 1990s, the CDO market has moved from one dominated by balance sheet transactions to one dominated by arbitrage deals, as the market found enough appetite for all tranches of CDOs to make the process of arranging and/or managing them to be attractive. Within the CLO market today, arbitrage issuance activity has slowed considerably owing to very expensive senior liability costs relative to the price of the underlying assets. But despite the mega-slowdown in arbitrage activity, we have witnessed \$47 billion of issuance in balance sheet deals this year, which matches the full year 2005 total balance sheet issuance and is on pace to match the 2006 total of \$66 billion (2007 totals were \$80 billion). While collateral varies, many of the deals are "Middle Market" in nature, otherwise known as SME loans in Europe. The European dominance in issuance is partly because many middle-market CLOs in the US are actually arbitrage transactions.

What is the motivation for these balance sheet deals? While it is not always crystal clear, we believe there are two: namely regulatory capital relief and funding. Since 2003,

Europe has dominated in issuance of balance sheet transactions, and we believe that recent transactions support both goals. Many deals are in Spain, where government supported programs are in place to create efficient financing frameworks for SME loans.

We note only a handful of synthetic deals in the 2008 totals (\$3.5 billion, or about 7%, but in synthetic deals the full notional amount is generally not considered). We believe that the synthetic deals are likely purely related to regulatory capital, given that the ECB will not accept synthetic deals (or CLNs) as collateral. While there are many arguments for transacting synthetic securitizations, including flexibility and scale, cash securitization in this context has the advantage of true sales, no P/L volatility associated with derivatives hedges, and no counterparty charges.

SECURITIZATION CAPITAL EFFICIENCY – EXAMPLE

Capital efficiency is a key reason behind the use of securitizations over direct loan exposure; we provide two illustrative examples in Exhibit 2 based on Basel II risk weightings. We use two generic examples of a full capital structure CLO as well as a two-tranche securitization that is in vogue today and show the impact of doing/not doing securitization as well as the kind of risk being divested. In the first example (full capital structure CLO), one can clearly see that today's pricing makes holding most kinds of senior risk attractive (spread per unit of capital charge is high) and mezzanine risk very unattractive (equity is an IO trade). Even in the two-tranche securitizations (the second example), retaining the junior piece makes for poor economics. In the traditional balance sheet world (under Basel I or from a funding perspective), banks faced with overburdened balance sheets are doing the exact opposite of what rational securitization suggests under Basel II. The focus on lowest absolute \$ amount of capital as opposed to the most capital-efficient strategy is creating lopsided pricing across the capital structure, with senior tranches trading very wide.

SECURITIZATION FOR FUNDING PURPOSES

Balance sheet securitizations have also been motivated by funding needs through secured borrowings from central banks. There was much concern in the market about changes the ECB would make with respect to eligible asset-backed securities as collateral in money market operations. The ECB did announce several changes, including effectively raising haircuts, stressing the importance of outside ratings, and leaving the door open for any specific exclusion of collateral. While these measures certainly tighten liquidity using securitized collateral, they do not eliminate the practice, as some in the market feared. As currently defined, the assets in the securitization must be a true sale and the tranche in question must not be subordinated to other tranches in the same deal. This means that only cash super senior tranches would be eligible for use as collateral.

CRE SECURITIZATION FOR CPM

CPM activity to date has been very limited in scope within the commercial real estate world, for a variety of reasons. Before the credit crisis, many banks were essentially intermediaries in the CRE debt world, focusing on warehousing whole loans for purposes of creating and distributing securitizations (CMBS and CRE CDOs generally). As such, unlike the corporate credit markets, there was not much need to either hedge such exposure or transform it into capital-efficient assets to retain on balance sheets.

But the credit crisis has indeed re-written the script within commercial real estate finance, and with CMBS issuance at a standstill, banks clearly are motivated to manage current real estate loan exposures and perhaps even create a new business model for the future underwriting of commercial real estate loans, where direct loan investments by banks could be pooled together in securitizations with certain tranches being retained in banking books.

exhibit 2

Balance Sheet CLO Capital Efficiency (Basel II)				
	Tranche Thickness	Spread	Capital Charge	Spread/Capital
Full Capital Structure Securitization				
AAA	70%	220	0.6%	3.9
AA	9%	450	0.6%	7.0
A	5%	650	1.0%	6.8
BBB	4%	900	4.8%	1.9
BB	4%	1400	34.0%	0.4
Equity	8%	3000	100.0%	0.3
Portfolio	100%	559	11.5%	0.5
Two Tranche Securitizations				
A	84%	270	1.0%	2.8
D	16%	2075	100.0%	0.2
Portfolio	100%	559	11.5%	0.5

Source: Morgan Stanley

As with corporate credit, there are two CPM motivations, regulatory capital and funding. There has been one publicly disclosed European CRE CDO issued for funding purposes this year, but beyond this transaction, we do not believe there has been much in the way of CPM activity, although one-off synthetic securitizations for regulatory capital relief purposes would not necessarily be publicly disclosed. Nevertheless, we believe that there is sufficient scope for CPM activity in the CRE loan space. The challenge for the market will be to digest the risk, especially regulatory capital motivated transactions that, in today's market, would require the sale of equity and mezzanine tranches of CRE loan portfolios.

exhibit 3

Capital Efficiency for Securitized CRE Portfolios (Basel II)

	Tranche Thickness	Spread	Capital Charge	Spread/Capital
AAA-SS	70%	290	1.6%	1.81
AAA-M	10%	390	1.6%	2.44
AAA-J	8%	625	1.6%	3.91
AA	3%	850	2.0%	4.25
A	2%	1300	2.8%	4.64
BBB	4%	2100	6.0%	3.50
BB	1%	2630	34.0%	0.77
B	1%	2830	100.0%	0.28
NR	2%	3000	100.0%	0.30

Source: Morgan Stanley

In Exhibit 3 we show the impact of CRE securitization under the standardized approach and yet again divesting junior tranches stands out as the optimal strategy. We are a bit ahead of the market here, but make the hypothetical assumption that a bank manages its CRE loan exposures in a manner similar to corporate loans, using tranches as hedges for inefficient parts of the portfolio, thereby creating long exposure to efficient parts of the portfolio. We describe this methodology in detail in Chapter 26. The conclusions are very similar to what we see in corporate credit portfolios, which should be no surprise since the dislocations in the market that cause senior risks to be cheap are common to both markets. Our point is to highlight efficiency in CRE portfolios, even though the practice of such hedging or securitization strategies is far less common in the CRE world than in the corporate credit world.

BASEL II – SECURITIZATION IS NOT ALWAYS BETTER

While the argument for using securitization as a tool to create more efficient exposures to credit from a regulatory capital perspective under Basel II is quite strong, it is by no means a truism. When we consider the impact of significant credit deterioration, as in subprime, “de-securitization” of second order securitizations like ABS CDOs might actually provide regulatory capital relief. This is in contrast to first order securitizations such as corporate CDOs and CLOs where securitization is capital enhancing.

Several super senior tranches of ABS CDOs have been downgraded multiple times and by multiple notches – some from as high as AAA at trade inception to BB and below currently. Downgrades below BB carry a full capital charge of 100% (i.e., full deduction) under Basel II external ratings approaches. There are many instances where a significant portion of bonds in the underlying CDO portfolio (especially RMBS securitized by prime/alt-A loans as well as CMBS bonds) have not experienced a similar magnitude of downgrades as the super senior tranches of the ABS CDOs (particularly in high grade ABS CDOs). It is worth noting that in most cases, the deals are already in EOD status and only the super senior tranches have any positive market value remaining. Under such circumstances, decomposing super senior tranches into their underlying collateral components has the potential to be more optimal from a regulatory capital perspective. This can be accomplished either synthetically or through a cash unwind process as the controlling class of a CDO.

IS THERE CAPACITY IN THE MARKET?

To the degree that CPM activity motivates the creation of balance sheet CLOs, and the sale of at least some part of it into the market, the question of market capacity is certainly a logical one. Traditional balance sheet CLOs have involved retaining junior risk while selling the bulk of the notional exposure via senior tranches, but under Basel II, such strategies will not provide much regulatory capital relief, and we expect the opposite flow to occur.

In our view, the problem with securitization/CDO appetite today is really at the senior level, since funding costs for financial institutions in general are too high to justify deployment of high-quality assets, even though they are likely the cheapest parts of the market. The difference between where unfunded super senior and funded super senior trade in the market (quite a large basis) demonstrates how important capital is to financial institutions. Equity tranches are perhaps a bit easier to distribute in this market environment, as long as the underlying collateral is transparent, since there is generally a clearing price given the convex characteristics and implicit leverage (as opposed to financing type of leverage). But this flow is very dependent on a healthy levered investment community.

CONCLUSION – SECURITIZATION AS A SOLUTION

A confluence of factors – Basel II's fairer treatment for senior risk, today's attractive pricing on senior tranches relative to junior tranches, and a recessionary backdrop that hurts junior tranches – should encourage investors to start retaining/buying senior tranches. This should restore the balance in senior tranches over time, which is the key to the securitization market.

We expect a sea change in banks' approach to securitization. Much of securitization today is viewed as a funding source due to balance sheet constraints – which is fair since that is

the need of the hour. But we expect securitization to be increasingly approached from a regulatory capital and economic risk management perspective. This should make the pricing of risk across the capital structure more rational and viable.

One of the criticisms of Basel II is that it is pro-cyclical and reinforces financial cycles. However, Basel II is actually a good middle ground between capital based on MTM accounting (as is in effect happening today – losses dictate capital raising efforts) and the other extreme such as Basel I (which barely differentiates credit quality at all). Basel II penalizes banks holding inferior assets by reflecting higher capital, driven by lower ratings (either internal or external) of

assets, without holding it hostage to significant market disruptions. Basel II strategies are valuation sensitive (unlike Basel I) and therefore anti-cyclical to some extent as they can identify relatively overvalued assets. For instance, Basel II would have clearly highlighted the richness of super senior in 2007 ahead of the credit crisis.

Much to the chagrin of many market doomsayers, securitization technology is in fact an important tool to repair the markets going forward, even though it was a central part of what went wrong as well. Securitization is part and parcel of the Basel II regime (the term appears 263 times in the 347 page accord), and as the regulatory capital environment shifts, securitization-based solutions can take a center stage.

Bank Loan Hedging Dynamics – CPM Update

June 19, 2009

Sivan Mahadevan
Phanikiran Naraparaju
Ashley Musfeldt

Bank loan hedgers are the natural buyers of protection in the market, and as the Basel II clock continues to tick, we note that the significant re-pricing of portfolio hedging opportunities, along with the changing regulatory environment, leave CPM practitioners with many choices and challenges. CPM (or credit portfolio management) is the industry term used to describe the practice of managing credit exposures in corporate lending institutions.

We initiated our research work on CPM about a year ago, in anticipation of the transition to Basel II over the then ensuing 18 months. Although we started our process a few months after Bear Stearns was rescued (and systemic risks felt like they were transforming to idiosyncratic ones), the Lehman collapse in September sent this issue to the back burner at banks for obvious reasons (see Chapters 25-27). However, the full implementation of Basel II is now under way in Japan and is only six months away in Europe, and despite other capital-related issues in the global banking industry, banks are indeed focused on efficient management of credit exposures under Basel II. One of the biggest challenges, though, is thinking about potential changes to regulatory capital frameworks given the expectation of significant financial regulation overhaul.

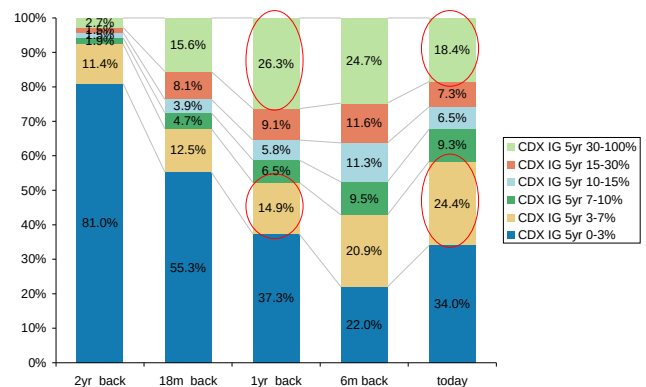
Indeed in the US, the Obama administration is proposing the biggest changes to financial regulation since the 1930s. We have some sense of the scope of these based on the published white paper, but it is impossible to predict specifics given all that now needs to happen, including debate and approval in Congress. But we take comfort from two points. First, securitization appears to be part of the solution (even though it might be rated significantly differently than today), and there is acknowledgement of systemic risks within senior tranches, something the rating agencies are already addressing. Second, the proposal states that any securitization solutions that reduce regulatory capital requirements must see a commensurate reduction in economic risk as well. This latter point fits in well with some of the Basel II efficient hedges that we see in the market and stands in contrast to the non-economically motivated Basel I hedges that are slowing fading into history.

In this chapter we review opportunities in the market to hedge or re-hedge portfolios under a Basel II framework, taking into consideration both the significant re-pricing of the investment-grade structured credit capital structure this year and the expected changes in ratings methodologies. In a nutshell, we now find that the hedging “sweet spot” in

investment grade has moved from mezzanine tranches down to equity, and more complicated solutions can be made even more efficient by combining equity hedges with some forms of senior-mezzanine protection.

exhibit 1

Risk Allocation – Efficient Hedges Will Shift with Market Pricing



Source: Morgan Stanley

PRICING SHIFTS – LESS LOPSIDED TODAY

Changes in rating agency methodologies have been important, but in the year since we started thinking about Basel II and hedging strategies, pricing has also changed dramatically. By this we mean both across assets (IG vs. HY) as well as within the capital structure – senior risk vs. junior risk. A year ago, a relatively obvious conclusion of our analysis was that the senior parts of the capital structure were less efficient as hedges than junior tranches (see Exhibit 1). With 30-100% accounting for 26% of the index risk, there was little incentive to hedge this senior risk (unlike Basel I). But as seen in Exhibit 1, this pricing has normalized somewhat during the year, with 3-7% absorbing a much higher proportion of risk to the benefit of the senior tranches and even equity. As a result, it is less obvious where the optimal hedging strategies lie today from a valuation perspective, and we dig into the details later on. Apart from valuation, risk transfer considerations also come into play, so efficient strategies would still require hedging some amount of junior risk (either equity or mezzanine risk).

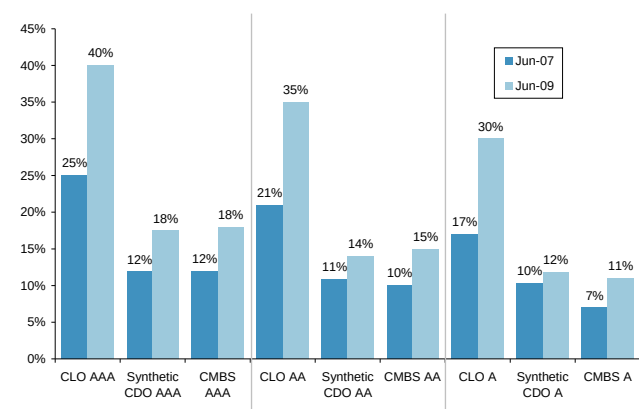
RATINGS IN FLUX – STEP CHANGE

The direct impact of the financial crisis in terms of defaults and rating migrations is certainly being felt with changing subordinations and portfolio quality. But perhaps the most important factor that has changed in the last year or so is that the rating agencies are recalibrating ratings thresholds across asset classes. The recalibration is clearly necessary in some

areas, but in others they are more precautionary and conservative. For example, the minimum subordination levels for a bespoke CDO portfolio and CMBS pre- and post ratings changes are up from the 12% context to the 18% context (see Exhibit 2). This sea-change in ratings thresholds is forcing the market to reevaluate hedging strategies, as the capital relief obtainable has changed dramatically. In Chapter 26 we discussed in detail the optimal hedging strategies involved in hedging junior mezzanine risk and retaining senior risk. For anyone following the standardized approach, the ratings shifts will at a minimum alter the optimal hedging strategy but may also influence the choice of approach in some cases.

exhibit 2

Estimated Subordination Requirements Before and After Criteria Changes



Note: These are expectations of generic levels and not based on actual deals.

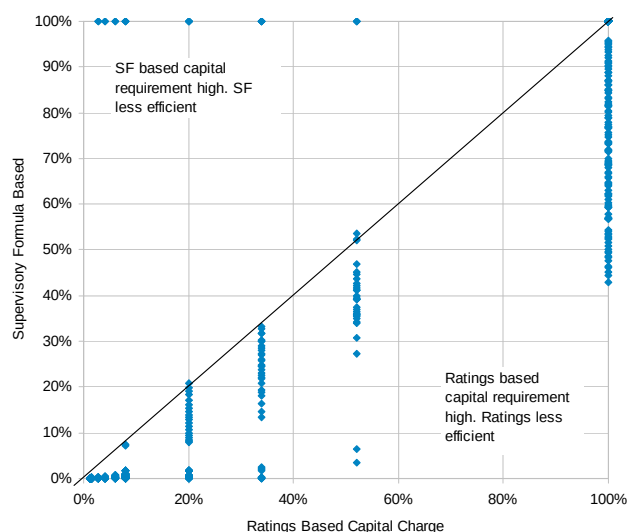
Source: Morgan Stanley, Moody's, S&P

THE LEVERAGED LOAN PROBLEM REMAINS

In the IG space the better liquidity in single-name CDS as well as an active index tranche market makes hedging solutions easier to implement. It is tougher to find solutions for leveraged loans, which are affected quite adversely by the unfavourable treatment of lower-quality assets under Basel II. Two-tranche securitizations focused on achieving funding through central bank programs are often seen, but as we highlighted earlier the ratings barrier is likely to get onerous. Ratings were already more punitive than the IRB approach for leveraged loans, and this gulf will only increase with the revisions of the past year (Exhibit 3). As such we see banks leaning toward the IRB approach.

exhibit 3

Leveraged Loans – Supervisory Formula More Efficient Generally



Note: Compares capital charge for different tranche solutions under the standardized and IRB approach.

Source: Morgan Stanley, BIS, Moody's

The relative preference for different approaches will be a function of how banks rework their loss assumptions after the crisis and where the gulf between the bank assumptions and rating agency assumptions narrows. We suspect rating agencies will generally err on the conservative side for most assets, and hence more banks will find it attractive to move to an IRB approach as opposed to the standardized approach, wherever possible. While the smaller banks may still find the standardized approach resource-efficient, IRB is likely to be the target for most banks with scale.

Another popular strategy is simply using LCDs to hedge – but this is feasible only for the larger liquid names. There is an additional factor that comes into play: Loans are not necessarily held in mark-to-market books, so hedging with an MTM instrument is an issue. Solutions to surmount this are being explored.

OPTIMAL IG HEDGING CHOICES TODAY

As noted above, Basel II strategies are likely to be more dynamic as shifts in pricing can alter the efficiency of specific solutions. In Exhibit 4, we show how the results of a two-tranche Basel II solution for the 5-year CDX IG index have changed over the past two years. We graph the spread/required capital (which is a measure of return on capital) on the Y-axis. On the X-axis we graph various detachment points between the equity tranche and senior tranche. The flat line represents the return on capital for simply holding the CDX portfolio with no hedges, so it is the return-on-capital bogey to beat. The two curves in the graph represent: a) retaining the equity tranche and hedging with

the senior tranche (the lower curve), (L-H) and b) retaining the senior tranche and hedging with the equity tranche (H-L) (the higher curve). See Chapter 26 for more details on the methodology.

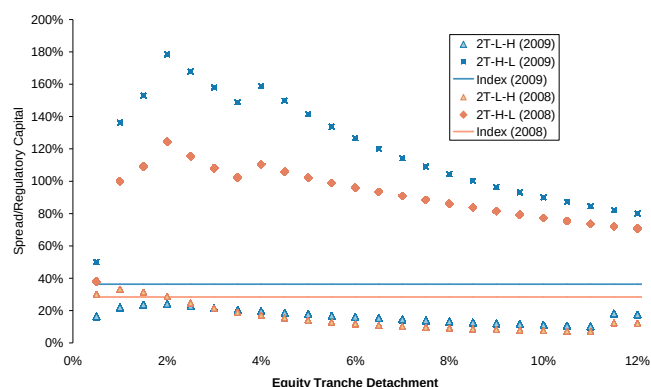
What has changed? Retaining the senior risk and hedging the equity risk was and continues to be the most efficient strategy, rather than the other way around. Within this there are other themes as well – equity tranche risk is less attractive today relative to the index for this portfolio, and there is no solution under which retaining equity is an improvement over retaining the portfolio. We think a significant theme is mezzanine cheapness and equity richness (from a seller of protection perspective).

Within the subset of solutions possible, there were few alternatives last year which improved the spread per capital over the index but did not involve retaining 3-100% type of risk. Today, there is more scope for divergent strategies, as mezz is pricing in a lot more risk and a number of solutions which involve retaining a part of the mezz slice show a significant improvement in spread per capital over just retaining the index portfolio. The point here is that while two-tranche strategies optimize the spread per capital, they may also reduce the absolute level of capital deployed. As

such, some CPM practitioners may prefer alternate three-tranche solutions which still earn above the index spread per capital. While not shown in the exhibit, three-tranche solutions which involve buying equity protection and retaining some amount of the mezzanine tranche risk while hedging senior mezzanine risk can also be viable.

exhibit 4

2009 vs. 2008: Equity Tranche Hedges Optimal



Source: Morgan Stanley

Basel Update and Structured Credit Implications

March 12, 2010

Sivan Mahadevan
Ashley Musfeldt
Phanikiran Naraparaju

There has been much focus in the market on the implications of new financial regulation on structured credit flows, in particular at banking institutions who use structured credit to both invest in credit and hedge credit exposures. Most banks outside of the US are on Basel II, and much of the rhetoric around changes to Basel II (i.e., Basel III) have not really addressed securitization or structured credit within banking books, as Basel II is reasonably comprehensive with respect to securitizations. There remains much uncertainty in the market about potential paths of regulation, but much of the uncertainty centers on the liability side rather than the banking book asset side at this point. As it stands now, the Basel II approach for securitization would govern most banking books from a structured credit perspective.

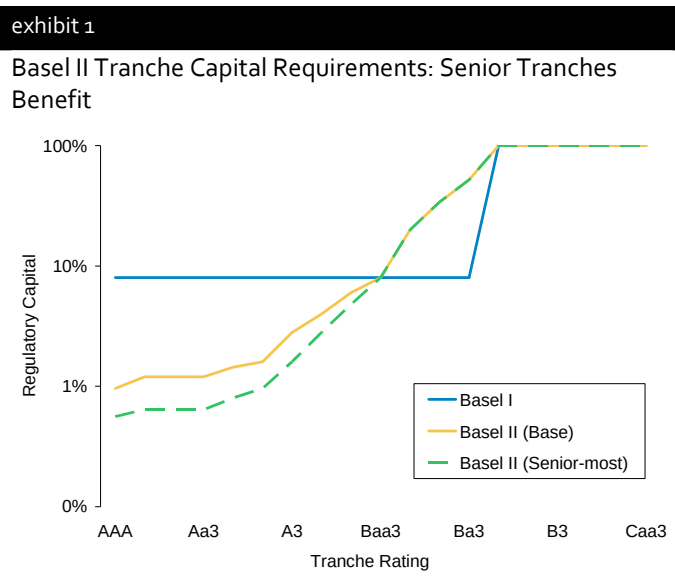
As such, the bottom line is that for corporate credit risk held in structured/secured form, some form of rating (either internal or external depending on the institution) will determine the risk weights, which when combined with market pricing will determine regulatory capital efficiency. The good news is that this approach is more aligned with economic capital than Basel I, and with the rating agencies' significant shifts in ratings methodologies last year (much more conservative), there is much less opportunity for any form of ratings arbitrage.

We have addressed structured credit capital efficiency quite extensively in the past. We published a number of research notes in summer of 2008, later compiled as: Bank Loan Hedging Today – CPM as a Market Force, Sep 12, 2008, and an update last year (see Chapters 25-28). Market participants have since moved from firefighting mode to investment mode, even as the key variables – outstanding ratings of such structures, as well as market prices – changed dramatically. We focus on three market themes in this chapter related to regulatory capital efficiency.

Legacy CSOs and unwinds: Much of the multi-hundred billion dollar universe remains rated below investment grade and is therefore inefficient to hold for most banking institutions. Firmer underlying credit markets have motivated sales, and we have already seen several billion in notional of sales and liquidations in 2010. The buyer base has viewed the opportunity as a distressed one, with mid-teen type IRRs, and so far there has been enough of a buyer base to absorb supply at these levels. Key risks are that valuations rise to a level where IRRs are much lower, or supply exceeds demand. In either case, the flow could pressure credit spreads wider as buyers look to unwind structures or delta hedge.

Basel II hedging solutions: For bank loan portfolios under Basel II, optimal hedging strategies will vary over time and are a function of market pricing. Hedging first-loss exposure and retaining the rest remains the efficient solution today, which makes sense to us from an economic perspective as well. For some portfolios, hedging junior mezzanine risk might also make sense.

Senior mezzanine is the sweet spot: Finally, from an investment perspective, senior mezzanine tranches are the most efficient under Basel II, and we find this to be the cheapest part of the market by our more economic valuation measures as well. Super senior can also be efficient to hold, although not as much as senior mezzanine.



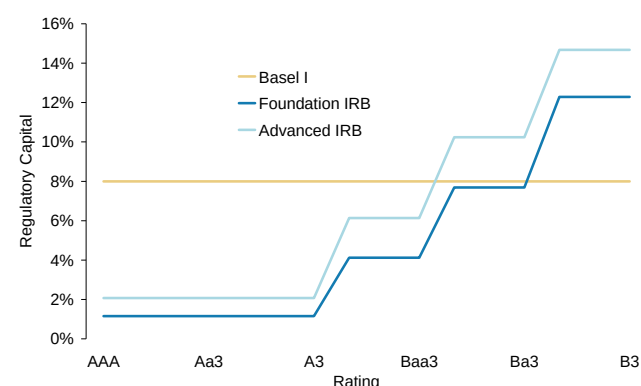
Source: BIS, Morgan Stanley Research

BASEL II RECAP AND LESSONS FROM EUROPE

To start, we provide a quick regulatory capital refresher for investors completely new to Basel. Basel I (introduced in 1988) was an extremely simple first attempt by the BCBS to strengthen the international banking system in a consistent way. Basel I introduced 'risk-weightings', whereby assets on a bank's balance sheet are separated into different classes, with prescribed risk-weightings. Basel I differentiated risk weights by asset class but without much regard to credit quality – for example all corporate credit got a 100% risk weighting irrespective of whether it was AA or BBB. Basel II is a significant step up wherein the amount of regulatory capital is a function of the credit quality (see Exhibit 2 for plain corporate exposures and Exhibit 1 for tranche exposures) bringing alignment in regulatory and economic capital.

exhibit 2

Basel II Corporate Risk Capital Requirements: Aligning Regulatory and Economic Capital



Note: IRB – Internal Ratings Based Approach.

Source: BIS, Morgan Stanley

While Basel II is new to US, European banks have already been under implementation of Basel II since 2008. As is clear from the discussion above, banks with a greater proportion of IG assets may actually end up reducing the risk-weighted assets. In a world where banks err towards more liquid and higher quality credit and have greater amount of buffers, Basel II could end up being more favorable from the asset perspective. There are also liability side implications (in terms of what constitutes capital) which we do not address here.

If the European Banks' experience is any indicator, most large banks will ultimately seek approval from regulators to use the Advanced Internal Ratings Based Approach (AIRB). AIRB specifies methods to calculate the risk weighting based on internal estimates of the Probability of Default (PD), Exposure at Default (EAD) and Loss Given Default (LGD).

Because Basel II regulatory capital is closely aligned to the economic capital, changes in market pricing can influence strategies to a greater extent than under Basel I, as we will discuss later on. While senior tranches generically benefit relative to junior tranches, relative value perceptions – for example, between junior mezz and senior mezz or between senior and super senior – can change based on the pricing.

STRUCTURED CREDIT SWEET SPOT

From an investment perspective, we update our analyses of the attractiveness of index tranches as measured by the spread per regulatory capital, keeping in mind the more stringent rating criteria (see Exhibit 3). For example IG13 10%-15% could potentially be a single 'A' rated tranche which attracts a risk weighting of 20%, implying regulatory capital of 1.6% of the notional (i.e., risk weight * capital ratio) of 8%. With current spread in the 100bp context, that translates into a spread per capital of 64%.

exhibit 3

Spread per Regulatory Capital for Index Tranches

S9	S9 3Y	S9 5Y	S9 7Y	S9 10Y
3-6%	4%	4%	5%	6%
6-9%	26%	29%	15%	4%
9-12%	21%	69%	38%	11%
12-22%	17%	30%	47%	22%
22-100%	8%	25%	46%	53%

IG13	IG13 3Y	IG13 5Y
3-7%	4%	5%
7-10%	55%	30%
10-15%	32%	64%
15-30%	16%	26%
30-100%	9%	24%

IG9	IG9 3Y	IG9 5Y	IG9 7Y	IG9 10Y
3-7%	3%	8%	10%	11%
7-10%	50%	14%	4%	6%
10-15%	16%	35%	21%	5%
15-30%	7%	21%	34%	19%
30-100%	8%	22%	44%	59%

Bespokes*				
Rating	Tranche	Portfolio1~	Tranche	Portfolio2~
BB	4.5 - 6.5%	22%	4.2% - 6.2%	18%
BBB	6.5 - 8.5%	92%	6.2% - 8.2%	77%
A	8.5 - 11.0%	269%	8.2% - 10.5%	175%
AA	11.0 - 15.0%	217%	10.5% - 14.5%	92%
AAA	15.0 - 20.0%	125%	14.5% - 17.5%	52%

Note: * 100% IG portfolios 5yr maturity. ~ 80% US Name Portfolio. ~ 80% European name portfolio.

Source: S&P, Morgan Stanley Research

Since we last published a similar table, we have had a continuation of a recovery in the belly of the capital structure – earlier the most attractive tranches from a long perspective. Given today's changed pricing, the 9-100% part of the capital structure in iTraxx S9 7yr and 22-100% tranche in iTraxx S9 10yr seem to be the most attractive longs. iTraxx S9 7yr and 10yr 3-9% seem the least attractive tranches from a Basel II perspective. CDX IG13 5yr 7-15% seems to be the most attractive from a long perspective whereas 3-7% seems to be the least attractive part of the capital structure.

Across CDX IG9 7yr and 10yr, the 15-100% part of the capital structure seems to be most attractive from a long perspective, whereas 7-10% seems to be the least attractive part of the capital structure. We exclude equity tranches from this analysis given their full deductions under the capital model.

Finally, we run the same analysis on two examples of a potential new issue bespoke transaction with 5yr maturity. Both these portfolios are 100% IG underlying, one with a Euro centric portfolio and the other with a US centric portfolio. As seen in Exhibit 2, A/AA tranches are the new

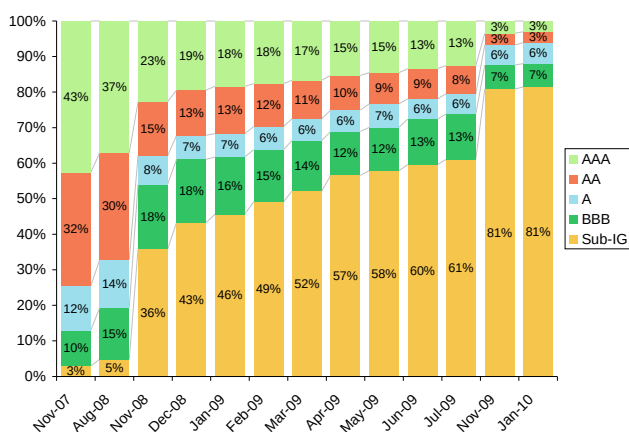
sweet spot from a long perspective. Between the super senior (AAAs) and junior mezz (BBBs), relative value is less obvious and is likely to be sensitive to changes in market pricing.

LEGACY BESPOKE CSO MARKET

As far as the bespoke CSOs from the last cycle are concerned, they are deeply underwater from a ratings perspective, as we have highlighted previously (see Exhibit 4). The sub-IG share jumped from under 5% at start of 2008, to 45% at the start of 2009 (post Lehman and Icelandics) and now stands at 81% (after the ratings methodology changes implemented in the second half of 2009). Clearly, these mostly mezzanine CSOs look unfavorable under Basel II. Despite their having the poor ratings problem for sometime now, unappetizingly low valuations were a key impediment for sellers early in 2009, as was a lack of an organized buyer base and lack of adequate depth.

exhibit 4

Legacy Bespoke CSOs – Largely Below IG Now



Source: S&P, Morgan Stanley Research

A year ago, there was perhaps more imperative for restructuring (to improve portfolios and subordination in a still wide spread environment with less stringent rating criteria), but with stronger markets today, selling is now the more relevant theme. As we approach the tightest spread levels since the crisis, there seems to be greater willingness by sellers to 'take profits' after hanging on through the crisis.

It is hard to estimate how much will eventually be sold, as data in this part of the market is weak and unreliable, but we do believe that the stronger the underlying credit performance (particularly with many of the distressed financials that are in these portfolios), the more sellers will emerge. All that said, we would not completely write off restructuring as a solution for investors. Certainly, retracing the ratings slide back to AAA/AA through restructuring is a

herculean task with the new stringent ratings criteria. Nevertheless, restructuring is a viable alternative as long as the goals are not too ambitious. Just targeting a move of few notches from B to BB- or from BB to BBB- makes a material difference to the amount of capital held in absolute terms under Basel II, but would not cost that much in today's tighter spread environment.

The predominant nature of this risk was originally mezz like but is more equity/junior mezz like today. We expect to see widening pressure on the junior parts of the capital structure as a result of potential unwinds, but a maturing ratings-insensitive buyer base has developed in 2009 and should be able to absorb these sales as they occur. Orderly CSO BWICs are a regular occurrence today and appetite for secondary CSO paper remains fairly strong for now.

SECURITIZATION AND CAPITAL RELIEF STRATEGIES

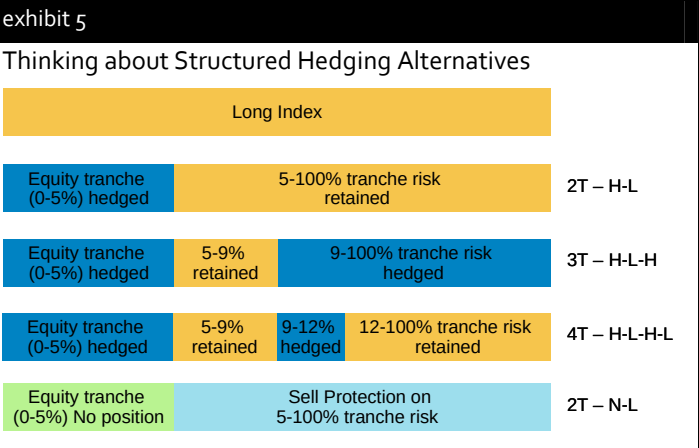
Last but not least, securitization and capital relief strategies have experienced a complete U-turn from Basel I to Basel II. The most common Basel I strategy involved retaining the yieldiest part of the capital structure, i.e., the equity tranche, and buying protection on 3-100% part of the capital structure. Under Basel II, the optimal strategies for Europe proved to be the exact opposite – the focus is on what part of the capital structure is most efficient to retain, and as we discussed above, it is really the senior, higher quality tranches that were the most attractive risk to retain. Investors have a number of types of hedging choices (see Exhibit 5).

Based on simulation of various hedging alternatives (see Chapter 26 for a detailed discussion of our methodology), we identified the following as the most optimal hedging strategies for CDX IG13 type portfolio (see Exhibit 6):

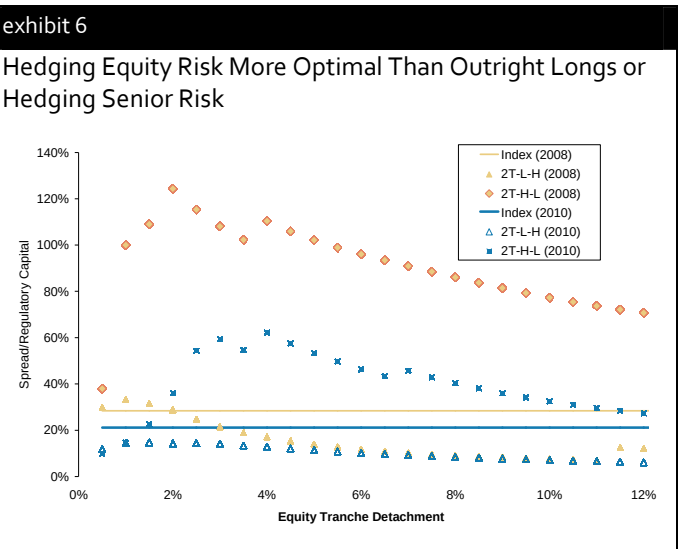
- Retaining senior risk (x-100%) is good, where the attachment is 4% based on today's market pricing for CDX IG type portfolios. The optimal attachment has drifted up from 2% since we last discussed this (see Chapter 28).
- This analysis is based on the relatively cleaner IG13 portfolio. As the quality of the portfolio becomes lower, the higher the optimal attachment moves. For example, when we first ran this analysis in 2008, retaining X-100% was optimal for both IG and LCDX portfolios. Whereas 'X' was around 2% to 4% for IG portfolios, it was 9% to 12% for leveraged loan portfolios.
- For the wider spread portfolios, both equity and junior mezz look good to hedge today. It seems the key hedging decision for many investors is whether to hedge equity tranches or mezzanine or even both. This decision will be a function of the portfolio, yield targets, and hedging capacity of investors.

CONCLUSION

As valuations reach the tightest levels we have seen in two years now, we see renewed focus and debate on longer term drivers and what will influence the fair value of credit. Given that banks and financial institutions are amongst the primary investors for credit, the regulatory regime and capital requirements across various types of assets will be an important driver of pricing of credit in both an absolute and relative sense within the tranche capital structure.



Source: Morgan Stanley Research



Note: 2T L-H means split portfolio in two parts and retaining equity (0-x%) and hedging x-100%. (See Exhibit 5).
Source: S&P, Morgan Stanley Research

Section E

The Market for Customized Credit

Distress, in an Unlikely Place

January 16, 2009

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The terms “distress” and “investment grade” generally do not go hand in hand. Even at the lows of the credit markets in late 2008, the average prices of investment grade corporate bonds did not fall below 85%, hardly a price that a distressed investor would consider appealing. The wides on CDX were sub 300 bp, also quite far from distressed levels.

Yet one corner of the market where the two terms collide is in bespoke mezzanine structures, where a combination of LBO and financials heavy portfolios, thin mezzanine tranches, subordination losses from the Lehman, Washington Mutual and Icelandic bank defaults, and funding collateral with credit issues has created a sizable distressed opportunity in a seemingly unlikely place. The size of this opportunity is quite substantial, upwards of \$300 billion notional, and the opportunities range from single-digit-priced long-duration bonds with poor credit composition and little remaining subordination to healthier structures with shorter durations and more subordination at substantially higher prices, but still distressed.

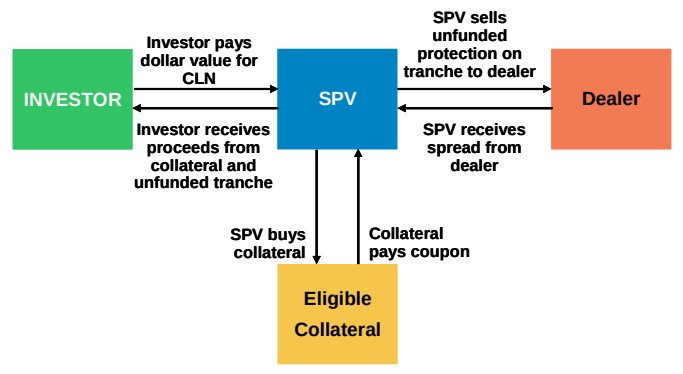
In our view, while a variety of skills is needed to prosecute this distressed opportunity, investment grade skills are perhaps the most important, which is part of the irony. We view the distressed CSO (single-tranche synthetics) opportunity as a significant one that plays to a healing credit market theme, with the prospect of owning bonds that can be valued anywhere from IO-value to option-value to hold-to-maturity value. This chapter serves as an introduction to the opportunity, and Chapters 31-34 will build on this.

Compared to the many distressed opportunities out there, tranches are among the simpler ones to evaluate across many variables, although the funding component can be more complicated. We categorize the variety of opportunities available into three groups: IO-like tranches priced and traded purely on coupon value; tranches with some option-value on principal recovery; and lastly, tranches that have a reasonable chance of being “money good” with moderate dollar prices.

We discuss the two components – the unfunded tranche and the funding collateral – essential for evaluating this distressed opportunity.

exhibit 1

Typical Bespoke Mezzanine CLN Structure



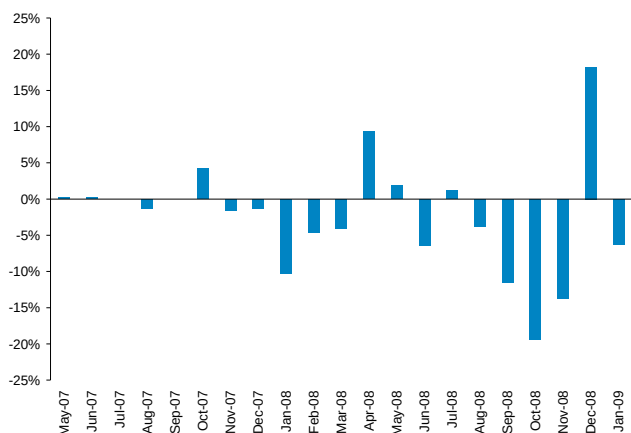
Source: Morgan Stanley

STEEP PRICE DECLINES OF Q4 OFFER AN OPPORTUNITY

The last quarter of 2008 was horrendous for most credit asset classes, and synthetic CDOs were no exception. The 5-year benchmark index dropped double-digit points for three straight months (September through November) on the unexpected financial defaults and the subsequent mayhem (see Exhibit 2). Distressed bespoke CSOs suffered disproportionately due to several factors: first, the use of wider spread credits in the portfolio compounded by additional tail risk relative to the indices; the prevalence of defaulted financials in bespoke portfolios and subsequent loss of subordination (ironic considering financials were added for diversity and stability); and finally variation in duration due to maturity differences between bespoke structures and the indices. Keeping these factors in mind, we estimate that median distressed CDO prices based on the unfunded tranche component alone could be in the price range of \$30–35. That said, the variety of portfolios, maturities, and subordination implies that shorter-dated or more senior risk in the \$50–60 range can also be sourced.

exhibit 2

5yr Index Junior Mezz Monthly Performance



Source: Morgan Stanley Research

We broadly divide distressed opportunities into three groups:

IO-Value. These are the really low-dollar-price instruments, (below \$10), which are priced like IOs – typically equal to about 2-3 years of coupon. It is expected that the principal value will be little to nothing at maturity; the valuation is done based on expectations of how many coupons the structure will pay out before the notional goes to zero. The yield to maturity of these structures would be in the 60+ range.

Option-Value. These are mid-priced structures (\$5–30) that capture the interest component plus the small chance of principal recovery. Yields for these deals are likely in the 30-60% range.

Money-Good. These structures can have any or all of the following: shorter maturities, cleaner portfolios, higher subordination, and greater thickness. Consequently they have higher associated prices (\$20–60). These are categorized as money-good because while nowhere near a certainty, the likelihood that there will be at least a partial principal repayment is high. Yields here are much lower (25–50%) but still fairly attractive compared to the high-single-digit and low-to-mid-teen risk-adjusted returns for plain vanilla IG/loans or HY credit. There are obviously more risk remote tranches as well.

exhibit 3

Synthetic CDO Market Overview

	Nov-07	Aug-08	Nov-08
Deal Type			
Synthetic CDO of CDO	642	639	618
Synthetic CDO	2217	2,395	2072
Total	2,859	3,034	2,690
Management Type			
Managed	47.1%	49.9%	52.3%
Static	43.8%	40.7%	38.4%
Hybrid (lightly managed, etc.)	9.2%	9.4%	9.2%
Original Rating Category			
AAA	44.8%	44.9%	45.1%
AA	34.1%	33.2%	34.6%
A	11.7%	11.9%	11.4%
BBB	7.1%	7.6%	6.8%
Sub-Investment Grade	2.2%	2.4%	2.0%
Current Rating Category			
AAA	42.6%	37.1%	22.8%
AA	32.0%	30.1%	15.1%
A	12.5%	13.5%	8.1%
BBB	10.0%	14.5%	18.3%
Sub-Investment Grade	2.9%	4.8%	35.8%
Geography			
Europe	52.7%	51.6%	55.1%
Asia	11.9%	11.5%	12.6%
North America	35.4%	36.9%	32.3%

Source: Morgan Stanley Research

RATINGS BECOMING LESS RELEVANT

Tranche ratings, already under pressure from defaults and negative rating transitions in the underlying portfolios, are also being affected by more stringent ratings criteria announced in the last several weeks. However, ratings are becoming less relevant, given that the dollar prices have approached significantly distressed levels over the course of the last quarter. Furthermore, the credit universe has already transitioned dramatically from a median AA-rated universe in 2007 to a BBB/Sub-IG-heavy universe in the post-Lehman environment (see Exhibit 3). With the added impact of the rating criteria revisions, we expect the investor base to do one of three things: reconcile themselves to holding non-IG tranches given deeply distressed prices (most likely), restructure to enhance ratings (less likely) or attempt to liquidate at current prices (somewhat likely). Meanwhile, banks that were set up to inventory these products are faced with decreased risk-taking capacity of their own. We estimate at least \$300 billion notional is still outstanding and from Exhibit 3, we can see that the number of deals outstanding has decreased about 10% since August 2008. Some of this risk may be owned by dealers now, but we believe the bulk of this risk is still in the hands of structured investors.

STRUCTURAL VALUATION OF THE UNDERLYING TRANCHE

Taking advantage of this dislocation requires some background knowledge of the market, though we stress that sophisticated correlation models aren't as necessary here as they were earlier. We break down the tranche valuation aspect into several components.

Subordination. Many of the 2008 defaults were common names in bespoke CLNs, which tend to be entirely IG names. The GSEs, Lehman, Washington Mutual, and the Icelandic banks in particular were popular constituents of these structures, and thus many of the structures out there have had at least some write-down. What's important is what's left in the portfolio and how many additional losses can be sustained.

Thickness. Thin tranches were the dominant theme when these assets were originally structured, to increase structural leverage and thus spread. This matters because if the CLN is touched at all, it could require only one additional default to wipe out the entire tranche. This is very different from the index tranches, which are all at least 3% thick and can withstand a greater number of defaults once breached.

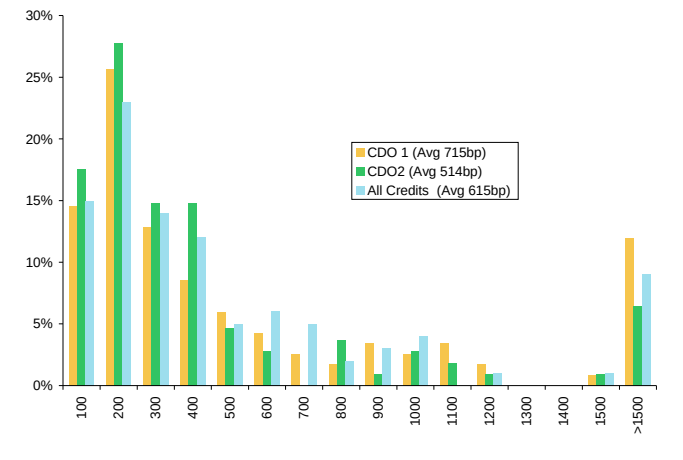
Portfolio composition. This is clearly important, particularly if there is little subordination left in the structure. We have written extensively about tails and portfolio dispersion in the past, usually concerning the various index series. It is no less important here when evaluating bespoke structures and indeed the valuation methodology is largely the same.

While it is hard to generalize the different portfolios out there, S&P recently published a list of the top 100 corporate obligors out of approximately 450 corporate obligors referenced in roughly 1,000 US synthetic CDOs, giving the market a very good sense of what a typical portfolio could look like (see Exhibit 4). These 100 names represent about half of the total corporate exposure in US deals. While the tail risk for different CDO portfolios would vary significantly, it is fair to say that the belly and the left tail have similarities across portfolios. Often the portfolios would be wider-spread portfolios than the index at inception and that continues today. A typical CDO portfolio spread currently could be in the 500-700bp range, i.e., almost 2x the spread of the CDX index, with the 10% tail contributing 40% or more of the index spread (though tight spread portfolios are common). The exact attachments vary but an original AAA tranche would likely attach in the high single digits and have a 1% thickness, with a price under 20%.

To combat some of the extreme dispersion present in these portfolios, we like paying for risk management and recommend using some amount of the coupon income to pay for a manager and/or buying protection on tail names.

exhibit 4

What Do Typical Portfolios Look Like?



Source: Morgan Stanley Research

Recovery. The majority of these structures are floating recovery, and as we've seen lately, particularly with respect to Tribune, recovery can be very low on certain names. Some, however, are fixed recovery, generally 40%, and this added certainty regarding number of defaults needed to impact the tranche can be very valuable.

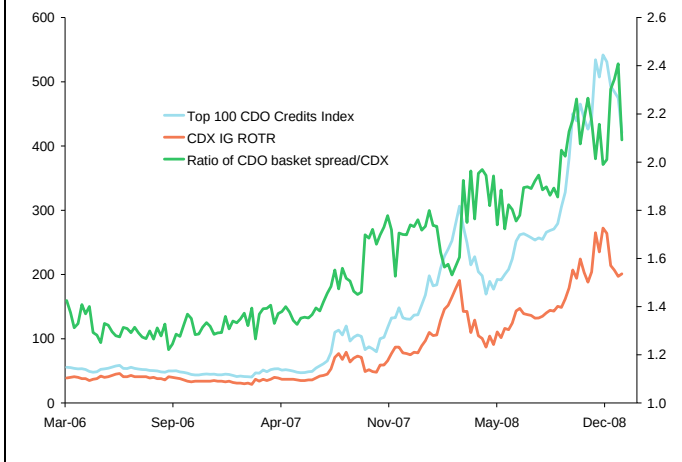
Managed or static. Some CLNs are managed synthetics, and the added presence of a manager can be valuable, depending on the manager.

Maturity. The five-year maturity deals done in 2004/2005 have very short maturities and yet some are still trading at highly distressed levels. Alternatively, in 2006/2007 most CLNs were structured with 7- and 10-year maturities, as investors shunned the shorter maturities in a reach-for-yield environment characterized by steep spread curves. Today, with curves flat or inverted, we argue that maturity is far less important than the first five components listed above, though still worth noting, particularly in the very short end.

Coupon. While we argue that the deeply discounted dollar price is the more eye-catching price component, for those bespokes we consider to fall under the category of "IO opportunity," coupon is important.

exhibit 5

Top 100 Names vs. CDX



Source: Morgan Stanley Research

COLLATERAL VALUATION

While analyzing the underlying corporate portfolios is something that most investment grade investors ought to be comfortable with, the collateral component may be a little less familiar.

In these synthetic funded CDO structures, a wide variety of collateral has been used over time, ranging from money market instruments, to asset-backed paper (credit cards), to covered bonds and beyond. There is some discrepancy in collateral performance with some trading at a significant

discount, while others are still trading at or near par. MMF and covered bond collateral both score high in our view, as they have performed reasonably well so far, whereas we are wary of GICs or other structured products.

Credit card AAAs are common collateral and are structurally well protected, despite the stresses in this form of consumer credit today. We also note that there is a significant variation in the composition of different credit card ABS trusts and this creates significant performance variation. A strong positive factor for this asset class is the potential support from TALF. Fed chairman Ben Bernanke, in a recent speech, suggested the Fed will begin a program to bolster securitization markets for consumer credit. The Term Asset-Backed Securities Loan Facility will finance up to \$200 billion in securities backed by loans to small businesses, students, credit-card holders and car buyers. We hope to discuss different collateral types and valuation implication in subsequent research.

CONCLUSION

With our views on a healing investment grade credit market globally, we feel that there is a significant opportunity today to prosecute what is perhaps the most distressed corner of the investment grade market, the several hundred billion dollar market for distressed synthetic CDOs. The opportunity is less about structure and more about credit, both the corporate credit backing the tranche and the collateral backing the credit-linked notes, and it is ultimately this skill set that is necessary to evaluate deals. We will devote additional research to evaluating these components together to find the best value among the opportunities out there.

Getting a Handle on Distressed Mezz

January 30, 2009

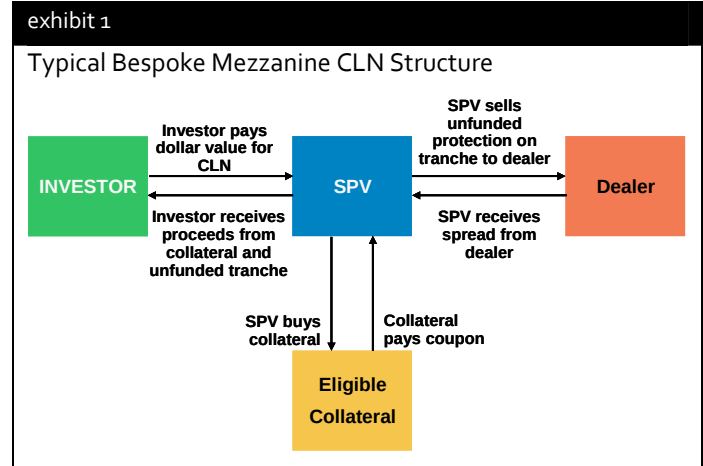
Sivan Mahadevan
Ashley Musfeldt
Phanikiran Naraparaju

In Chapter 30, we began our discussion of the current opportunity for investing in distressed bespoke mezzanine structures. This opportunity is several hundred billion dollars in size, and a good amount of risk needs to change hands given price and ratings actions. Since most deals have predominantly investment grade credit as the primary building block, we have argued and continue to stress that fundamental IG corporate credit analysis capabilities are probably the most important skill required to take advantage of this opportunity. Here, understanding jump-to-default and liquidity risks can be useful in bucketing companies into high-, medium- and low-risk credits and then assessing the potential residual life of the unfunded tranche component, given subordination and other structural features.

The ultimate potential gains from this distressed opportunity can be quite significant, especially considering the notion that many structures may ultimately mature with no realized losses given subordination levels and high quality funding collateral. However, picking the winners from the losers is challenging, and the cost of being wrong can be high, given the thinness of some of these tranches.

In this chapter, we dig deeper into the fundamental analysis, and combine this with a closer look at the funding component. When evaluating these two components together, it can be helpful to think of the entire structure as a first-to-default on the collateral and tranche together. If we evaluate the two parts individually, it is possible to come up with a conservative estimate for the ultimate value of the asset, since the likelihood of survival of each is probably related. To evaluate these assets, we break down our analysis into four parts.

Collateral: The wide variety of collateral available for funding these structures makes this a difficult component to generalize, but we highlight some of the important factors in the most common types of collateral in the market today. We stress that deal-specific factors can vary.



Source: Morgan Stanley

IO structures: In these assets, timing of defaults in the underlying portfolio and longevity of the collateral are the most important considerations. We highlight some of the unique features of an asset that is expected to have a low to zero maturity value.

Potentially money-good assets: Here, collateral is of utmost importance, and the investor should be much more comfortable with the portfolio and/or subordination. Hedging strategies for protection against extreme scenarios should be much cheaper.

Option-value assets: These are somewhere in between the two extremes; since they are still trading at highly distressed levels, overall survival of both the collateral and unfunded tranche should be more likely than for IO structures, though default timing and probabilities are less certain. Coupons, too, can vary in this type of asset.

GOOD COLLATERAL, BAD COLLATERAL

Recall that when a credit-linked note (CLN) is created, the proceeds from the sale of the note are used to set up a special purpose vehicle (SPV), which purchases high quality collateral for funding purposes. The SPV then sells protection on the reference entity (for our purposes, a tranche) in unfunded form to the swap counterparty (dealer or CP). When these structures were originally created, the collateral was generally AAA-rated, with ratings cushion, and priced at par with a Libor-flat-like coupon. However, as the market in general has suffered, so have the specific underlying assets that generally make up the funding component. These encompass a variety of assets, including but not limited to: covered bonds, money market funds (MMFs), credit card ABS, GICs, corporate bonds, sovereign bonds (Treasury, JGBs), agencies, and dealer repos. While we stress deal-specific factors are important, these assets are currently

valued in varying degrees of distress (prices range anywhere from 60-100 cents). We offer our general views on these as follows:

Credit-card ABS. Credit-card ABS are one of the most common types of funding for bespoke structures. The main considerations here are the issuer, composition of the security, and structural features. For instance, since many of these have a certain percentage of sub-prime borrowers, knowing what percentage is the key to evaluating the quality of the overall collateral. We estimate this collateral could be trading from the mid-60s to the mid-80s.

Guaranteed investment contracts. GICs can vary widely in terms of quality and price, depending on a number of factors, including who the issuer is, what thresholds are required to be breached before additional collateral is posted, and what additional collateral is permitted. Typically there is a certain ratings threshold that must be breached before the posting requirement comes into effect, so for example a GIC backed by GE would not currently have anything additional posted, since GE has maintained its AAA-rating. A GIC backed by a monoline, on the other hand, would have already dropped below the ratings threshold. The main consideration here is jump risk – the risk that any issuer would be downgraded faster than they could post the additional collateral required. While these benefit from the additional collateral requirements and triggers, they are considerably more opaque and difficult to price.

Corporate bonds. These are theoretically the easiest and most transparent to evaluate, though they introduce an element of idiosyncratic risk to the structure. They would have had a rating greater than or equal to the tranche rating at inception, so the majority were AAA-rated corporate bonds, such as GECC, Berkshire Hathaway, or a monoline. We would argue that duration is a key consideration.

Covered bonds. Covered bonds are somewhere in between agencies and senior financials in terms of the risk profile. These assets continue to be rated well (AAA/AA) and we are generally constructive on this type of risk due to government support for financial institutions. Nevertheless, there is a fair amount of dispersion in prices (from above par to 75 cents) depending on the duration of the bond and issuer.

ONE ASSET, TWO VALUATIONS

Because collateral default leads to liquidation of the collateral and distribution of proceeds to the swap counterparty and the investor in that order, the first step in the valuation of these structures involves assessing which of the two elements – the tranche or the collateral – will outlast the other. In many cases, such as covered, agency, or sovereign bonds, the comfortable answer is that the collateral is highly likely to make it to tranche maturity. Most assets, however, are less clear and are highly deal specific. If the tranche is the weaker link, the valuation focuses on tail risk

vs. remaining subordination in the tranche. If the collateral is the weaker link, the valuation focuses on average life of the collateral and potential liquidation value of the collateral vs. the MTM on the unfunded tranche.

exhibit 2

Sample Distressed Bespoke Mezzanine Assets

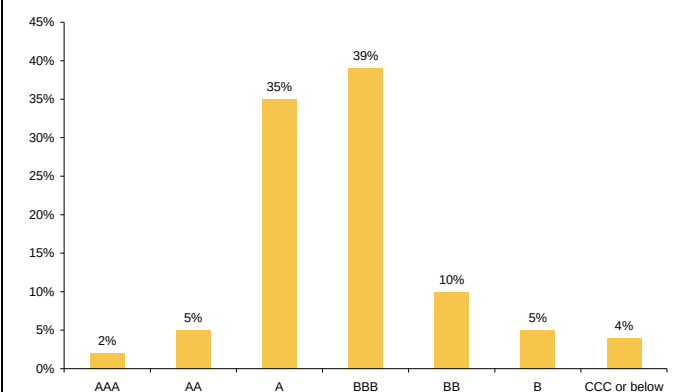
Type	Tranche Type	Cpn	Maturity	Price (%)	Yield to Maturity	Price Coupon Ratio
IO-Value	Mezz	L+250	2014	3.0%	-	0.8
IO-Value	Mezz	L+160	2017	4.6%	61%	2.0
IO-Value	Mezz	L+65	2014	5.4%	70%	3.1
IO-Value	Mezz	L+430	2017	7.5%	77%	1.3
IO-Value	Equity	L+1200	2013	11.0%	-	0.8
IO-Value	Equity	L+1600	2011	25.0%	-	1.4
Option-Value	Mezz	L+150	2012	8.0%	84%	3.1
Option-Value	Mezz	L+210	2013	13.3%	63%	4.2
Option-Value	Mezz	L+100	2013	16.0%	53%	7.7
Money-Good	Senior	L+140	2009	59.0%	66%	23.7
Money-Good	Senior	L+121	2011	61.2%	26%	16.7
Money-Good	Senior	L+121	2010	62.0%	28%	31.9
Money-Good	Mezz	L+30	2010	70.0%	28%	52.4

Source: Morgan Stanley Research

In Exhibit 2, we provide a sampling of distressed bespoke mezzanine tranches in the market today. These vary in portfolio and collateral on a broader level, and across several more specific structural features as well. For instance, there are a few that have an off-market fixed recovery, making them more attractive than they would initially appear, while others have third-party managers, making the portfolio risk harder to determine upfront. From this information alone, it is difficult to assess relative value, but it does highlight the variety and depth of the opportunity.

exhibit 3

Ratings Distribution of 100 Popular Names in CDOs



Source: Morgan Stanley

PORTFOLIOS, SUBORDINATION AND PRICING

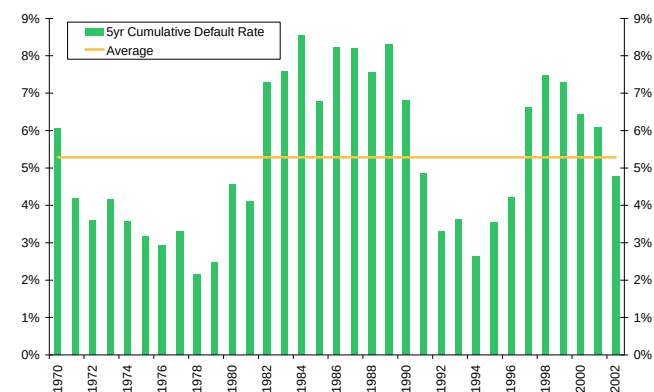
Along with collateral analysis, assessing the portfolio risk is another crucial step in evaluating this opportunity. While

each asset has its own unique underlying portfolio composition, we do get some clarity on the general makeup of these assets by looking at a list published by S&P of the 100 most common names in bespoke mezzanine CDOs. We analyzed the portfolio characteristics of these credits for clues about the expected default rates for these portfolios and hence subordination requirements. Almost a fifth of the portfolio of these 100 common names is now rated below investment grade (Exhibit 3). Furthermore, sector composition is skewed heavily towards financials (including Insurers [26%] and Consumer Discretionary [24%]) followed by Telecom (13%). Many of the non-financials were bull-market-era LBO names. We applied the historical default rate for similar corporate portfolios (in terms of ratings distribution) and found that the worst case 5-year cumulative default rates for a similarly rated portfolio would be in the 8-9% context (Exhibit 4), nearly double that for CDX/iTraxx type portfolios. This puts credit losses in the 5-6% range, if we assume a conservative recovery of 30%.

Clearly the experience of individual deals would vary, but deals with less than five years of residual maturity and subordination in excess of 6% could expect at least some recovery of principal. The timing of subordination erosion also matters. For the deals in Exhibit 2, we calculate risk-adjusted IRRs assuming the majority of the expected defaults in the 5-year period occur in 2009 and 2010. These conservative estimates give IRRs in the 20-50% range.

exhibit 4

Historical Default Rates for a Typical CDO Portfolio



Source: Morgan Stanley

Further, we would divide portfolios into three stratifications: 1) wide-trading names that are difficult to hedge and very binary; 2) wide-trading names where the outcomes are more skewed toward survival; and 3) healthier names that are more default remote. We believe that credits in the first camp should be “written-off” in a worst-case analysis to determine how much subordination is left in the structure and that credits in the second camp are candidates for active hedging in any distressed mezzanine investment strategy. For the third camp, these are credits that should be left unhedged and

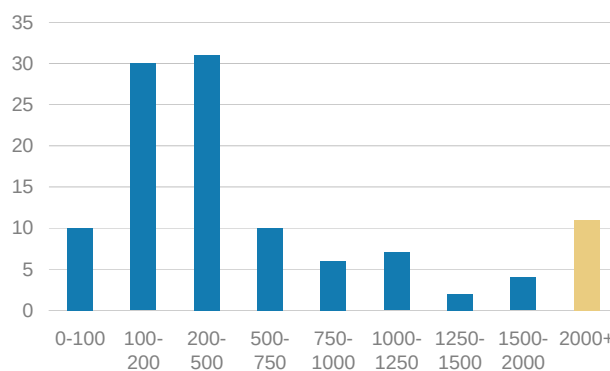
represent the “hope” of the investment strategy. Investors will find that some of the IO-value tranches are timing plays on the wide end of this tail, and the option-value plays are more a survival bet on the tail.

VALUING IO-LIKE TRANCHES

IO-like tranches, to reiterate, are those that derive the majority of their value from the coupon stream, since they are highly unlikely to return any principal. We pick a sample trade from the list in Exhibit 2, priced at 3% and a coupon of L+250bps with the maturity at least 5 years away. The asset is clearly not expected to survive until maturity, and the extreme tail risk in this sample portfolio is quite high. Nonetheless, this asset will break even in a year and half, and thus any additional coupon payments or value at maturity is excess return for the investor. The main focus is to collect the coupon for as long as possible, on the highest notional possible.

exhibit 5

Sample IO-Value Portfolio Spread Distribution



Source: Morgan Stanley

Clearly, given a certain amount of subordination, portfolio composition is a key factor here, followed by timing of defaults. It is already assumed that there will be several defaults in the portfolio, and for most IO-value tranches, it is assumed that the tranche will be wiped out. This is where the fundamental credit skills become paramount – a view on the tails is critical. Since yields here are in the 60%+ range, we only need one out of five deals to survive longer than the breakeven period to achieve mid-teen returns and compare favorably to single-digit returns on IG credit or mid-teen returns for the HY universe.

POTENTIALLY MONEY-GOOD TRANCHES

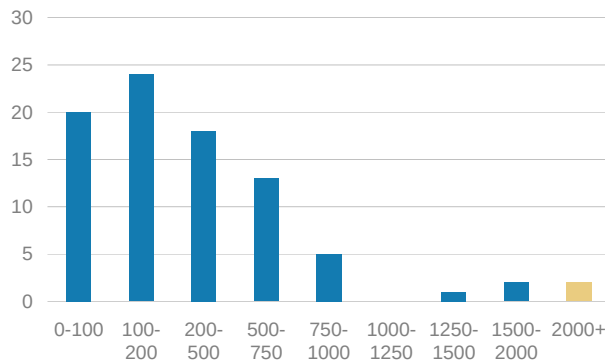
At the other end of the spectrum are higher-dollar-price tranches that are expected to be money good in most reasonable scenarios. These are often thicker tranches with higher subordinations, and cleaner portfolios or shorter maturities. For these assets, coupon and price take a backseat to the ultimate survival of both the tranche and the collateral. This is not to say that price and coupon are unimportant;

clearly they are, but the bulk of the return is expected through redemption at maturity.

In this case, probability of survival is the name of the game rather than timing of default. In Exhibit 5, we show a sample spread distribution for an asset we consider to be distressed, yet money good. Compare this with the IO-value example in Exhibit 3, and the difference in tail risk is readily apparent. Since the primary focus here is asset value at maturity, and given that yields are in the 25%-50% range, we would recommend paying some of the carry to hedge the truly extreme tail scenarios by buying protection on XOver or HVol for example, since we envision a HY collapse more readily than a scenario in which 15% losses are reached on an IG portfolio. Detailed collateral analysis is important here, since the expectation is that the tranche will mature at or near par.

exhibit 6

Sample Money-Good Portfolio Spread Distribution



Source: Morgan Stanley

OPTION-VALUE TRANCHEs

Somewhere in between these two extremes are tranches that we consider to be “option-like” – meaning that they are cheap

enough such that the downside is limited, yet a proper breakeven would require that they mature at a level greater than zero. The primary difference between the IO-value deals and these is that these option-value tranches generally have a better quality portfolio and tail. Alternatively, they may differ from the IO-value assets in that these may have very low coupons, making them more akin to a PO structure. Here, because the price is so low and downside so limited, they are comparable to very out-of-the-money options on tranche survival. Either way, the value at maturity is nowhere near as certain as the money-good assets. This type of asset is as difficult to analyze as it is to classify, a situation which is further compounded by the need for the collateral to survive to maturity as well.

Here we would define the note value as the lesser of the expected value of either the tranche or the liquidation value of the collateral, less potential MTM on the swap. An example of this from Exhibit 2 is the tranche priced at 16 cents, with a subordination of 7.25% and a coupon of L+100bps. In this particular example, the coupon adds some value, though a large part of the value here is the limited downside (16%) vs. the higher subordination. We would expect the portfolio on this asset to have a number of extremely distressed names to price in that range, but again, this is where the option-like value comes into play.

CONCLUSION

As we have previously mentioned, the variety of the distressed bespoke mezzanine opportunity is vast. There is an array of assets at different prices and risk levels, suitable for investors with various risk appetites. The vital thing is being comfortable with the portfolio, structural features, and ultimately the quality and longevity of the collateral.

Restructuring Renaissance

May 8, 2009

Sivan Mahadevan
Phanikiran Naraparaju
Ashley Musfeldt

The speed at which markets and psychology have turned recently should not be ignored, and many in the investment community are willing to look beyond near-term issues to an environment of slow economic growth and fundamental valuation opportunities. Investment grade credit will continue to measure up well by these metrics, and if companies have good business models and access to liquidity, then “large cap” default risk ought to shrink substantially as we move forward. The structured credit mezzanine machine of 2004-2007 was all about this theme, as investors bought levered tranches that could withstand a few defaults and safely return principal. This credit cycle has clearly thrown that investment thesis out the window, even though most of the several hundred billion dollars of mezzanine tranches in the market have yet to realize principal loss. So what should investors in this now-distressed part of the investment grade market do?

With the new CDS SNAC, we thought we’d gotten rid of the “Restructuring” lexicon from our vernacular, but restructuring of a different sort is back in vogue. In the last credit cycle (2001-2002), synthetic CDOs backed by combinations of IG and crossover credits were stressed, and while some investors sold, most ultimately hung on to their risks and restructured in many cases to ride out the storm. A high “fallen angel” default environment was responsible for triggering the stress, but post Sarbanes-Oxley, investment grade default risk fell materially, and a credit recovery ensued.

In this cycle, we have replaced stress with distress in bespoke mezzanine structures, and that difference has forced many to sell, along with other factors including a significantly weaker environment for financial institutions (the large buyers of synthetic CDOs) owing to systemic risk and capital needs. As such, we have been fairly vocal about our expectation of more selling this time around, which combined with valuations is one of the reasons we continue to believe that bespoke mezzanine is among the most interesting distressed opportunities in the market. However, we are also seeing motivation to restructure deals, as many investors turn more constructive on risk markets and feel that adding capital may result in a healthier total return rather than selling today.

We believe the market is big enough to make both opportunities (buying distressed and restructuring existing transactions) feasible and attractive. The opportunity to restructure is perhaps better in the current environment (with spreads and distressed tails still wide) than it would be in a

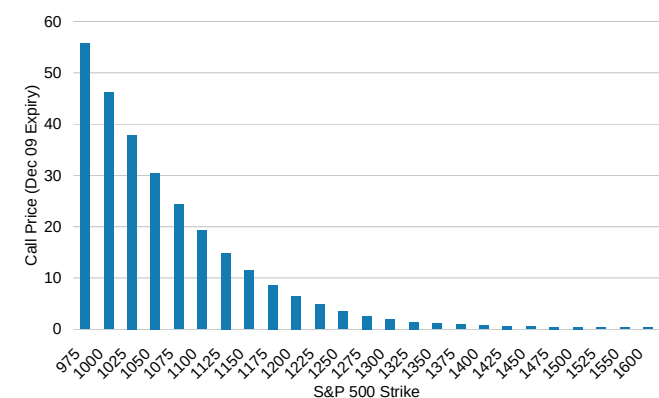
much healthier environment, so timing is important. We devote this chapter to better understanding how restructuring is less expensive than one would guess, and how increasing ratings from B-rated to BB or BBB is both attractive price-wise and valuable from a capital perspective (for banks and insurance companies). We caution that one potential downside of restructuring is that the portfolio could continue to deteriorate meaningfully from a ratings or default perspective, resulting in a loss not only of original principal, but additional restructuring capital as well. Thus, having a constructive view on a large part of the portfolio is important. Furthermore, we highlight that any capital spent on restructuring is effectively realizing a cost upfront that won’t be returned, even if the tranche matures at par.

WHY IS NOW A GOOD TIME TO RESTRUCTURE?

Why is an environment of wide spreads and a still-wide tail of credits a good time to restructure? The answer is that market-implied losses on credit are still high, but an inflection in the cycle now gives us more confidence in stating that actual losses from default may be significantly lower than what is implied.

exhibit 1

S&P 500 – Deep OTM Calls Have Similar Prices



Source: Morgan Stanley, S&P

We use a simple options example to illustrate. The S&P 500 index is at about 900 today, and the market peak was just above 1500. If an investor had purchased an ATM 1500 strike call option near the peak, it would have been very expensive. Today that option is very cheap, but more importantly, an option with a 1200 strike is also very cheap (assuming a Dec-09 expiry). So an investor with a bullish view who does not believe that 1500 is attainable could restructure the 1500 strike call to a 1200 strike for a relatively small cost and own an option that seems more valuable, even though a risk-neutral derivatives pricing

model would not agree. If the S&P 500 were to rally to 1100, the 1200 strike option would be very expensive even though the 1500 strike option would likely still be cheap.

RATINGS, REAL-WORLD LOSSES AND PRICING

We cite pricing from the equity market's most liquid options to illustrate a complex point. If a credit portfolio has a 22% expected loss (implied by CDS curves and spreads), a 5-6% tranche is very OTM from a survivability perspective. So is an 8-9% tranche, but one's real-world view of probability of loss can be meaningfully different for an 8-9% tranche than a 5-6% tranche, and the cost of moving up to it is not that significant today. It will be significant if spreads rally, curves steepen, and tails compress and the expected loss falls from 20% to 10% (which is still wider than when the mezzanine tranche was likely purchased).

exhibit 2

Required Attachment Points for Sample Portfolios

Rating	Maturity: Jun 2014		Maturity: Jun 2016	
	New S&P Model Attach	Old S&P Model Attach	New S&P Model Attach	Old S&P Model Attach
AAA	17.5%	11.9%	20.2%	13.0%
AA+	15.0%	11.5%	17.5%	12.6%
AA	14.0%	10.9%	16.4%	12.1%
AA-	13.0%	10.7%	15.3%	11.8%
A+	12.3%	10.5%	14.6%	11.5%
A	11.8%	10.3%	14.0%	11.3%
A-	10.9%	10.0%	12.9%	10.9%
BBB+	10.3%	9.5%	12.1%	10.5%
BBB	9.8%	9.0%	11.5%	9.9%
BBB-	9.1%	8.2%	10.7%	9.1%
BB+	8.5%	7.9%	10.1%	8.8%
BB	8.1%	7.5%	9.5%	8.3%
BB-	7.5%	7.1%	8.9%	7.9%
B+	7.0%	6.8%	8.4%	7.5%
B	6.6%	6.4%	7.9%	7.1%
B-	6.0%	6.0%	7.3%	6.6%
CCC+	5.5%	5.1%	6.7%	5.7%
CCC	5.0%	4.5%	6.1%	5.2%
CCC-	4.7%	3.7%	5.8%	4.3%

Source: Morgan Stanley, S&P

We also note that the ratings of an 8-9% tranche can be much better than a 5-6% tranche, given that rating agencies consider real-world losses, not risk-neutral. Restructuring adds value due to the disconnect between ratings, capital charges, real-world loss expectations vis-à-vis the market models and pricing.

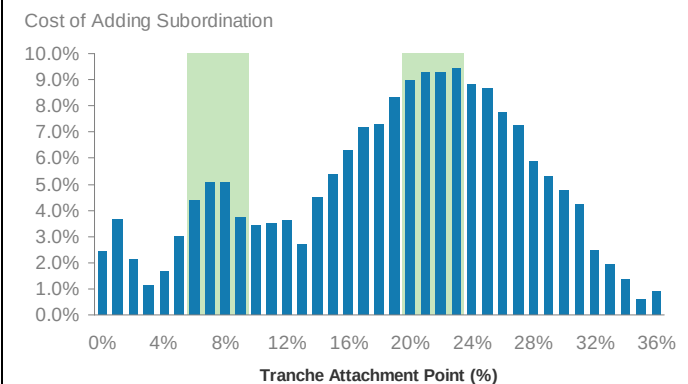
ADDING SUBORDINATION IS NOT AS EXPENSIVE TODAY

At its most basic, adding subordination simply involves taking the PV of the original tranche, subtracting from it the PV of the tranche with higher subordination, and paying the difference. Taking the example of a portfolio with a high expected loss (say 20-25% range), we examine the mechanics of an actual restructuring. Say the investor had a

tranche that was originally 6.5-7.5% and AAA-rated at inception. After several defaults and downgrades on the portfolio, this tranche is now 5-6% and rated CCC- (see Exhibit 3). Clearly this tranche is substantially underwater, with an attachment point of less than a quarter of the expected loss on the portfolio. If the investor has a market view on overall losses, or even a view on the tail names, one solution is to add subordination – not to anywhere near the expected loss, but simply a few percentage points higher, where the investor sees a significantly reduced probability of impairment.

exhibit 3

Cost of Adding Subordination



Source: Morgan Stanley

Note: Change in PV of the tranche by adding 1% of additional subordination, divided by the PV of the original tranche

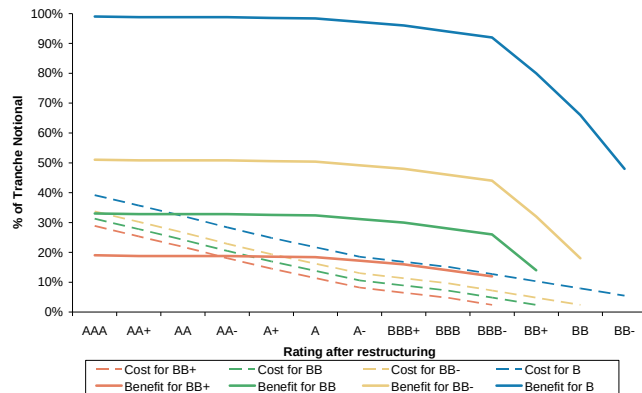
In Exhibit 3, we illustrate that as subordination is added closer to the expected loss (around 20% attachment) the cost gets higher, but that further away from the expected loss (around 7% attachment), the cost is much more manageable. This is shown by calculating the PV difference of the original tranche and the new, higher tranche, as a percentage of the original tranche's PV. One interesting thing to note is that the distribution is considerably less predictable at the very low part of the capital structure – this is due to the lumpy tail distribution that disproportionately affects the equity tranche. In this particular portfolio, approximately 10% of the names are wide enough to be considered tail names, and the effect of these names on the equity tranches is quite significant.

In fact, it is not until we get to around 11%, or half the expected loss, that we start to see the distribution even out meaningfully. In addition to shoring up additional protection from portfolio losses at a relatively low cost, there is an ancillary benefit to additional subordination: improved tranche ratings. Continuing with our previous sample portfolio with a 22% expected loss, the 5-6% tranche is now CCC- by the old S&P model, and using the new criteria, would not even hit the CCC- threshold (see Exhibit 3). If the investor chose to add, say, 3% subordination and make the tranche an 8-9% tranche instead, this would boost the rating

from nonexistent to B in the new model and from CCC- to a BB- in the old model. For some investors, this jump can have real economic implications in the form of reduced capital requirements.

exhibit 4

Cost vs. Capital Benefits of Adding Subordination (Basel II)



Source: Morgan Stanley

WEIGHING BENEFITS VS. COSTS UNDER BASEL II

The benefits of the restructuring are two-fold, one being the reduced riskiness in a real-world sense for the investor (as we discussed earlier) and the second is reduction in actual capital requirement. In our example, we go from a 5-6% tranche, which is below CCC-, to an 8-9% tranche, which is B rated. This is valuable subordination in the investor's eyes, although it does not result in reduction in capital under Basel II (see Exhibit 4). For this additional subordination and ratings jump, the investor would pay about 10% of the tranche notional, so \$1mm for a \$10mm tranche. Adding a bit more subordination (1-2%) to go from B/CCC to a BB-tranche, however, can significantly reduce the capital required to be held against such an asset, up to half the tranche notional (Exhibit 4). This would cost another 5% in tranche notional or \$0.5MM for \$10MM tranche. While these amounts seem large, they are dwarfed by the benefit of capital requirements and improved likelihood of the tranche's survival.

While it does cost a bit more, the capital benefits of improving a BB- rated tranche to a BBB are quite valuable – as much as 44% of the tranche notional in additional capital relief. If the investor were improving a B-rated tranche to an investment grade BBB- (8% attachment to 11% in our scenario), the charge would be roughly equivalent (usually, it would be more except for the lumpy tail in our sample portfolio) – or \$1mm to go from a B to a BBB-. To go from an 8% attachment to a 14% attachment, the charge would be about 16%, or \$1.6mm on a \$10mm portfolio. In all these cases, the benefit outweighs the cost.

SUBORDINATION VS. PORTFOLIO AND MATURITY CHANGES

Aside from adding subordination, another way to improve the PV of a tranche along with its ratings is to eliminate some of the widest or most concerning tail names. In a way, this is similar to the ratings optimization exercises that may have taken place when these transactions were originally structured. The issue today is that depending on the spread of the name to be removed compared to its replacement, the cost is quite a bit higher today, given the extreme levels of some tail names. In our sample portfolio, removing one of the widest names and replacing it with a name on the tighter part of the spread distribution would cost a little under \$150k for a \$10MM notional tranche. Substituting two names would end up, in this very specific instance, costing roughly the same as buying 1% additional subordination, and the benefits of doing one type of restructuring over the other would be portfolio, tranche, and name specific. Furthermore, it is possible that there are some names that could be swapped for a higher-rated name for little cost, though we do see investors moving away from looking at ratings as the sole measure of portfolio riskiness. The important thing to highlight in this round of restructurings is that we no longer see investors relying as heavily on name substitutions as in the past.

Restructuring in most cases involves additional cost, which is usually plugged by adding cash but also by generating value through maturity extension. In today's inverted curve environment, extension is less juicy. On the contrary, investors might explore adding cash to reduce the maturity of the transaction in line with the better sentiment/confidence for near-term default risk generally in this rally.

From Distressed to Just Stressed

January 22, 2010

Sivan Mahadevan
Ashley Musfeldt
Phanikiran Naraparaju

A year ago, we called it “Distress, in an Unlikely Place” (see Chapter 30). The legacy bespoke mezzanine portion of the structured credit market was the victim of a combination of LBOs and financials, thin tranches, losses from actual defaults, and funding collateral concerns. Deals structured with mostly investment grade credit risk had morphed into a \$300 billion distressed opportunity, and few investors had the skills to dig through the wreckage to sort out risk and find value. Structures with coupons ranging from L+30bp to L+500bp and maturities from a few months out to about 8 years were trading at levels as low as 1% to as high as about 70%.

In the year hence, several things happened. More defaults forced additional losses of subordination in some mezzanine tranches with a handful experiencing a total loss. However, at the same time, the most powerful rally in credit market history along with an outperformance of mezzanine-type risk caused some assets to enjoy price jumps of 10 to 60 points, with some shorter-dated deals even maturing at par. A buyer base for distressed CSOs emerged as well, equipped with simple methodologies for identifying value and forming a view on the tail credits.

Today, the distressed opportunity has not disappeared as many sellers in the market remain, but pricing now has forced the opportunity up the valuation curve, to a point at which we might label the opportunity simply “stressed.” While some of the most obvious trades and opportunities were snatched up early on and today there is a healthier level of participation in auctions, we still see interesting value in the market. We continue to categorize the market into the following classifications:

Option-value. With very low prices, thin tranches and little subordination remaining relative to portfolio tail risk, we consider these structures as option-like for their reduced cost and low downside, as well as their somewhat binary maturity value.

IO structures. These also have little subordination left and at least a handful of stressed credits where the likelihood of further erosion is quite high. The key distinction between these structures and those that are more option-like is that more of the value in these assets is derived from the coupon, either because it is high, or because the price is so low that even if the structure is wiped out, the investor should have profited from the coupon payments.

Potentially money-good. These structures have good amounts of subordination for the remaining maturity and portfolios that, while they may still contain very risky names, are not a cause for concern given the healthy subordination. These deals tend to be priced in the 60-90% range, and still have very good returns to maturity, but they also have higher deltas to the market, so investors need to consider market-to-market fluctuations.

We continue to find value in the stressed legacy deals to be attractive on average, but every deal is different, and the opportunity requires more credit work today than it perhaps did a year ago. We expect a broader participation base as more sellers come to the market, motivated by both higher valuations and the turn in the calendar. We provide detail on performance of select deals in 2009, the drivers of this performance, ratings migration and tail risk analysis for those who are looking to value opportunities.

A LOOK BACK AT THE DISTRESSED MEZZANINE MARKET

In early 2009, this opportunity first started gaining traction as more and more traditional bespoke mezzanine investors were forced to liquidate their holdings, for a variety of reasons. Some couldn't hold assets that were rated below a certain threshold, and the rating agency model changes and portfolio deterioration caused their tranches to hit this threshold. Other investors had changes to their investing rules and were subject to risk managers who were growing increasingly wary of products with any amount of structuring in them. Still others were spooked by the markets in general and were fleeing to higher quality assets.

Whatever the reasons, we saw an unprecedented inventory of distressed assets at the beginning of last year presenting the perfect opportunity for anyone willing to sift through them. Initially, a majority of the deals available had very low dollar prices and were easily categorized as either option-value or IO-value. They were primarily static deals and some had fixed recovery, making evaluation more straightforward. Today, the inventory list has changed somewhat in terms of both pricing and structure. Many are now managed deals, and while some highly distressed option-like and IO-value assets remain, a majority of the deals we see today are in the higher-priced potentially money-good category.

exhibit 1

Sample Distressed Bespoke Mezzanine Assets

Type	Subordination	Coupon	Maturity	Jan 2009 Price (%)	Current Price
IO-Value	2.50%	L + 200	2017		13.0%
IO-Value	1.25%	L + 700	2017	13.0%	17.5%
Option-Value	0.75%	L + 120	2013		9.5%
Option-Value	1.25%	L + 75	2013	6.0%	13.0%
Option-Value	2.75%	L + 40	2013		27.0%
Option-Value	2.00%	L + 50	2013		21.5%
Option-Value	3.50%	L + 80	2016		20.0%
Potentially Money-Good	20.00%	L + 100	2014		72.0%
Potentially Money-Good	6.75%	L + 115	2013	16.0%	70.0%
Potentially Money-Good	15.00%	L + 121	2010	33.5%	94.5%
Potentially Money-Good	1.75%	L + 180	2013	19.0%	59.5%
Potentially Money-Good	5.00%	L + 210	2013	14.5%	63.5%
Potentially Money-Good	12.25%	L + 28	2013		91.0%
Potentially Money-Good	2.50%	L + 300	2013	29.0%	87.0%
Potentially Money-Good	9.25%	L + 45	2013		80.0%
Potentially Money-Good	12.00%	L + 50	2013		89.0%
Potentially Money-Good	6.50%	L + 500	2012		99.3%

Source: Morgan Stanley Research

A TAIL OF THREE ASSETS

In examining the changes to the inventory of bespoke mezzanine assets available today, it is clear that there are far fewer very low dollar price instruments. There are several reasons for this shift in inventory. First and foremost, the credit rally in 2009 was enormous and, as the general credit markets healed, so did the bespoke structured markets. Second, the very low dollar prices on some option-value and IO-value trades were the “low-hanging fruit” trades for many investors looking for assets with reduced downside and high upside, thus those were the first assets to go. Additionally the risk re-pricing we saw in the tranching markets this year, with mezzanine-type risk outperforming at the expense of much of the rest of the capital structure, could have easily been worth a few points on some of these assets. Finally, some technical aspects to these structures may have played a role as well, including time decay and rolldown, which would have impacted the more short-dated assets, as well as an overall drop in portfolio dispersion, which would have affected those tranches with lower subordination.

To get a better sense of the performance drivers over the course of the last year, we took a closer look at three assets for which we have pricing in both January 2009 and today, and see that they had highly divergent experiences. Asset A was priced in the mid-teens just a year ago and today is offered in the low 70s. Asset B also increased in value, though we would still consider it to be option-like, as it went from single-digit pricing to mid-teens. Finally, Asset C was always an option-value- / IO-value-type tranche that went from a low-single-digit price to defaulting completely after 5 additional defaults in the underlying portfolio.

exhibit 2

Portfolios Behind 3 Deals, With 3 Different Outcomes



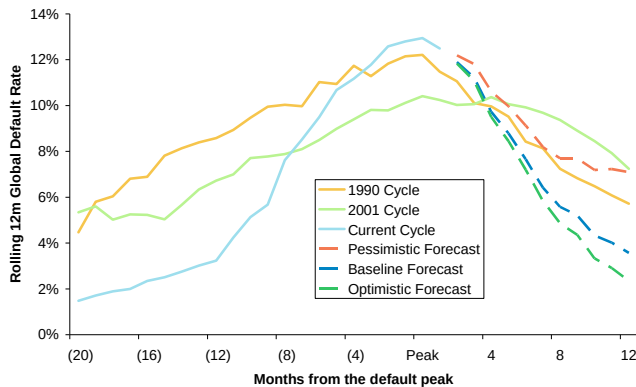
Source: Morgan Stanley Research

We begin by looking at the portfolio composition and see that the market rally, as well as a drop in dispersion, was significant (Exhibit 2). The first graph is Asset A, which went from being option-like to money-good, and the reduction in tail risk is pretty significant – from over 20%+ of the portfolio trading wider than 1000bp to only 10% trading wider than 800bps, and only one additional default. The second CLN had less tail risk to begin with, and though it was reduced, it was not reduced by as significant an amount. The final graph shows Asset C, and while there was a substantial reduction in tail risk and increase in the number of names that tightened to less than 100bps, the 5 additional defaults were the main (and really, only) driver of performance here.

Clearly, remaining subordination played an important role, as the higher tranches were able to withstand the high-loss environment. However, portfolio selection played a large role here as well, which brings us to another point – the default rate in 2009 was high and some deals suffered significant erosion in subordination. One-year default rates hit a peak late in 2009, although they are expected to decline in 2010.

exhibit 3

1yr Default Rates Hit a Peak in 2009



Source: Morgan Stanley Research, Moody's

UNDERLYING DEFAULT AND SPREAD TRENDS

While 2009 saw a number of low-recovery defaults early in the year, many of these had been sub-investment grade for a considerable period of time and thus unlikely to be popular in bespokes. Exhibit 4 highlights a list of credit events that settled via the ISDA auction protocol in 2009 and of these, only three were on the list of the 100 most popular names in synthetic CDOs (based on S&P's report from December 2008). Recovery for these three was 53% – above market average for the year. In general, we see a trend of rising recoveries for two reasons: (1) the credits defaulting later in the cycle are arguably more resilient, having lasted this far into the crisis, and (2) we have had a secular improvement in asset prices over the course of the year. The bottom line is that the default and recovery environment has turned out to be less onerous than was originally feared when the crisis began, and now seems to be improving into 2010.

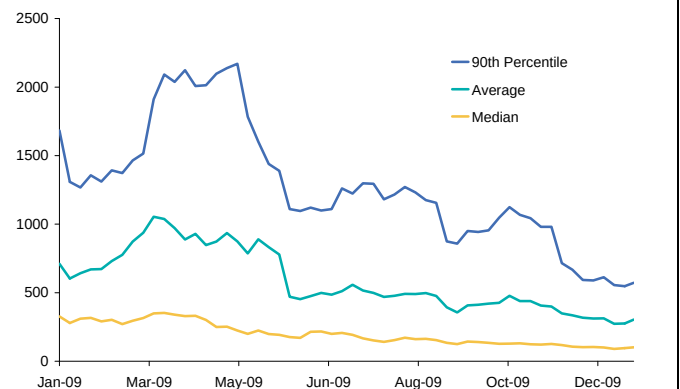
Nevertheless, tail risk is what makes the legacy mezzanine bespoke market. Looking at current spreads on the top 100 corporate obligors based on roughly 1000 US synthetic CDOs, we see that while the average spread for this basket of 100 names has tightened in line with the broader market, it is worth noting that there are still 8 names trading wider than 1,000bp and 11 wider than 500bp. This would suggest that many bespoke deals still contain at least some tail risk relative to the market, mostly in the form of financials and LBOs.

The point that value lies in the tail is also evident when looking at the performance of the 100 most popular names

over the course of 2009. The tail contributes such a large percentage of the index spread that the median spread is considerably lower than the average spread of the constituents. While tail credits tended to outperform in 2009, we still see further room to go and expect this theme of tail convergence to the rest of the portfolio to continue in 2010.

exhibit 4

Portfolio Spread Performance: Value Lies in the Tail



Source: Morgan Stanley Research

The distressed mezzanine opportunity will continue to be a levered play on spread compression and healing in this part of the portfolio, as we think these tail names are what give the legacy market its character and are the primary source of the stressed valuations. With above-market spreads in an improving default and recovery environment, these credits will likely be the source of value in the stressed legacy mezzanine deals of today. The focus of the new issue market (as and when it develops) is likely to be on improving portfolio quality from a sector and tail perspective, giving investors a different type of opportunity than those presented in the legacy market.

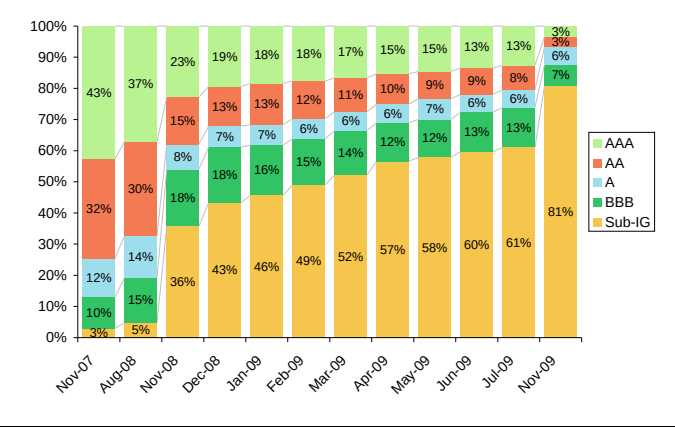
RATINGS AND VALUATIONS CATALYST FOR SALES

While valuation for the current distressed mezzanine opportunity is largely a function of underlying constituents, remaining subordination and time left to maturity, we expect ratings changes to continue to influence this market as well. Following the implementation of the changes in ratings methodology and the effect of default rates through 2009, a whopping 80% of the legacy market (consisting of 1500+ CDOs and 330+ synthetic CDOs) is now rated sub-investment grade compared to just 43% at the start of 2009. Within a matter of 18 months, the ratings profile of this market has been turned completely upside down (see Exhibit 5). For ratings-based investors, this ratings deterioration theme will be central when deciding whether to restructure the transaction or exit the asset class altogether. If it is the latter that happens, the market will continue to have inventory from which to choose. Poor valuations were an impediment to selling these assets earlier in 2009, but today the market has a more robust list of participants waiting for

auctions to provide a reasonable exit strategy for those wanting to sell.

exhibit 5

Distribution of Ratings for CDOs



Source: Morgan Stanley Research, S&P

We still think this asset class has yet to complete the transition from a ratings-focused investor base into one that is more distressed and high yield focused. While we still like synthetic mezzanine tranches as a viable investment platform for investors who use ratings, we think the needs of that investor base are probably better met from a new issue market that takes into account the radically altered ratings and valuation landscape.

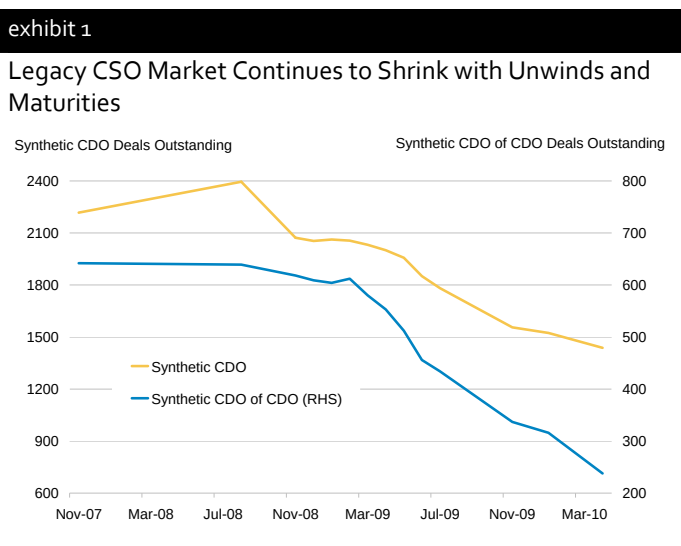
Distressed CSOs Decouple and Stabilize

June 9, 2010

Sivan Mahadevan
Ashley Musfeldt
Phanikiran Naraparaju

Despite significantly higher levels of credit volatility and negative performance of investment grade in general this year, the legacy market for bespoke mezzanine tranches (referencing investment grade credit portfolios and totaling roughly \$300 billion notional at the market peak) has witnessed both increased trading activity and somewhat higher price performance. Demand for such structures remains strong in the market as both buyers and sellers have their motivations, and while the core investment grade market remains weaker this year, the universe of more distressed investment grade and fallen angel credits that drives valuation of legacy CSOs has actually experienced more positive performance YTD.

While the increase in distressed CSO trading activity could be somewhat counterintuitive in an environment of poorer liquidity, it makes sense when we look at the development of these flows. The first wave of distressed CSO selling came in late 2008/early 2009, as downgrades and market fears in general and highly structured products specifically were dominant forces. When pricing on these distressed assets began to climb with overall market healing in early to mid-2009, some potential distressed sellers held back. Over this full period, a buyer base with the appropriate level of structural and credit skills has continued to develop to absorb supply.



Source: Morgan Stanley Research, S&P

Outright unwinding of legacy CSOs has continued and even gained momentum in 2010, and indeed less than 40% of original CDO²s remain outstanding as of April 2010, and

only two-thirds of the CSOs issued are still outstanding today (owing both to unwinds and maturities of many of the original 5-year deals). Admittedly, this still leaves a fair amount of CSO risk outstanding (about \$170 billion notional). From a seller's perspective, rising valuation is a key catalyst for pulling the trigger on sales, even though the pressure to sell from a regulatory capital perspective has been around for a considerable period. The transition of the product from having a predominantly investment grade ratings profile to one with 80% or more sub-investment grade and associated higher capital requirement weighs heavily on this still ratings-sensitive investor base.

The ongoing motivation for buyers continues to be a unique opportunity to take a levered long view of the tails of the CSO portfolios (which tend to be the previous cycle's distressed financials and LBOs) as well as absolute valuation, which can be quite compelling compared to standard index tranches and the broader high yield market. In the remainder of this chapter, we focus on following themes:

CSO valuations decouple and diverge: This is largely due to the stressed fallen angel names that drive mezzanine tranche risk having had mixed to positive performance of late, compared to a weaker broader credit market. This has resulted in a CSO universe with positive YTD performance as a whole, despite negative credit market performance. We also see stronger relative performance in CSOs referencing US credits vs. European credits, owing to healing tail risks in the US and an increase in some European tail risks.

Credits stabilize: These same credits have stabilized from a credit-migration perspective (based on Moody's analyses), and Moody's metrics suggest that they have the potential to improve CSO portfolios from a credit perspective as well.

There remains a valuation premium for complexity: We see typical distressed US CSOs with 2 to 3 years remaining until maturity, at prices in the 60% range, offering no-default IRRs in the high teens. Comparable index tranches or high yield assets are yielding in the low teens and high single digits respectively. Distressed CSOs with European-focused portfolios tend to trade at higher prices (high 70s).

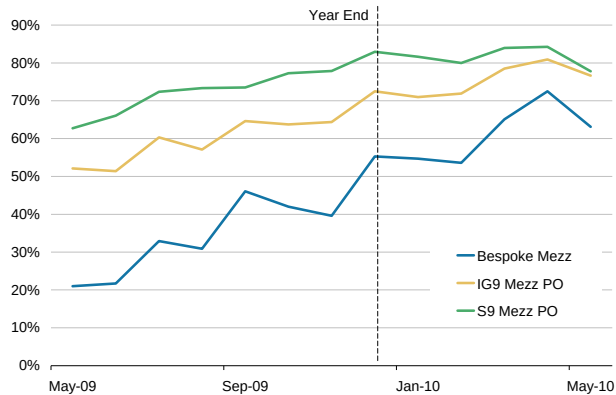
DIVERGING VALUATIONS TRENDS

Despite the market volatility and even the underperformance of junior mezzanine tranches recently, distressed mezzanine deals have had a mixed performance. We are seeing considerable divergence between US and European deals in line with the underlying spread performance. Short-dated US CSOs with distressed financials and HY exposure have done quite well YTD as short-end spreads continue to normalize. For example, some of the 2.5-3.5yr US CSOs are up as much

as 10-15% YTD (see Exhibit 2), which is better than the equivalent IG9 5-year mezzanine PO, which is up just under 6%. In contrast, some European CSOs are flat or down 10% YTD, especially the slightly longer-dated ones. This is reflected in the performance of the iTraxx S9 5-year mezzanine PO, which would be down about 5% YTD. Much of this outperformance by US CSOs is due to the meaningful decline in spreads for the insurers that essentially make up the tail for several distressed mezzanine deals (see Exhibit 3). Although performance of HY generally has been mixed, financials, including iStar Financial, PMI Group, Radian Group and MGIC, have performed quite well during this period.

exhibit 2

Price Evolution of Index Mezzanine and an Illustrative Bespoke Mezzanine Deal (\$ prices)



Notes: Bespoke Mezz – illustrative short-dated deal with US distressed financials and HY exposure.

Source: Morgan Stanley Research

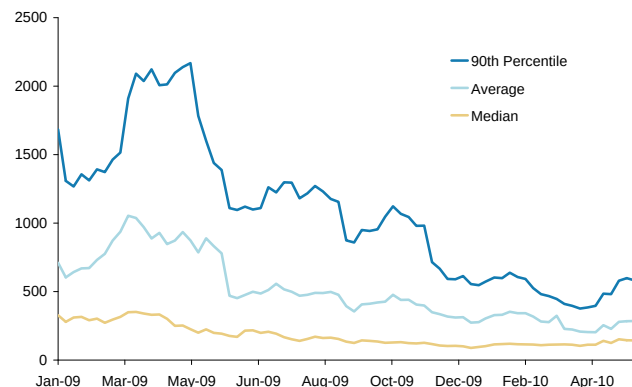
Moody's recently launched a new index tracking the 100 most commonly referenced corporate entities in CSOs that Moody's rates. The MCSO 100 Index (as it is called) has seen generally stabilized credit quality of late, after severely negative credit migration that began in early 2008. The credits in the index, which tend to be concentrated in the Financials, Telecom and Retail sectors, have also seen an improvement in Moody's measure of market-implied ratings. These market-implied ratings, which are directly related to CDS levels, indicate signs of stabilization and potential improvement in these CSO portfolios, according to Moody's.

PORTFOLIO TAIL DRIVING VALUATIONS AND PERFORMANCE

As we highlighted in Chapter 33, the value in distressed mezzanine lies in the tail and that has played out in performance in 2010. We think the theme remains intact.

exhibit 3

Portfolio Spread Performance: Value Lies in the Tail



Source: Morgan Stanley Research, S&P.

Looking at current spreads on the top 100 corporate obligors based on roughly 1,000 US CSOs, we see that while the average spread for this basket of 100 names has tightened in line with the broader market, there are still 5 names trading wider than 1,000bp and 14 wider than 500bp (see Exhibit 3). This would suggest that many bespoke deals still contain at least some tail risk relative to the market, mostly in the form of financials and LBOs.

LEGACY CSO SELLER MOTIVATION

Legacy CSO investors have had reasons to sell (ratings, regulatory capital requirements, etc.) for some time now. The constraint was low valuations and the ability of the market to absorb the supply. On both these counts, investors looking to exit long-held positions currently have a reasonably favorable environment in our view, despite wider market spreads. We already discussed the valuation picture, which we think is still reasonable for sellers. In terms of liquidity, the demand side of the market has developed rather well, with a large number of distressed investors set up to participate in potential sales and absorb the supply. The relative transparency of the underlying credits (typically the leading IG and HY credits) has helped grow the distressed investor base. As a result of the above factors, we are seeing considerable notional of risk changing hands even during what has been a volatile period for credit.

LEGACY CSO BUYER MOTIVATION

From a buyer's perspective, many distressed CSO opportunities still compare favorably with some of the alternatives in both the structured credit and high yield markets (see Exhibit 4). We estimate prices in the \$60-\$80 range for many 2- to 3-year CSO opportunities today, which can result in yields from high single digits to nearly 20%. By comparison, the US HY CCC-rated bond universe offers yields of about 12.3% with an average price of \$86.8 today. And in the structured credit world, PO forms of index mezzanine tranches (with 3-year maturities) have yields in

the 10% range. The distressed mezzanine opportunity should continue to be a levered play on spread compression and healing in the portfolio tails, as we think these tail names are what give the legacy market its character and are the primary source of the stressed valuations. Thus an improving default and recovery environment, in conjunction with still wide differentials between the tails and the averages, supports this opportunity and the still very engaged buyer base.

exhibit 4

Distressed Mezzanine CSOs vs. Alternatives

	Maturity (yrs)	Price	Yield (%)
US HY CCC	4.5	86.8	12.3
US HY B	5.7	96.5	9.3
US HY BB	5.5	98.7	7.6
IG9 5yr Mezz PO	2.5	76.2	11.5
S9 5yr Mezz PO	3.0	77.0	9.1
Distressed Mezzanine	2.8	61.0	19.3

Note: Mezzanine pricing is illustrative only

Source: Morgan Stanley Research, Yield Book

CONCLUSION

We continue to see the potential for flows and capital gain opportunities in the legacy market for CSOs. The key opportunities now are in deals that are more “potentially money-good” in nature, but these deals tend to be priced in the \$60 range and higher with recent spread compression. As such, we remind investors that the credit market beta of such deals may in fact be higher going forward than in the recent past.

The Market for Customized Credit Solutions

January 18, 2012

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While much of the recent activity and liquidity within the structured credit markets has been focused on index-based instruments, it is important not to ignore the market for more customized credit risk, which began 15 years ago with the market's first synthetic balance sheet transaction. In fact, much of the initial flow in the CDS market was in the form of risk transfer from bank lending books (where hedging solutions were in demand) to end investors looking for alternative forms of long credit investment. In the early days, customization was one of the main motivations for investors considering structured credit solutions for their credit portfolios, whether for hedging or risk taking.

Even when the index tranche market began in 2003-2004, much risk taking continued in customized form, with the index tranche market providing important liquidity, transparency, price benchmarking and hedging opportunities. In fact, "bespokes" became a synonym for new risk within the synthetic structured credit world. However, both during and following the financial crisis, the balance of market activity swung in favor of index-based solutions given investor demand for standardization over customization in a very challenging liquidity environment.

Today, the liquidity challenges in the market certainly remain, and there are other challenges to bespokes in a Dodd-Frank world as well, including efficiency from a trading perspective (relative to trades cleared through central clearing houses) and a capital perspective (which is still a work in progress for banking institutions and insurance companies).

Nevertheless, we see ongoing market activity in customized structured credit in two forms. First, the secondary market for legacy instruments (mostly those issued from 2005 to 2007 in managed or broadly distributed form) persists, given a risk transfer from many of the original bank buyers to other end investors today. We have done quite a bit of work on this risk-transfer theme since early 2009, and today we find many structures have healed in pricing terms and have either been unwound or have matured (with some experiencing write downs) while those outstanding tend to track the market movements of a common set of tail names including distressed financials and yesteryear's LBOs.

Second, and perhaps less known, is new risk activity. This new risk market looks perhaps more like it did over a decade ago than just prior to the crisis, as investors seek specific

opportunities rather than broadly marketed or syndicated transactions. Banks are considering first-loss hedges and super senior risk overlays as a way of reducing idiosyncratic risk funded with increased market risk. Other investors are looking for low dollar price long convexity plays, and implementing these in equity and mezzanine tranches. Still others are considering less risky portfolio replication trades, where investors seek safer senior and thicker tranches as alternatives to owning delta-one credit.

In this chapter, we provide an update on the state of the legacy bespoke market using a valuation approach we first utilized when opportunities were the most distressed, and we provide an overview of some of the new risk in the market. Key themes are as follows:

Healing in legacy bespokes: Through a credit rally, a reduction in tail risk, maturities and various unwinds, the amount of credit risk held in legacy bespoke structures has fallen quite substantially. The reduction has been greatest in structures with European exposures and static deals as they are the most likely to have matured or been unwound.

Watch the three Ts: Nevertheless, secondary market opportunities remain in the legacy market, with a wide range of prices depending on the term, tranching, and portfolio tail risk.

New customized solutions – what are the trends? First-loss hedging and systemic risk overlays are still the norm in bank lending portfolios, while many investors look for credit convexity in long equity tranche structures and yield generation through senior tranche portfolio replication strategies.

New customized solutions – what do we like? We favor short-dated equity tranches that mix investment grade names with some healthy high yield credits as call options on credit quality. We also like senior thick bespoke tranches that trade at multiples of the market spread today (compared to fractions of market spread prior to the financial crisis).

STATE OF THE LEGACY MARKET

We have written much on the legacy market for bespokes since the financial crisis, covering the distressed opportunity in 2009 through a significant healing of the market by 2010 (see Chapters 30-34 in this handbook). There continues to be a significant amount of de-risking in the market, both in the form of maturing deals and those being unwound. We see the number of total synthetic structures falling from a peak of over 3,000 in 2008 to under 1,000 today. The bulk of the decline has been in static transactions (fixed portfolios as

opposed to those that have been managed) largely because they are simpler to unwind and were older, and thus more likely to have simply matured. From a geographic perspective, structures with European exposures have been more commonly unwound or have matured, partly because of the European bank deleveraging theme. European structures made up 55% of the total exposure at the peak of the market, but account for only 42% today.

exhibit 1

Legacy Rated Synthetic CDO Deal Summary

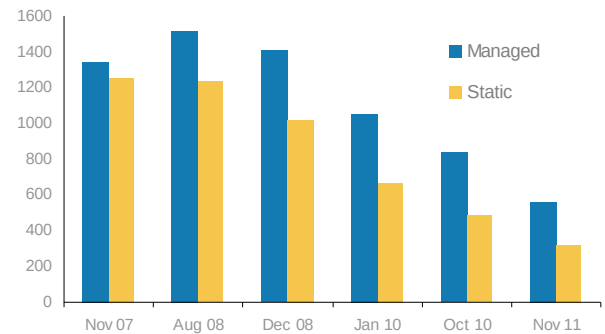
	Nov-07	Aug-08	Nov-08	Jan-10	Oct-10	Current
Deal Type						
Synthetic CDO of CDO	642	639	609	316	152	73
Synthetic CDO	2217	2,395	2054	1524	1285	859
Total	2,859	3,034	2663	1840	1437	932
Management Type						
Managed	47.1%	49.9%	52.7%	57.1%	58.3%	60.5%
Static	43.8%	40.7%	38.1%	36.3%	33.5%	34.1%
Hybrid (lightly managed)	9.2%	9.4%	9.2%	6.6%	8.2%	5.5%
Original Rating Category						
AAA	44.8%	44.9%	45.2%	45.9%	46.0%	47.5%
AA	34.1%	33.2%	34.5%	34.5%	33.4%	34.6%
A	11.7%	11.9%	11.1%	10.5%	9.7%	8.2%
BBB	7.1%	7.6%	7.0%	6.6%	7.4%	7.3%
Sub-Investment Grade	2.2%	2.4%	2.2%	2.6%	3.5%	2.9%
Current Rating Category						
AAA	42.6%	37.1%	19.4%	3.0%	4.0%	3.4%
AA	32.0%	30.1%	12.7%	3.1%	3.9%	3.4%
A	12.5%	13.5%	6.7%	5.9%	5.4%	5.5%
BBB	10.0%	14.5%	17.8%	6.6%	6.5%	5.5%
Sub-Investment Grade	2.9%	4.8%	43.3%	81.5%	80.2%	82.2%
Geography						
Europe	52.7%	51.6%	55.2%	49.9%	49.2%	42.4%
Asia	11.9%	11.5%	11.7%	10.1%	8.3%	6.5%
North America	35.4%	36.9%	33.1%	40.0%	42.6%	51.0%

Source: S&P, Morgan Stanley Research

Even though most synthetic CDOs outstanding today are rated below investment grade (see Exhibit 3), we believe looking at tail names should provide some insight into actual default risk in some of these deals. S&P published a report in September 2011 detailing the 100 most widely referenced obligors in US synthetic CDOs. Exhibit 4 shows the 8 widest trading names from that universe. While this is clearly a “market-wide” number that may not apply to most deals on a stand-alone basis, it is interesting that the tail appears less or equally stressed compared to the on-the-run CDX HY index and even the legacy CDX IG9 index.

exhibit 2

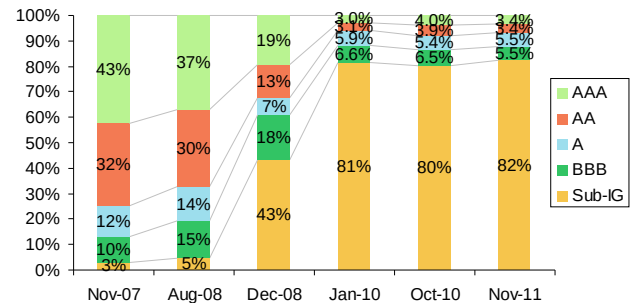
Legacy Market – Proportion of Static Deals Decreasing



Source: Morgan Stanley Research, S&P

exhibit 3

Legacy Synthetic CDO Ratings Migration



Source: Morgan Stanley Research, S&P

This suggests to us that while many outstanding deals are rated below investment grade, is it arguably attributable to the thin tranching and low subordination of many of these deals, rather than a real distressed portfolio tail (the new ratings methodology is more punitive towards thin tranches). Given our constructive view on default risk in 2012, these structures could be interesting longs.

While most of the legacy market is no longer distressed, significant risk-transfer opportunities in the structured credit markets remain, driven by, among other things, European bank deleveraging. We see opportunities in three classes of valuation – IO Value, Option Value and Potentially Money Good – and summarize structure and pricing below.

IO Value. The most distressed structures are those that we classify as IO Value, where there is little if any subordination remaining, thin tranching, a portfolio with stressed names, and a high likelihood of further losses. Deals like this tend to price in the single digits to low teens.

exhibit 4

Most Widely Referenced Names in Legacy Portfolios –
(Spreads wider than 700bp)

Entity	5Y CDS Spread
Radian Group Inc.	2,548
MGIC Investment Corporation	1,612
MBIA Insurance Corporation	1,238
iStar Financial Inc.	889
Assured Guaranty Municipal Corp.	866
Sprint Nextel Corporation	856
MGM Resorts International	747

Source: Morgan Stanley, S&P

Option Value. We classify deals with thinner tranches, stressed tail names yet higher levels of subordination as more “option-like” in nature. These previously would have priced at very low dollar prices as well, but with healing in the credit markets, many now have prices in the 30% context. They can still be somewhat binary in their outcome, but we see them as less risky than the IO-value trades.

Potentially Money Good. Finally, the healthier deals in the legacy market might be some of the best buy-and-hold values, as they contain significant subordination or less tail risk in the portfolios. Many of these opportunities have less time to maturity (two years and under). Prices for such structures can range from 40% to 90%, depending on term, tranche thickness, and the tail names.

exhibit 5

Sample Legacy Bespoke Opportunities

Type	Subordination	Coupon	Price (%)	Maturity (yrs)
IO - Value	0.0%	L + 50	<10	1.9
IO - Value	0.0%	6.65%	Low Teens	5.1
Option-Value	2.3%	3mL + 220	Mid 30s	5.7
Potentially Money Good	8.5%	3mE + 80	Mid 90s	1.2
Potentially Money Good	4.2%	L + 210	Low 80s	1.2
Potentially Money Good	2.4%	3mL + 30	Mid 40s	1.9
Potentially Money Good	4.2%	3mL + 120	High 50s	4.9

Source: Morgan Stanley, 3mL = 3 month Libor, 3mE = 3 month Euribor

NEW CUSTOMIZED SOLUTIONS – WHAT ARE THE TRENDS?

We see three relatively strong trends in the market with respect to demand for customized credit solutions, two that are primarily long in nature, and a third that is more hedging focused.

First loss protection. The natural regulatory and economical hedge for many bank lending portfolios is to buy first loss protection, and this trade must be done in bespoke form almost by definition to address the specific exposures in a given bank lending book. While the market has seen some

flows here, we anticipate additional activity as banks adjust their exposures and we get more clarity on Dodd-Frank capital rules and their inter-operability with the Basel framework. One consequence of this is that we may see more lending books use senior tranches (and in many cases, super senior tranches) as an overlay to CDS hedges to effectively reduce idiosyncratic exposure and add back in market exposure.

First loss longs. Many investors are looking for convex opportunities in credit given relatively strong corporate credit fundamentals despite a market backdrop of elevated volatility, reduced credit liquidity, and regulatory and European sovereign uncertainty. One way to implement this long credit view in a way that limits downside in spread widening scenarios is by taking first loss exposure on a bespoke portfolio. While we’ve seen activity in the index markets as well, for those investors who emphasize the importance of credit selection and maximizing portfolio overlap, a bespoke portfolio may be a better fit. Additionally, some investors may have plenty of exposure to the narrower list of index constituents through various products, and a bespoke structure can help diversify away from this.

Portfolio replication. These trades tend to be thicker and have significant subordination, yet some pick-up over the average spread of the underlying portfolio. In this way, these mezzanine structures are more similar to early structured credit flows from a decade ago than to the 2005-2007 period, when structures were highly optimized, ratings driven (as opposed to being investor driven), and thinner. In the 2005-2007 period, tranches were very thin (1%), low attaching (3% to 7% range) and longer maturity (7- to 10-year risk), with spreads in the 150 to 200 bp range. Today, these portfolio replication trades are thicker (5%-10%), higher attaching (7% to 20%) and yet retain attractive spread levels.

The motivation here is simply diversification from the corporate bond market for investors who have a continued appetite for credit risk, and perhaps less appetite for direct interest rate risk. The natural sellers of protection are insurance companies, in our view at least.

NEW CUSTOMIZED SOLUTIONS – WHAT DO WE LIKE?

So among these new risk opportunities, what structures do we prefer and where do we see value? While it is more difficult to have a specific call when the portfolio and structure is so customized, we do have certain preferences depending on investor motivation, and most of our preferred trades here mirror our preferences in the more liquid index tranche markets.

First loss POs – call options on credit quality. In a risk-neutral framework, credit spreads compensate investors purely for future losses due to defaults. Today, credit spreads are priced such that they compensate investors well in excess of worst case historical losses, while corporate fundamentals

remain very positive. In this kind of environment, instruments that isolate default risk from spread volatility premium and systemic risk appeal to us. Equity tranche risk in corporate credit index tranches is one way to position for a healthy default view, and our preferred trade is long-dated equity tranche POs in legacy CDX IG and iTraxx Main indices, given low dollar prices that put an “option-like” floor on trade losses. The key risk to this trade is an increase in defaults and/ or lower than anticipated recoveries.

exhibit 6

PO IRRs versus Number of Defaults

				IRR (by portfolio defaults)						
Index		Mty	Price	0	1	2	3	4	5	6
Legacy Equity POs										
IG9 7y	0-3	Dec-14	31	46%	35%	21%	4%	-22%	FL	FL
IG9 7y	0-7	Dec-14	59	20%	16%	13%	9%	6%	1%	-20%
IG9 10y	0-3	Dec-17	17	35%	30%	23%	14%	-1%	FL	FL
IG9 10y	0-7	Dec-17	37	18%	17%	15%	13%	11%	9%	-3%
S9 7y	0-3	Jun-15	29	45%	37%	29%	19%	6%	-13%	-64%
S9 7y	0-6	Jun-15	45	27%	23%	20%	16%	13%	8%	4%
S9 10y	0-3	Jun-18	16	36%	32%	28%	22%	15%	4%	-35%
S9 10y	0-6	Jun-18	25	24%	23%	21%	19%	17%	14%	12%

Source: Morgan Stanley Research; Note: Recovery Assumption = 40%

For this type of risk to price at reasonably good risk/reward and convexity levels, a large portfolio tail is necessary. In the index market, the IG9 series has names that may be too binary for some investors, and as an alternative, we think bespoke that mix in healthy high yield names (where we believe near-term default risk is low given the debt maturity profile) are good alternatives to index equity tranche POs. While pricing can vary, we see opportunities for 3-year risk (and shorter) offering no-default IRRs in the 30% to 40% range.

exhibit 7

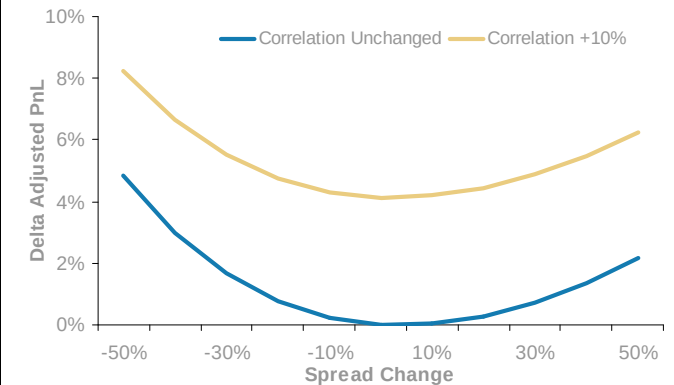
Correlation – US Has Repriced Lower



Source: Morgan Stanley Research

exhibit 8

Equity Tranches Benefit From Convexity



Source: Morgan Stanley Research

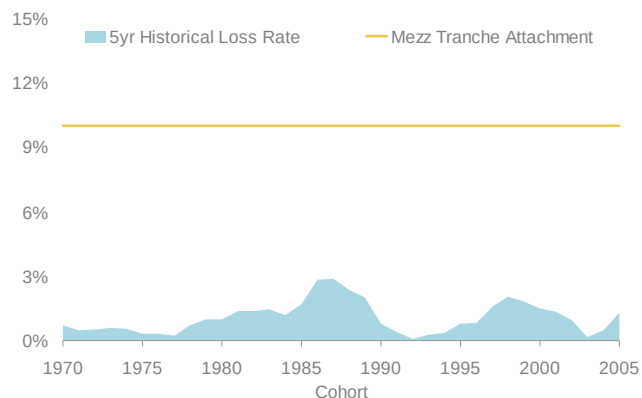
Credit replication – from fractions to multiples. We like the defensive nature of senior tranches in this market environment, given their lower overall exposures to the market (i.e., relatively low deltas) and significant subordination, which can easily protect investors through a full credit cycle. Pre-crisis, these tranches offered spreads that were a fraction of the index level, while today, given wider spreads overall and a significant repricing of the tranche capital structure, now trade at multiples of the index spread.

Furthermore, tranches in newer CDX IG indices are thicker, which also mitigates some of the negative convexity that mezzanine tranches of yesteryear experienced. Additionally, we believe these types of tranches fit in nicely within the replication framework, which we recommend for insurance companies, where CDS (or other synthetic risk) acts as an overlay to a fixed rate Treasury portfolio as a source of additional yield.

For a longer-term-focused investor, the benefit of a bespoke over the index would be credit selection and the ability to target a certain risk characteristic (like a rating) that can be beneficial from a regulatory capital perspective, and we find many opportunities of 5-year risk in market with 10%+ attachment points on investment grade portfolios that would yield in excess of 300 bp. The key risk to this trade is a sharp widening in credit spreads, in which case the risk allocated to this tranche across the capital structure increases significantly.

exhibit 9

Senior Tranche Subordinations at Multiples of Historical Losses



Source: Morgan Stanley Research, Moody's

CONCERNS WITH CUSTOMIZED SOLUTIONS

As with any decision to purchase an off-the-rack product versus one customized to suit certain needs, there is a

tradeoff in secondary market liquidity and other factors. In the case of bespokes, these other factors can include clearing (in the future) and perhaps capital treatment (in the future for financial institutions).

With Dodd-Frank implementation, at some point potentially as early as this year, certain standardized CDS transactions will have to be centrally cleared, where margining and collateral will be determined by a central counterparty instead of a specific bank. Bespokes in derivative form would most likely not be centrally cleared, which means that margining would be non-standard (as it is now), but the capital treatment for bank counterparties to bespoke trades would likely be more onerous than for cleared trades. This can raise the cost of such transactions and potentially lower the liquidity as well relative to cleared transactions.

Also for end investors who are banks, the capital treatment of bespokes and more broadly securitized credit is still uncertain under Dodd-Frank. As of the writing of this chapter, this is a Notice of Proposed Rulemaking (NPR) outstanding for risk-based capital guidelines for corporate credit and securitized credit that will address bespokes as well.

Section F

Munis and Sovereigns

Municipal CDS – The Next Stage

March 7, 2012

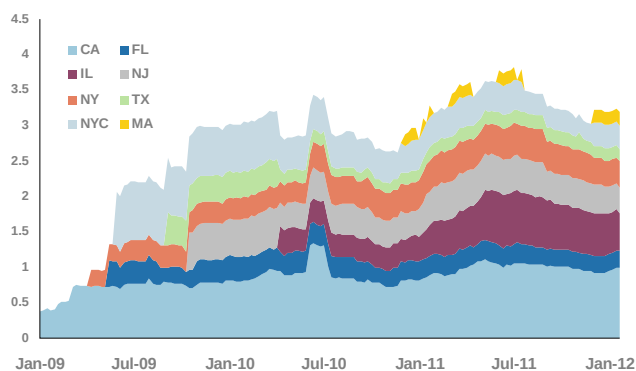
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The municipal market, one of the older corners of the bond market, has undergone some dramatic changes in the last several years. The emergence of BABs transformed this market from one in which only very few local investors and funds could capture the full yield available by realizing the tax benefits to one in which a broader base of investors could invest in the debt of a particular state. This development of the taxable market coincided with a global rise in credit risk premiums and an increase in liquidity in non-corporate credit derivatives, particularly sovereign European debt.

Additionally, starting in 2007, the municipal market culture underwent a shift from being thought of purely as a rates product to one that was more credit driven on the back of stress in the monoline insurers and subsequent decline in the usage of municipal bond insurance. Suddenly, many municipal bonds had to stand on their own (albeit primarily strong) creditworthiness.

exhibit 1

Municipal CDS Net Notional Outstanding



Source: DTCC

Against this backdrop, we saw the emergence of the municipal CDS market, which began trading prior to the crisis, but really only began to develop in a meaningful way in the past 3 years or so. As this market has grown, so has the attention it has gotten from traditional municipal tax-exempt investors to broader, more macro focused accounts. Along

the way, one question we hear is whether this product is necessary, primarily given the historically low default rates of municipal general obligation (GO) bonds. Indeed we agree that from a pure default protection standpoint, this product may appear expensive, especially when looking at historical data.

However, default protection is not, and should not be the sole motivation for any CDS participant, to which even investment grade CDS hedge investors can attest. Indeed the corporate CDS market grew out of a need to hedge loan exposures for banks, but it grew into a thriving, liquid two-way market because additional motivations brought other investors to the table. For us, the impetus behind this market feels somewhat opposite from the corporate market, in that in this nascent stage, the long investor motivation feels just as strong as that of the hedger, given current technicals in the underlying cash market.

As municipal market participants know, the BABs program was discontinued in 2010, making tax-exempt issuance the overwhelming vehicle for GO issuers. However, during its brief tenure, the BABs program, for the first time in a scalable, systematic way, created a product that didn't advantage local investors over others, and by extension, broadened the investor base. Suddenly, out-of-state and even foreign investors had a way to gain exposure to the municipal market, in a way they really hadn't before, at least not in this size.

With the BABs program gone, at least for the foreseeable future, and with lingering global risk premiums and ongoing focus on governmental entities, we see the municipal CDS product in today's environment as a way to access an otherwise inaccessible asset class. Taxable municipal investors, which are a small but growing component of the broader municipal investor base, can use CDS to source risk that would be otherwise unattractively low yielding on a name-by-name basis, or use the MCDX index to get broader market exposure, like any other ETF or index product.

Today, flows in the municipal CDS market remain high, and market participants are focused on making the contract as standard and fungible as the older, more developed corporate CDS market. Indeed, a process similar to the corporate SNAC (Standard North American Contract) process is set to be implemented for municipal CDS on April 3, 2012, and should add the same standardization to municipal CDS as it did for corporates back in 2009.

EVOLUTION OF THE CORPORATE CDS MARKET

In order to understand the municipal CDS market we first look to the corporate CDS market for lessons learned. In 1998, the corporate CDS market began as a way for banks to hedge corporate loan exposures. By 2002, a two-way market had emerged on a broader array of names, and the cash / CDS basis began to emerge as a fair value metric (see Chapter 6). At this point, two factors contributed to the growth in liquidity of the corporate CDS market: contract standardization and structured CDS products.

Contract standardization. Initially, corporate CDS products were bilateral OTC derivative products that were settled individually when credit events were declared. As the number of credit events picked up, and outstanding notional amounts on CDS trades climbed, the market became increasingly alarmed by a lack of a standard settlement mechanism, out of which the auction settlement protocol was born (see Chapter 5). More recently, in 2009 the SNAC protocol standardized coupons for single-name CDS transactions, made adherence to the auction a requirement, and institutionalized the regional credit event determination committees, so deliverable obligations and credit event timing would be the same for all market participants.

Product growth. The other factor that contributed to the explosive growth of the CDS market in 2002-06 was the emergence of products built with CDS as the underlying building blocks. The CDX indices were certainly part of this, and once these became standard across dealers, they became both a market barometer and focal point of liquidity for the broader CDS and corporate bond markets. Around the same time, market participants began trading smaller baskets of credit, followed by levered forms of these baskets, known as FTDs (see Chapter 4).

Next came the liquid, standardized tranche market, which provided benchmark models and metrics for the structured products market, allowing for growth in bespoke structured products. Post-crisis, volumes in bespoke products have waned, though index tranches remain liquid, and more recently, the credit index options market has grown, given investors' desire to hedge not only defaults, but also volatility.

Both structured product growth and market standardization have contributed to the maturation of the corporate CDS market. The municipal market, on the other hand, is still in the somewhat nascent stage, and to us, feels like it more closely resembles the corporate CDS market of 2002. However, with the benefit of hindsight, we anticipate the municipal market will develop much more quickly.

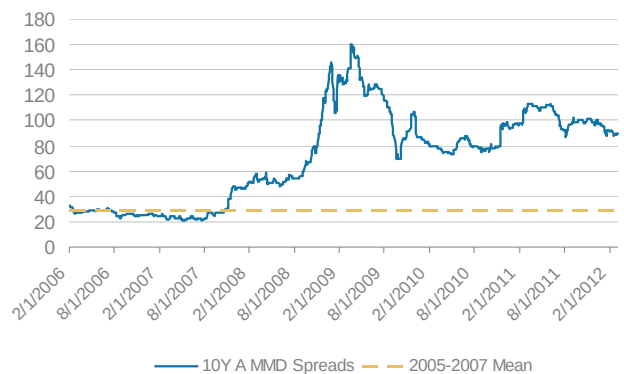
MUNICIPAL DEFAULTS: A BRIEF HISTORY

As the municipal market transitioned away from being a purely rates-driven market and toward one that is more credit-driven, the concept of municipal CDS as an investment

tool became increasingly relevant. This transition was born out of not only the early phases of the credit crisis but also the coincident decline of municipal bond insurance. In 2007, monoline insurers began to show weakness in their insured books of mortgage-backed and other asset-backed securities, leading the rating agencies to downgrade them, which in turn led to a vote of no confidence from the municipal market regarding the ability of the insurers to make good on their municipal bond guarantees. Consequently, the influence of municipal bond insurance on the market waned sharply.

exhibit 2

10-Year, A-Rated Muni Credit Spreads



Source: Morgan Stanley, Municipal Market Data

As a result, municipal spreads widened considerably as the market grappled with pricing the credit component of municipal bonds. This re-pricing led to permanently higher spreads across all sectors and qualities, in contrast to the historically tight spreads that enabled municipals to more closely track rates.

Credit quality, and accordingly credit spreads, thus gained substantially greater influence on the market. However, despite heightened credit quality concerns, available default and recovery data continue to paint the picture of municipal GO bonds as a comparatively low-risk asset class. Historical data demonstrates a low default rate for most types of municipal debt, including investment grade and high yield, in both GO and non-GO issues (see Exhibit 3), particularly when compared to its corporate counterparts.

exhibit 3

Average 10-Year Cumulative Issuer Weighted Default Rates, Moody's-Rated Only, 1970-2009

	IG	HY	All
GO	0.01%	0.01%	0.01%
Non-GO	0.13%	7.37%	0.19%
Corporates	2.50%	34.01%	11.06%

Source: Moody's

Because historical defaults have been rare, there is a great deal of uncertainty regarding the likelihood of a credit event trigger on a municipal CDS contract and the subsequent recovery value of the contract. Municipal CDS credit events are triggered when there is a failure to pay or a restructuring (see Appendix A for more details), and bankruptcy is notably *not* included on the list of credit event triggers. We look at several situations that have arisen in the municipal market over the years to determine what historical events, if any, would have triggered a CDS contract, had such a market existed at the time.

State of Arkansas. The State of Arkansas defaulted on and ultimately restructured a portion of its debt in 1934, the last state default on record. The credit event mirrors those defined within muni CDS contracts when the State missed coupon payments.

Orange County, CA. When Orange County declared bankruptcy in 1994, it continued to make bond payments on long-term debt. However, the county's short-term debt was restructured, with lenders agreeing to extend the maturity of the debt by one year. Upon exiting Chapter 9 after 18 months in bankruptcy, bondholders were not subject to a further restructuring. Given that the obligation characteristics to trigger a municipal CDS are "borrowed money", the short-term restructuring could have triggered the CDS.

Vallejo, CA. The Chapter 9 bankruptcy of the city of Vallejo in 2008 provides a more recent example. Though the city had filed for Chapter 9 bankruptcy, it continued making debt service payments for several months. Though the city eventually missed a payment, there was a significant period when its bonds were trading at distressed levels before the coupon payment was missed and the CDS would have triggered. Additionally, the City's debt was eventually restructured in the bankruptcy process, with bondholders taking a haircut and maturities being extended. In this case, any *new* CDS contracts written after the missed coupon would have triggered in the restructuring, though there would likely have been far fewer contracts to settle. Though rare, there is some precedence for the double triggering of the CDS in the corporate market, most notably Thomson in Europe, in which a restructuring credit event in August 2009 was followed by a bankruptcy credit event in December 2009.

exhibit 4

Recoveries in Selected Municipal Defaults

Obligor	Year	Type	Amount (\$MM)	Recovery
Washington Public Power Supply (WA)	1983	Electric Enterprise	\$2,250	40%
Belfield (ND)	1987	GO	\$2.38	55%
Vanceburg Electric (KY)	1987	Electric Enterprise	\$124.5	100%
Baldwin County (AL)	1988	GO	\$6-\$8	100%
Polk County (IA)	1991	Lease	\$39	100%
Orange County (CA)	1994	GO	\$124.5	100%
Vallejo (CA)	2008	GO	\$6-\$8	5%-53%
Connector 2000 (SC)	2010	Toll Road	\$200	5%-66%

Source: Moody's, Bloomberg, US Bankruptcy Court

Jefferson County, AL. The case of Jefferson County, Alabama, reflects another instance of municipal distress and default. The county's sewer system came under stress when yields on its debt, which was primarily in auction-rate and variable-rate mode, spiked after the auction-rate market froze in 2008. This was compounded when the county lost use of an occupational tax as a source of significant revenue. The cost burden led to the county filing for bankruptcy in November 2011 and missing payments on its general obligation warrants, which would have likely triggered CDS.

Connector Highway Bonds. Connector 2000 was a toll-road project that was constructed by a public-private partnership in South Carolina. The traffic on the road failed to generate expected toll revenues, which eventually led Connector to announce in 2009 that the association would miss its January 2010 payment. In this instance, a CDS contract would have triggered when Connector defaulted on its revenue bonds in January 2010.

More detail on municipal defaults and bankruptcies can be found in [Municipal Bankruptcy: 3 Lessons and a Strategy](#), February 7, 2012.

MUNICIPAL CDS – THE NEXT STEPS

The municipal CDS market continues to develop much like the corporate market did in 2003 and again in 2009, with the SNAC protocol, and today a similar protocol for municipals is set to go into effect on April 3, 2012. We anticipate several changes that should help make the market more standardized and fungible, including:

- **Fixed Coupons.** The municipal CDS market will move to fixed coupons (of 100bp and 500bp, similar to the corporate CDS market) to make the unwinding and netting of risk simple, and make contracts fungible. While contracts may be quoted on a spread basis, they will effectively trade on an upfront price basis, much like the MCDX index already does (see Appendices A and C). To calculate the upfront, a recovery value of 75% will be used.

- **Incorporation of the Americas Determinations Committee (DC).** The DC will make binding decisions on whether a credit event has occurred, whether or not an auction will be conducted, and what is deliverable into the auction. A subset of the Americas DC, known as “Muni Dealer Institutions” will be participants in the auction settlement process.
- **Hardwired Credit Event Settlement.** The now well-established ISDA process for credit event settlement via an auction will become standard. We cover the auction process in Appendix B, as well as Chapter 5. This eliminates the need for physical settlement of US Municipal CDS transactions.
- **Credit Event and Succession Look-Back Provisions.** The effective date of CDS contracts will be changed to a rolling 60-day period prior to the current date. The change narrows the window in which credit events can be triggered and succession events can be declared.

One area that we think merits attention with regard to the municipal CDS contract is the credit event triggers themselves. Municipal CDS credit events include failure to pay and restructuring, and because most municipal deals do not specify a grace period, a failure to pay could potentially be triggered after only three days of payment lapse, whereas the corporate market allows for a 30-day grace period. Furthermore, restructuring is actually “OldR”, much like

corporate CDS prior to the Xerox restructuring in 2001. What this means is that any debt, up to 30 years, is deliverable into the contract in either a restructuring or a failure-to-pay credit event scenario.

Furthermore, and unique to the municipal markets, debt does not accelerate in a default, making every credit event similar to a corporate restructuring. This makes the distinction between a failure to pay and restructuring less relevant, as in neither scenario would the issuer owe par to the investor. While we just explored the remoteness of such defaults in the municipal market, particularly in state GO debt, it is still a worthy exercise to consider what would actually happen if one of these CDS contracts were to experience a credit event.

Most notably, and most impactful on CDS pricing, is the concept of a “cheapest-to-deliver” (CTD) bond in a credit event settlement, which can be a particularly important driver of valuation, especially under OldR. What this means is that any eligible *pari passu* bond, out to 30 years, including both taxable and tax-exempt, can be used in the auction. For many issuers, particularly GO issuers, debt that is 20-years to maturity and beyond would be conceivable to source, and there is a good possibility that this debt would be trading at a discount. So even in a scenario in which the issuer were to rectify the failure-to-pay in a very short time frame, at par, discount bonds could conceivably be available for use in the auction.

Appendix A: Municipal CDS Primer

Chapter 1 contains introductory materials on the CDS market, primarily referencing the corporate CDS market. Here, we aim to provide a similar overview, albeit with a focus on municipal CDS. A single-name municipal CDS is an OTC contract between the seller and the buyer of protection against the risk of default on a set of debt obligations issued by a specified reference entity. The protection buyer pays a premium to the protection seller and is, in turn, covered for the period. A municipal CDS is triggered if, during the term of protection, an event that materially affects the cash flows of the reference debt obligation takes place.

Credit events in the muni CDS market include failure to pay and restructuring. A CDS contract specifies the precise name of the legal entity on which it provides default protection. Thus, a credit default swap referencing a specific state or GO obligation does NOT cover other entities within that state; i.e., a failure to pay on a New York City bond would not trigger New York State CDS, for example, and vice versa.

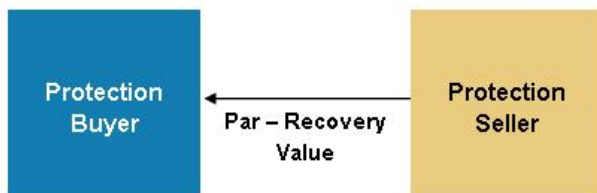
exhibit A.1

CDS Mechanics

CDS Cash Flows Before Maturity / Default



Cash Settlement in Case of Default



Source: Morgan Stanley

CREDIT DEFAULT SWAP CASHFLOWS

Prior to SNAC implementation, municipal CDS traded with no upfront payment, and the coupon was simply the quoted premium. Post-SNAC, a municipal CDS trade could contain three cash flows: an upfront amount paid at inception to either the buyer or seller of protection, the coupon paid quarterly to the seller of protection and the possible loss payout if a credit event occurs, illustrated in Exhibit A.1.

Cash flow dates are the 20th of March, June, September and December of every year, and the maturity of the contract is

typically 5 or 10 years from trade inception, rounded to the nearest cash flow dates. If an investor enters a transaction in between the payment dates, the protection seller would make a payment of accrued premium to the protection buyer, to reflect the fact that although the protection buyer would pay premium for the full quarter on the next payment date, the protection is only for part of the quarter.

The equivalent CDS premium strives to capture the probability-weighted expected loss on a particular reference entity in a risk-neutral sense – thus, the PV of the annuity stream should be the same as the expected loss on the contract. To calculate the CDS premium we first need to estimate the probability of default and expected loss given default. The following equations summarize the pricing approach:

$$PV \text{ of CDS Spread} = PV \text{ of Expected Loss}$$

$$\text{Expected Loss} = \text{Loss Given Default} * \text{Probability of Default}$$

Where expected loss given default equals

$$\text{Protection Notional} * (1 - \text{Estimated Recovery Rate}).$$

While these calculations can clearly get quite onerous once we calculate them over several periods with curve shape taken into account, most market participants use either internal models or CDSW on Bloomberg.

As we mentioned, after SNAC, municipal CDS will trade with a standard coupon and upfront payment, similar to how corporate CDS contracts have traded since 2009. The upfront payment is made to compensate either the buyer or the seller for the fact that the coupon is not equal to the fair value spread. Upfront payments are calculated by taking the difference between the trading spread and the coupon and multiplying this difference by the duration. So for instance, if an investor were to buy protection on a 5-year CDS with a duration of 4 at 150bps, the buyer would pay 2% $((150 - 100) * 4 = 200\text{bps}, \text{ or } 2\%)$ of the notional upfront and pay 100bps running. Note that if the quoted spread is higher than the coupon, the buyer of protection pays the upfront, while if the quoted spread is less than fixed coupon, the seller of protection pays the upfront.

WHY COUPONS ARE FIXED

Prior to SNAC in 2009 in the corporate markets, and prior to SNAC in municipal CDS, most CDS contracts traded on an all running basis. Thus, if an investor were to buy protection on a name that was quoted at 110bp, the trade would simply have a coupon of 110bp. At low spread levels, the difference is minimal and thus not much of a risk issue. During times of credit distress, however, this can be an issue, as unwinding a CDS contract is complicated when the strike price is very far out of the money, a theme we have highlighted in numerous pieces (see [LCDS, After the Trade](#) from June 1, 2007). In the past, investors who wanted to unwind a trade done without a fixed coupon often found large penalties

associated with the unwind due to the mismatched coupon streams. With a fixed coupon this legacy annuity risk is significantly reduced, as any difference in risk premium is paid in the form of an upfront payment. To illustrate this concept, we offer the following example.

Consider an example, where an investor purchased 5-year protection on credit “X” at 100bp and then subsequently elected to monetize the trade when spreads to the same maturity widened to 1,000bp. To monetize this widening, the investor could do one of two things. The first would be to sell protection at a strike of 1,000bp, and then hold two offsetting trades, one that is paying 100bps and another that is receiving 1,000bps. In this scenario, the investor would be hedged from a default risk perspective but would now have IO risk in this credit in the event of default.

Why? The key here is the uncertainty of the total cash flows and P&L for the investor, depending on the whether or not a credit event is triggered. The investor now receives 900bp (the difference between these two coupon streams) until maturity or a credit event. If a credit event were to occur, both contracts would trigger, the default loss settlement of the two contracts would offset each other, and *both* coupon streams would stop. Thus, the 900bps the investor had been receiving on a default “riskless” basis is now gone. And while the default settlement is hedged, the coupon stream is still exposed to default risk. In this example, if the default occurs one year later, the investor would have received \$0.9mm on a \$10mm position (900bp * 10mm). If the default is in four years, the investor would have received \$3.6mm in total cash flow from the coupons.

To eliminate this risk, the investor can instead unwind the original 100bp swap and receive the PV of the 900bps difference between the 100bps coupon and the 1,000bps of credit risk the swap is now worth to monetize the P&L immediately. The investor unwinds the original swap struck at 100bp and receives an upfront amount of \$2.9mm, which is the PV of the 900bp stream assuming a remaining duration of 3.2. In addition to having hedged the default risk on the reference entity, the investor has now eliminated the credit risk inherent in the risky coupon (as well as counterparty exposure, which is a separate issue). From the investor’s perspective, the trade is completely closed and monetized. However, the uncertainty in cash flows is not eliminated, but instead transferred to the dealer, because the dealer now has to hedge the position with the market standard contract. If floating coupons were market standard, the dealer would have paid out \$2.9mm today to buy protection at 100bp running from the investor, but can only sell protection to another investor on the now current running coupon – in this case 1,000bp.

If there is an immediate default, the dealer loses the \$2.9mm paid to the investor but has not received any of the 1,000bp coupon from the protection sold. The dealer has effectively

become exposed to the jump risk that the investor just got rid of. In the past, the dealer could buy short dated protection (say 1-year protection) to hedge these early default scenarios, but would most likely pass this additional cost, or at least some fraction of it, on to the investor who was unwinding the trade. This resulted in different spreads shown depending on whether the investor was unwinding an off-market coupon or putting on a new trade.

COMMON METRICS AND MODELS

When trading CDS, an investor goes long risk (sells protection) and earns a “spread” to compensate for the probability of default of the underlying entity in a risk-neutral framework. As a result, the present value of all these incremental cash flows should equal the present value of expected losses. The probability of default of an entity is determined using market CDS spreads across the maturity spectrum.

PV01. The PV01 of a CDS contract (referred to as risky duration) is simply the present value of 1bp of the cash flow at every coupon payment date. It can also be interpreted as the mark-to market sensitivity of expected losses in a CDS contract to a 1bp parallel shift in the spread term structure of the underlying entity. In contrast to the calculation for the duration of a bond or risk-free interest rate swap, the discount factors for CDS duration are higher, as they have a survival probability term associated with them. This additional discount factor reduces the present value of cash flows, and consequently the risky duration for similar cash flows in CDS vs. less risky assets.

Duration. The duration of a CDS contract depends on the shape of the maturity structure as it directly impacts survival probability. Spread delta or DV01 is a more pertinent metric to look at today, given the advent of fixed coupons. Unlike duration, which is the sensitivity of the contract to a change in expected loss assumptions, the spread delta captures *just* the mark-to-market sensitivity of the contract to a 1bp parallel shift in the credit curve. Thus, we get this equation:

$$\text{Upfront payment} = (\text{Par spread} - \text{Fixed Coupon}) * \text{PV01}$$

DV01. DV01 is essentially the first derivative of the upfront payment with respect to par spread. The above equation shows that when a contract trades at par, the spread delta is the same as duration. However, as the coupon differs from the par spread, the change in upfront payment also depends on the change in PV01 with respect to spread (i.e., convexity).

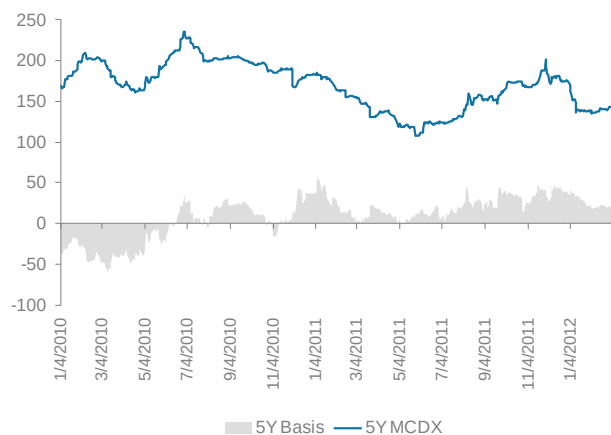
MUNICIPAL CDS VS CORPORATE CDS

While municipal CDS and corporates are mechanically similar, there are some differences we think are worth noting, particularly given that this market is relatively newer, and most credit derivatives market participants are more familiar with corporate CDS.

Tax Implications and cash / CDS comparisons. Given the tax-exempt nature of the majority of the market vs. the taxable nature of the CDS and index products, basis trading is generally unfeasible for out-of-state investors (doubly so for offshore investors). The BABs market somewhat mitigated this, and any taxable issuance on behalf of issuers will help, but given the majority of issuance and outstanding notionals today are tax-exempt, we think this is quite relevant.

exhibit A.2

Basis Historical Data

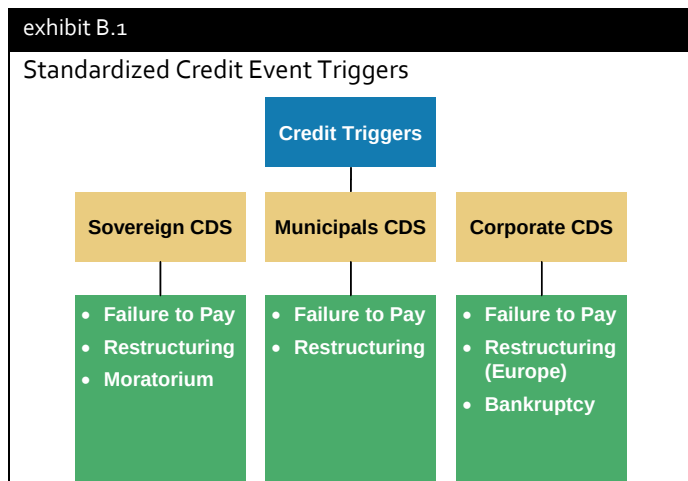


Source: Morgan Stanley

Credit event deliverables. Because debt does not accelerate upon default like corporate CDS, a credit event, even a failure-to-pay would be more analogous to a restructuring in the corporate market. The implications of non-accelerating debt in the context of CDS is that if a coupon payment is missed, the municipality is only responsible for paying the bondholder the missing coupon, and par is still not due until final maturity. Furthermore, unlike most corporate bonds, a bondholder has no claim to any of the municipality's assets, only the payability of the taxpayers or future revenue streams from the entity.

Appendix B: What Happens in a (Municipal) Credit Event

In a risk-neutral framework, the primary value of a CDS contract is derived from the credit events that can trigger it, the instruments that are deliverable upon this event, and their expected recovery. In the municipal market, failure to pay and restructuring are the only two triggers that apply, with the notable omission of bankruptcy (see Exhibit B.1).



Source: Morgan Stanley Research

Confirmations for US municipal CDS transactions are governed by both the 2003 ISDA Credit Derivatives Definitions and an ISDA supplement that covers US municipal reference entities. We endeavor to provide a brief overview here, but we do recommend readers visit the ISDA and MarkIt Partners websites for further information.

Failure to pay. For bonds that do not specify a grace period in the bond documents, a failure to pay credit event is triggered if an entity doesn't make payments within 3 days. This would apply to most GO and revenue bond issues, very few of which specify a grace period. Payment does not have to be on the specified reference obligation, but on any debt *pari passu* to the reference security. Failure to pay does *not* trigger if a municipal entity fails to make non-debt related payments such as wages to employees or payments to contractors.

Restructuring. The ISDA 2003 credit derivatives definitions characterize restructuring credit events as a reduction in coupon or principal, a deferral of an interest or principal payment, a change in the ranking of priority of payments, or changes to the currency of the interest and principal payments. Additionally, the restructuring must be coercive to trigger the CDS – i.e., it must be agreed upon by a sufficient number of holders before it becomes binding on all the holders. Furthermore, restructuring is not applicable if any of the above events has not been the result of deterioration in

credit quality. Under old-style restructuring, which applies to municipal CDS, in the event of a restructuring, bonds of any maturity less than 30 years can be delivered. This can introduce a significant amount of interest rate risk into a CDS contract upon a restructuring credit event (i.e., a 30 year fixed-rate bond could trade at a significant discount to par in a high interest rate environment). This is particularly important given that the municipal market tends to be long in duration, and in a credit event, bonds do not accelerate (i.e., become due immediately, as they would in a corporate bankruptcy).

THE CREDIT EVENT SETTLEMENT PROCESS

The municipal credit event settlement process, once the municipal SNAC protocol goes into effect on April 3, will make municipal CDS similar to corporate CDS. While this is by no means a comprehensive explanation of what would happen in the event that a municipal credit event were to be actually triggered, we highlight three things to be aware of once the contract has undergone the SNAC process:

Determination Committee (DC). The Americas DC will be the governing DC for municipal credit events, and will have the right to vote on whether a credit event has taken place. A 6-member subset (identified by trading volume), known as “Muni Dealer Institutions” will participate in the auction.

Declaring a credit event. To initiate the credit event process, market participants submit a request to the DC who will decide whether a credit event has occurred and what type. Finally, the DC determines whether there will be an auction. In case the outstanding CDS in a name is very small, no auction may be conducted.

Auction process. The auction is meant to concentrate trading in the bonds in a short window of time, and is then applied to all outstanding credit derivative products, including both single-name CDS and indices, as well as any other derivatives products (see Chapter 5 for a complete description of the credit event auction process). The important thing to note is that the auction will now be standard for all municipal CDS contracts, and thus one standard recovery will be in place for all market participants.

Appendix C: MCDX Primer

Launched in May 2008, MCDX is a portfolio of 50 single-name credit default swaps, serving both as an investment vehicle and a barometer of market activity. It was set up to roll biannually and pays quarterly, exactly like the corporate indices. MCDX comprises 50 municipal reference entities, each adhering to a stringent set of criteria:

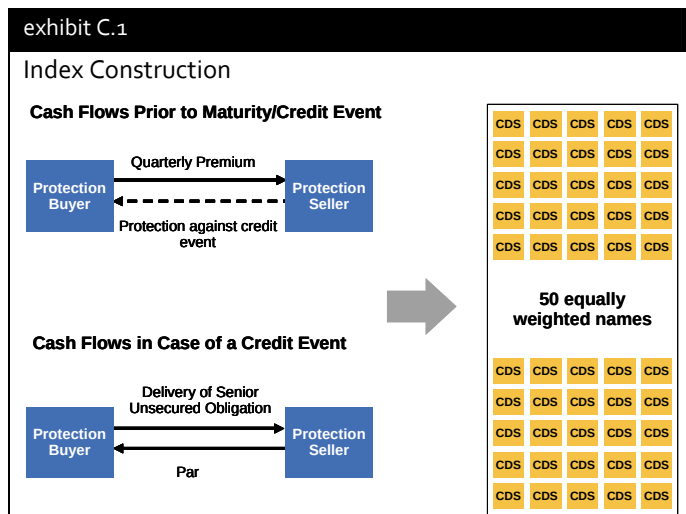
- \$250mm debt outstanding, must be uninsured parity debt
- Credit events are failure to pay and restructuring
- Minimum rating of Baa3 (Moody's), or BBB- (S&P and Fitch) that are not on negative watch for downgrade below investment grade
- 3 obligation types; Revenue Obligation, Full Faith and Credit Obligation and General Fund Obligation
- Excludes healthcare and tobacco issues

By buying protection on an index, an investor is protected against defaults in the underlying portfolio. In return, the buyer makes quarterly premium payments to the protection seller, just like a single-name CDS. If there is a default, the protection seller pays par less recovery determined in the auction settlement. Exhibits C.1 and C.2 show the cash flows in an index before and after a default. Some key characteristics that are unique to the indices are as follows:

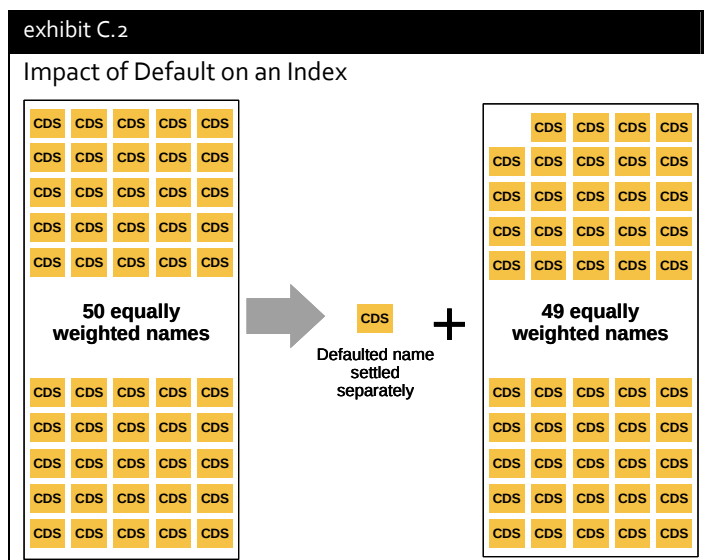
Static underlying portfolio. Once the MCDX series composition is determined it remains static unless a credit event occurs. Reference entity changes are made in new index series rather than the current index. All names are equally weighted.

Rolling to new series. As time passes the index maturity remains the same while the 5-year maturity for new CDS contracts gets longer and eventually the single-name CDS and the index maturities are no longer the same. To rectify this, new index series are introduced periodically and the latest series of the indices represents the current on-the-run index. When a new MCDX series is launched every 6 months (on April 3 and October 3), the maturity is extended and the portfolio constituents are reviewed and can be changed from one series to another.

Quoting convention. MCDX, like CDX IG and other indices, is quoted on a spread basis, though it trades with a standard fixed coupon with an upfront payment to account for the difference between the market spread and the coupon.



Source: Morgan Stanley



Source: Morgan Stanley

Cash flows at trade inception. The coupon amount is fixed for all trades on a particular series and an upfront payment is made to compensate either the buyer or the seller for the fact that the coupon is not equal to the fair value spread. Upfront payments are calculated by taking the difference between the trading spread and the coupon and multiplying this difference by the duration. So for instance, if an investor were to buy protection on the 5-year index with a duration of 4 at 150bps, the buyer would pay 2% $((150-100)*4 = 200\text{bps}, \text{ or } 2\%)$ of the notional upfront and pay 100bps running. Again, if the quoted spread is higher than the coupon, the buyer of protection pays the upfront, while if the quoted spread is less than fixed coupon the seller of protection pays the upfront.

Cash flows upon default. When an underlying single name defaults, it is removed from the index and settled separately. If one of the 50 underlying names in MCDX were to default, the remaining index (known as version 2) would have 49

names and the same fixed coupon. The seller of protection would owe the buyer of protection par minus recovery (going forward, determined at auction) on $1/50^{\text{th}}$ of the index trade notional. After a default, the premium payments for the index would be $(49/50) \times \text{index coupon} \times \text{original notional}$.

Recovery Assumptions. Unlike the corporate market, which has robust historical default and recovery data, the muni market has had very few defaults historically, when looking exclusively at general obligation and revenue bonds. Recovery assumptions are therefore largely theoretical. The recovery assumption used for calculating the upfront payment exchanged when putting on an MCDX trade will be 75%, however, the recovery as a result of default will still be “floating”, and determined at auction.

Constituent liquidity. Unlike CDX IG and iTraxx Main, a large portion of MCDX constituents are still illiquid. LCDX and to a certain extent CDX HY in the corporate space are

similar in this regard, and still function well as indices, but index arb trading is not possible on indices of this nature, and thus in certain environments the index and the underlying CDS can trade out of sync.

Correlation. A point of differentiation on MCDX is the high degree of default correlation in the underlying names. Indeed, a municipal default is a highly remote occurrence. An economic downturn is the most likely reason for a precipitous decrease in tax revenues, and while economic distress can be regional in some circumstances – i.e., loss of manufacturing jobs in Detroit, natural disasters in the coastal regions, etc. – economic downturns are often nationwide. Therefore, regional or tax source diversification will do little to help in periods of a US recession, and there are no “countercyclical” municipalities with which to diversify a portfolio.

Muni Mania

February 1, 2008

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One of the strongest themes that emerged in the latter half of 2007 was the significant pickup in systemic risk across the global financial system. The impact had and continues to have huge negative implications on risky credit assets, but it has also led to growth and participation in new credit market instruments. In Europe, there has been an explosion in the usage of sovereign CDS as a tool to hedge systemic risk, position for sovereign stress and capture the unprecedented wide spreads on sovereign entities in leveraged and structured products. CDS on European sovereign names now trade in the 10-40 bp range.

These same systemic market themes have played a role in the explosive recent growth of credit default swaps on US municipal issuers. US housing stress, state-specific economic conditions and the monoline debacle have clearly played a role as well. Up until last summer the over \$2.2 trillion US municipal market was, despite its size, a quiet market. States, counties, cities, school districts and other entities that could borrow at tax-exempt rates would bring their issue to market, where much of it would be sold to individual investors and nearly all of it was purchased with the expectation that it would be held to maturity.

exhibit 1

Current Indicative Levels for Muni CDS
(As of January 31, 2008)

Issuer	State	Moody's	S&P	10yr CDS
California	CA	A1	A+	58/65
New York, NY	NY	Aa3	AA	46/53
Florida	FL	Aa1	AAA	44/52
Michigan	MI	Aa3	AA-	43/51
Ohio	OH	Aa1	AA+	37/43
New Jersey	NJ	Aa3	AA	36/42
New York State	NY	Aa3	AA	35/41
Illinois	IL	Aa3	AA	25/31
Massachusetts	MA	Aa2	AA	22/28
Texas	TX	Aa1	AA	21/27

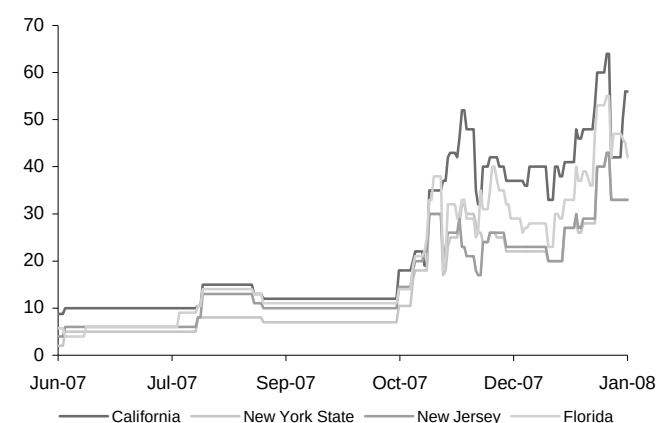
Source: Morgan Stanley Research

When the first signs of the impending housing issues began to surface in February, market participants began to connect the dots. If foreclosures and bankruptcies really were on the rise, this could lead to a dramatic reduction in tax revenues. The virtually non-existent municipal CDS market suddenly seemed like a good place to speculate on the state of the US

housing market. We estimate that the volume of muni CDS trading has gone from several hundred million total in the 5 years prior to July 2007 to over \$50 billion in the 6 months since then.

exhibit 2

Historical Spreads on Muni CDS



Source: Morgan Stanley Research

Further compounding the trouble in the municipal market is the uncertain monoline situation. Though the monolines have run into trouble insuring ABS CDO paper, their original business plan was insuring the municipal bond market (indeed, MBIA initially stood for Municipal Bond Insurance Association). With the monolines' future ability to pay out claims or even write new business in question, the municipal market was threatened with the notion that hundreds of billions of dollars of insurance wraps are deemed much less valuable than just a few months ago. While most of the general obligation (GO) debt that is referenced in the CDS market isn't wrapped by monolines, the deteriorating credit quality of the monolines was causing more uncertainty in an already shaken market. Many new issues were put on hold, while the market waited for some resolution.

Finally, many investors realized that municipalities' expenses tend to be rather fixed, while tax revenues can vary widely year to year based on economic conditions and that municipalities can be reluctant to raise taxes prematurely.

Meanwhile, as the new issue market was coming under pressure, the CDS volumes were increasing. Buyers of protection further pushed the credit spreads wider as more non-traditional funds entered the market from the short side. Exhibit 2 shows that the CDS spreads, historically in the single digits, widened to 40-60 bps, and Exhibit 1 shows indicative levels on 10 liquid names as of January 31, 2008.

One reason the muni CDS market had been slow to develop up until summer 2007 is that munis have largely been thought of by traditional municipal bond investors as tax and interest rate products. Since most issuers had AA or AAA credit ratings, the interest rate environment or a potential change in the tax code was of much greater concern than the credit quality of the municipality. When the twin storms of poor monoline health and the collapse of the subprime mortgage market set in, the credit aspect of muni bonds began to take on more significance. Since the muni market was already familiar with a wide array of interest rate and tax derivatives, CDS was an easy next step in muni derivative products.

As the current US credit and real estate cycle plays out, we expect interest in the US municipal CDS market to continue to grow, with broader participation among investors and investor segments. This chapter represents our first thoughts on the municipal CDS market, and we provide some general considerations for trading muni CDS, as well as some of the fundamental and technical aspects of the market, including the following.

Ratings. Municipalities and other tax-exempt issuers are rated somewhat more conservatively than their corporate counterparts. We explain the mapping from one to the other.

CDS Contract. Municipalities have only two triggers; failure to pay and (old-style) restructuring. We explore the technical side of this largely untested contract and its specific terms.

Tax-exemption. Municipal bonds are exempt from state, local and federal taxes for residents of the municipality. This has important implications for basis trades.

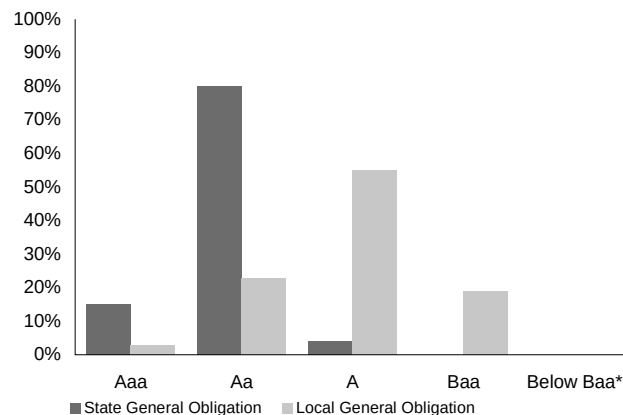
Market Participants. Hedge funds as well as dealers, insurance companies and banks have been responsible for the early growth in the market. As more investors enter the market and some structured products develop, we expect liquidity and the number of traded entities to increase. We highlight some structured ideas.

RATINGS

Of the 10-15 liquid muni CDS names, all of them reference state general obligation bonds and one city general obligation bond (NYC). Though the muni market as a whole has a range of credits, in this chapter we focus on the larger, more liquid and more highly rated general obligation debt and largely ignore the school district, housing finance, utility and revenue credits such as toll roads and stadium financings, as CDS on these entities is still not liquidly traded.

exhibit 3

Moody's Municipal Scale Rating Distribution



Note: For obligors with multiple ratings on different security pledges (senior, subordinated, etc.), the percentages are based on the obligor's highest rating.

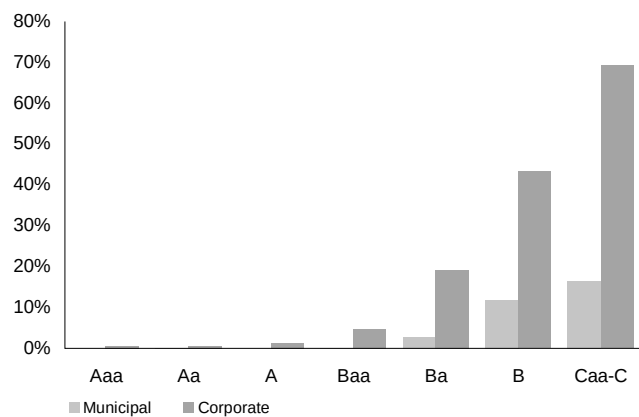
** 0.003% of local government GO's are rated below Baa*

Source: The US Municipal Bond Rating Scale: Mapping to the Global Rating Scale and Assigning Global Scale Ratings to Municipal Obligations. Moody's Public Finance Credit Committee

Before we delve into some of the details of muni CDS, it's important to review the rating agency approach to examining credit quality. One of the main reasons munis haven't developed into credit instruments is their historically stable ratings and, at least in the GO category, almost total lack of defaults. In fact, when looking at general obligation debt, which is guaranteed by tax revenues of the issuing authority, since 1970, there has been only one "credit event", Baldwin County, Alabama, and the bonds recovered at par 15 days later – an event that would not have triggered a CDS contract.

exhibit 4

Average 10-year Cumulative Issuer Weighted Default Rates, Moody's-Rated Only, 1970-2006



Source: The US Municipal Bond Rating Scale: Mapping to the Global Rating Scale and Assigning Global Scale Ratings to Municipal Obligations. Moody's Public Finance Credit Committee

So from a historical default perspective, much of this GO debt would be indistinguishable. To solve that, muni issuers and investors sought a separate municipal ratings scale to be able to differentiate the fine gradations of taxing ability from one credit to the next. For issues being marketed to non-traditional investors, an issuer can request a “global scale” rating, formerly called a corporate equivalent rating. We include in Exhibit 5 this global scale conversion mapping. The first column shows the muni rating and the next two columns show the global scale rating that corresponds to the muni rating for the specific issuer type. For example, a state GO credit with a muni rating of A1 would be comparable from an expected loss perspective to a global Aaa. Since municipal general obligation bonds both tend to default less often and recover higher, this mapping has some credits jumping several notches.

exhibit 5

Moody's Global Scale Ratings

Municipal Scale Rating	State GO	Local GO; Water / Sewer; State Revolving Fund; State Lease Obligation and Special Tax
Aaa	Aaa	Aaa
Aa1	Aaa	Aaa
Aa2	Aaa	Aaa
Aa3	Aaa	Aaa
A1	Aaa	Aa1
A2	Aa1	Aa1
A3	Aa1	Aa1
Baa1	Aa1	Aa2
Baa2	Aa2	Aa3
Baa3	Aa2	Aa3
Ba1	Aa3	A1
Ba2	Aa3	A2
Ba3	A1	A3
B1	A2	Baa1
B2	A3	Baa2
B3	Baa1	Baa3
Caa1	Baa2	Baa3
Caa2	Baa3	Ba1
Caa3	Baa3	Ba2

Note: The chart above is a conversion scale from the municipal rating awarded to an issuer in the far left column to the Global Scale Rating in the middle and right-hand column. For instance, if a State GO bond received a municipal rating of A1, it would have a Global Scale Rating of Aaa. Likewise, a Local GO bond that had a municipal rating of A1 would have a Global Scale Rating of Aa1.

Source: The US Municipal Bond Rating Scale: Mapping to the Global Rating Scale and Assigning Global Scale Ratings to Municipal Obligations. Moody's Public Finance Credit Committee

THE MUNI CDS CONTRACT – IMPORTANT DETAILS

While history tells us that default likelihood for the state GO municipal issuers that are commonly traded in the muni CDS market are effectively zero, there is some amount of uncertainty related to how features of muni CDS contracts

would behave when tested. This uncertainty should and does add some amount of risk premium to CDS contracts.

Generally, confirmations for US municipal CDS transactions are governed by both the 2003 ISDA Credit Derivatives Definitions and an ISDA supplement that covers US municipal reference entities. Failure to pay and restructuring are the two standard credit events, and it is important to note that restructuring is “old style”, i.e., not ModR that is used for corporate reference entities. Old-style restructuring implies that, in the event of a restructuring, bonds of any maturity less than 30 years can be delivered, which can introduce a significant amount of interest rate risk into a muni CDS contract upon a restructuring credit event (i.e., a 30-year fixed-rate bond could trade at a significant discount to par in a high interest rate environment). This is particularly important given that the muni market tends to be long in duration. In corporate credit, the restructuring of a Xerox obligation in 2002 brought interest rate risk to light, which was a key driver in the introduction of ModR to the 2003 ISDA definitions.

With effectively no default history on state GO's, the CDS market needs to make other assumptions as well. As of now, 80% recovery appears to be a market standard, which will impact CDS unwinds as well as any fixed-recovery products, including first-to-default baskets (see later section).

With the explosion of CDS volumes over the past few months, there is a bit of segmentation among investor bases, based on our experiences. Many of the protection buyers are hedge funds who are motivated by the story we describe in the introduction, as well as a rising risk premium play (similar to sovereign CDS). Protection sellers tend to be insurance companies, banks and dealers, all of whom are attracted by the unfunded nature of derivative contract, despite the lack of any tax-exempt treatment for muni CDS. Standard CDS maturity in the market is ten years.

CASH / CDS BASIS

We've written extensively about the cash / CDS basis in the corporate market (see Chapter 6). In munis, however, basis packages are much more complicated, stemming from their tax-exempt status. A basis package on a Florida GO bond will look most attractive to a Florida-based investor, less so to a non-Florida-based US investor, and worst to a non-US domiciled investor. This is because a Florida bond is tax exempt on a state and federal level, while CDS does not benefit from any tax-exempt status. As such, there are really 3 basis packages for every bond. For this reason and others, there has been very little overlap between muni CDS users and muni bond holders. Nevertheless, basis packages represent interesting opportunities for those who can claim one or more tax-exemptions. The potential tax-exempt treatment of GO bonds (for a given investor) impacts the funding side of the trade as well, and the market uses different BMA and MMD curves instead of Libor for this reason.

STRUCTURED SOLUTIONS

If muni CDS continues to trade at the wide levels, we see interesting ways of playing the market from a structured perspective. Prior to this year there have been just a handful of muni cash CDOs, and this has largely been the extent of munis in structured finance land.

To take a structured view on the muni market we prefer first-to-default (FTD) and more generally nth-to-default trades. Current market pricing suggests that implied correlation on these baskets is very high, reflecting the systemic nature of underlying risk. As such, from a correlation perspective, good long credit positions would be 2nd to default or 2nd-to-nth trades, while FTDs might look more attractive on a short credit basis. However, the biggest issue for these trades isn't so much the correlation as the assumed recovery. With almost no historical loss data to draw from, recovery could easily be as high as 95% or even 100% (rating agency assumptions). Though if this market truly represents an unprecedented and highly correlated paradigm shift, recovery could be much lower. The market standard right now is to use 80% as the floating recovery amount, which makes pricing look very unusual for those used to playing in the corporate FTD space. Exhibit 6 shows pricing on a 5-name muni CDS basket with a 10-year maturity and a sum-of-spreads of approximately 220 bps. In this exhibit, we show the impact of fixed vs. floating recovery on FTD pricing.

exhibit 6

Sample Muni FTD Pricing (Pricing as of January 31, 2008)

Recovery	Indicative FTD Level
40% Fixed	225
80% Floating	83

Note: Pricing assumes a 5-name basket with a 10yr maturity and a 220bps sum of spreads

Source: Morgan Stanley Research

We've heard some investors explore the idea of using 40% fixed recovery to make pricing on muni FTDs more attractive. In our hypothetical basket, this would give an FTD price of roughly 225bps and would look much more like the corporate FTD pricing many investors are used to.

CONCLUSION – INVESTMENT STRATEGIES

Thanks in part to regional economic stress, falling real estate prices, poor health in monolines and generally high systemic risk, the municipal market is now one that must consider the implications of credit risk. Muni CDS is a burgeoning but still niche sector of the credit derivatives markets, but it does offer important tools to position for higher or lower levels of credit risk, hedge existing portfolios of muni risk, or participate in the market through simple structured solutions. We expect greater two-way liquidity and a wider variety of reference entities to choose from, but we also caution that low levels of actual default risk may leave the finer details of the CDS contract untested.

MCDX: Credit Risk or Risk Premium?

May 21, 2008

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Following the global crisis led by the collapse of the securitization markets that began in the second half of 2007, events in the first half of 2008, particularly post-Bear Stearns, have been characterized by a paradigm shift in the market from systemic to cyclical risk. In the corporate credit space IG spreads have retreated from their recent highs, though they remain significantly elevated compared to pre-crisis levels.

Indeed a portion of the risk currently priced into the credit market in the corporate space is uncertainty and MTM pressures, as opposed to a pure default risk premium. The corporate credit market is adapting to this new environment, as we highlighted in *Credit Calm or Credit Storm?* May 16, 2008.

In the municipal market, the shift from a systemic to a cyclical market environment has different implications. In late 2007, when the muni CDS market began trading, initial concerns related to many technical aspects of the muni market, including the stress in the monoline sector and failed auctions in the short-term auction rate securities market. As a result, municipal issuers and investors alike have adapted to the monoline issue by becoming more familiar with municipal credit risk, and to the auction rate issue by issuing more long-term debt and moving away from ARN products.

Unlike in past recessions, the current economic vulnerabilities are characterized less by excess leverage in corporate sectors than by increased pressure on the American consumer and homeowner – known in muni-land as the American taxpayer, and the only source of revenue for municipalities.

It is in this environment that MCDX was launched. MCDX is a 50-name index referencing the credit risk from municipal debt. Initial trading volumes have been strong, indicating a willingness on the part of investors to view munis as a credit product, in addition to the tax and rate product it has long been considered. As the market continues to gain traction, we hope to see greater liquidity and transparency, and a broadening of the applications for single-name muni CDS, from extended use in FTD products to inclusion in corporate bespoke portfolios as they become more ratings efficient relative to corporates.

In this chapter we highlight some of our preliminary thoughts regarding opportunities in this market. Some are the same as one would expect when considering any credit index product, while others are unique to the municipal space.

exhibit 1		
Moody's Global Scale Ratings		
Municipal Scale Rating	State GO	Local GO; Water/Sewer; State Revolving Fund; State Lease Obligation and Special Tax
Aaa	Aaa	Aaa
Aa1	Aaa	Aaa
Aa2	Aaa	Aaa
Aa3	Aaa	Aaa
A1	Aaa	Aa1
A2	Aa1	Aa1
A3	Aa1	Aa1
Baa1	Aa1	Aa2
Baa2	Aa2	Aa3
Baa3	Aa2	Aa3
Ba1	Aa3	A1
Ba2	Aa3	A2
Ba3	A1	A3
B1	A2	Baa1
B2	A3	Baa2
B3	Baa1	Baa3
Caa1	Baa2	Baa3
Caa2	Baa3	Ba1
Caa3	Baa3	Ba2

Source: *The US Municipal Bond Rating Scale: Mapping to the Global Rating Scale and Assigning Global Scale Ratings to Municipal Obligations*. Moody's Public Finance Credit Committee

Index Mechanics. The mechanics of this market are largely the same as they are for the existing indices, including roll dates and settlement mechanics. We caution investors to pay close attention to deliverability and settlement issues pertaining to individual reference entities.

Index Composition. This index comprises 50 names, representing 23 of the 50 states in addition to Puerto Rico. Nearly half are general obligation issues, and the rest are backed by revenues. None are wrapped by the monolines, and all are IG rated on the Moody's Municipal Ratings Scale and above A1 on Moody's Global Scale Ratings ("GSR"). The main considerations in terms of composition are spread distribution and correlation of the reference entities, which we discuss further.

Market Participants. There is currently little overlap between traditional muni bond investors, most of whom are wealthy individuals and P&C insurance companies, and muni CDS investors, who tend to be hedge funds. This is primarily

because traditional muni investors tend to view muni bonds as a tax product that has a certain degree of interest rate risk. CDS investors are isolating the credit component, something most cash investors aren't yet considering. Basis trading is still a long way off, as tax implications prevent most basis packages from being attractive except to a small subset of in-state investors.

Trading Strategies. We compare MCDX to IG10, 30-100%, sovereigns, and the RPX index and highlight a few trade ideas we like in this environment.

INDEX MECHANICS AND TECHNICALS

When considering any index trade, an investor must look at the mechanics and logistics of the index as a whole in addition to the specific underlying reference entities. In the case of MCDX, the index rolls biannually and pays quarterly, exactly like the corporate indices. While compositional changes at the next roll date are unclear, having the opportunity to change the names should only bolster liquidity as we learn what names are most representative and which names gain the most traction in the relatively new single name market. Other specifics when considering MCDX are:

Recovery Assumptions. Unlike the corporate market, which has robust historical default and recovery data, the muni market has had very few defaults historically, when looking exclusively at general obligation and revenue bonds. Recovery assumptions are therefore largely theoretical. Right now the assumption used for calculating the upfront payment exchanged when putting on an MCDX trade is 80%, but until we have more data, this number could vary widely in the event of an actual default.

Initial Liquidity. Unlike CDX IG and iTraxx Main, a large portion of MCDX constituents are still illiquid. If the index continues to gain momentum, single-name liquidity could pick up, but in the meantime, index arb trading is unlikely, much like we see in LCDX and CDX HY.

Tax Implications. Given the tax-exempt nature of the underlying cash asset vs. the taxable nature of the CDS and index products, basis trading is generally unfeasible for out-of-state investors (and of course offshore investors). See a report written by our colleagues in Interest Rates Strategy from May 16, 2008, titled [Municipals: Muni Bond CDS Basis: MCDX, BMA/Libor Ratios and Muni Cash Bonds](#) for further explanation on this concept.

Untested Credit Events. A credit event in the muni space is a largely untested occurrence, and since a municipality cannot file for bankruptcy, a credit event would be more analogous to a restructuring in the IG corporate space. Most municipal bonds are non-accelerating, meaning that if a coupon payment is missed, the municipality is only responsible for paying the bondholder the missing coupon, and par is still not due until final maturity. Therefore, a seller

of protection could be required to take delivery of a bond, and have to hold it to maturity to realize par.

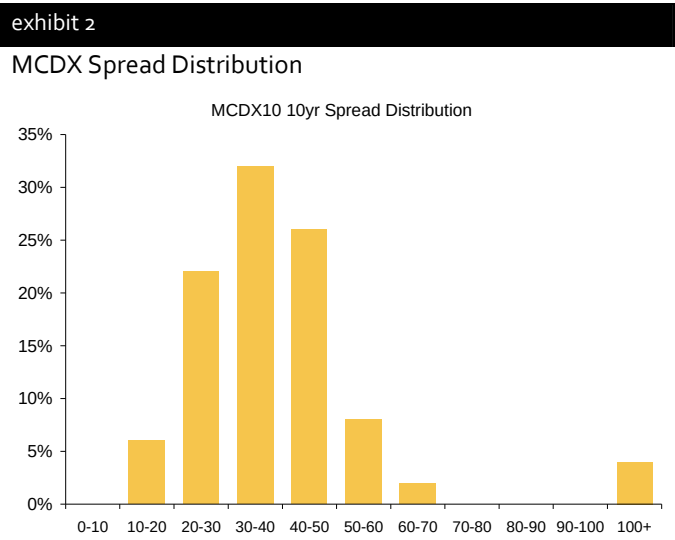
Furthermore, unlike most corporate bonds, a bondholder has no claim to any of the municipality's assets, only the payability of the taxpayers or future revenue streams from the entity.

Deliverability and Settlement. Market participants are looking to adopt an auction protocol as the standard credit event settlement mechanism, but the documents currently state that physical settlement will be used. This raises issues of deliverability and eligibility in the event of default. While a seller of protection would only be required to pay the accreted value if a zero-coupon bond were delivered, many muni bonds are issued at a discount, with lower coupons, creating some cheapest-to-deliver issues.

INDEX COMPOSITION AND FUNDAMENTALS

MCDX comprises 50 municipal reference entities, each adhering to a stringent set of criteria:

- \$250mm debt outstanding
- Credit events are failure to pay and restructuring
- IG rated on the municipal scale; equivalent to single-A and above on Moody's Global Scale Ratings
- 3 obligation types: Revenue Obligation, Full Faith and Credit Obligation and General Fund Obligation
- No healthcare, tobacco or housing bonds
- All uninsured – no monoline wrapped bonds



Source: Morgan Stanley

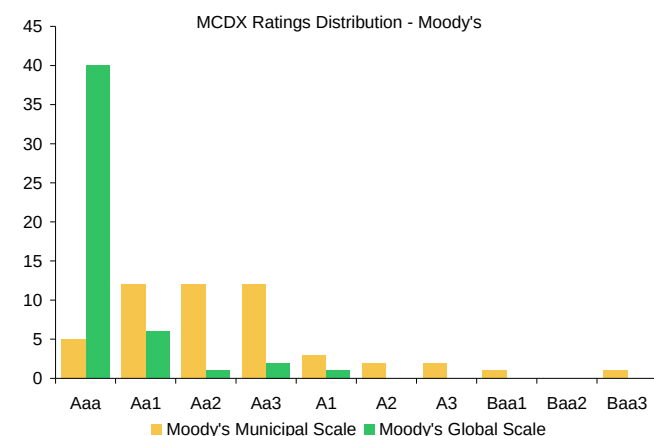
All of these criteria suggest that the index is highly representative of the largest municipalities and therefore a good proxy for the health of the municipal market as a whole.

The last three criteria in particular ensure that some of the technical factors that have weighed on the muni market in recent months, such as the monoline situation, will not cloud trading levels on MCDX.

Two other factors come into play when considering an index trade position: correlation of the underlying names and spread distribution.

exhibit 3

MCDX Ratings Distribution – Moody's



Source: Morgan Stanley, Moody's

In the case of MCDX, there are two reference entities, both in Puerto Rico, trading above 100 bp, and the remaining are between 10 bp and 70 bp, with the bulk of the names in the 30-40 bp range, making this a fairly uniformly distributed index, in comparison to the more disparate CDX indices. In this respect, it is much more akin to iTraxx, both from an overall credit-quality perspective and from a tail perspective.

A point of differentiation on MCDX is the high degree of default correlation in the underlying names. Indeed, a municipal default is a highly remote occurrence. An economic downturn is the most likely reason for a precipitous decrease in tax revenues (aside from fraud). While economic distress can be regional in some circumstances, i.e., loss of manufacturing jobs in Detroit, natural disasters in the coastal regions, etc., economic downturns are often nationwide. Therefore regional or tax source diversification will do little to help in periods of a US recession, and there are no "countercyclical" municipalities with which to diversify a portfolio.

TRADE OPPORTUNITIES

At the conclusion of the second week of trading on the MCDX index, we have already begun to see an increase in volumes traded. As this index gains traction, we see some potential trading strategies, especially when paired with the corporate indices or other asset classes.

Expected Loss Trade. This trade works in both 5yr and 10yr. The liquid point for single-name muni CDS is 10yrs, though MCDX has good liquidity in 5yrs as well. Currently, MCDX10 10yr is trading at 47 bp and CDX 10 is trading at 93 bp in 10yrs. The corresponding expected loss, assuming a duration of 7.7 for both, is 3.6% for MCDX and 7.2% for CDX IG. When you assume a standard recovery of 80% for munis and 40% for IG corporates, this implies an 18.1% default rate in the muni market and an 11.9% default rate in corporates!

exhibit 4

MCDX10 vs. CDX IG10

	MCDX10	CDX IG10
Spread	47 bp	93 bp
Recovery	80%	40%
10yr Expected Loss	3.6%	7.2%
10yr Implied Default Rate	18.1%	11.9%

Source: Morgan Stanley

In light of this, we like going long MCDX and shorting CDX against it, either by being carry neutral or slightly carry positive.

Risk Premium Trade. Another way we like thinking about MCDX is as a pure risk premium trade. Given that historical default data for municipal risk are nearly non-existent (once housing and hospital revenue bonds are removed), we consider much of the spread on MCDX to be compensation for spread volatility as opposed to actual default risk. In this way we consider MCDX to be somewhat analogous to the pure risk premium 30-100% tranche in CDX IG10. We like going long MCDX vs. shorting CDX IG10 30-100%, equal notional. Given the super senior tranches' poor technicals, this tranche could widen dramatically again in a weakening environment. Assuming current market levels, an investor would receive \$52k upfront to go long \$10mm MCDX and pay nothing to get into a CDX IG10 30-100% tranche trade at 27 bp. Rolldown for MCDX is much steeper than the super senior tranche and this trade is carry positive.

Another way to trade MCDX as a pure risk premium play is to compare it to European sovereign CDS. Both muni CDS and sovereign CDS are a good way to express a view on the local economies of either the US or the EU region. EU sovereigns have never had a default, and munis have had relatively few since 1970, and the few that have occurred had extremely high recoveries, many at par.

Fundamental Credit Trade. Property taxes are an important contributor to the coffers of municipalities and are dependent on the level of home prices. We find that changes in the state revenues and local home prices are positively and significant correlated (Exhibit 5).

It is reasonable therefore to conjecture that MCDX spreads and measures of home prices such as the RPX index and others affected by the trajectory of home price appreciation such as the ABX index (particularly AAA ABX sub series) are likely to be correlated. This suggests that there is the potential for hedging between these alternative indices. We like going long RPX and shorting MCDX, given that RPX is pricing in significantly greater weakness in local housing markets, and thus by extension, property tax revenues.

CONCLUSION

If it feels like we’ve covered a lot of ground in this chapter we have our reasons. In our view, the community of investors with significant experience in *both* credit derivatives indices and municipal bonds is small, and therefore there are learning curves that many need to climb. Furthermore, credit derivatives tied to municipal bonds have unique issues that should result in some interesting tests of

contract language over time. We are indeed excited about strategic opportunities in the muni credit space involving single-names and indices. However, we caution that we are in the early days of a market that will need time (and increased credit risk) to mature.

exhibit 5		
MCDX10 vs CDX IG10		
	Portion of MCDX	Correlation of Change in Taxes and Home Prices
Arizona	6.0%	52.9%
California	12.0%	46.2%
New York	16.0%	53.5%
Texas	10.0%	42.0%

Source: Morgan Stanley, OFHEO, U.S. Department of Commerce: Census Bureau

Munis vs. Corporates in Recession: CDS Considerations

October 24, 2008

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Both municipalities and corporations are already seeing pain in this economic cycle, though structural differences indicate strengths and weaknesses for each. To start, a municipality has taxing authority and thus arguably has greater autonomy on the revenue side of the budget than does a corporation, since, in the end, a corporation cannot force the public to buy more widgets. On the other hand, corporations have an easier time on the expenditure portion of the budget, since often budgetary decision making lies with one person or a few people. A municipality by nature is a representative government of the people, and may have less ability to cut large expenditures without going through several lengthy approval processes by many people.

With this in mind, we take a fresh look at the municipal CDS market, taking into consideration the unprecedented events of September in the financials sector specifically and the credit markets more broadly.

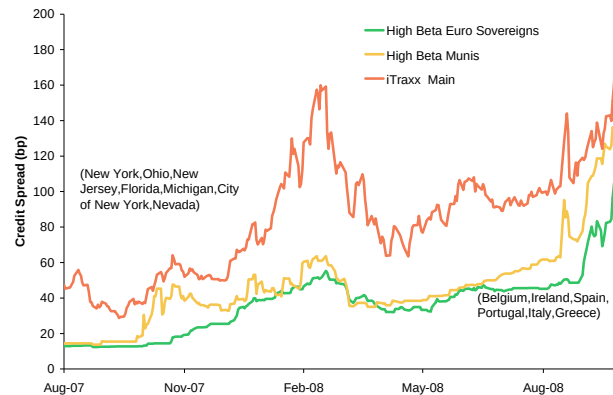
Corporate credit has priced in a more severe default cycle and indeed many sectors are starting to look cheap. Our IG strategy team highlighted the opportunities in going long the financial sector in their [Credit Basis Report: Banking on the Government](#) (Oct. 17, 2008).

Because the corporate credit and CDS markets are further into this economic cycle than their municipal counterparts, we expect to see certain sectors of the corporate market recover sooner. Indeed, even if the financials sector begins a slow recovery, there are a number of economic indicators that imply that the real economy could continue to be weak for the near future, and we acknowledge the possibility that some of these could prove to be leading indicators for the muni CDS market. We point out, however, that these are unprecedented times for many asset classes, and the very new muni CDS market is no exception.

We note that our thoughts and ideas illustrated in this chapter relate to the muni CDS market only. On the cash side of the municipal bond market, the real risk of an actual default remains low, and credit risk is only one of many factors influencing bond prices. Interest rates and tax policy are also key components, and indeed the tax-exempt status offered on most municipal bonds can make some of these assets quite attractive.

exhibit 1

Munis, Sovereigns and Corporates



Source: Morgan Stanley

Throughout this chapter, we aim to highlight several considerations when thinking about muni CDS, and offer some updated thoughts on the market in light of recent economic events, including:

CDS Contract. CDS triggers, restructuring language, and non-acceleration of the debt during a credit event are our main focus here, particularly given lessons learned in the FRE and FNM recovery auction process.

Recovery. One can envision a variety of scenarios under which a CDS could trigger, and we highlight the fundamental, mechanical, and technical aspects that could ultimately impact the CDS recovery price.

Economic Cycle. We explore our view that many corporate sectors are well into the downturn, and offer our thoughts on how to execute this view with the CDX IG and MCDX indices.

Localization. We've primarily focused our attention on the muni market as a whole, but we see compelling reasons to drill down to the local and regional levels to examine how local industries could potentially impact the CDS on certain municipalities more than others.

THE MUNICIPAL CDS CONTRACT

We first highlighted some of the unique technicals inherent in the muni CDS contract in our *Muni Mania* piece from earlier this year (see Chapter 37). We discussed then and reiterate now that the muni CDS contract has only two credit event triggers: failure-to-pay and restructuring. Restructuring here is not the same as corporate CDS today, which uses

ModR restructuring, but rather muni CDS contracts can be triggered by OldR restructuring. The significance here is that a 30-year bond could hypothetically be delivered if a credit event were triggered, introducing significant interest rate risk into the final recovery price.

Furthermore, municipal bonds are non-accelerating in the event of a default, meaning that if a credit event were triggered, the principal amount on the bond would not become due. This phenomenon can occur in the corporate market as well, and though it is rare, it was recently brought to light when Fannie Mae and Freddie Mac were placed under conservatorship by the U.S. government. This triggered the CDS contracts and resulted in some initial confusion as to deliverability and created some uncertainty surrounding recovery. While these issues were eventually resolved in the Fannie Mae/Freddie Mac auctions in an orderly fashion, they did highlight the unique issues that can arise when debt does not accelerate in a default scenario.

While restructuring language and non-acceleration are two unique issues to the municipal CDS market, another thing to consider is the economic circumstances under which the contract is triggered. We envision two very distinct scenarios:

exhibit 2		
Moody's Rated Municipal Defaults 1970-2006		
Type of Obligation	1970-2000	2000-2006
General Obligation	1	0
Non-General Obligations of State and Local Governments	3	1
Electric Power	2	0

Source: Moody's

The first is a situation that is not so much a true default as a technical default – a restructuring or a failure to pay that is rectified in a short period of time, but not short enough to avoid triggering the CDS contract, i.e., a failure to pay that is resolved shortly thereafter. In this scenario, the bonds are essentially money good and recovery could be very high, up to par + interest. This happened in 1987 in Baldwin County, Alabama, where the county was unable to make a principal and interest payment. Recovery on this was 100% of principal and interest 15 days later. Had the county not been able to pay this debt back for 30 days, this would have been a scenario in which a credit event would have been triggered. In scenarios like these, the biggest concern here is not recovery, but non-acceleration of debt, as discussed above. This is an example of a scenario in which the CDS contract could trigger despite the fact that the cash bonds would have been a fundamentally sound investment.

The second scenario is one in which the municipality is truly distressed and more akin to an actual default. In this situation, the bonds would essentially be worth very little, since

bondholders have no claim to a municipality's assets. The biggest concern here is recovery risk.

RECOVERY VARIABLES AND CONSIDERATIONS

One of the uncertain aspects of a hypothetical municipal default is recovery. Currently, the assumption used when pricing single-name CDS contracts, MCDX trades, FTDs on munis and any CDS unwinds is 80%. Moody's indicates that the average loss given default on state and local government general obligation credits is between 5 and 10%, implying recoveries in the 90s, though they caution (as do we) that this is based on limited available data.

exhibit 3	
Average Loss Given Default (LGD) Rates for Selected Municipal Sectors	
Type of Obligation	Loss Given Default (LGD)
State Government General Obligation	5%
Local Government General Obligation (including Instrumentalities, Territories and Commonwealths)	10%
Water / Sewer Enterprise	10%
State Revolving Funds	10%
State Lease Obligation and Special Tax	10%
Local Government Lease Obligation and Special Tax (including Instrumentalities, Territories and Commonwealths)	15%
Mass Transit	15%
Toll Roads and Bridges – Established	15%
Toll Roads and Bridges – Start Ups	55%

Source: Moody's

Again, we highlight the issue of non-acceleration. When the CDS contracts of Fannie Mae and Freddie Mac were triggered in September 2008, this issue introduced a degree of recovery uncertainty leading up to the auction settlement, since there were a number of longer-dated bonds in the market. If principal were accelerated and interest payments were to cease, as in the case in most defaults in the CDS market, recovery rightly tracks the expected amount recouped to the bondholder. In a situation in which the principal on the debt does not become due, duration takes on a larger role in final loss determination.

While an orderly recovery was eventually determined in the Fannie Mae and Freddie Mac auctions, we note that we are currently in a very low interest rate environment. If a similar situation were to arise and a CDS contract were triggered with non-accelerating debt in a high rate environment, especially in the longer duration municipal market, interest rates would play a much larger role in the recovery determination process.

exhibit 4

Recoveries in Selected Moody's Rated Municipal Defaults 1970-2006

Obligor	Year	Defaulted Debt	Recovery
Baldwin County (AL)	1988	\$6-\$8mm	Full P & I
Belfield – Special Assessment Bonds (ND)	1987	\$2.38mm	~55%
Polk County – Sports Facility Rev Bonds (IA)	1991	\$39mm	Called @ par
Orange County – Pension Ob (CA)	1994	\$110mm	Full P & I
Cicero Local Development Corp (NY)	2003	\$15.3mm	~10.3%
Washington Public Power Supply System (WA)	1983	\$2.25bn	40% (after class action suit)
Vanceburg Electric System – Greenup Hydro Project (KY)	1987	\$124.5mm	Principal plus accrued interest

Source: Moody's, Bloomberg

CORPORATES AND MUNIS IN THE ECONOMIC CYCLE

In Chapter 38 we highlighted a number of trade opportunities we saw in the municipal and corporate sectors via the CDS market. At that time we advocated going long 10yr MCDX 10 at 47bps and shorting 10yr CDX IG10 at 93bps by being carry neutral or slightly carry positive. Today we advocate closing out this pair trade.

Today we like going long (selling protection) CDX IG11 vs. shorting (buying protection) MCDX11, notional flat (or at a ratio of 1:1) for a variety of reasons. First, we believe that a lot of economic pain is already priced into the corporate market. We highlight that economic weakness is not limited to one asset class or another in this environment, and that the credit markets will have a negative impact on both corporations and municipalities alike; however, we do feel that corporates are further into this economic cycle than their municipal counterparts, and thus they have a greater degree of pain priced in already.

Further, we like the CDX IG11 portfolio, and think that a number of tail risk names that had plagued earlier series were removed at the roll, making this a more attractive portfolio. MCDX11 is compositionally the same as MCDX10, with a large percentage of revenue bonds that do not have the same taxing authority as their general obligation counterparts.

We like executing this trade in the 5yr maturity for the shorter duration. We acknowledge that there is still a great deal of uncertainty in the corporate CDS market that could cause this trade to underperform.

LOCALIZED MUNICIPALITIES VS. LOCAL INDUSTRY

For the most part, we've focused our municipal CDS coverage on the technicals, mechanics and broad fundamentals both on an outright basis and as a vehicle for cross-asset comparison. Municipalities, and by extension muni CDS, taken individually can be an interesting way to look at and express a view on local economic strengths and weaknesses. We've started to see CDS market participants differentiate individual credits, whereas just months ago muni CDS traded largely within a small range of one another.

exhibit 5

Indicative Muni CDS Trading Levels

Reference Entity	10yr CDS Spreads
California	175
New Jersey	125
New York State	135
Florida	125
Illinois	95
Massachusetts	109
Texas	85
Ohio	95
New York City	175
Michigan	155

Source: Morgan Stanley
As of October 23, 2008

For instance, municipalities in the Northeast would be more affected by a Wall Street downturn or eventual recovery. Energy corporates could be more correlated with Texas municipalities, and certain retailers and industries could track the CDS spreads of other regions as well. In an interesting parallel, we're starting to see some of this differentiation in the European sovereign market as well

Sovereign CDS Markets – A Corporate Perspective

January 29, 2010

Sivan Mahadevan
Phanikiran Naraparaju
Ashley Musfeldt

Excessive debt (relative to GDP), growing fiscal deficits (relating to the repair of the financial system), and fighting off a global recession have reduced the credit quality of and increased funding risks for many OECD member countries. As investors struggled with these risk factors during the past year, a sovereign CDS market for many of these countries has rapidly expanded and today, this market serves as an important way to implement hedges.

CDS spreads on European sovereigns have widened to the point where they are on par with, or even exceed corporate credit spreads in aggregate. While the market's concerns over Greece certainly contribute to this pricing, we note that sovereigns are a high "correlation" market, which has forced spreads wider across many other OECD sovereigns. As such, today's sovereign CDS market levels are now causing many investors to examine the likelihood of an OECD sovereign "default." In our view, an actual OECD sovereign default seems a less likely credit event trigger than a "restructuring." This stands in contrast to corporate CDS contracts where bankruptcy is the most common credit event.

Corporate CDS markets benefited from an evolution that started in the late 1990s, when both natural buyers and sellers of protection emerged to balance flows over time, and as such there is much useful history. In this chapter, we use this corporate CDS experience in an attempt to better understand how the sovereign CDS market might evolve over time as well as hypothesize on the impact a more active sovereign CDS market could have on the underlying cash markets. The presence of two disparate investor bases – one focused on rates and cash and the other on credit – creates a greater chance of dislocations in the sovereign CDS market, which can sometimes be a source of opportunities for those who know where to look. We summarize our key themes below and provide more detail in the remainder of this chapter.

What does "default" risk mean? "Default" risk is somewhat of a misnomer for OECD sovereign CDS, as both a "failure to pay" and "repudiation/moratorium" are unlikely (much less so than in emerging market sovereigns and also less likely than corporate bankruptcy). Restructuring as a credit event, and the deliverability of bonds and loans in this context, should be a bigger driver of OECD sovereign CDS valuation.

High correlation market. As credit quality in sovereigns and financials and systemic risks are all interconnected, we believe that sovereign CDS is a highly correlated market,

which can add a risk premium to CDS levels. While FTDs on sovereign-only baskets don't trade often, the limited color we do have is that correlation on these products can be in the low-70s to mid-80s.

Who is the natural seller of protection? We believe that the sovereign CDS market lacks a natural seller of protection today, and without this community of investors, flows can be less balanced. We see evidence of this in both absolute spread levels and cash vs. CDS basis relationships. Sellers of protection in corporates generally come from portfolio flows (indices, baskets and structured credit), and the sovereign CDS market would stand to benefit from these flows as well. To do this, long investors must get comfortable with credit event and deliverability language and consider the value of selling protection (away from storied sovereigns) in the context of corporate credit risk. We think there are several interesting ways to play the OECD sovereign CDS space:

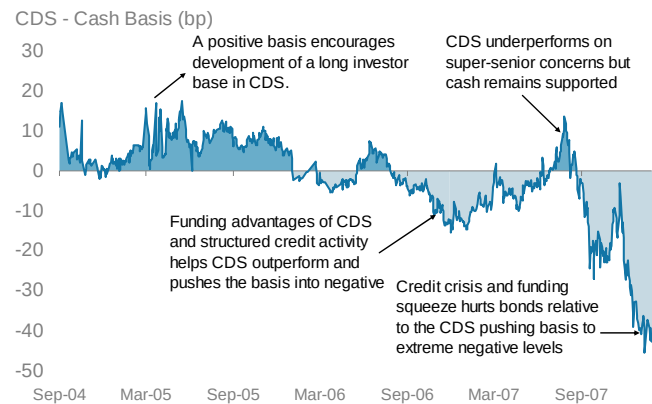
- **Basis trades:** Pair the long-maturity cheapest-to-deliver bonds with the shorter-dated CDS to position for extreme moves in either direction.
- **Single-name credit-linked notes:** We like these for their ability to de-link the rate and credit components of risk in a cleaner way, and take advantage of the positive basis.
- **2nd-nth-to-default baskets:** Due to their inherent exposure to systemic risk, sovereigns are a high correlation market. However, high correlations are already priced into the sovereign CDS markets, which is a compelling reason to take a look at selling 2nd-nth-to-default protection.
- **Adding sovereigns to corporate bespoke tranches:** Including OECD sovereign risk in corporate portfolios can benefit bespoke tranche investors, due to the increased spread value and added diversity.
- **Buy SovX receiver spreads:** We like these as a play for both convergence and a market rally, as market differentiation is already in place in the index spread and implied volatility levels.

A BIT OF HISTORY OF CORPORATE CDS FLOWS

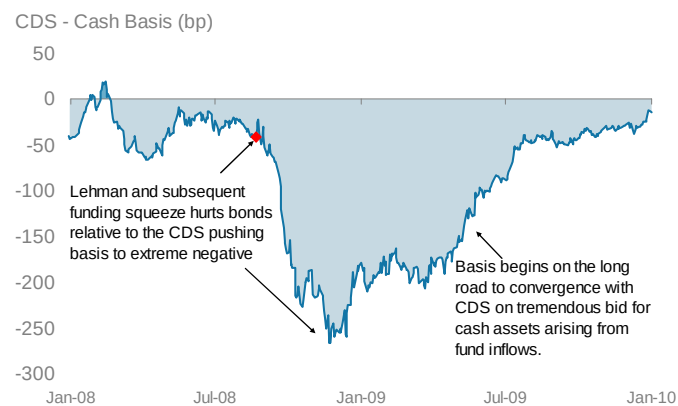
Corporate CDS markets were established in the mid-1990s, mainly as a way for banks to hedge their loan exposures, and much of the initial shape of the contract (restructuring risk, shorter maturities) was influenced by this community. But a one-way trade never makes markets, so it was critically important for the market to embrace CDS from an investment perspective as well. As such, in the late 1990s and early 2000s, CDS took a form that was more "bond-like" and subsequently the contract became more investor friendly.

exhibit 1

Evolution of Corporate CDS-Cash Basis (2004-07)



Evolution of Corporate CDS-Cash Basis (2007-10)



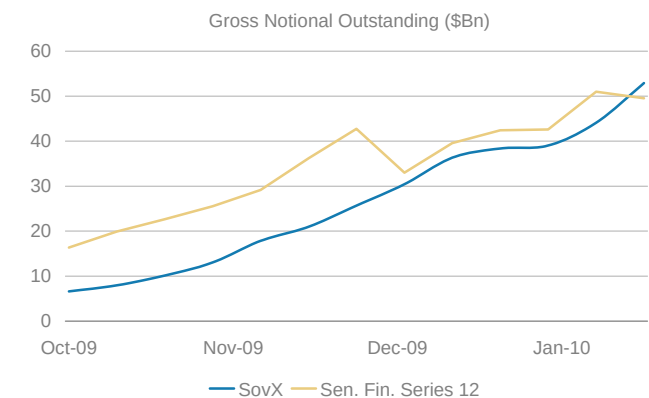
Source: Morgan Stanley Company Research

What followed were large investment flows through the sale of protection in corporate CDS contracts, mainly in portfolio form through structured credit and market indices. This natural seller of protection was an important factor in market growth by providing a balance in the market. Eventually this flow became a dominant force for CDS market performance, at least until the systemic risk period of the recent crisis, though even during the systemic crisis the cash products suffered more from their inherent funding disadvantage. The most recent changes to the corporate CDS market – the SNAC protocol – made the contract more fungible by standardizing coupons, requiring adherence with ISDA protocols, and addressing the concerns surrounding restructuring risk.

As the sovereign CDS market develops, we argue that it needs to find the same balance. As with corporate CDS, the first phase of activity seems to have been skewed towards an investor base motivated by hedging. The launch of SovX (the European index of 15 equally weighted sovereign entities) has helped improve liquidity in the sovereign space by increasing synthetic risk-taking that had been lacking up until then.

exhibit 2

Market Volumes: SovX Overtaking Financials



Source: Morgan Stanley Research, DTCC

SovX has been seeing a steady pickup in activity relative to corporate CDS, and is now ahead of Financials (see Exhibit 3). Although volumes are still low compared to iTraxx Main and the sovereign cash bond market, we expect volumes and liquidity to grow throughout the year. Increased volumes and liquidity in the underlying index should bolster flows and risk taking in the SovX options product, providing yet another avenue for investors to express views on the sovereign market. Though CDS volumes are certainly increasing, the small size of the long investor base in sovereign CDS (relative to that in the cash market) is one reason why the basis between sovereign CDS and bonds is so positive (CDS trading wide to bonds) and perhaps will remain so. A positive basis was common to the corporate CDS market in the early days due to a similar imbalance between buyers and sellers of protection. The next step on the path to market maturity for sovereign CDS is to find the natural seller of protection, which we explore in detail further on. First, however, we address elements in the CDS contract that are unique to sovereigns.

exhibit 3

Overview of Standard Credit Events for Various CDS Contracts

	Bankruptcy	Failure to Pay	Failure to Pay (Grace Period Appl.)	Obligation Acceleration	Repudiation/Moratorium	Restructuring (Old R)	Restructuring Maturity Limitation and Fully Transferable Obligation (Mod R)	Modified Restructuring Maturity Limit and Conditionally Transferable Obligation (Mod Mod R)	Multiple Holder Obligation Required? (If Restructuring)
Asia Sovereign	N	Y	N	N	Y	Y	N	N	Y
Singapore Sovereign	N	Y	N	N	Y	Y	N	N	Y
Latin America Sovereign	N	N	Y	Y	Y	Y	N	N	N
Emerging European & Middle Eastern Sovereign	N	N	Y	Y	Y	Y	N	N	N
Western European Sovereign	N	Y	N	N	Y	Y	N	N	Y
Japan Sovereign	N	Y	N	N	Y	Y	N	N	N
Australia Sovereign	N	Y	N	N	Y	N	Y	N	Y
New Zealand Sovereign	N	Y	N	N	Y	N	Y	N	Y
U.S. Municipal Full Faith And Credit	N	Y	N	N	N	Y	N	N	N
U.S. Municipal General Fund	N	Y	N	N	N	Y	N	N	N
U.S. Municipal Revenue	N	Y	N	N	N	Y	N	N	N
North American Corporate	Y	Y	N	N	N	N	N	N	Y
European Corporate	Y	Y	N	N	N	N	N	Y	Y
Australia Corporate	Y	Y	N	N	N	N	Y	N	Y
New Zealand Corporate	Y	Y	N	N	N	N	Y	N	Y
Japan Corporate	Y	Y	N	N	N	Y	N	N	N
Singapore Corporate	Y	Y	N	N	N	Y	N	N	Y
Asia Corporate	Y	Y	N	N	N	Y	N	N	Y
Subordinated European Insurance Corporate	Y	Y	N	N	N	Y	N	N	Y
Emerging European Corporate LPN	Y	N	Y	N	Y	Y	N	N	Y/N
Emerging European Corporate	Y	N	Y	N	Y	Y	N	N	Y/N
"Latin America Corporate B"	Y	N	Y	Y	Y	Y	N	N	N
"Latin America Corporate BL"	Y	N	Y	Y	Y	Y	N	N	Y

Source: ISDA, Morgan Stanley Research

WHAT IS “DEFAULT” RISK FOR A SOVEREIGN?

The ultimate value of a CDS contract is derived from the events that can trigger it, the bonds and loans that are deliverable upon this event, and their expected recovery. Market sentiment and technicals can greatly influence CDS spreads, but it is important to consider contract language to determine value. The vast majority of credit events in the corporate world are bankruptcies, and the CDS contract is now well tested for such events. For most OECD sovereigns, the credit events are failure to pay, repudiation/moratorium and restructuring. We think restructuring is the primary trigger event that should drive CDS valuation.

The ISDA 2003 credit derivatives definitions characterize restructuring credit events as a reduction in coupon or principal, a deferral of an interest or principal payment, a change in the ranking of priority of payments, or certain changes to the currency used to pay interest and principal. The currency criteria states that a redenomination of debt into G7 currencies is permitted, as is the redenomination into any OECD currency as long as the local-currency long-term debt rating is AAA or higher by S&P, Moody’s, or Fitch. This latter point is important for EMU members with sub-AAA ratings as an event where they leave the EMU and redenominate their debt into a new local currency would trigger a CDS credit event via restructuring. We note that for most OECD sovereigns, these credit event tests are permissible on all borrowed money (including bi-lateral lending agreements).

Beyond credit events, the other important factor is deliverable obligations. Most OECD sovereigns use “Old R” restructuring under which maturities of up to 30 years are deliverable if a restructuring credit event occurs, allowing interest rate levels to determine the cheapest-to-deliver bond if bonds do not accelerate (i.e., become due immediately as they would in a corporate bankruptcy). Thus it is conceivable for a CDS credit event to trigger from a restructuring where all the bonds of the sovereign remain outstanding and continue trading based on their interest rate risk. This issue is not new to the CDS market. In 2002 the restructuring of a Xerox obligation brought this type of interest rate risk to light, driving the introduction of ModR to the 2003 ISDA definitions. This is an issue for the US muni CDS market as well (Chapter 37).

WERE THE GSES A RELEVANT EXAMPLE?

In September 2008, the US government put both Fannie Mae (FNM) and Freddie Mac (FRE) into conservatorship, triggering a bankruptcy credit event, as they were public corporations that traded with standard corporate CDS language. However, the bonds and loans neither defaulted nor accelerated, so the credit event acted more like a “benign” restructuring. As a result, there was considerable discussion in the market about bond deliverability afterwards, given that duration and coupon structure could determine the cheapest-to-deliver (zero-coupon instruments in particular).

In the end, ISDA and the dealer community limited deliverables to prevent significant interest rate risk exposure (recoveries were above 90%), and the event ultimately helped form the standardized ISDA determination committee that exists today for any credit event. For sovereign CDS, we note that the ISDA Determination Committee could also limit extreme cheapest-to-deliver opportunities in a credit event situation, but it is not possible to predict the potential outcome.

CORPORATE BASIS VS. SOVEREIGN BASIS

With this background, we jump into the most basic measure of CDS value: the CDS-cash basis. The sovereign basis highlights the many subtle but important differences between the cash and CDS market in this asset class, as the completely disparate investor bases on each side have led to different market technicals. While the sovereign cash product is dominated by interest rate focused investors, the sovereign CDS product has an investor base that is much more credit focused. Unlike corporate bonds, which underperformed CDS due to the withdrawal of leverage from the system and lack of funding, sovereign debt has benefitted substantially from a flight to quality bid, even as sovereign CDS widened from historically tight levels. We don’t expect the sovereign CDS flows to challenge the domination of the cash-focused investor base, but from a corporate credit investor perspective, the positive basis highlights the attractiveness of Sovereign CDS as a credit product.

In the corporate world, an improvement in liquidity and funding conditions has helped the cash-CDS basis normalize from extreme negative levels. In contrast, the sovereign basis has been positive for much of the crisis and remains so. There are some complexities to be considered in calculating the sovereign basis stemming from the correlation between bond performance and the currencies of the bonds and/ or the currencies of the CDS contract – an issue one generally does not have to worry about in the corporate market. We note that a CDS contract is “specified” in a given currency, and all cash flows into and out of the contract are in that currency, even if bonds that are delivered are in a different currency (i.e., the face amount of bonds delivered into the contract must be exchanged at the then currency rate). As such there is little settlement risk with respect to currency fluctuations, but currency movements can help determine value for the holder of a basis package. A typical sovereign CDS could end up with three types of deliverables, though home currency bonds would constitute the bulk of the deliverable debt.

- **Home currency bonds.** There is a strong correlation between the currency and prices of deliverables. This is an issue for sovereigns with their own currencies but also for EMU countries. Deterioration in performance of bonds is also likely to be accompanied by an underperformance of the currencies. The CDS contract is usually in a different currency than the home currency; thus home currency

bonds would likely be the cheapest-to-deliver in a credit event, all else being equal.

- **Bonds with same currency of the CDS contract.** A small portion of sovereign debt is sometimes in the currency of the contract (often the USD or EUR), which would still be deliverable. If the home currency bonds are cheapest-to-deliver and closely track CDS, bonds in the same currency as the CDS contract should trade at a premium, or at a more positive (or less negative) basis to the CDS than home currency bonds.
- **Deliverables with currencies uncorrelated to the price of the bonds or the contract currency.** These bonds likely trade with a basis that lies between the home currency bonds and the CDS contract basis.

Given most sovereign debt is denominated in the home currency, we adapt the CDS-cash basis model we use for corporates to take into account currency forwards to calculate a fair basis. We will discuss the sovereign basis in more detail in a future note.

GETTING LONG ASYMMETRY: MOVING FROM A YIELD MINDSET TO A PRICE MINDSET

Although the basis is positive, there are situations in which an investor might prefer taking risk via the cash bond rather than the CDS. One of the earliest trades we recommended in the nascent corporate CDS world should appeal to sovereign investors today (see *Getting Long Asymmetry*, April 14, 2003). The trade involves a long-short package: we buy a low dollar-price, longer-maturity bond (often amongst the cheapest-to-deliver bonds) paired with shorter-dated protection.

exhibit 4

Hypothetical Long CTD Bond vs CDS

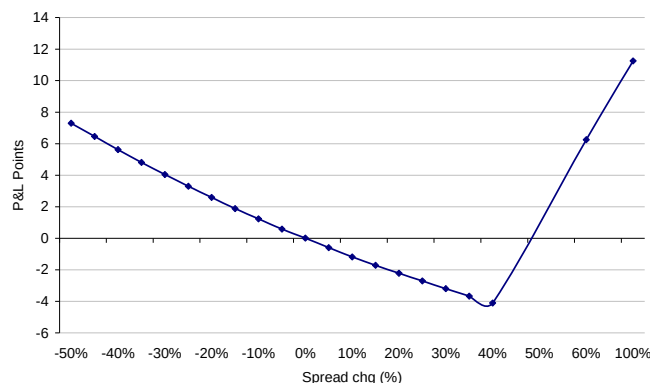
	Cpn	Maturity	Price	Z-spread	Duration
Bond	4.0	Dec-35	78.0	250	13.0
CDS	1.0	Mar-15	89.3	350	3.9

Source: Morgan Stanley Research

We take a generic example of some of the cheapest priced bonds, which typically have a maturity of 15-30 years and are therefore deliverable into the CDS contract. The key is the transition of the investor from pricing bonds on a yield basis to a price basis, limiting downside in bear scenarios. In a rally, the duration difference between the bond and CDS would limit the downside. In an extreme widening/credit event the package pays ~11.5 points and there could be a convergence of the CDS upfront to the bond par discount. The impact of convergence of the CDS (350bp) to the Z-spreads (250bp) in a rally should be offset by the duration differences between the two instruments.

exhibit 5

Hypothetical P&L Profile of the Asymmetry Trade



Source: Morgan Stanley Research

In this uncertain environment, where sharp spread tightening is just as likely and may immediately follow spread widening, the trade is a good way of positioning for upside in either extreme but is a negative carry trade with a negative P&L for small moves wider and exposed to rate moves in the scenario in which spreads rally considerably. Note that the bond (EUR) and the CDS (USD) are denominated in different currencies and as such run currency risk. We highlight that bonds have interest rate risk, which can be hedged.

CORRELATION LESSONS AND FTDS

Apart from this, most of the ideas from here try to take advantage of today's elevated sovereign spreads in the context of corporate credit portfolios. In the nascent phase of the corporate CDS market, one product that was popular was the first-to-default trade on small baskets (see *Levering the Lightly Levered*, April 14, 2003), the predecessors to today's large basket synthetic CDOs.

Sovereign credit risk has both an idiosyncratic and systemic component to it, as does corporate credit, which we learned in the last cycle. We believe the systemic component is even greater in the sovereign market, as many investors use sovereign CDS as a systemic hedging instrument. As such, portfolios of sovereign credit risk can be perceived to be highly correlated, which can impact pricing by allocating more risk to the "senior" tranches. If this phenomenon is accurately reflected in market pricing for baskets, it is negative for the FTD protection seller, but positive for the seller of 2nd-nth-to-default protection, as higher correlations will put disproportionate risk in the non-first-loss part of the trade.

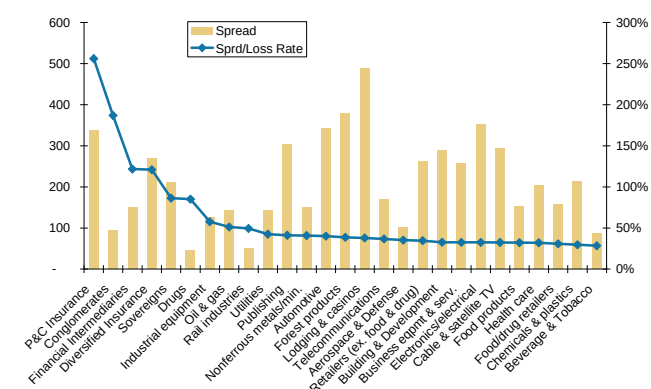
FINDING A BID IN CORPORATE BASKETS

Beyond small basket structures, sovereign CDS could also enjoy a long base from a return of bespoke structured credit flows. Due to their formerly tight trading ranges, they were not very popular constituents of the 2005-2007 vintage bespoke deals, and in fact didn't become very attractive until

late 2007, early 2008. While new structured credit flows have been fairly miniscule, if and when these flows recover, sovereigns could find a natural seller of protection. Exhibit 6 shows that sovereign CDS, following their recent underperformance, are now much more attractive for inclusion in new bespoke portfolios. We highlight that even if sovereigns faced a 2-notch downgrade from current levels they would still look attractive in a portfolio context, indicating how far sovereigns have widened in anticipation of credit deterioration.

exhibit 6

Sovereigns in a Portfolio Context



Source: Morgan Stanley Research

In Exhibit 7, we show a sample list of standard A / BBB credits in the iTraxx Main portfolio, and contrast these with sovereigns, indicating the pickup in spread *and* rating that can be achieved by including sovereign credit in a portfolio. While there are certainly other credits that may be more optimal than this, these are illustrative of credits that would be included to show that sovereigns are worthy of consideration.

exhibit 7

Sovereign vs. Corporates: Spreads and Ratings

Sovereign			Corporate		
Reference Entity	Rating	Spread	Reference Entity	Rating	Spread
Kingdom of Belgium	AA+	64	Carrefour	A	64
United Kingdom	AAA	82	E.ON AG	A	63
Republic of Austria	AAA	88	Solvay	A	60
Republic of Italy	A+	121	Publicis Groupe SA	BBB+	75
Kingdom of Spain	AA+	138	Sodexo	BBB+	62
Ireland	AA	149	Linde AG	BBB+	59

Source: Morgan Stanley Research

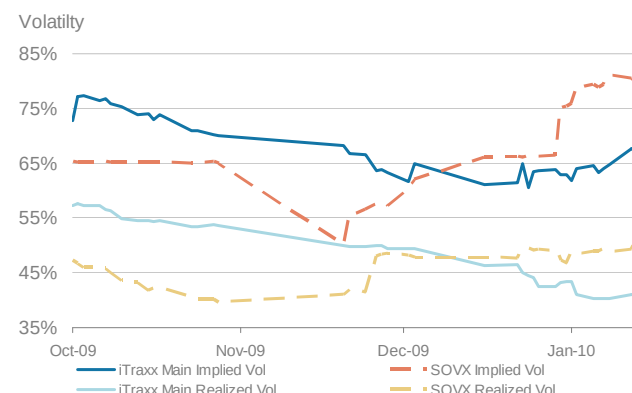
Additionally there are further benefits of including sovereigns in bespoke portfolios as they can increase diversification and have lower correlation with corporates under the new S&P ratings criteria, though their sometimes lower recovery can be a detracting factor sometimes. To demonstrate, we looked at a standard new issue bespoke in the market today and swapped out one of the names for one of the sovereigns listed in Exhibit 8. One name swap was worth around 0.05% in terms of subordination, which, while small, could be meaningful if done several times.

SOVX OPTIONS

The uncertainty in sovereign valuation is also reflected in the options market, where the implied volatility for SovX is now at a 15pt premium to iTraxx Main vol, (although both indices trade largely in the same range). Given this backdrop of MTM risk in either direction, along with implied volatility levels that are still well above recent realized vols, we believe vol neutral directional strategies will have the most appeal for investors. In particular, we think that bullish capped downside strategies, like receiver spreads for instance, make the most sense. We like receiver spreads as a low cost way of taking long exposure with minimal volatility exposure in a whipsaw market. This strategy would be more cost effective than being outright long the index (or receivers) and would provide a good payoff in case of a modest move tighter in indices, capping downside at the same time.

exhibit 8

SOVX Implied: Premium to Implied and Realized Vol



Source: Morgan Stanley Research

Section G

Glossary

Glossary

ADJUSTED BASIS

Difference between the adjusted Z-Spread of a bond and a CDS premium. The Z-Spread is adjusted for bonds trading above or below par.

ALPHA

Refers to the non-systematic return that an investor adds to a portfolio, relative to its benchmark.

AMERICAN-STYLE OPTION

An option that can be exercised by the holder at any time from the date of purchase up to the expiry date. American-style option premiums are typically greater than European-style options due to the increased flexibility afforded to the option holder.

AMORTIZING TERM LOAN

A term loan is a funded loan type where once borrowed funds are repaid, the funds cannot be borrowed again. Under an amortizing term loan agreement, the borrower pays back the principal over the life of the loan – amortization is usually back-ended.

ARBITRAGE CDO

CDO designed to take advantage of the funding gap (i.e., earn the spread between the yield of the underlying assets and the cost of CDO liabilities).

ASSET SWAP SPREAD

The spread to the swap curve on the floating-leg payments of an asset swap, in exchange for fixed cash flows.

ASSET SWAP

A swap of the returns or cash flows of two assets. They are commonly used to convert fixed cash flows to floating.

ASSIGNMENTS

A secondary trading convention in the loan market. When a loan is traded on an assignment basis, the assignee (buyer) becomes the direct signatory to the loan and receives payments directly from the administrative agent. In the event of default, assignees will have complete rights and access to private information. Assignments usually require the consent of the borrower and the agent on a not-to-be-unreasonably-withheld basis. See also “Participations”.

ATTACHMENT POINT

Defines the amount of losses in the reference pool of assets that needs to occur before the tranche starts to experience losses. Can be expressed as a percentage or as an absolute value.

AT-THE-MONEY

Refers to when the strike spread is equal to the market spread for both payer and receiver options.

AVAILABLE FUNDS CAP

The interest on US home equity loan securitizations is capped at a certain pre-specified rate called the available funds cap rate. This feature introduces interest rate risk to such securitizations and credit default swaps where the reference obligation is a home equity loan.

BACK-END DEFAULT RISK

The default risk of a portfolio that is weighted towards the maturity date of the tranche, rather than in the near future. Opposite of Near-Term Default Risk.

BALANCE SHEET CDO

CDO created to allow the sponsor (typically a bank) to remove credit exposure from its balance sheet in order to achieve regulatory capital relief. Balance sheet CDOs are typically static.

BASE CORRELATION

Base correlation methodology considers all non-equity tranches as a portfolio of long 0% to the detachment point tranche and short 0% to the attachment point tranche. The implied correlation figures calculated in this fashion from given tranche prices are referred to as base correlation.

BASIS (OR CDS-BOND BASIS)

Difference between CDS premium and a selected spread measure of a bond. A positive basis implies that the prevailing default swap premium is greater than the spread on the bond.

BERMUDAN-STYLE OPTION

An option that can be exercised by the holder only on a series of preset dates up to a final maturity date (similar to a hybrid of American- and European-style options).

BERNOULLI DISTRIBUTION

A probability distribution of an event with two possible outcomes. Defaults can be represented by a Bernoulli distribution, where p equals the probability of default and $1-p$ equals the probability of a no-default scenario.

BESPOKE

Typically refers to a customized portfolio/collateral pool (may be customized by either the buyer or seller).

BETA

Measures the price sensitivity of a security to movements in the broad market. If beta is greater than one, any change in the market will result in a magnified price move (multiplied by the beta) of the security.

BINOMIAL EXPANSION TECHNIQUE (BET)

Valuation methodology used by Moody's to rate cash CDOs. Reduces a given portfolio to a number of uncorrelated assets (see Diversity Score) and calculates the expected loss on each of the tranches by varying the number and probability of defaults. The expected loss values are then used to assign ratings to the various tranches.

BULLET BOND

A bond where the repayment of principal occurs only at the maturity date. Also known as a Non-Callable Bond.

CALLABLE BOND

A bond where the issuer holds the option to redeem the bond before its maturity date, typically at a preset price and at/after a preset date.

CANCELABLE CDS

CDS in which the protection buyer is long the option to cancel protection, similar to a callable bond

CANCELABLE SWAP

A CDS where the protection buyer has an option to cancel the CDS, usually after an initial period when the swap cannot be cancelled. For example, a five-year CDS that can be cancelled anytime after two years.

CARRY

The differential between the yield on an investment and the cost of funding the position. Also the difference between buy and sell legs in a relative value relationship. Typically measured in basis points.

CASH CDO

A stand-alone special purpose vehicle that invests in a diversified pool of assets in a funded fashion, as opposed to using credit derivatives (unfunded). Investments are funded through the issuance of multiple classes of securities.

CASH FLOW CDO

A CDO whereby the principal and interest of the liabilities of the structure are paid using the cash flows generated by the underlying collateral.

CASH FLOW METHODOLOGY

Values a CDO by discounting the scheduled cash flows of the various tranches over one or multiple paths. Also incorporates default, recovery and reinvestment assumptions as potential impacts on the expected cash flows.

CASH FLOW TRIGGERS

Structural features of a cash CDO that can determine and change payouts to different tranches based on the performance of the underlying collateral.

CASH SETTLEMENT

The settlement of a CDS contract whereby the protection buyer pays an amount equal to the market value of a deliverable obligation and receives par upon the occurrence of a credit event.

CDO EVALUATOR

Proprietary S&P model used to rate various CDO tranches. Uses a Monte Carlo simulation to generate a probability distribution of defaults for each asset, taking individual credit ratings, number of assets, industry concentration and default correlation into consideration.

CDO SQUARED

A CDO whose underlying collateral pool includes tranches of other CDO structures.

CDX HVOL INDEX

A sub-index of CDX NA, comprising 30 of the highest-volatility credits in the index, as determined by a consortium of dealers.

CDX NA HY INDEX

The synthetic index of 100 non investment grade entities domiciled in North America. Composition of the index is rules based and determined by a dealer liquidity poll. CDX HY index is rolled over every six months in March and September.

CDX NA IG INDEX

The synthetic index of 125 investment grade credits in North America, as determined by a consortium of dealers. Rolls over on September 20 and March 20 of each calendar year.

CHEAPEST-TO-DELIVER OPTION

The option held by the protection buyer to deliver the cheapest deliverable obligation available to the protection seller when a credit event occurs.

CMCDS

Constant Maturity CDS is a default swap where the premium is reset to equal a fixed percentage of the prevailing plain vanilla default swap for a certain term.

COLLATERAL MANAGER

Party responsible for making the investment and trading decisions in the reference portfolio. Decisions are typically made within pre-designated guidelines.

COLLATERALIZED BOND OBLIGATIONS (CBO)

A securitized pool of assets comprised predominantly of bonds. Examples of collateral may include investment grade bonds, high yield bonds, convertible debt, mezzanine debt or emerging market debt.

COLLATERALIZED LOAN OBLIGATIONS (CLO)

A securitized pool of assets comprised predominantly of loans, which may be investment grade, high yield or both.

COLLATERALIZED MORTGAGE OBLIGATIONS (CMO)

A securitized pool of assets comprised predominantly of mortgages.

COMPOUND CORRELATION

The implied correlation of a tranche, given its market price. Certain tranches may have more than one solution for a given price.

CONSTANT PROPORTION DEBT OBLIGATION (CPDO)

A spread and market value based rated structure, where the proceeds from the note are leveraged and invested in risky assets (mainly credit default swaps) to generate par plus coupon. The structure unwinds when present value of all future cash payments is generated (i.e., the “cash-in” date) or at maturity. The entire CPDO portfolio can also be unwound if the net asset value falls below a predetermined trigger level.

CONVERTIBLE BOND

A bond where the bondholder has the option to convert the bond into a preset number of a company’s shares.

CONVEXITY

The changes in delta given a change in underlying spreads. Typically measured as a ratio of the mark-to-market of a tranche, given a move of 100 bp in underlying spreads to the PV01 of a tranche multiplied by a factor of 100.

COPULA FUNCTION

Function used to transform a given set of loss distributions for individual credits into a joint loss distribution for a portfolio of credits.

CORPORATE LEVERAGE

A financial ratio that captures the amount of debt vs. the size of the company’s ability to pay, for example, a firm’s total debt divided by its EBITDA (earnings before interest, taxes, depreciation and amortization over the last 12 months). Typically, the higher the leverage ratio, the riskier the credit will be, given the insufficiency of the firm’s operating earnings to support its debt load.

CORRELATION SENSITIVITY

See RHO.

CORRELATION SKEW

When different tranches referencing the same underlying portfolio trade at different implied correlations. Also refers to the difference in correlation for attachment and detachment points of a tranche.

COVERED CALL

A call option sold by an investor who is already long the underlying asset.

CPDO

Constant Proportion Debt Obligation gives leveraged exposure to credit portfolios with a fixed coupon and no upsides and without offering principal protection. CPDO increases leverage when the strategy suffers losses and reduces leverage when the strategy has made gains.

CPPI

Constant Proportion Portfolio Insurance is a technique for levering up investments and provides full or partial principal protection by investing in a combination of high-risk and low-risk products (zero coupon bond or cash deposit). CPPI reduces leverage when the strategy suffers losses and increases leverage when strategy has made gains.

CREDIT ENHANCEMENT

Feature of a CDO designed to decrease the credit risk of the structure. Forms of credit enhancement may include overcollateralization, guarantees from insurers (see Wrapped Tranches) and letters of credit.

CREDIT EVENT

An event that materially affects the reference entity and triggers the termination of the CDS. Examples of credit events can include bankruptcy, failure to pay, obligation acceleration, repudiation, moratorium and restructuring.

CREDIT I/O

The residual default risk (in the form of credit risky residual coupon streams) resulting from an unwind of off-market credit default swap with a standard par default swap.

CREDIT-LINKED NOTE (CLN)

When credit risk is embedded into a note that pays a fixed or floating coupon reflecting the riskiness of a credit. See Funded Form.

CROSS SUBORDINATION

Provision of a CDO-Squared that allows the unused subordination of an inner tranche to be transferred to another inner tranche for the purpose of absorbing losses from the underlying inner CDO.

CURVE-ADJUSTED BASIS

Basis calculation that considers the full CDS curve, as opposed to just the CDS corresponding to the bond’s maturity.

CURVE-ADJUSTED PAR SPREAD (CAPS)

A spread measure that extends on curve-adjusted Z-Spread and adjusts for any discount/premium on the bond by making a specific recovery assumption.

CURVE-ADJUSTED Z-SPREAD

An adjusted Z-spread measure that takes into consideration the shape of the credit curve over Libor, instead of assuming a flat credit curve (as per the generic Z-Spread metric).

DEFAULT PROBABILITY

The probability that an issuer will default on its debt obligation. Default probabilities calculated using bond or CDS pricing are market implied or risk-neutral default probabilities, and are usually different from empirical probabilities.

DEFAULT PV

The present value of all expected default losses to the maturity of a credit derivative contract.

DEFAULT THRESHOLD

Used in structural models to signal the limit to which a firm's assets can decline before defaulting. This limit is equal to the firm's liabilities, and can be modeled to be static or dynamic over time.

DEFAULT-ADJUSTED CASH FLOWS

The expected cash flows of a CDO that are modified to reflect the likelihood of default. Multiply the probability of default by the expected cash flow and discount using LIBOR to derive a present value.

DELEVERING

In a CDO structure, paying down of liabilities with excess spread.

DELIVERABLE DEBT OBLIGATION

Bonds or loans that are eligible for delivery by the protection buyer when physical settlement is specified.

DELTA MIGRATION

Refers to the instability of tranche deltas with respect to movements in index spreads, changes in correlation and the passage of time.

DELTA

The resulting change in a derivative's price given a change in the underlying security's price. Also referred to as Hedge Ratio. Tranche sensitivity to changes in underlying portfolio spread, measured as a ratio of tranche PV01 to portfolio PV01. The delta of a tranche generally increases as the subordination of the tranche decreases.

DELTA-NEUTRAL

When the value of a tranche is unaffected by small changes on underlying spreads in the underlying index because of an offsetting hedge using the index or other tranche.

DESIGNATED PRIORITY

Designates the lien status of a loan, i.e., whether the loan is first lien or second lien. Claims on second liens rank behind those of the first-lien loans in an event of liquidation.

DETACHMENT POINT

Defines the amount of losses in the reference pool of assets that need to occur for a complete loss of principal for the tranche. Can be expressed as a percentage or as an absolute value.

DISTANCE-TO-DEFAULT

The difference between a firm's asset value and its liabilities, measured in units of standard deviation of the asset value. Effectively represents the number of standard deviations that a firm is from default and can be used to compute the probability of default.

DIVERSITY SCORE (N)

Moody's-developed numerical statistic (N) reflecting the number of uncorrelated homogenous assets with identical default probabilities and equal par values, reduced from a given portfolio.

DIVIDEND-DISCOUNT MODEL

An equity valuation model that compares the current stock price to the present value of all future expected dividends from a company, using a discount rate reflecting the risk-free rate and the appropriate risk premium for the company.

DOUBLE BINOMIAL METHOD

Used to value CDOs comprised of two distinct pools of underlying assets. Each pool is assumed to be uncorrelated and each has a distinct default probability and diversity score.

DURATION-WEIGHTED

The size of the offsetting positions of a trade determined by the duration of each position.

EQUITY IO

Interest-only structures, taking exposure to equity tranches by receiving a coupon in exchange for an upfront payment (which is never paid back). As the underlying incurs losses, the notional on which the large coupon is based gets written down, reducing the coupon.

EQUITY PO

Principal Only structures where an investor pays an upfront price (say X% of notional) and receives 100% of notional less any principal losses (on an equity tranche) due to defaults at maturity. There are no interim cash flows.

EQUITY

The most subordinated tranche of a CDO. Also known as a first-loss piece, as any losses experienced by the underlying collateral pool will reduce the notional of the tranche before flowing upwards to the other tranches. Typically trades on a points upfront basis for synthetic indices.

EUROPEAN-STYLE OPTION

An option that can be exercised by the holder only on the expiry date. European-style option premiums are typically lower than American-style options due to the reduced flexibility afforded to the option holder.

EXPECTED DEFAULT FREQUENCIES (EDFTM)

A predictor of issuer default over a specific term, generated by Moody's proprietary KMV Model. See Structural Model.

EXTENSION RISK

The risk that a callable security's duration is increased, due to the lack of prepayment by the issuer (typically driven by increased interest rates or spreads).

FACTOR MODEL

A statistical model that uses regression to quantify the contribution of various characteristics of the issuer/bond to the total spread of the bond. Sample characteristics may include Debt/EBITDA, duration and stock volatility.

FALLEN ANGEL

A bond that was originally issued an investment-grade rating but has since been downgraded to below investment-grade due to deteriorating credit quality. Opposite of a Rising Star.

FIRST-LOSS PIECE

See Equity.

FIRST-TO-DEFAULT BASKET

A levered investment on the default risk of a reference portfolio where the buyer of protection is protected against the first default of the basket, at which point the contract is settled and terminated. Basket size typically ranges from four to ten credits and can be funded or unfunded.

FUNDED FORM

When credit risk is embedded into a note that pays a fixed or floating coupon reflecting the riskiness of a credit and the protection seller pays the full principal amount at the inception of the trade. If a credit event occurs, losses result in a writedown of the principal; otherwise known as a Credit-Linked Note.

FUNDING GAP

The difference between the yield of the assets in the reference pool and the yield of the CDO's liabilities.

HAZARD RATE

The forward probability of default over a specified time horizon. Can be inferred from CDS premiums or asset swap spreads.

HEDGE RATIO

See Delta.

HOLDING COMPANY (HOLDCO)

A company that holds a controlling interest (usually has voting control) of another company. Also referred to as the parent company. Typically has a lower recovery relative to the operating company (see OPCO).

IDIOSYNCRATIC RISK

Risk that is specific to a security or issuer/company and unrelated to market risk. Such firm-specific risk can be diversified away. Opposite of Systemic Risk.

I-GAMMA

Sensitivity of a tranche value to jump-to-default risk or changes in the spread distribution of the underlying portfolio.

IMPLIED EQUITY VOLATILITY

The standard deviation of a stock's return, as implied by its option premiums. Often calculated using the Black-Scholes model.

IMPLIED FORWARD CDS

Default swap rates between two future dates implied by the current CDS curve.

IMPLIED OPTION VALUE

The value of the embedded option in a bond, as implied by the bond's price, a volatility assumption and the risky swap curve (thus reflecting the default risk inherent in the bond's cash flows).

IMPLIED SPREAD VOLATILITY

The standard deviation of a corporate security's spread changes, as implied by its spread option premiums.

INDEX ARBITRAGE

Difference between the traded index and the intrinsic index (calculated from single names). The intrinsic value of the index is calculated using a duration-weighted average, rather than a simple average of the underlying single-name spreads. A negative index basis implies that single names are trading cheaper than the index, and vice versa.

INNER CDO

Refers to the CDO tranches comprising the reference portfolios in a CDO-Squared.

INSTITUTIONAL TERM LOAN

A term loan is a funded loan type wherein once borrowed funds are repaid, the funds can not be borrowed again. Institutional term loans are usually taken out by leveraged borrowers (i.e., non-investment grade borrowers with debt/EBITDA greater than 2.0x) and repaid at maturity (there might be some minimal, back-ended principal repayments). They are normally longer dated compared to amortizing term loans (five to seven years) and may be prepaid at any time at par. Multiple tranches with varying maturities can co-exist within a facility.

INTEREST COVERAGE TEST

Ratio of the expected interest generated by the underlying collateral pool to the interest due on the given tranche, plus the interest due on all tranches senior to it.

INTERPOLATED SWAP SPREAD (I-SPREAD)

Also known as yield-on-yield spread. The spread of a security relative to the swap curve, calculated by taking the yield to maturity of a bond less the interpolated yield on the swap curve.

IN-THE-MONEY (OPTIONS)

For payer (payor) options, it refers to when the strike spread is less than the market spread on an underlying reference entity. For receiver options, the strike spread is greater than the market spread on the entity. In either case, the option holder will be incentivized to exercise the option.

IN-THE-MONEY (TRANCHES)

Refers to tranches where the expected loss on the index is above the tranche detachment point.

INVERTED CURVE

When the short end of the curve is at a higher level than the long end of the curve, such that the curve has a negative slope. Typically signals near-term risks with positive expectations in the long-term.

ISDA

The International Swaps and Derivatives Association is the trade association representing participants in the derivatives industry, covering swaps and options across all asset classes (interest rate, currency, commodity and energy, credit and equity). Its publications include credit derivatives definitions, which have improved standardization of CDS contracts.

ITRAXX EUROPE

The synthetic index that consists of 125 equally weighted credit default swaps on European entities. Composition of the index is rules based and determined by a dealer liquidity poll. iTraxx Europe is rolled over every six months in March and September.

JUMP TO DEFAULT RISK

The risk of a credit spread gapping significantly wider to imply a high probability of default in the near term, as opposed to a gradual spread widening.

JUNIOR TRANCHE

Tranche that is subordinated to the senior tranche in terms of its claim on the underlying collateral pool. Ratings and spreads on junior tranches will reflect the increased credit risk of the securities. Junior tranche investors are typically long correlation.

KMV MODEL

A Merton-based quantitative structural model proprietary to Moody's rating agency. Analyzes Expected Default Frequencies (EDFTM) by comparing the value of a firm's liabilities to its assets, using equity value and equity volatility as inputs. See Structural Model.

LEAPS

Acronym for Long-term Equity Anticipation Security. LEAPS are equity options with maturity dates of up to 36 months.

LEVERAGE

With respect to CDOs and other securitized assets, leverage refers to the size of the equity tranche relative to the total size of the structure.

LEVERED SUPER SENIOR

A strategy that gives exposure to super senior tranches through a non-recourse leveraging mechanism, often in unfunded form.

LIBOR

An acronym for London InterBank Offered Rate. LIBOR is the interest rate at which banks borrow funds from other banks and is commonly used as a benchmark for short-term interest rates.

MAKE-WHOLE PREMIUMS

When a security or debt tranche is called by either the issuer or the equity tranche holders, a premium payment (in addition to the par payment) may be required to compensate the debt holders for the early retirement of debt.

MANAGED

When the reference portfolio may be traded by the collateral manager according to designated guidelines and restrictions.

MANAGEMENT OPTION

The option held by management as to when and how to change a firm's capital structure. The probability of this option being exercised is typically derived subjectively, via fundamental analysis.

MARKET VALUE ASSET SWAP

An asset swap converting fixed cash flows to floating, with the original notional based on the original market value of the bond (not trading at par) and amortizing/accreting to par at maturity.

MARKET VALUE CDO

Cash CDO in which the principal and interest of the structure are paid using the proceeds generated by the sale and trading of the underlying collateral.

MARKET VALUE METHODOLOGY

Values a CDO by equating the market value of the assets (the underlying collateral) to the market value of the liabilities (comprised of the debt equity and management fees).

MARKET-IMPLIED DEFAULT RATE

The likelihood that an issuer will default, as implied by the spread of the issuer. Can be approximated by dividing the spread by the expected loss (par less recovery value).

MARKET-IMPLIED RECOVERY RATE

The expected value of the deliverable obligation (either market-value or its claim on the firm's assets), as implied by the spread of the issuer.

MARK-TO-MARKET

The current market value of a security or a position that reflects any unrealized gains or losses since inception.

MERTON-BASED MODEL

A structural model premised on the concept that a firm's equity is synonymous to a call option on the residual value of a firm's assets, once all liabilities have paid off. From the value of this call option, we can calculate the firm's distance to default, which reflects the likelihood of the firm defaulting.

MEZZANINE TRANCHE

A Junior Tranche that has some subordination.

M-GAMMA

Sensitivity of a default-neutral position to parallel shifts in spreads of underlying credits.

MM CLO

CLOs that invest primarily in middle-market loans.

MODIFIED MODIFIED RESTRUCTURING (MOD-MOD-R)

Under this definition, the main difference from Mod-R is that the protection buyer can deliver a deliverable obligation with maturity up to 60 months after restructuring (in the case of the restructured bond or loan) and 30 months in the case of all other deliverable obligations.

MODIFIED RESTRUCTURING (MOD-R)

Prior to the SNAC process in 2009, most North American IG CDS contracts had restructuring as a credit event. In the case of a restructuring credit event, the protection buyer must deliver obligations with a maturity date prior to a) 30 months following the restructuring and b) the latest final maturity date of any restructured bond or loan, but not shorter than the CDS contract.

MONTE CARLO SIMULATION

Technique that uses randomly generated values as inputs for given variables to construct a probability distribution of the resulting outcomes.

MULTISECTOR CDO

CDO whose underlying collateral pool references a combination of bonds and loans from different credit market sectors (e.g., corporate credits, ABS, MBS, CMBS, Emerging Markets, etc.).

MULTIVARIATE GAUSSIAN COPULA

The correlated multivariate normal distribution used to generate a portfolio loss distribution given single-name loss distributions.

NEAR-TERM DEFAULT RISK

Risk that the underlying portfolio/tranche is expected experience defaults in the near future. Typical of equity tranches. Opposite of Back-End Default Risk.

NEGATIVE BASIS TRADE

The purchase of bonds and protection on the same issuer in order to isolate the negative basis that exists between both securities (when the CDS premium is tighter than the spread of the bond).

NET COUPON

The difference between the bond coupon received (adjusted for the interest rate hedge) less the CDS premium paid out, when an investor buys bonds and CDS simultaneously.

NO RESTRUCTURING (NO-R)

Provision within a CDS contract which does not consider restructuring to be a valid credit event. No-R protection typically trades cheaper than Mod-R protection.

NON-CALLABLE BOND

See Bullet Bond.

NOPS

Notice of physical settlement – a required condition to settlement in transactions where physical settlement is applicable. Typically a NOPS should be delivered within 30 days of a credit event.

NOTIONAL-WEIGHTED

When the size of the legs of a trade are determined by the notional amount of each leg.

NTH-TO-DEFAULT BASKET

Similar to a First-to-Default basket, it is a levered investment on the default risk of a reference portfolio except it is the Nth credit event that triggers the contract. May be funded or unfunded.

OBLIGATION ACCELERATION

Credit event whereby the default of the reference entity causes the reference obligation to be due and payable, in lieu of the reference obligation's original maturity date.

OPERATING COMPANY (OPCO)

A company that is majority-owned by another company (the holding company). OpCo debt is often considered to have a higher recovery value because it has a closer claim to operating assets, relative to holding company debt.

OPTION-ADJUSTED SPREAD (OAS)

The spread of a corporate security relative to Treasuries or Libor, adjusted for embedded options.

OUTER CDO

Refers to the CDO whose underlying portfolio references other CDO tranches in a CDO-Squared.

OUT-OF-THE-MONEY (OPTIONS)

For payer (payor) options, it refers to when the strike spread is greater than the market spread on an underlying reference entity. For receiver options, the strike spread is less than the market spread on the entity. In either case, the option holder will likely let the option expire.

OUT-OF-THE-MONEY (TRANCHES)

Refers to tranches where the likelihood of default losses penetrating the tranche notional is low, as the expected loss on the index is below the attachment point of the tranche.

OVERCOLLATERALIZATION TEST

Valuation metric comparing the value of a CDO's underlying collateral to the structure's liabilities. A ratio greater than 1 represents an over-collateralization of the CDO's liabilities. See Par Coverage Test.

OVERLAP RATIO

Ratio of bonds overlapping between a CDO's collateral portfolio and a standardized synthetic index's reference portfolio, assuming the CDO investor uses that synthetic index as a hedge.

OVER-THE-COUNTER (OTC)

The market for securities that are not listed on one of the major exchanges.

PAIR TRADE

A combination of a long protection position and a short protection position. If implemented on individual credits, this trade may mitigate market risk and isolate credit risk.

PAR ASSET SWAP

An asset swap converting fixed cash flows to floating, with the notional of the swap based on par value.

PAR COVERAGE TEST

An overcollateralization test used to measure the overcollateralization of senior and junior tranches. To calculate, divide the sum of the par value of performing assets and the expected recovery value of defaulted assets by the par amount of the tranche plus the par amount of any tranche senior to it.

PAR SPREAD

The periodic, typically quarterly, premium that the protection buyer pays to the protection seller on a CDS contract so that the contract has a zero market value at inception.

PARI PASSU

Latin term meaning at an equal pace or without partiality. In the event of a liquidation, creditors that rank pari passu would have equal entitlement to the assets and hence would be paid pro rata in accordance with the amount of their claim.

PARTICIPATION RATE

In case of a CMCDs, the proportion of the current reference CDS index premium that a CMCDs protection buyer pays to the protection seller is referred to as the participation rate. The participation rate is fixed for the term of the CMCDs.

PAY-AS-YOU-GO

In the context of structured finance CDS, the settlement mechanism that replicates the exact economics of the underlying cash instrument. The buyer and the seller of CDS make "floating payments" to account for features unique to the structured finance instruments, such as principal writedowns, interest caps (AFC risk), and payment-in-kind (PIK) option.

PAYER (PAYOR) OPTION

The option to buy protection at a preset strike on an underlying reference entity. Also known as buyer options or puts.

PERFORMANCE ASYMMETRY

When the risk/reward ratio of an asset is skewed, either towards the upside or downside.

PHYSICAL SETTLEMENT

The settlement of a CDS contract whereby the protection buyer will deliver any deliverable debt obligation to the protection seller upon the occurrence of a credit event. After SNAC this is now non-standard.

PREMIUM PV

The present value of all expected premiums to the maturity of the CDS contract.

PREPAYMENT

When an issuer partially or completely pays down the debt ahead of the scheduled maturity date, thus shortening the weighted average maturity of the debt.

PROBABILITY-WEIGHTED BREAK-EVEN RECOVERY VALUE

The expected recovery value in the event of default, weighted using scenario analysis.

PRO-RATA PAYDOWN

A structural feature (found most commonly in high-grade ABS CDOs) that allows all classes of debt in the capital structure to be paid down in proportion to their outstanding balances as long as there has not been any breach of a coverage test and a certain proportion of the collateral balance is still outstanding.

PROTECTION BUYER

The counterparty of the CDS contract that pays premiums to the seller in exchange for protection against a credit event by the issuer. The buyer will either deliver a deliverable bond or make a payment equal to the market value of the bond to the seller and will receive par in exchange in case of a credit event.

PROTECTION SELLER

The counterparty of the CDS contract that receives premiums in exchange for guaranteeing the payment of par to the buyer in the event of default by the issuer. The seller will receive the deliverable bond or a payment equivalent to market value of the bond from the buyer.

PULL-TO-PAR

For a bond trading at either a discount or a premium, it is the tendency for the bond's price to converge to par as it approaches maturity.

PV01

The change in the price of a tranche given a 1 bp change in each of the underlying credits in the portfolio.

RAMP-UP PERIOD

Period of time where the CDO collateral manager invests in assets using the proceeds from the issuance of debt tranches from the CDO.

RATED EQUITY

A structure where excess spread from standard tranches is used to build subordination and may earn investment grade ratings in some cases.

RATINGS MIGRATION

The shift of a security or a group of securities from one rating class to another.

RAW BASIS

The difference between the 5-year CDS and the Z-spread of the bond.

RECEIVER OPTION

The option to sell protection at a preset strike on an underlying reference entity. Also known as seller options or calls.

RECOVERY LOCK

A form of recovery swap that isolates recovery risk by pairing a standard (floating recovery) default swap with a fixed recovery default swap. The recovery lock is quoted at a market-implied recovery value, i.e., the fixed recovery rate that is required to make the premiums of both legs the same.

RECOVERY RATE

The value of the deliverable obligation received by the protection seller when a credit event occurs, calculated as a percentage of par.

RECOVERY SWAP

A contract between two parties where the fixed recovery payer agrees to receive the difference between the predetermined recovery rate and the actual recovery rate on the reference obligation in case of a default.

REDUCED FORM MODEL

A credit valuation tool that models a firm's forward probability of default (hazard rate) over any time horizon, typically calibrated to a term structure of credit spreads.

REFERENCE OBLIGATION

The bond or loan specified in a CDS contract, used to determine other deliverable obligations if the reference entity defaults.

REINVESTMENT PERIOD

Length of time over which the CDO manager may reinvest the proceeds of matured or traded securities. When this period ends, the underlying portfolio will become a static one.

REPLACEMENT LANGUAGE

Provision within a Nth-to-default basket where the buyer of protection has the right to replace a reference entity that merges with another entity in the basket. Replacement entities must typically meet certain guidelines, such as sector and spread.

REPUDIATION/MORATORIUM

Credit event typically found in CDS contracts referencing sovereigns, whereby the government challenges the validity of one or more of its obligations or temporarily stops making payments on the reference obligation.

RE-RATING METHODOLOGY

Values a CDO by inferring a rating for a CDO tranche by calculating its expected loss and considering its risk characteristics. The tranche is then compared to similar structures in the market to determine a comparable spread.

REVOLVER/REVOLVING CREDIT FACILITY

The lender commits to make loans to a borrower up to a specified amount for a specified period. The borrower can draw down and repay at its discretion during this time. Revolvers are generally unfunded and mainly used by investment grade borrowers.

RHO

Change in tranche value due to changes in correlation.

RISING STAR

A bond that was originally issued a below investment-grade rating, but has since been upgraded to investment-grade status due to improving credit quality. Opposite of a Fallen Angel.

RISK-NEUTRAL DEFAULT PROBABILITY

Probability of default implied by CDS pricing for a given time period. Said differently, risk neutral default probability results in expected losses that match the present value of CDS premium. See Market-Implied Default Probability.

ROLL DOWN

The return generated solely due to the passage of time for a seller of protection on an upward-sloping curve, assuming no change in the curve. The steeper the curve, the higher the roll down return will be.

RUNNING PREMIUM

The spread paid periodically (typically quarterly) by protection buyers in addition to points upfront. Commonly used in default swaps on issuers with high default probabilities.

SAMPLING RISK

The risk that a sample does not reflect the true characteristics of the population. For example, a credit portfolio, while having the same expected losses as the entire universe of credits, might have a more fat-tailed distribution.

SELF-MANAGED SYNTHETICS

A structure that allows investors to take exposure in tranche form to a portfolio that he or she manages. Typically, no other exposure in this portfolio would be distributed by the dealer.

SENIOR DEBT

When the debt holder has a senior claim (relative to subordinate claims) on the firm's assets, in the event of default. Can be secured or unsecured.

SENIOR TRANCHE

Tranche bearing the least credit risk in the structure as it has a priority claim on all cash flows generated by the collateral pool. All other tranches are subordinated and their principal remains outstanding until the senior tranche is fully paid off. Generally carries the highest rating in the structure. In synthetic indices, senior tranche investors are typically short correlation.

SHARPE RATIO

Excess return on an investment (i.e., return over the risk-free rate) per unit of risk (as measured by standard deviation of the returns).

SHORT SQUEEZE

A situation where short sellers rush to cover their short positions by buying back the asset. The excess demand drives the price of the asset higher, making it more costly for other short sellers to close out their positions.

SINGLE TRANCHE SYNTHETIC CDO

Form of arbitrage CDO that allows the investor to customize the reference pool of assets by selecting the credits, as well as the attachment and detachment points.

SLR

Scenario loss rates, used by S&P to arrive at tranche ratings, are a quantile (points taken at regular intervals from a cumulative distribution function) of the portfolio loss distribution consistent with a given rating and maturity.

SPECIAL PURPOSE VEHICLE (SPV)

A stand-alone, bankruptcy-remote entity whose purpose is limited to the acquisition and financing of specific assets. Cash and synthetic CDOs, as well as other securitized asset pools, are typically issued from an SPV.

SPREAD DURATION

Sensitivity of the price of a corporate security to changes in the underlying credit spread.

SPREAD DV01

The change in value of a corporate security for a 1 basis point change in its spread.

SPREAD PER UNIT LEVERAGE (SPL)

A credit's spread divided by its leverage (Debt/LTM EBITDA). A simple measure of risk compensation in corporate spread products.

STATIC

When the reference portfolio is fixed at inception.

STEP-UP SUBORDINATION

Step-up subordination has a fixed notional, but its attachment and detachment points increase over time. So instead of leaving subordination fixed throughout the trade, a step-up tranche will increase subordination over time, more closely resembling the path of expected losses in a portfolio.

STRADDLE

A combination of options to buy and sell protection at the same strike spread. A long straddle is a view on rising volatility or wide moves in spread.

STRANGLE

This option combination is similar to straddle, but involves out-of-the-money options to buy and sell protection. A long strangle benefits from wide moves in spread, while a short strangle benefits if spreads stay within a narrow range.

STRATEGIC DEBT SERVICE MODEL

A modified structural model that incorporates the equity holders' option to voluntarily default and renegotiate the terms of debt with debt holders to their benefit, if the costs of firm liquidation are high.

STRUCTURAL DELEVERING

Refers to the decrease in overall leverage when two entities in a Nth-to-Default basket (without replacement language) merge.

STRUCTURAL MODEL

A credit valuation model premised on the concept of default occurring when a corporation's assets fall below its liabilities. Can be used to infer default probabilities and fair market spreads. See Merton-based Model.

SUBORDINATE DEBT

When the debt holders' claim on a firm's assets in the event of default ranks below the senior claims. Can be secured or unsecured.

SUCCESSION EVENT

According to 2003 ISDA Credit Derivatives Definitions, a succession event means an event in which one entity succeeds to the obligations of another entity, such as a merger, consolidation, transfer of assets or liabilities, spin-off or a similar event. However, if the exchange of obligations does not occur in connection with mergers and acquisitions activity as outlined above, that exchange does not represent a succession event.

SUCCESSOR

The new legal entity that a CDS references in case of owner changes for the original reference entity of the CDS.

SUM OF SPREADS

The sum of the premiums on the underlying credits in a basket. A buyer of protection on an Nth-to-Default basket will pay a percentage of that sum of spreads, based on the default correlation assumption.

SWAPTION

An option to enter into a swap. See Payer Option and Receiver Option.

SYNTHETIC CDO

A pool of credit default swaps that is tranching, creating synthetic exposure to multiple reference entities. Effectively, a CDO investor acts as a seller of protection to one or more counterparties.

SYNTHETIC RATED OVERCOLLATERALIZATION (SROC)

A measure of the sensitivity of a tranche in a synthetic CDO to future rating action, expressed as a ratio. 100% SROC implies an exactly sufficient credit enhancement to maintain rating on a tranche. Very loosely, SROC can be defined as $(1 - \text{Scenario Credit Loss}) / (1 - \text{Credit Enhancement})$.

SYSTEMIC RISK

The risk inherent in the entire market. Also known as market risk, it cannot be diversified away. Opposite of Idiosyncratic Risk.

TARGET LEVERAGE MODEL

A modified structural model that considers that a firm's capital structure can change over time, such that debt level changes in response to changes in the firm's asset value. Empirical studies show that a firm tends to issue more debt as asset values rise.

TERM LOAN

A term loan is a funded loan type wherein once borrowed funds are repaid, the funds cannot be borrowed again. Institutional term loans are usually taken out by leveraged borrowers (i.e., non-investment grade borrowers with debt/EBITDA greater than 2.0x) and repaid at maturity (there might be some minimal, back-ended principal repayments). They are normally longer dated compared to amortizing term loans (five to seven years) and may be prepaid at any time at par. Multiple tranches with varying maturities can co-exist within a facility.

THETA

Change in tranche value due to the passage of time.

TIME DECAY

Refer to Theta.

TRANCHE THICKNESS

The difference between the attachment and detachment point for a given tranche. Defines the maximum amount of losses that the tranche can experience.

TRANCHE

A portion of a securitized portfolio of assets. A group of assets are pooled together and then structured to create various securities (tranches) of different maturities and risks. In the credit derivatives market, index tranches are frequently traded. Losses are prioritized by the most-subordinated tranche up to the least-subordinated (most senior) tranche.

TREASURY SPREAD

The spread of a corporate bond relative to its underlying benchmark government bond, calculated by taking the yield to maturity of the corporate bond and subtracting the yield to maturity of the government bond.

TRUE FORWARD STARTING TRANCHES

Tranches where the notional amount of subordination is fixed until the forward starting date, independent of actual losses on the portfolio.

TRUSTEE

Party responsible for ensuring compliance with pre-designated trading and structural guidelines within a CDO structure. Periodically provides reports to investors detailing the status of the structure's assets and liabilities, as well as compliance updates.

UNFUNDED FORM

When no upfront payment is made by the seller of protection at the inception of the trade and no principal payment is paid at maturity. Payments between the buyer and seller of protection are comprised of premium payments and a termination payment should a credit event occur. Typical form of a plain vanilla credit default swap.

UPFRONT PREMIUM

The portion of the protection premium paid upfront, separate from the coupon. The present value of upfront and running payments is theoretically equal to the present value of par spread.

UPGRADE-DOWNGRADE RATIO

Compares the number of ratings upgrades versus the number of downgrades in a class of CDO tranches to determine the direction and magnitude of ratings migration.

UPWARD-SLOPING CURVE

When the short end of the curve is at a lower level than the long end of the curve, such that the curve has a positive slope. This shape reflects the increasing risk premium of the security over time.

VINTAGE

The year that the CDO was originally issued.

VIX

CBOE Volatility Index. A measure of equity volatility.

VOLATILITY SKEW

When options on the same underlying security trade at different implied volatilities. Types of skew can include horizontal skew (when near-term options trade at different implied volatilities than longer-dated options) and vertical skew (when options with different strikes trade at different implied volatilities).

VOLATILITY SMILE

A type of volatility skew where out-of-the-money options and in-the-money options trade at higher implied volatilities than at-the-money options, forming a "smile" shape.

WEIGHTED AVERAGE COST OF CAPITAL (WACC)

The weighted average expected cost of funding from all the sources of a firm's capital. Sources of capital typically include common equity, preferred equity and debt.

WEIGHTED AVERAGE LIFE (WAL)

The weighted average period of time over which the principal of a tranche is expected to remain outstanding since the time of issuance.

WEIGHTED AVERAGE MATURITY (WAM)

The weighted average period of time until the final principal payment of each tranche, weighted by the size of each tranche.

WEIGHTED AVERAGE RATING FACTOR (WARF)

A numerical metric indicating the credit quality of a portfolio (which can vary across ratings agencies). Typically, the higher the WARF, the lower the credit quality of the portfolio. Most CDO structures will designate a maximum WARF value as guidelines.

WEIGHTED AVERAGE SPREAD (WAS)

The weighted average spread of the underlying collateral/portfolio in a CDO structure.

YIELD-ON-YIELD SPREAD

See Interpolated Swap Spread (I-Spread).

YIELD-TO-WORST

The yield on a bond reflecting the most undesirable repayment schedule for a bondholder of a callable bond. Typically, will either be equal to the yield-to-call (to the earliest call date) if market yields are lower than the coupon rate, or equal to the yield-to-maturity if market yields are greater than the coupon rate (no prepayment).

ZERO COUPON EQUITY

See Equity PO.

Z-SPREAD

A constant spread over the Libor zero curve that equates the present value of a bond's cash flows to its market price.

Z-SPREAD-TO-WORST

A constant spread over the Libor zero curve that equates the present value of a bond's cash flows to its market price, given the most undesirable repayment schedule from a yield perspective for a callable bond. Similar concept to Yield-to-Worst.

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)		
	Count	% of Total	Count	% of Total IBC	% of Rating Category
Overweight/Buy	1120	38%	461	44%	41%
Equal-weight/Hold	1229	42%	449	42%	37%
Not-Rated/Hold	105	4%	24	2%	23%
Underweight/Sell	464	16%	124	12%	27%
Total	2,918		1058		

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